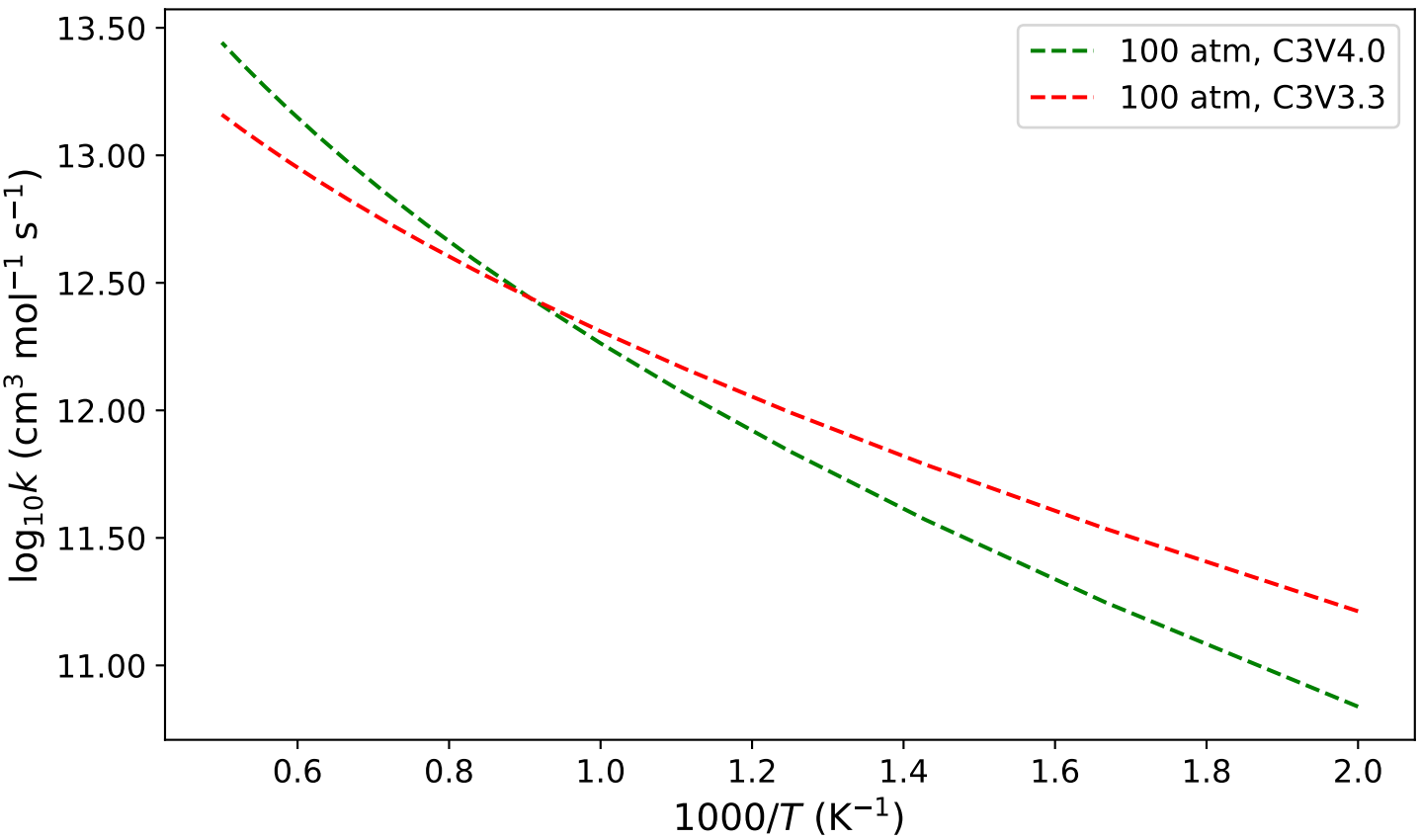
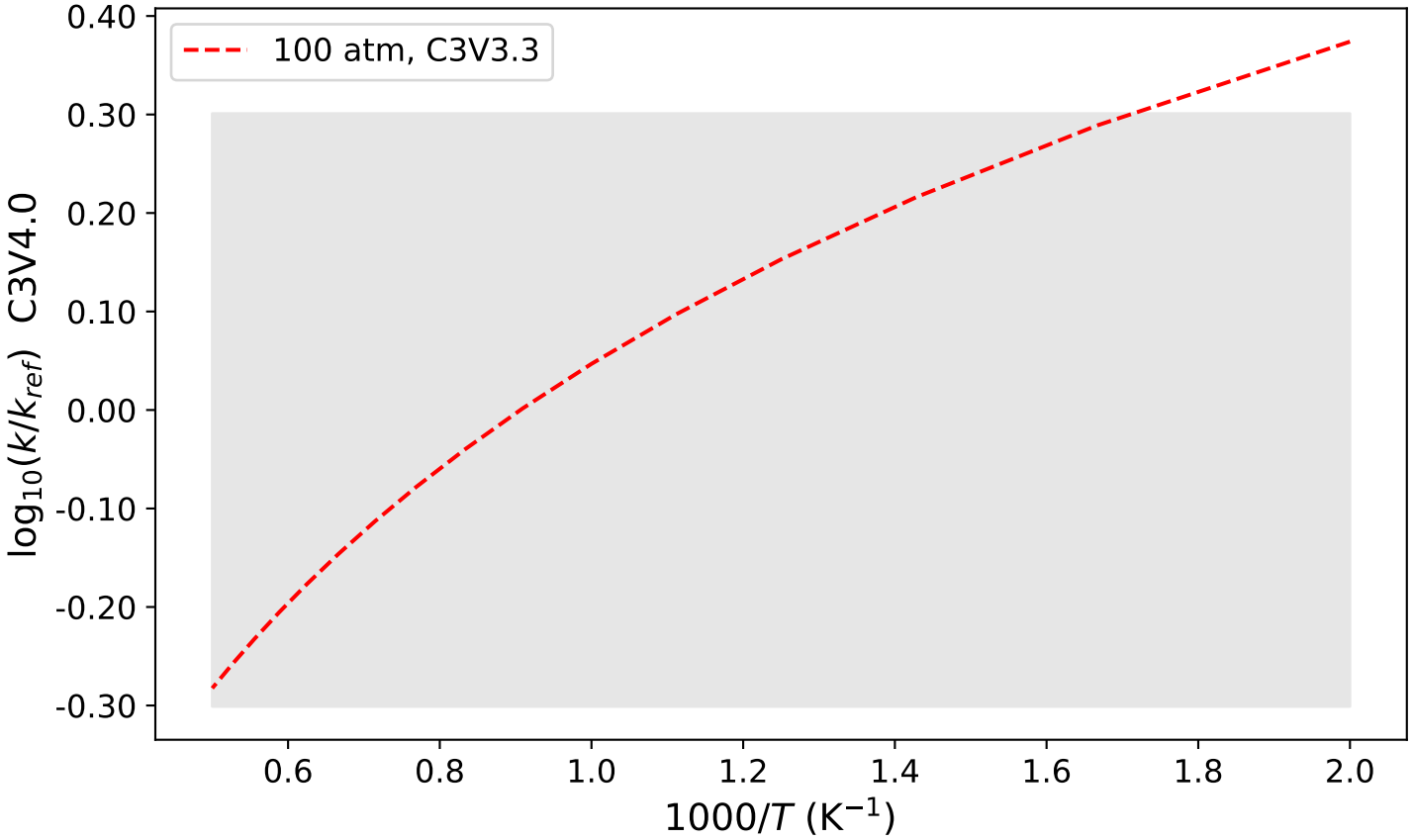
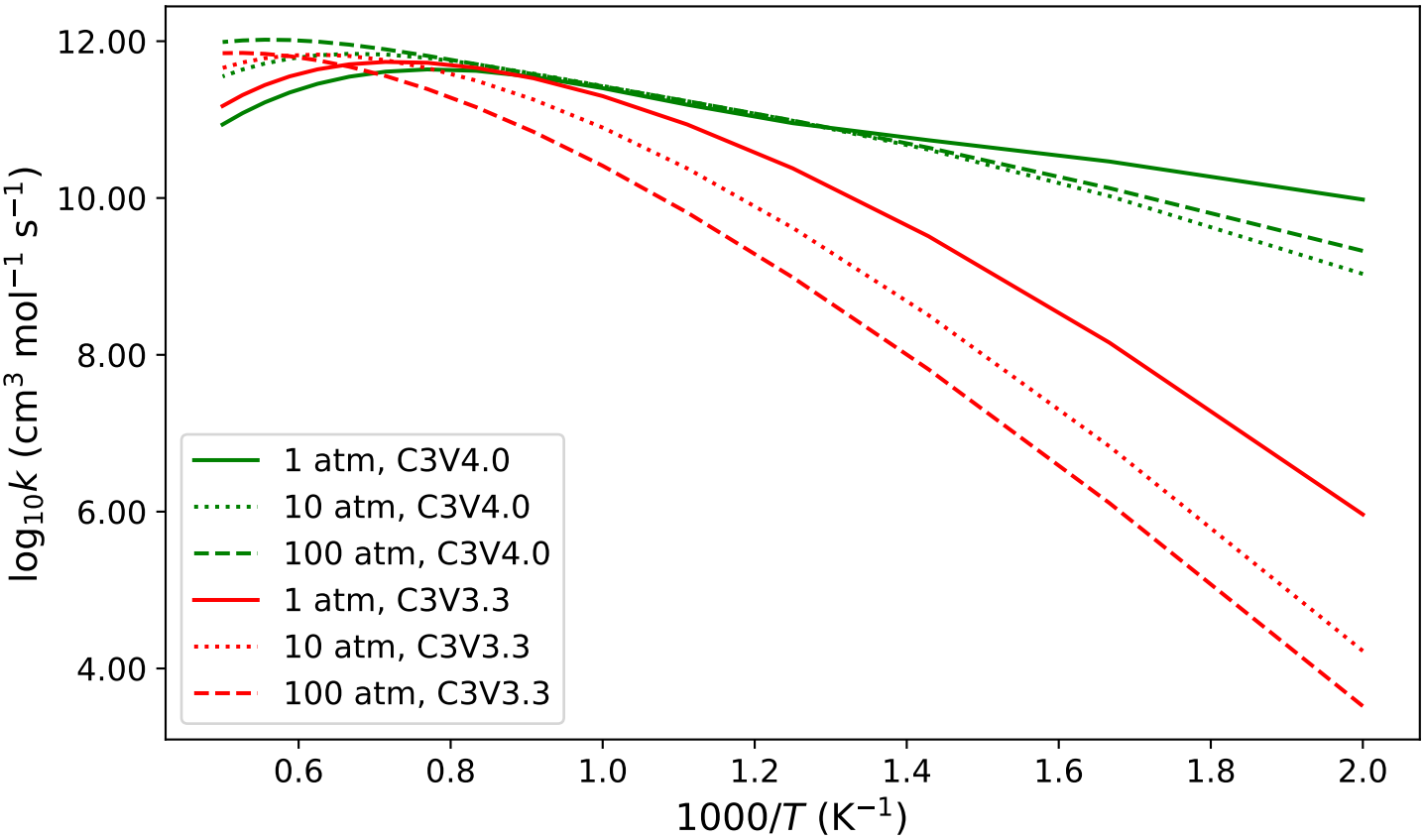
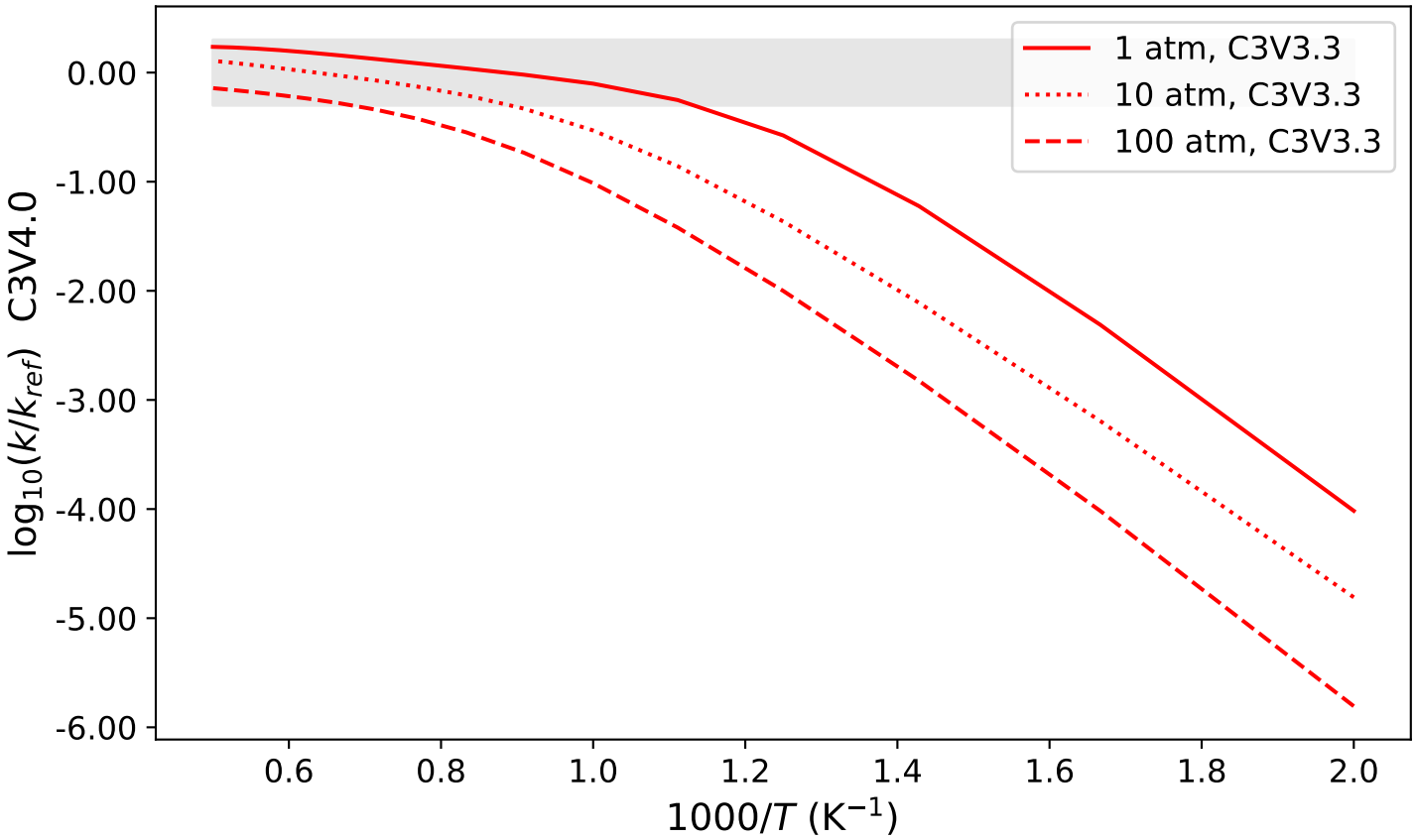
OH + C12H8 = H2O + C12H7



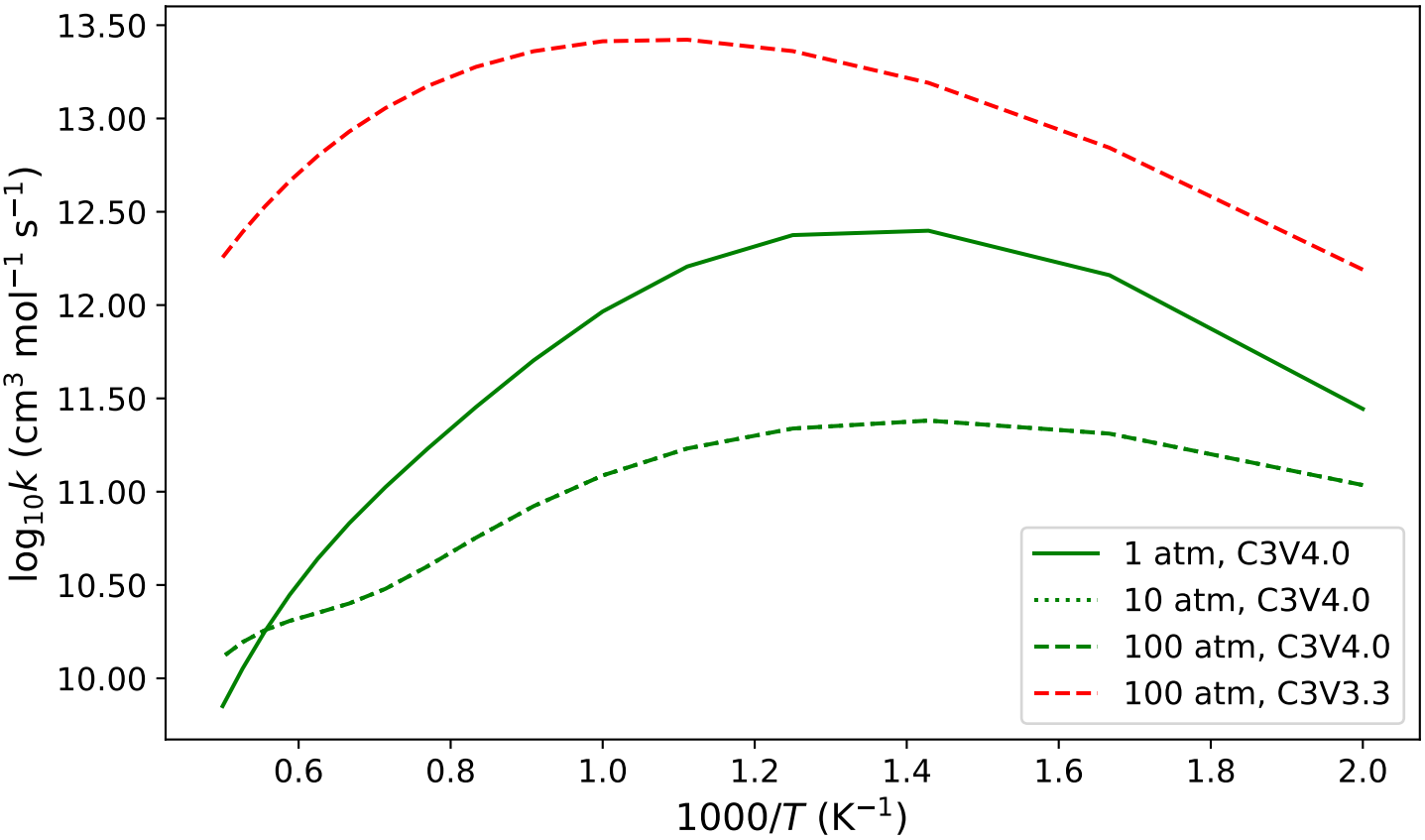


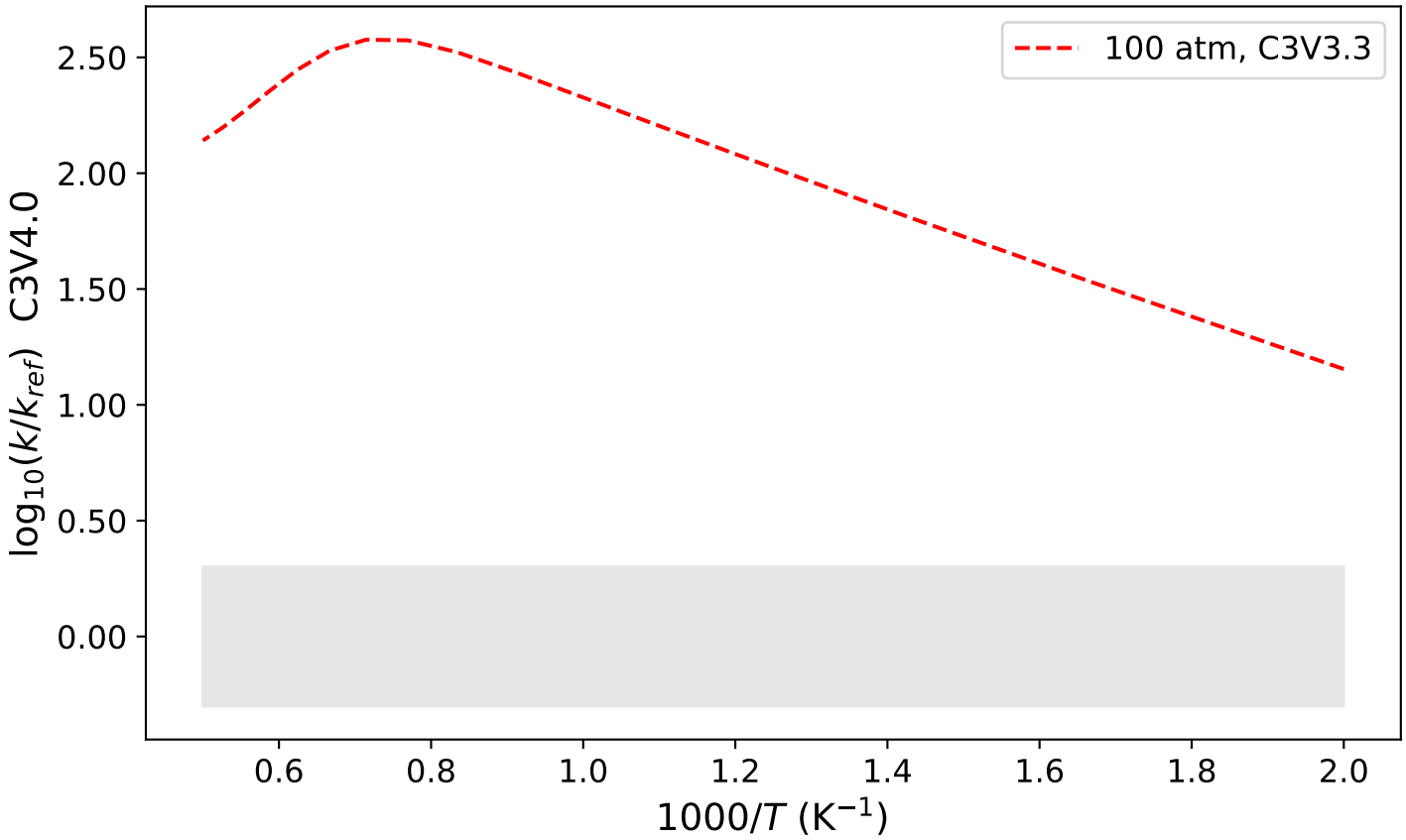
C2H2 + C6H4C2H = NAPH-A

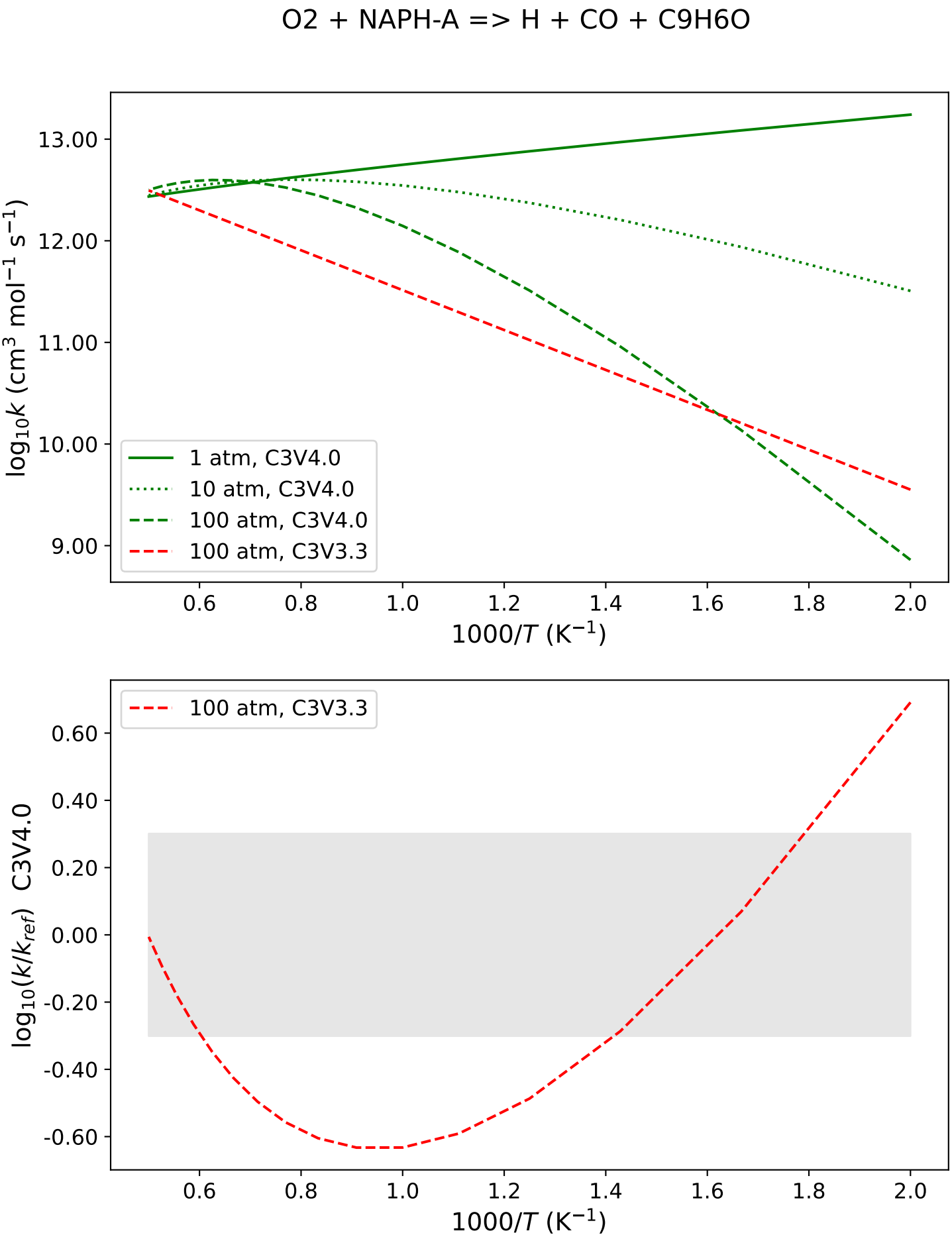




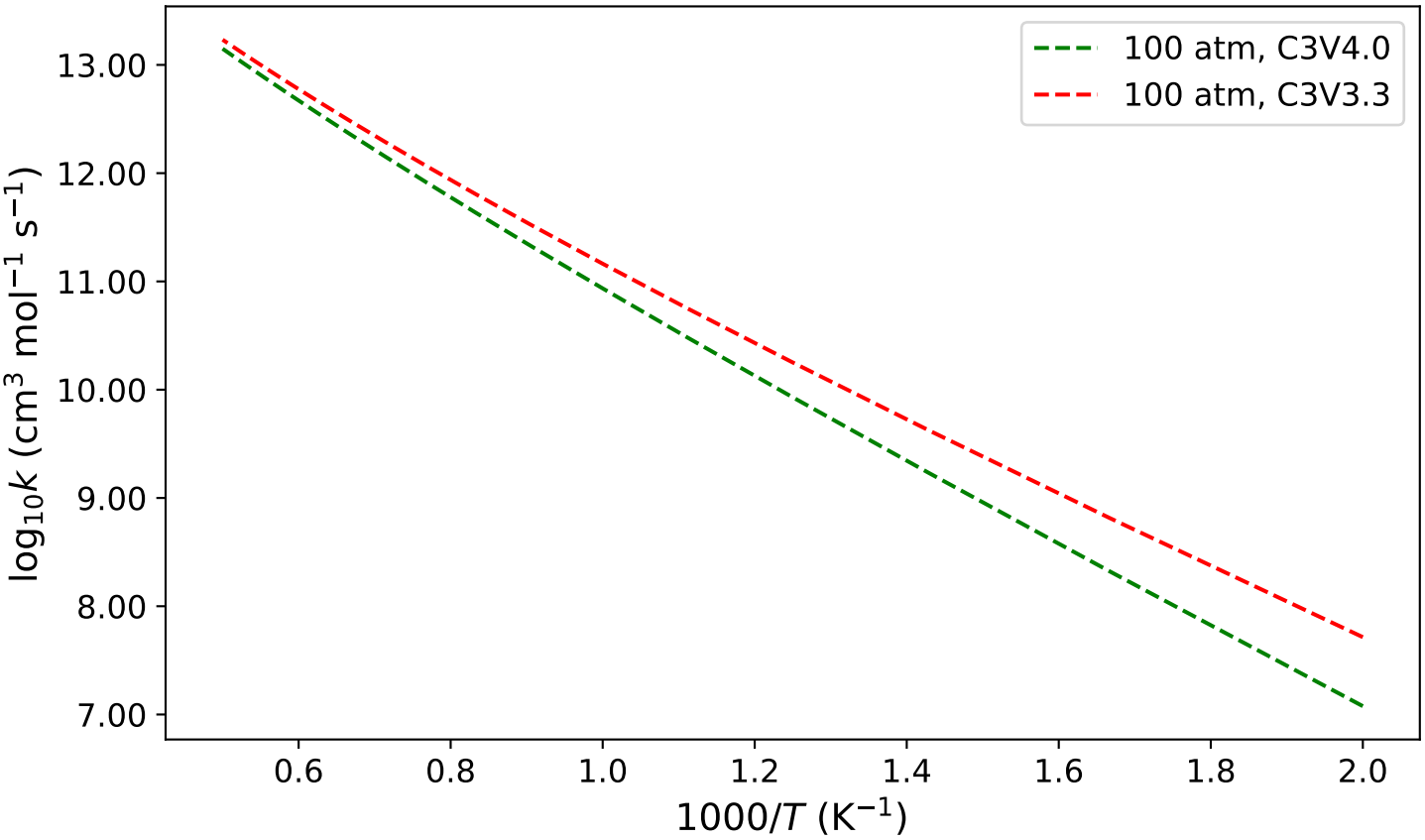
C3H3 + C7H5 = NAPH

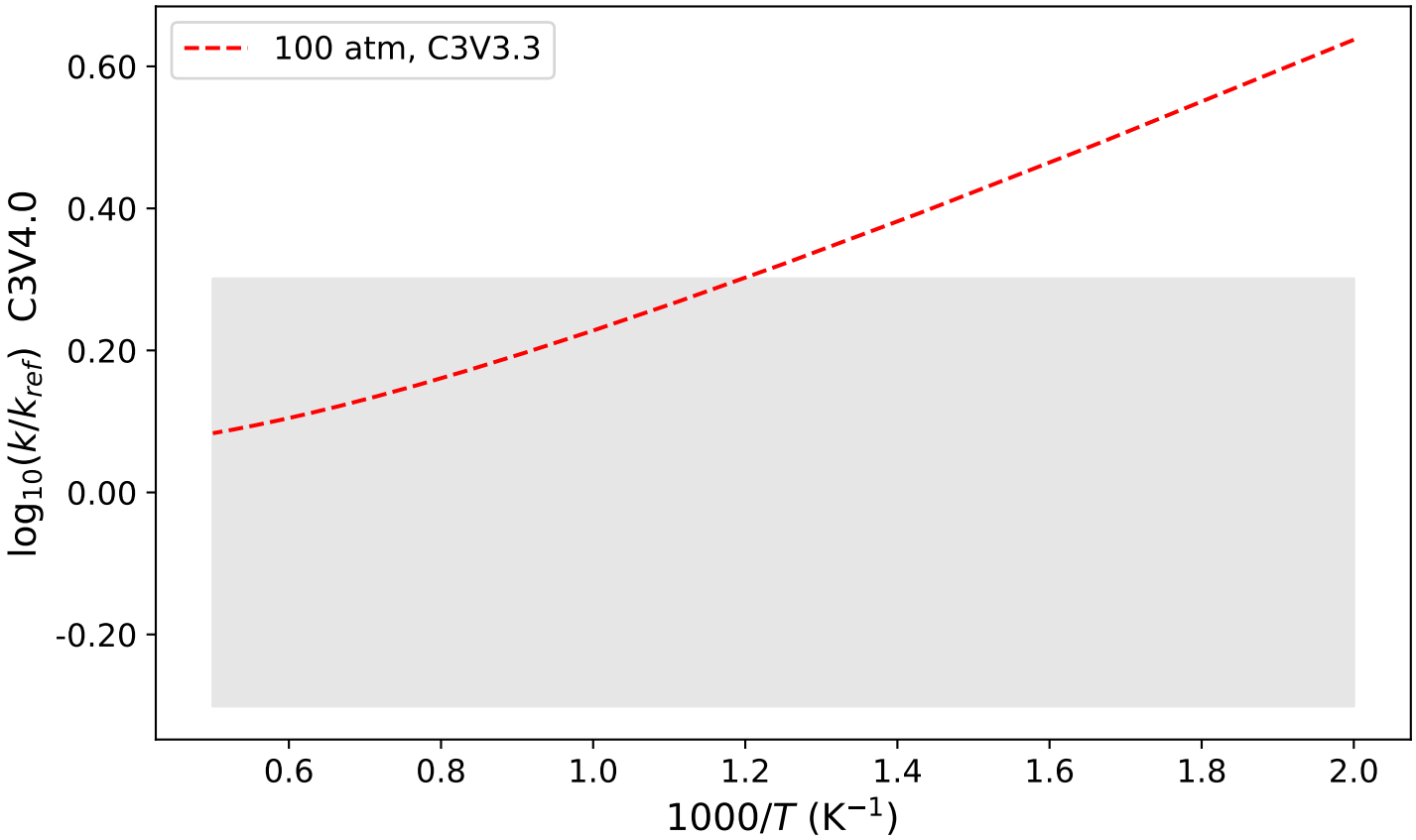




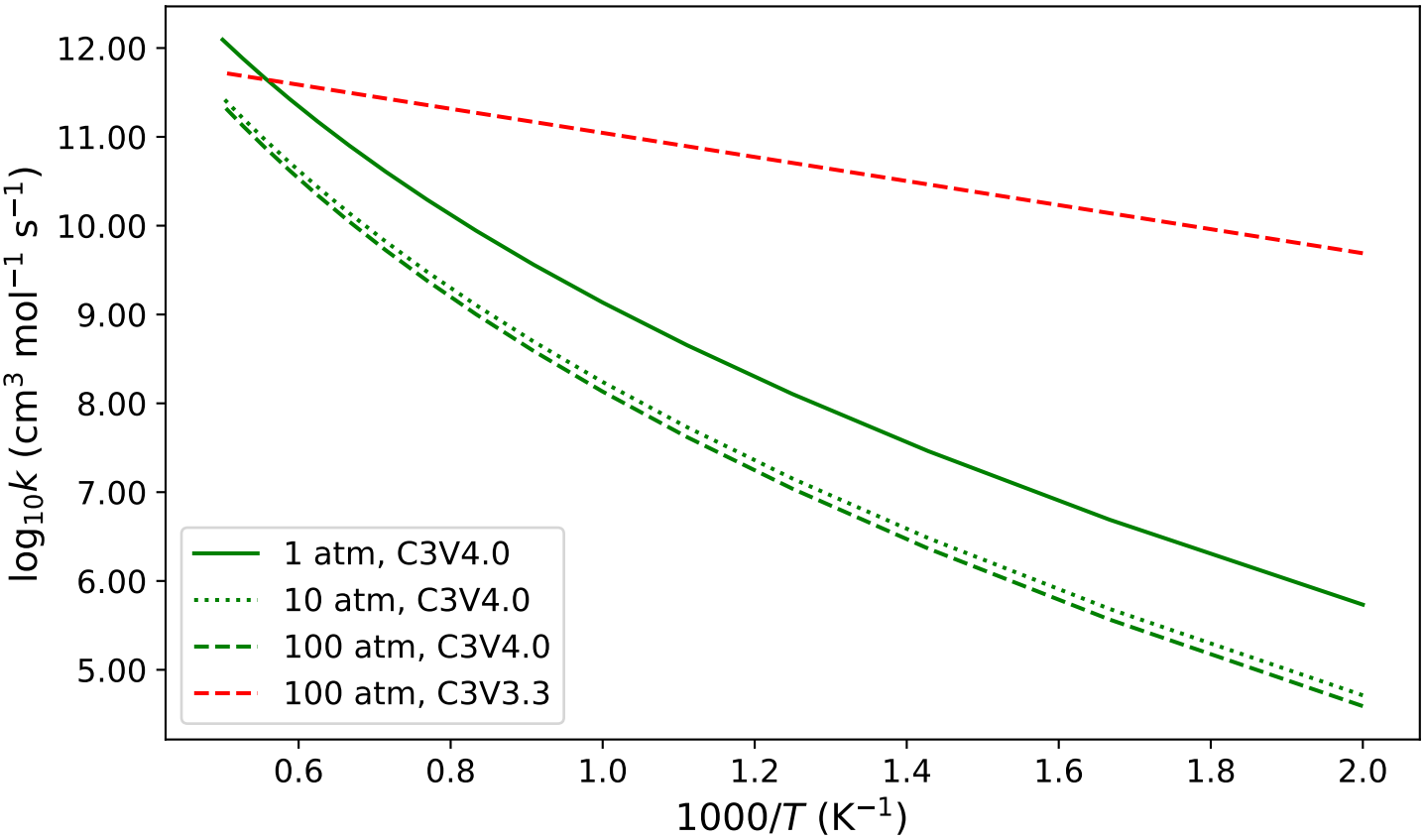

O2 + NAPH-A => H + CO + C9H6O
13.00
12.00
11.00
10.00
9.00
$\log_{10}k$ (cm^3 mol^{-1} s^{-1})
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K^{-1})
100 atm, C3V3.3
0.60
0.40
0.20
0.00
-0.20
-0.40
-0.60
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K^{-1})

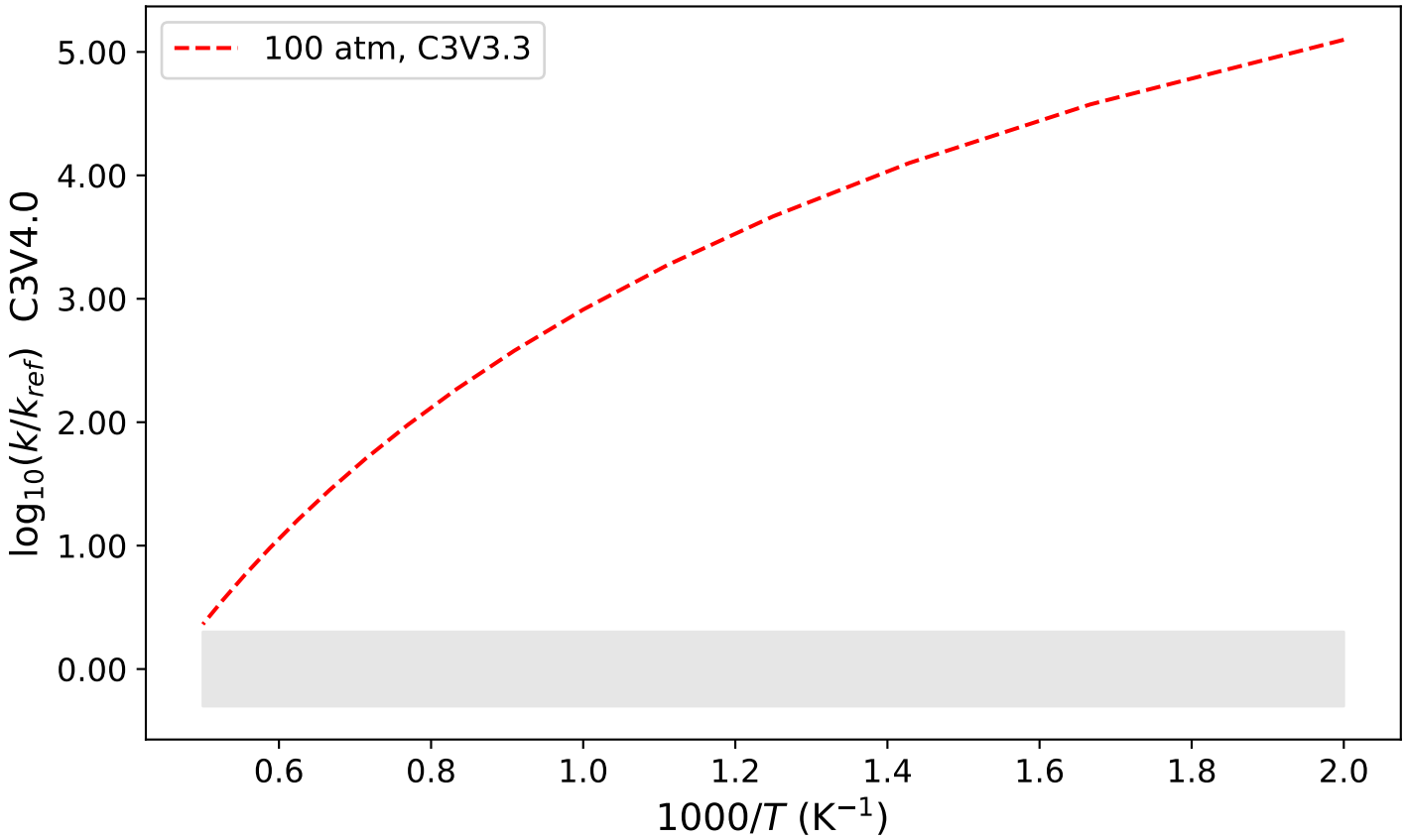
H + C6H6 = H2 + C6H5



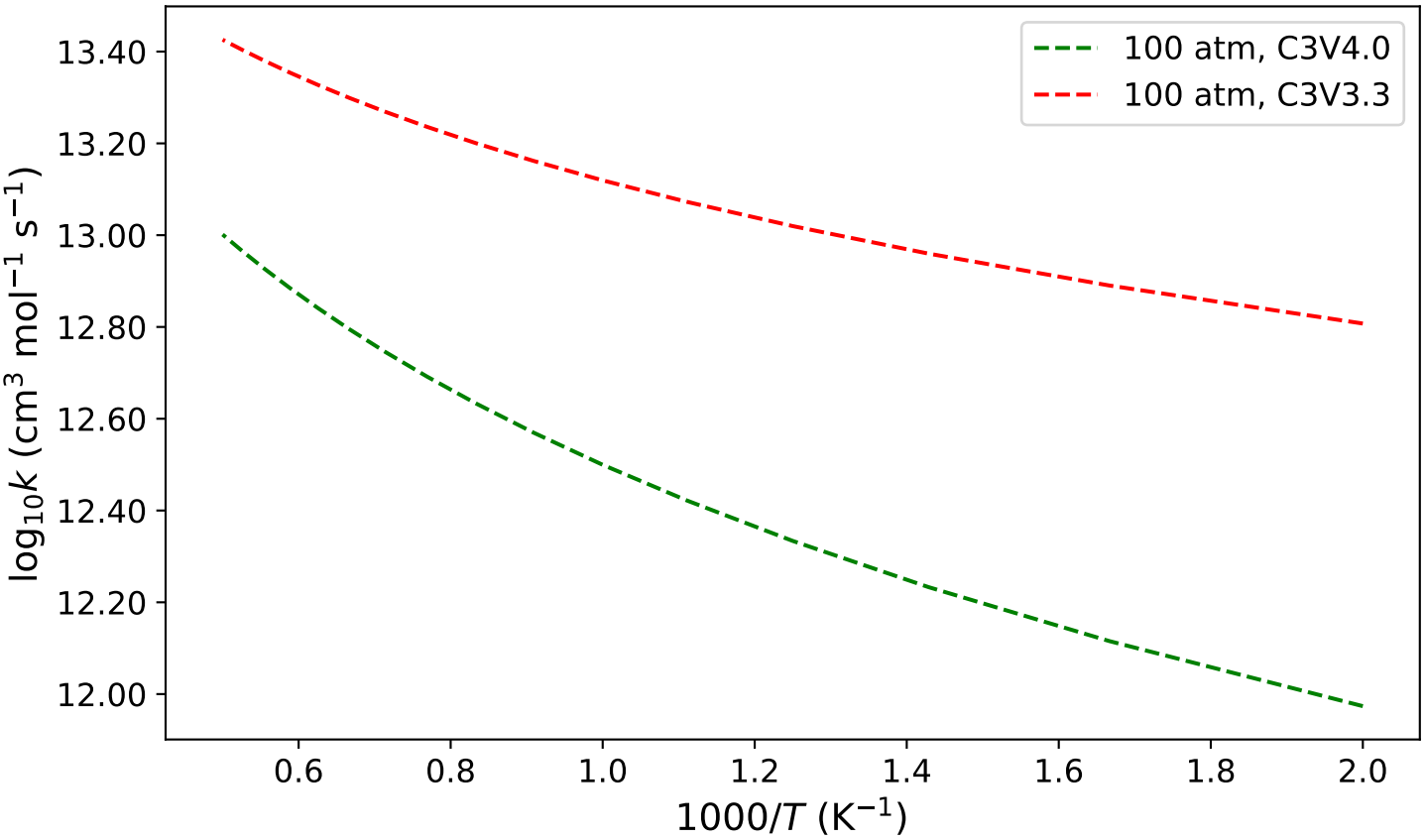


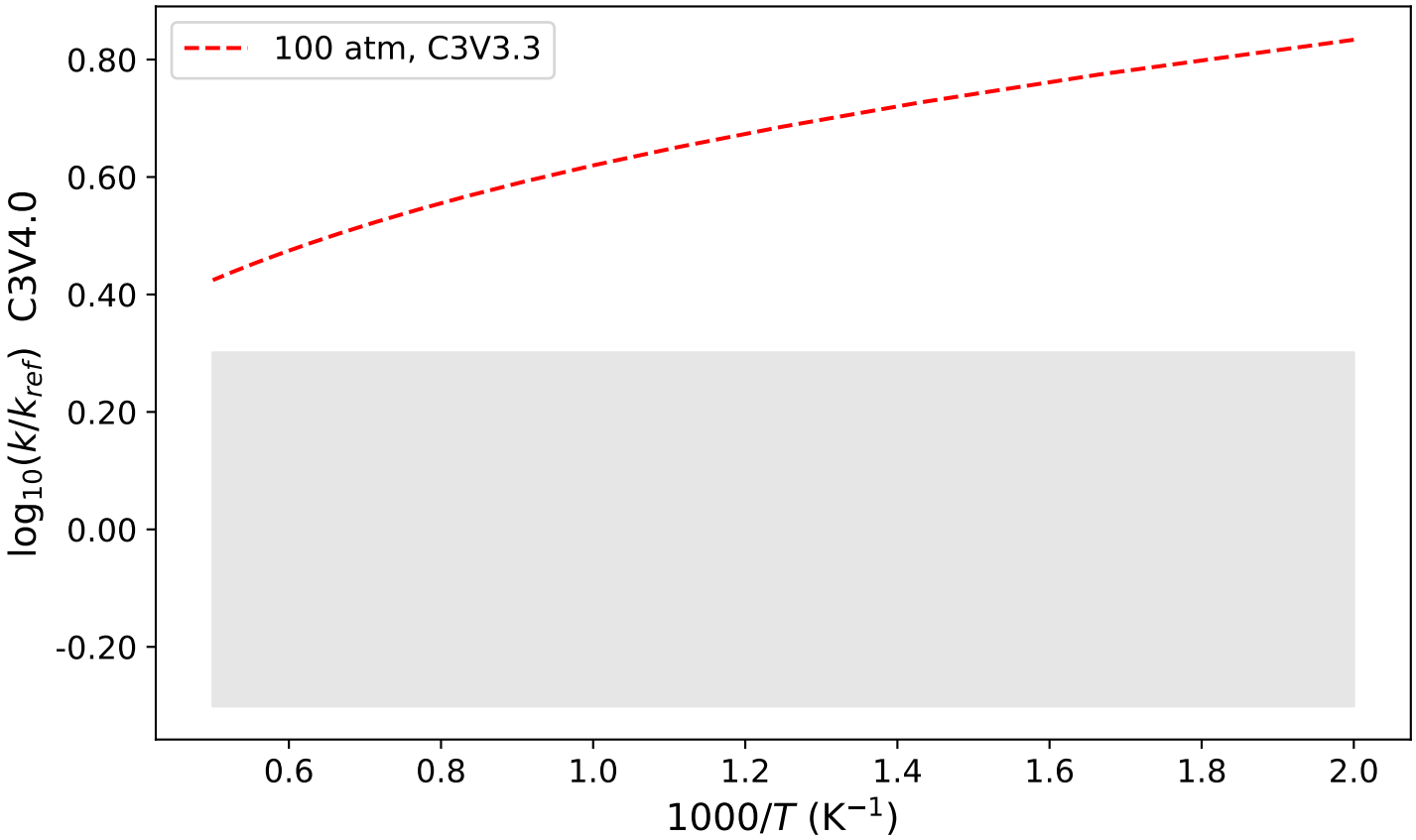
C6H5 + C2H4 = H + C6H5C2H3



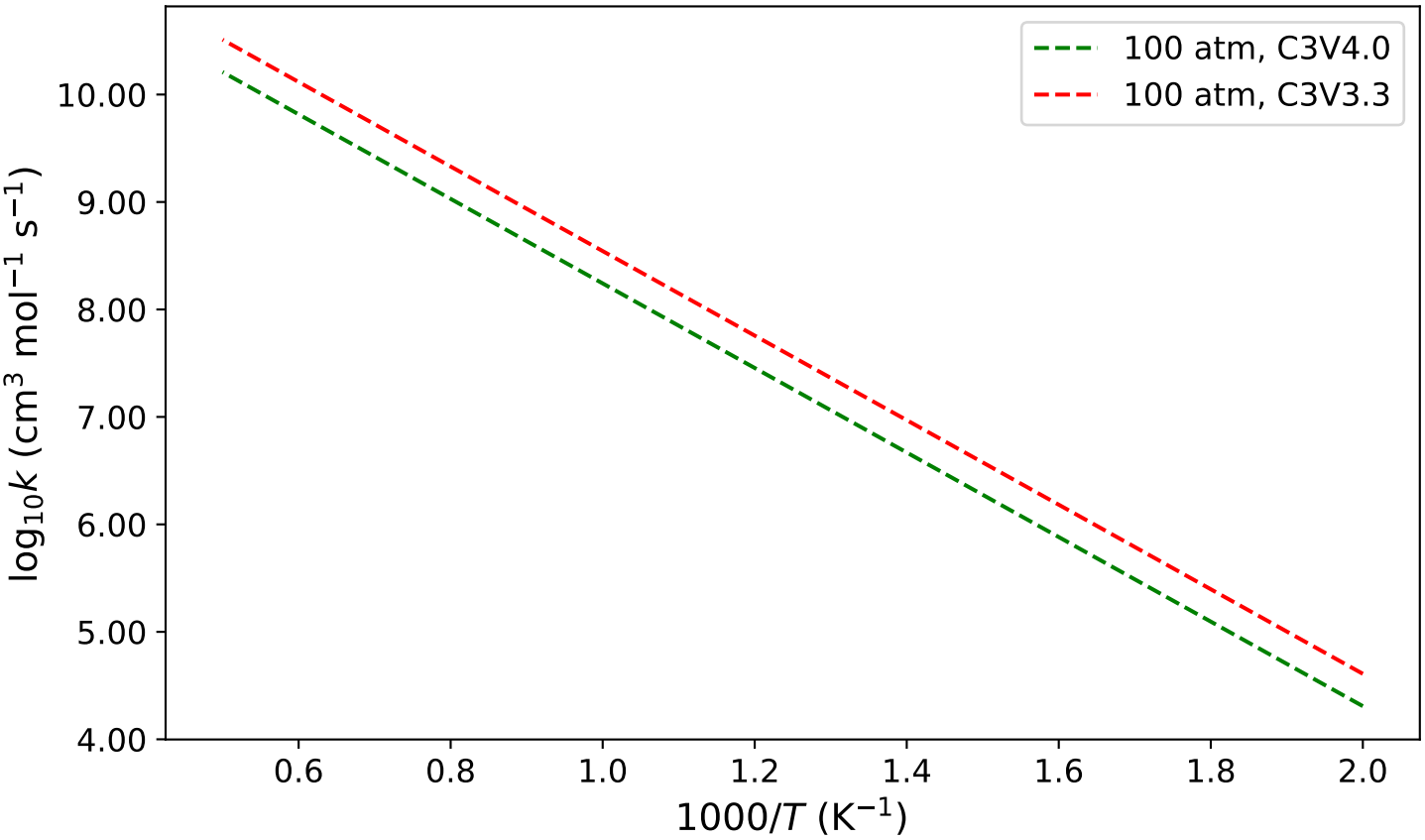


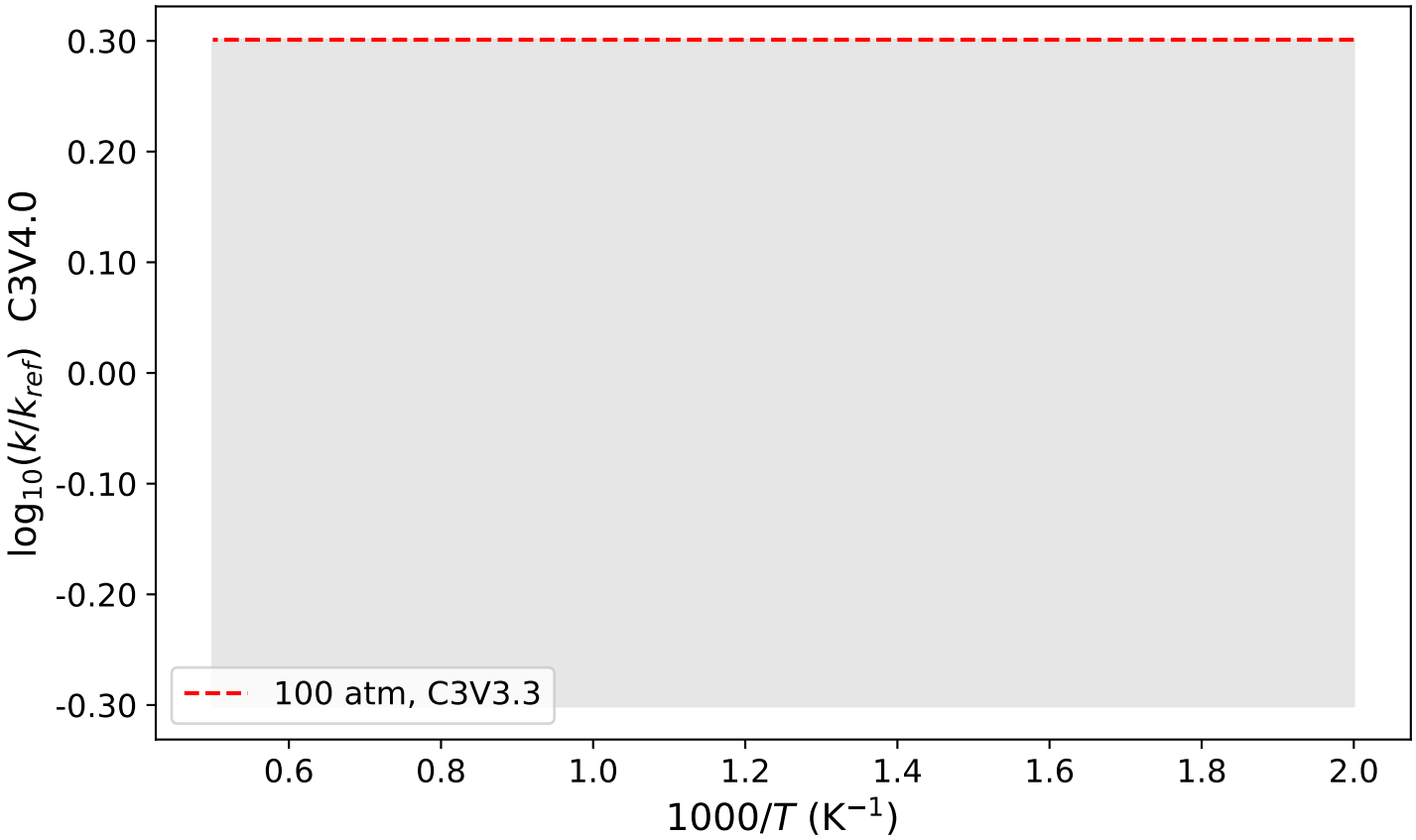
OH + C14H14 = H2O + C14H13



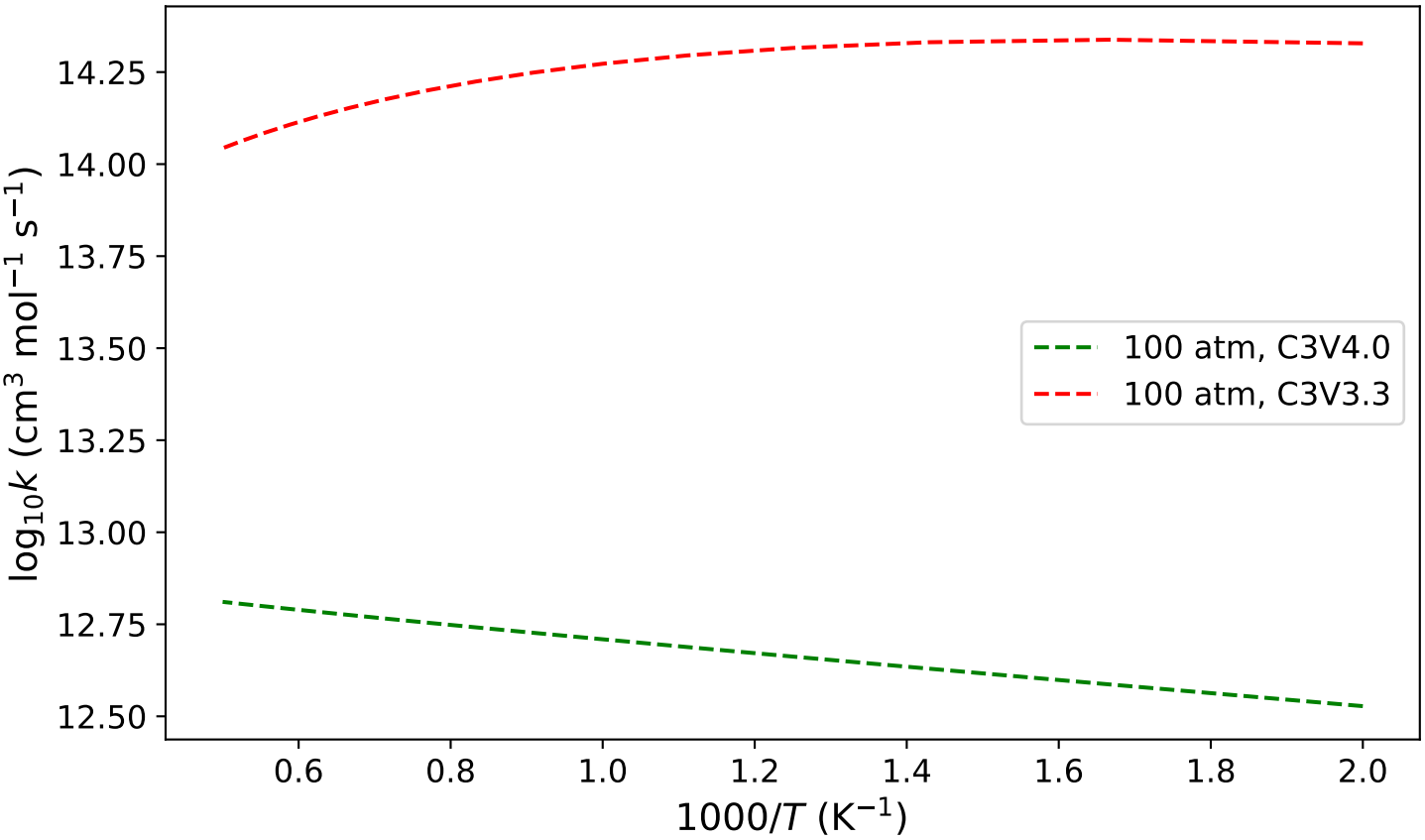


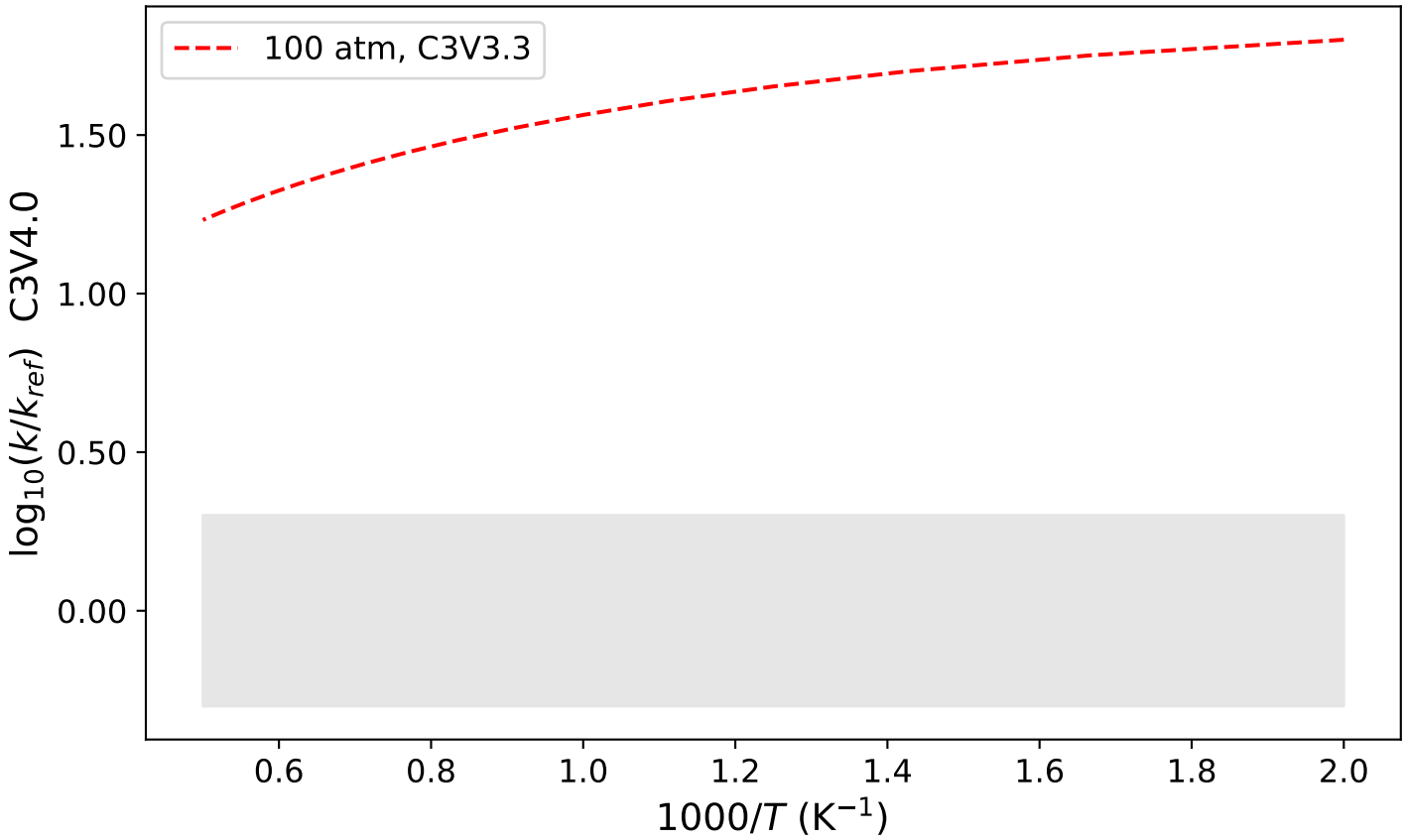
C3H4-P + C6H5CHCH3 => H + H2 + C10H7CH3-A



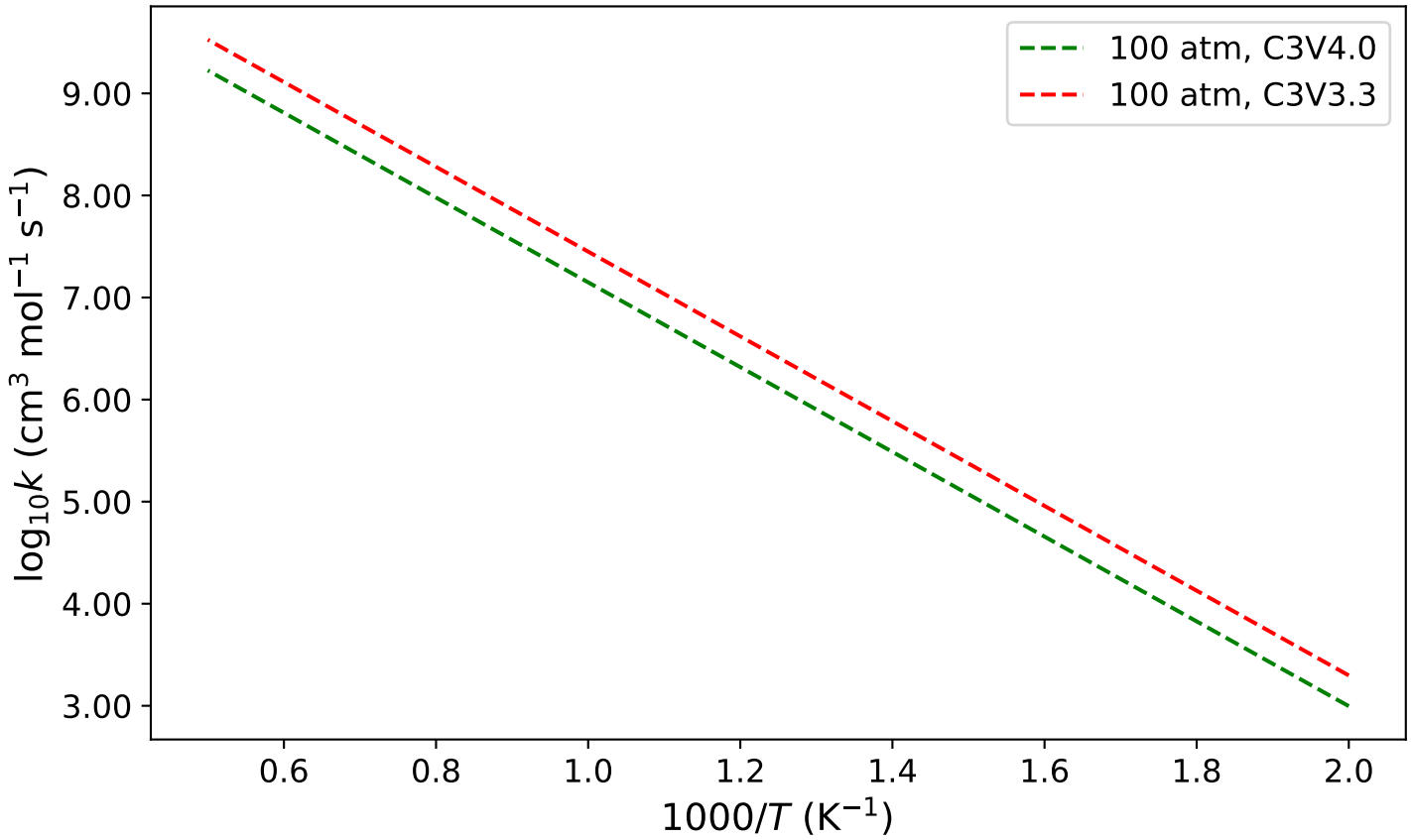


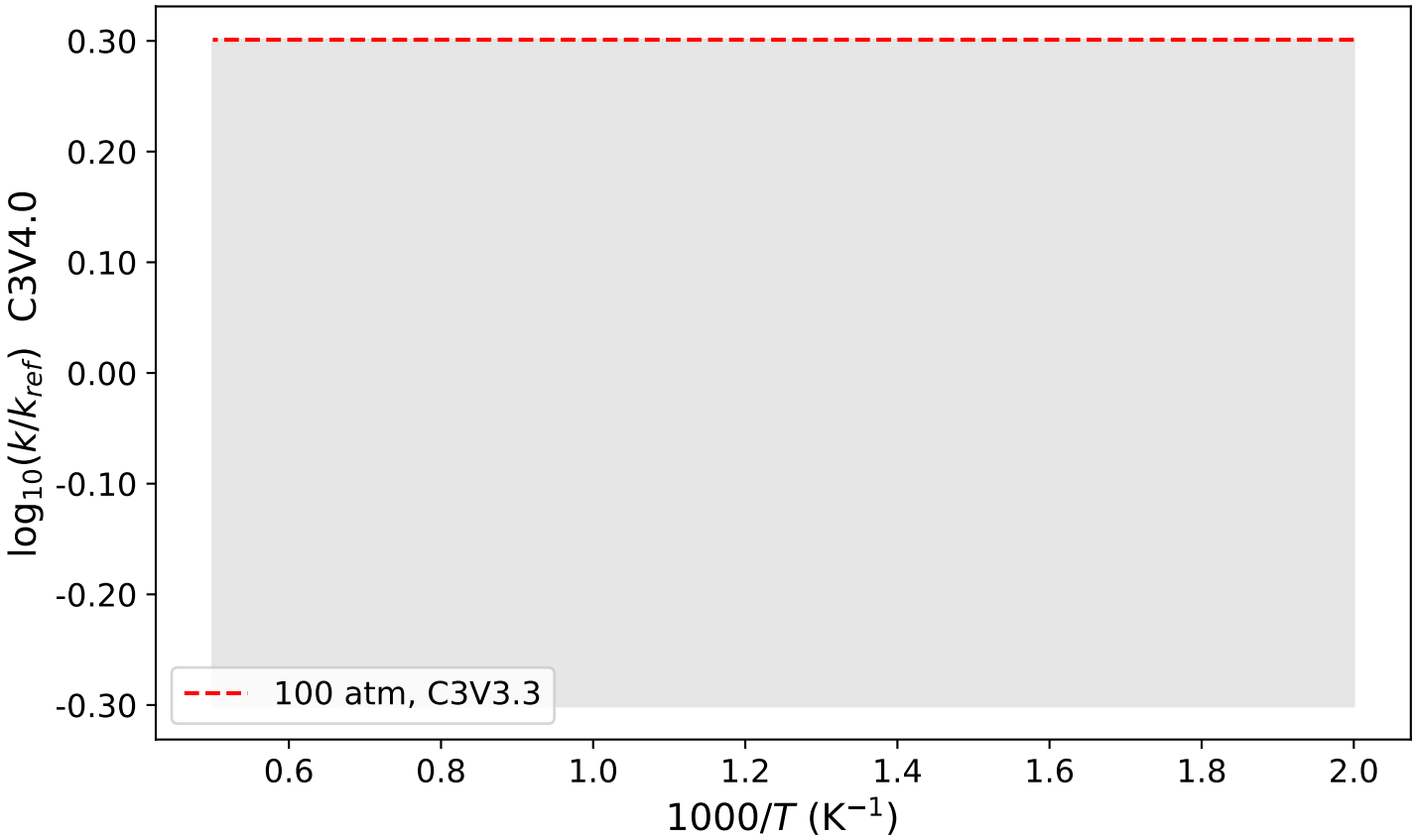
O + RXYLENE = H + C6H4CH3CHO



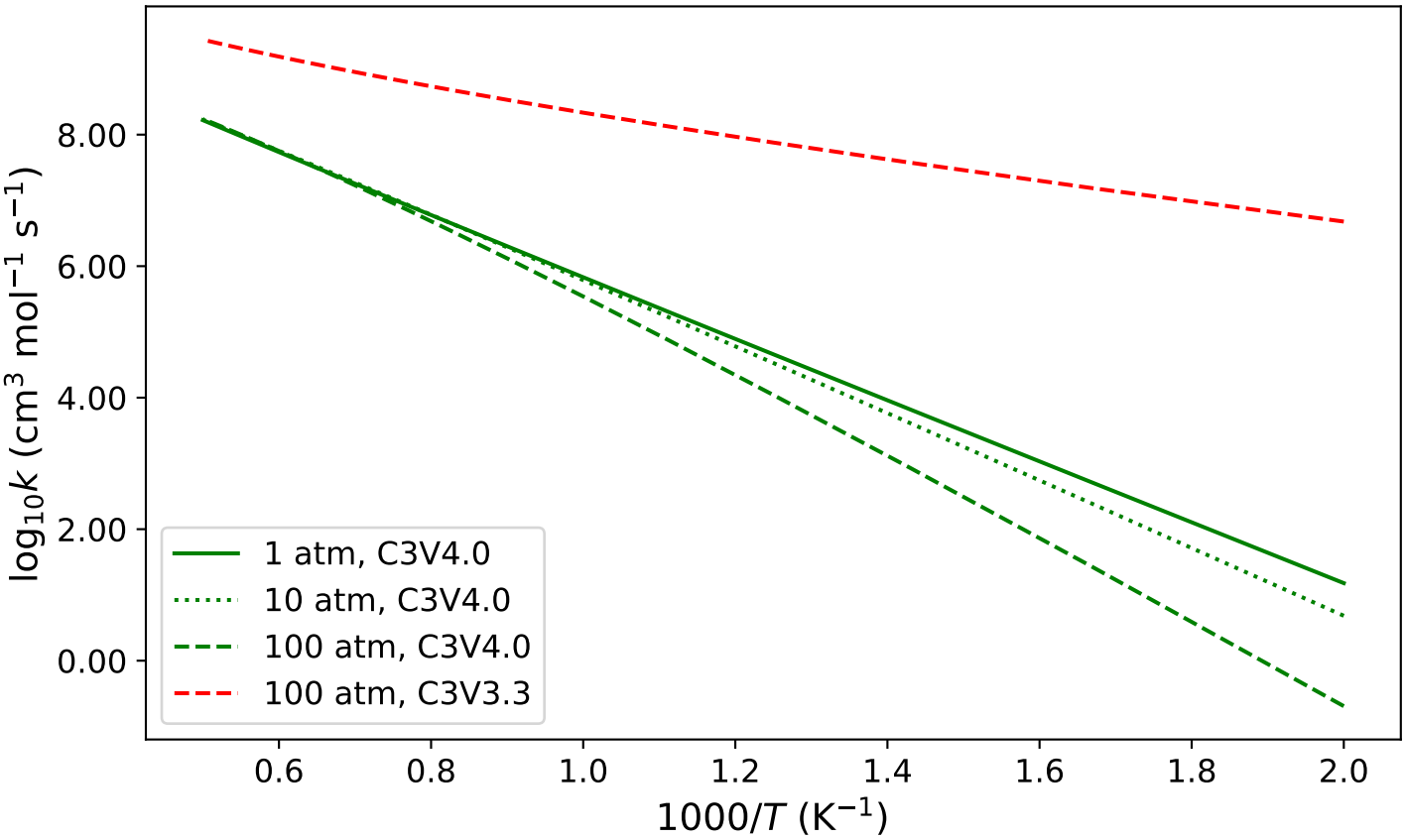


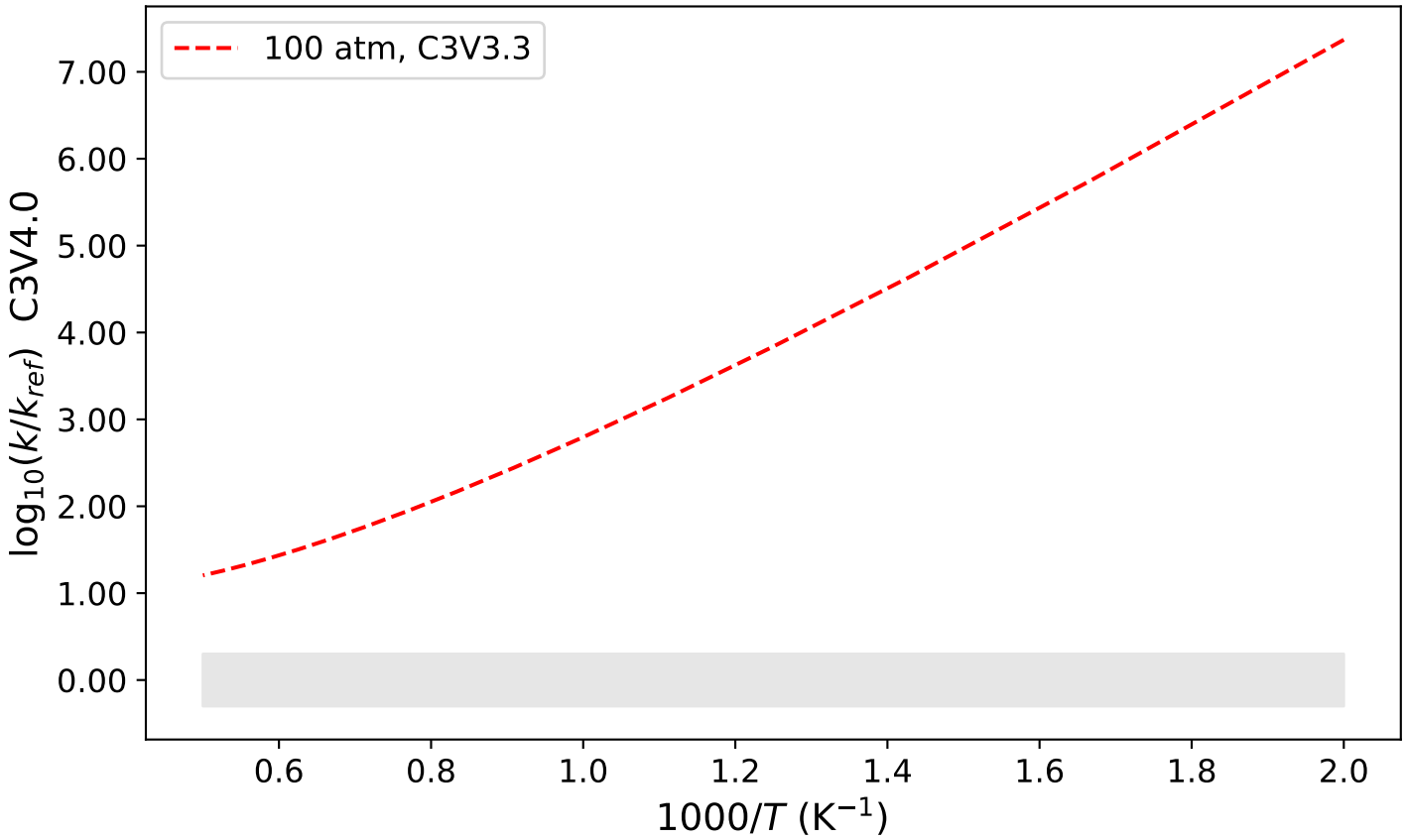
C3H3 + C6H5C2H5 => H + H2 + C10H7CH3-A



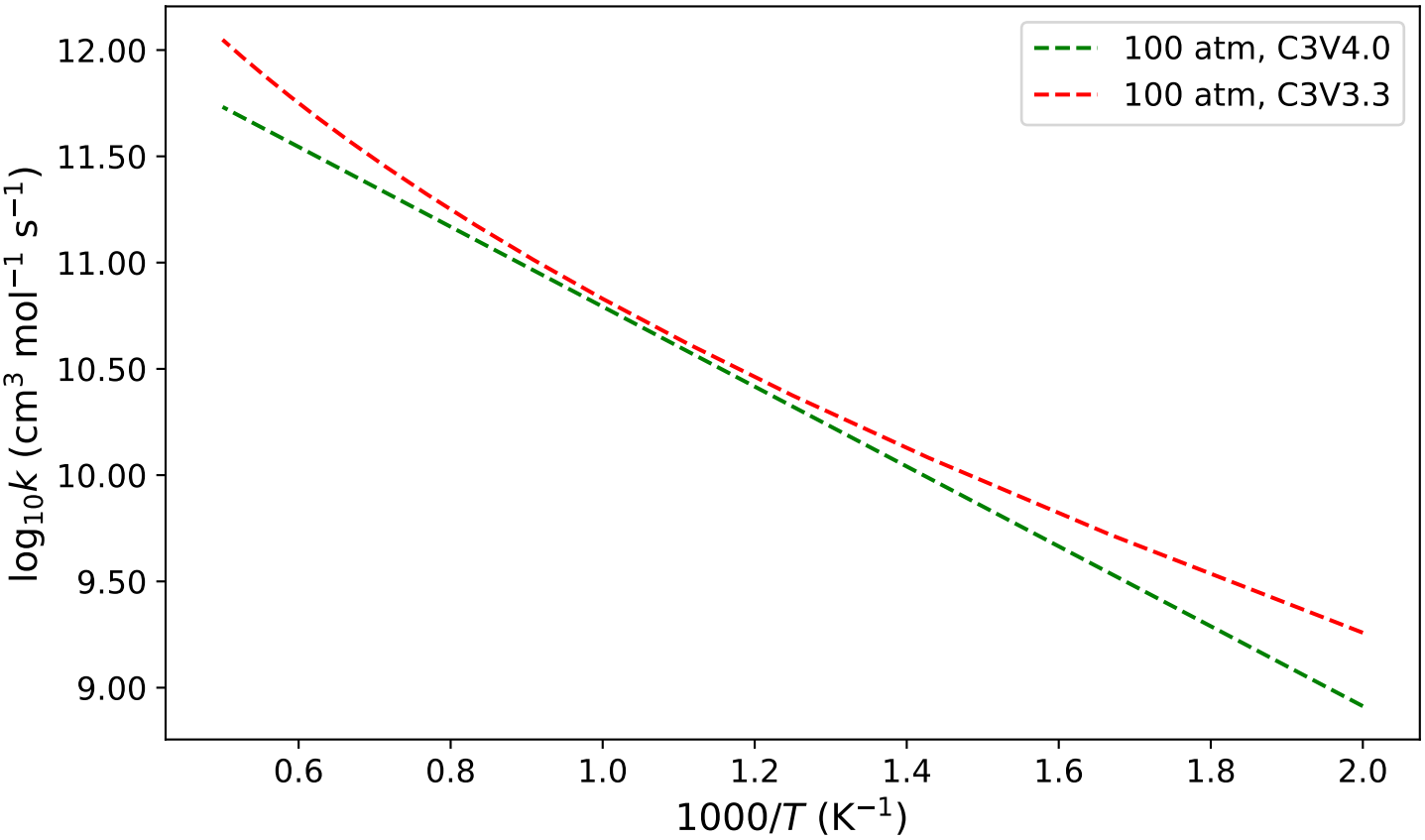


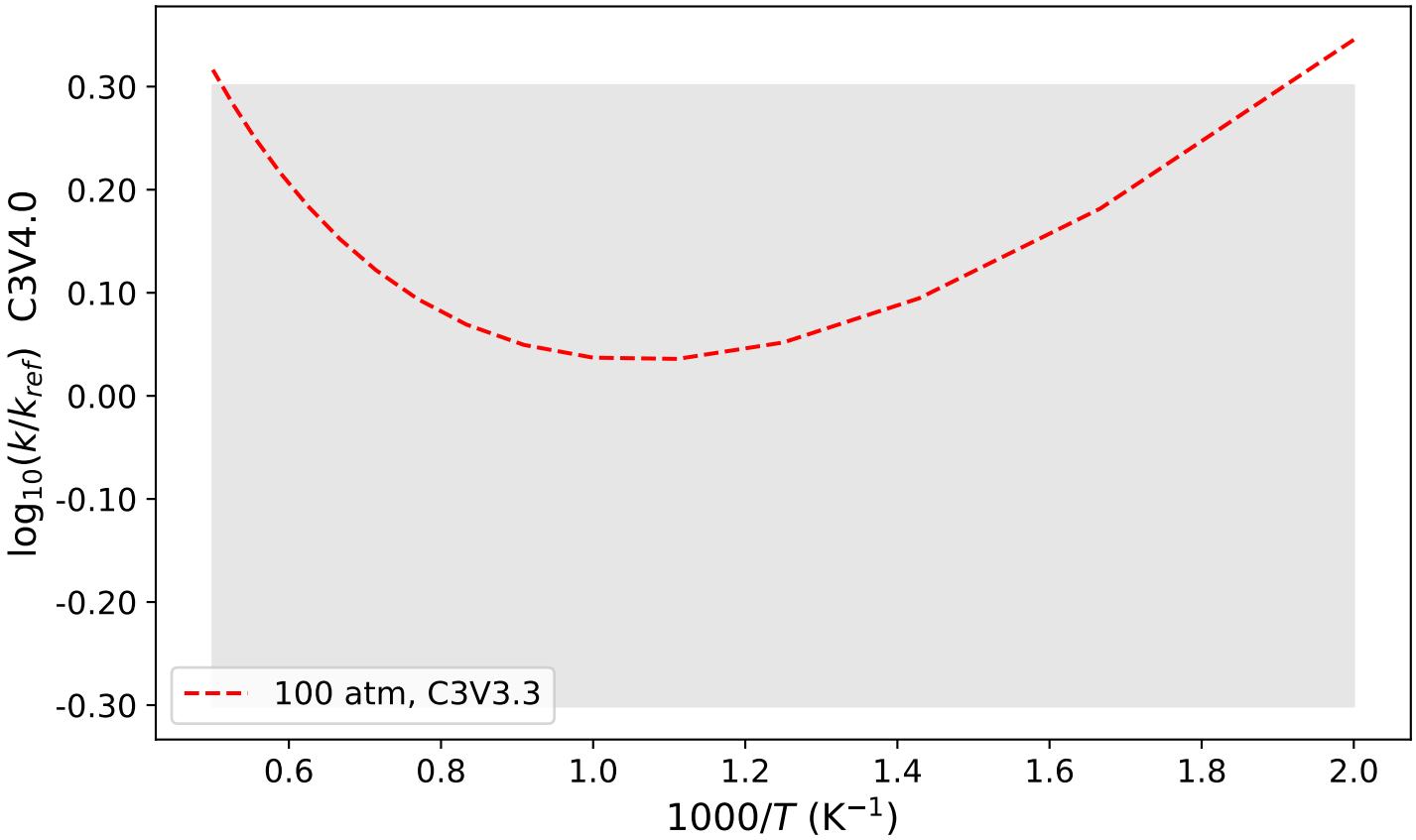
O2 + RXYLENE = CH2O + OC6H4CH3



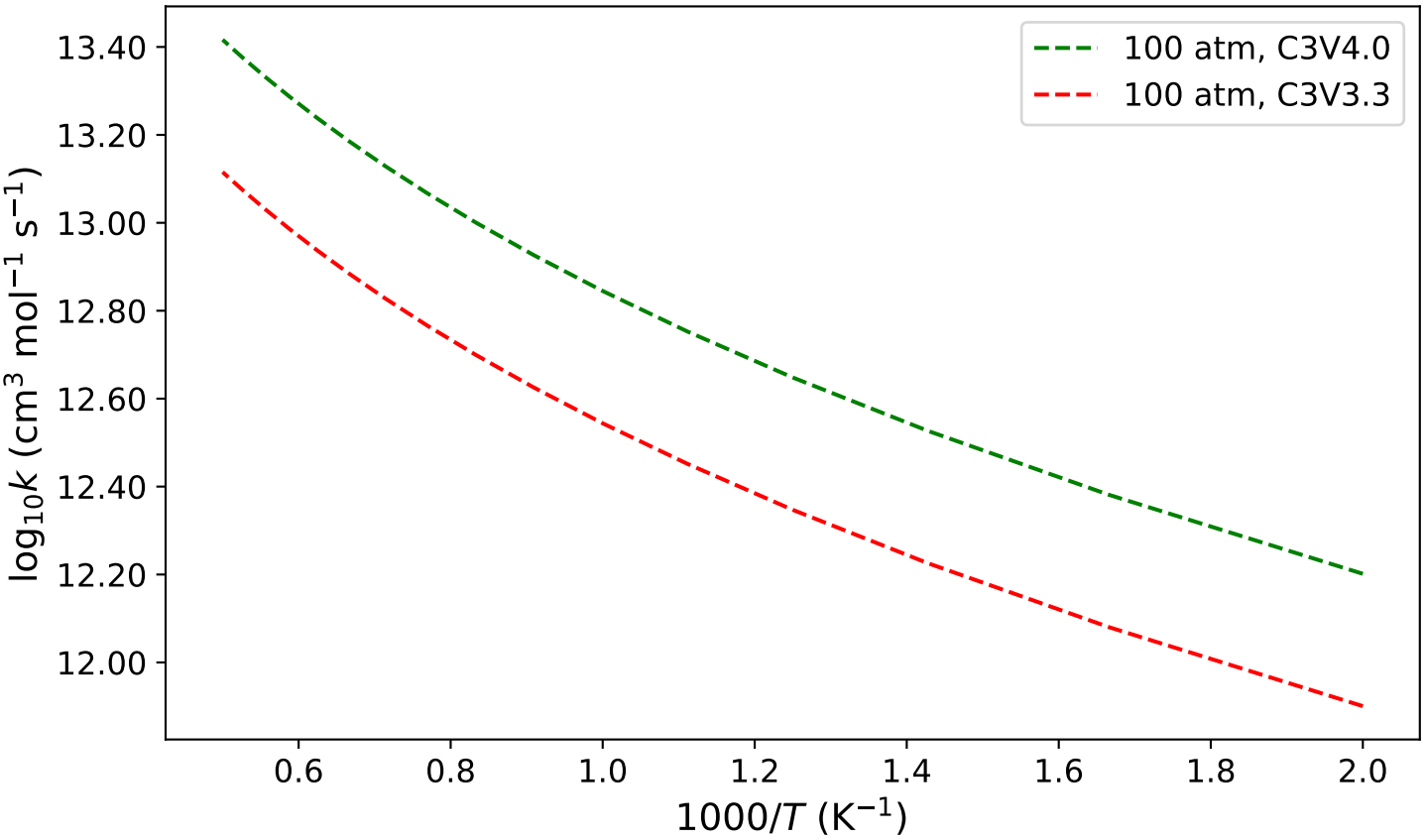


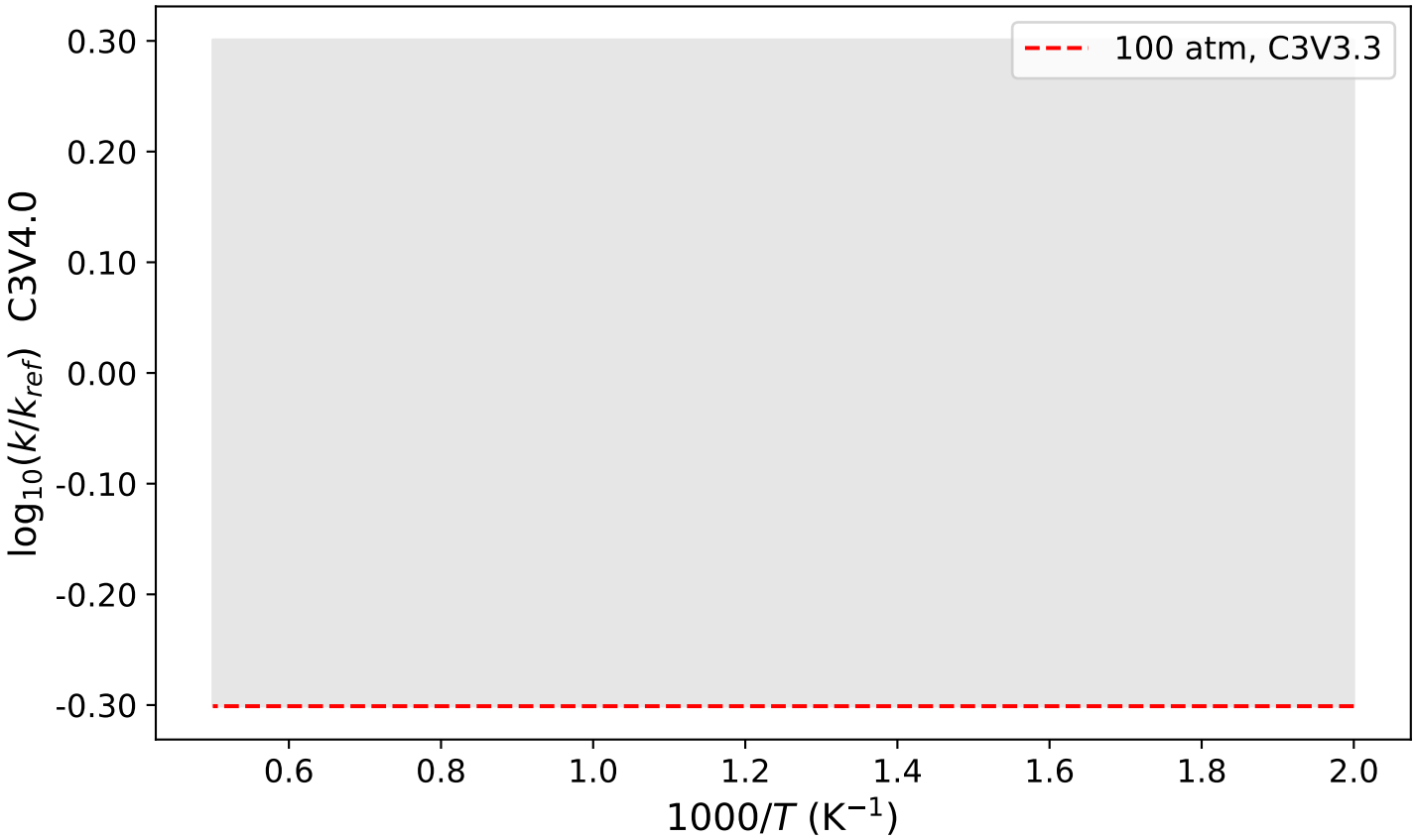
OH + C6H4CH3CHO = HCO + HOC6H4CH3



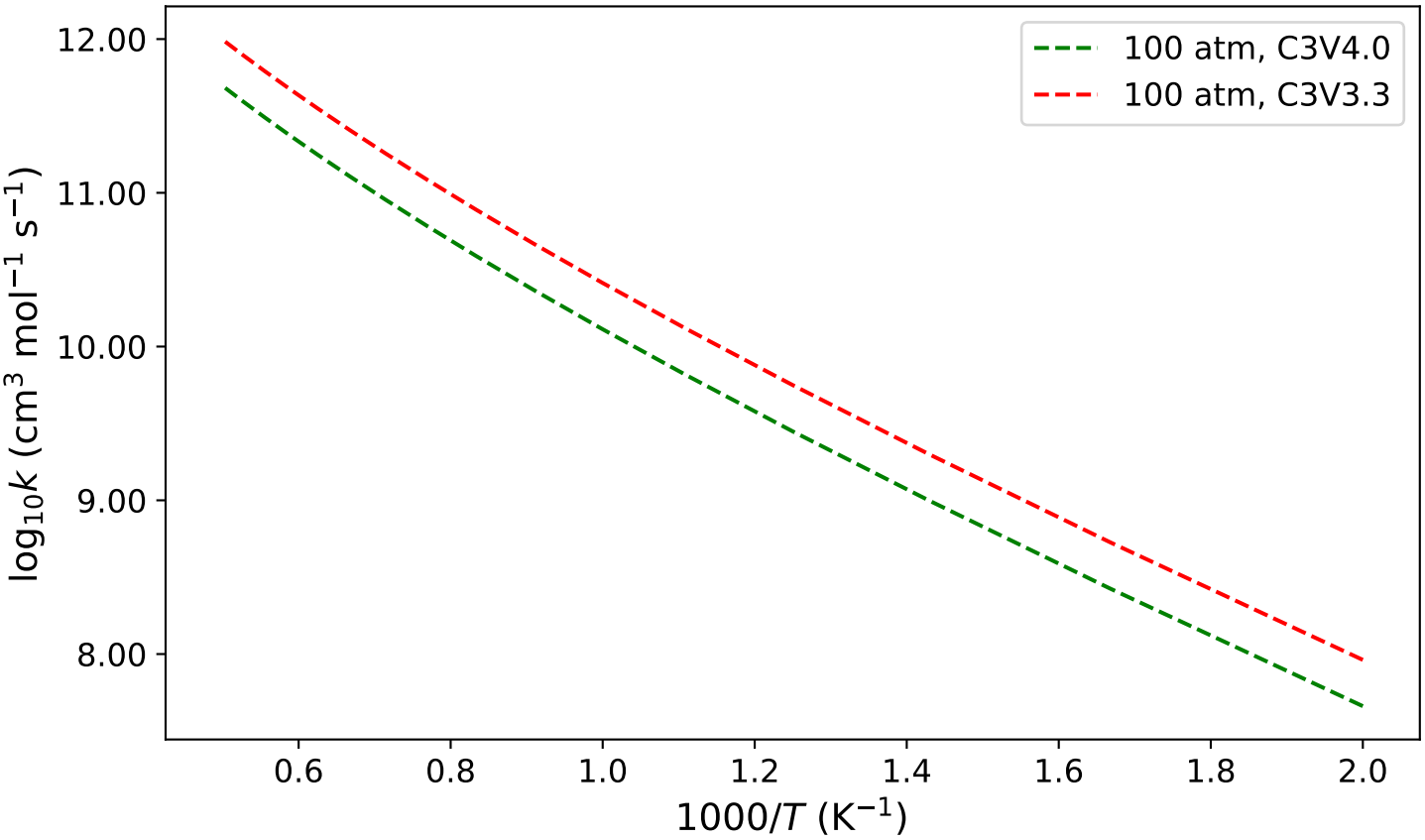


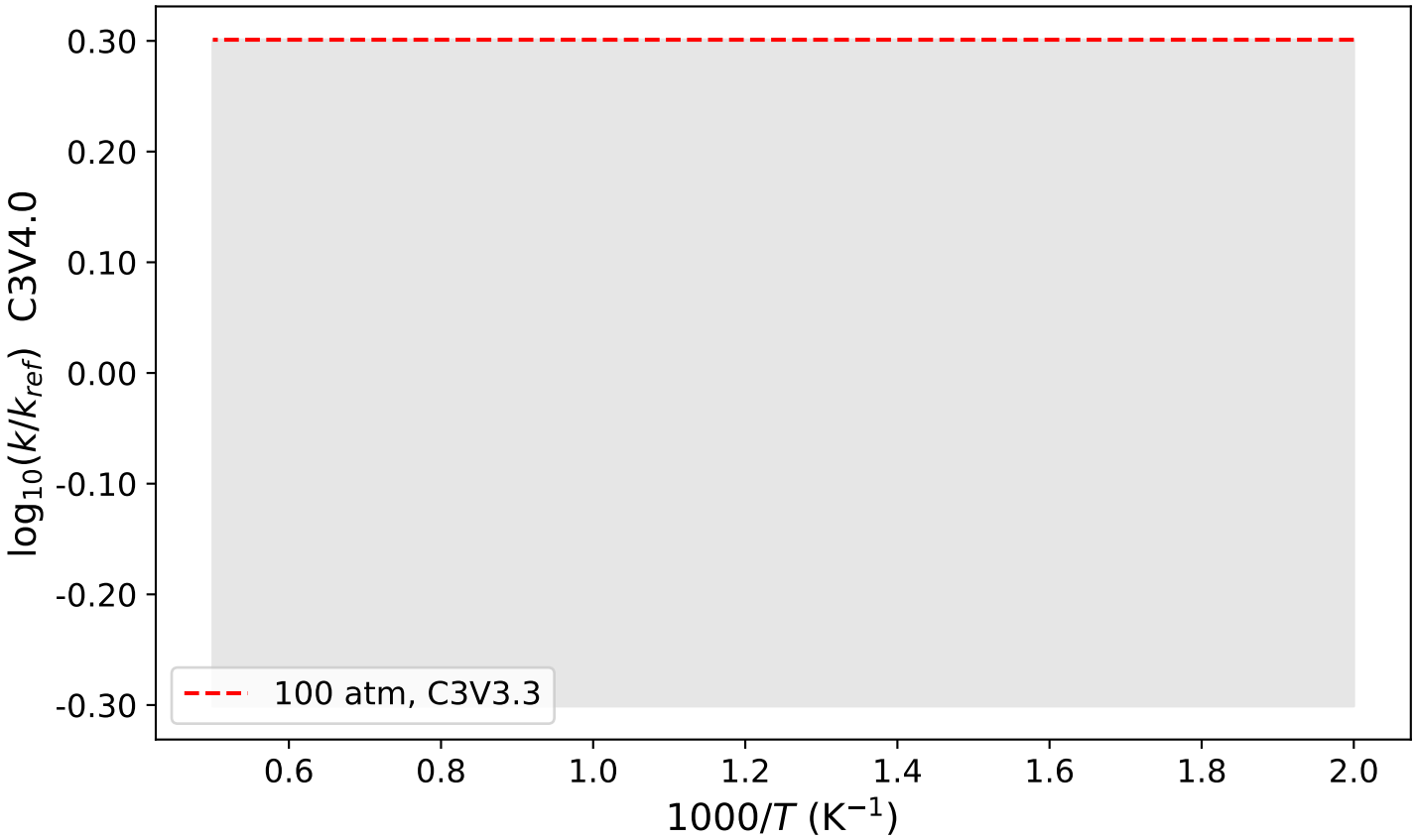
O + C6H5C2H3 = HCO + C6H5CH2



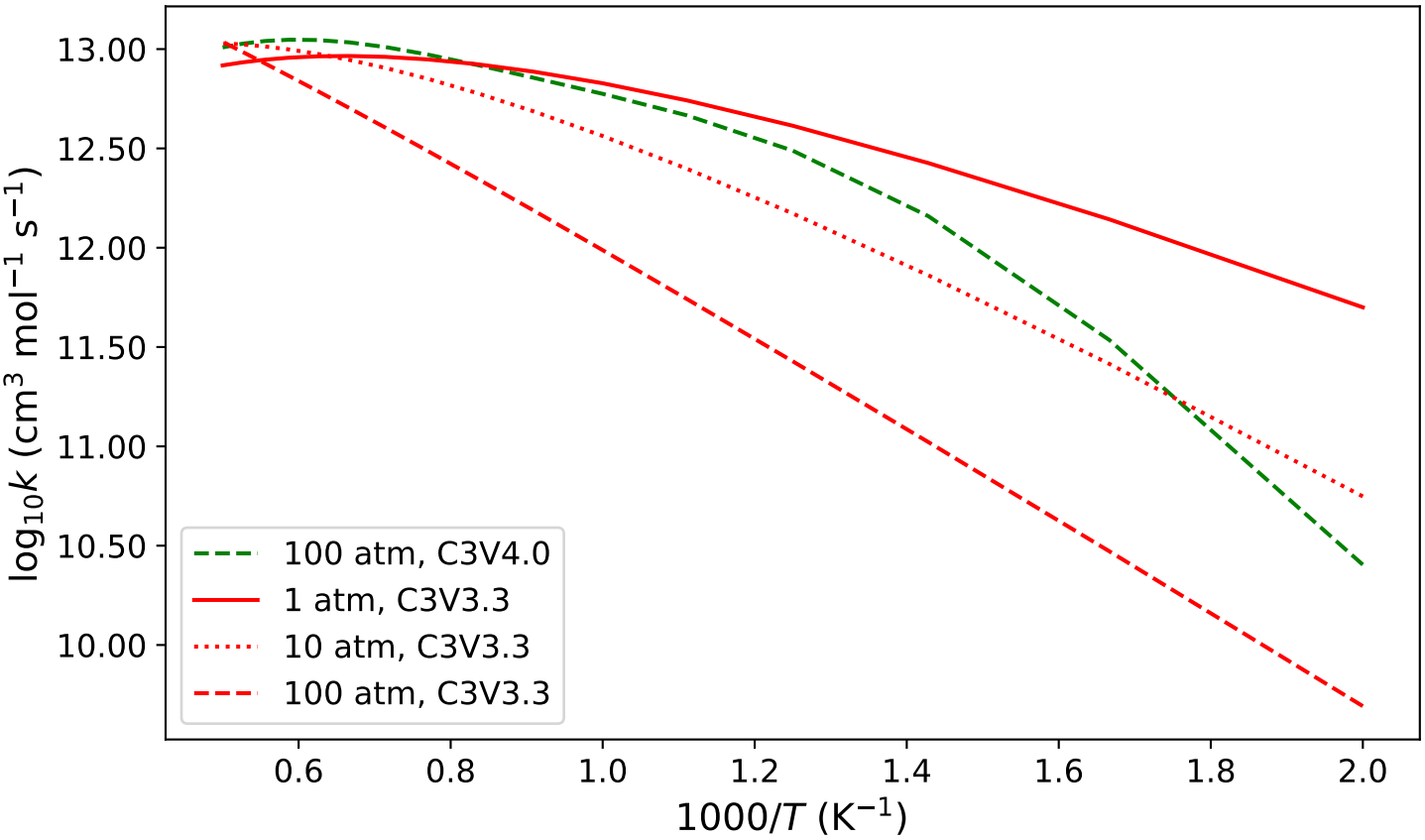


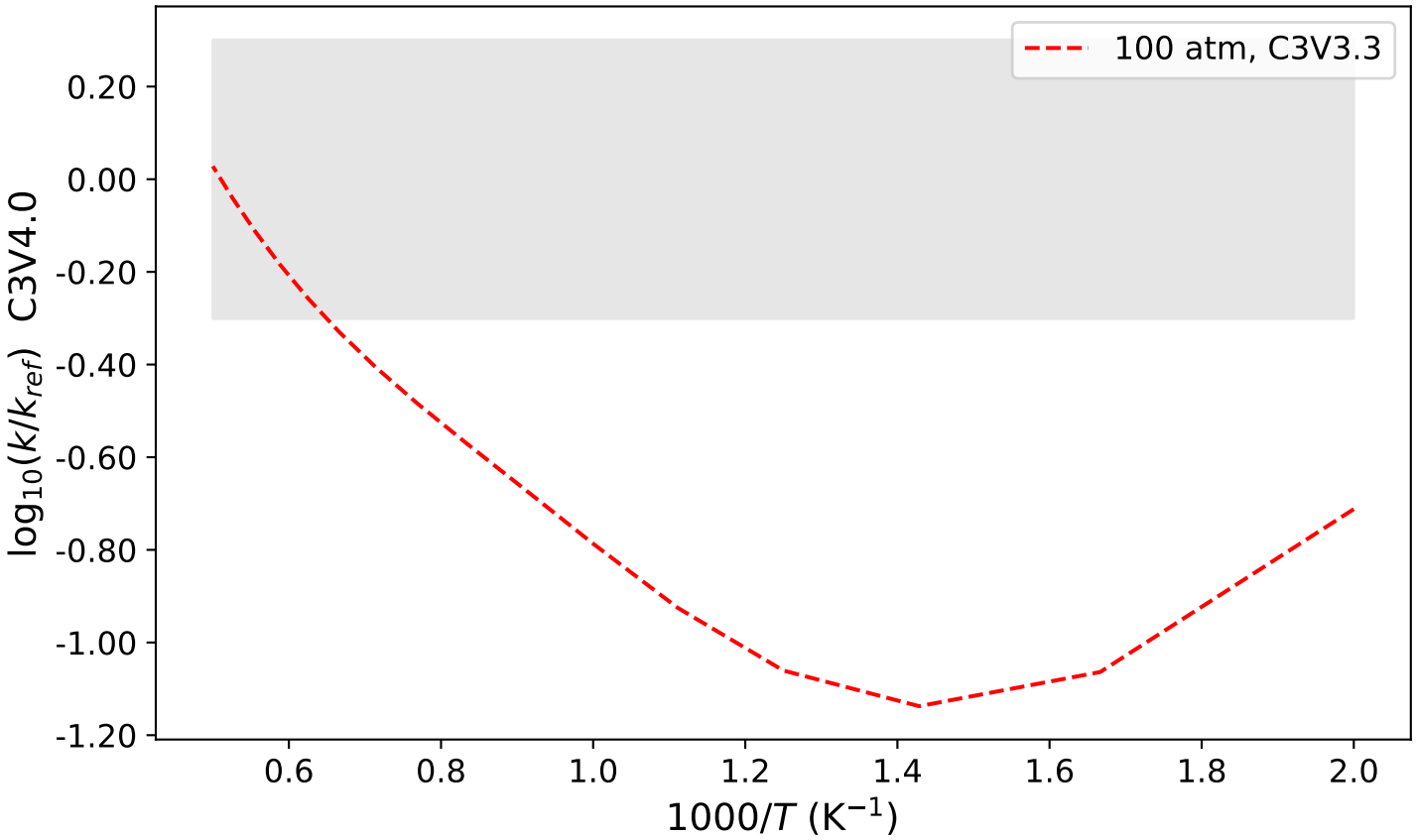
OH + NAPH = H + NAPHOH-A



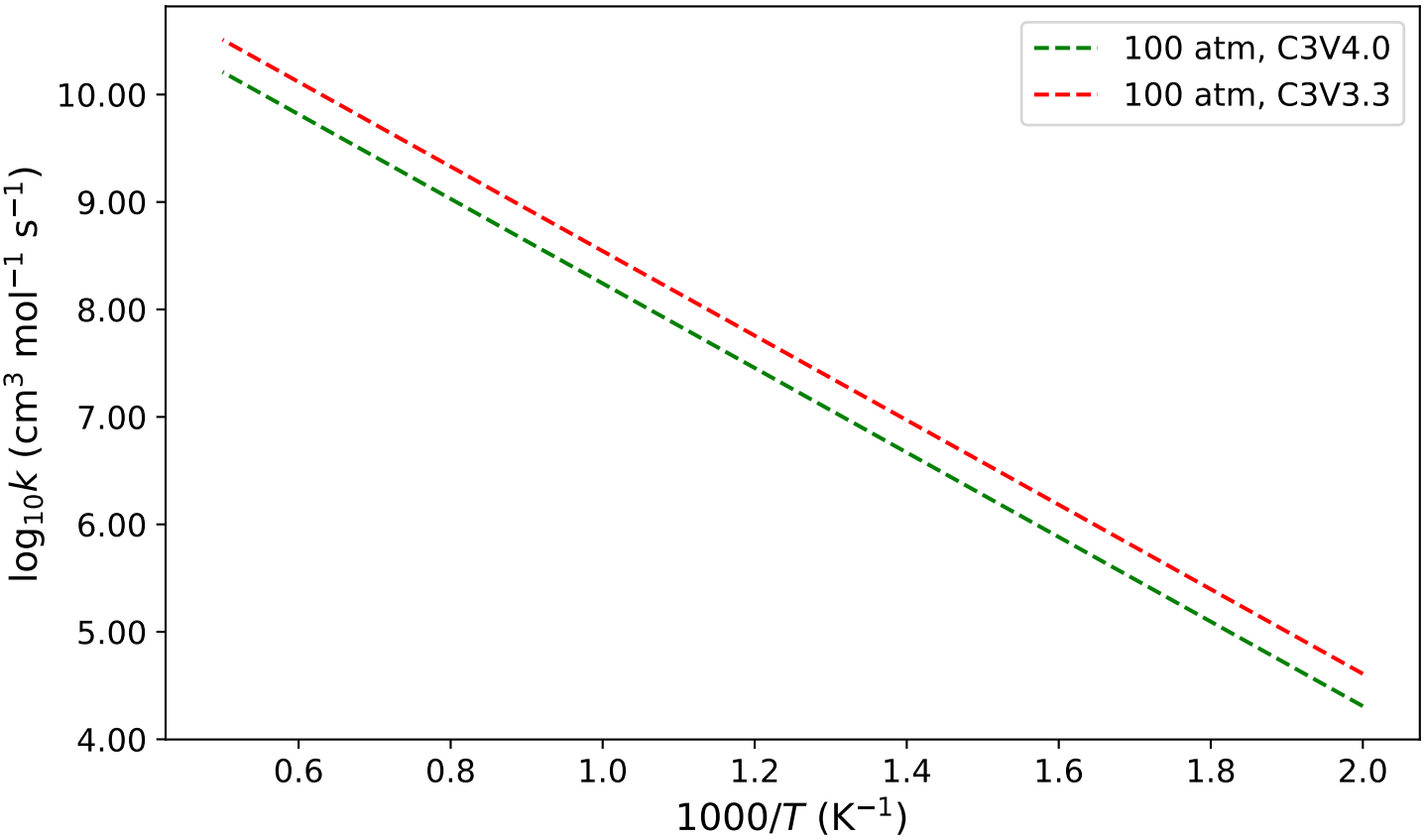


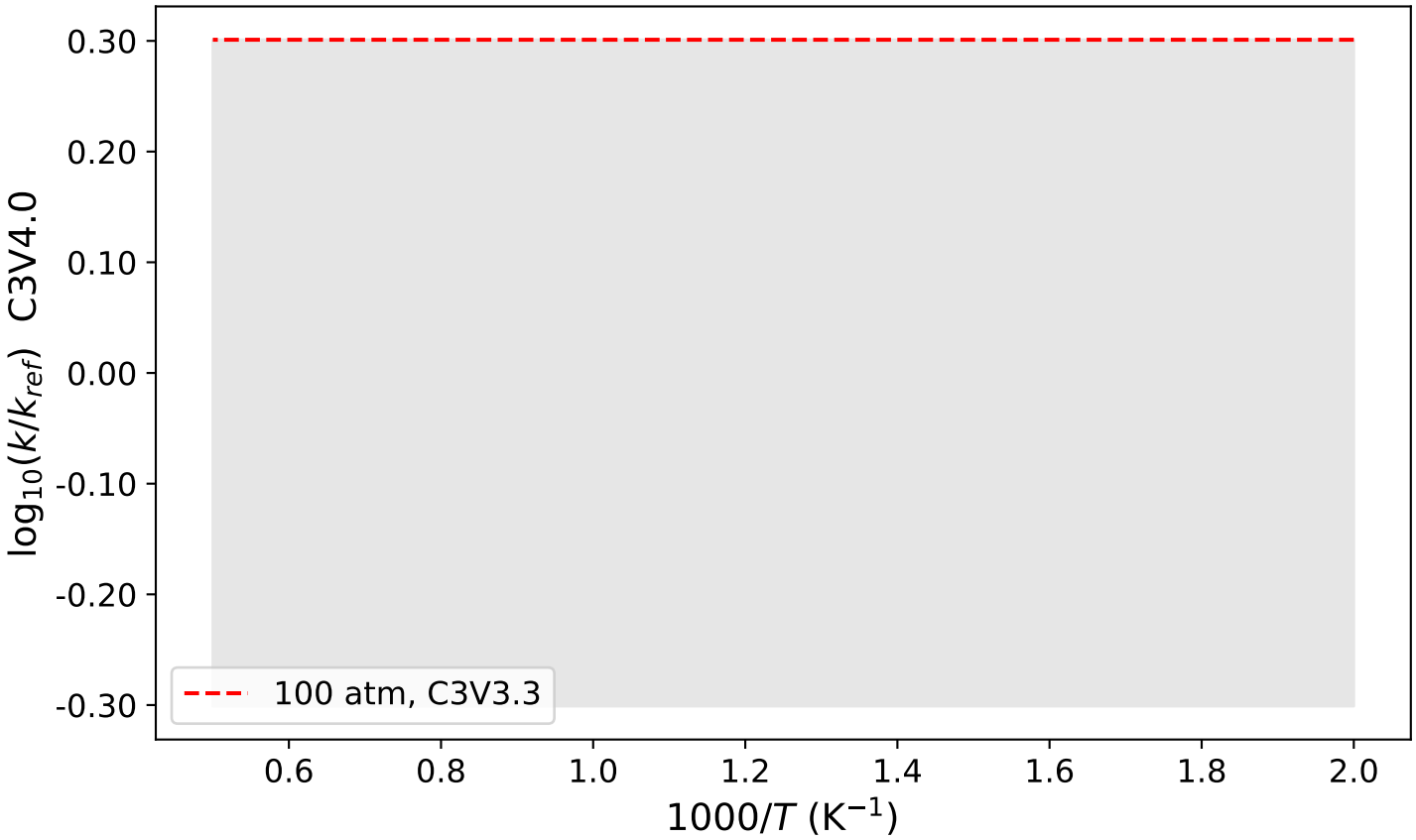
HO2 + C6H5CH2 = OH + C6H5CH2O



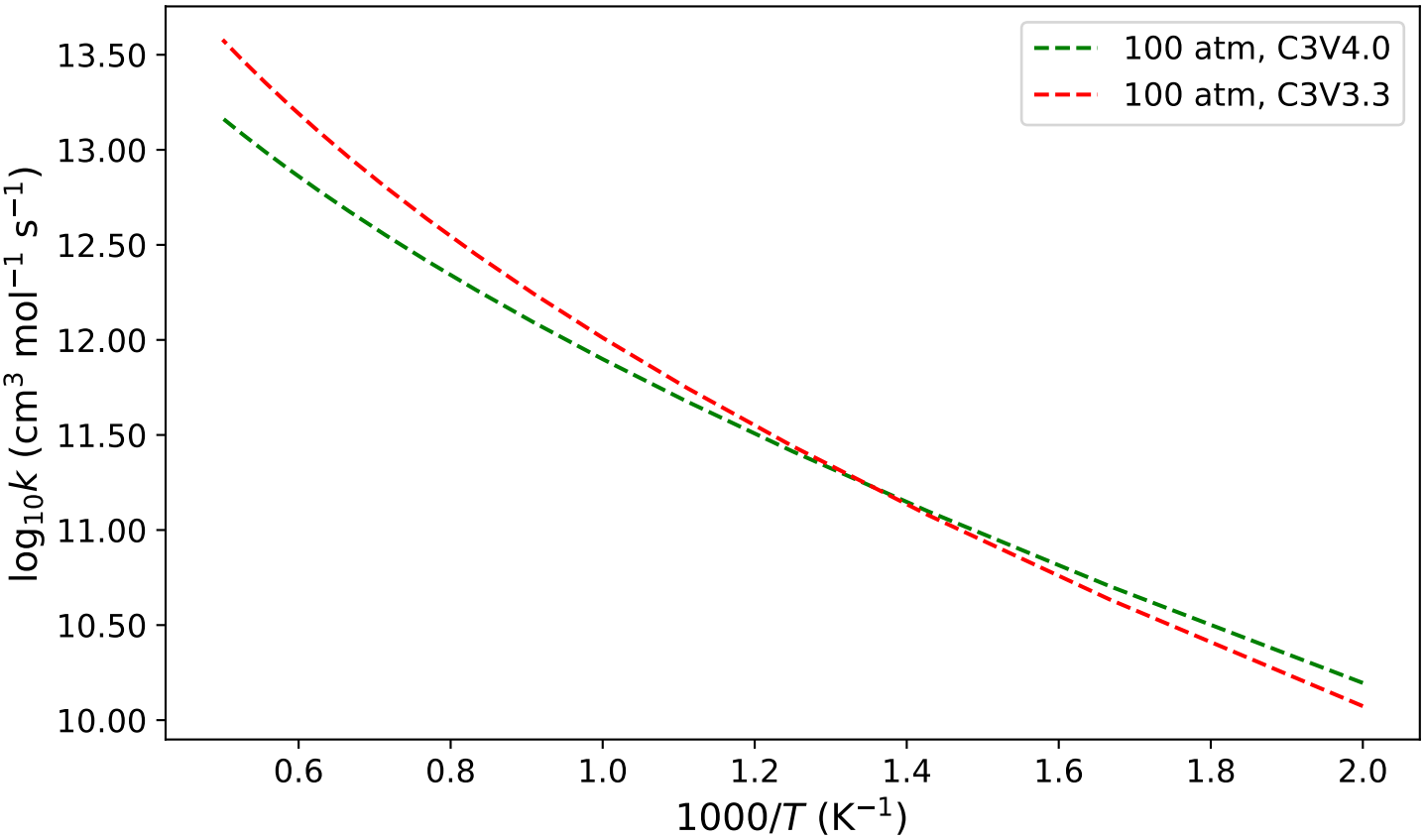


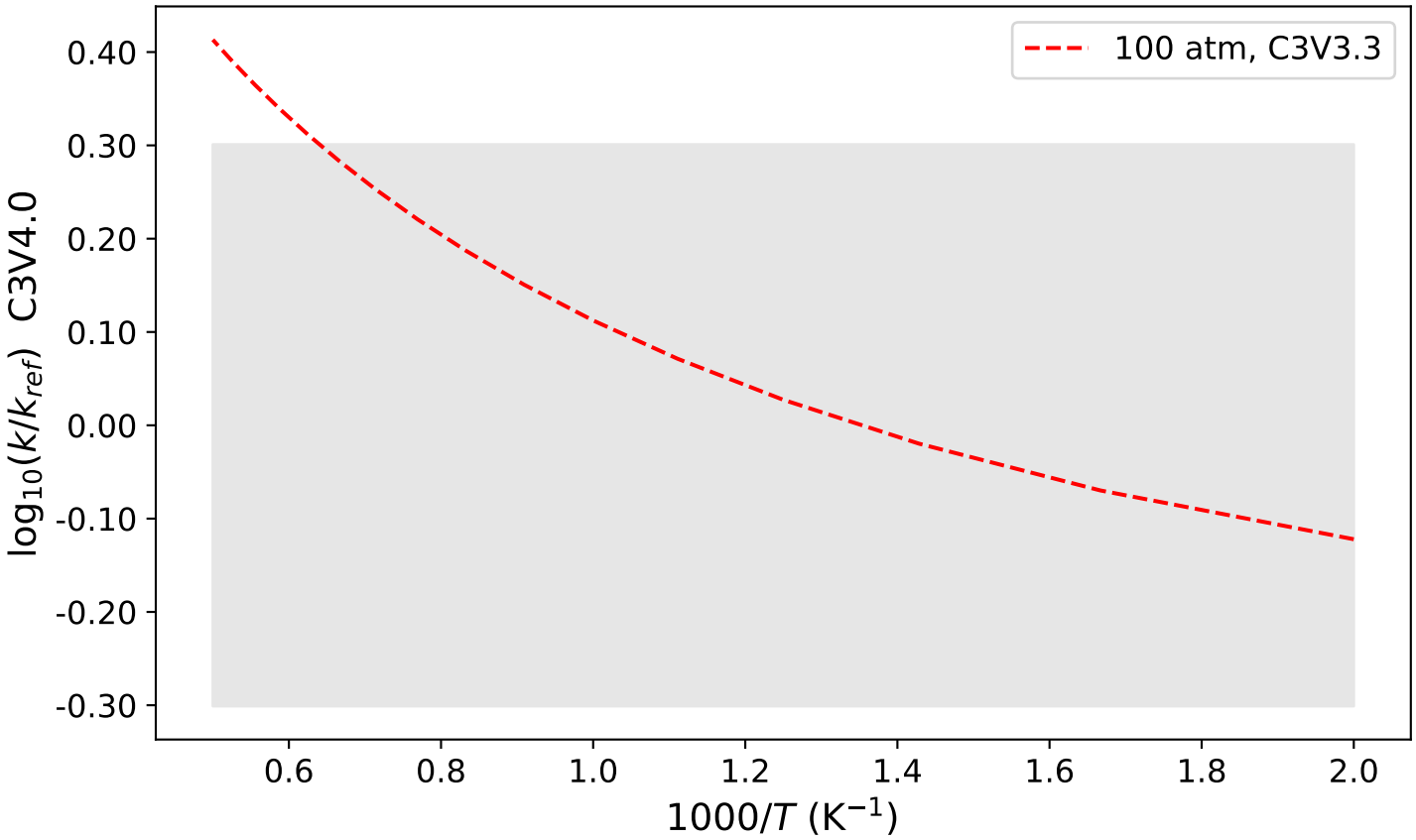
C3H4-A + C6H5CHCH3 => H + H2 + C10H7CH3-A



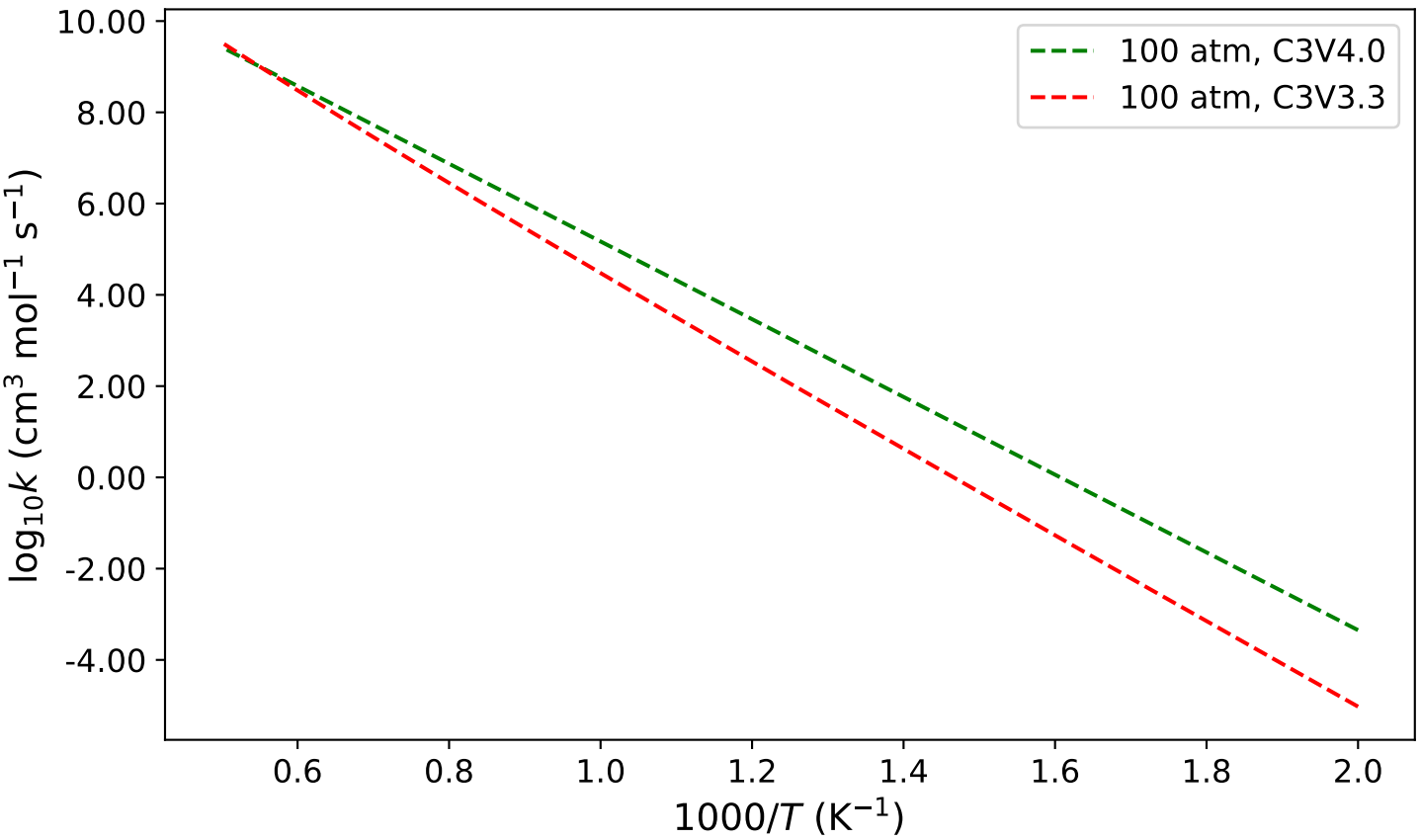


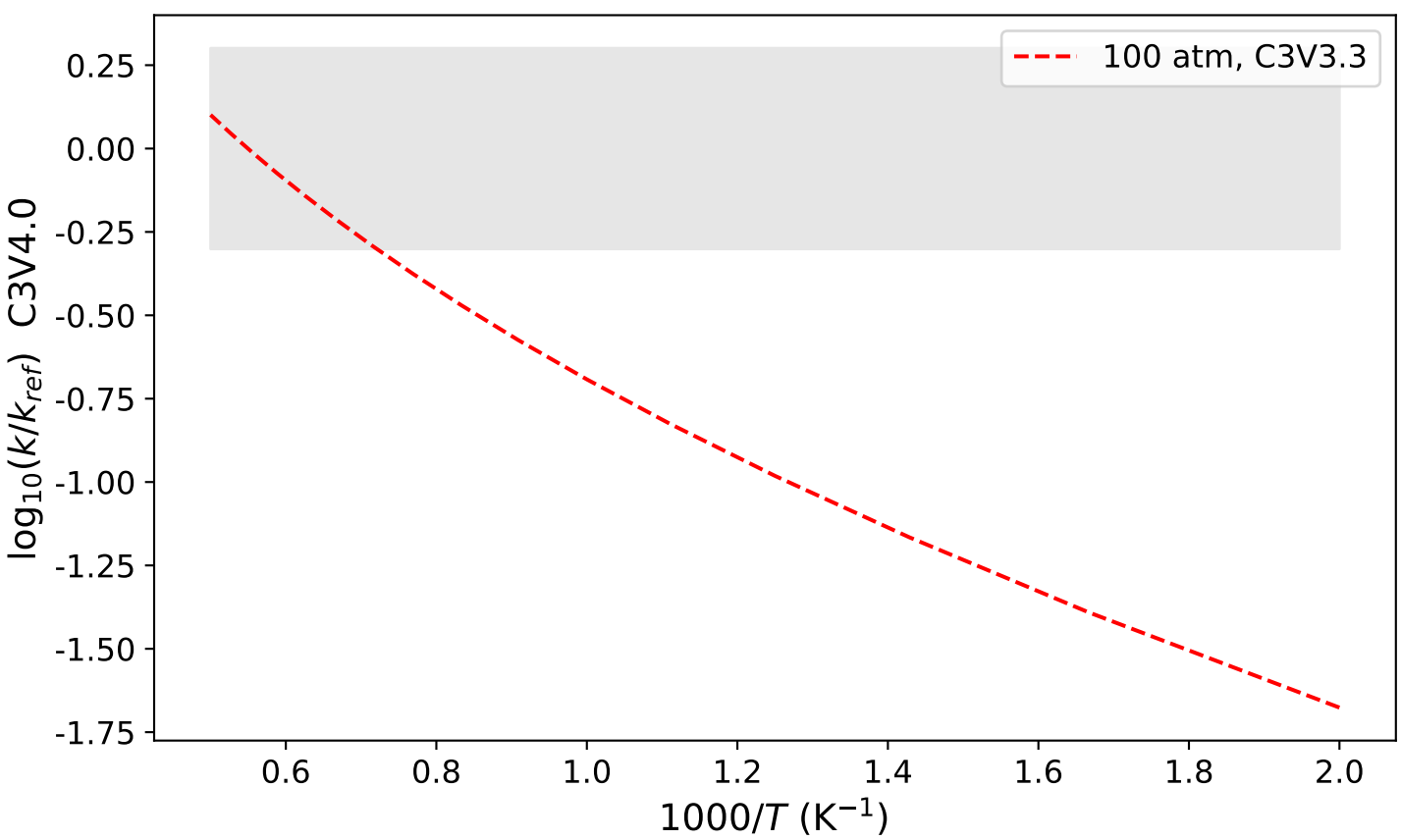
H + C10H7CH3-A = H2 + C10H7CH2-A



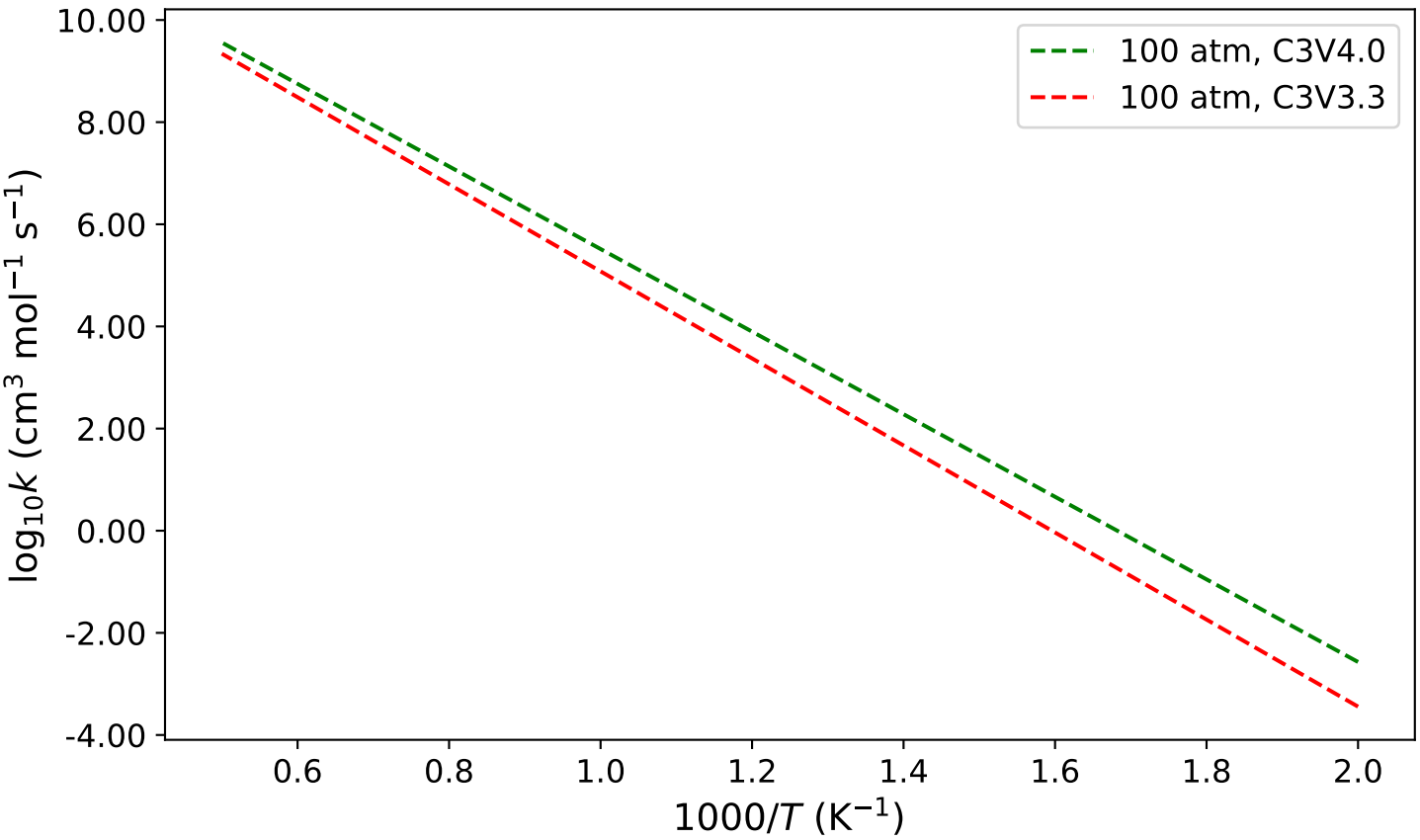


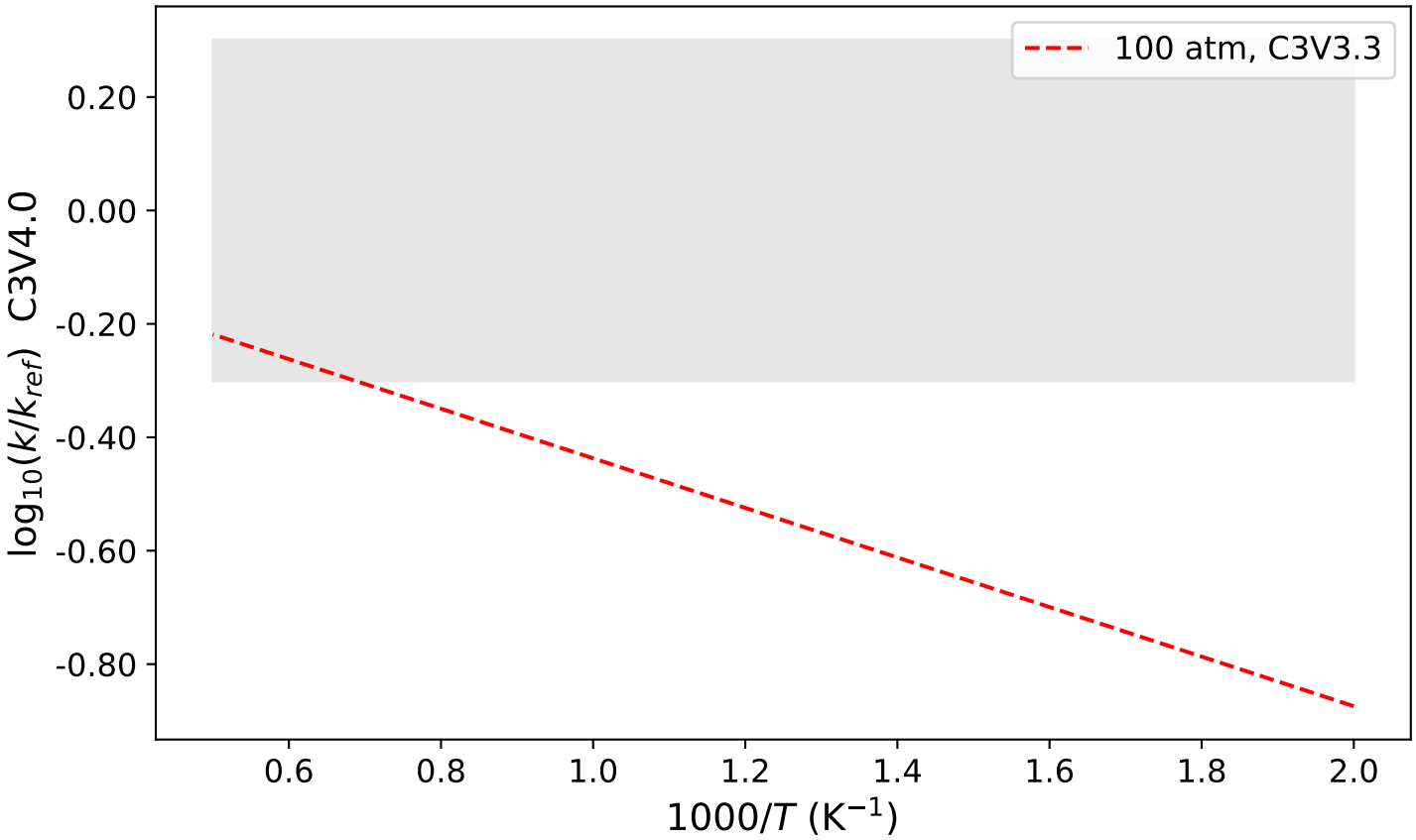
O2 + CYC5H8 = HO2 + CYC5H71-3



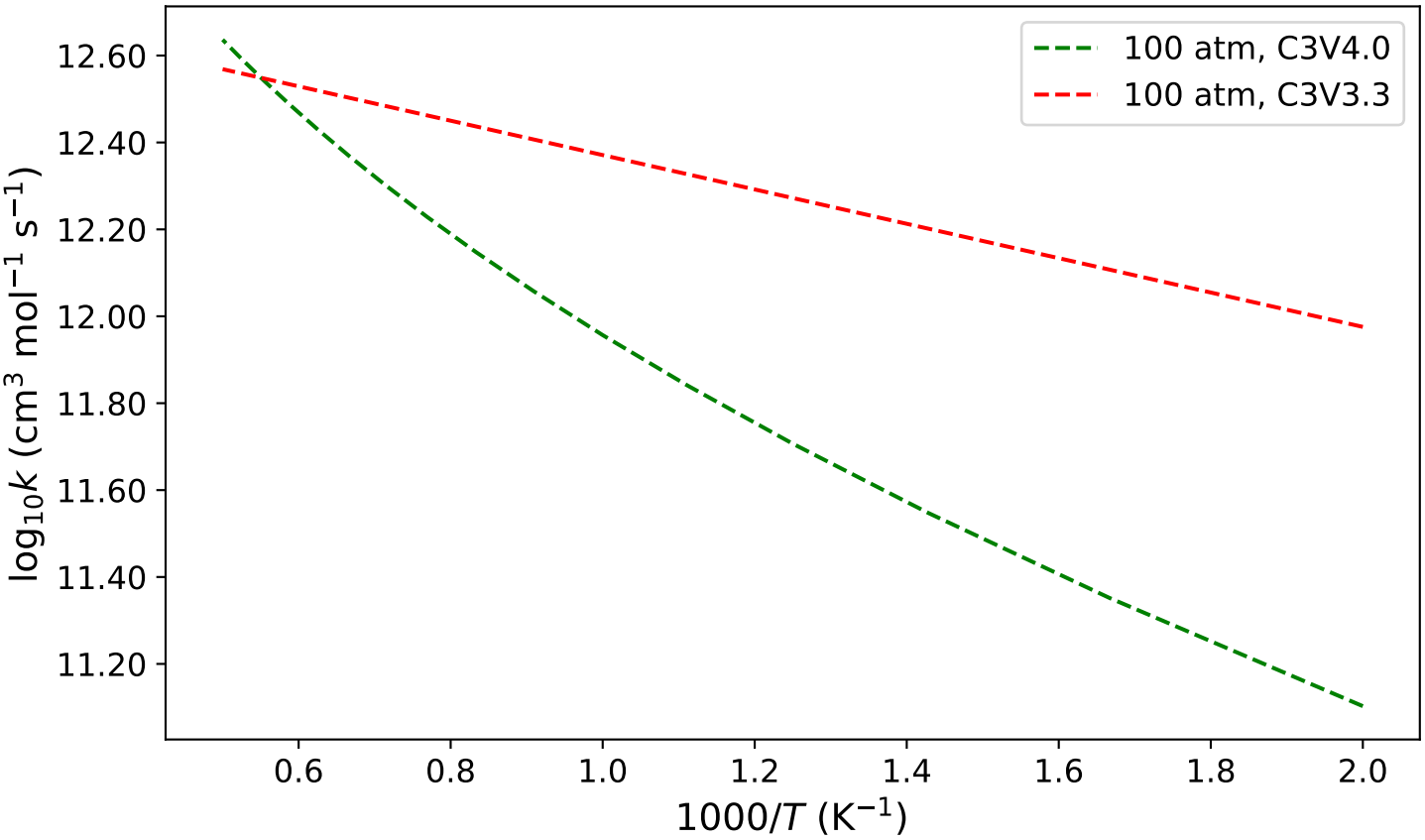


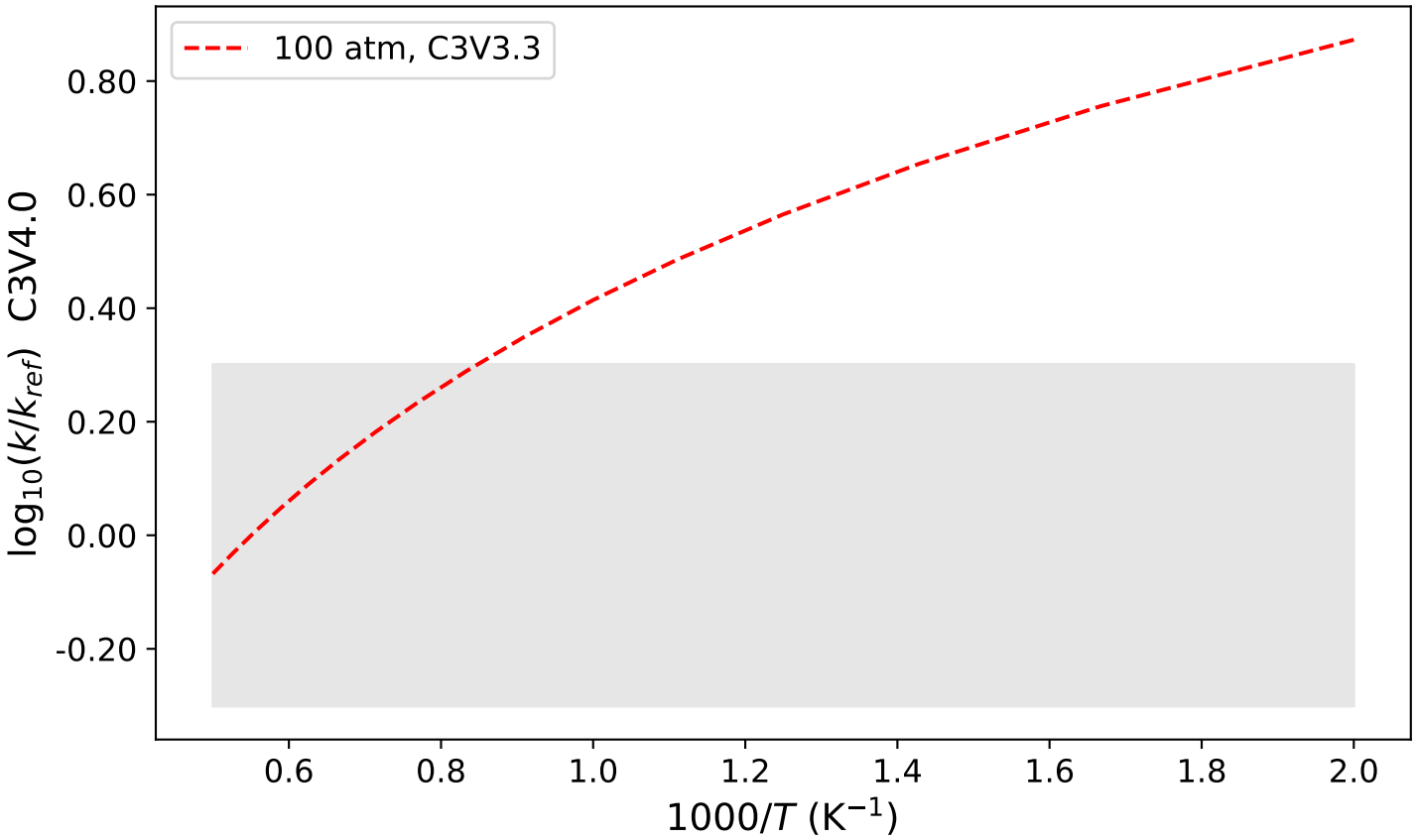
C5H6 + O2 => C3H5-A + HCO + CO



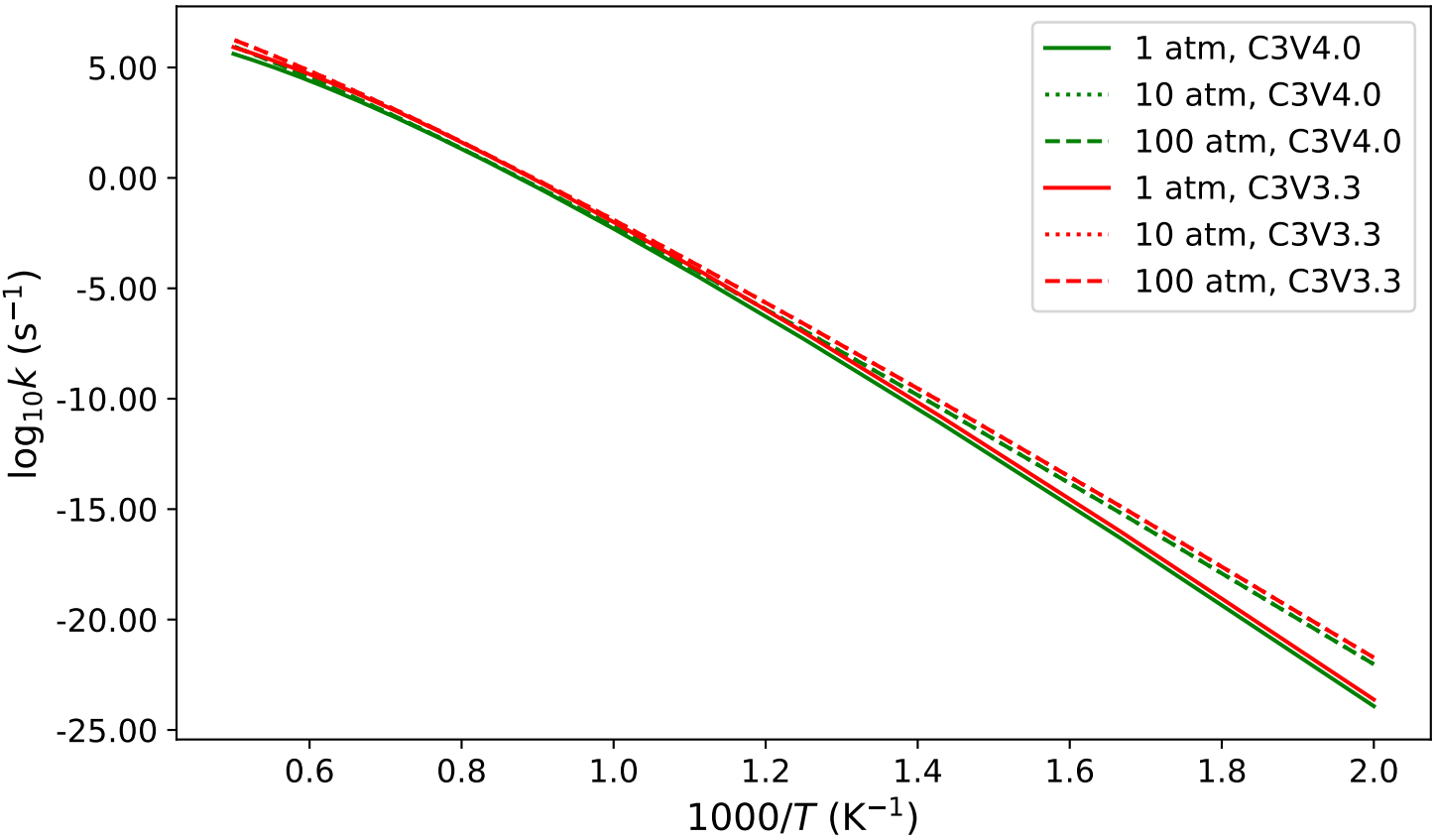


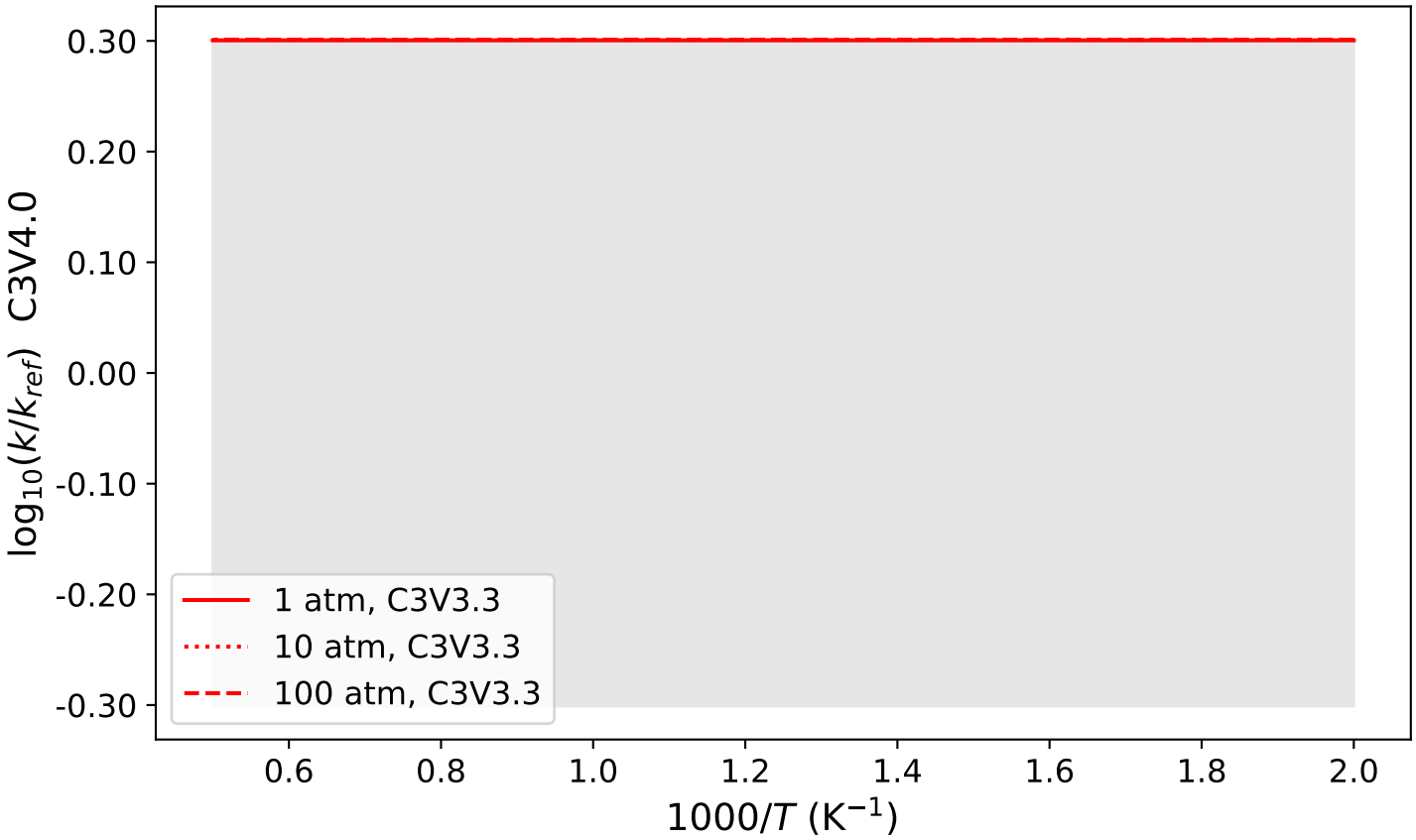
O + C6H4CH3CHO => OH + CO + C6H4CH3



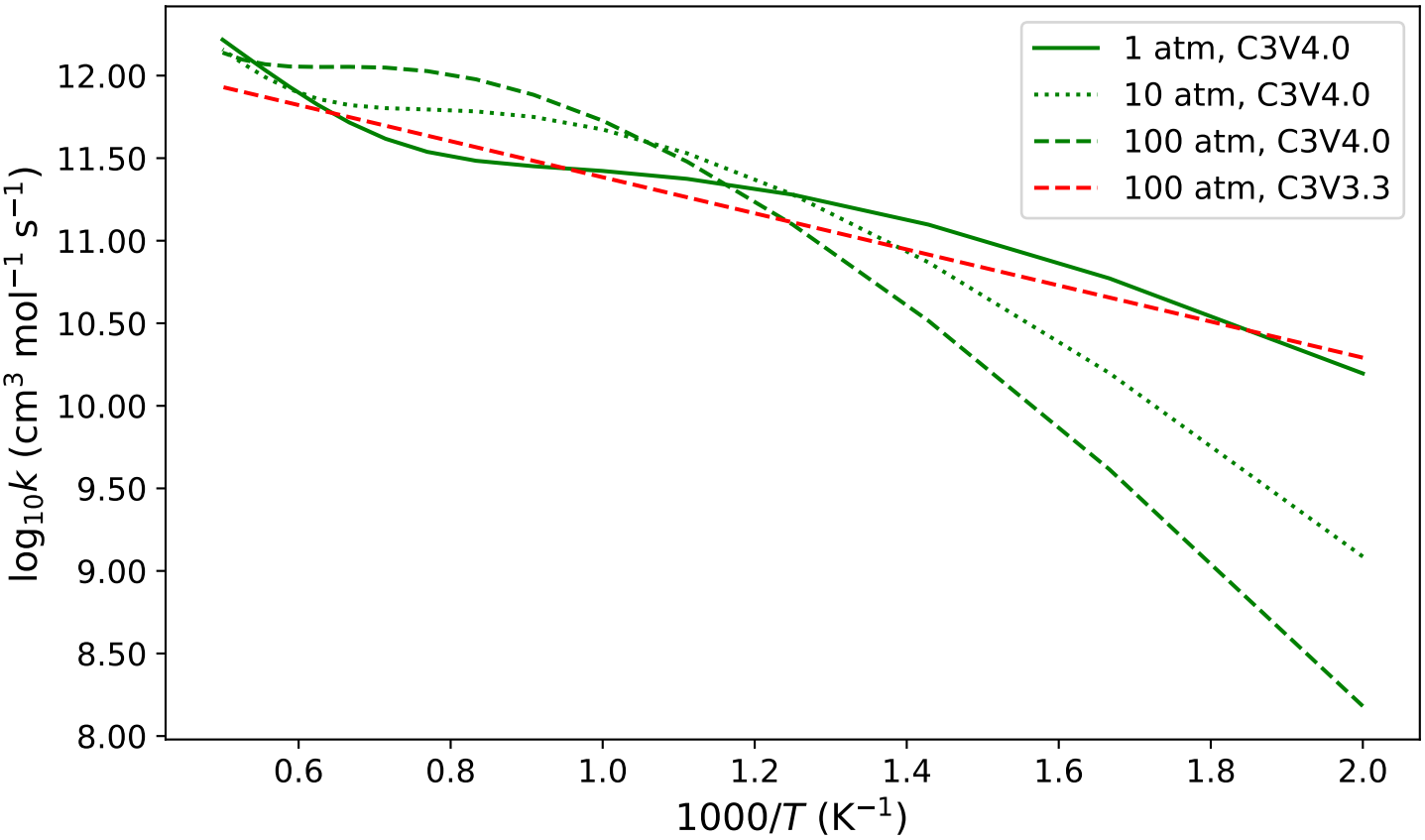


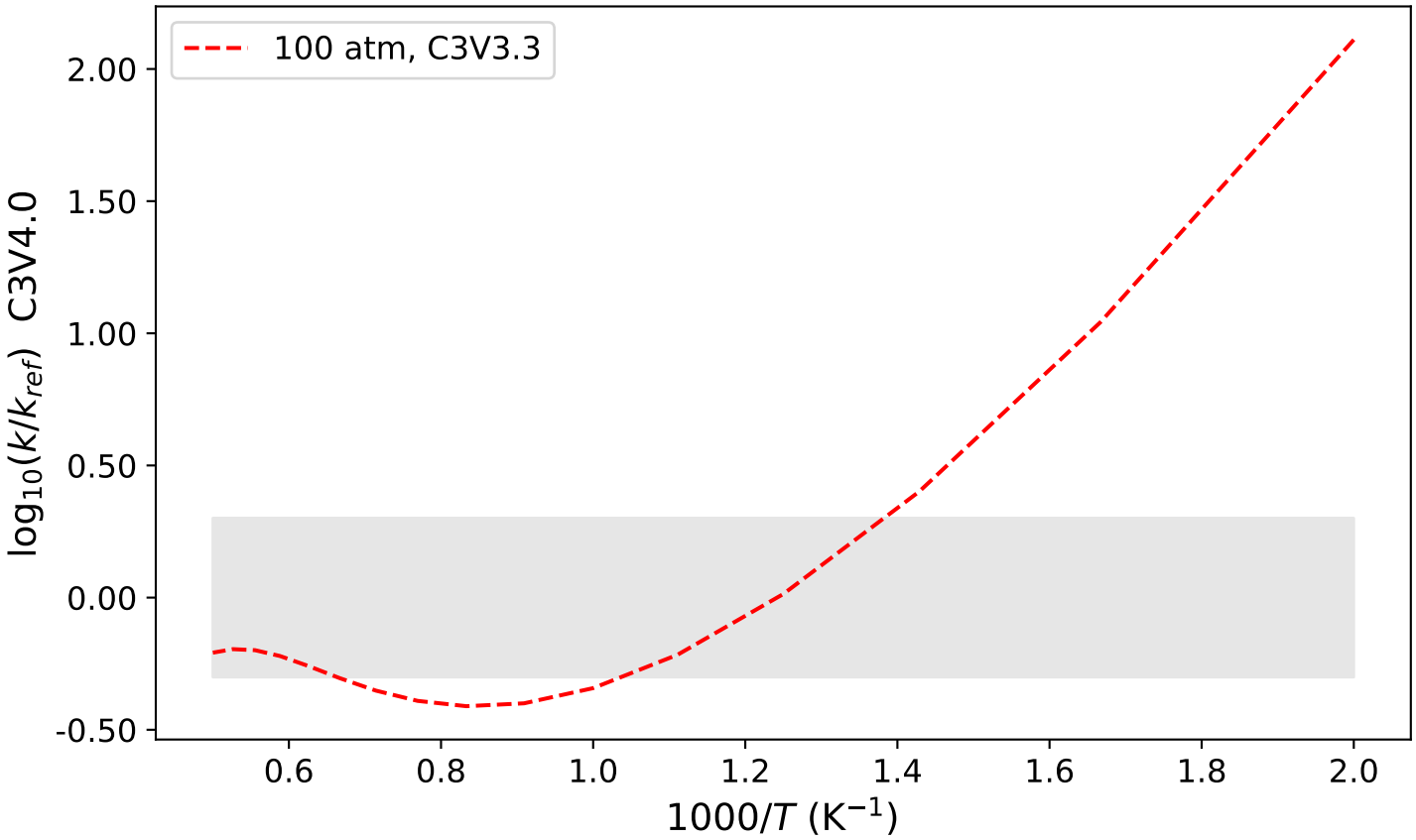
C7H5 = C3H3 + C4H2



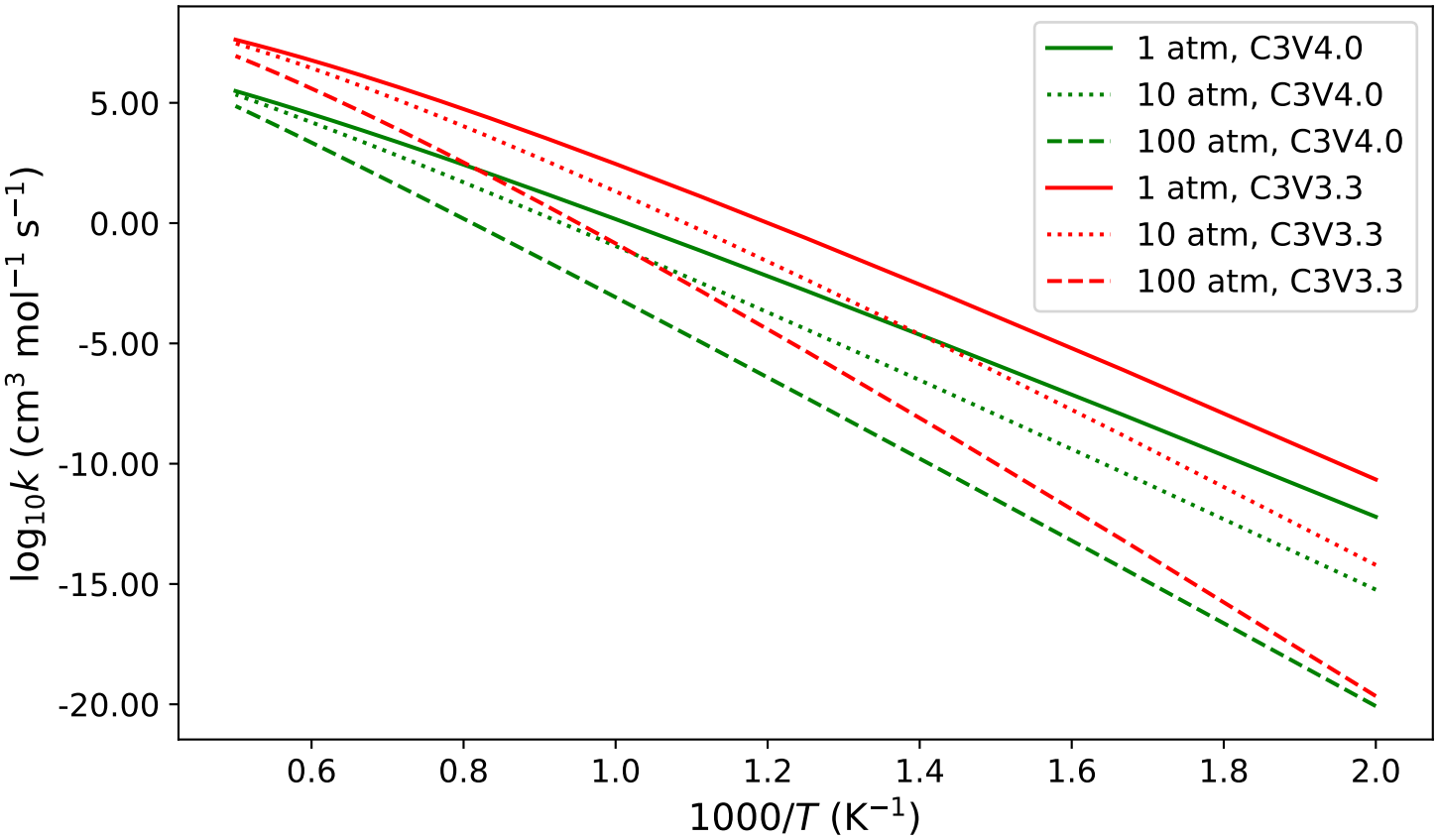


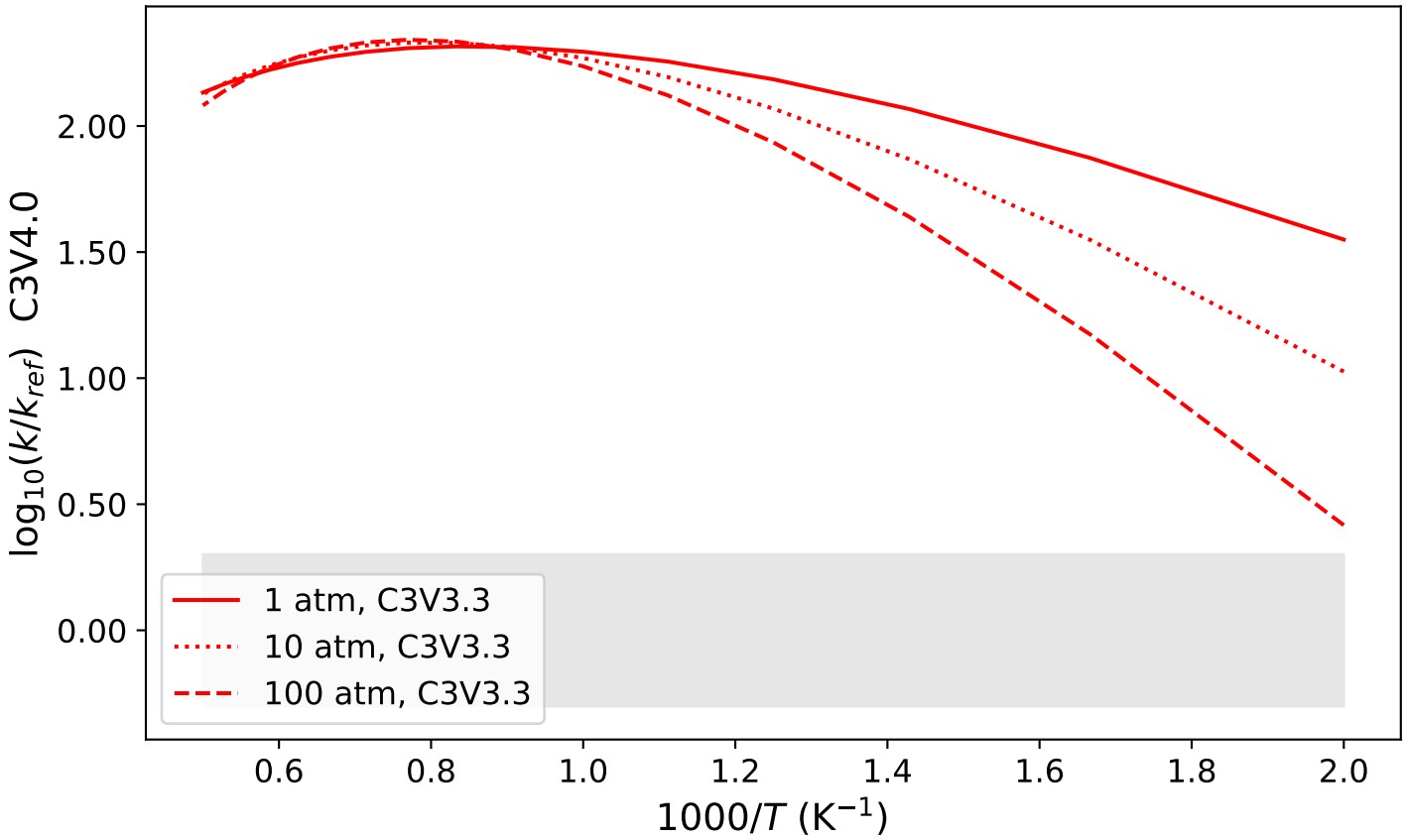
C3H4-A + NAPH-A = H + FLUORENE



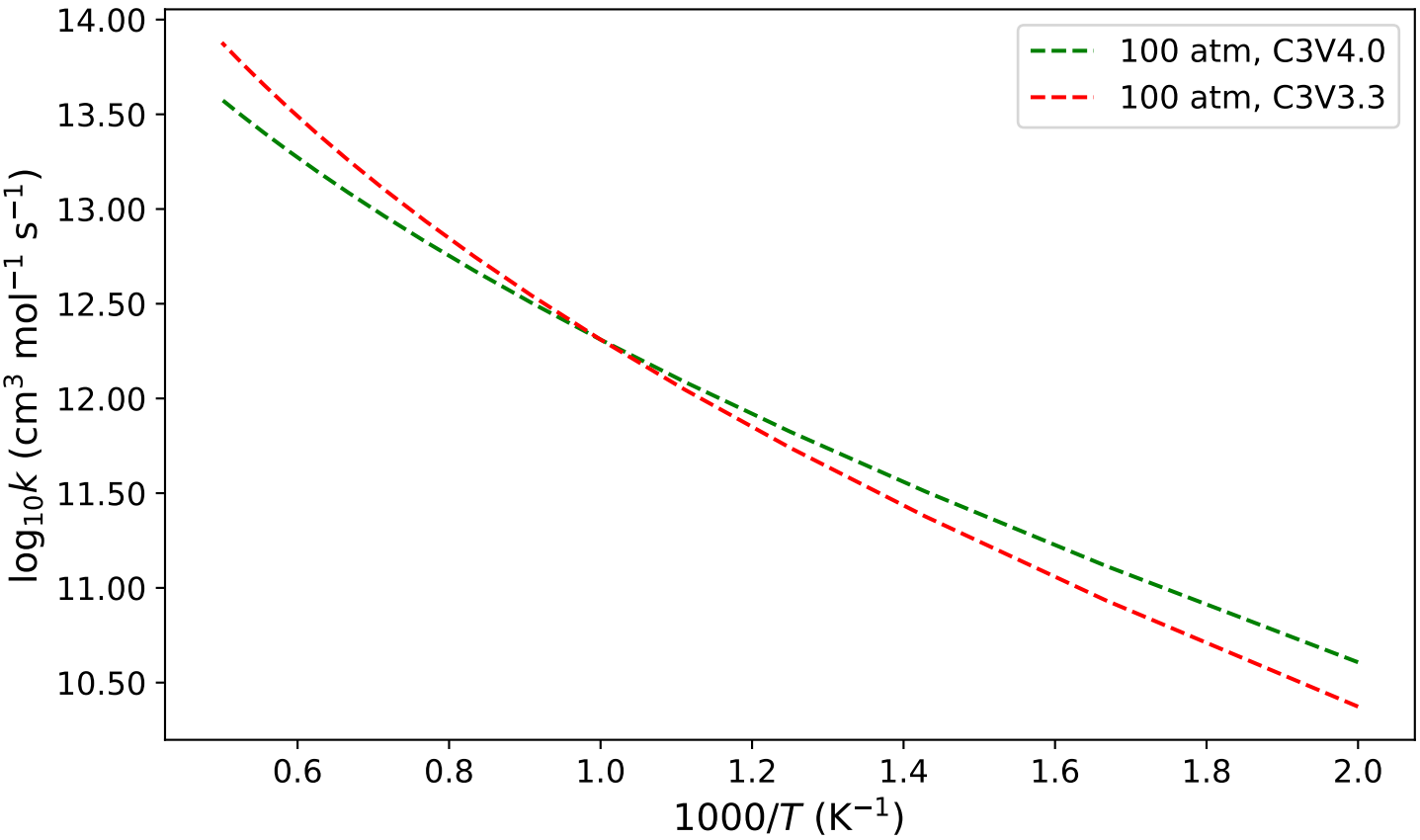


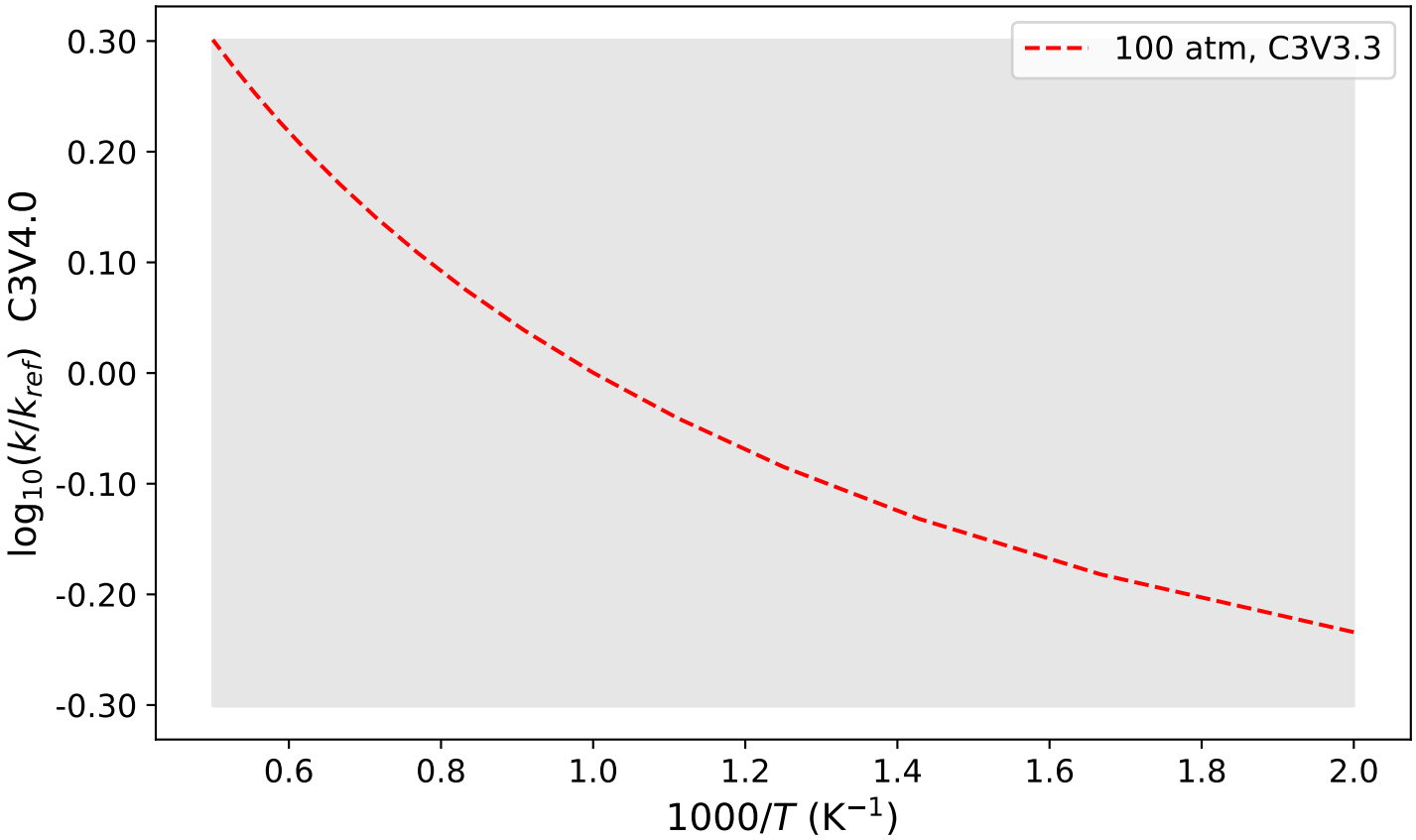
H + C5H6 = C2H3 + C3H4-P



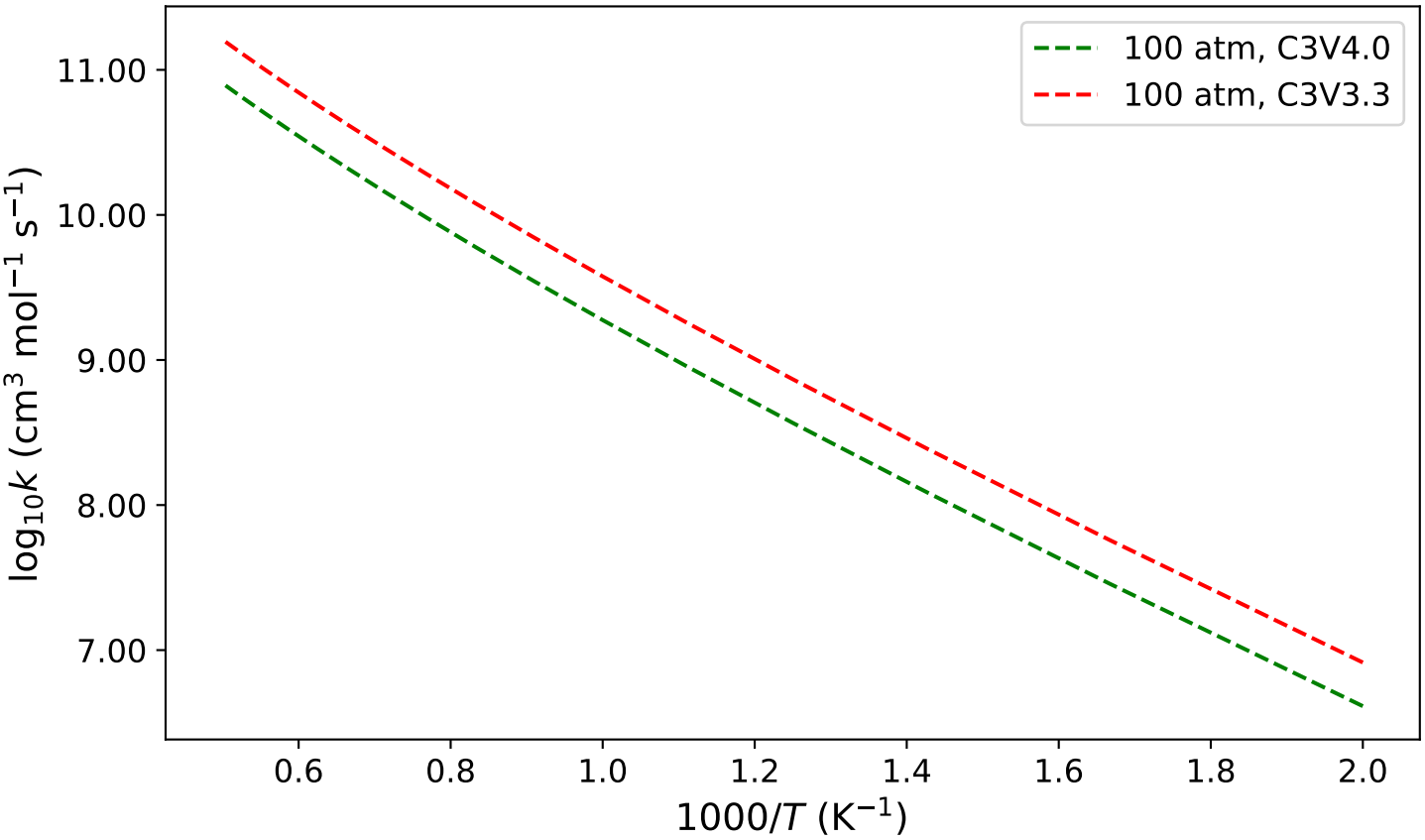


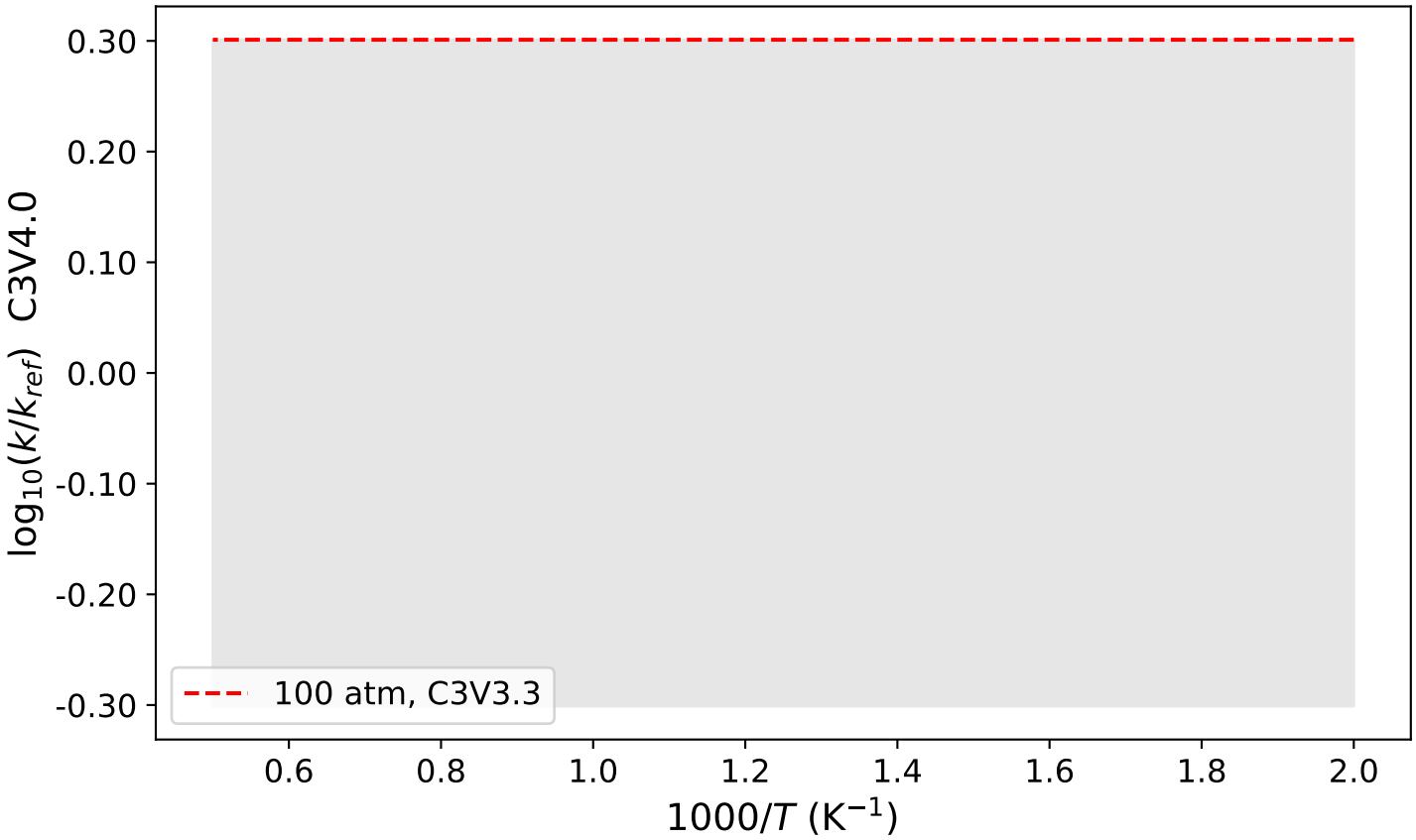
H + XYLENE = H2 + RXYLENE



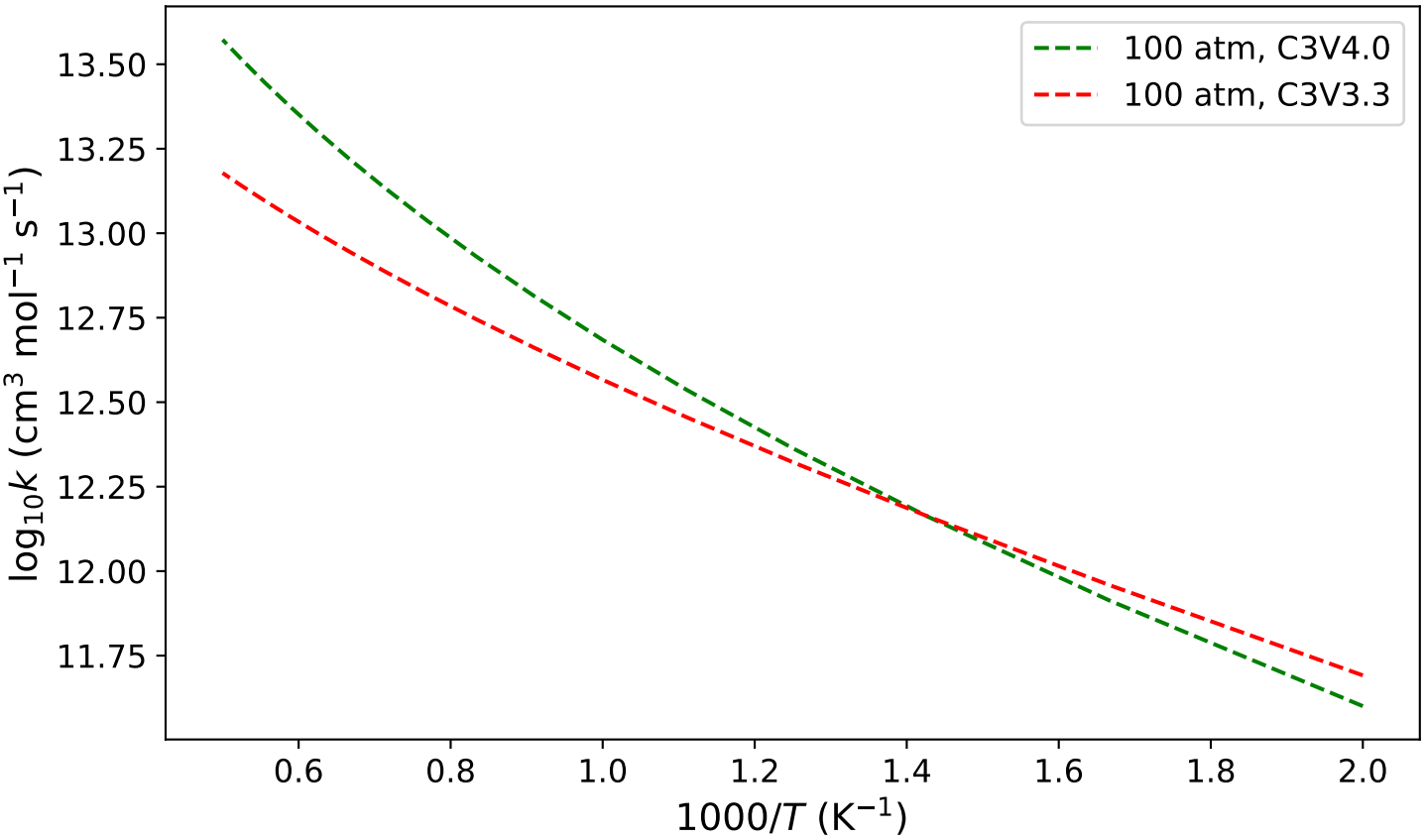


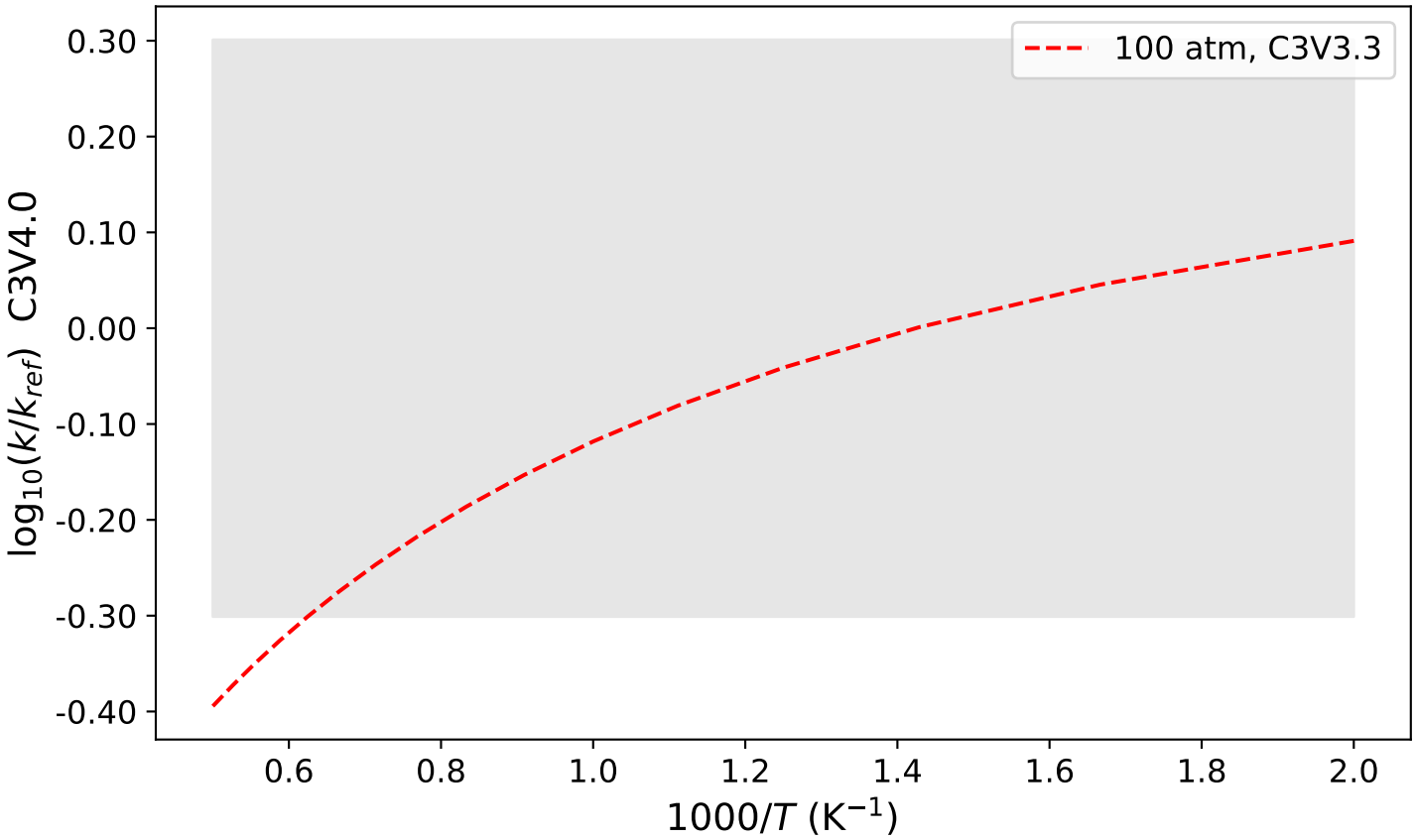
NAPH + C3H5-S = C3H6 + NAPH-A



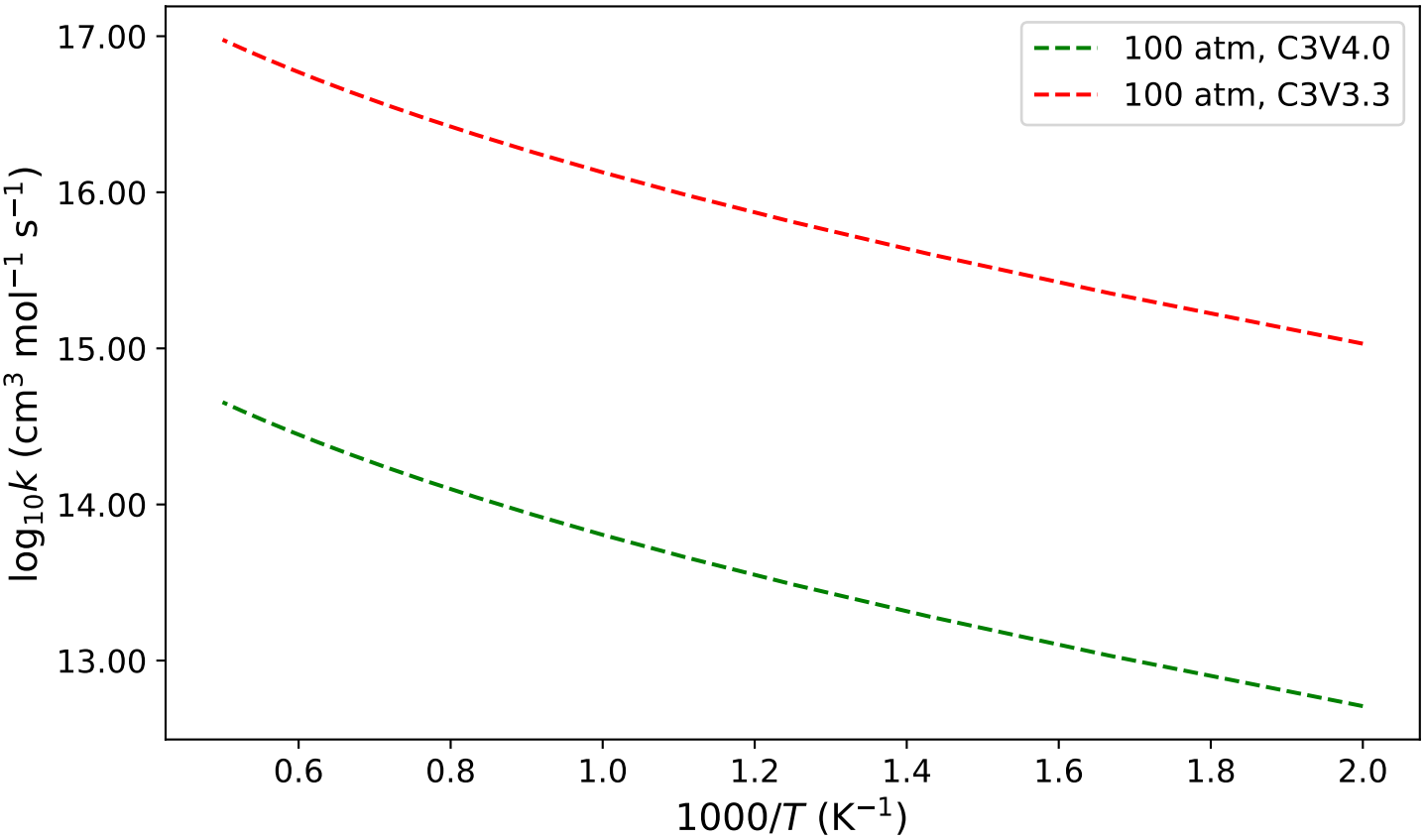


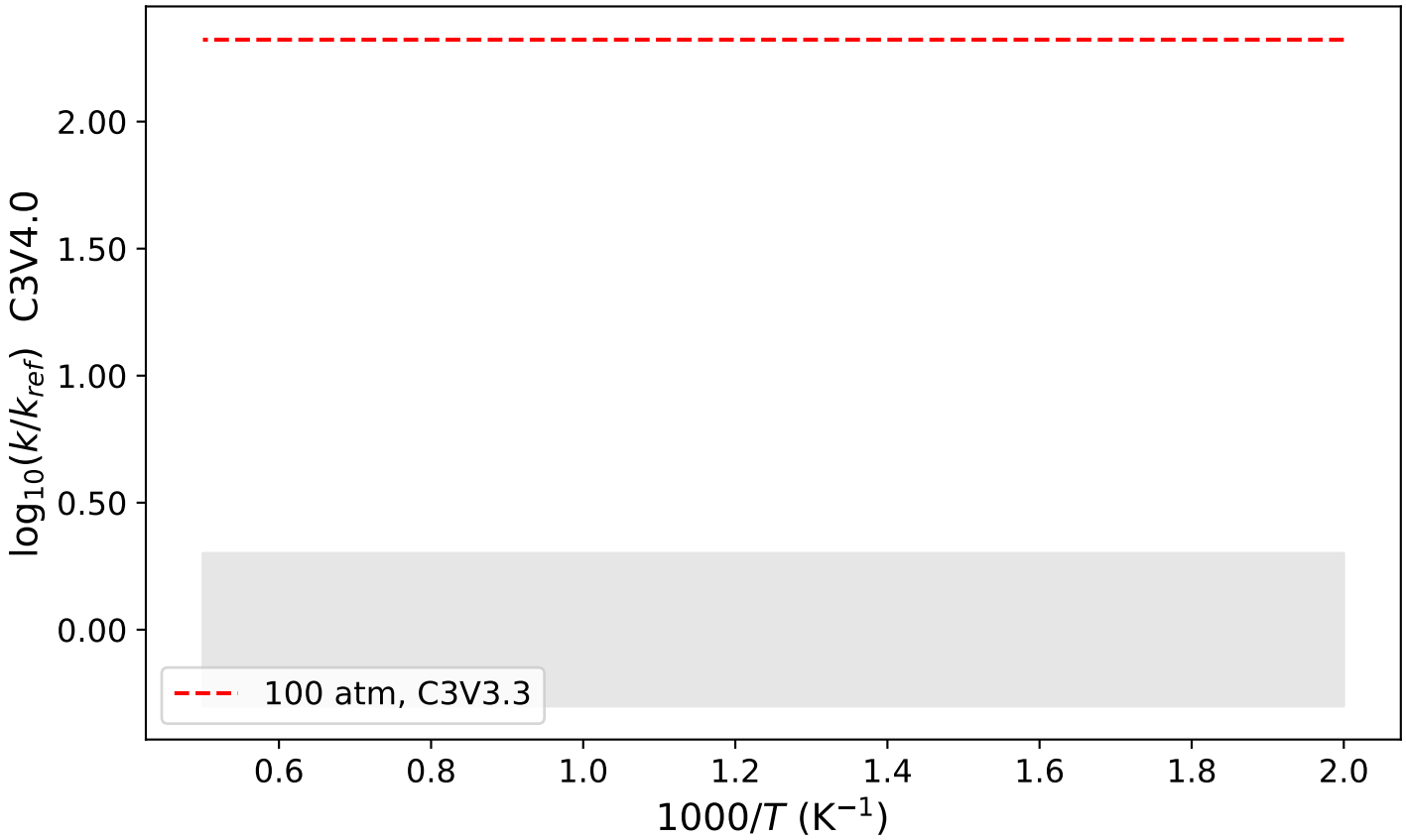
H + C6H4CH3CHO => H2 + CO + C6H4CH3



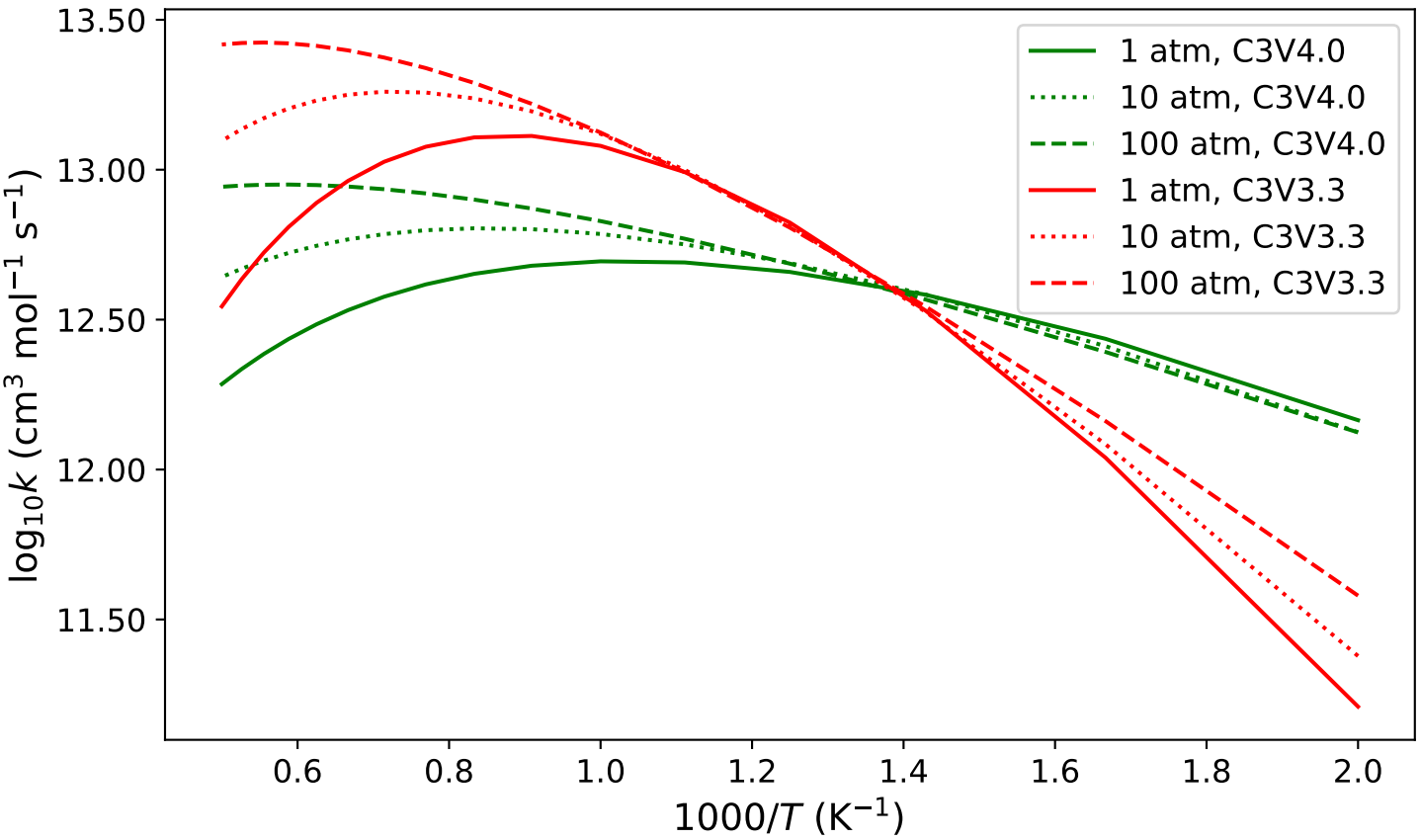


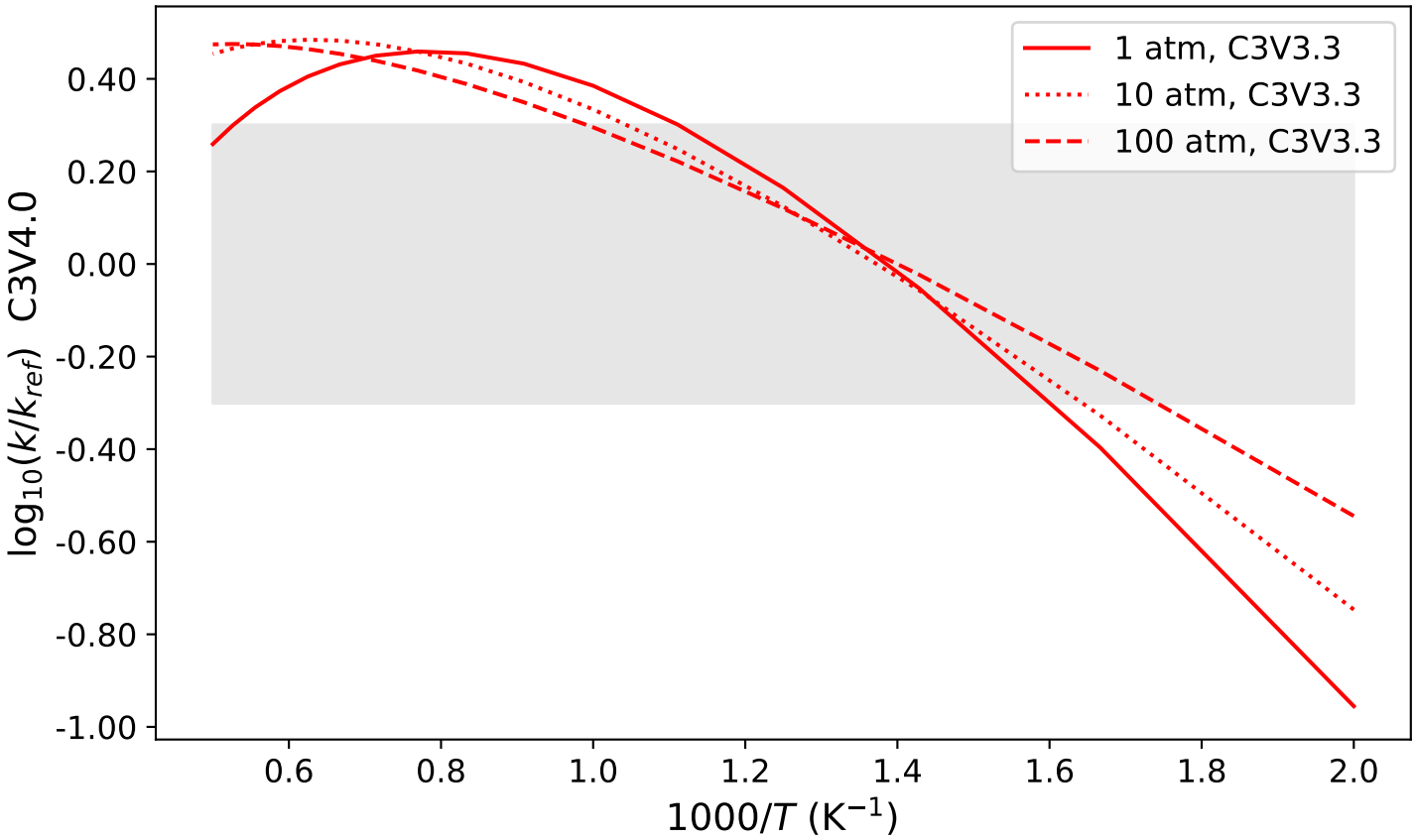
OH + C18H10 = H2O + C18H9



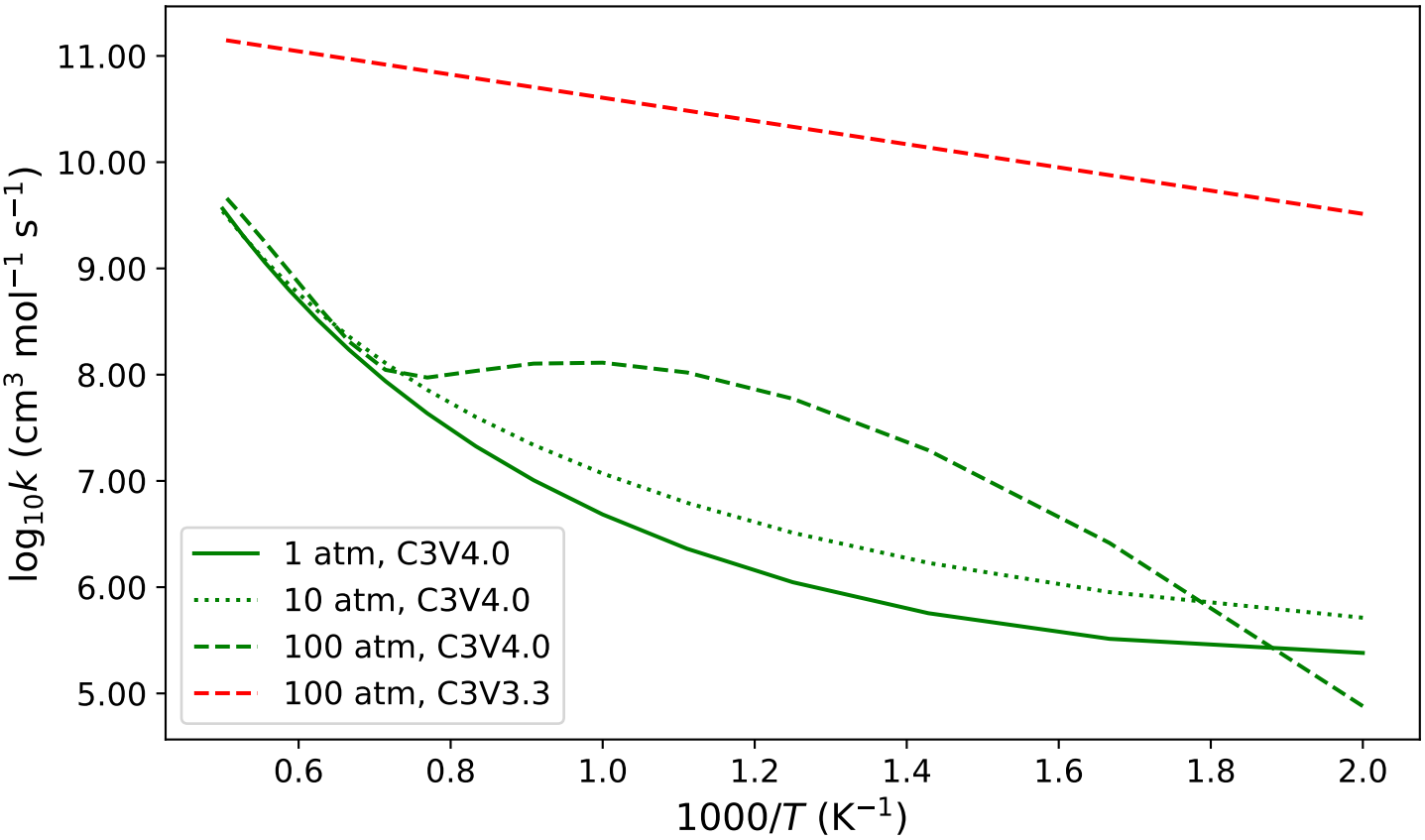


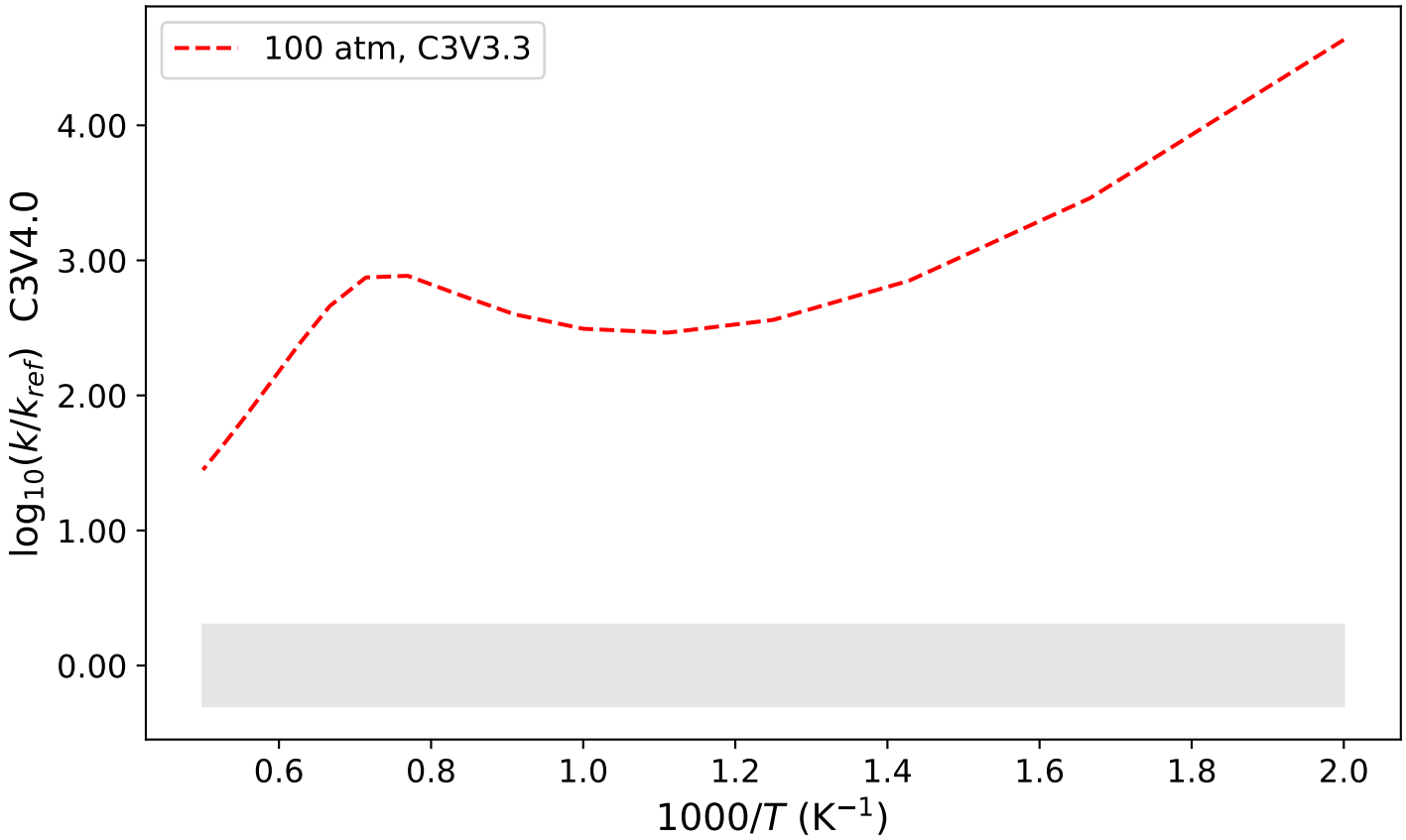
H + C5H6 = CYC5H71-3



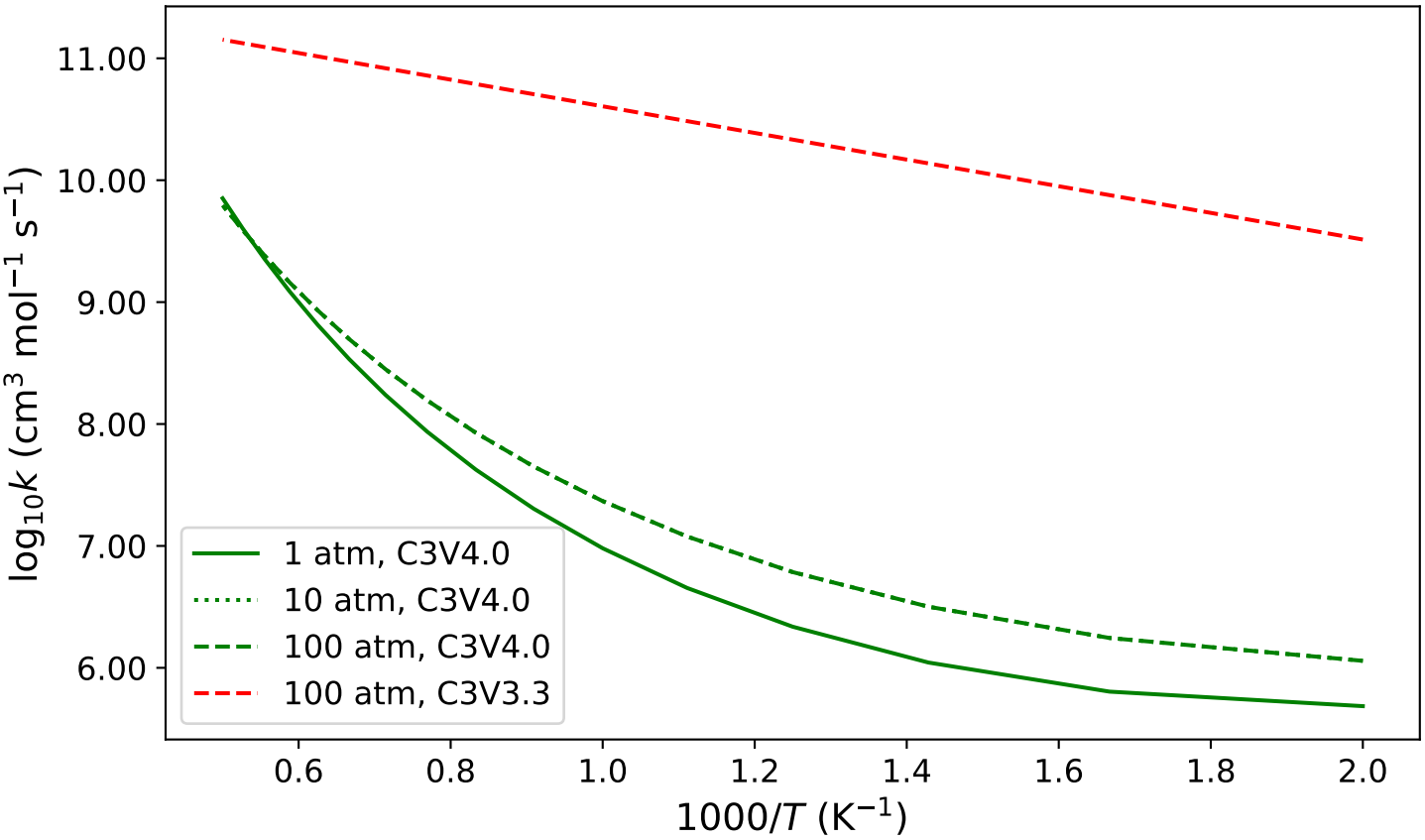


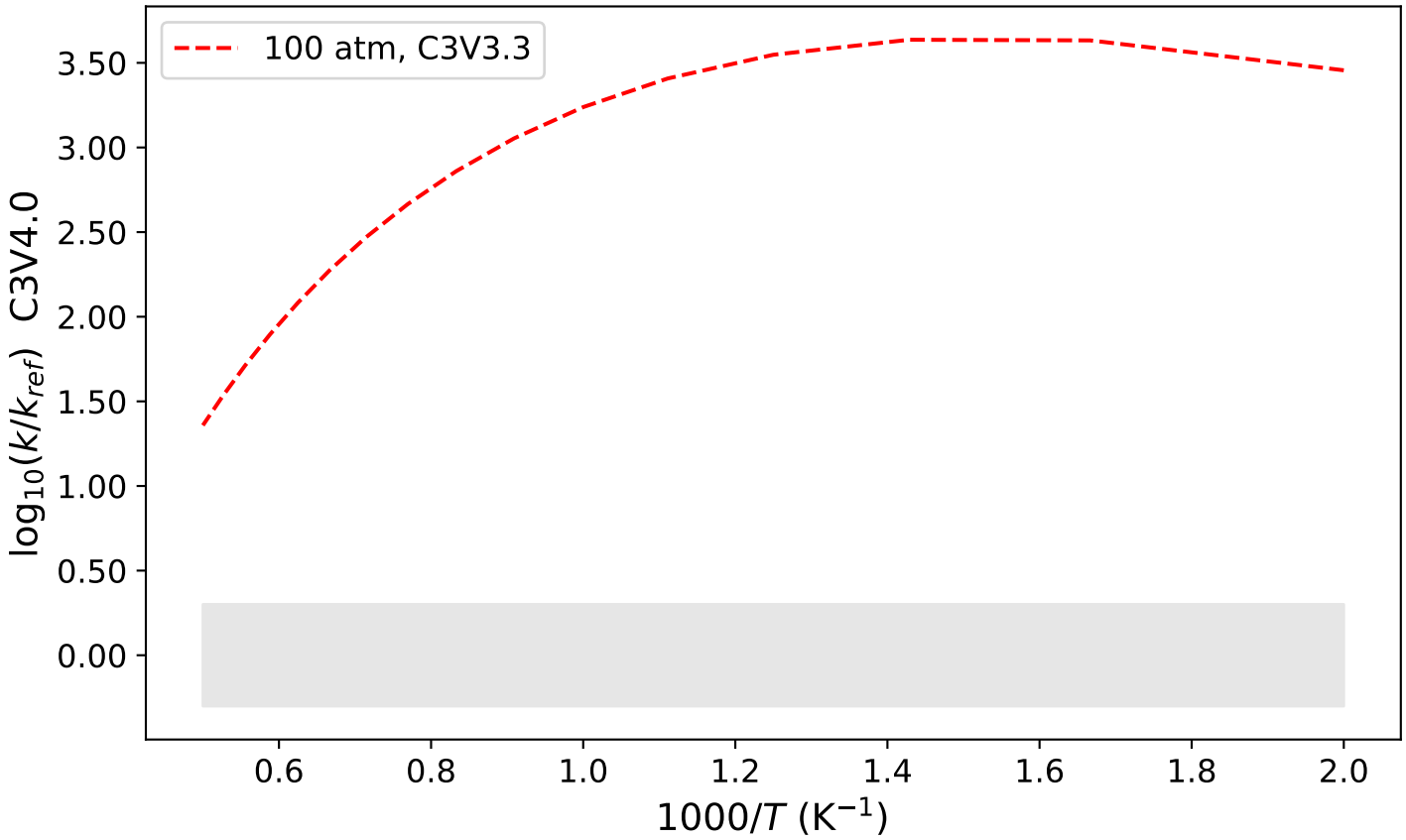
C5H6 + C4H3-I = H + IND



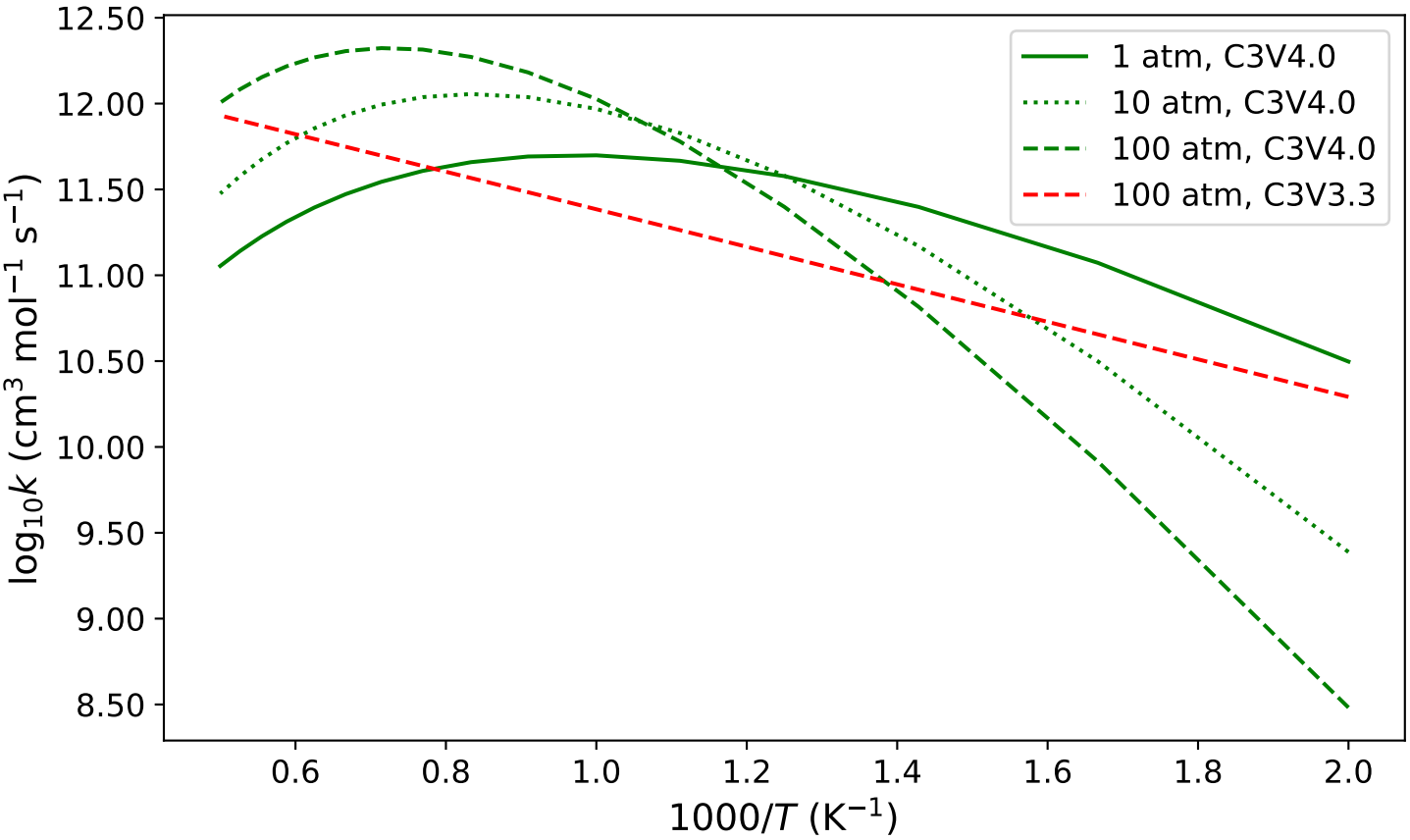


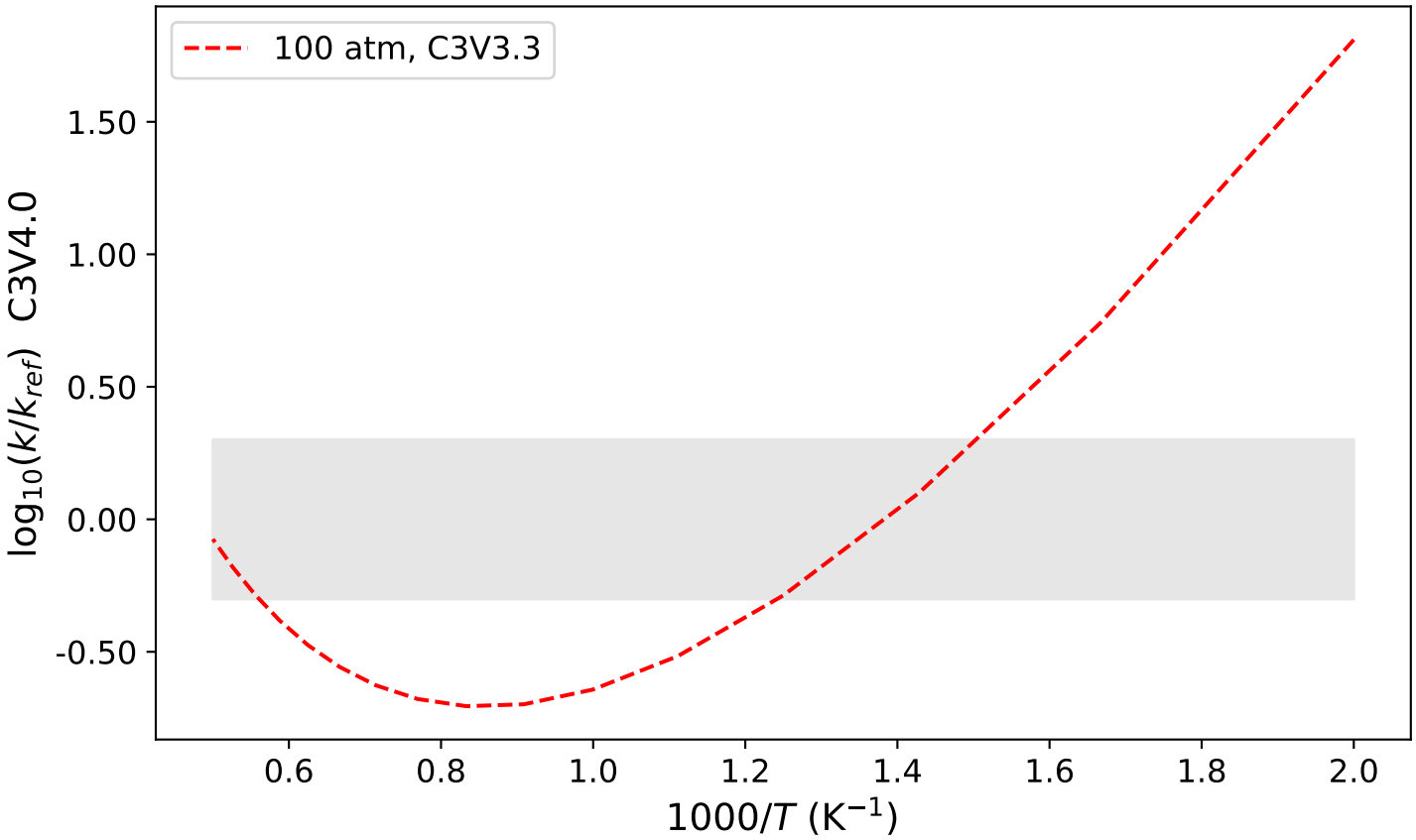
C6H6 + C4H3-I = H + NAPH



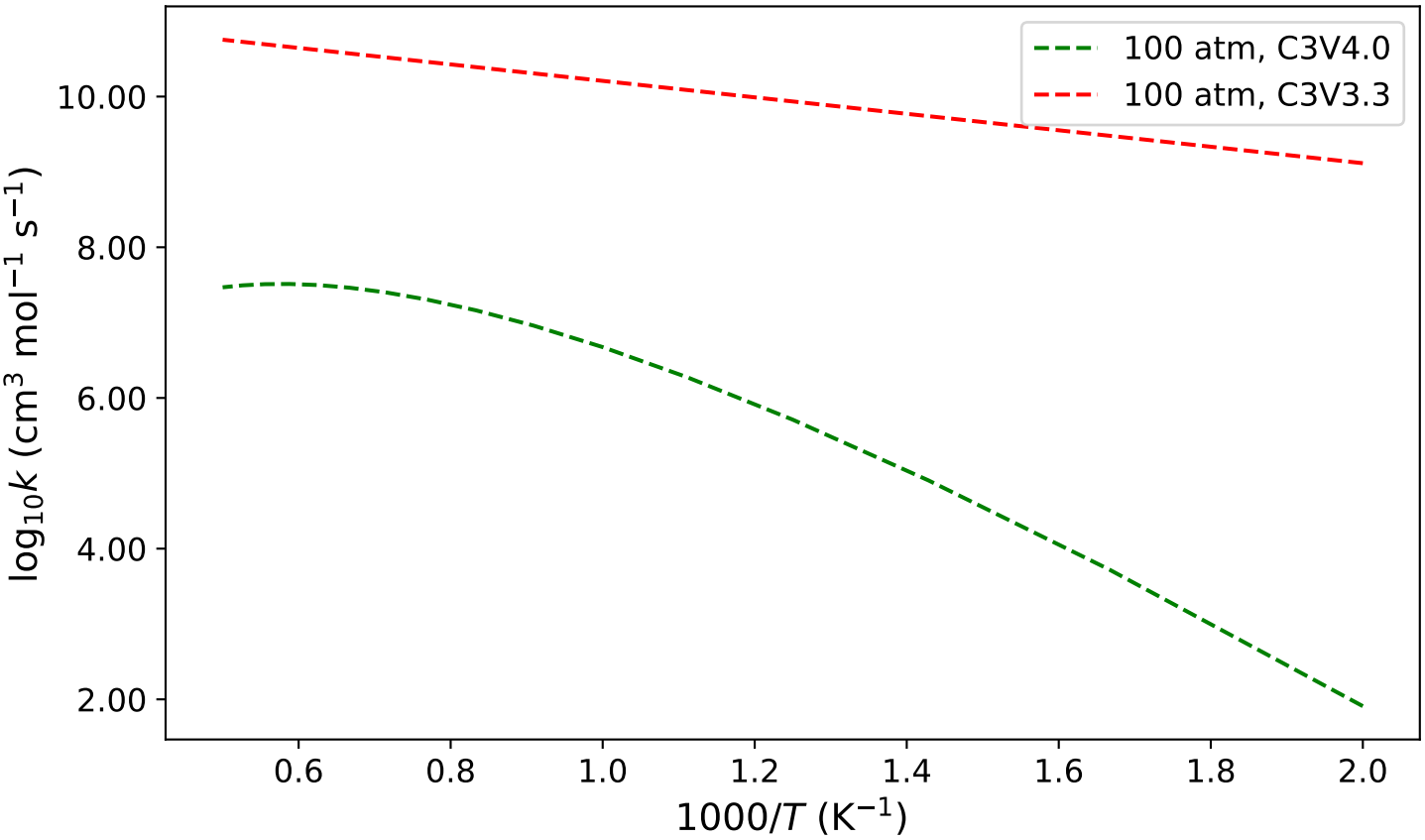


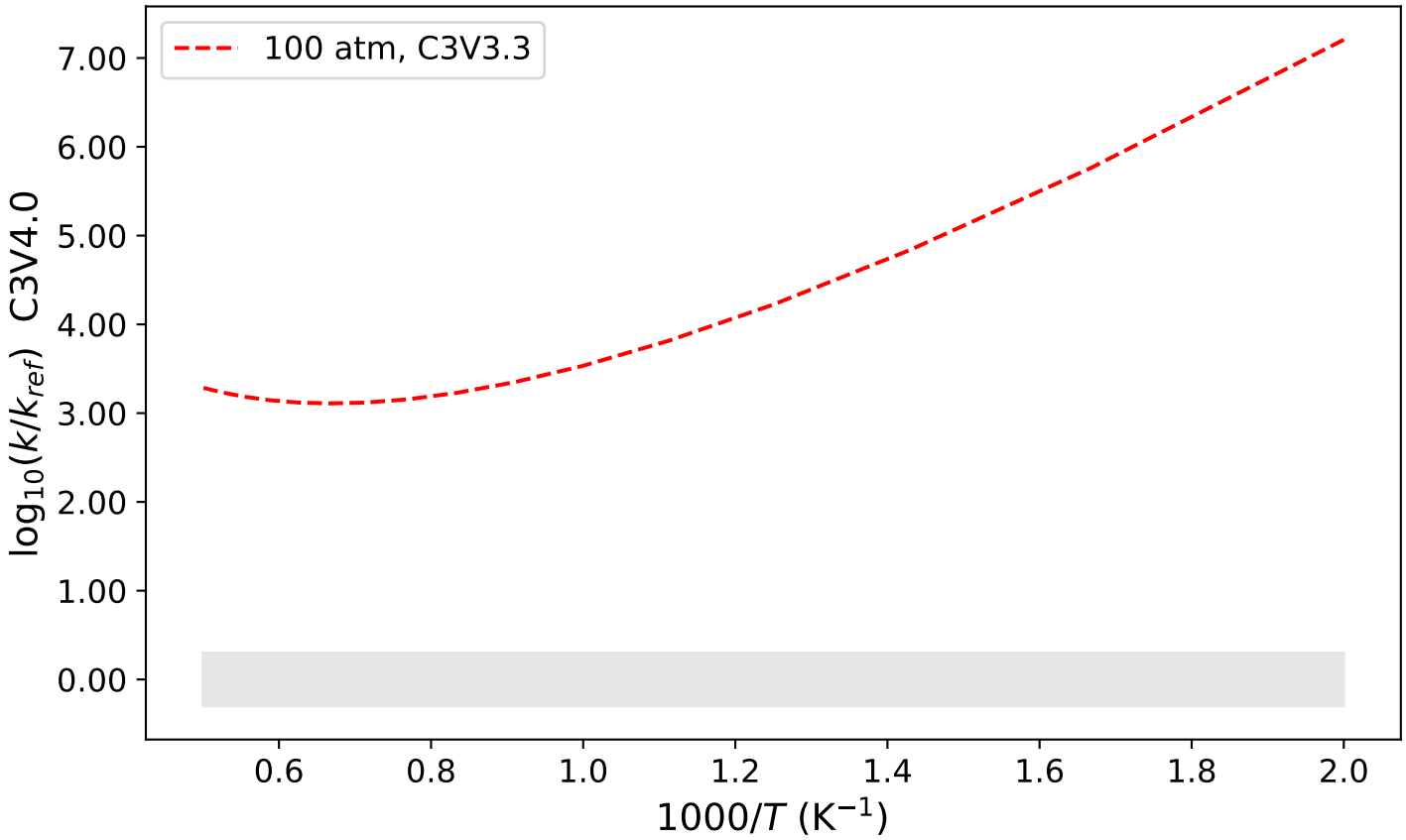
C6H5 + C3H4-A = H + IND



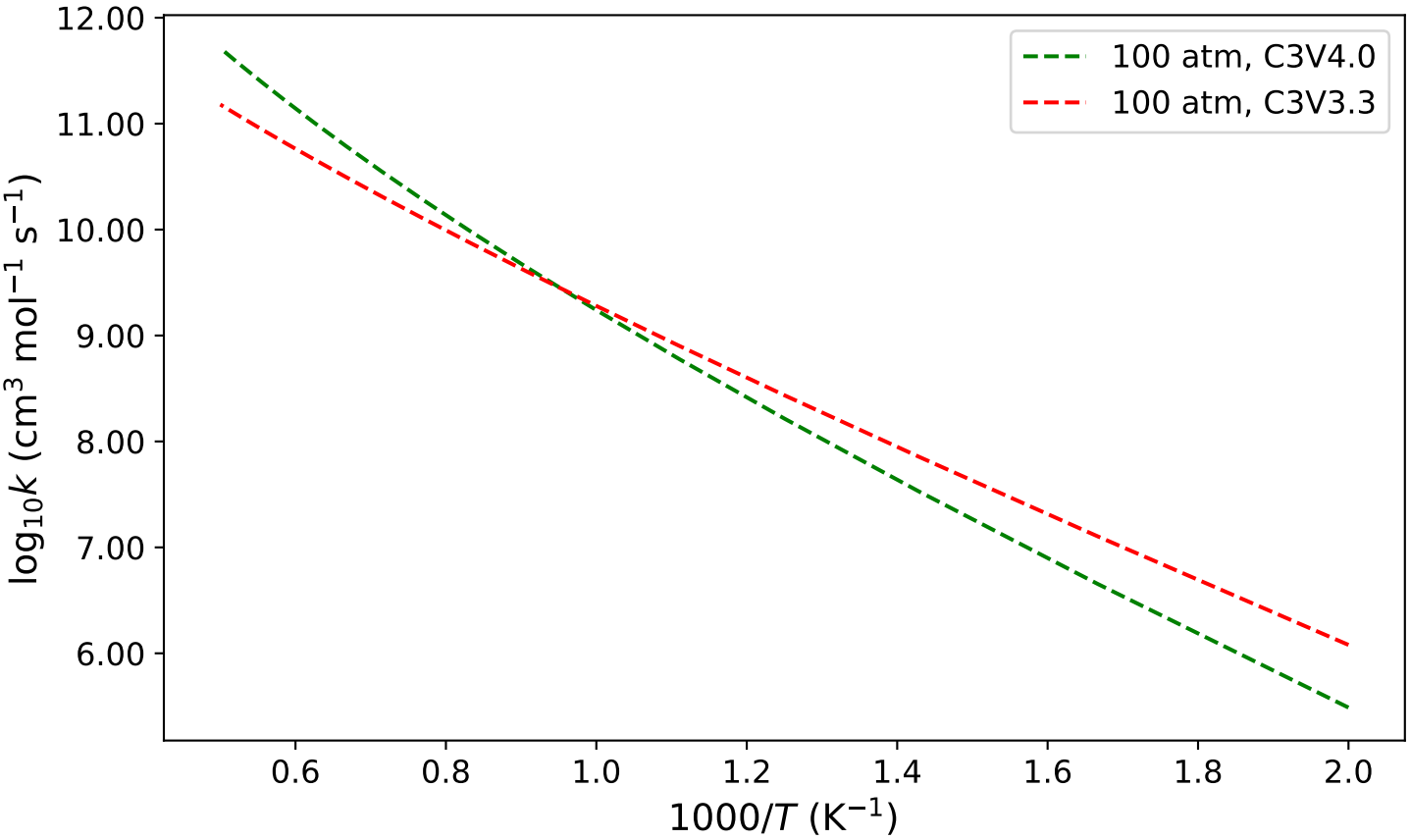


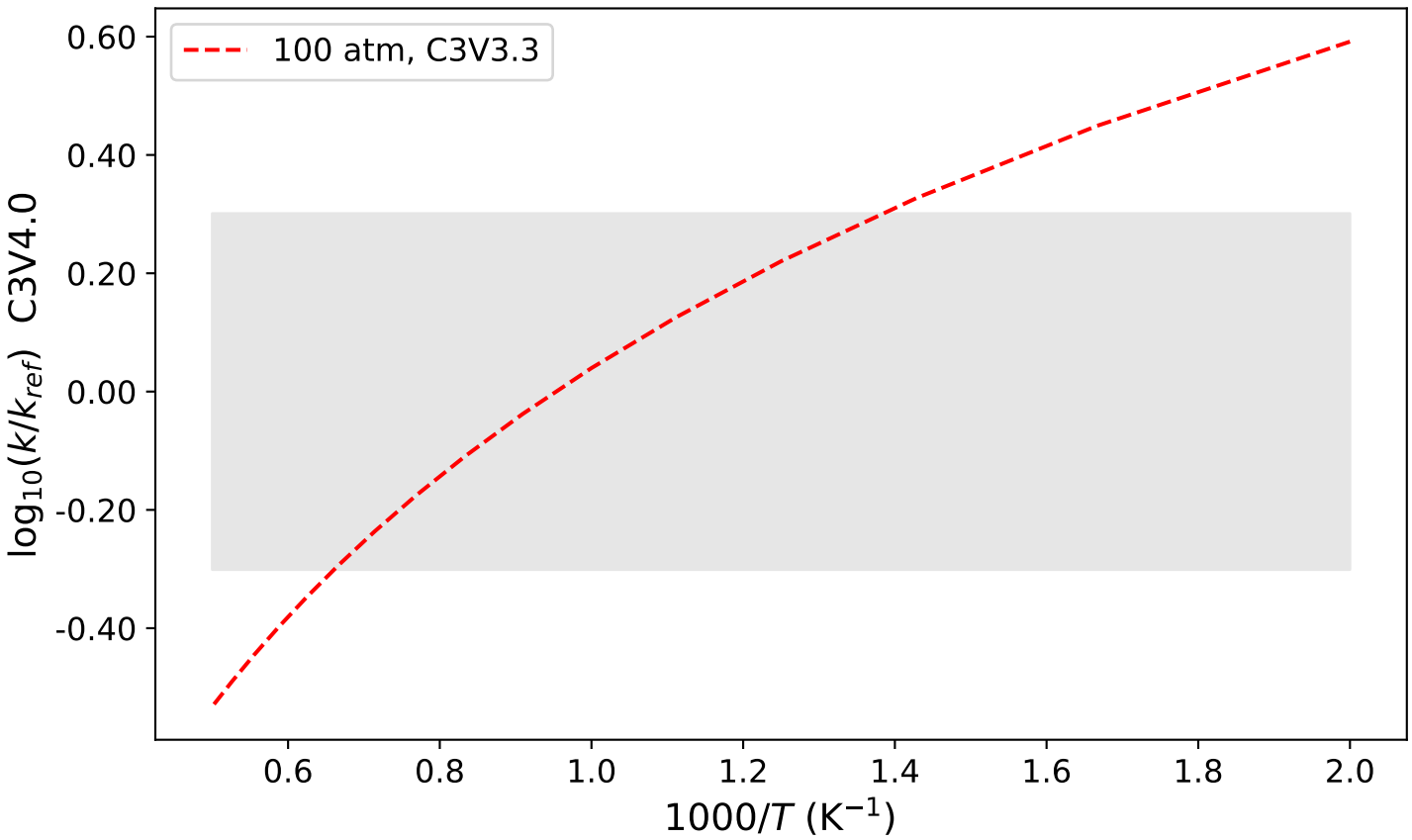
C3H3 + C4H6 = H + C6H5CH3



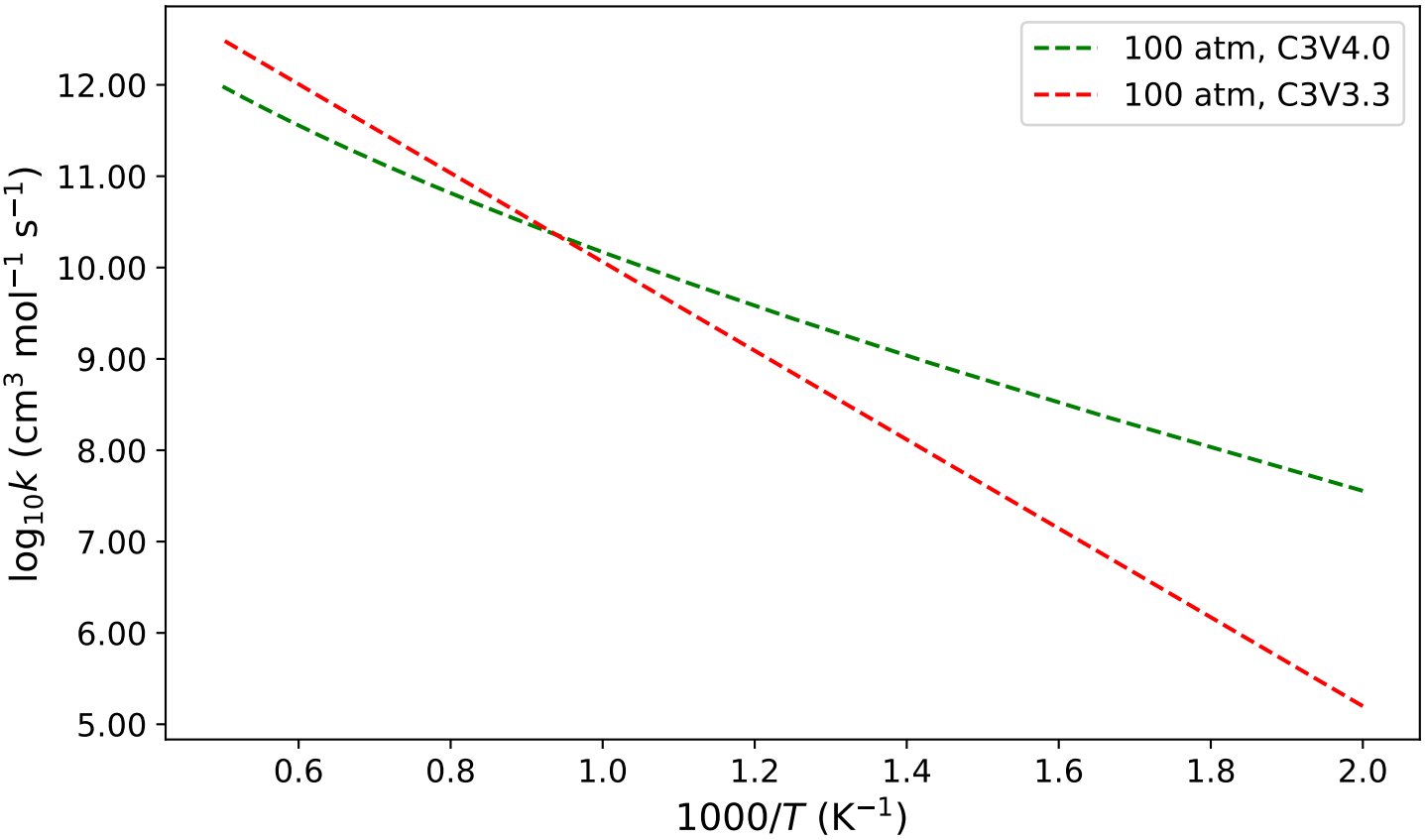


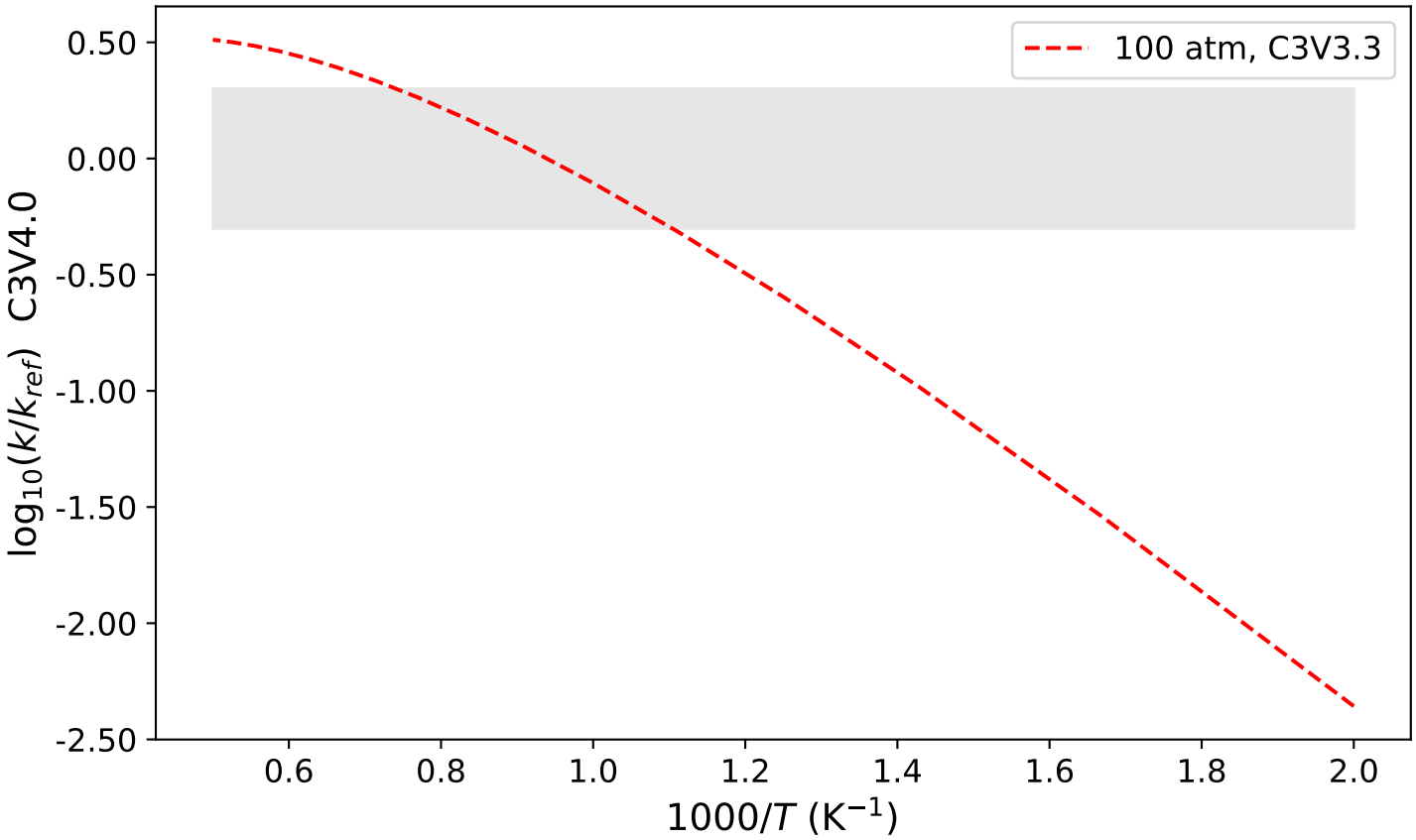
HO2 + CYC5H8 = H2O2 + CYC5H71-3



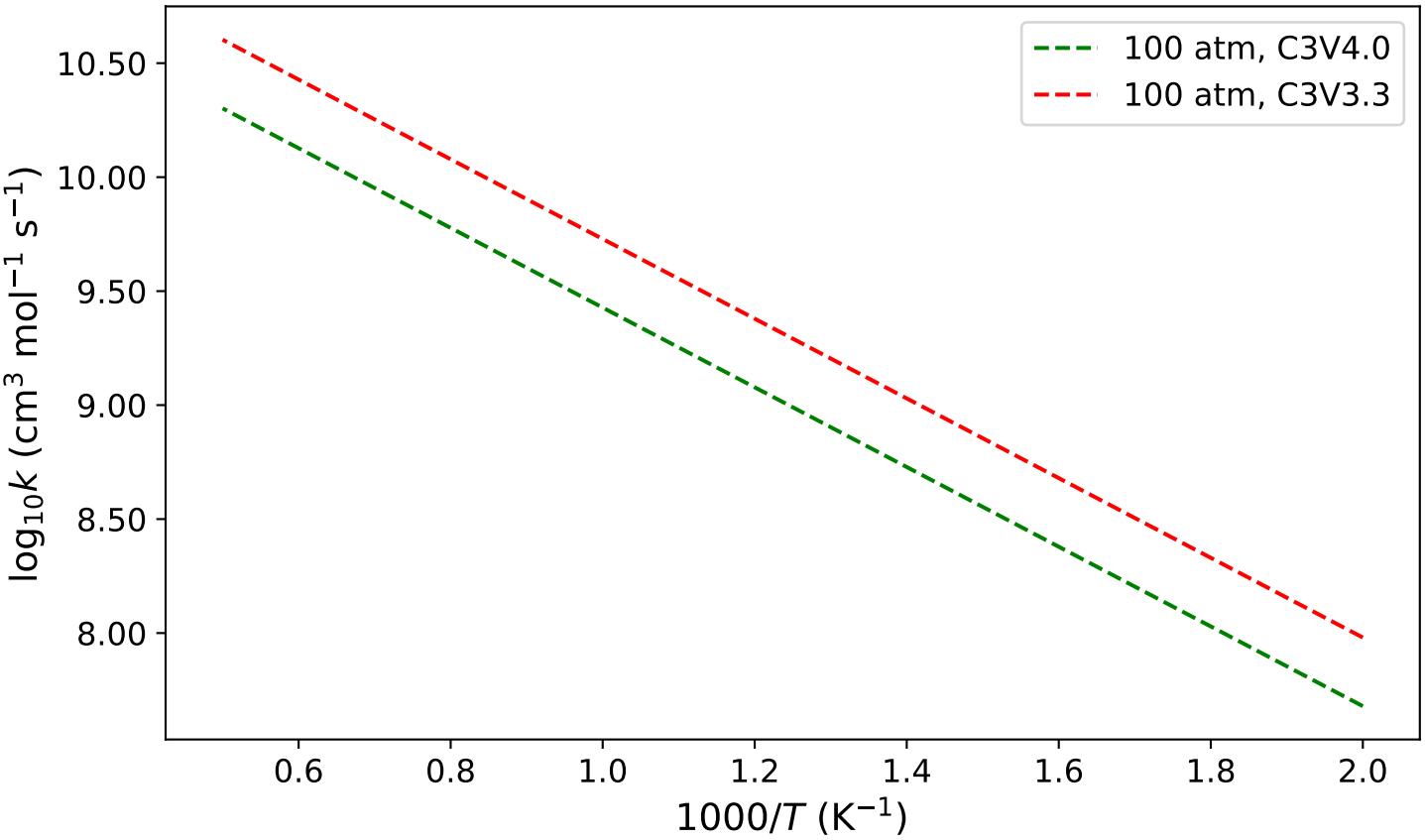


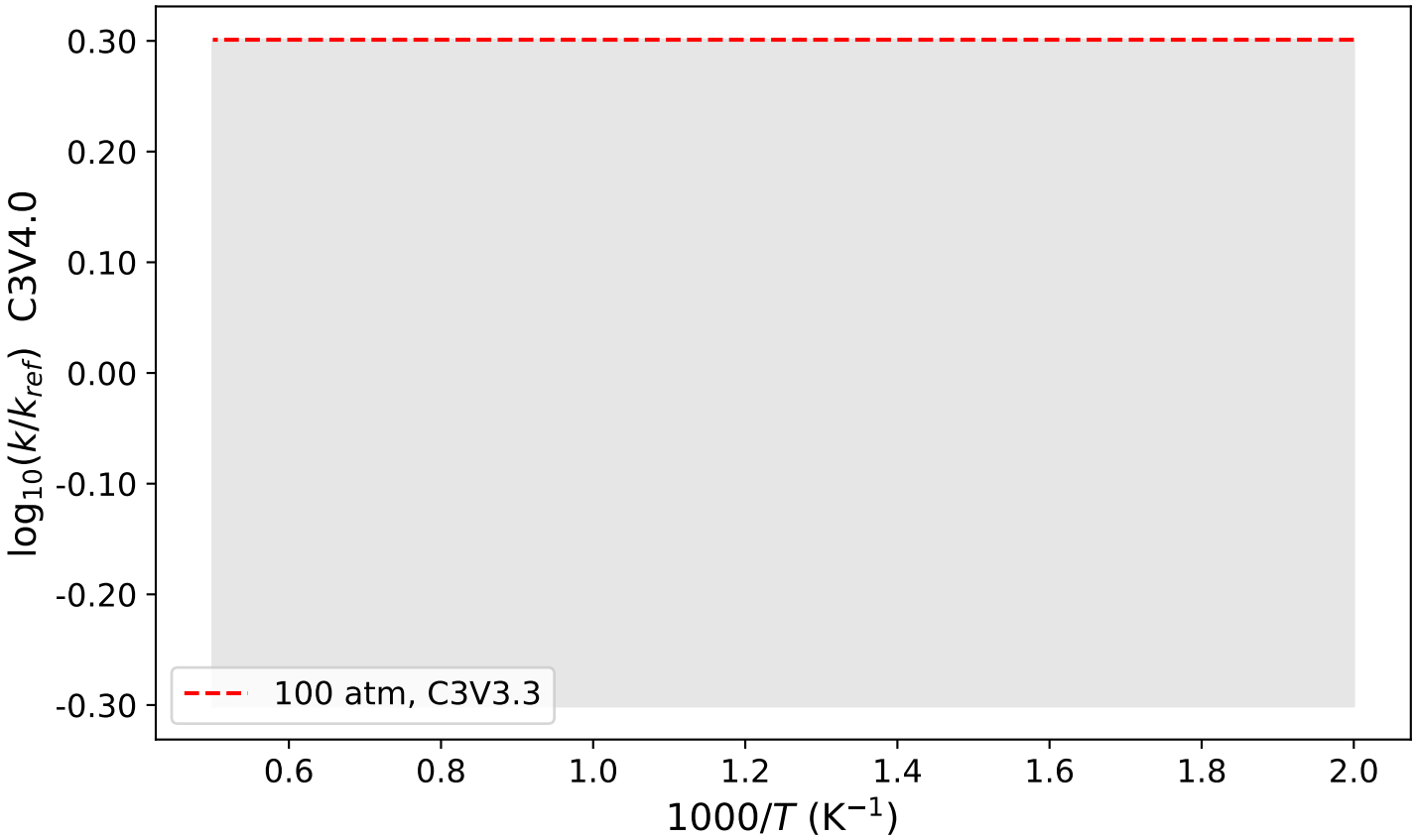
CH3 + XYLENE = CH4 + RXYLENE



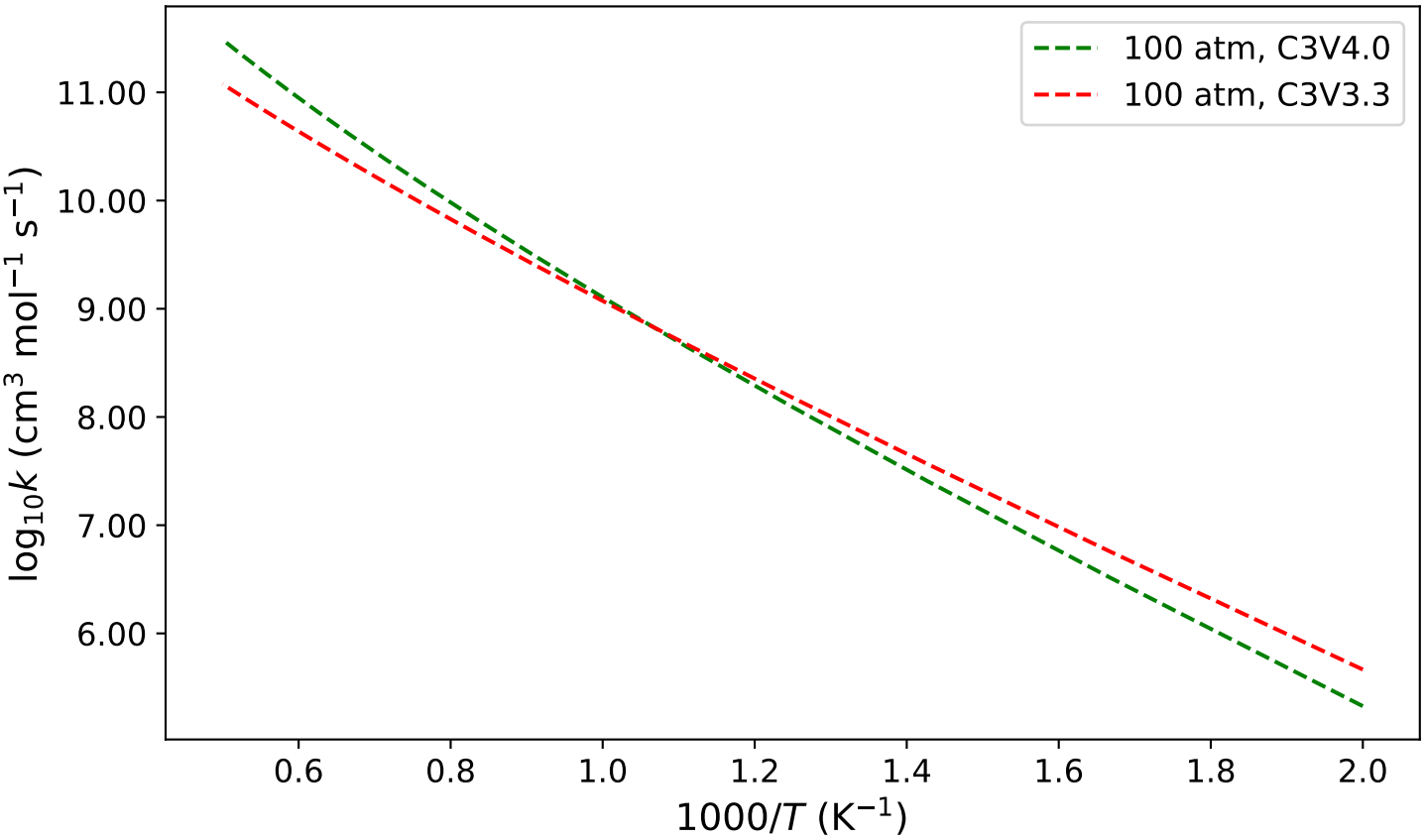


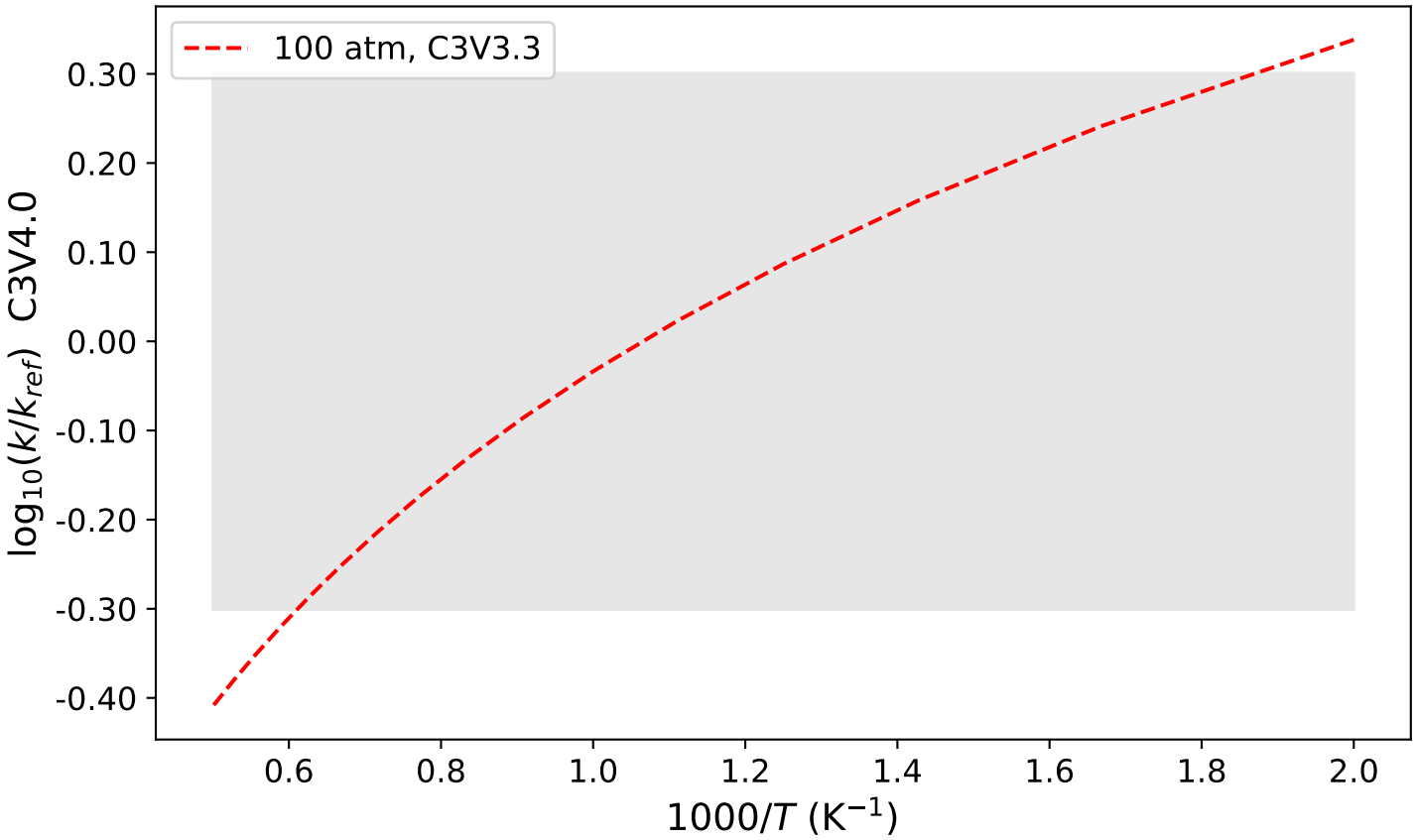
C6H5C2H + C2H = NAPH-A



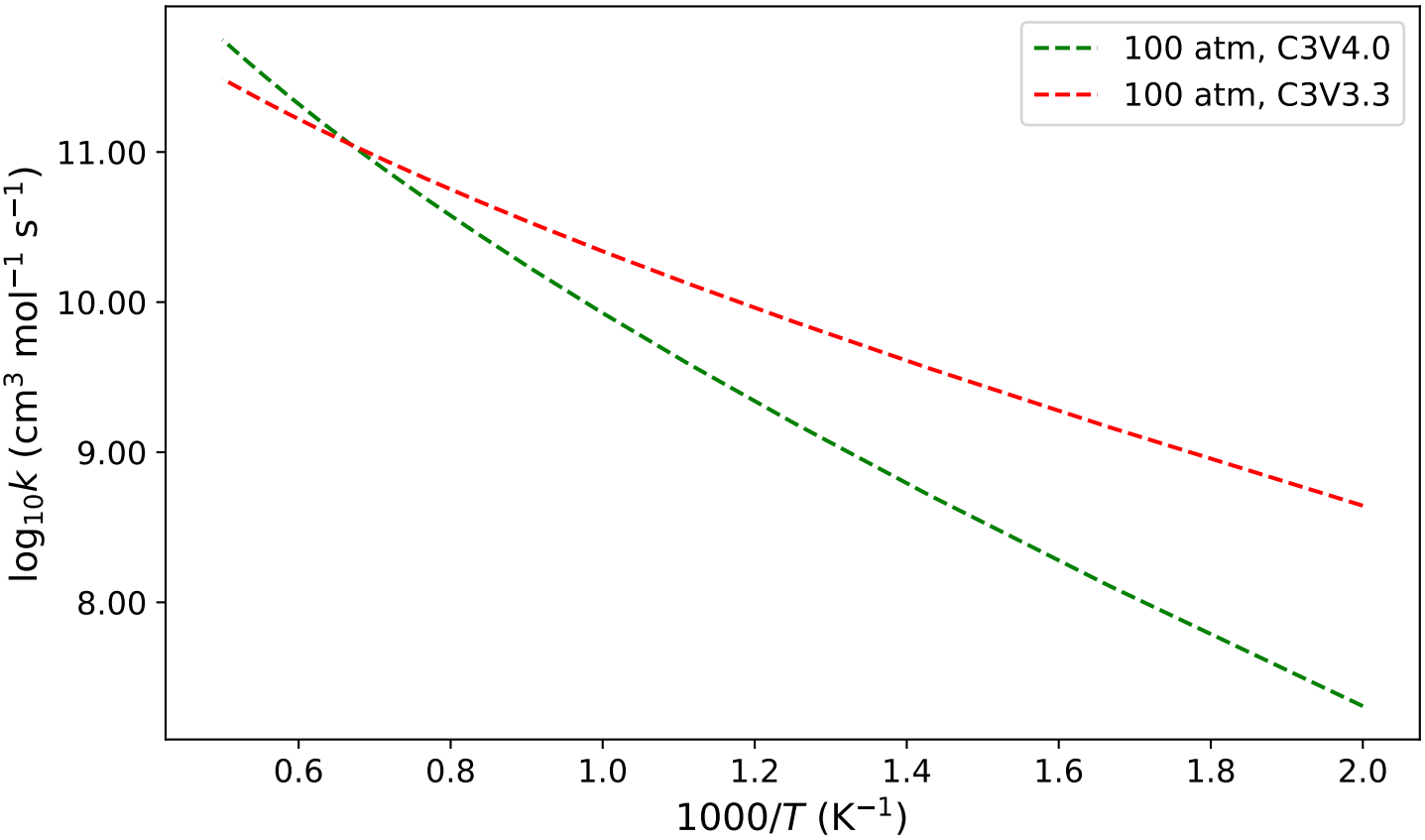


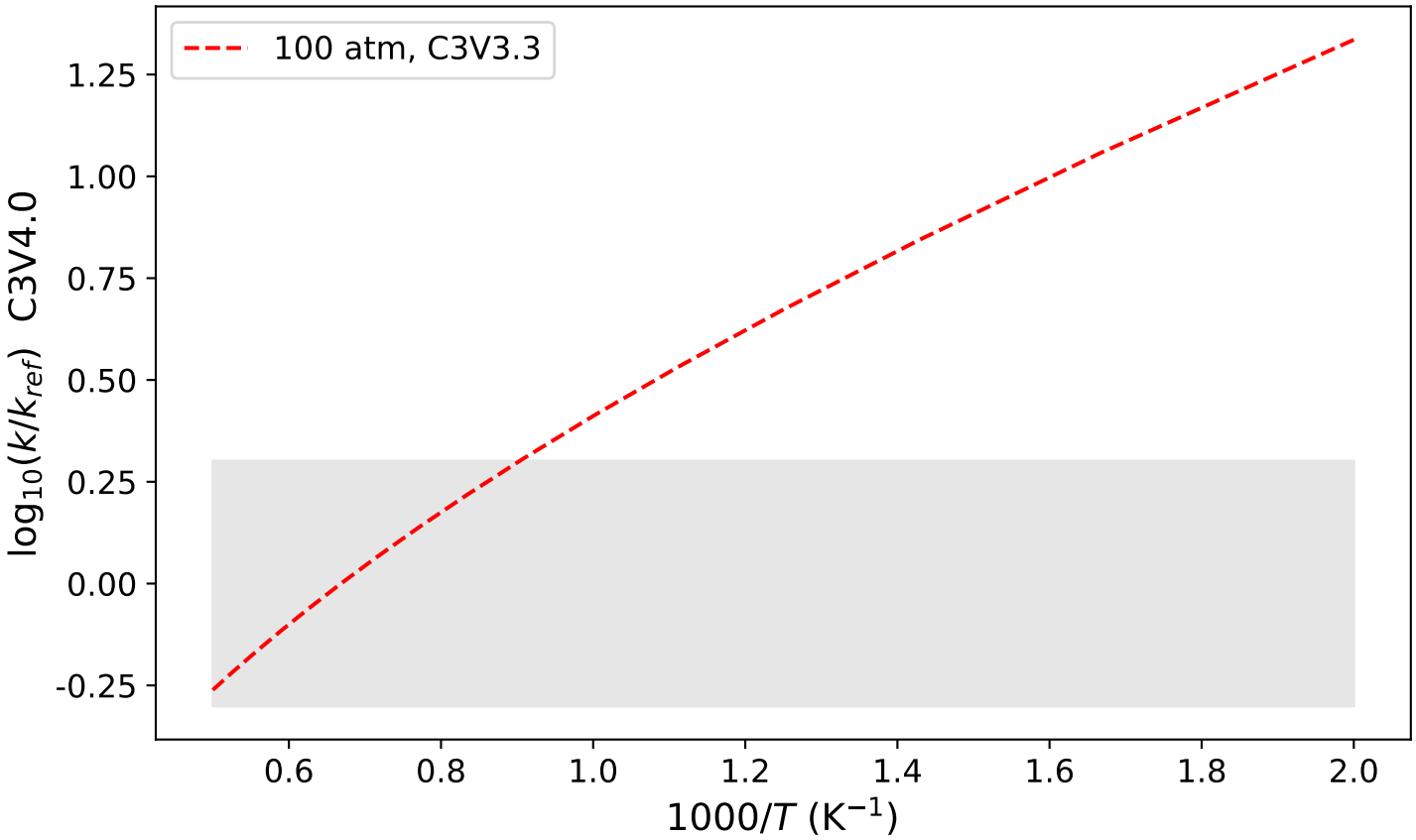
C3H5-A + CYC5H8 = C3H6 + CYC5H71-3



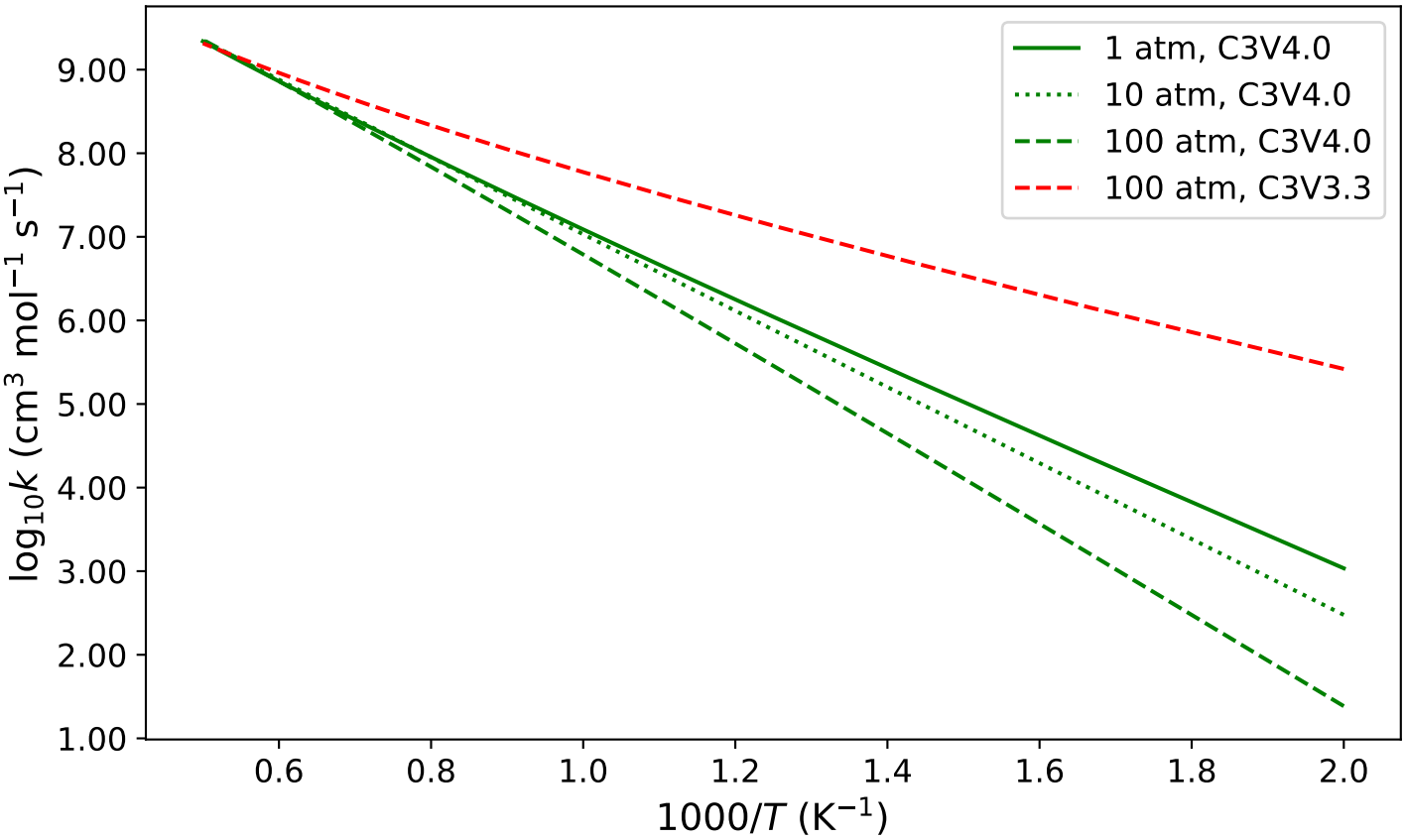


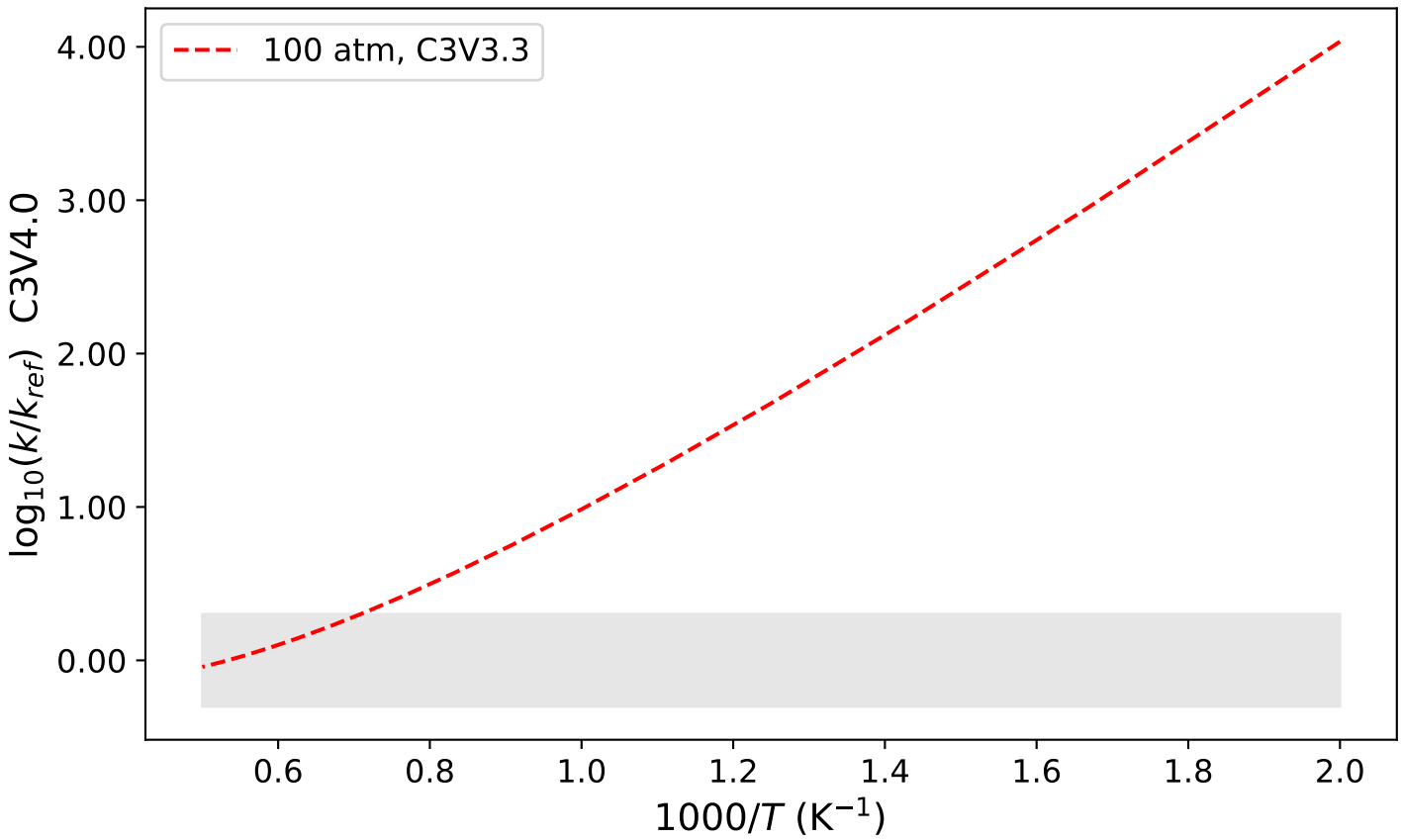
CH3 + C14H14 = CH4 + C14H13



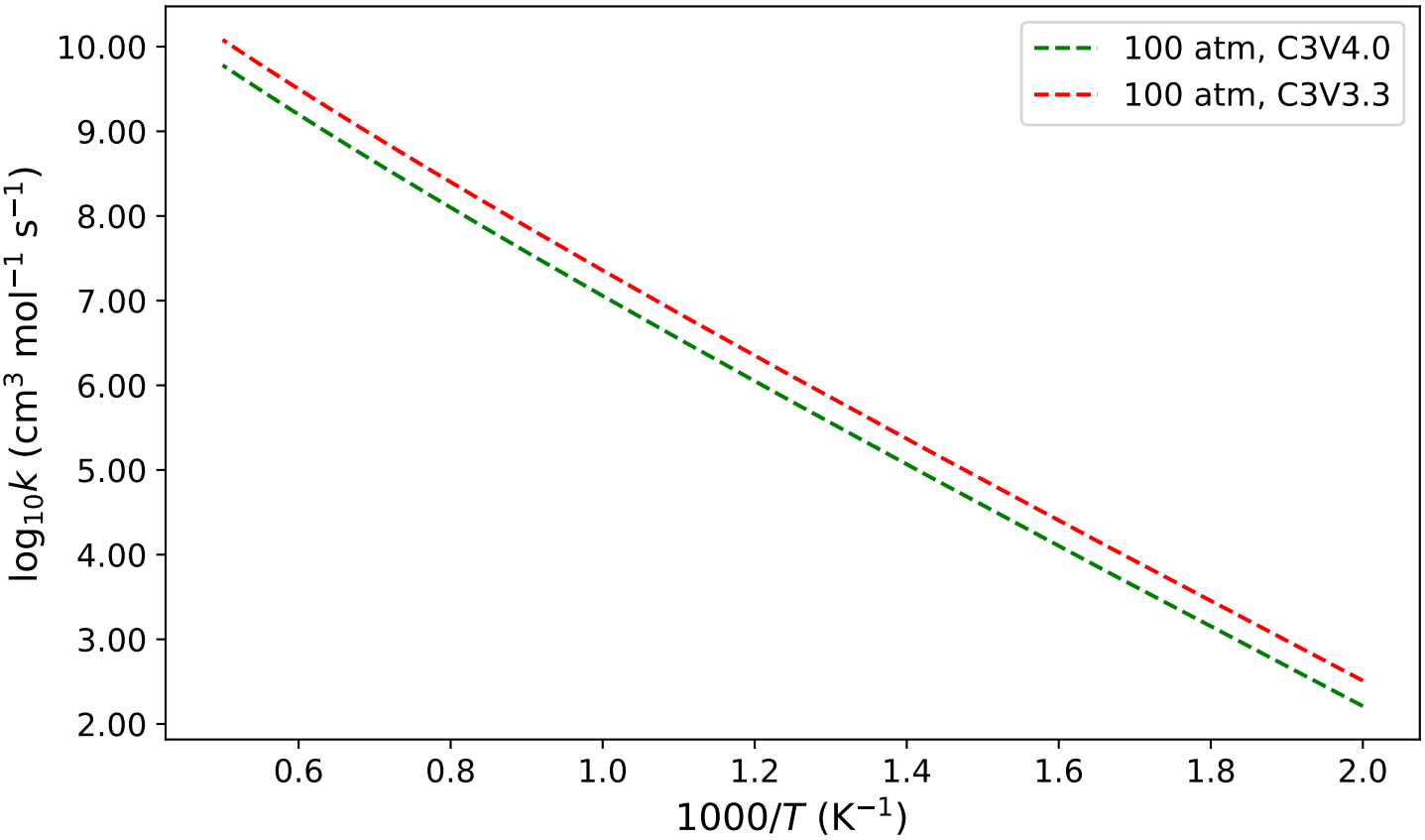


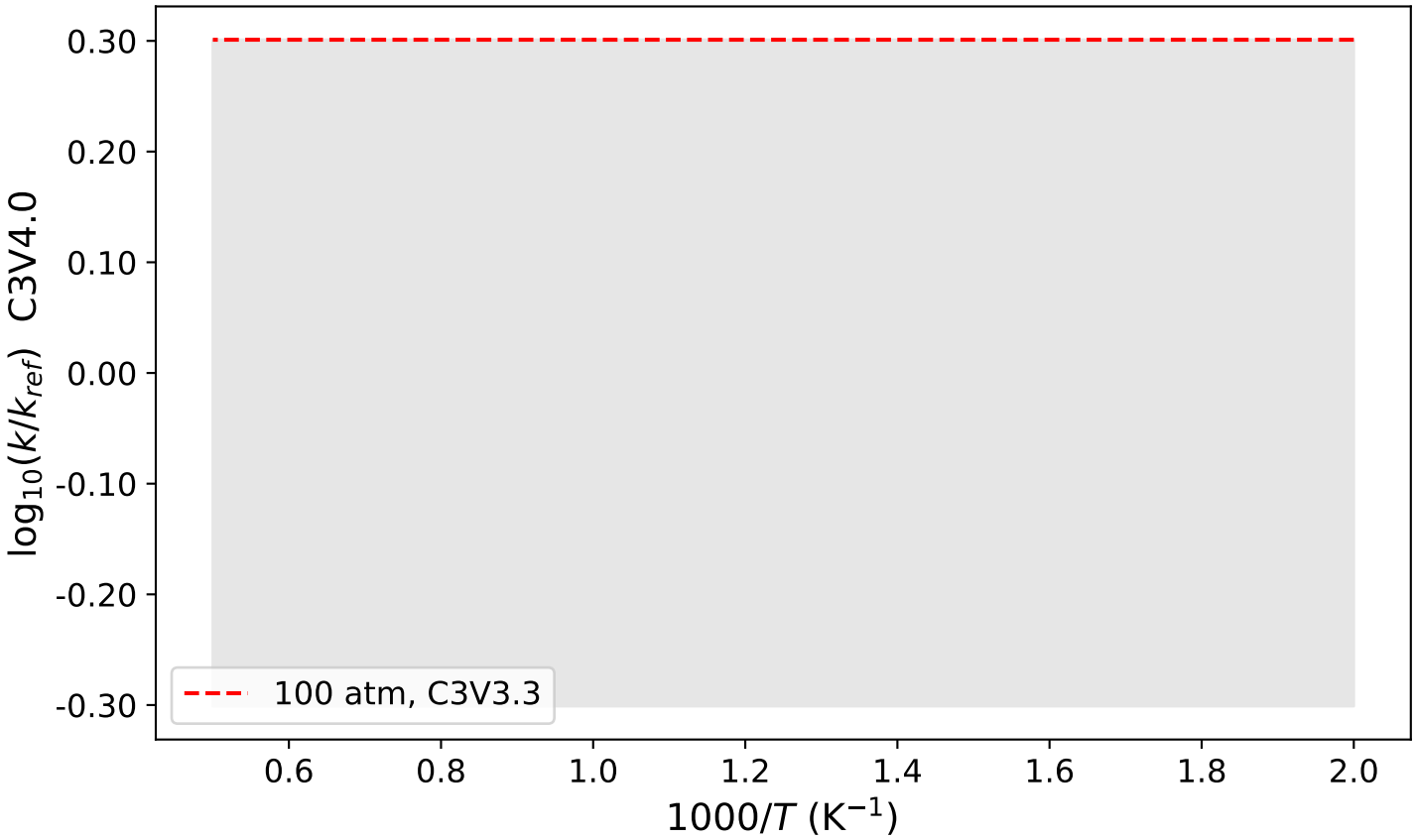
O2 + RXYLENE = OH + C6H4CH3CHO



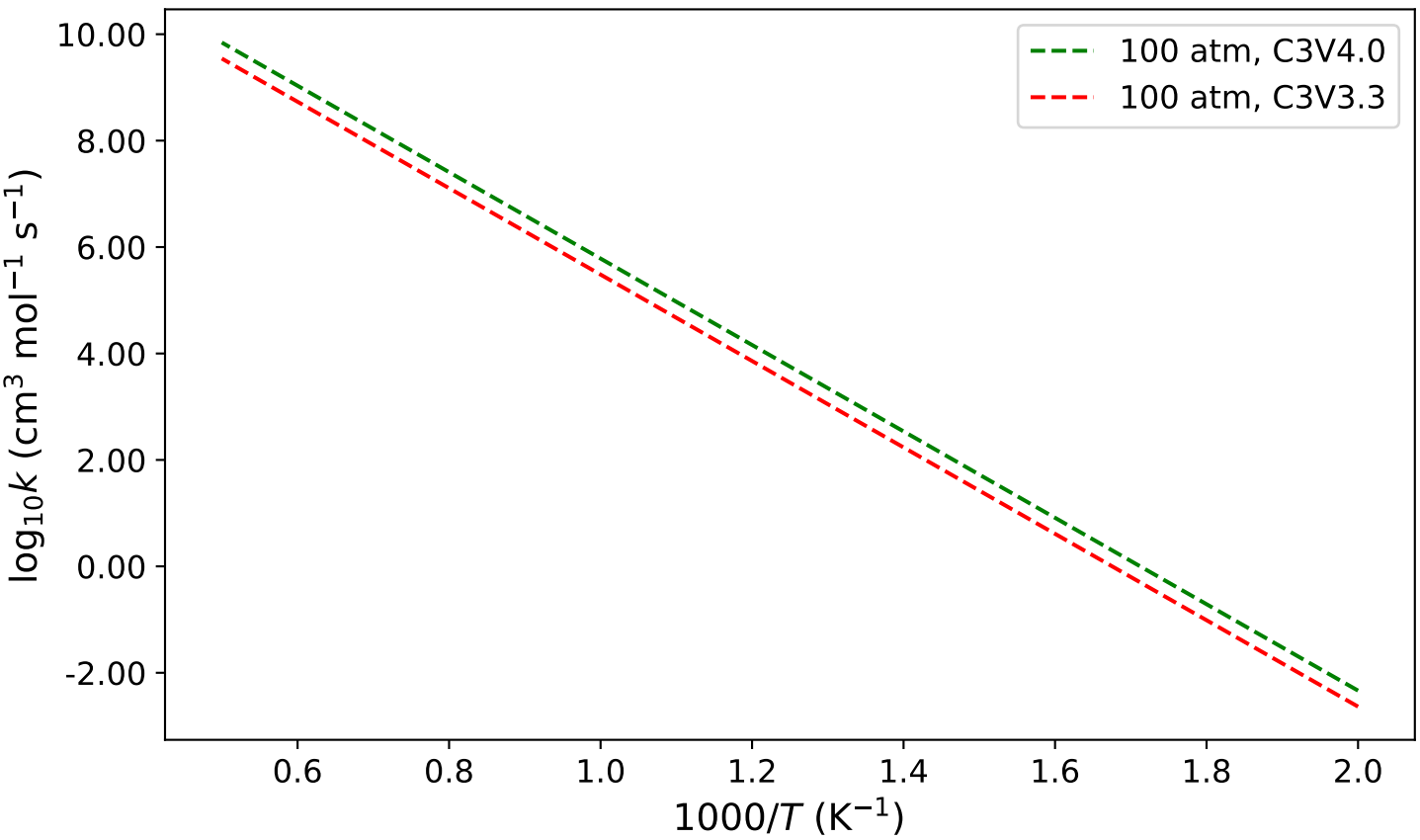


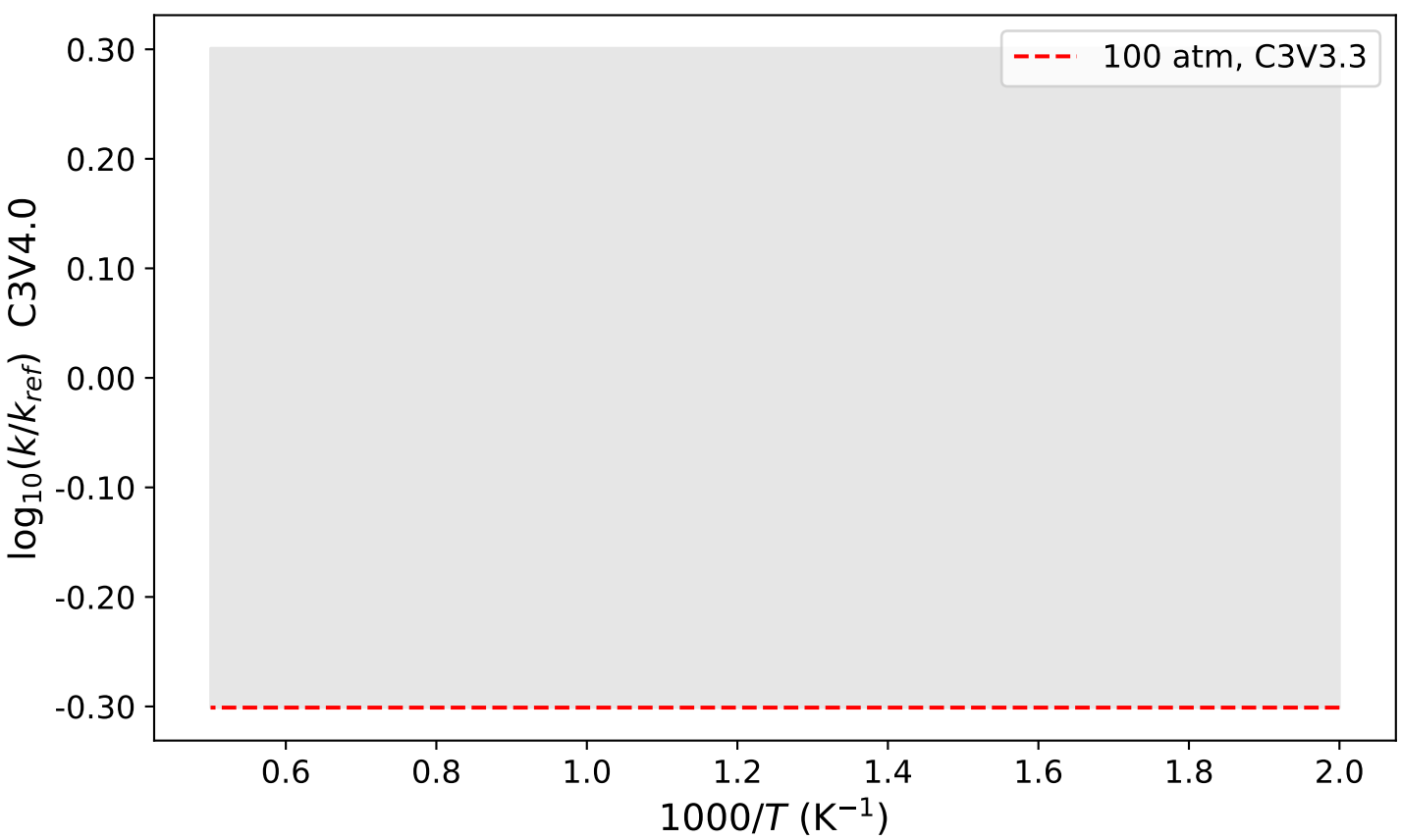
NAPH + C6H5CH2CH2 = NAPH-A + C6H5C2H5



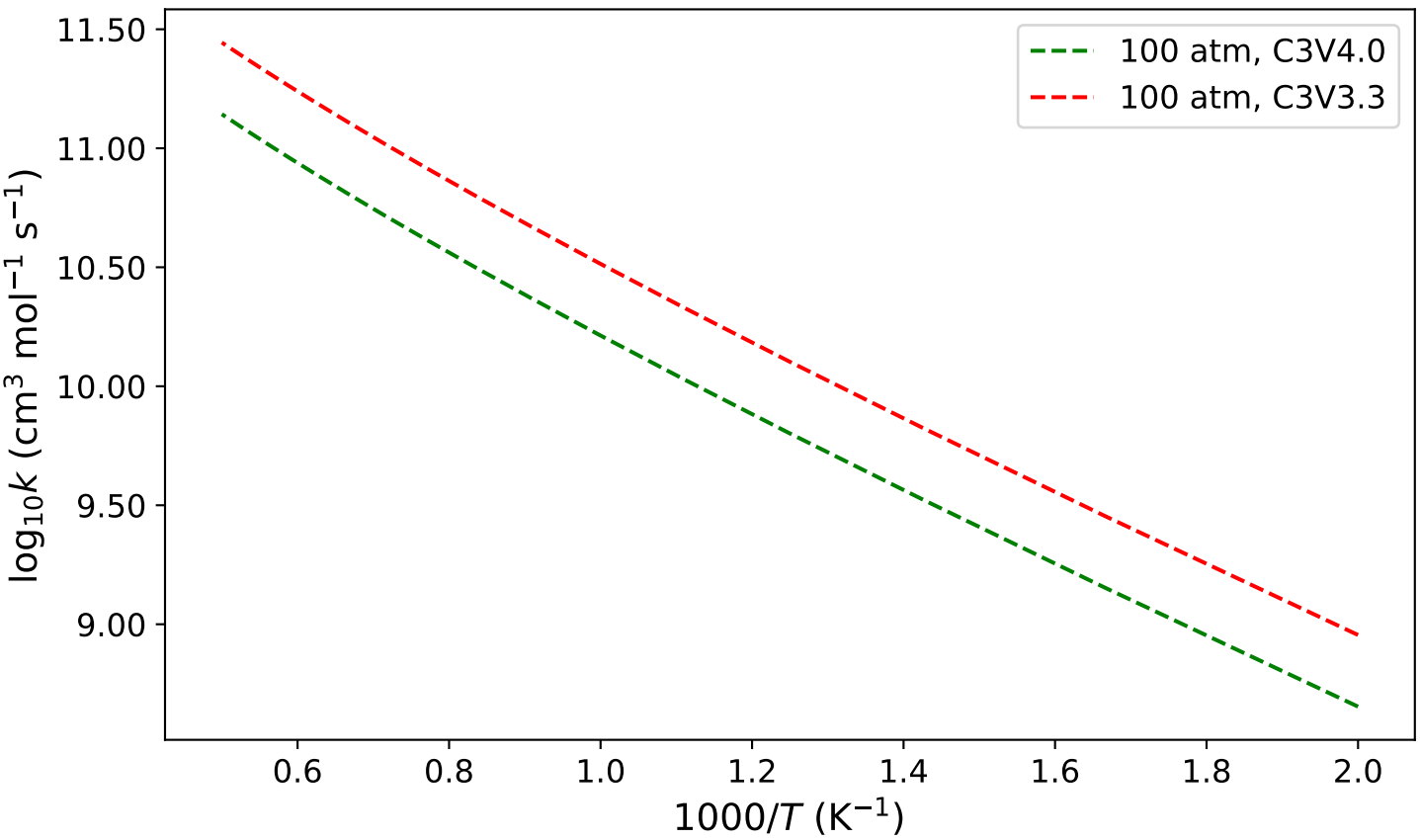


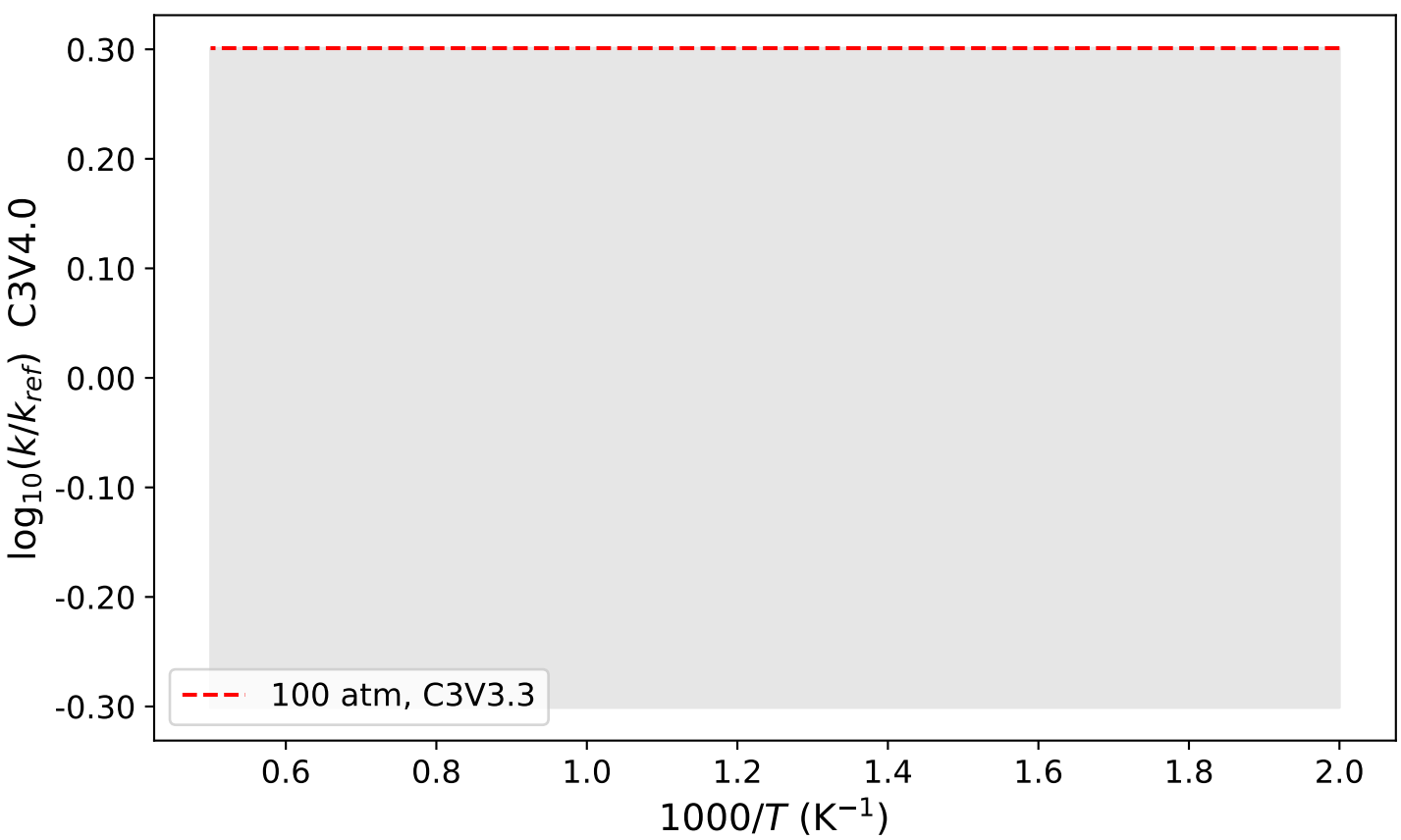
C5H6 + O2 = C5H5 + HO2



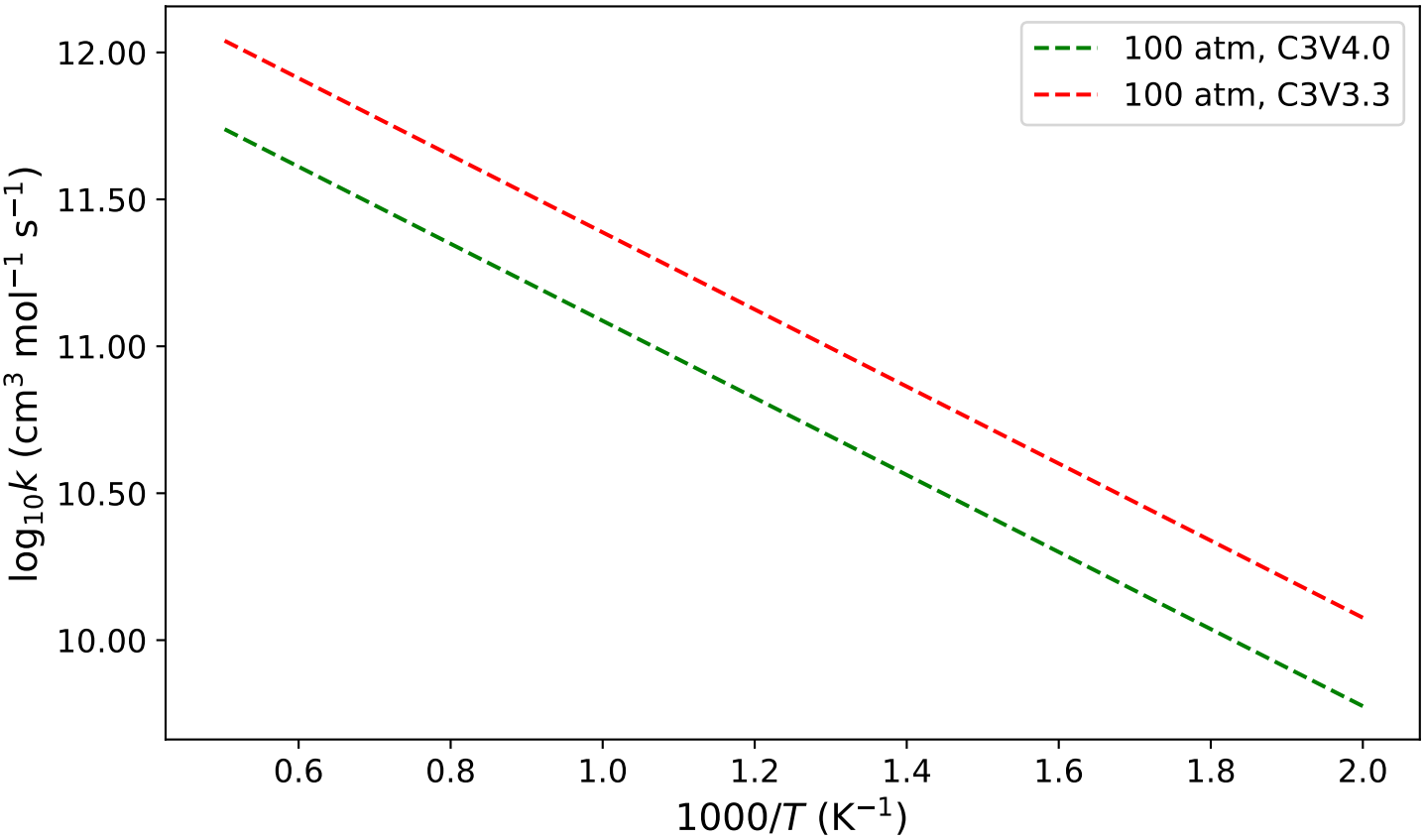


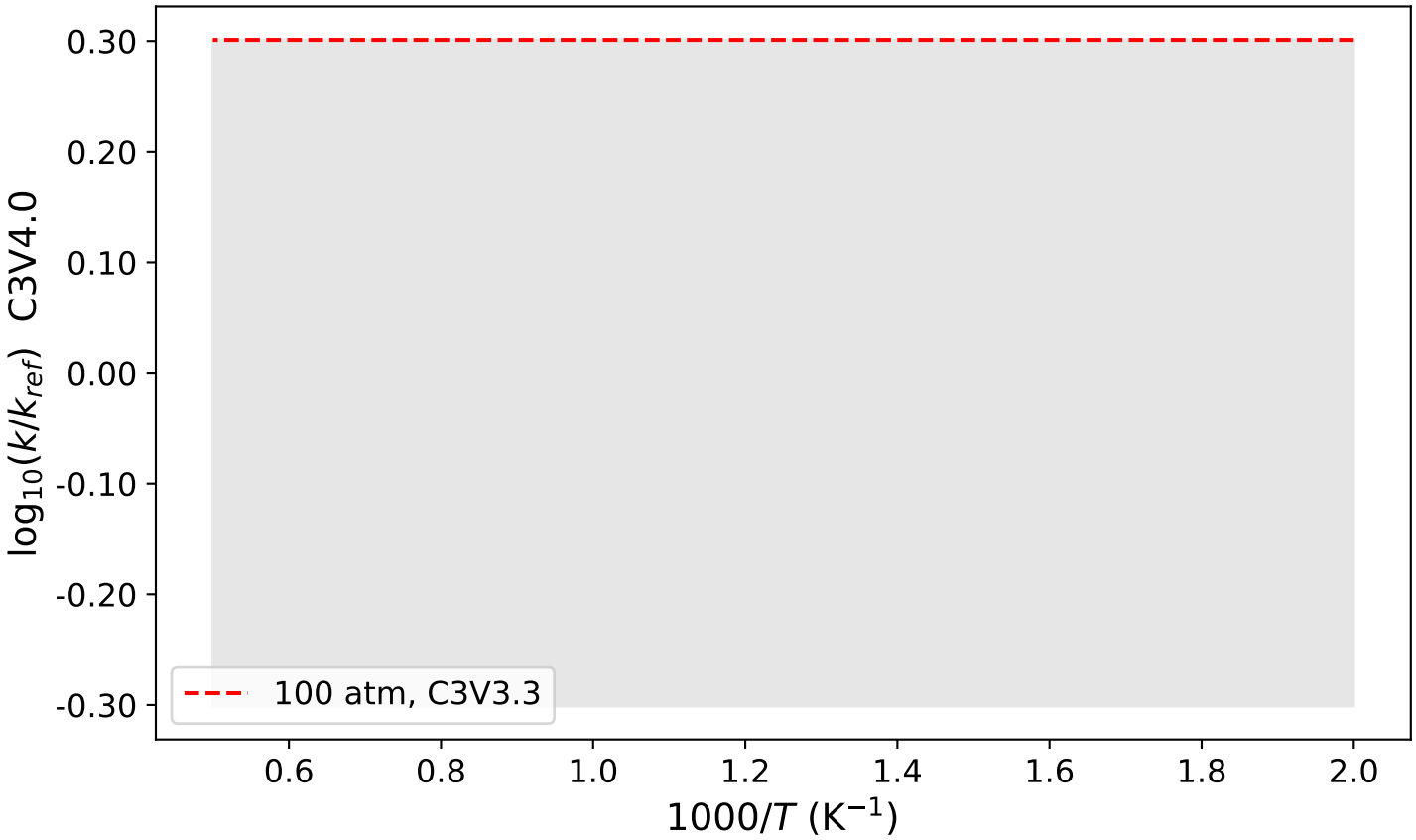
NAPH + C6H4OH = C6H5OH + NAPH-A



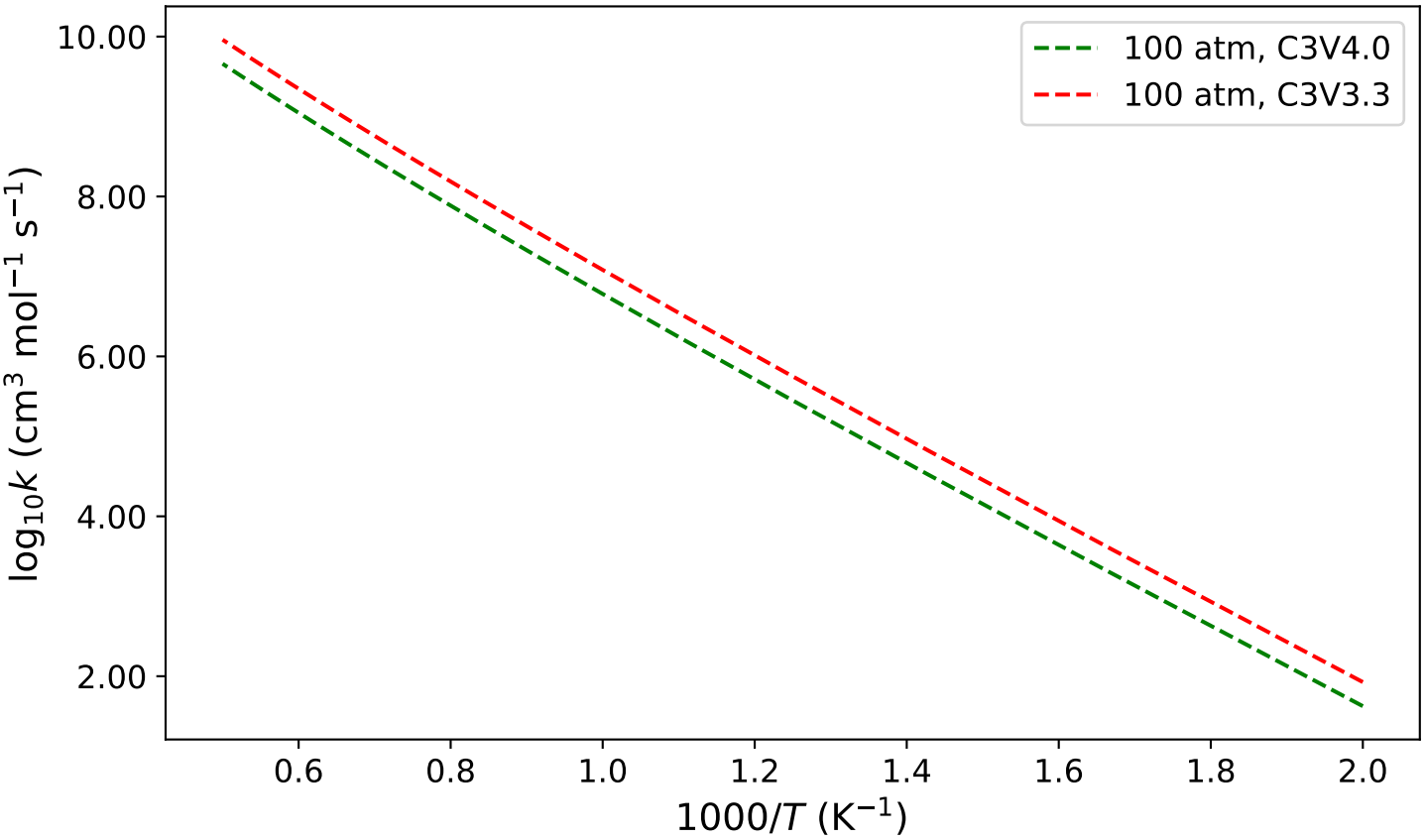


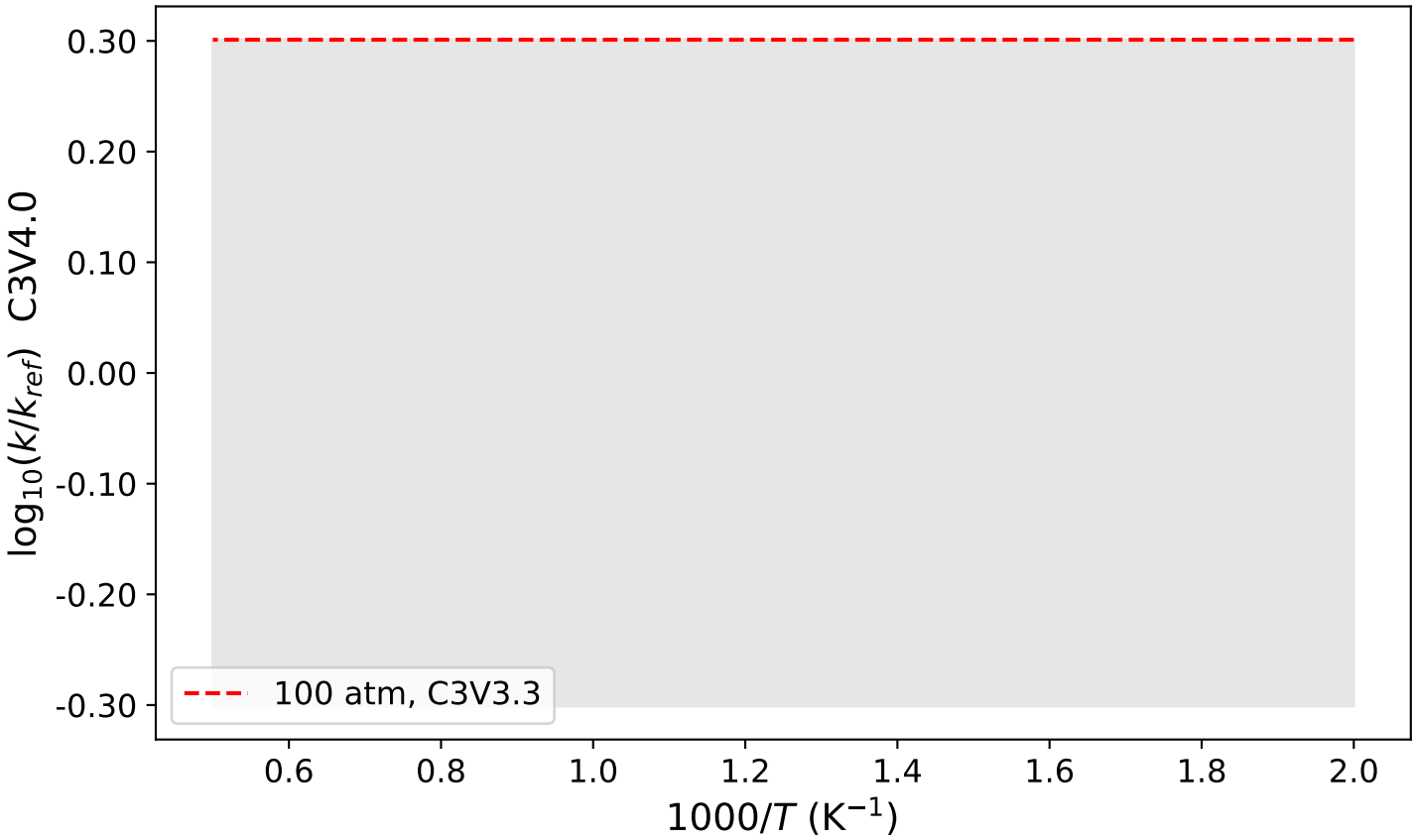
C3H3 + HOC6H4CH2 => H + H + NAPHOH-A



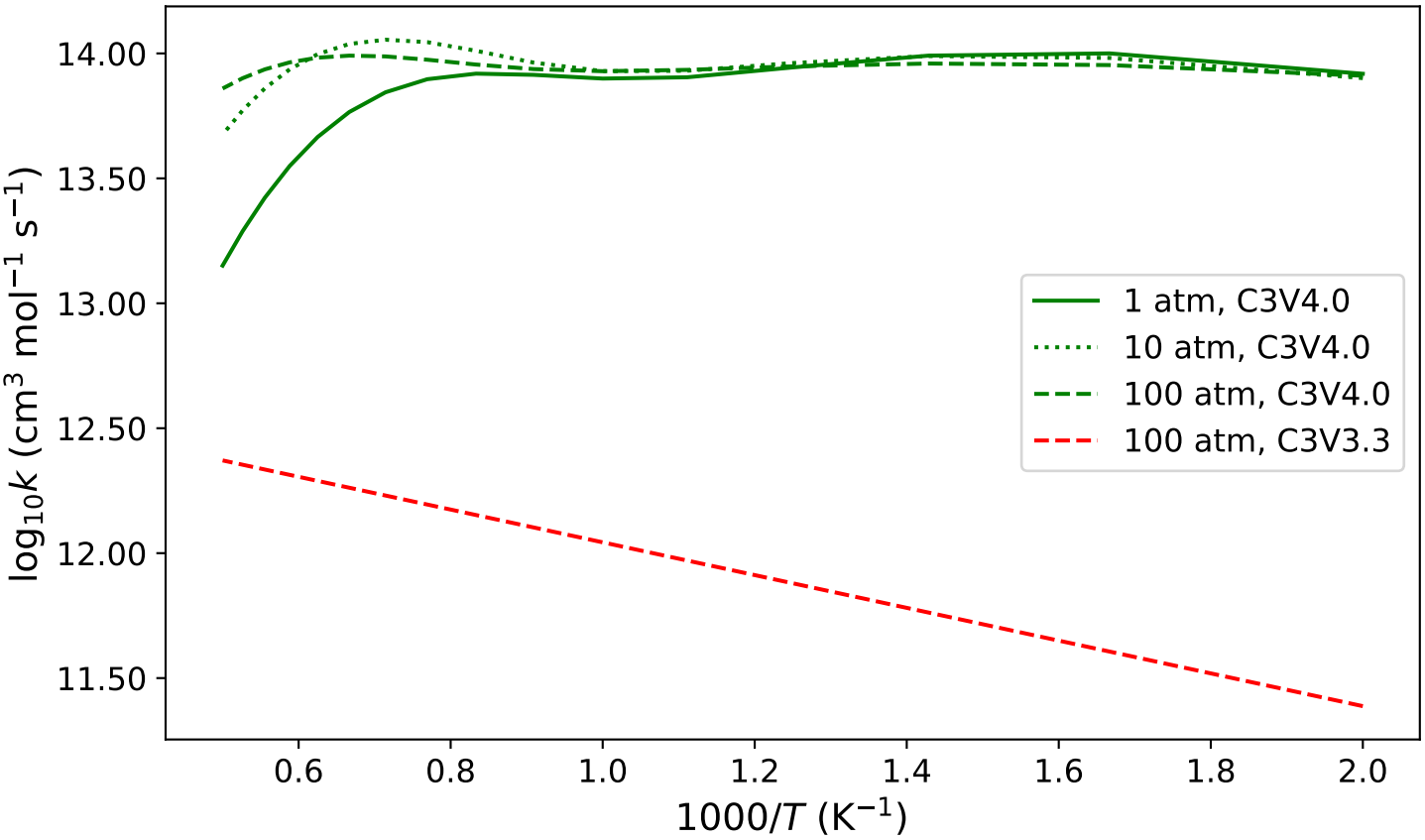


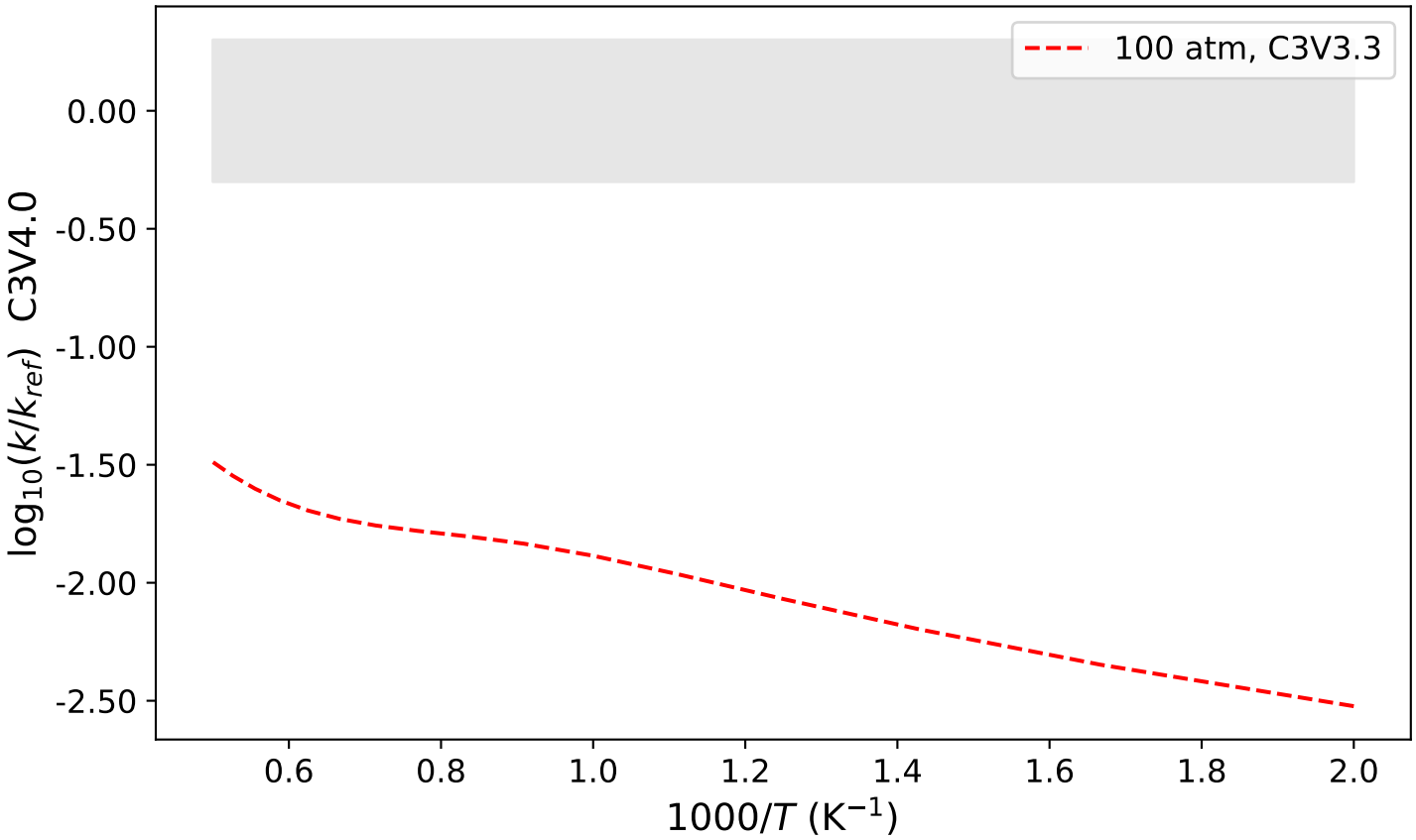
NAPH + IC4H7 = NAPH-A + IC4H8



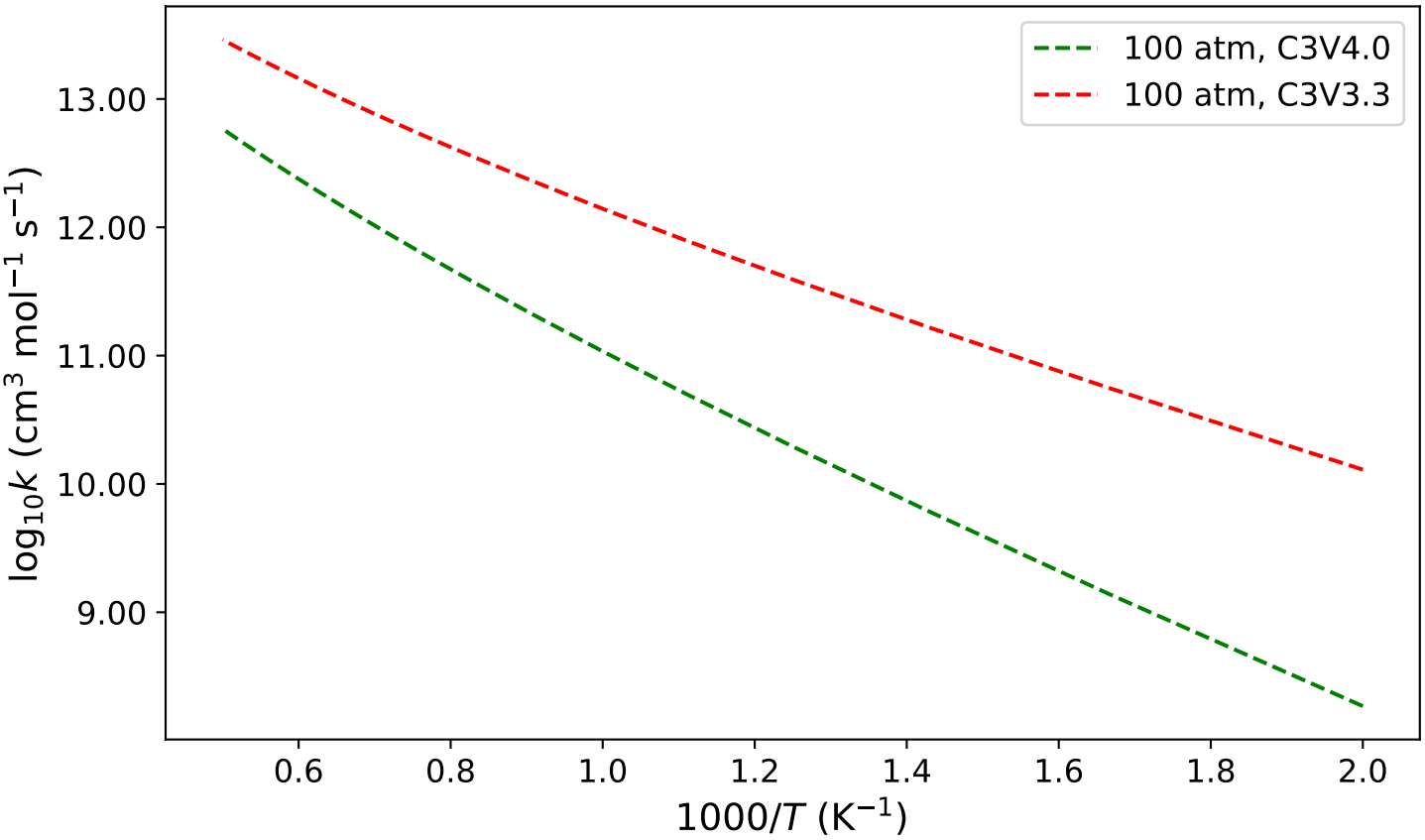


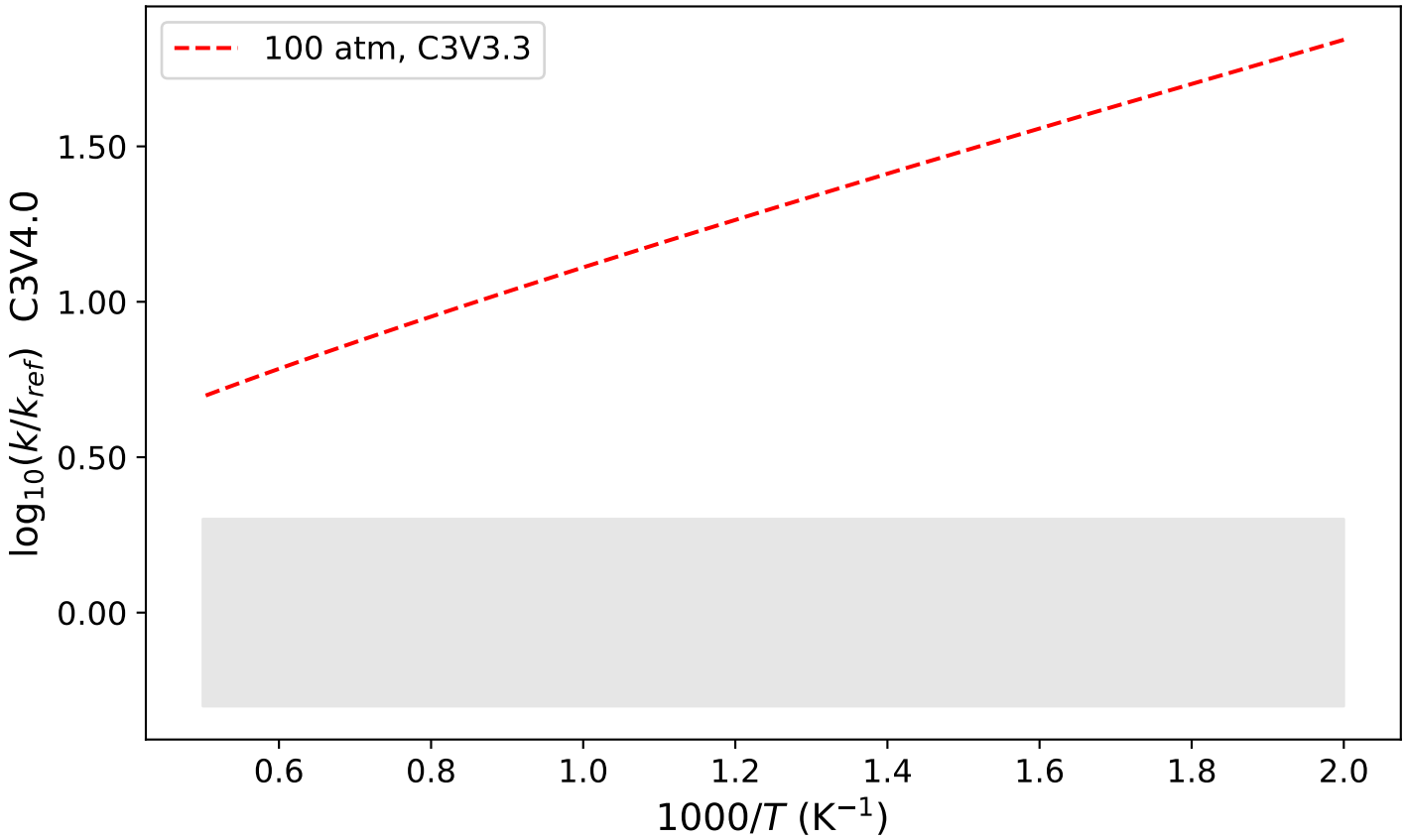
C3H3 + C6H4C2H = 0.5NAPH + 0.5C12H8



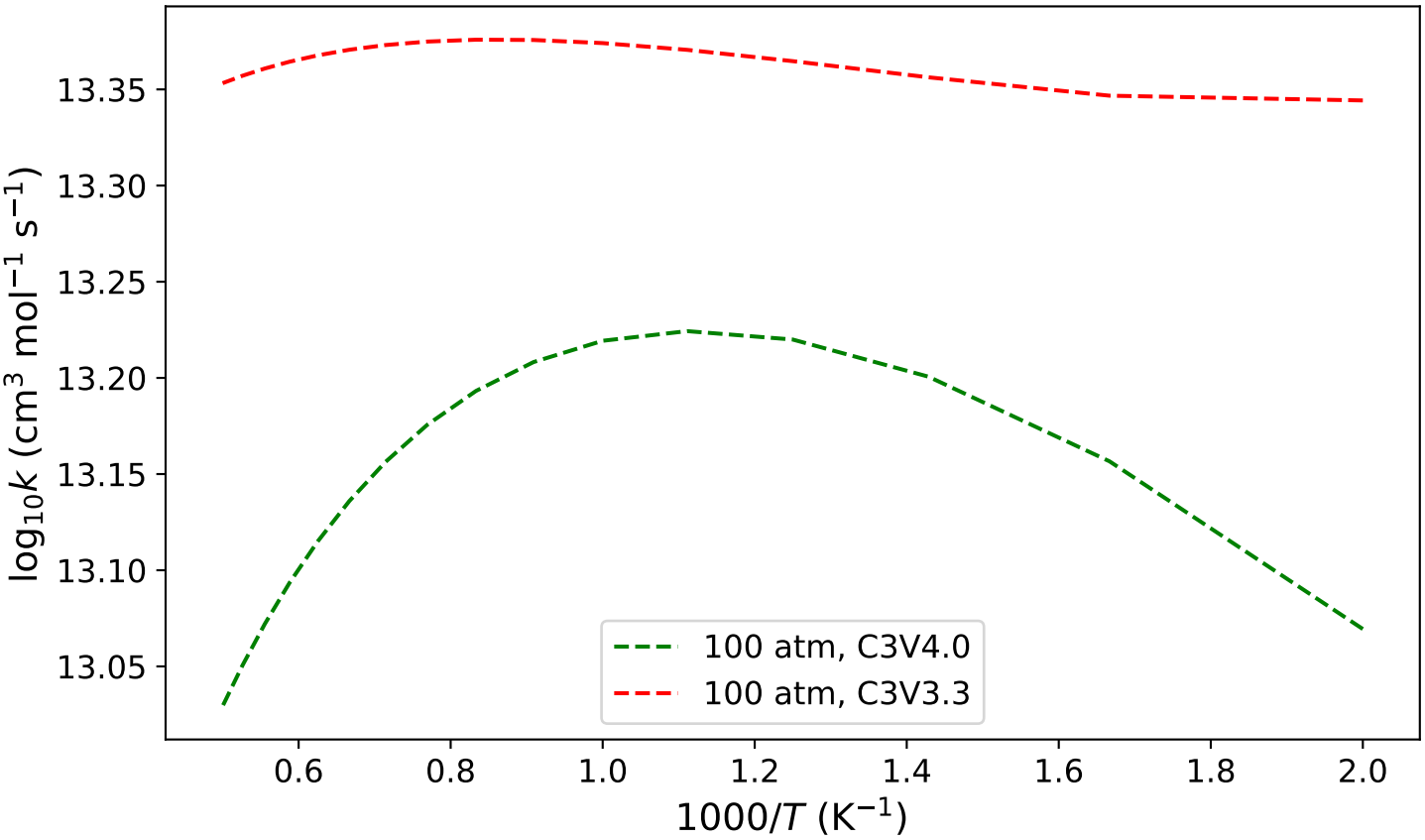


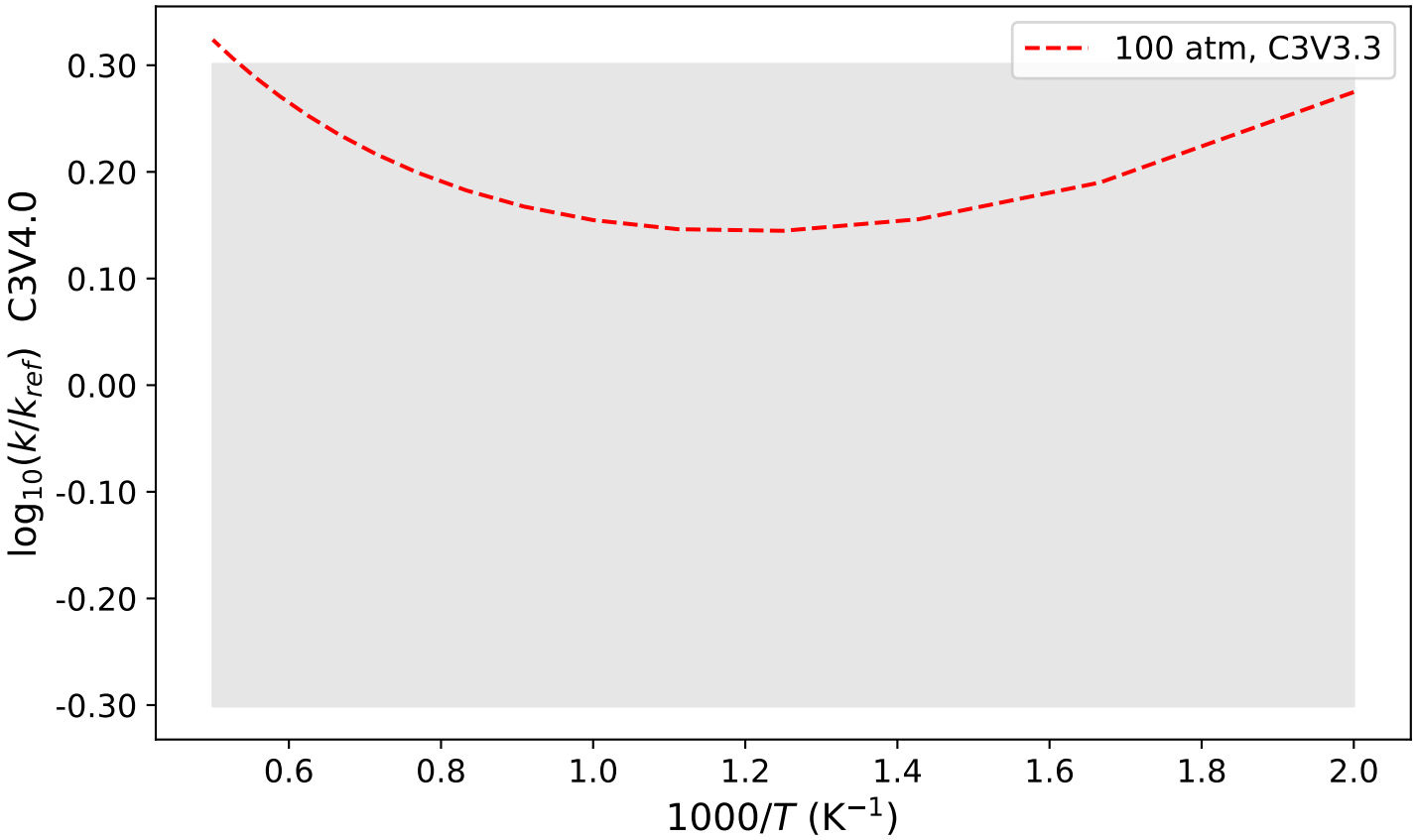
O + NAPH = OH + NAPH-A



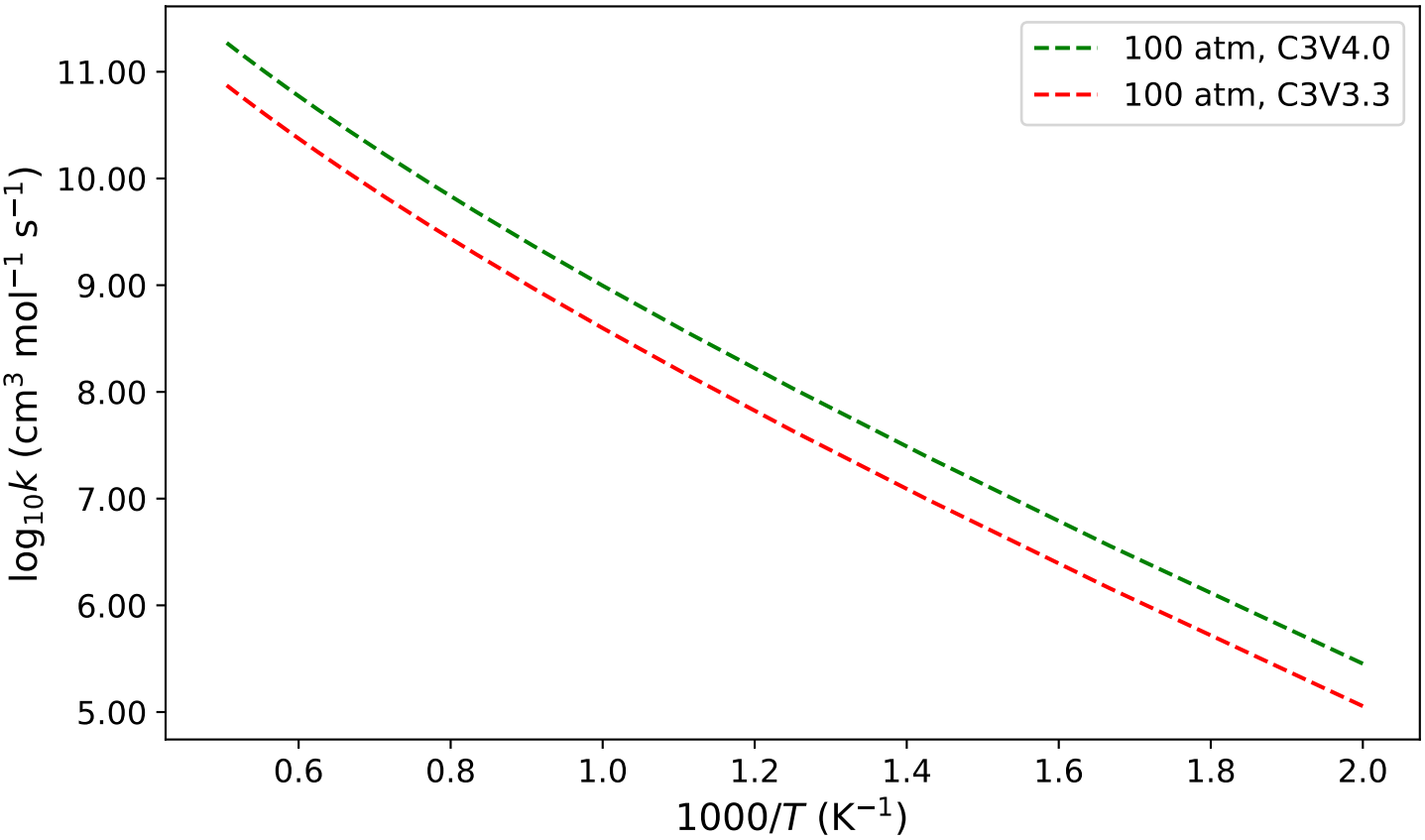


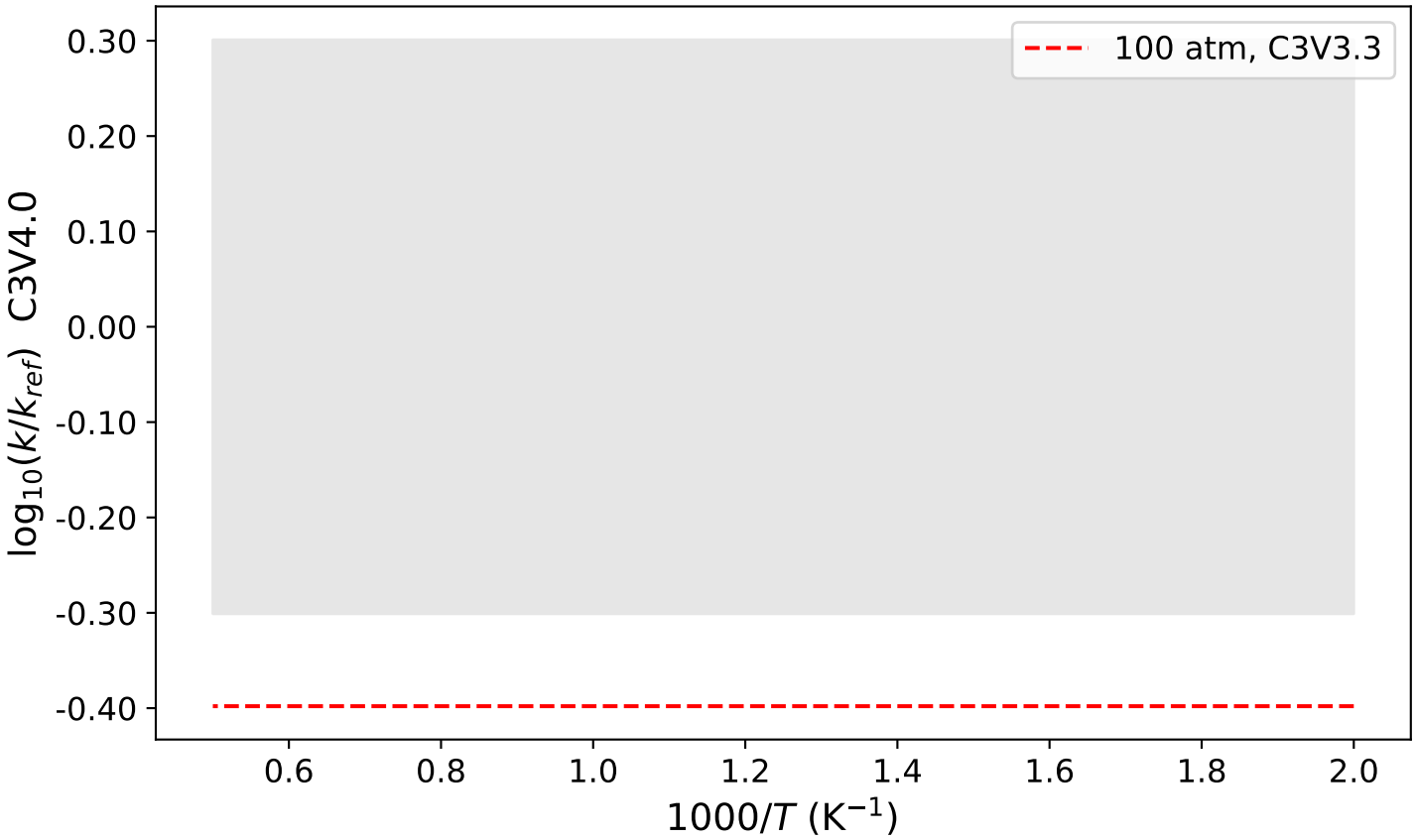
O + C9H7 = CO + C6H4C2H3



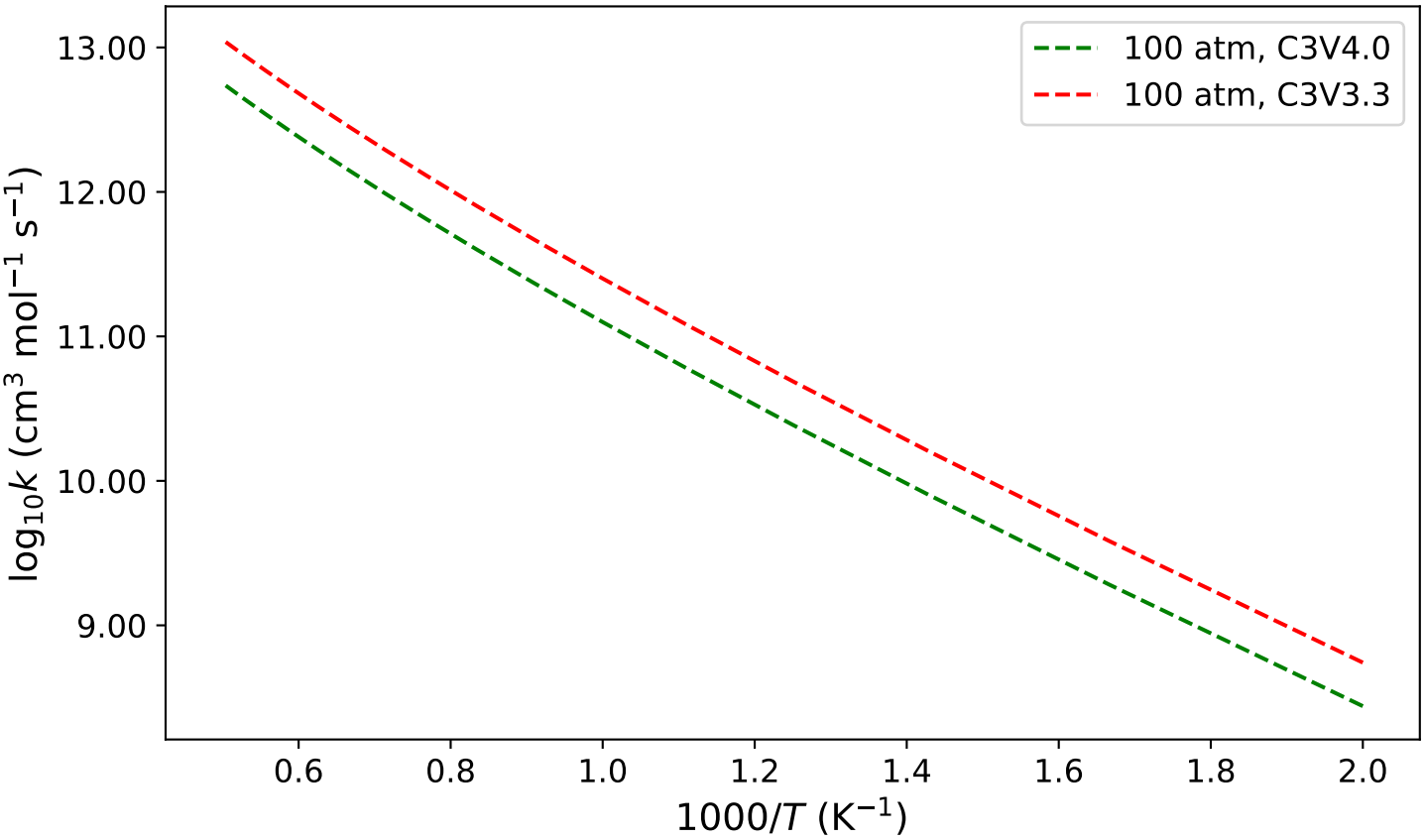


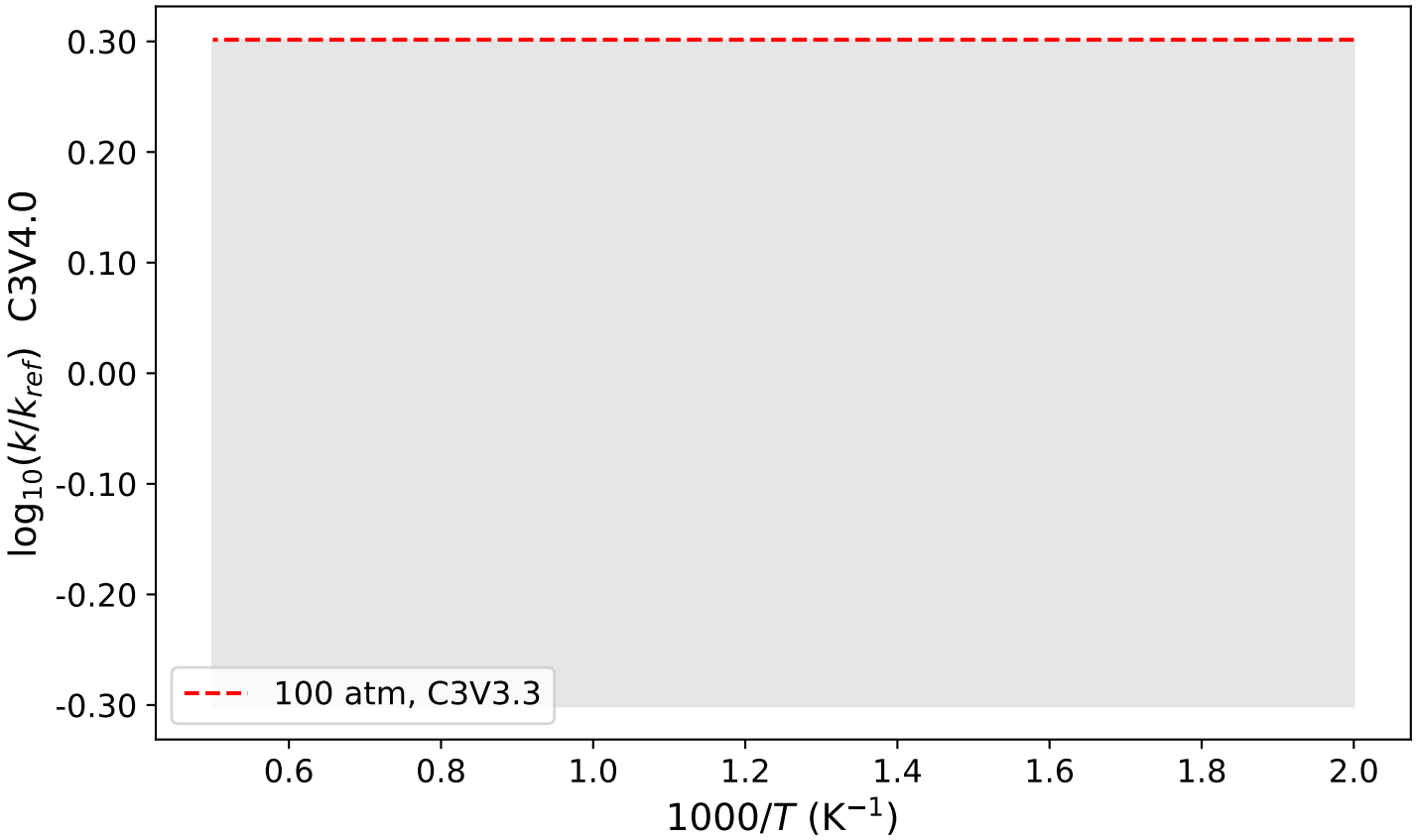
HO2 + C6H5CH3 = H2O2 + C6H5CH2



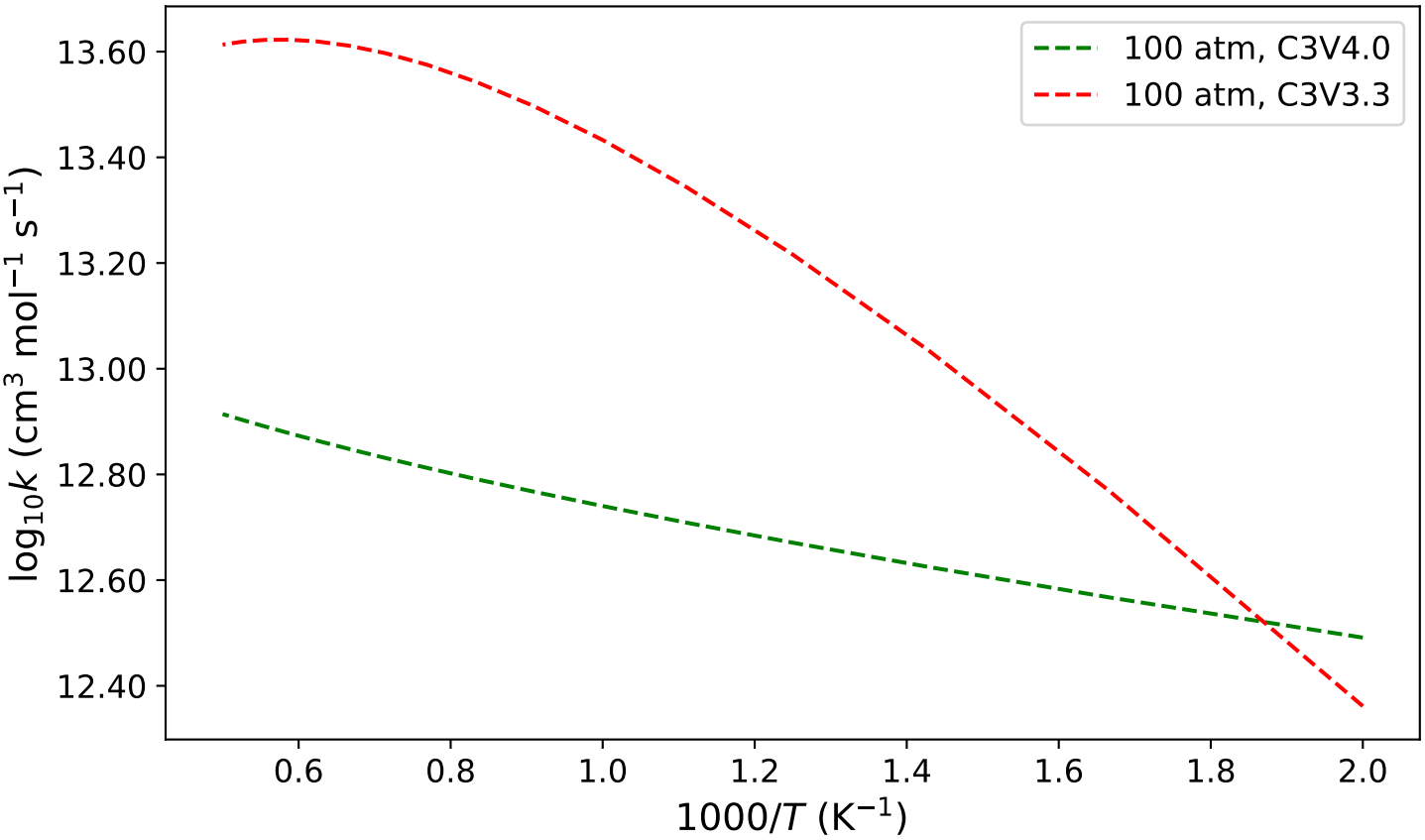


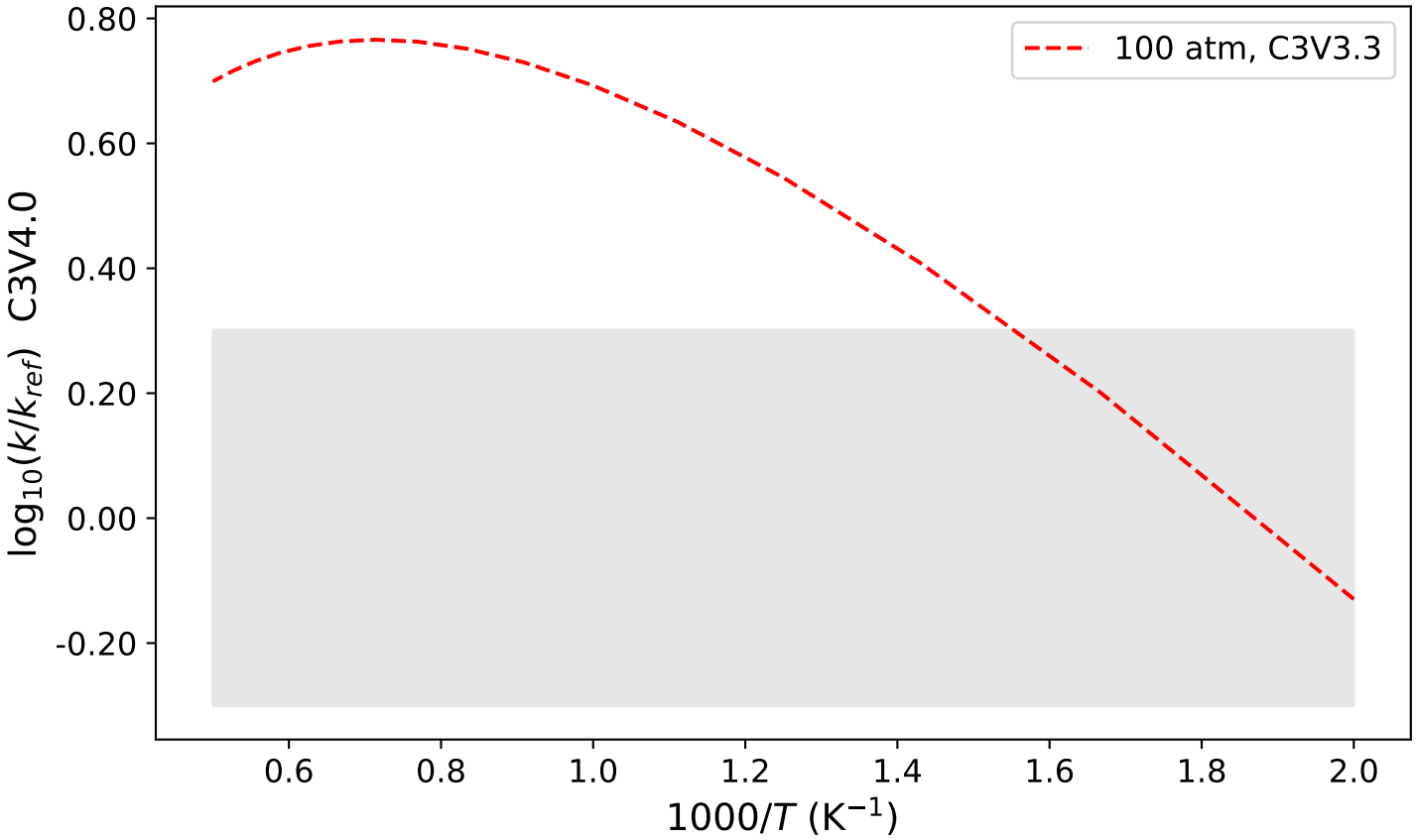
O + C10H7CH3-A = OH + C10H6CH3-A



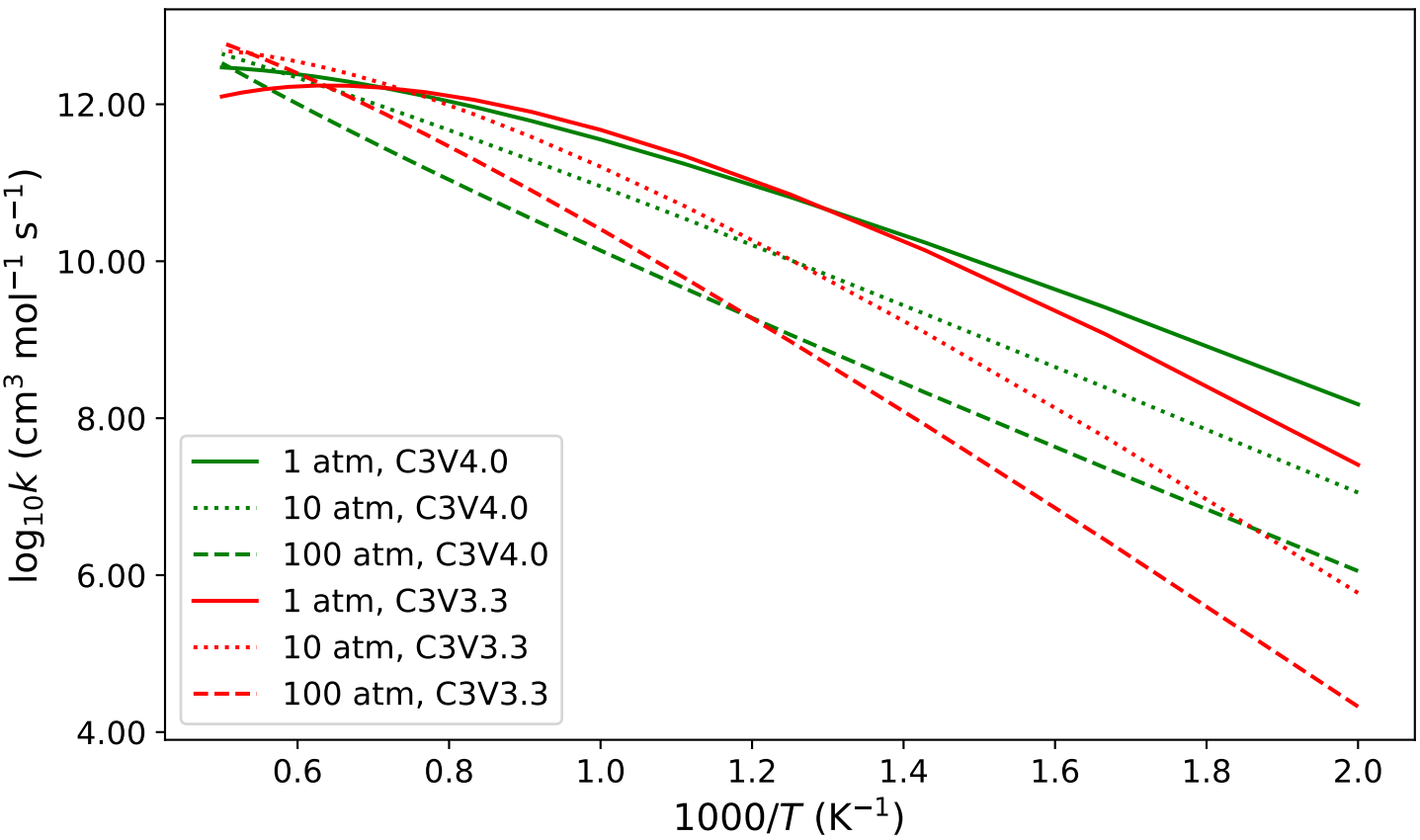


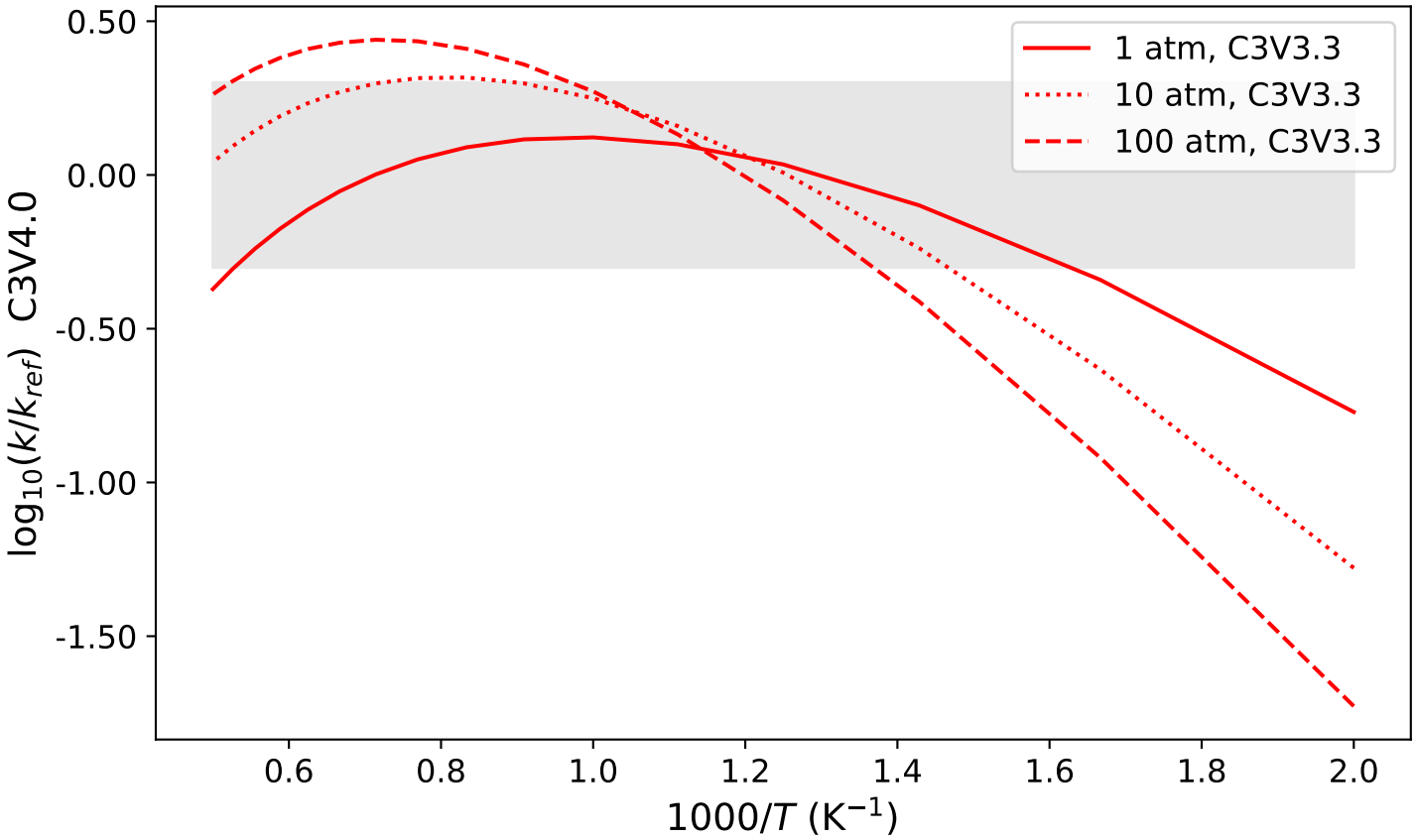
O + RXYLENE = HCO + C6H5CH3



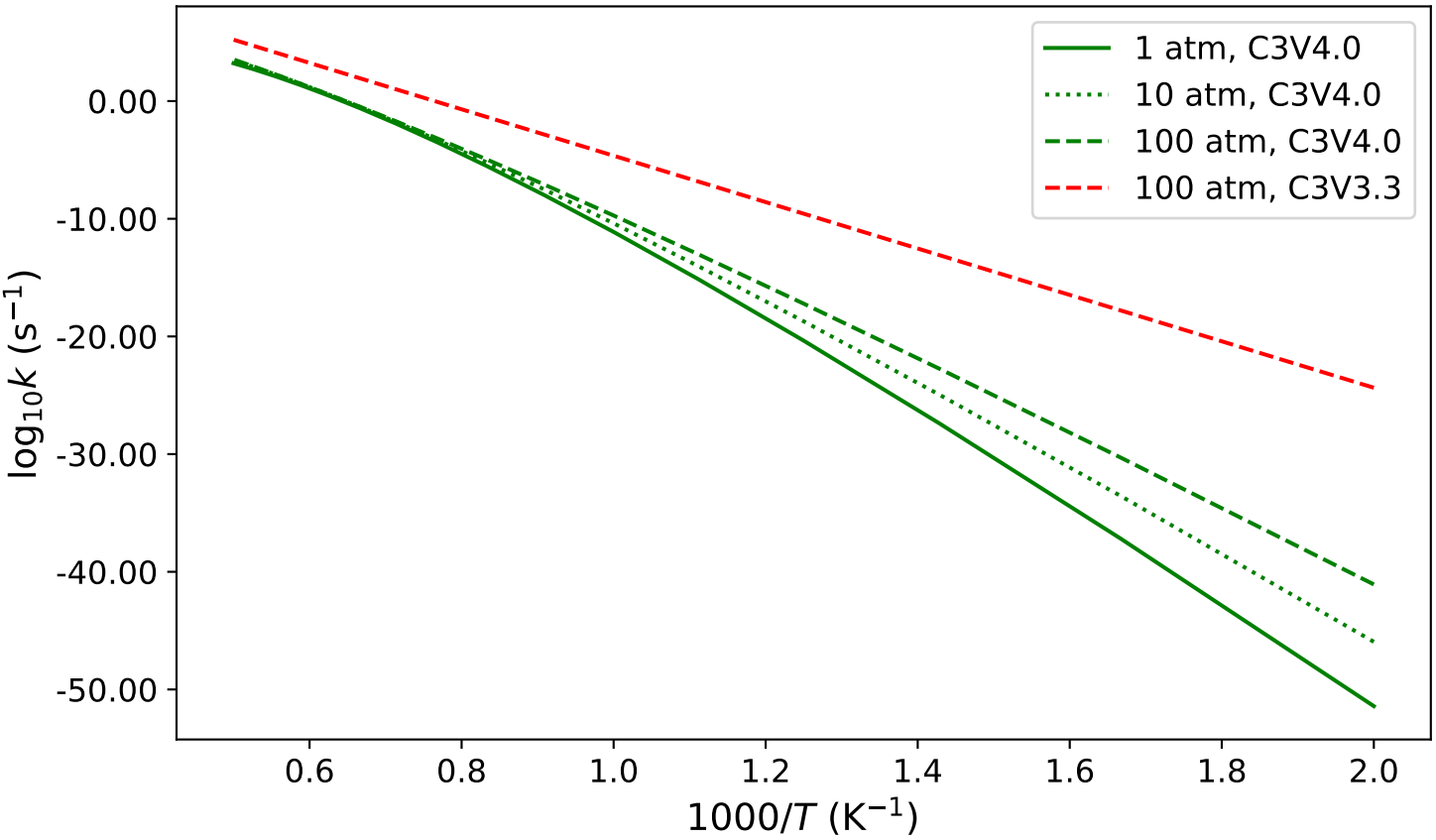


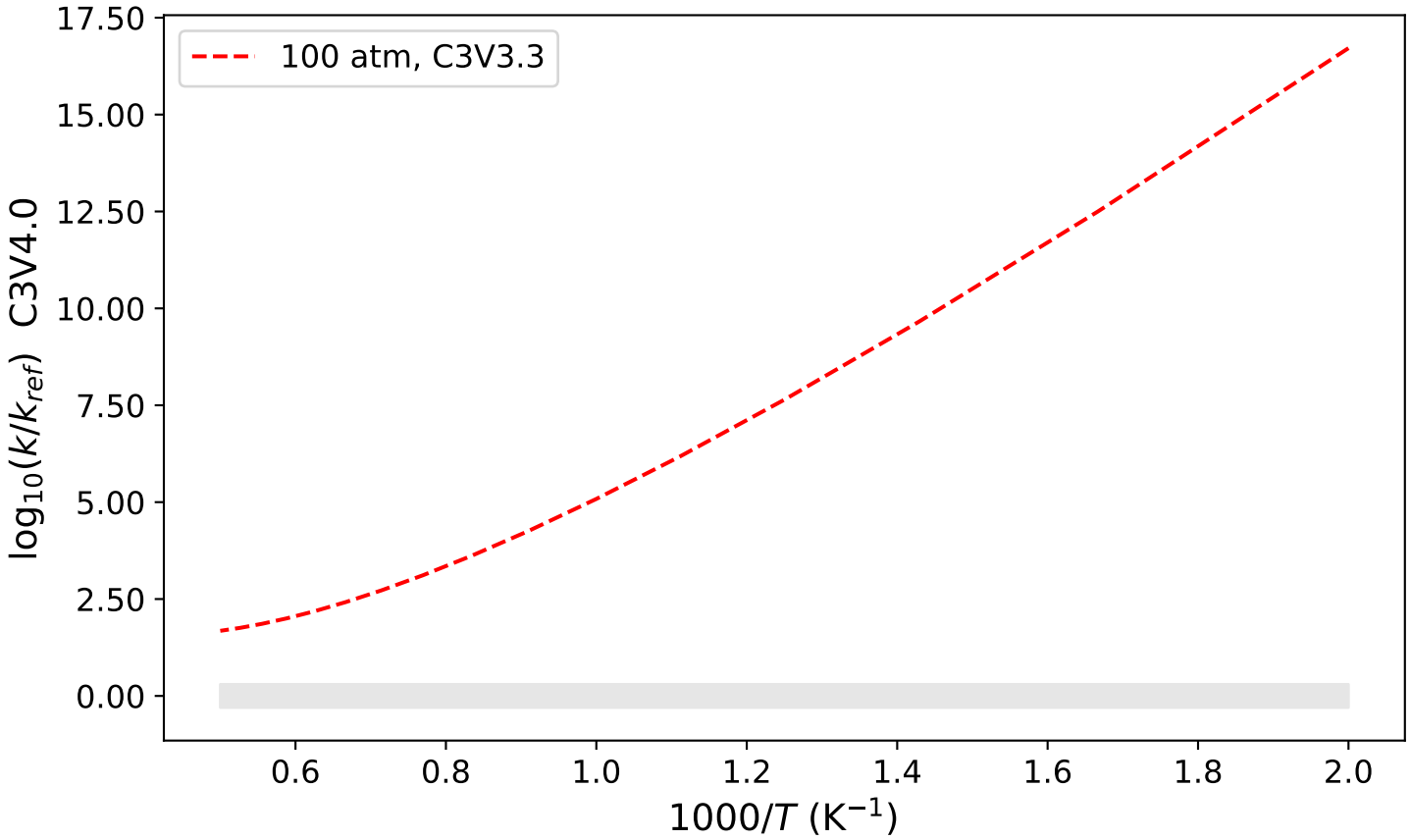
H + C5H6 = CVCCJCVC



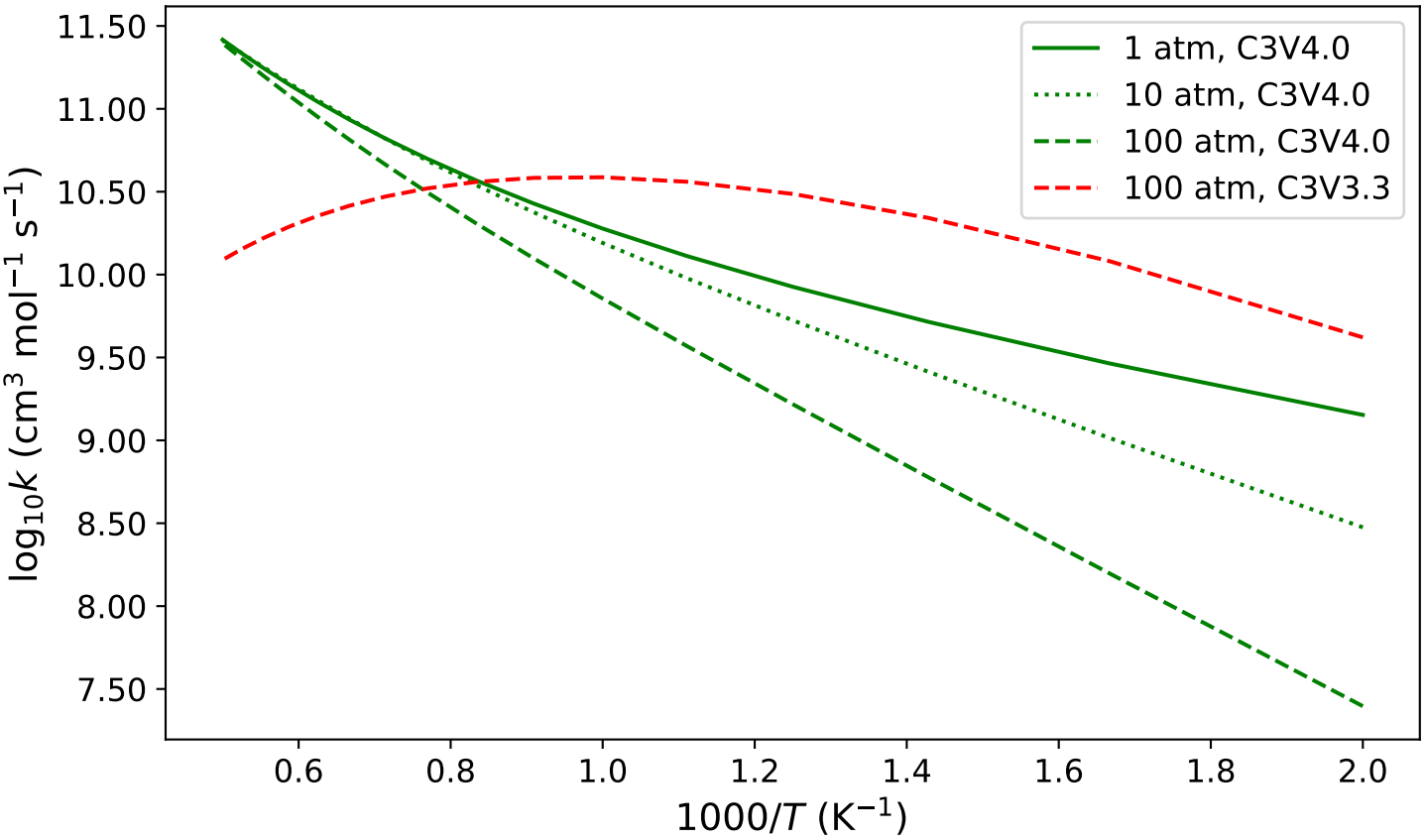


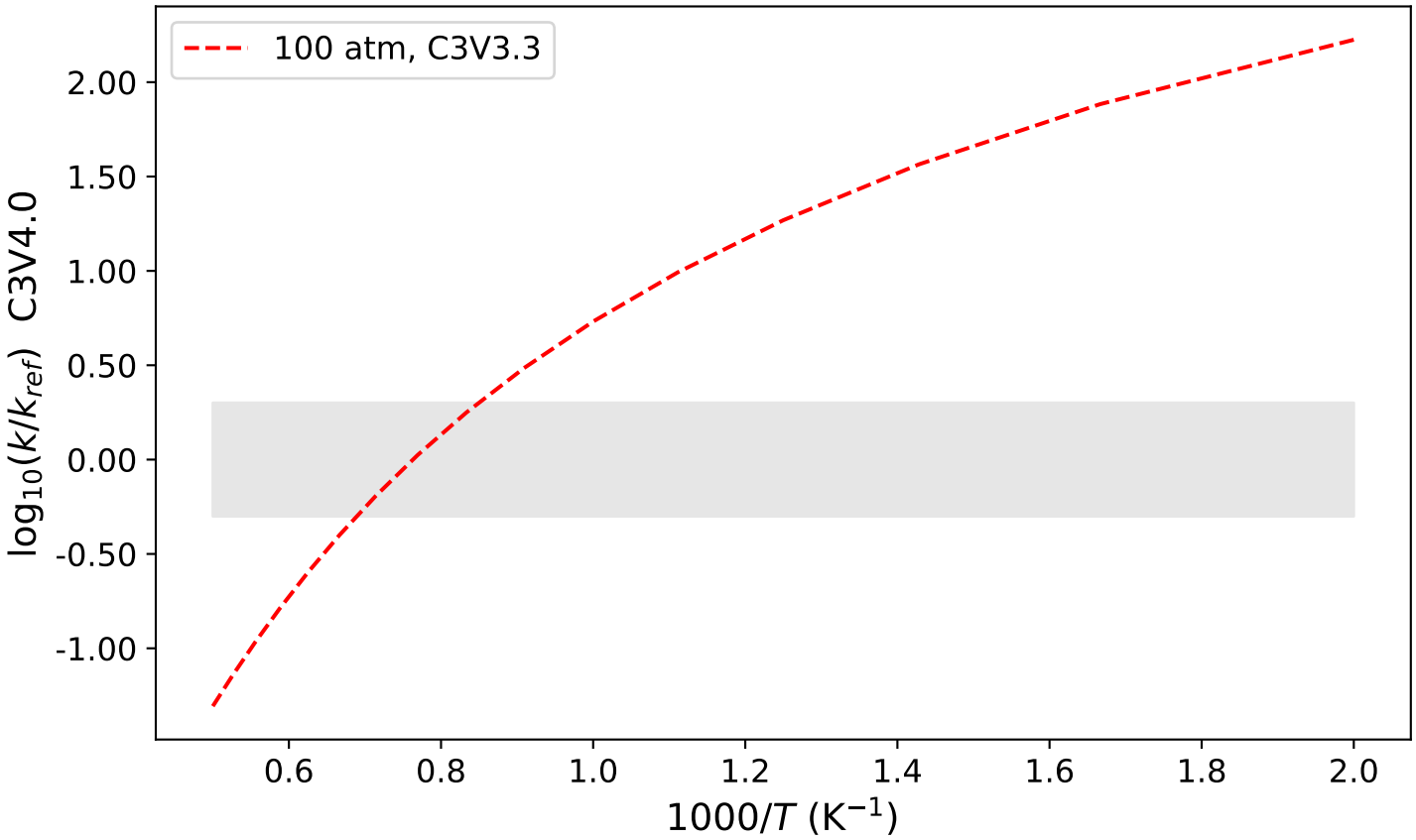
C9H7 = C5H5 + C4H2



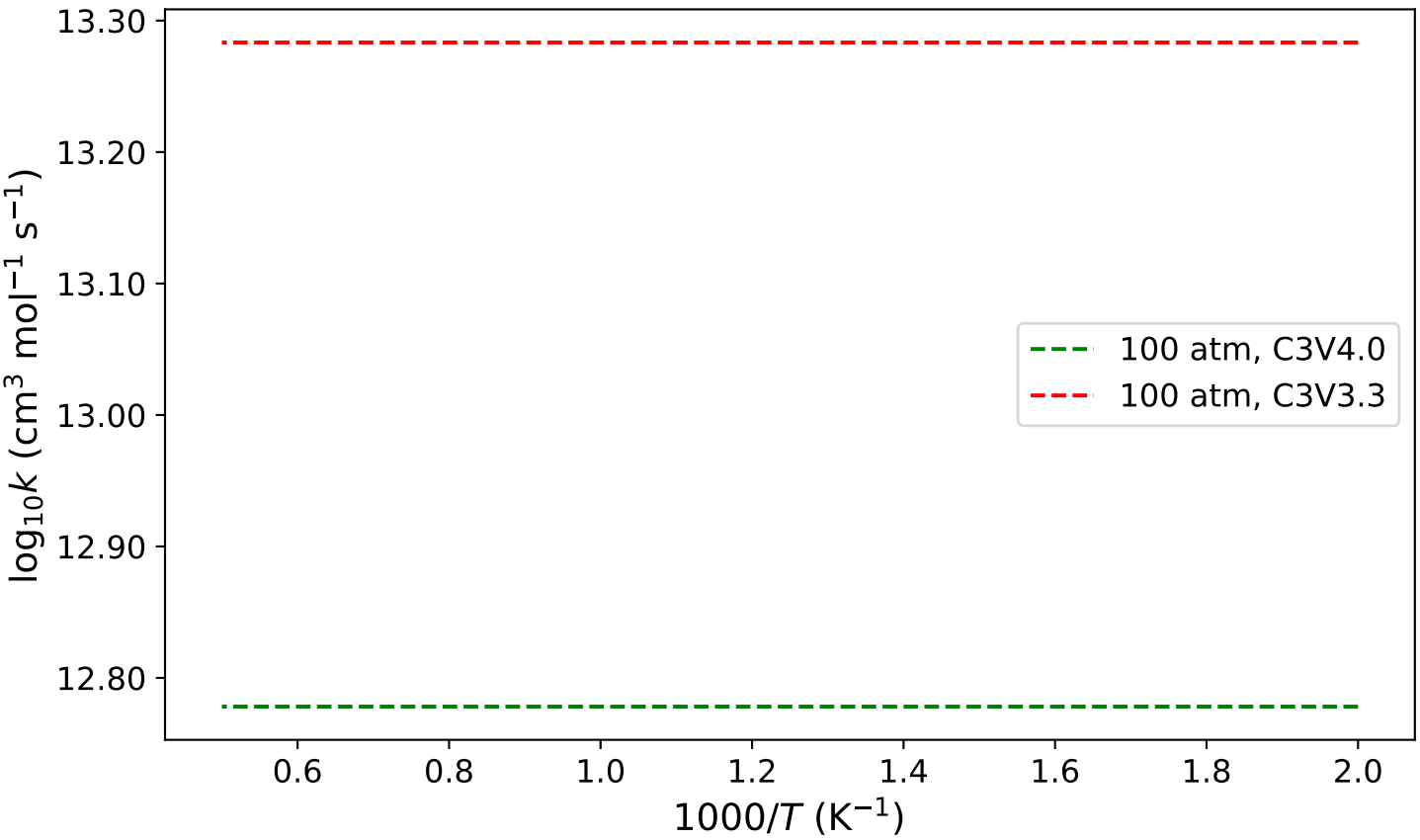


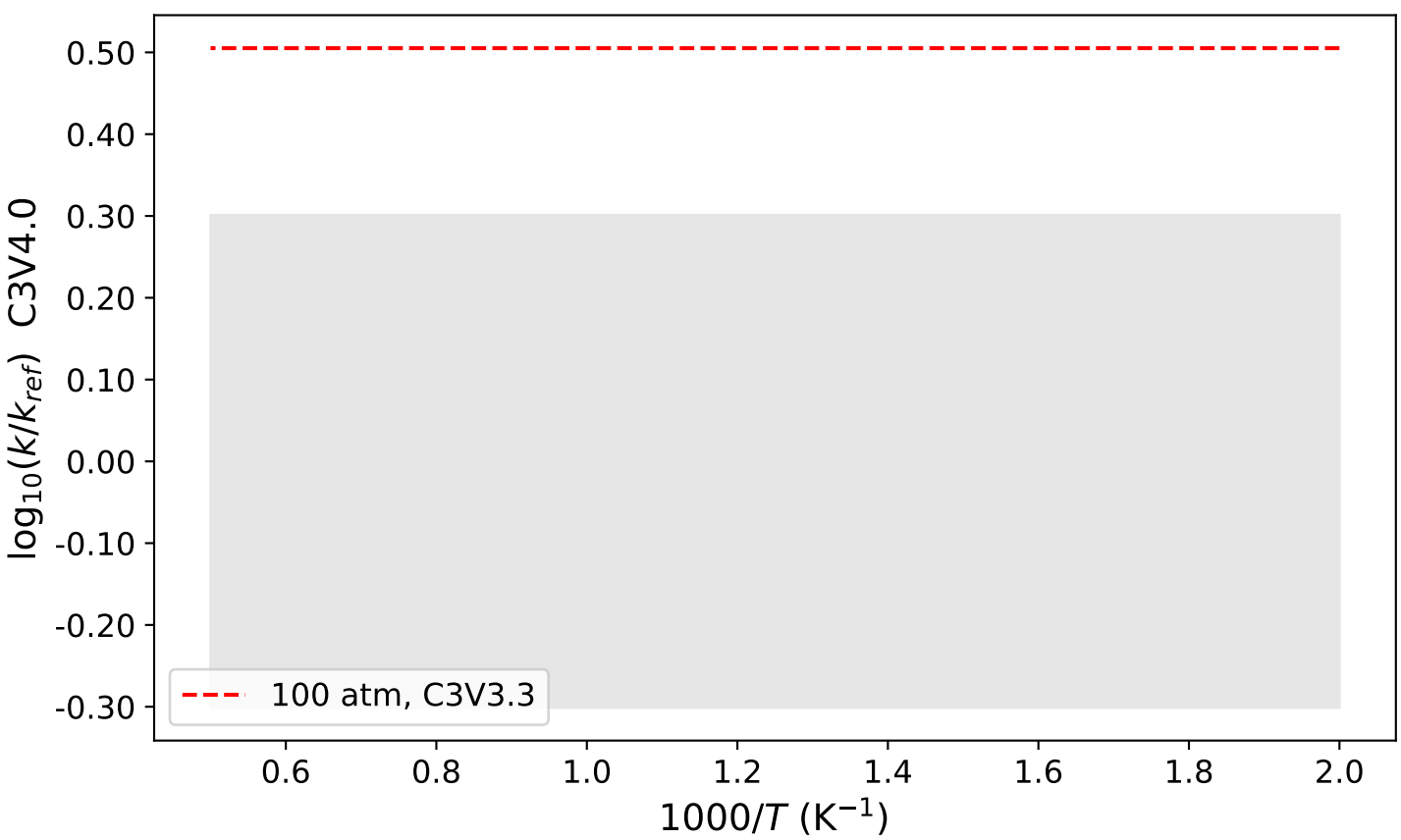
C5H5 + HO2 = H2O + C5H4O



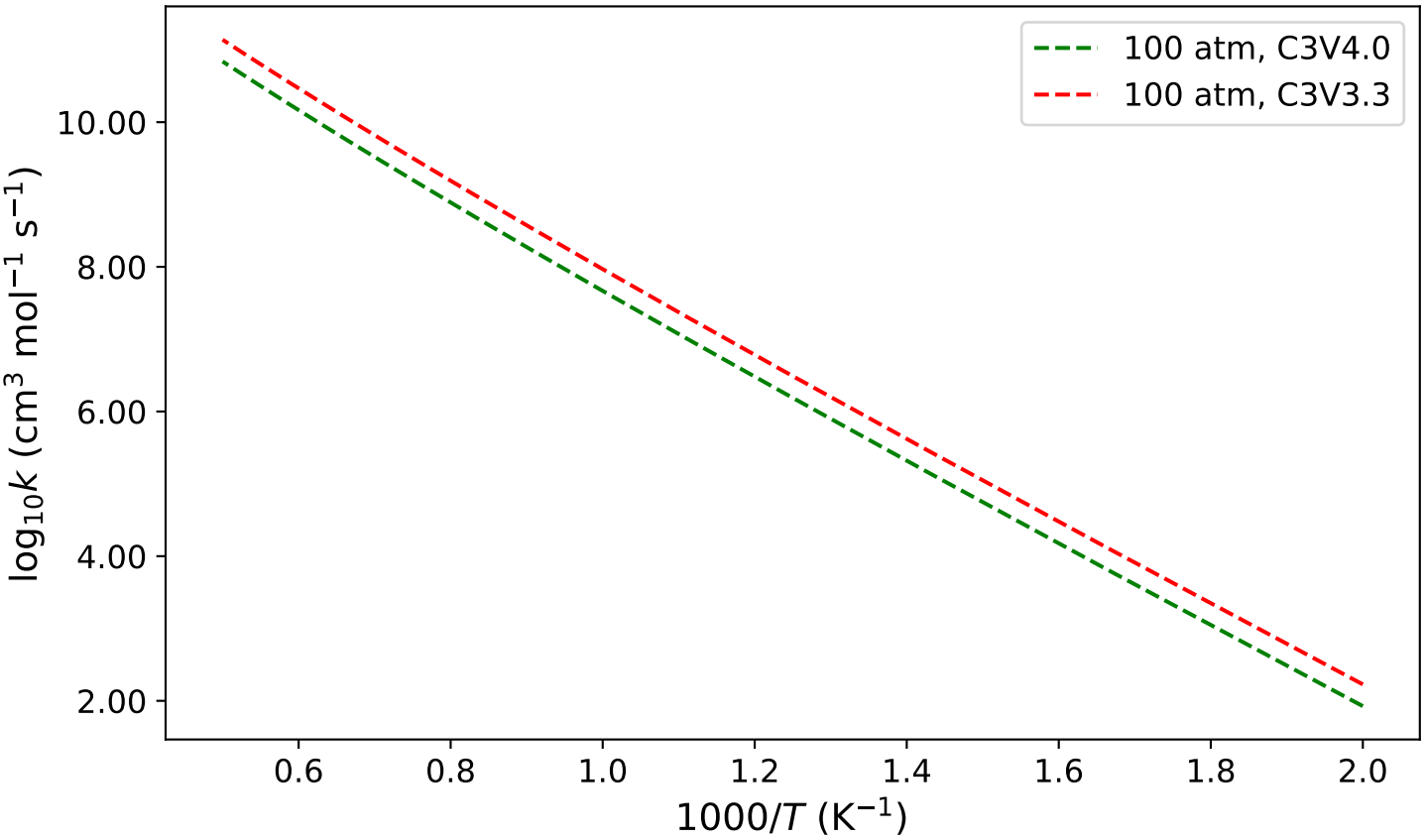


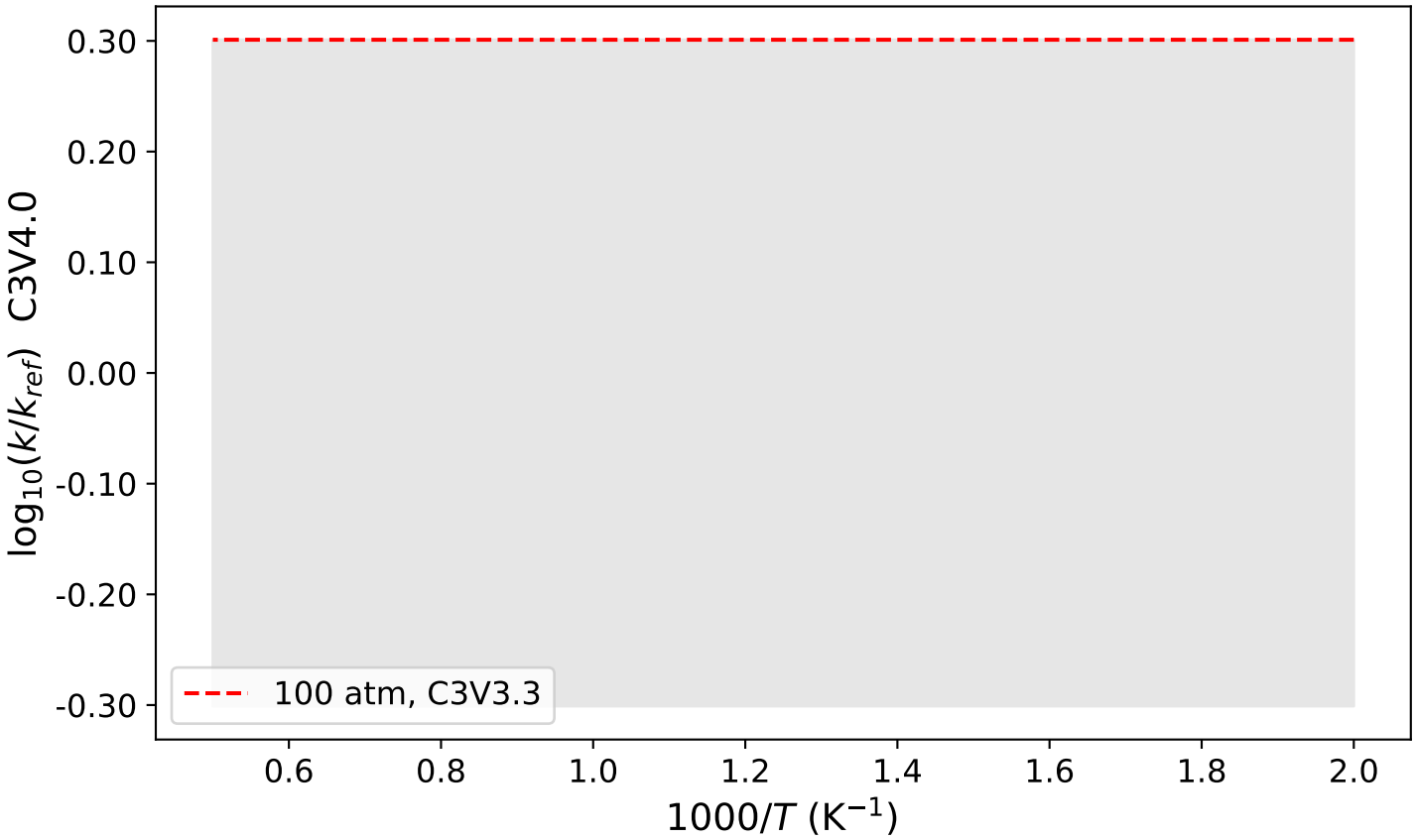
HO2 + C14H13 => OH + C6H5CH2 + C6H5CHO



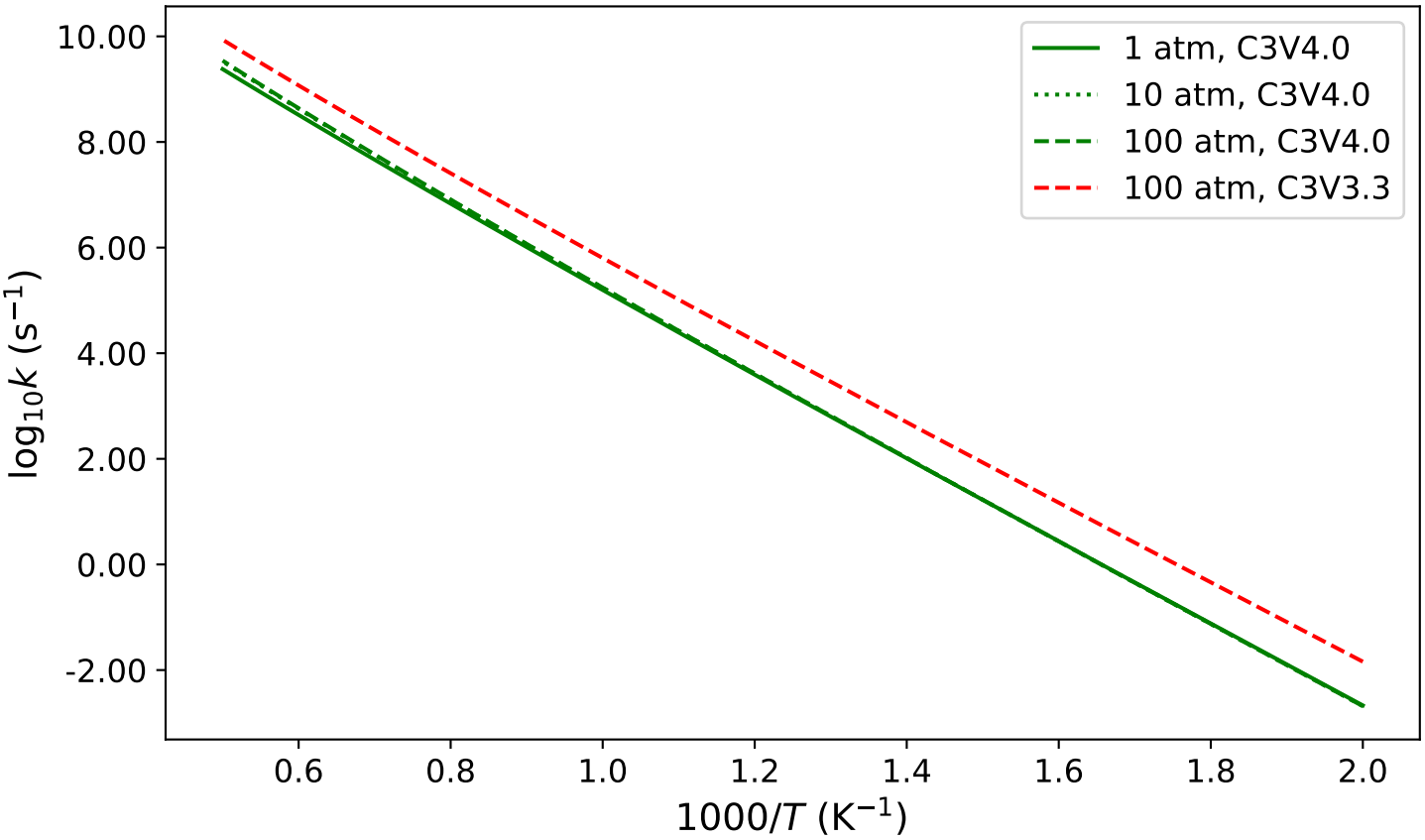


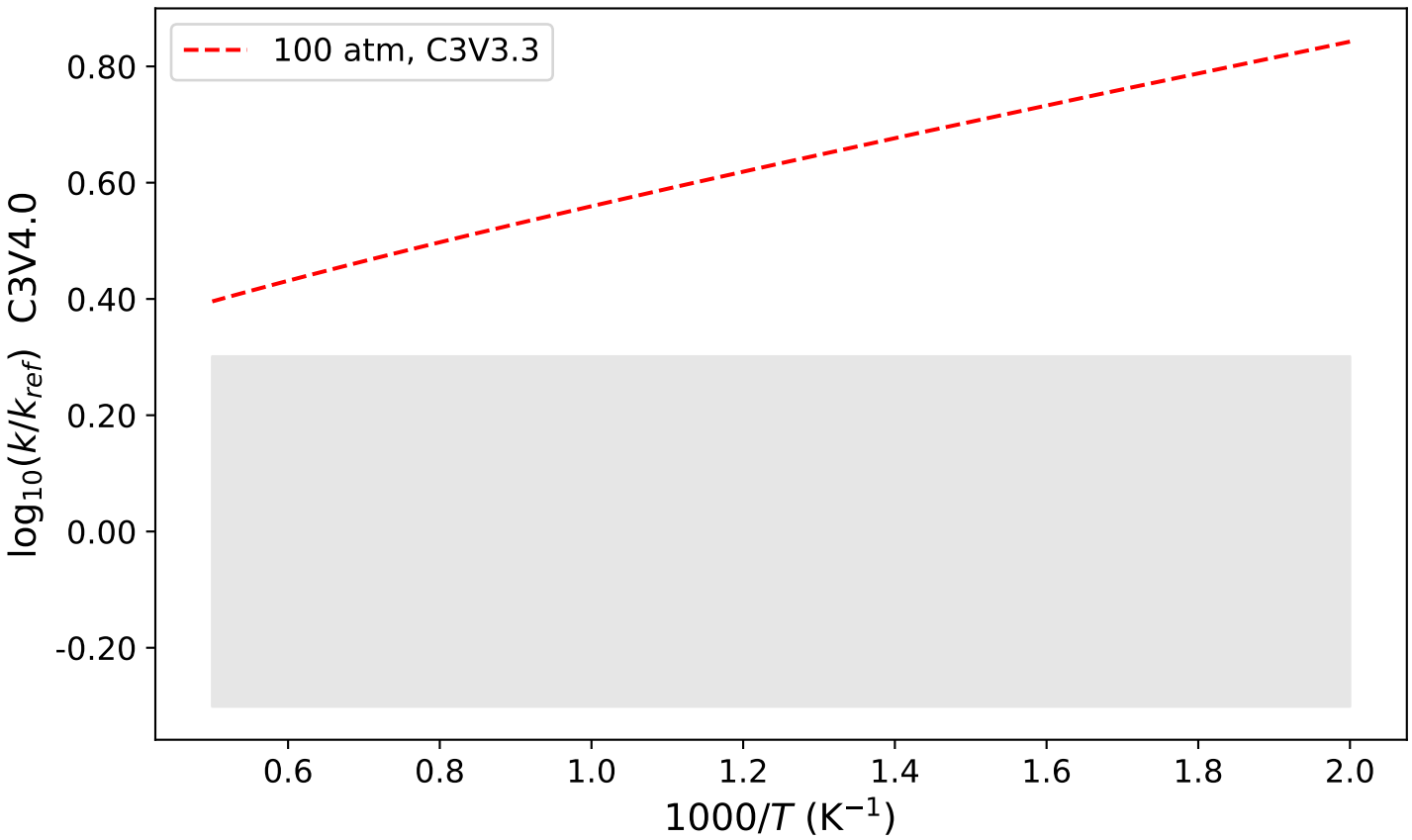
HO2 + NAPH = H2O2 + NAPH-A



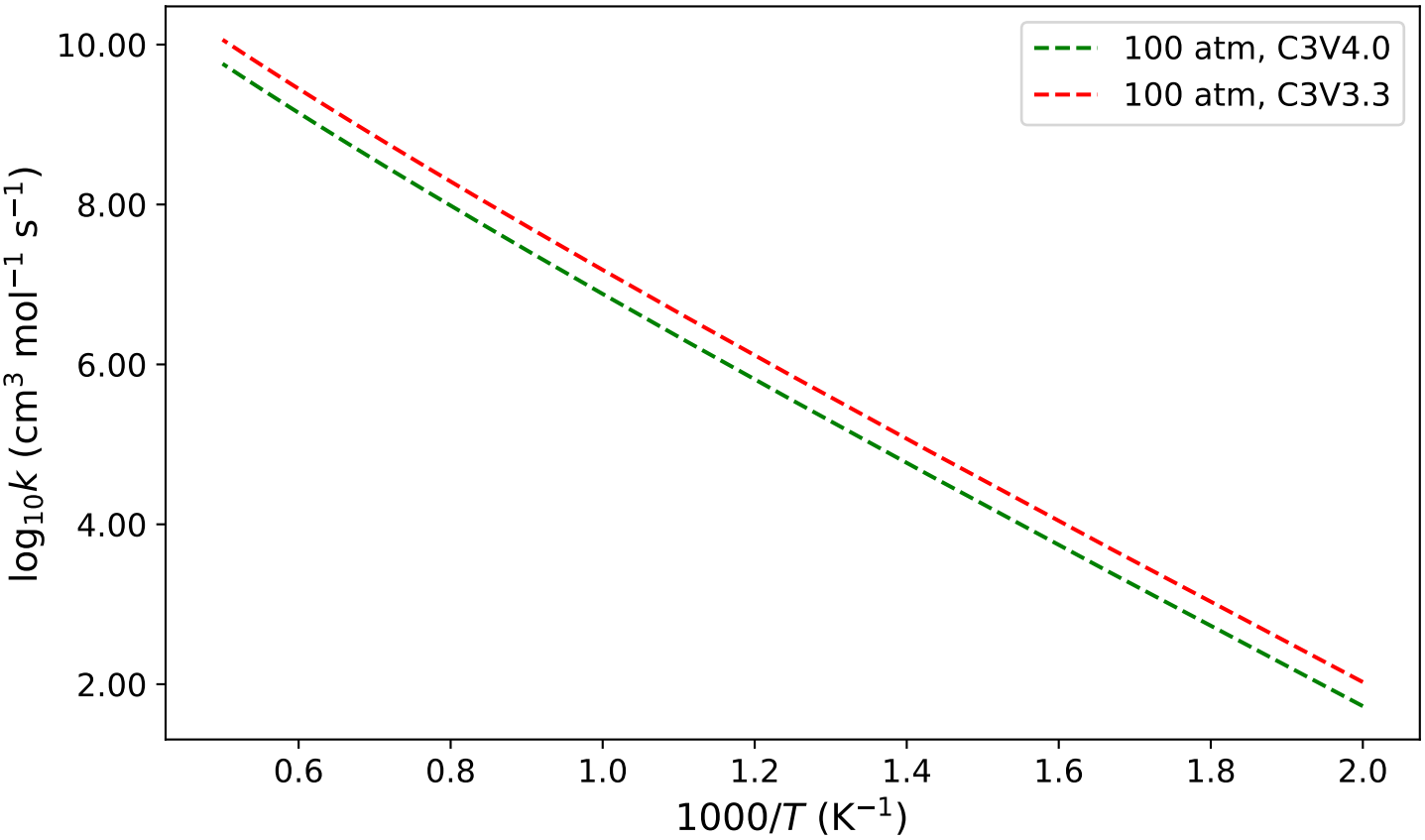


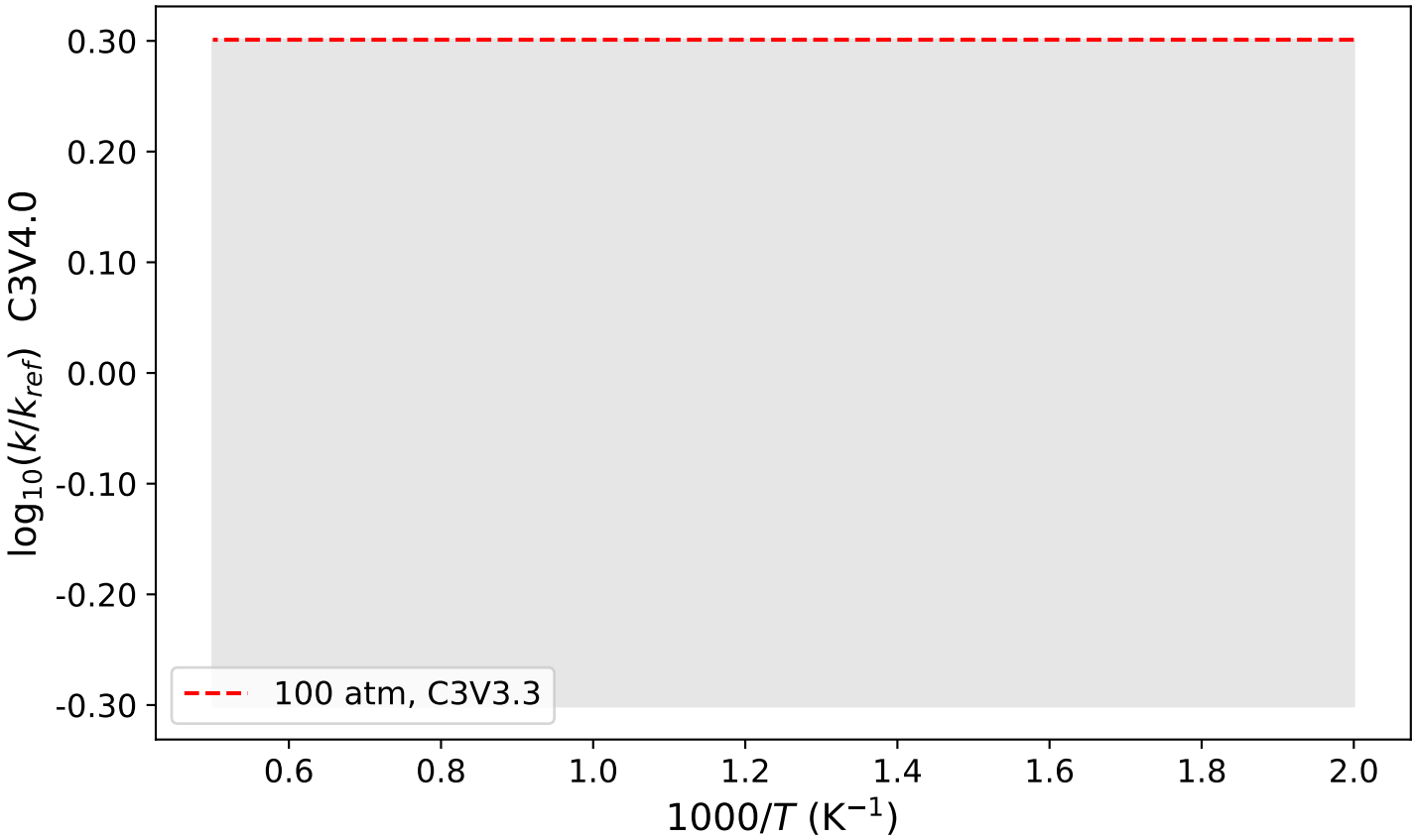
C6H5CH2CH2 = H + C6H5C2H3



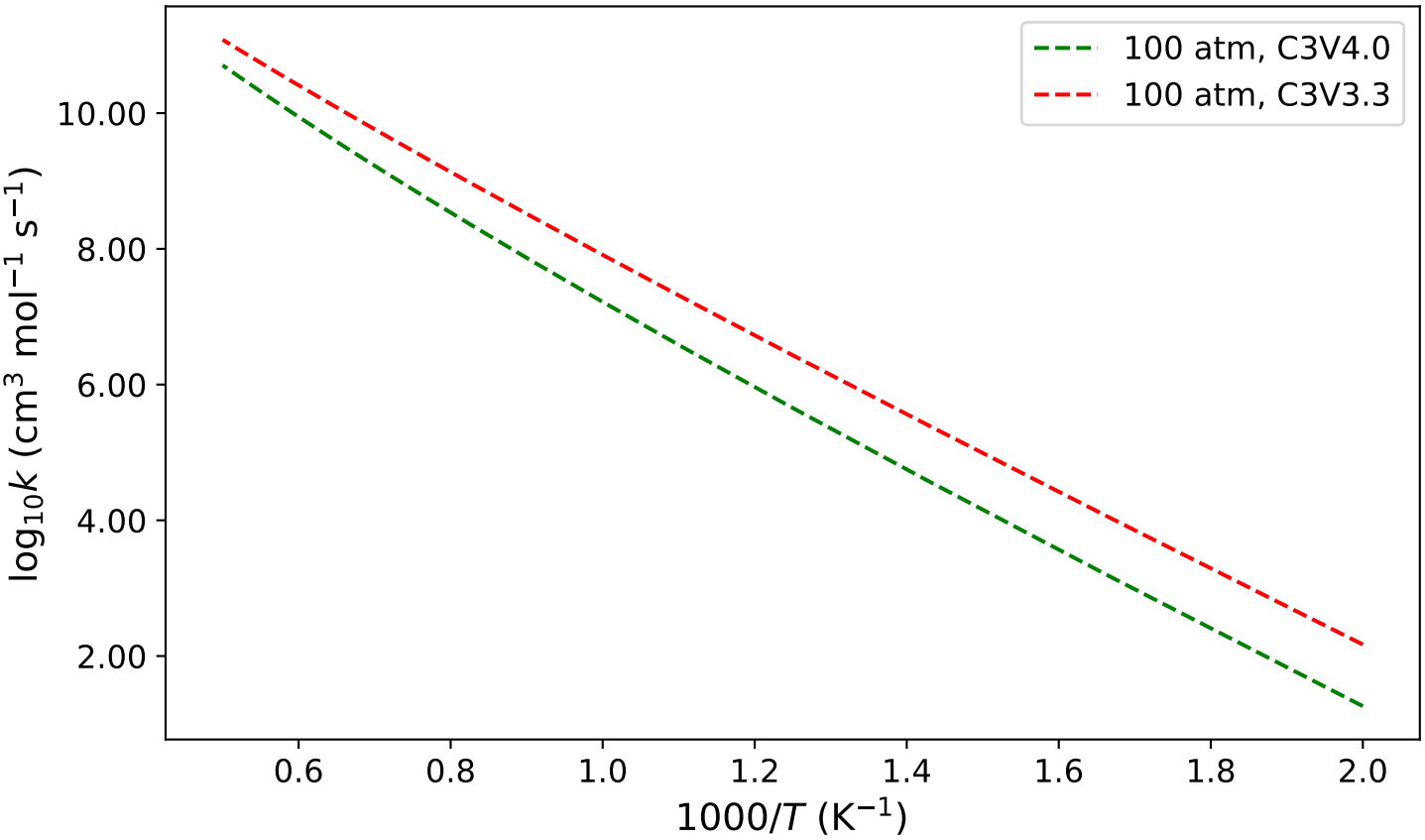


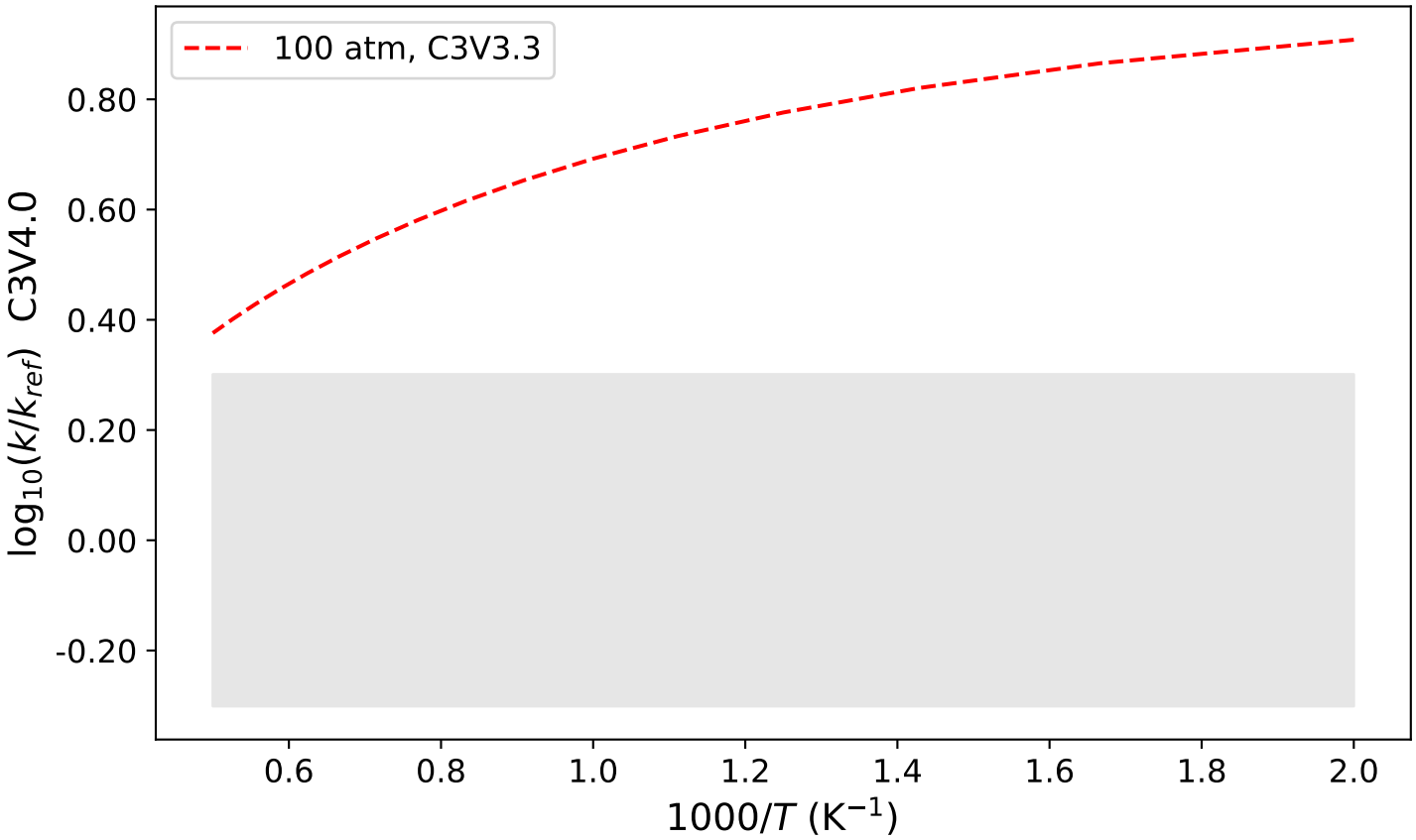
NAPH + C4H71-3 = C4H8-2 + NAPH-A



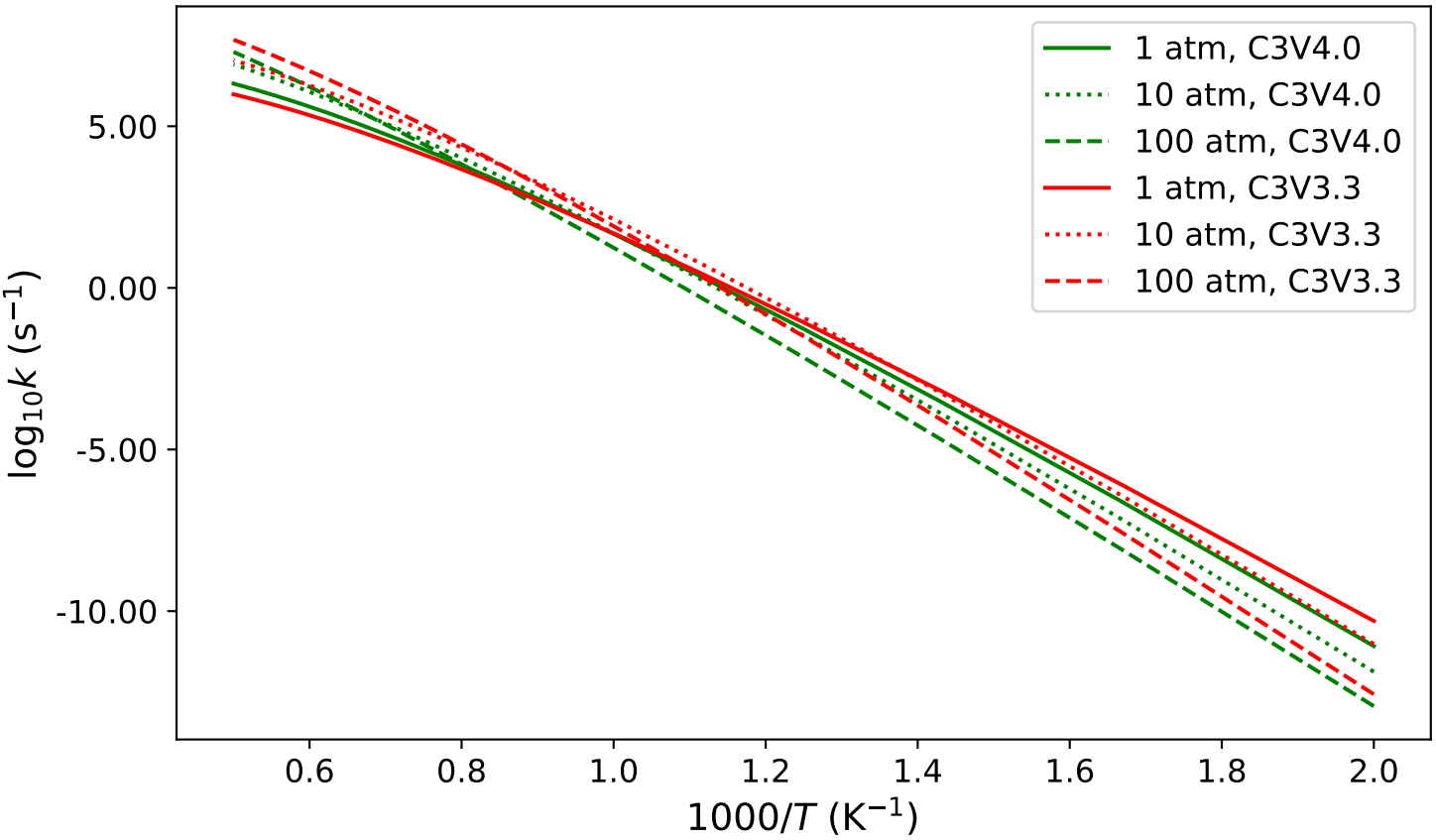


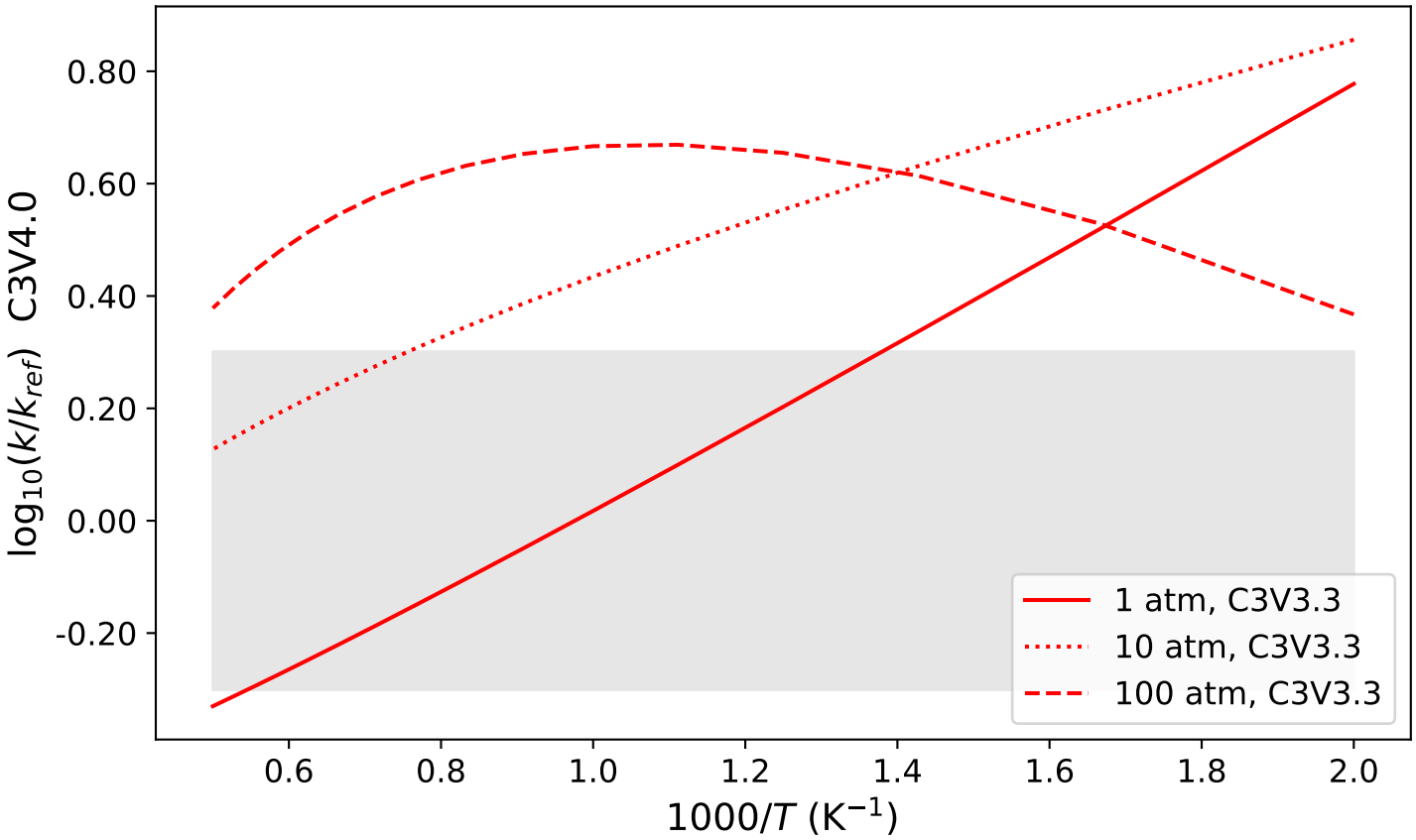
HO2 + C10H7CH3-A = H2O2 + C10H6CH3-A



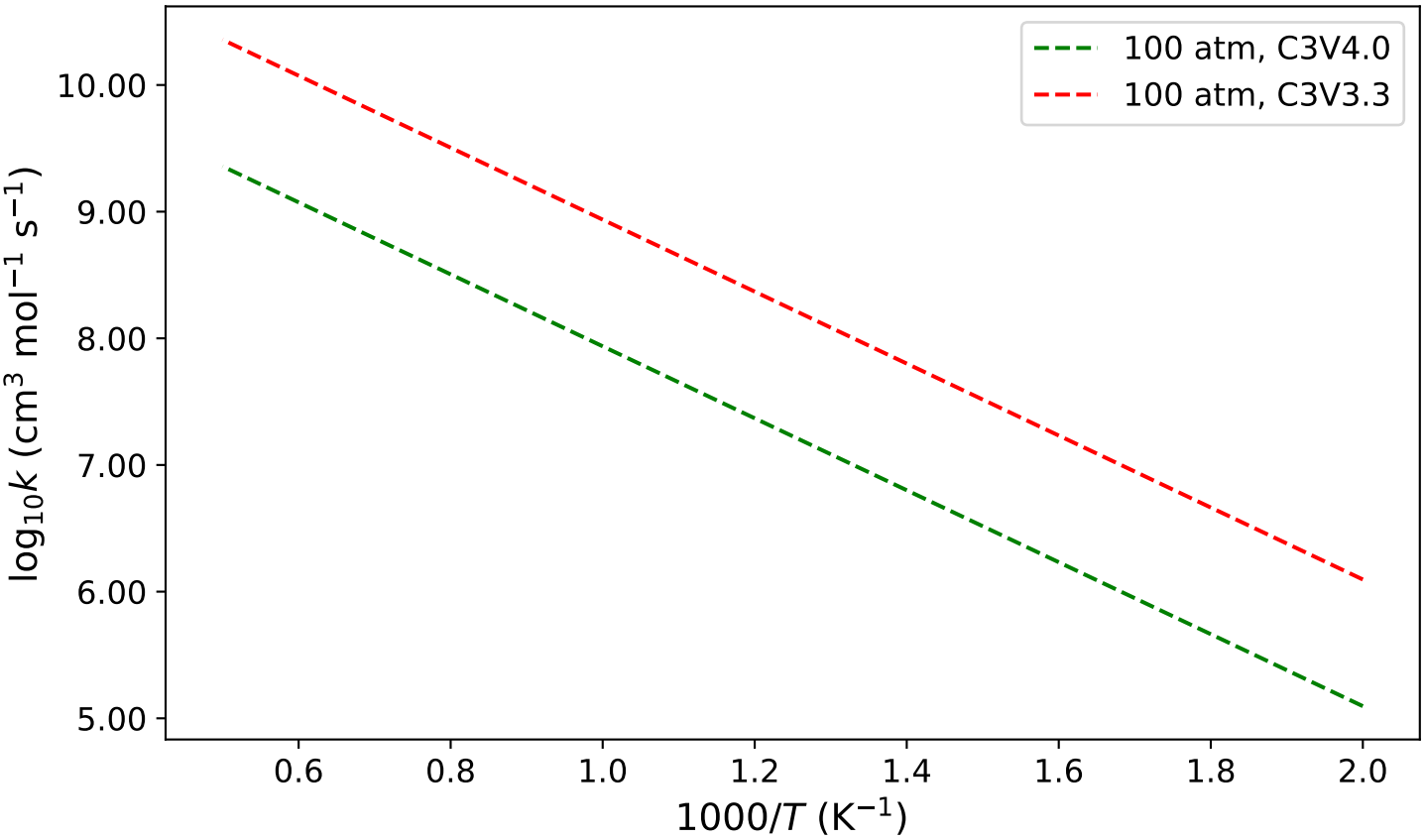


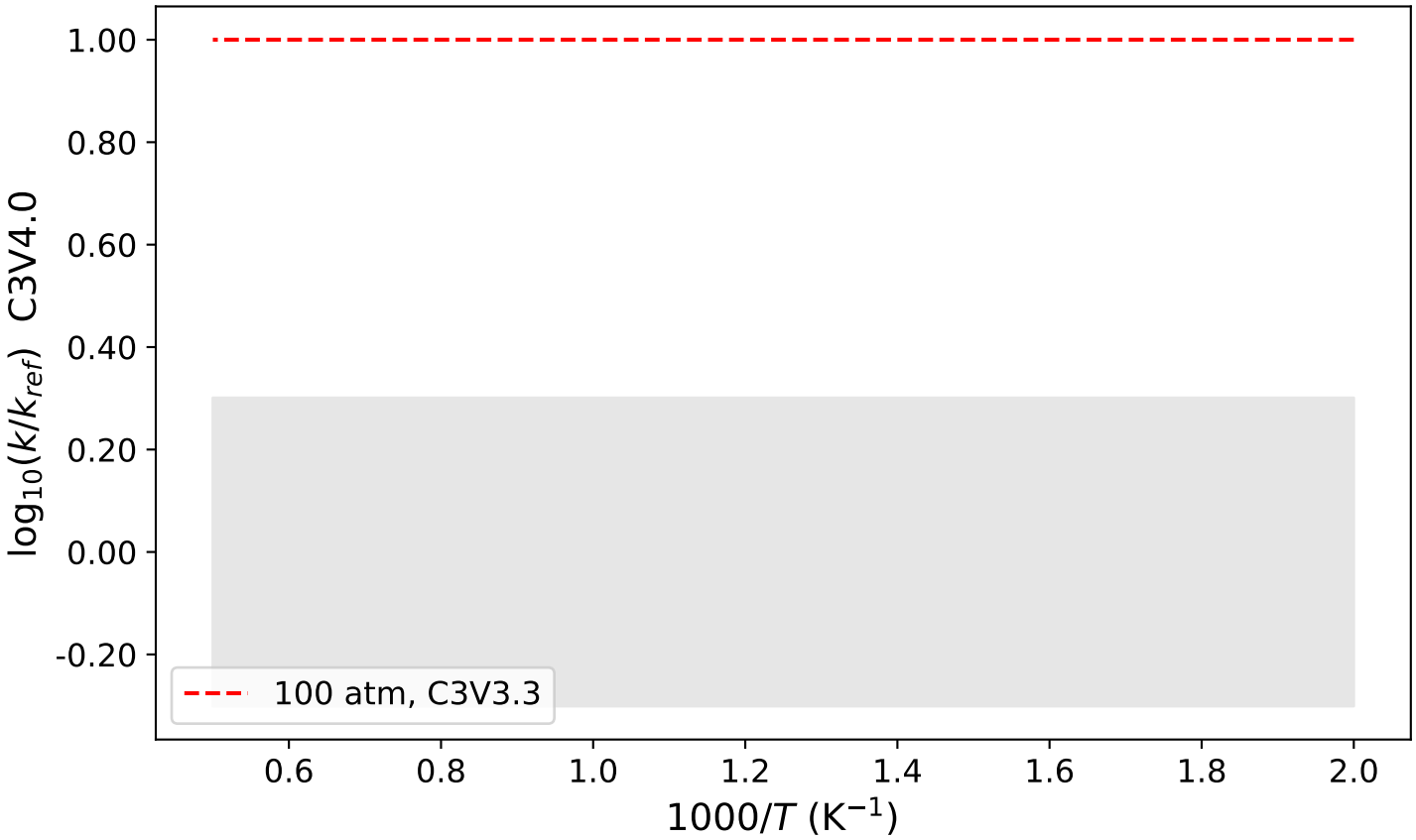
CVCCJCVC = C2H2 + C3H5-A



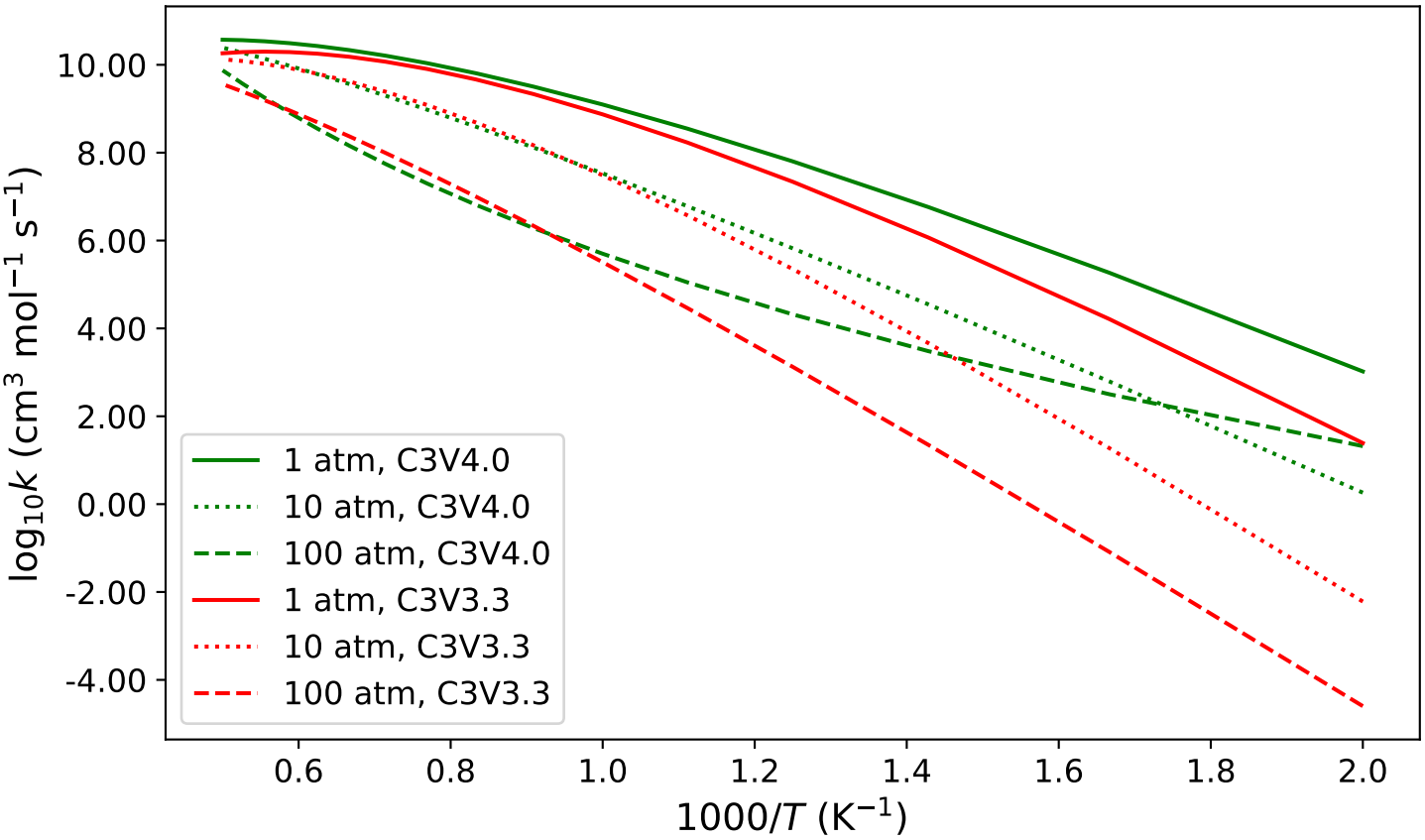


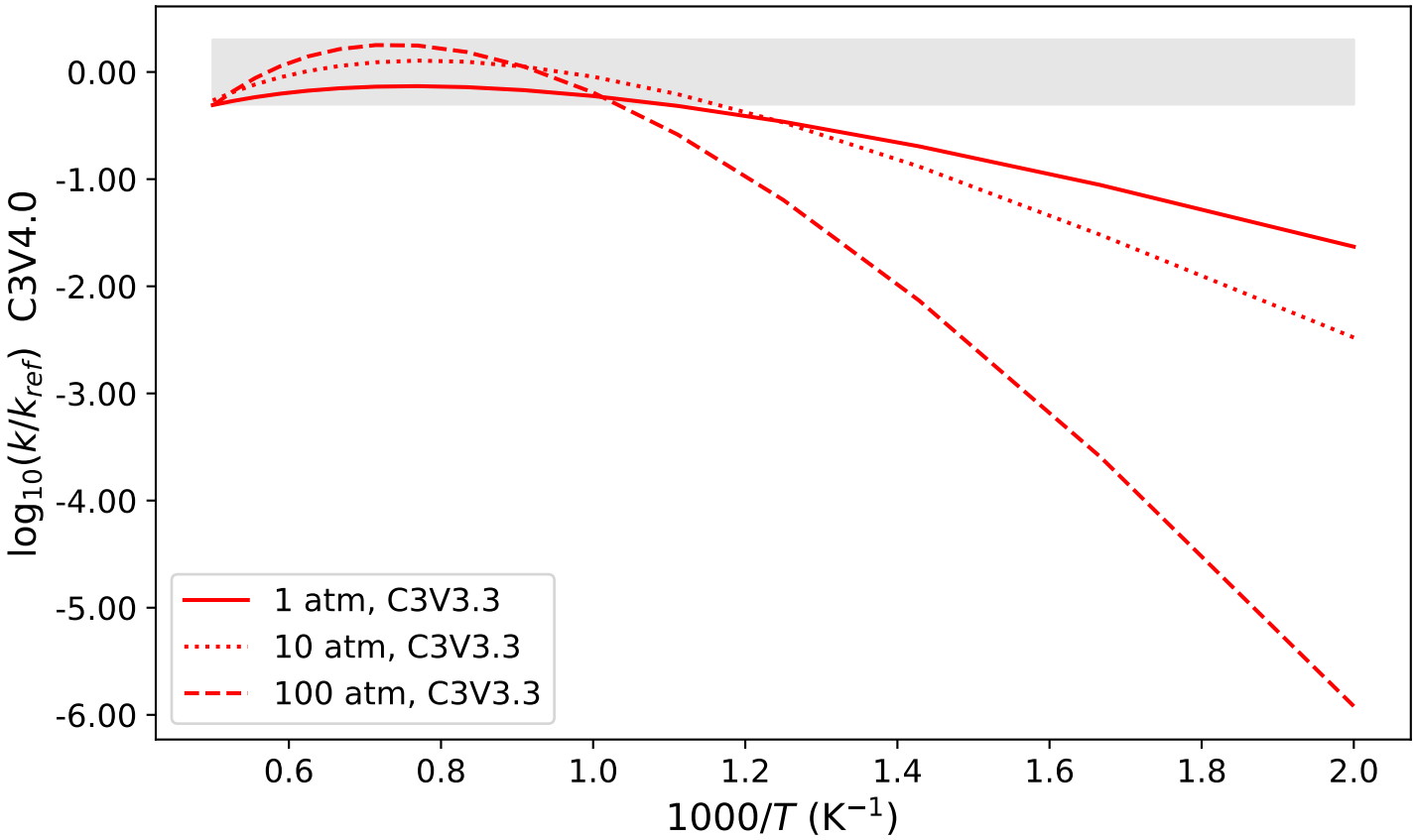
C4H6 + C4H71-3 => CH3 + H2 + C6H5CH3

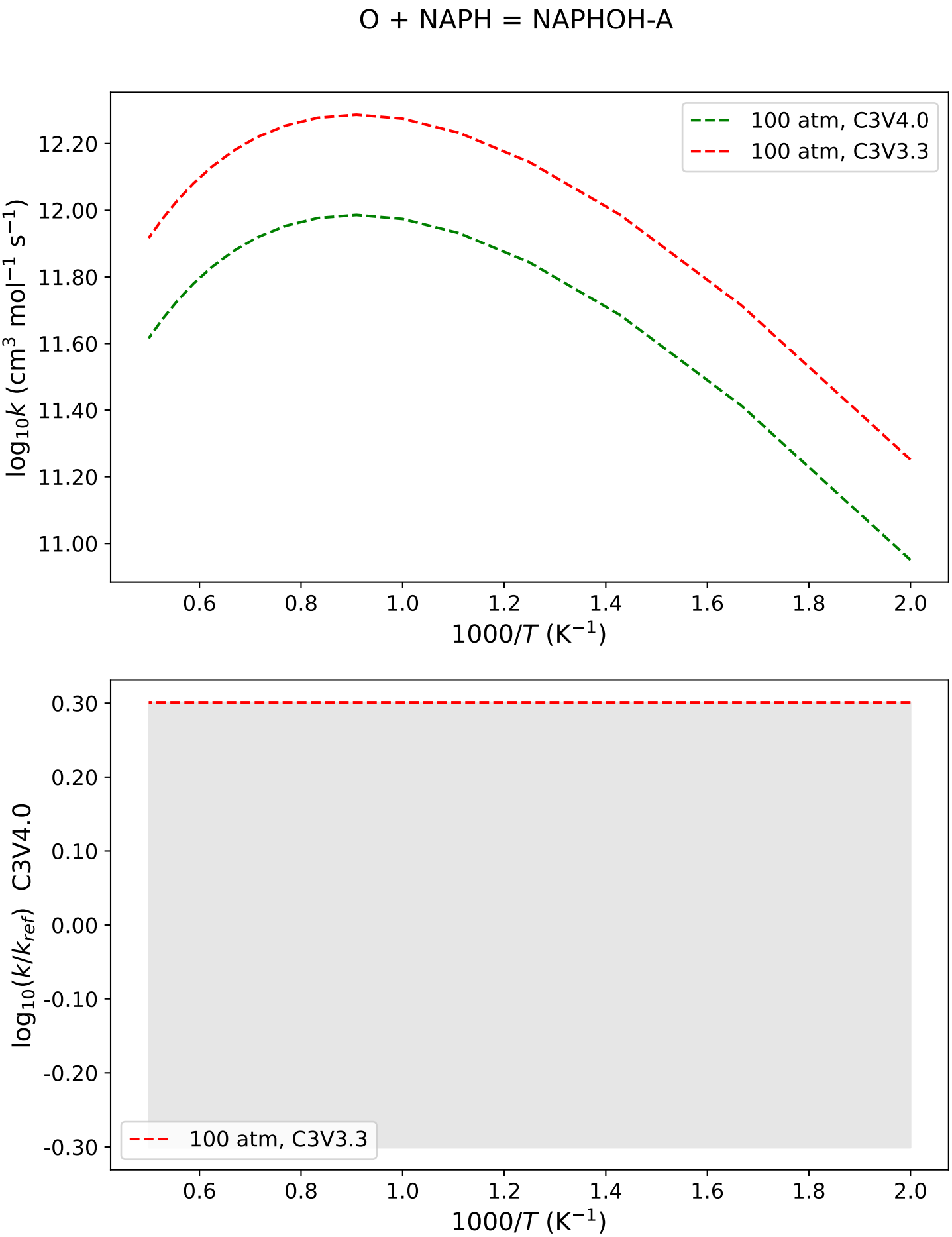




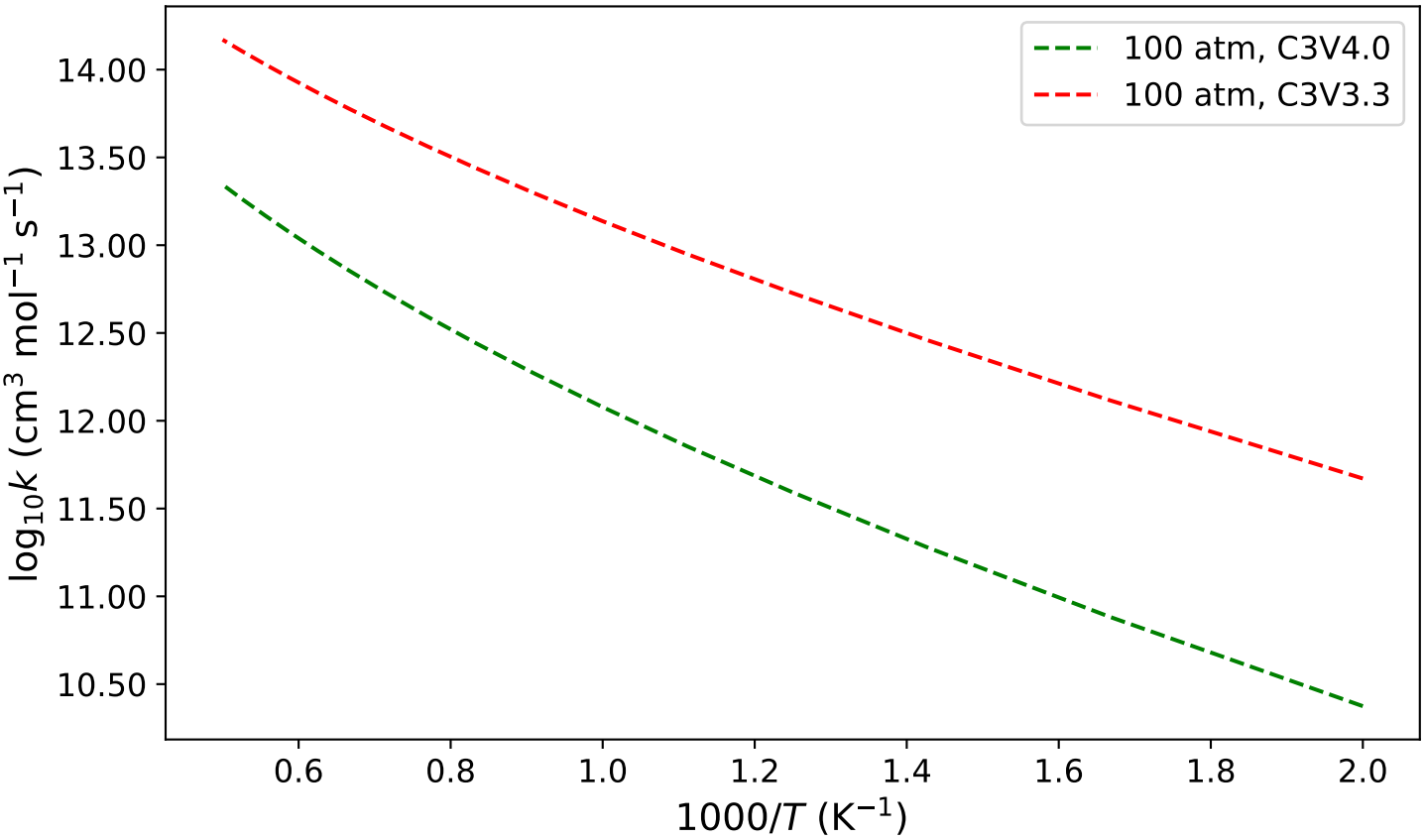
C2H4 + C6H4C2H = H + NAPH

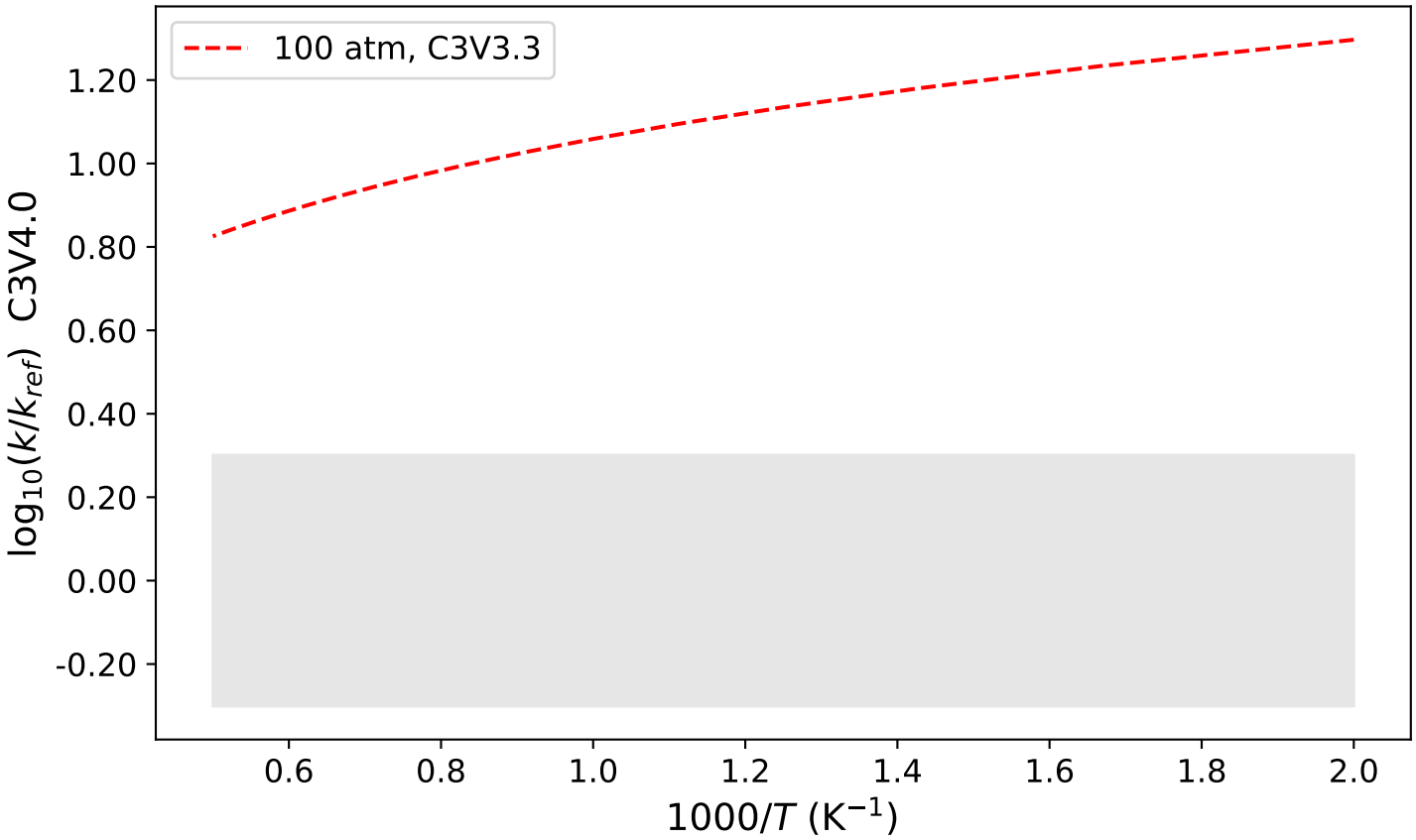




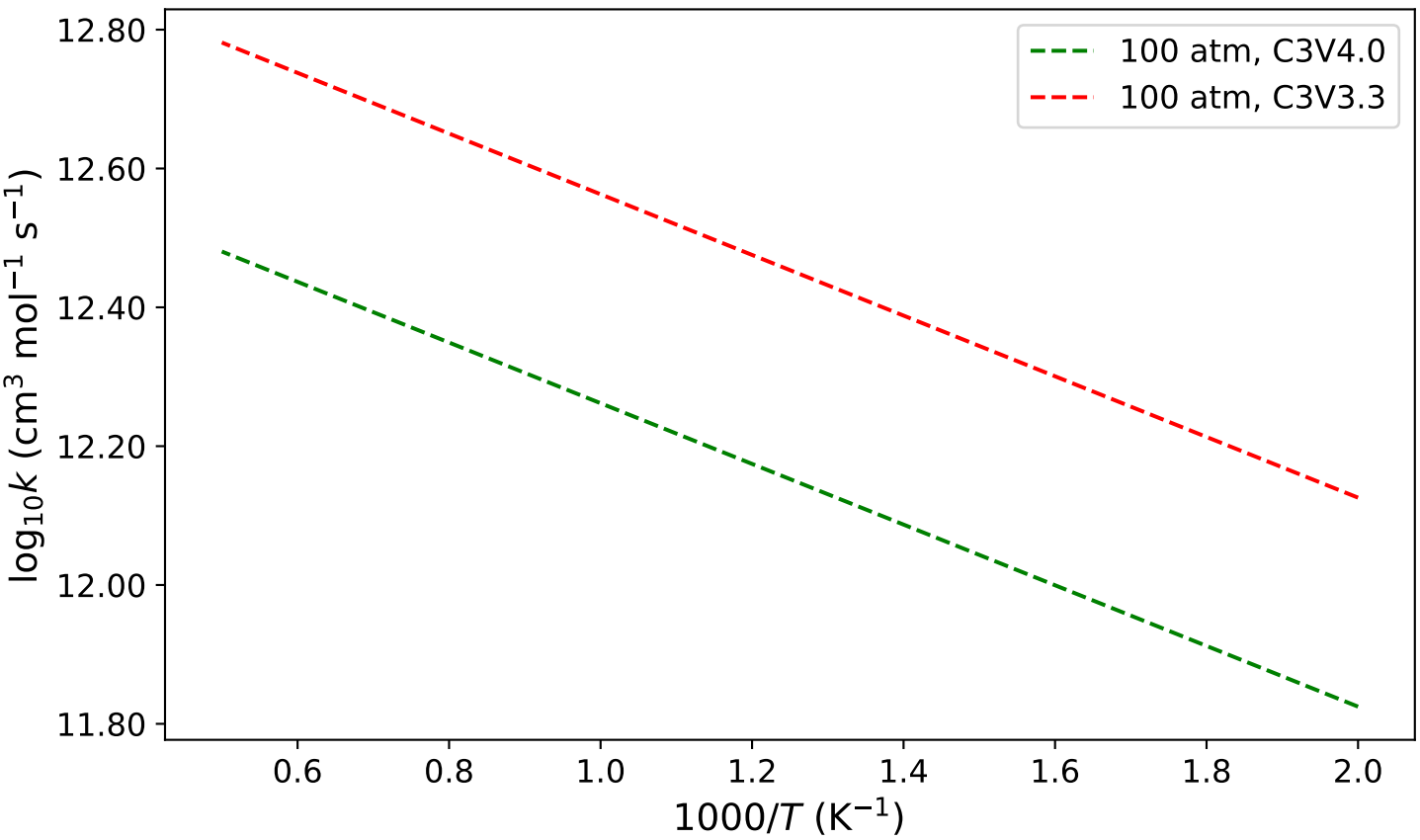

O + NAPH = NAPHOH-A
100 atm, C3V4.0
100 atm, C3V3.3
12.20
12.00
11.80
11.60
11.40
11.20
11.00
$\log_{10}k$ (cm^3 mol^{-1} s^{-1})
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K^{-1})
0.30
0.20
0.10
0.00
-0.10
-0.20
-0.30
$\log_{10}(k/k_{ref})$ C3V4.0
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K^{-1})

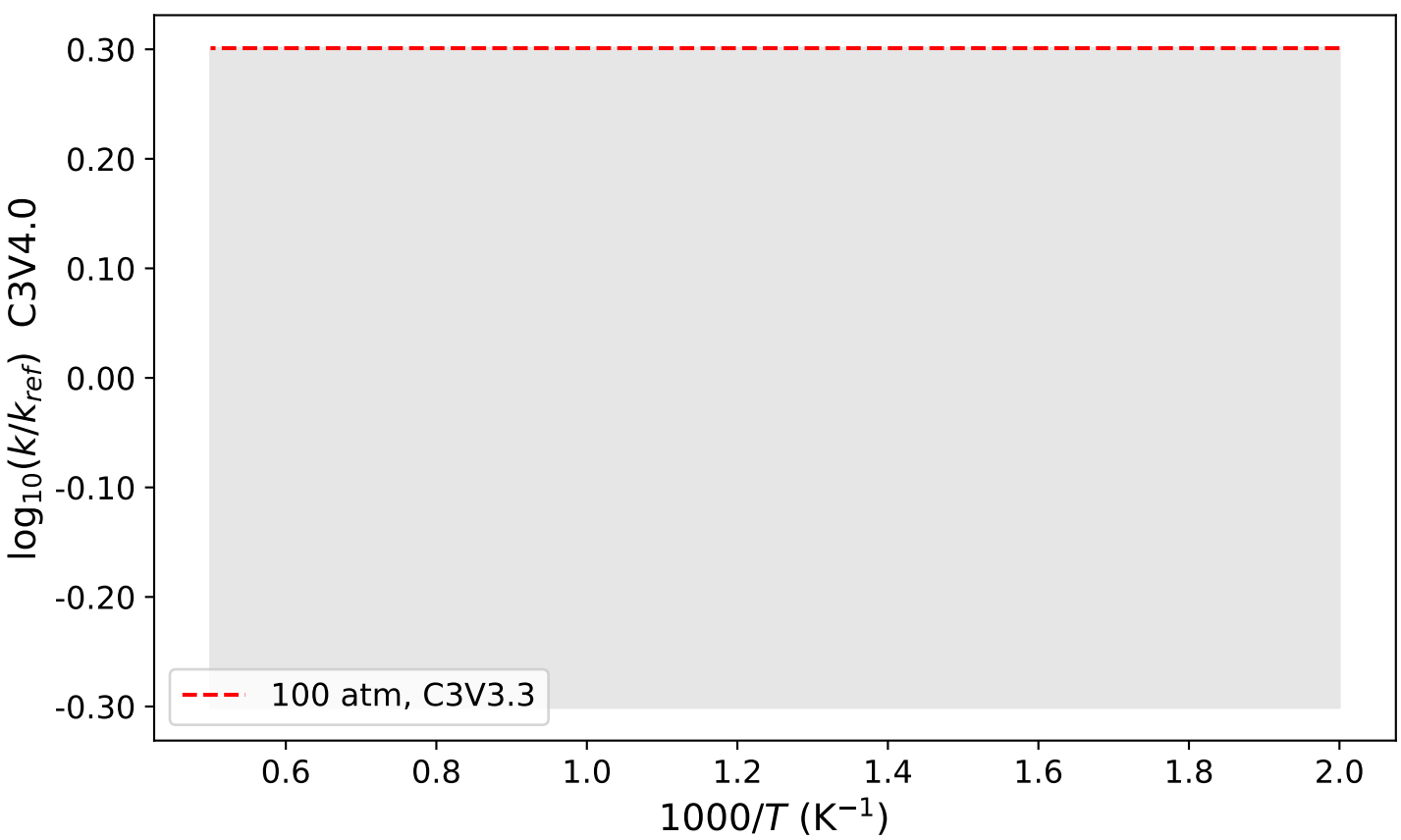
H + C14H14 = H2 + C14H13



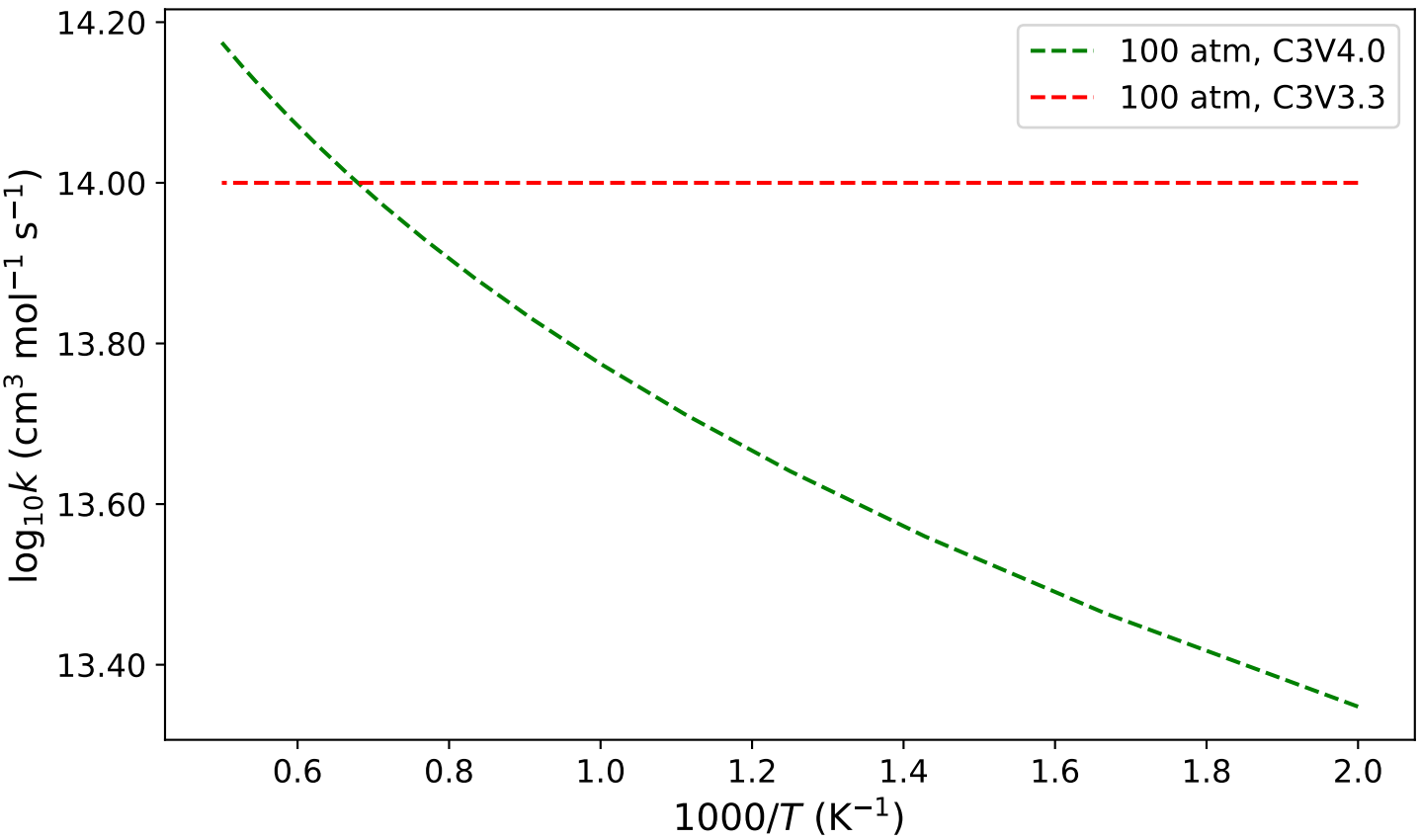


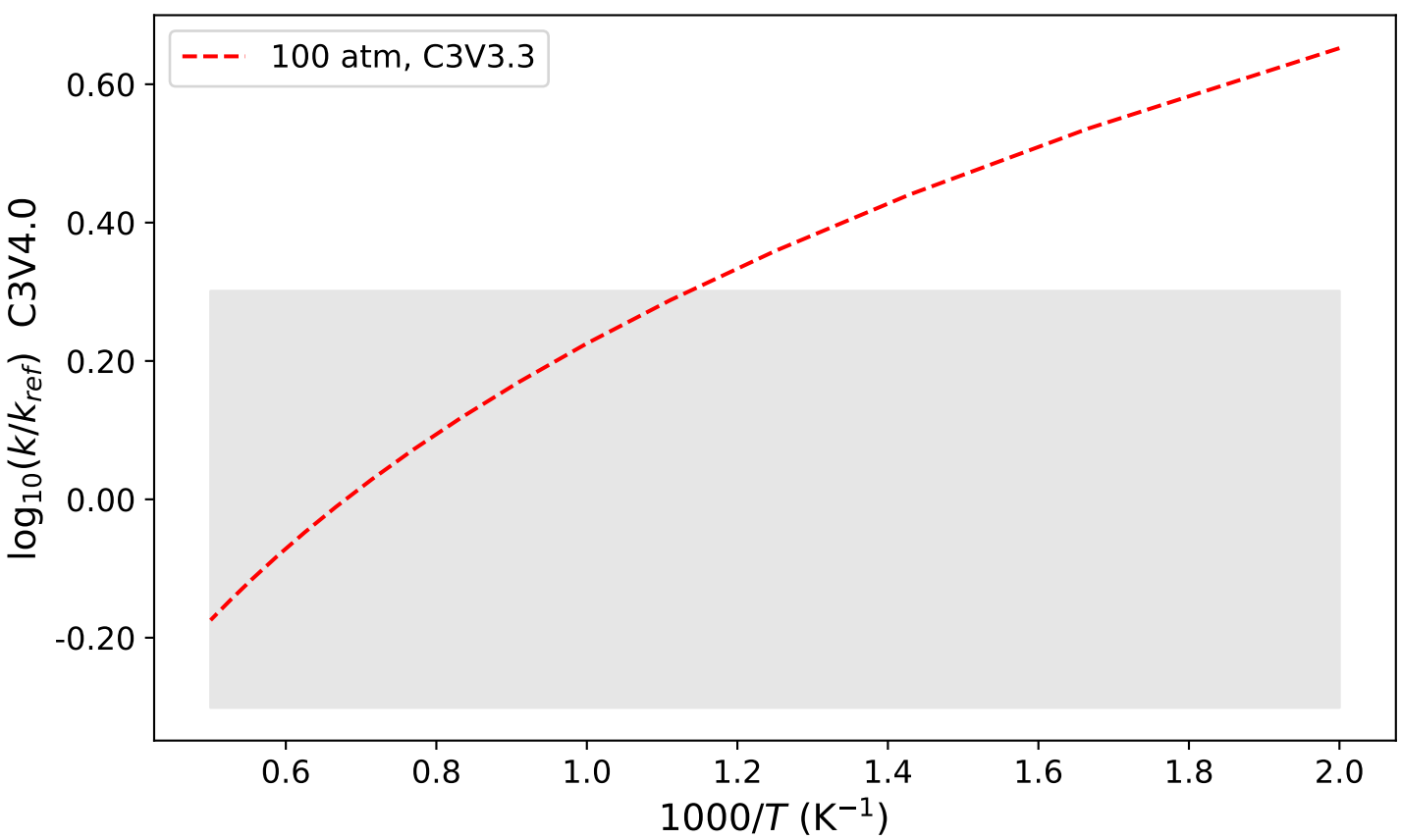
O + NAPH = H + NAPHO-A



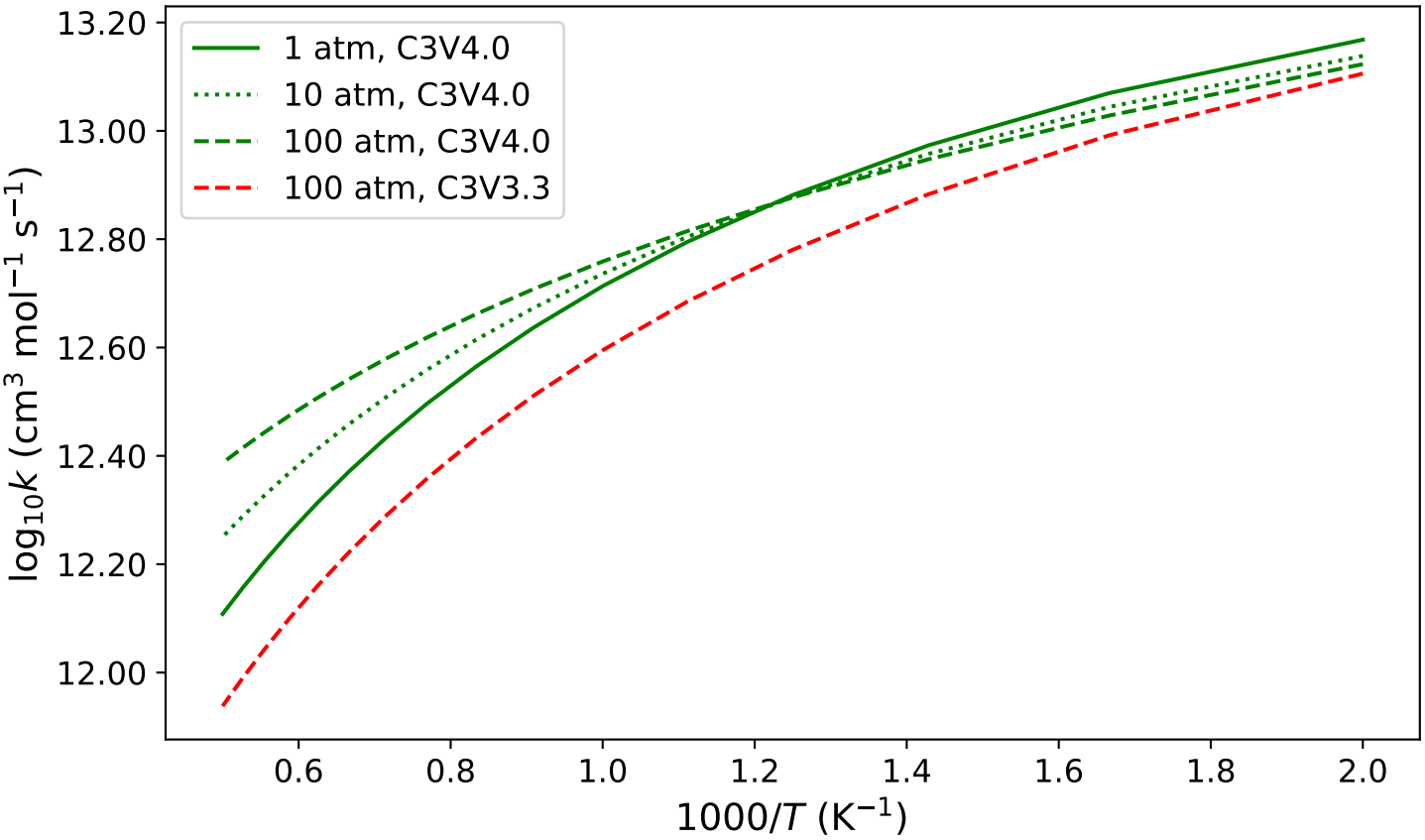


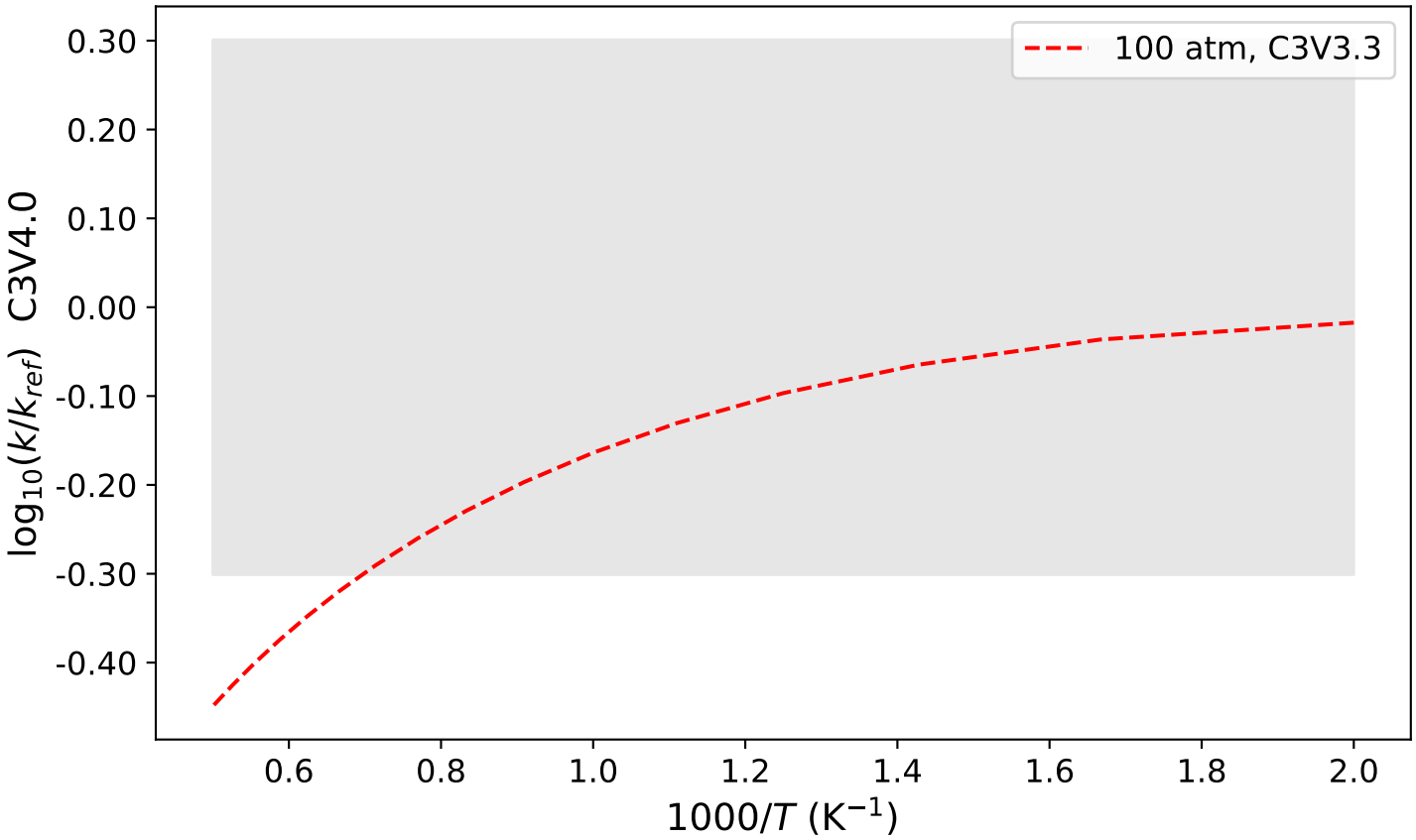
H + C6H5CHCH3 = C6H5C2H5



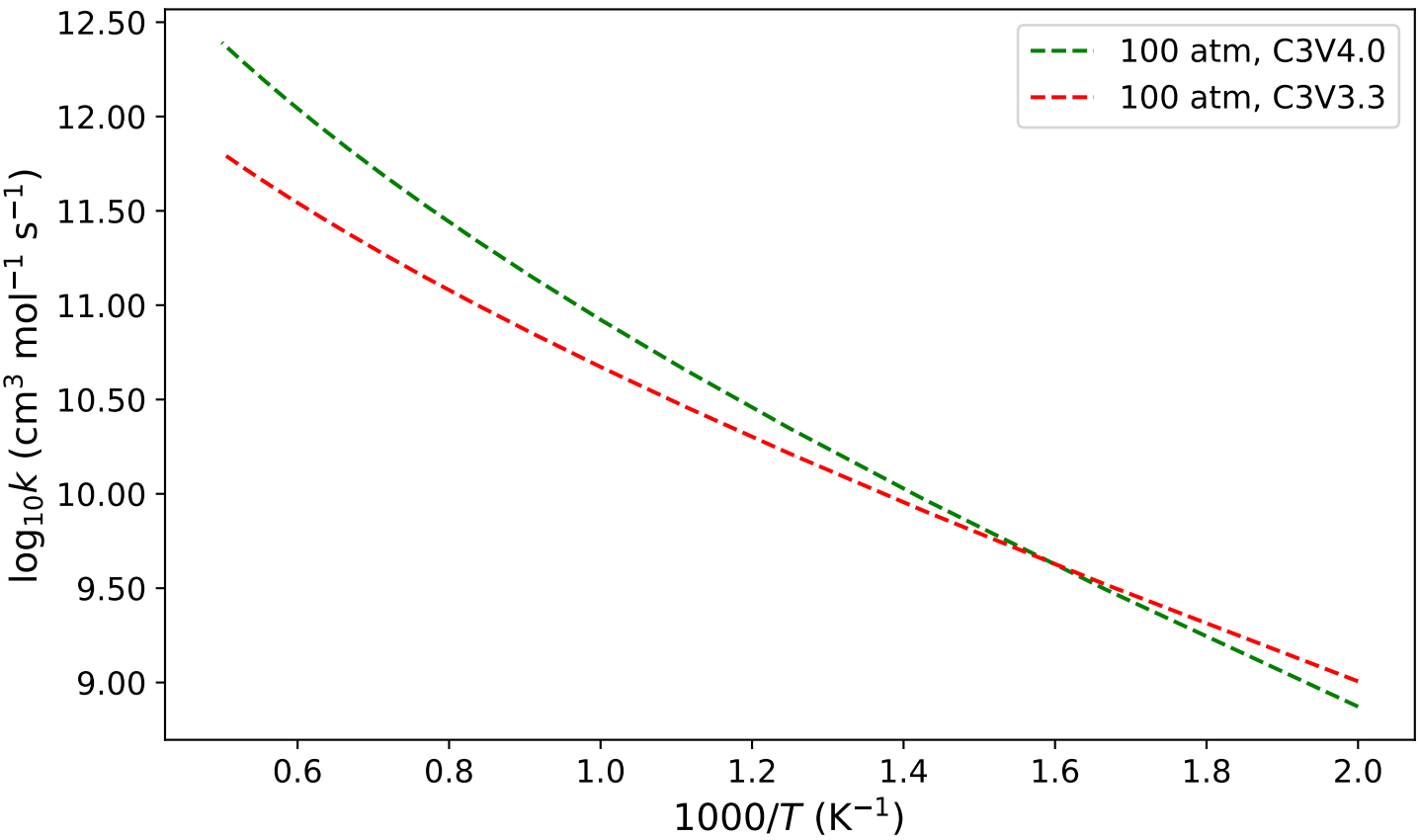


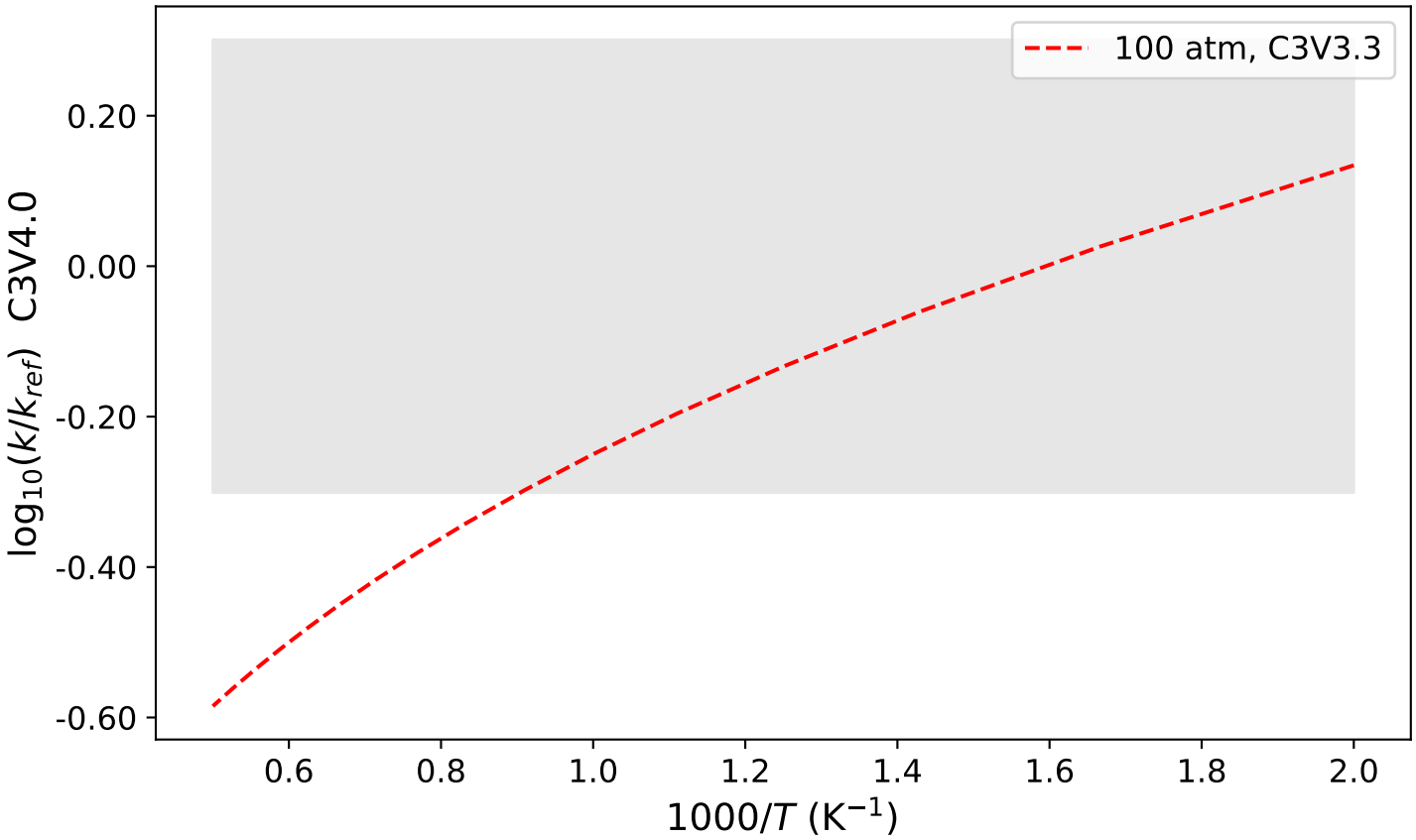
C6H5CH2 + C6H5CH2 = C14H14



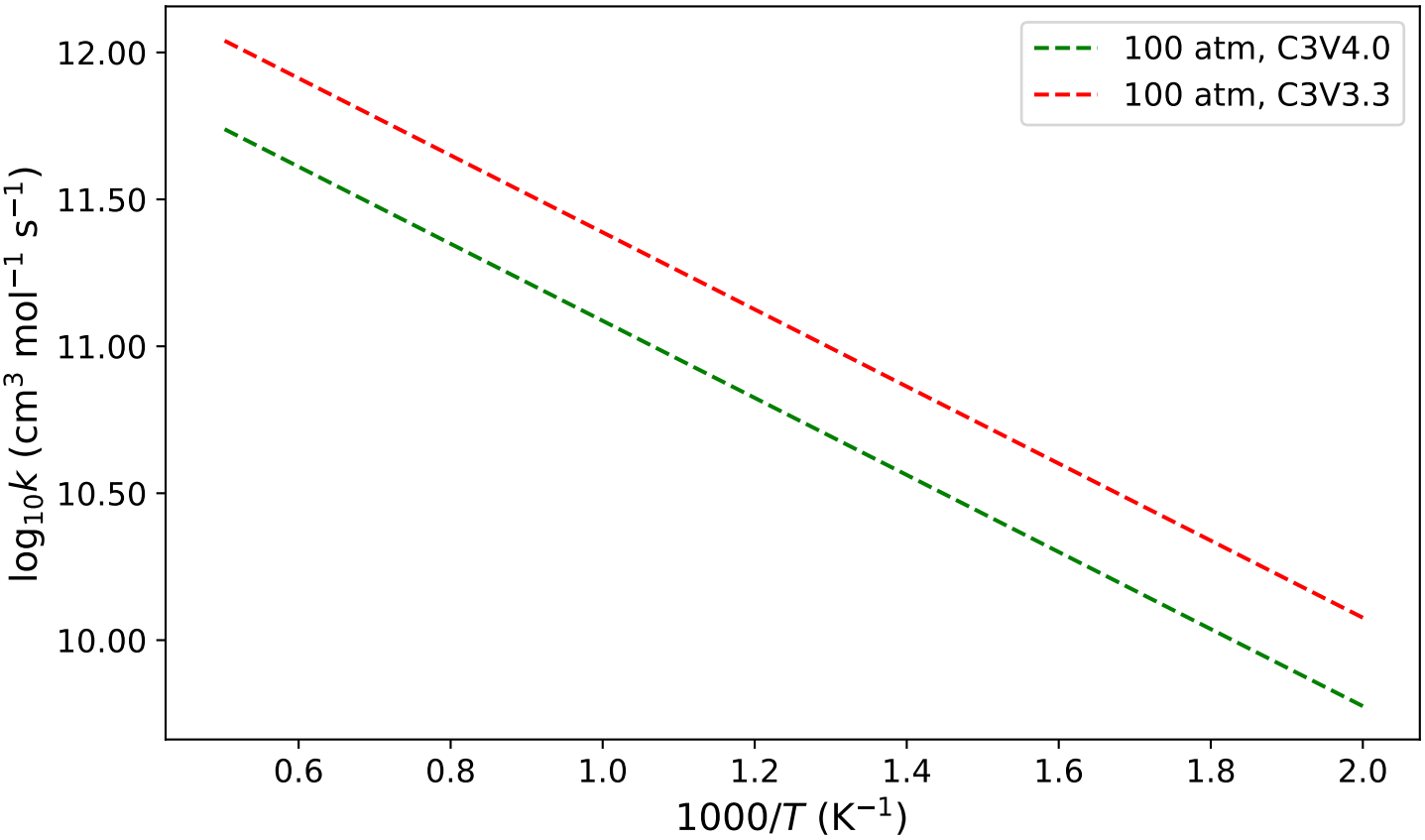


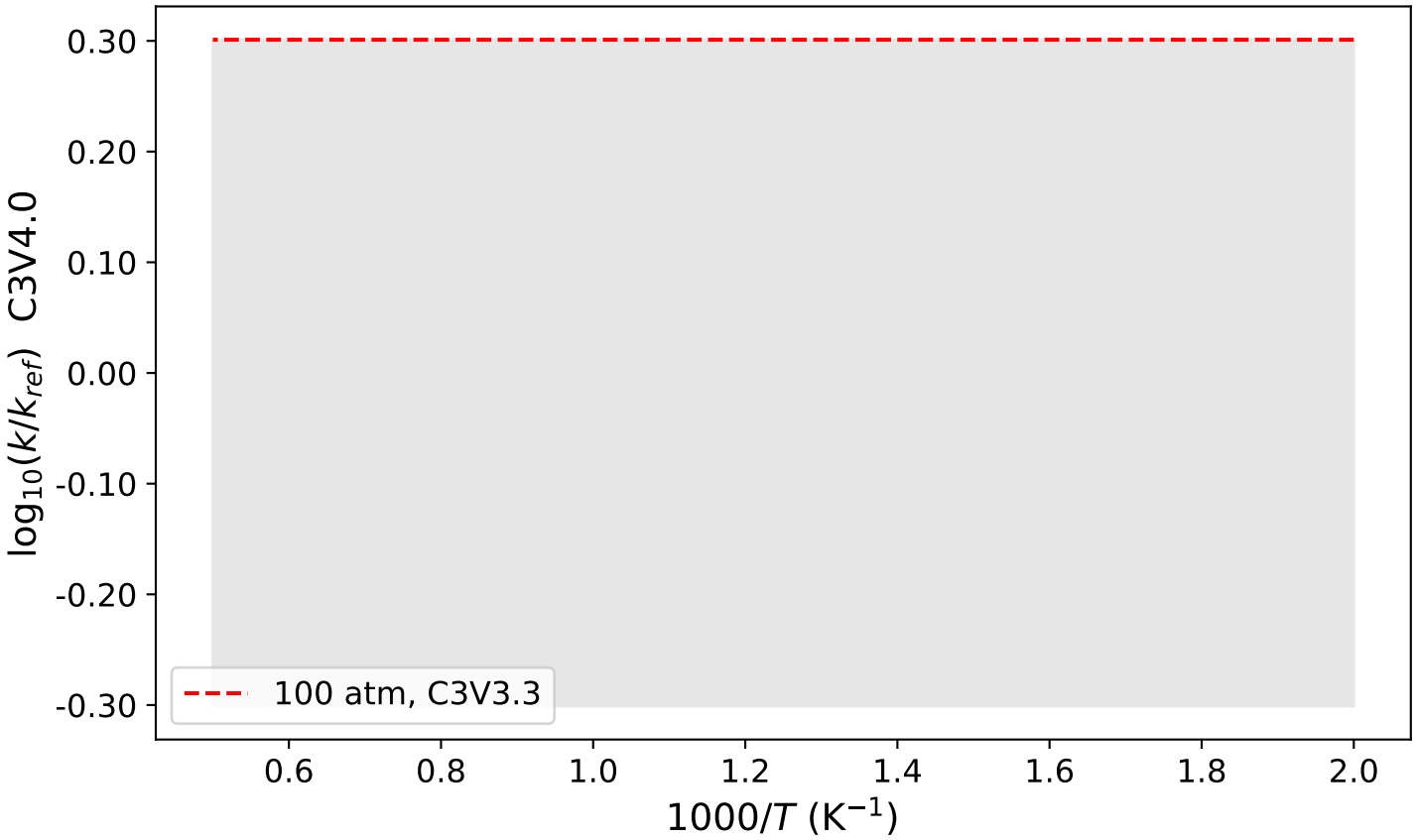
CH3 + CYC5H8 = CH4 + CYC5H71-3



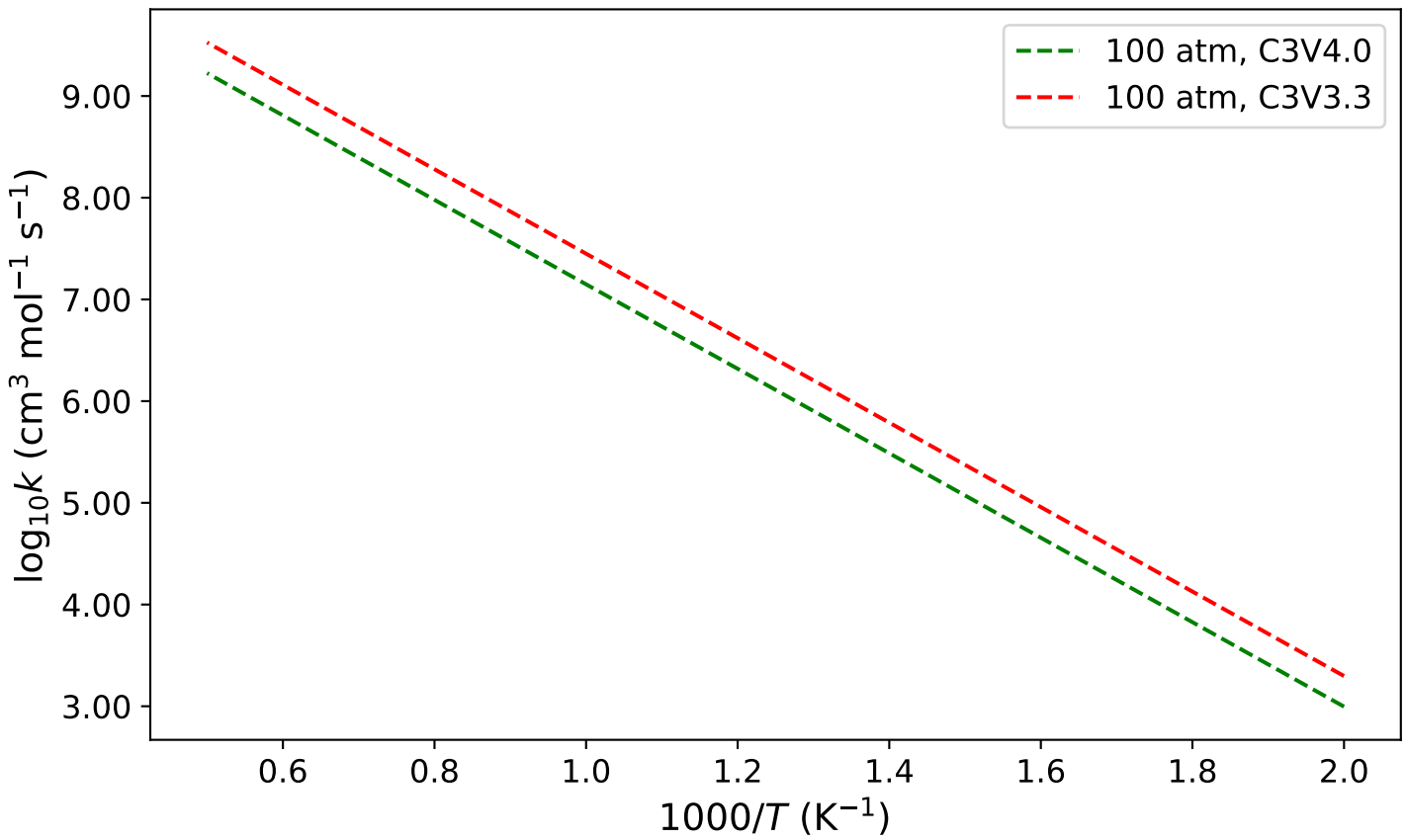


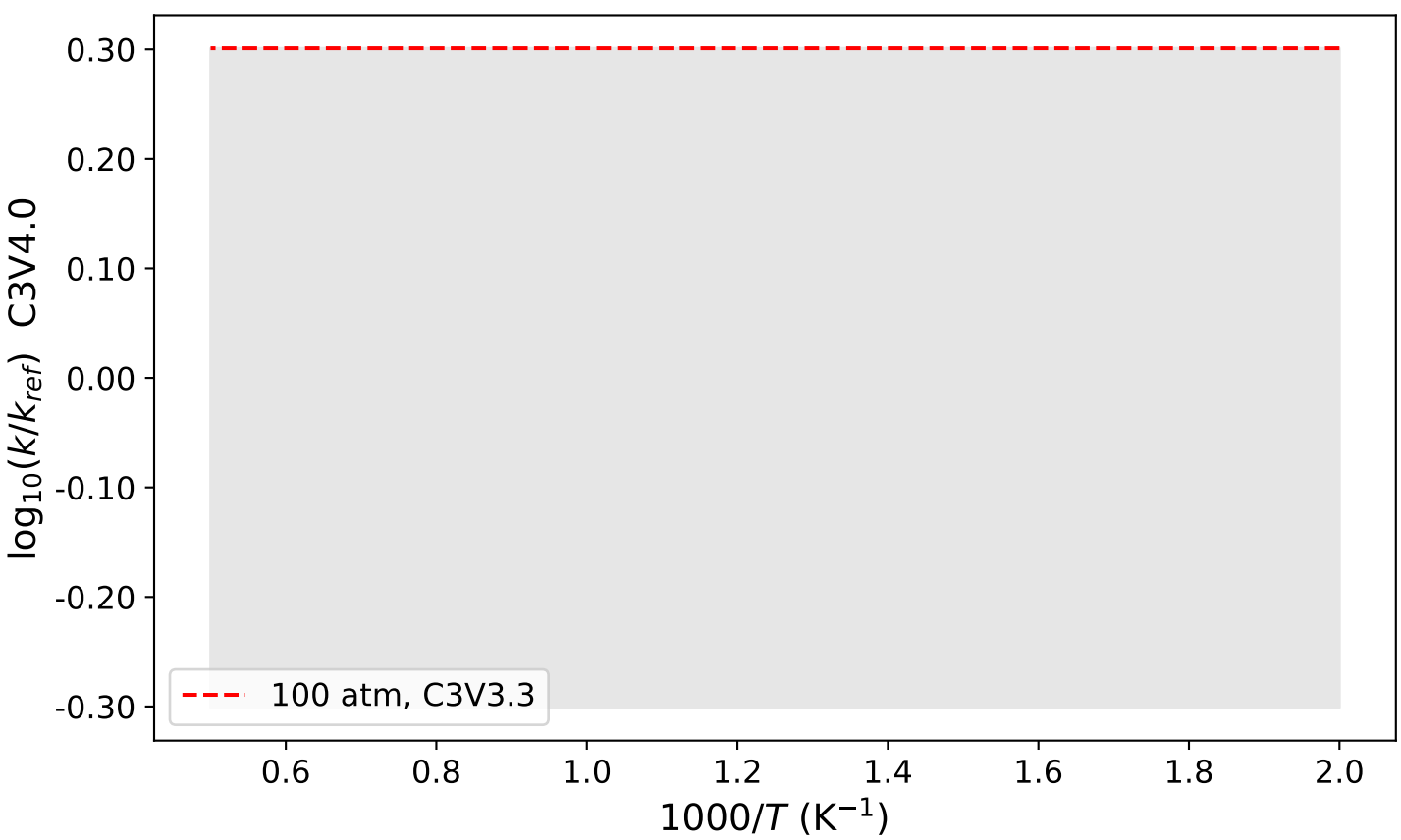
C5H5 + OC6H4CH3 => H + H + CO + C10H7CH3-A



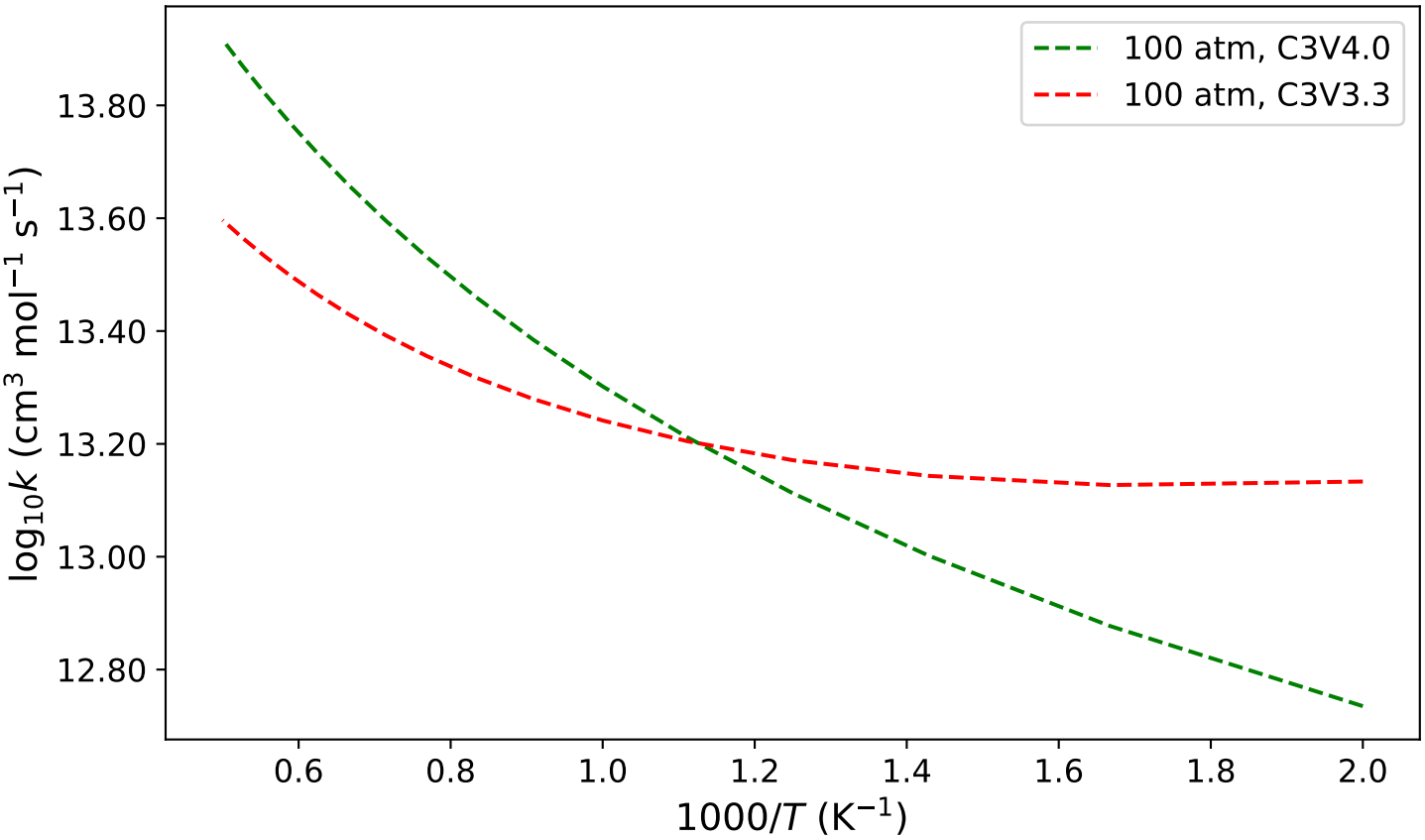


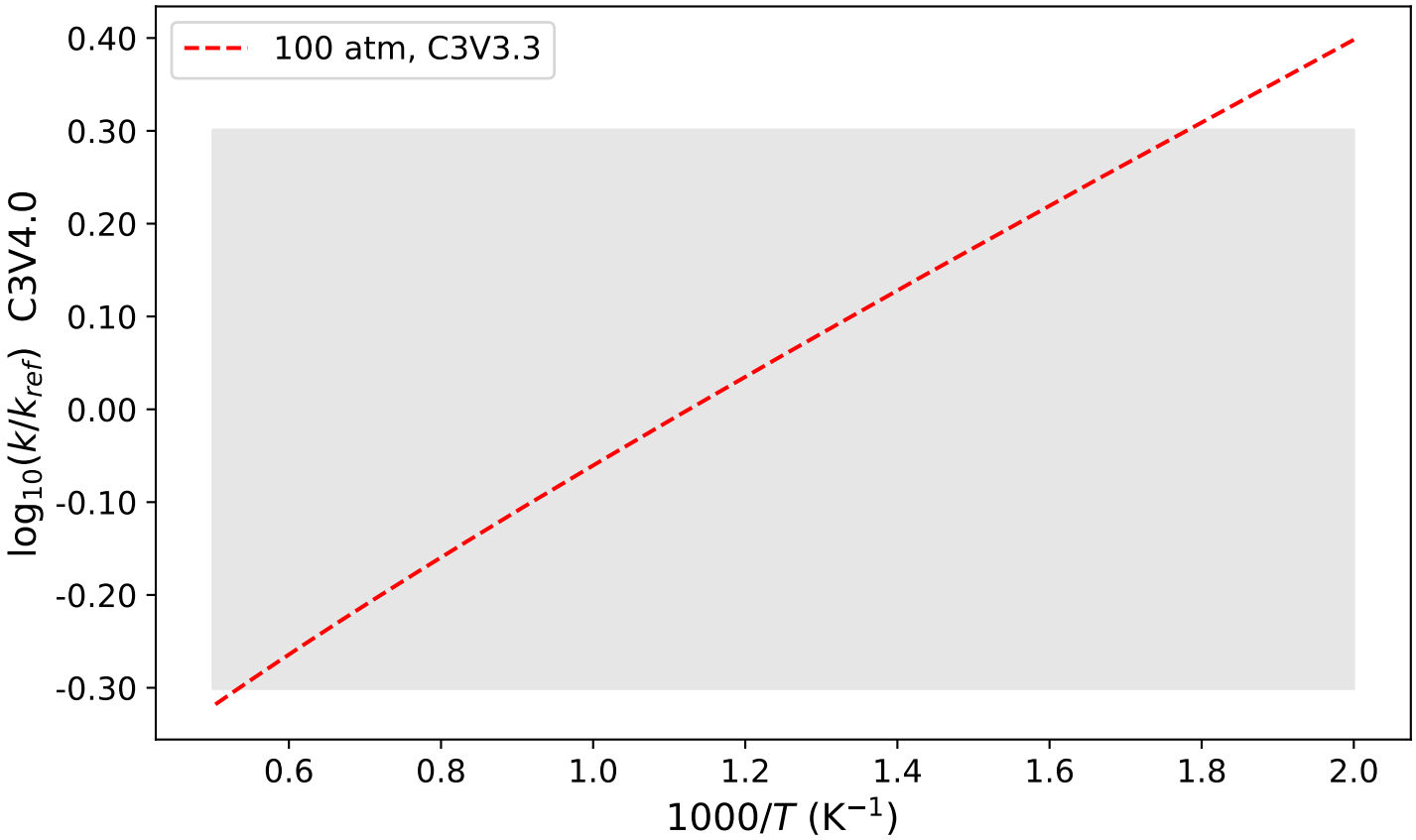
C3H3 + C6H5C2H = C10H7CH2-A



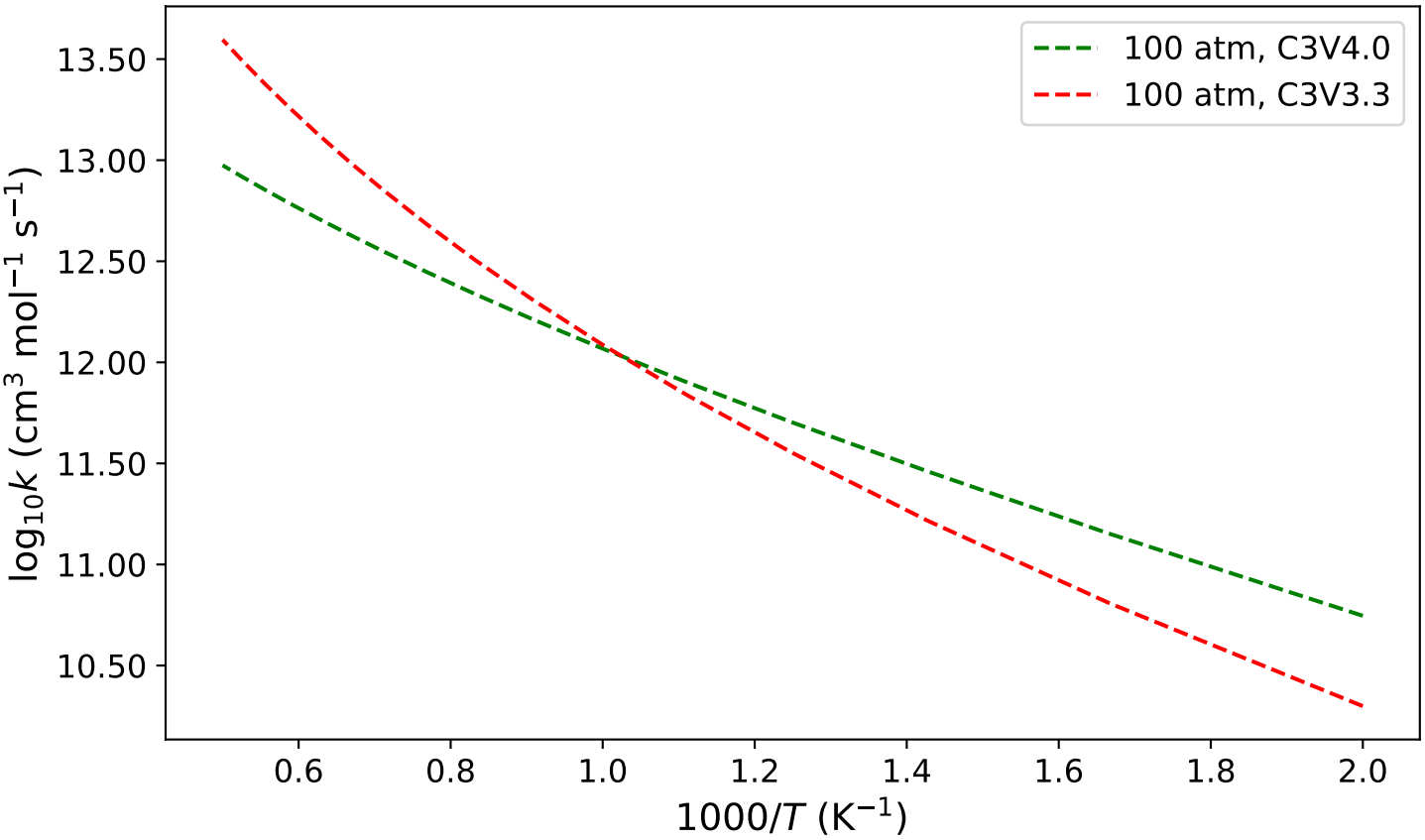


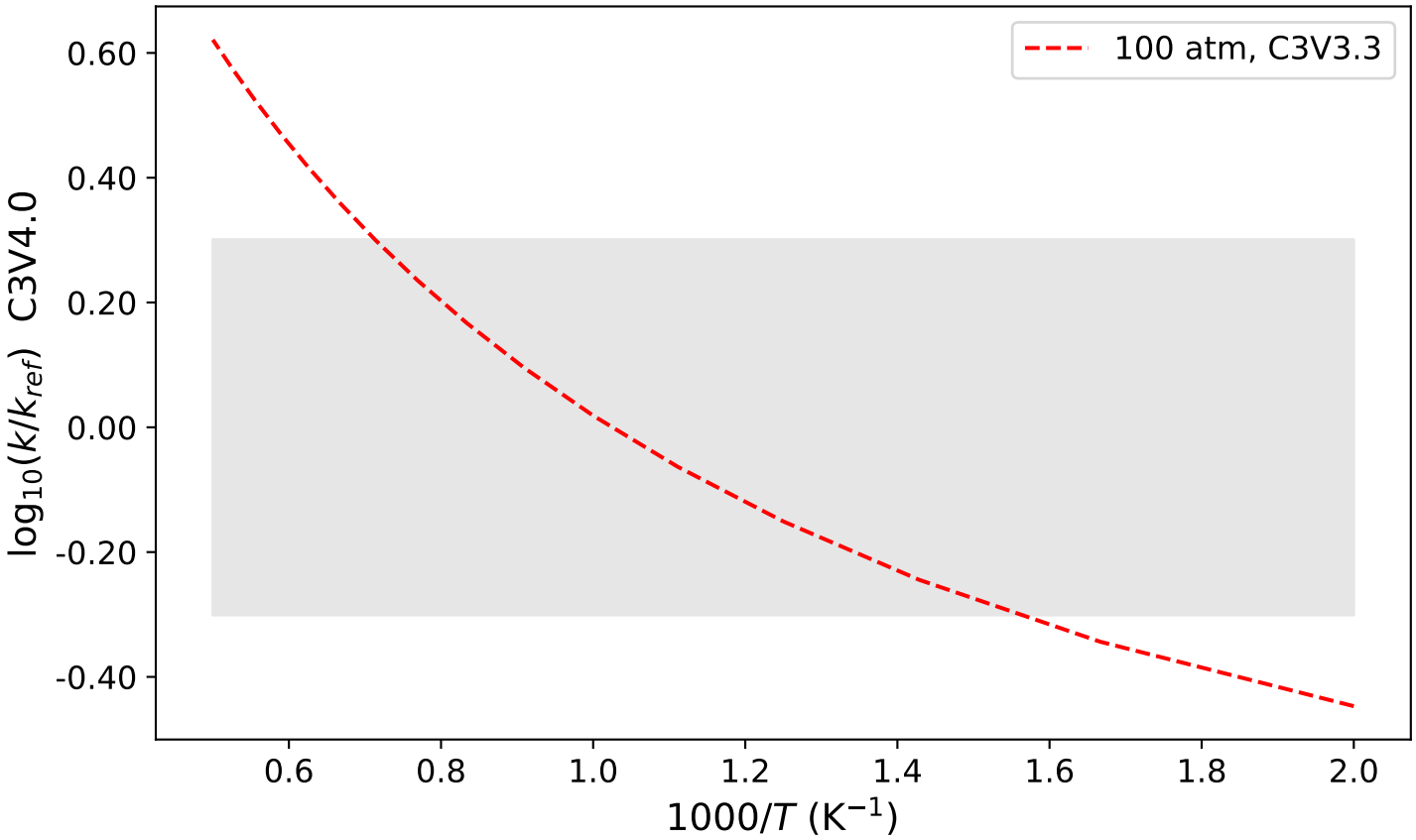
OH + CYC5H8 = H2O + CYC5H71-3



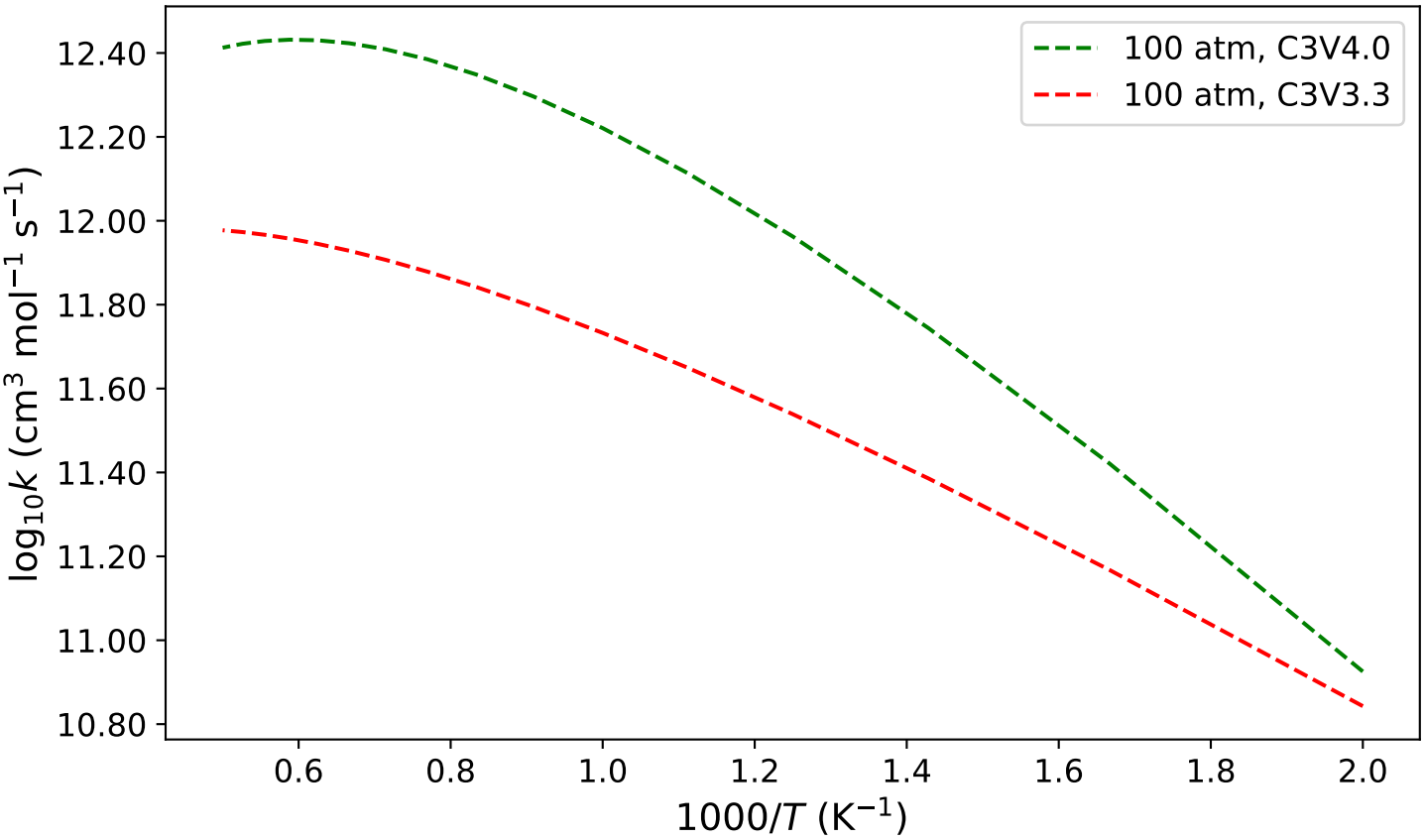


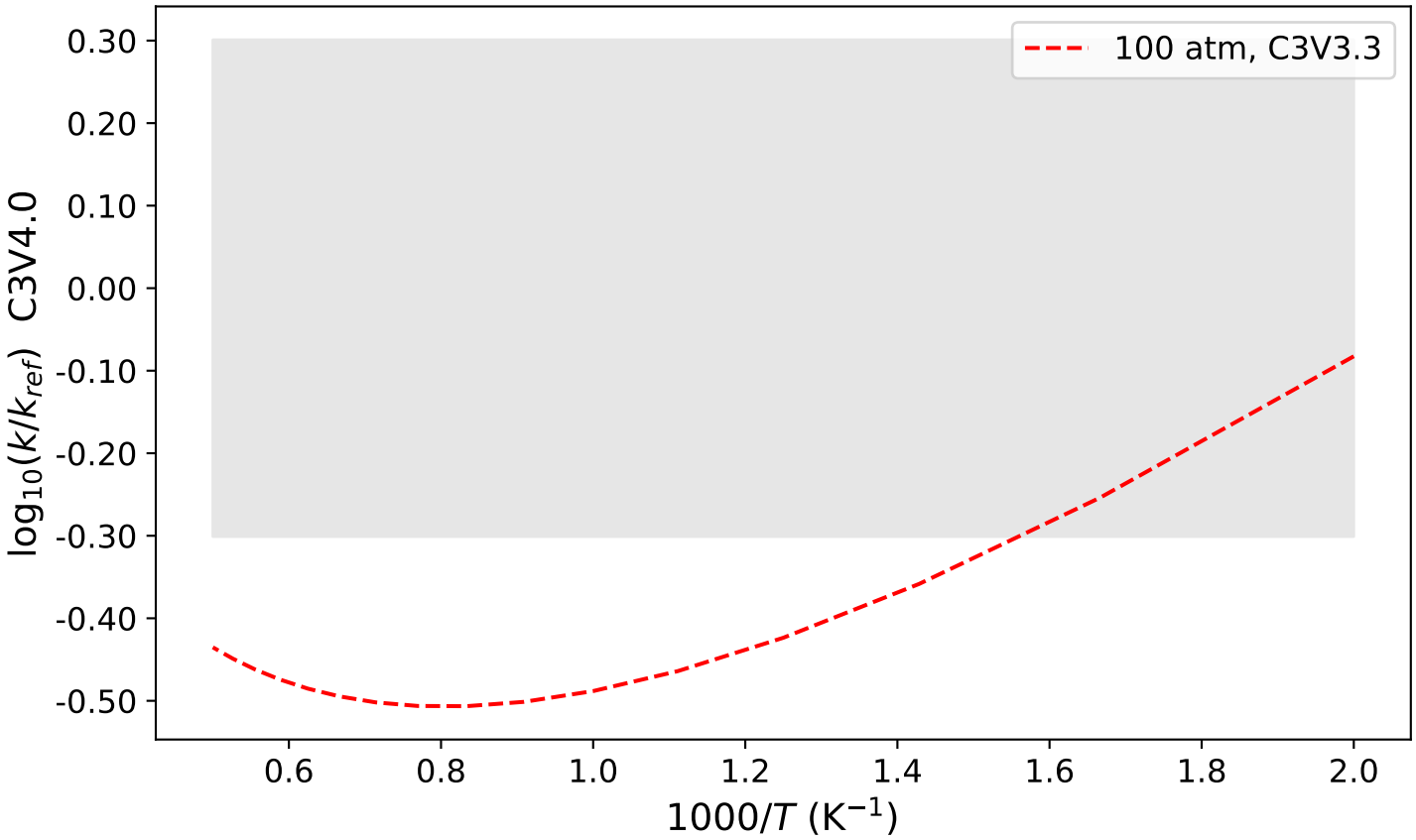
O + XYLENE = OH + RXYLENE



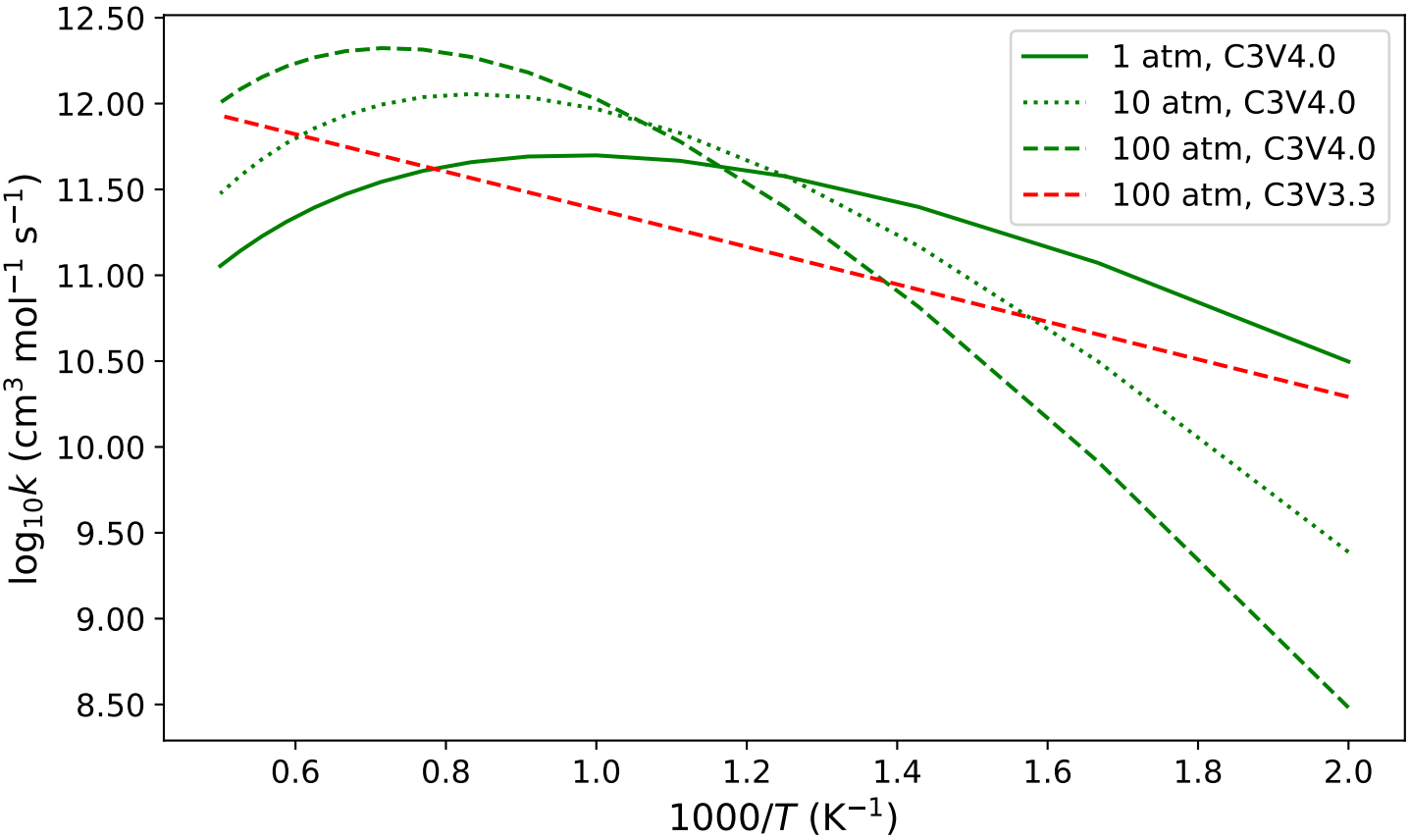


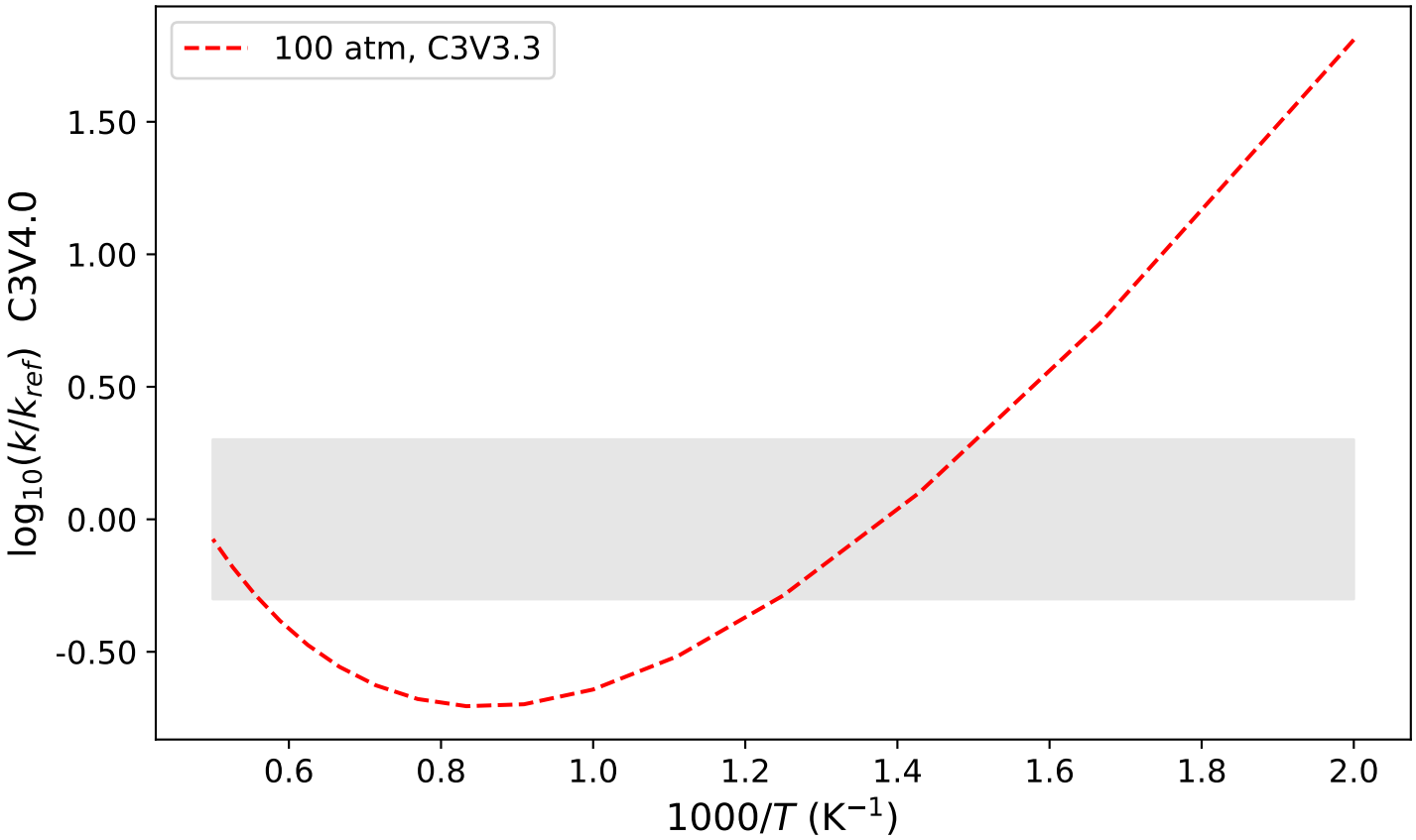
HO2 + RXYLENE => OH + HCO + C6H5CH3



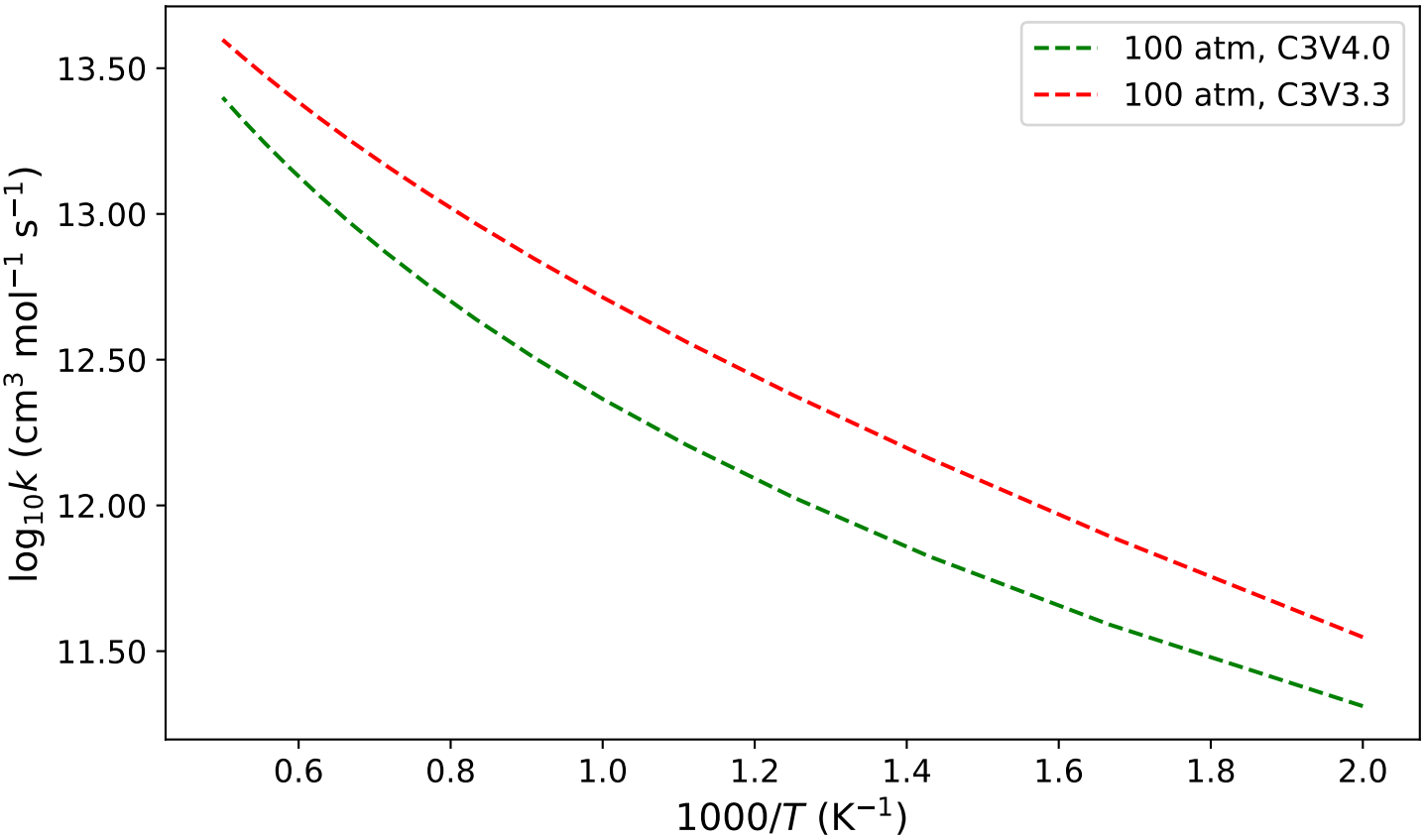


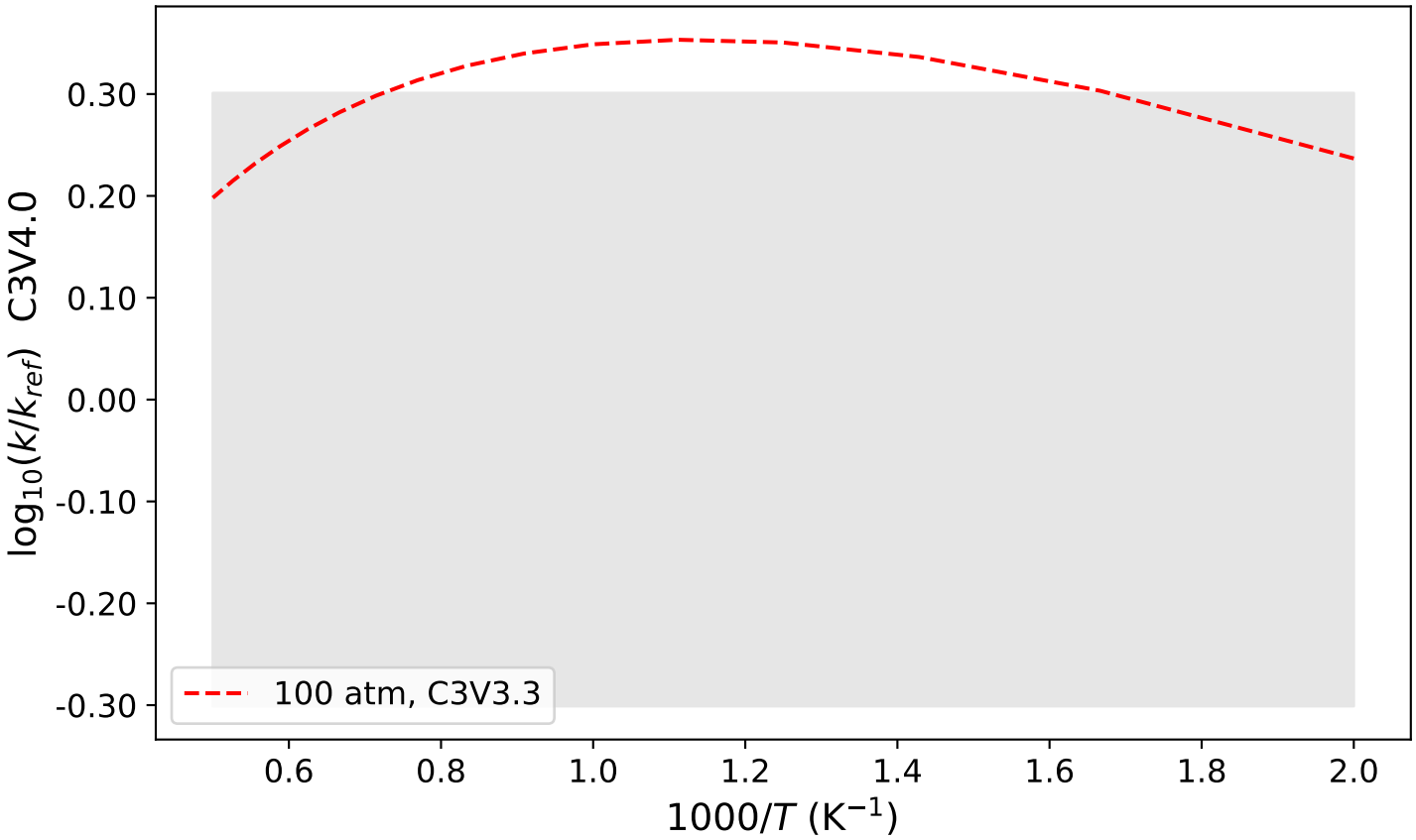
C6H5 + C3H4-P = H + IND



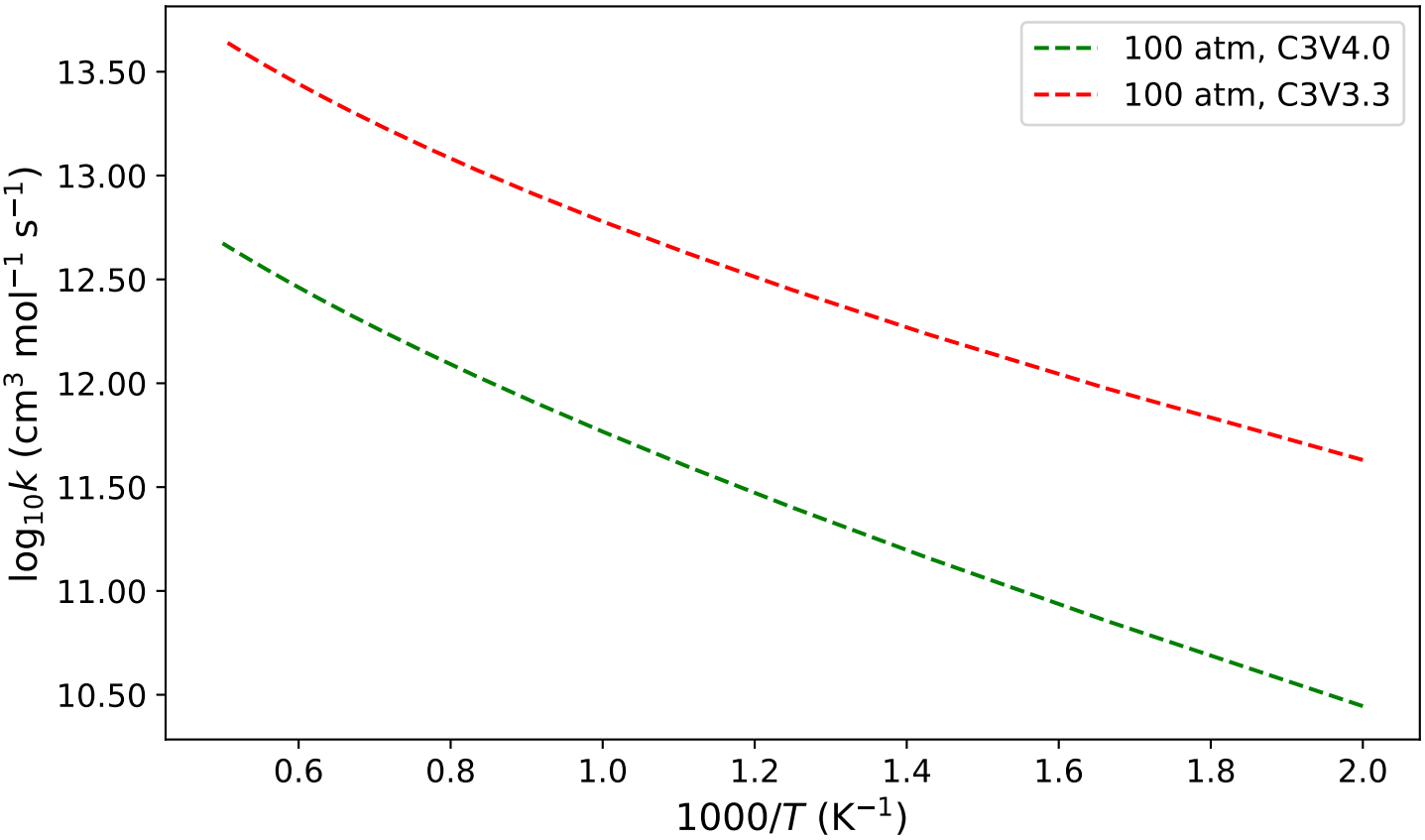


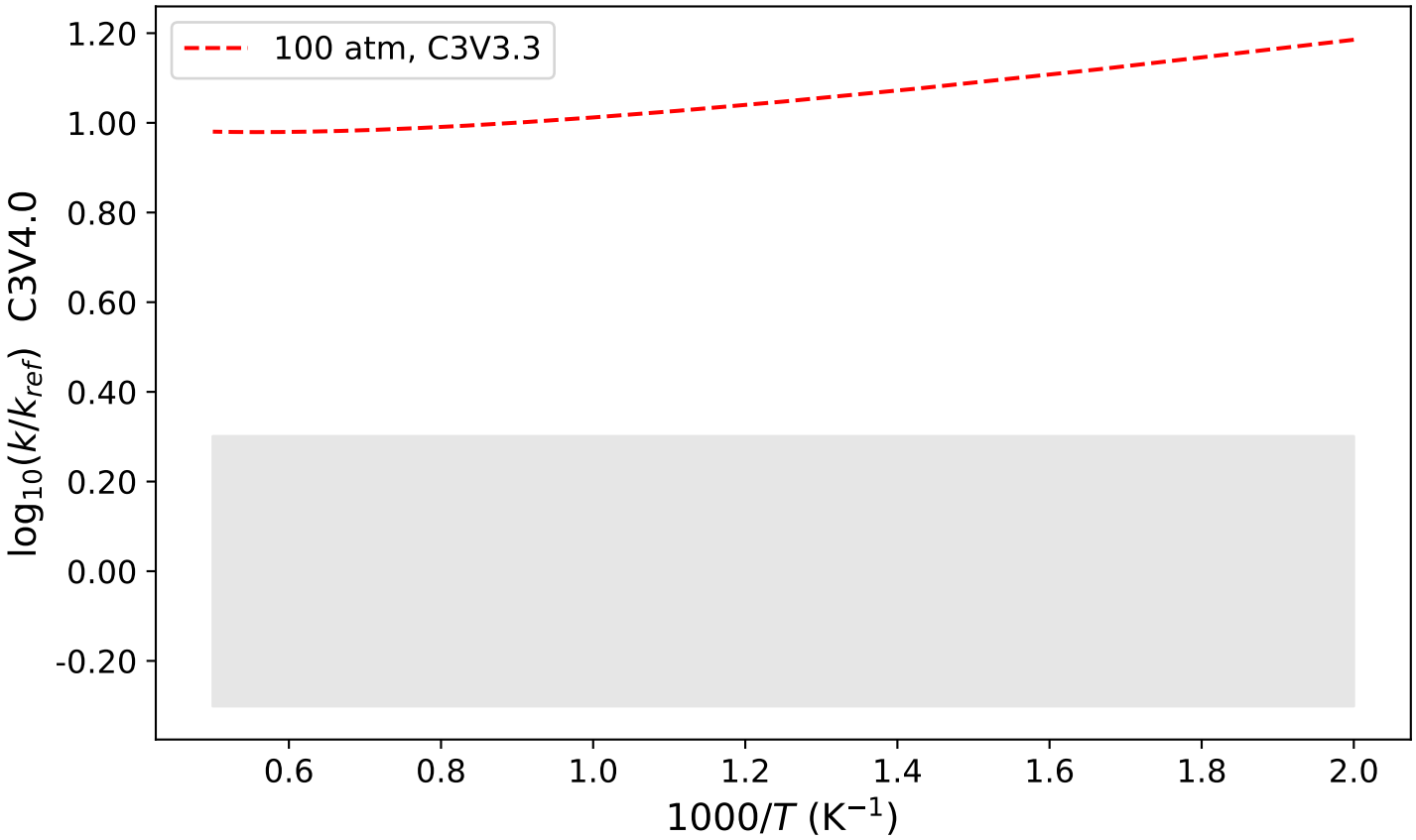
O + CYC5H8 = OH + CYC5H71-3



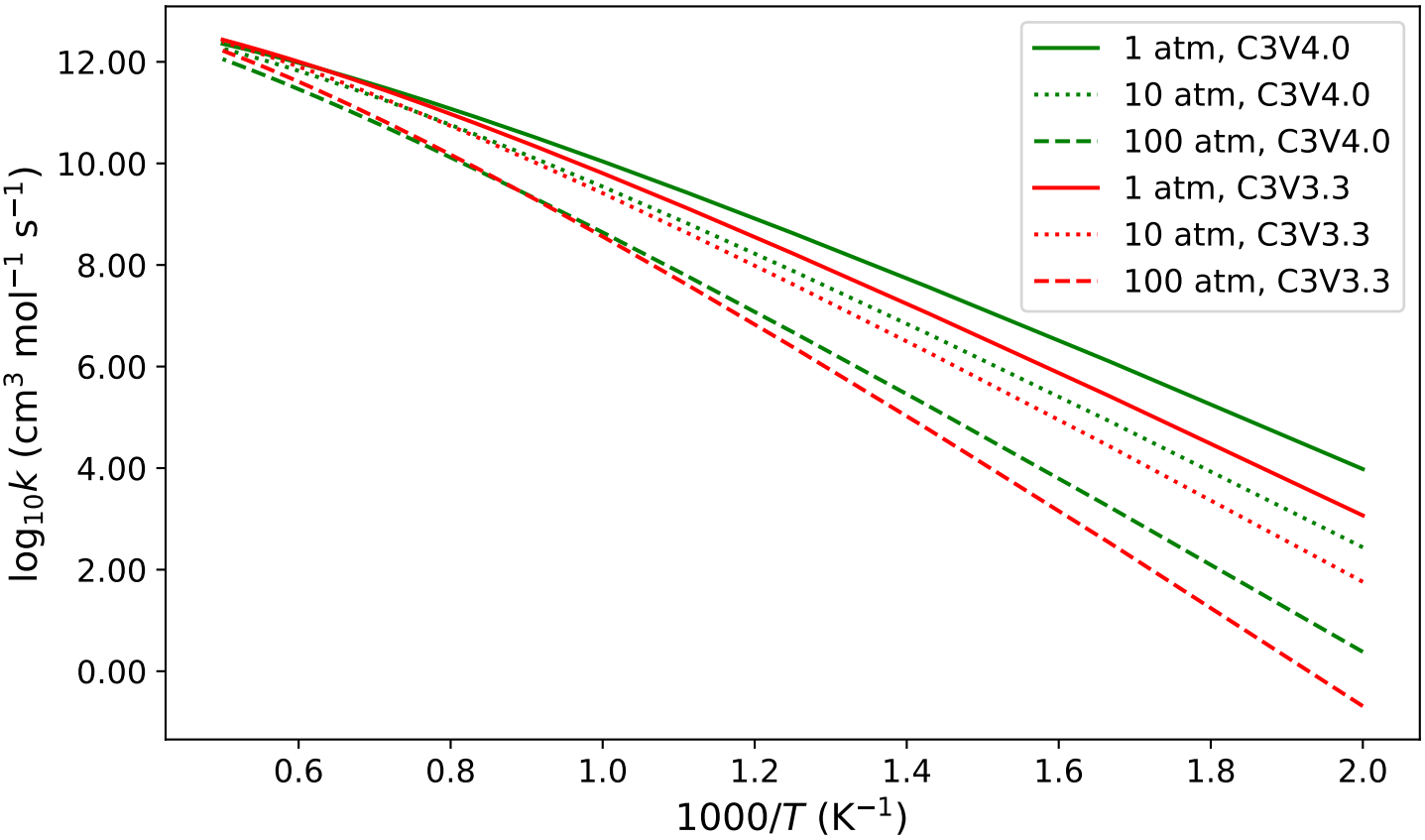


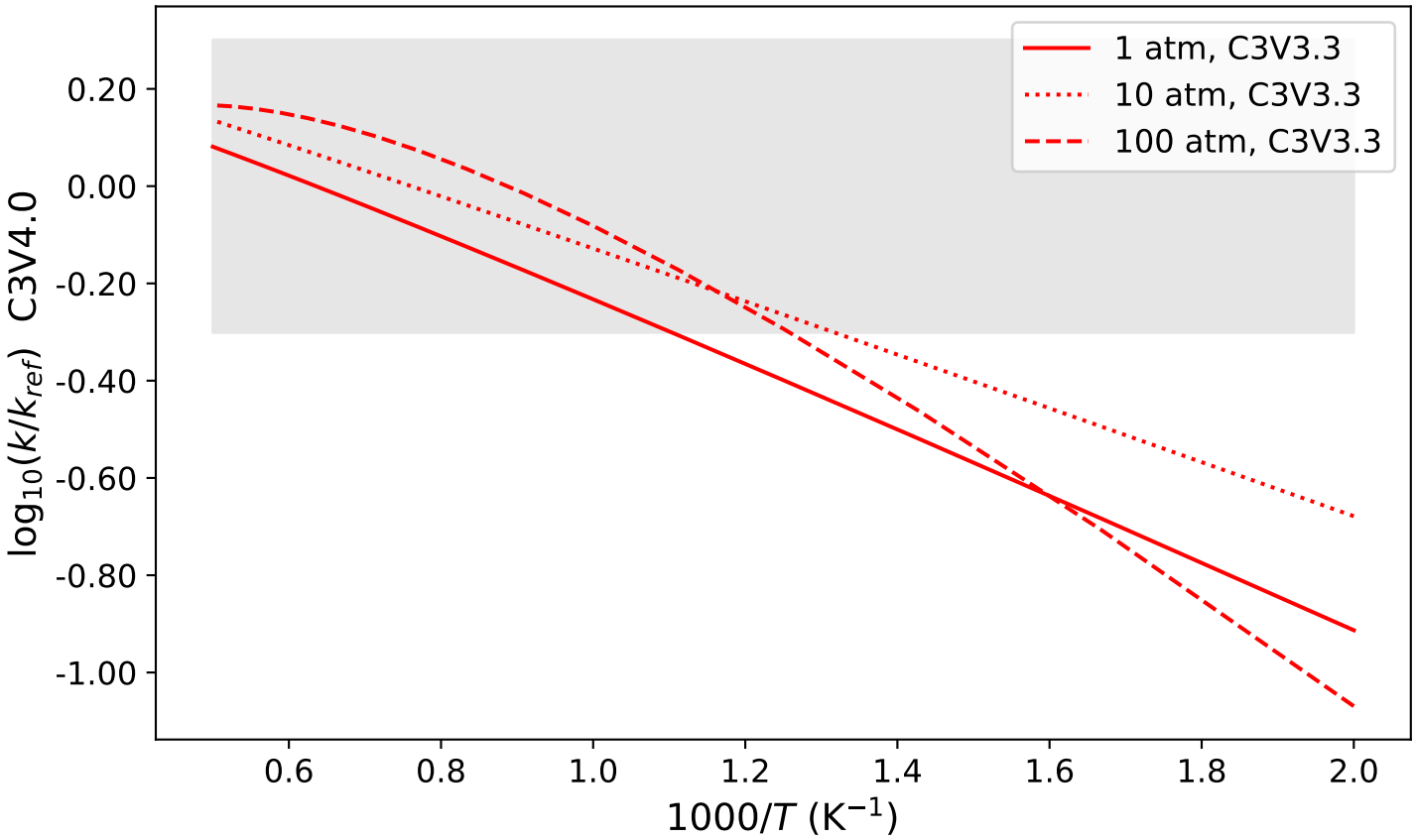
O + C14H14 = OH + C14H13



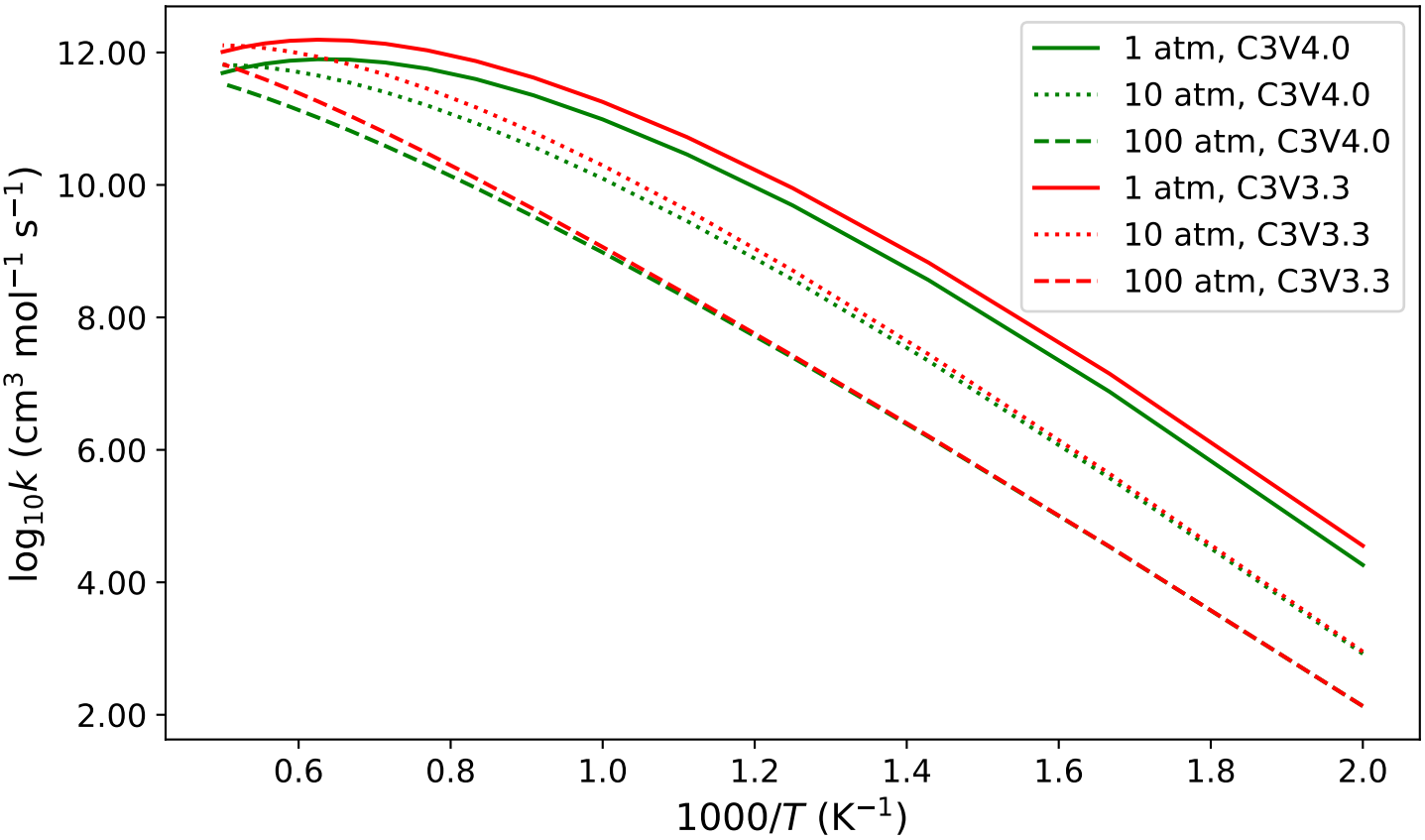


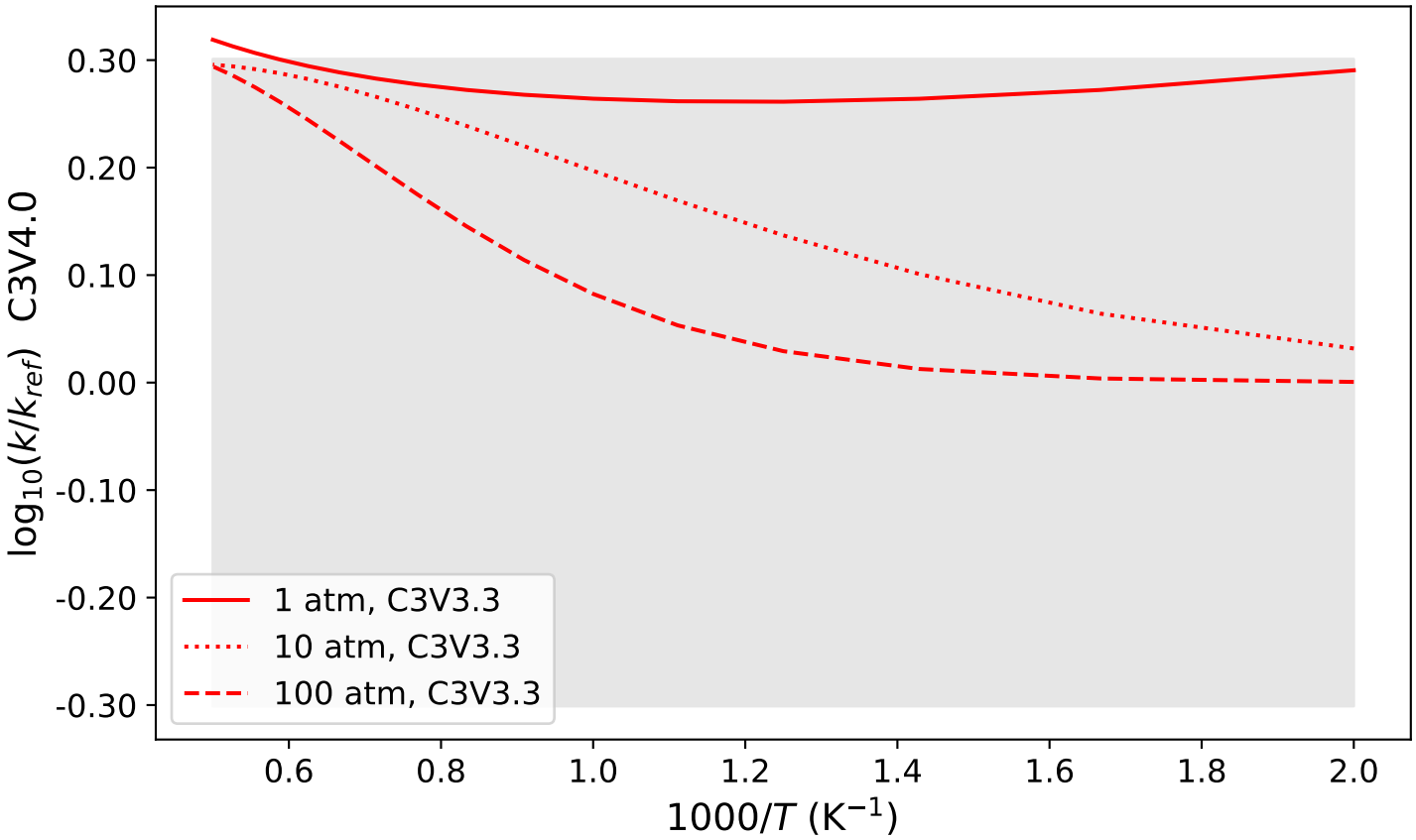
H + C5H6 = C2H2 + C3H5-A



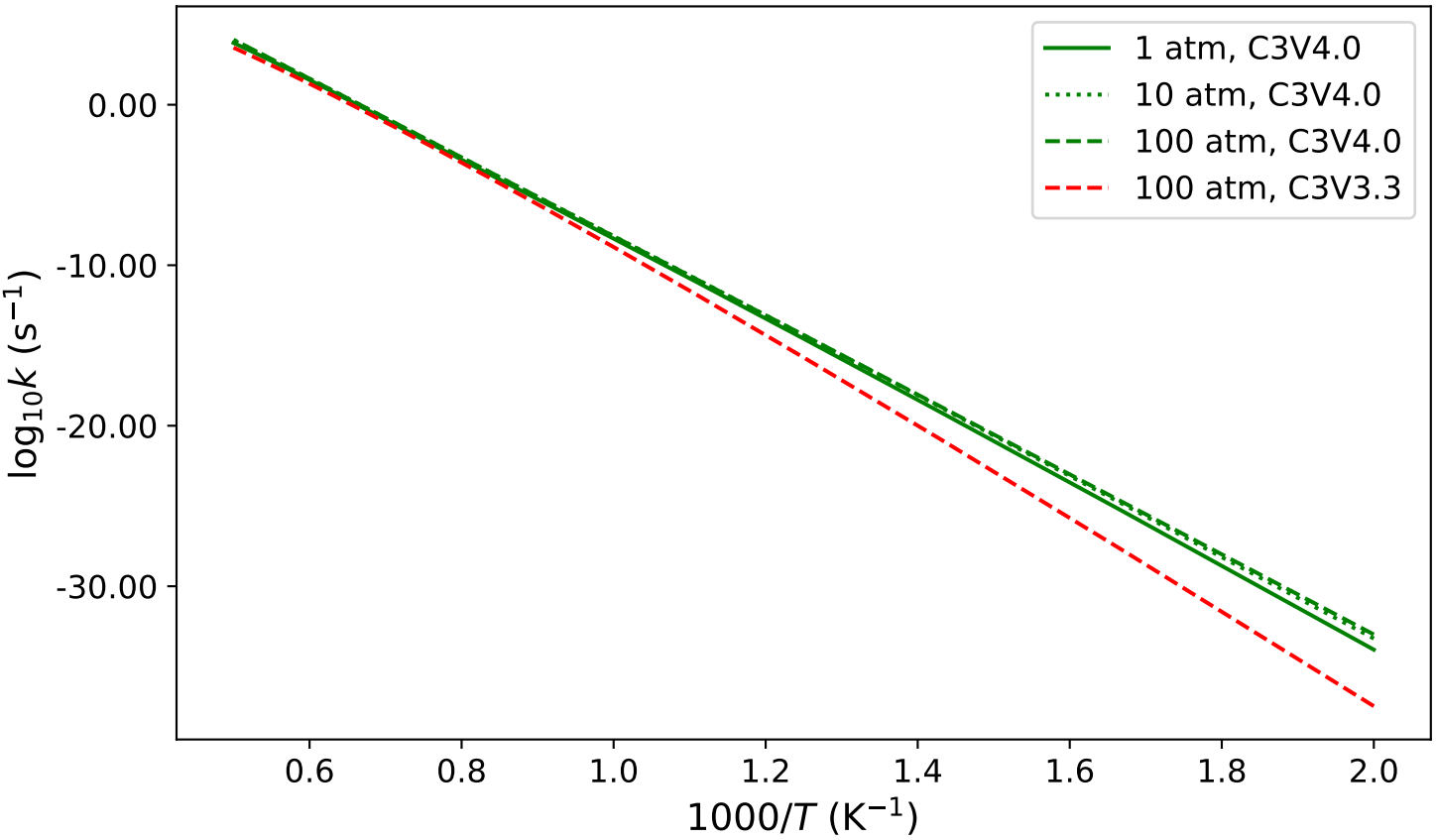


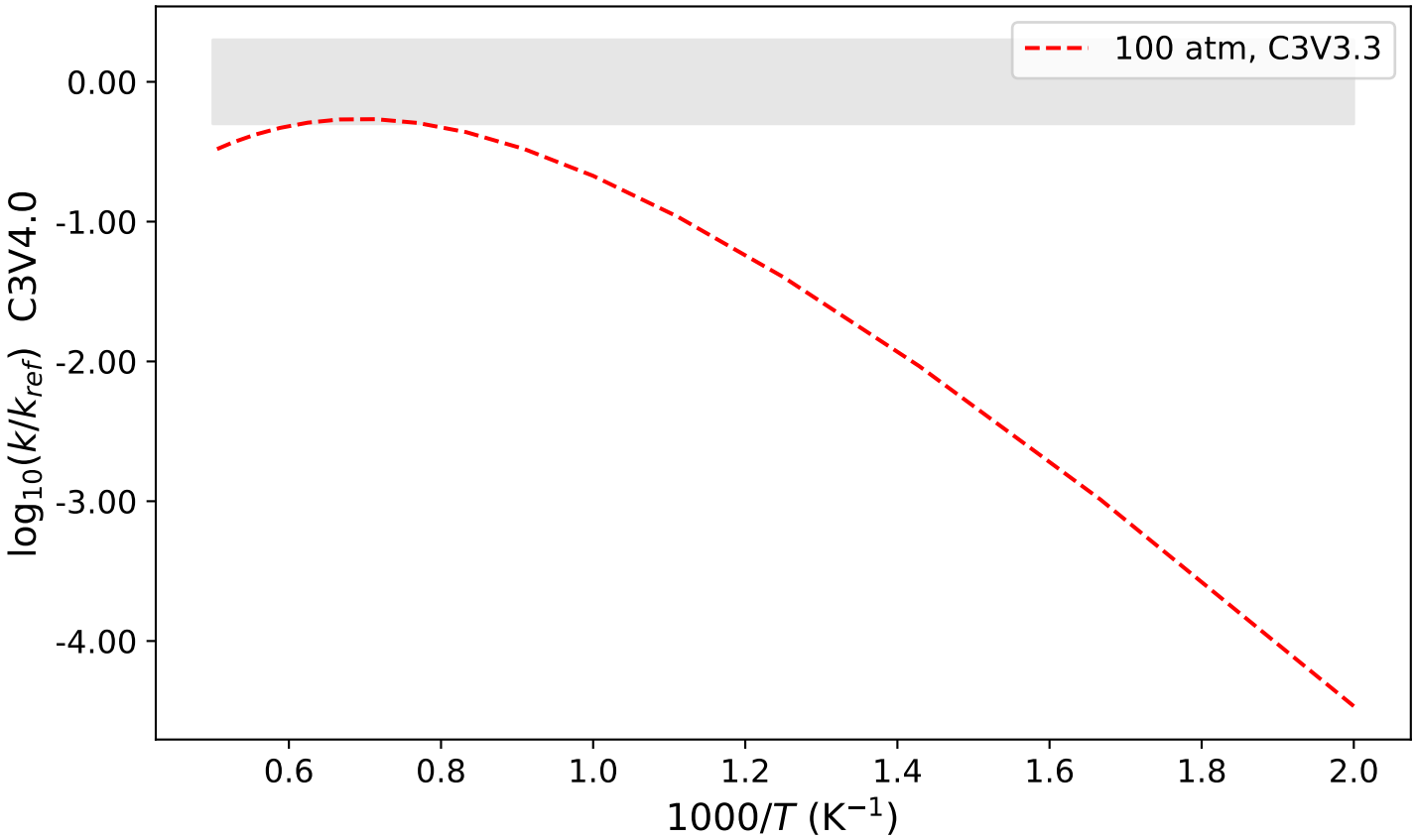
C6H5 + C4H2 = NAPH-A



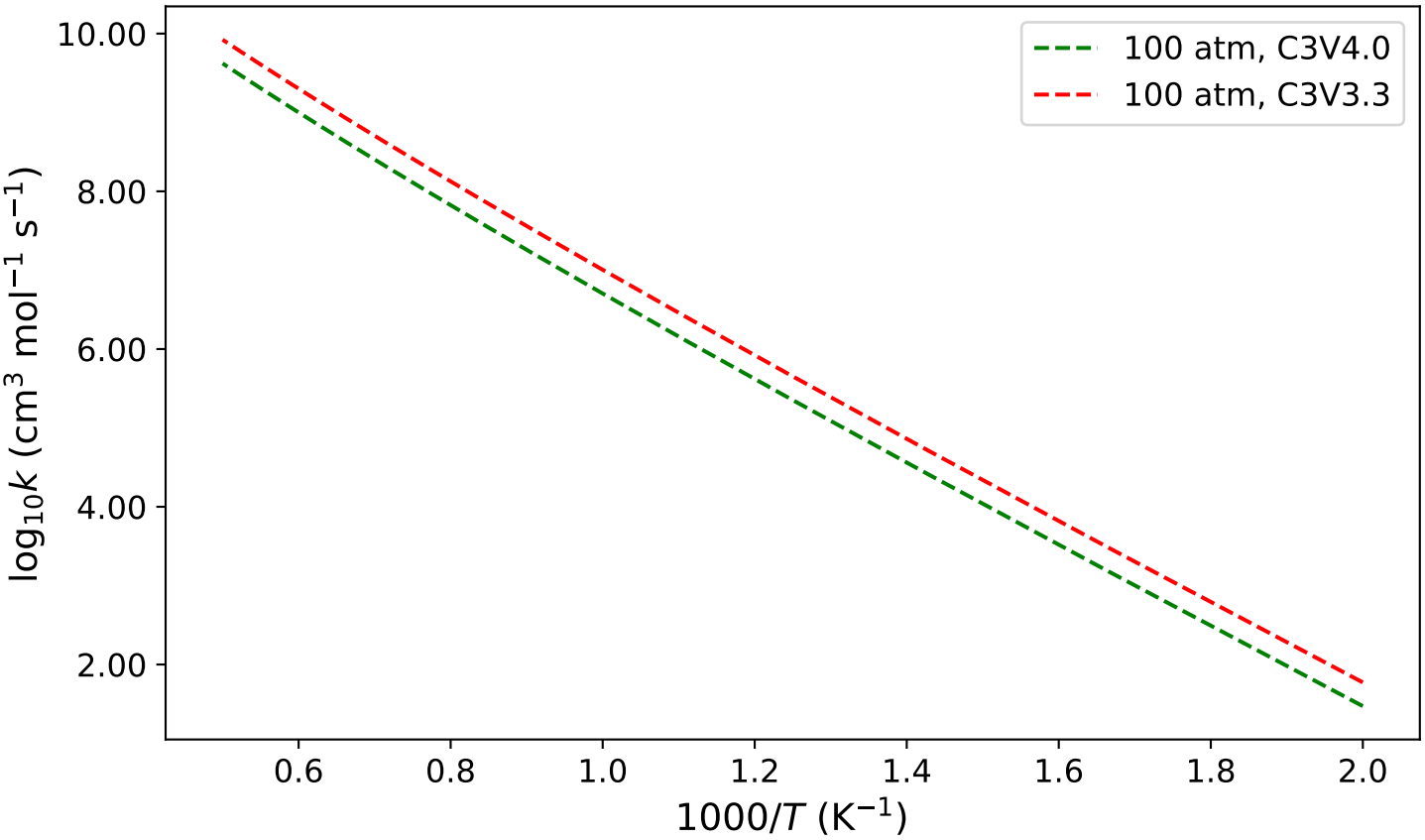


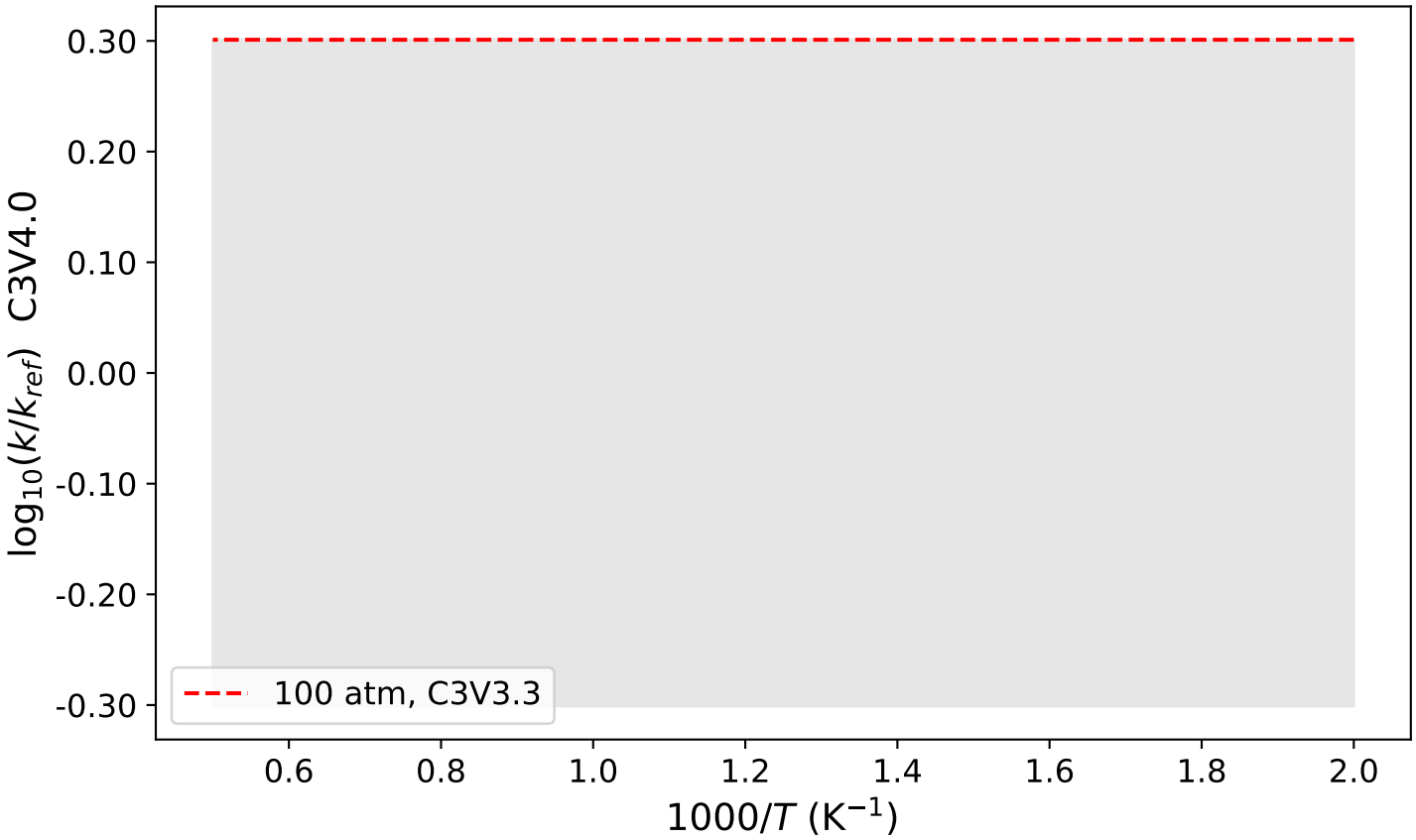
NAPH = H + NAPH-A


1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
0.00
-10.00
-20.00
-30.00
$\log_{10}k$ (s^{-1})
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ (K^{-1})

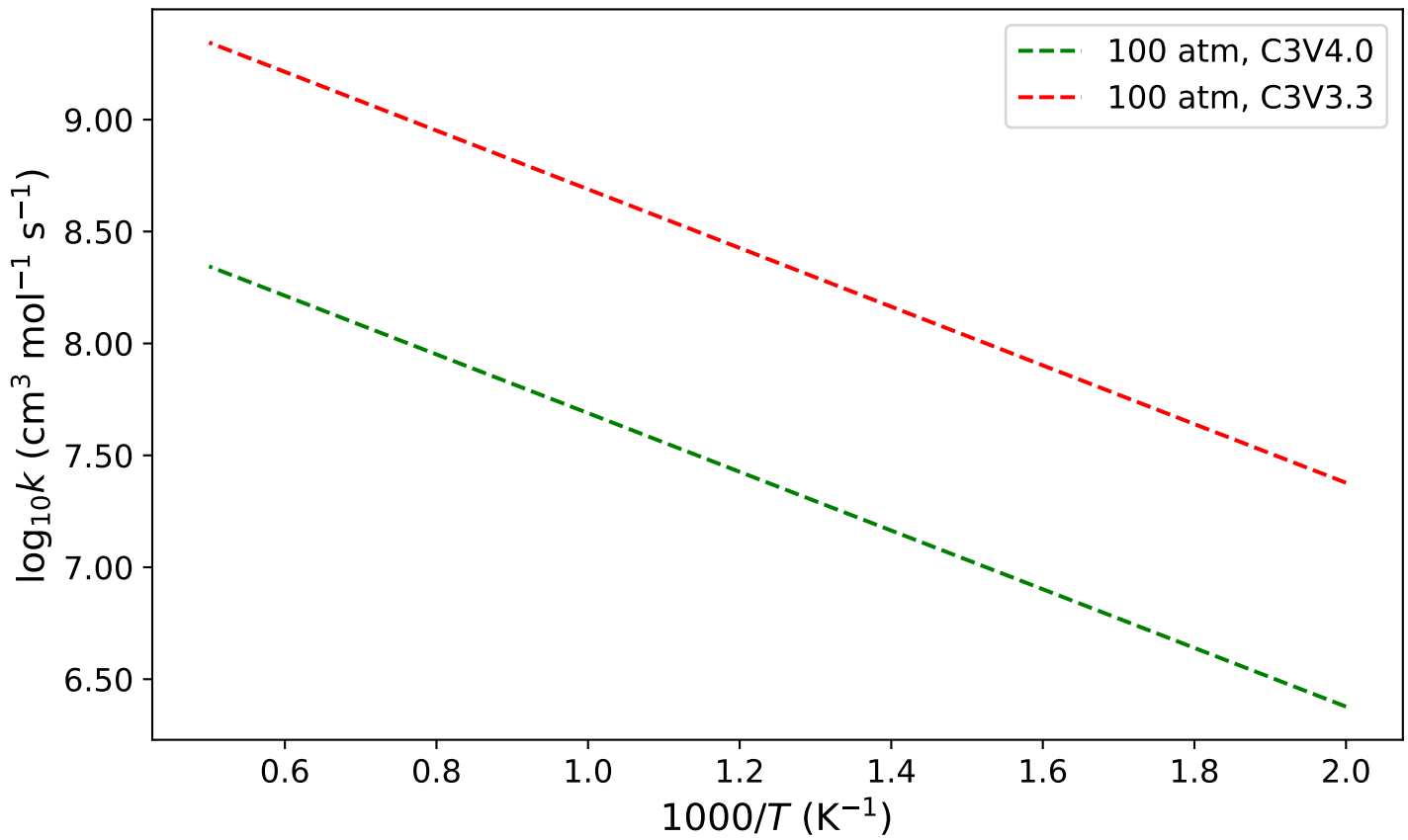

100 atm, C3V3.3
0.00
-1.00
-2.00
-3.00
-4.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ (K^{-1})

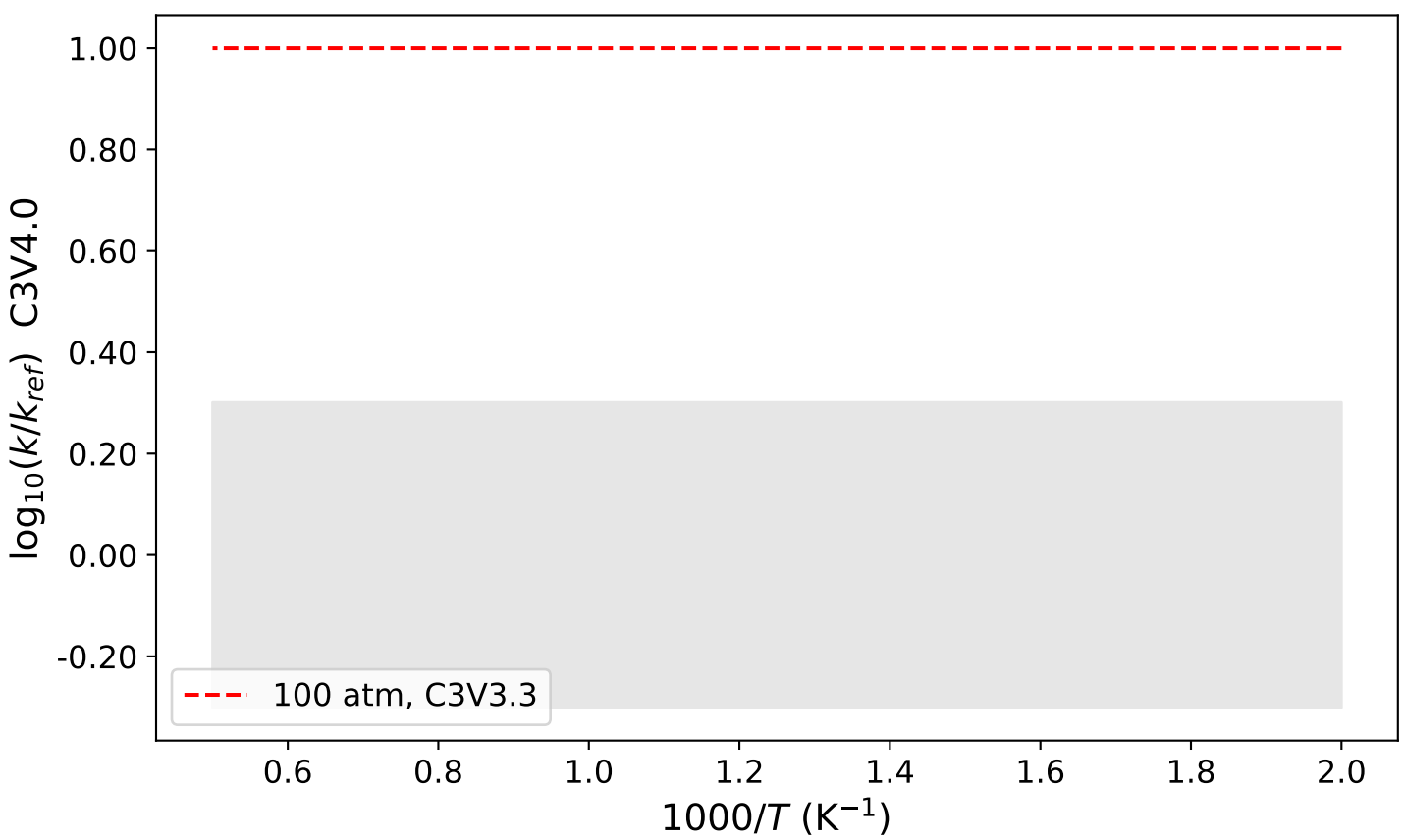
C3H3 + NAPH = C3H4-A + NAPH-A



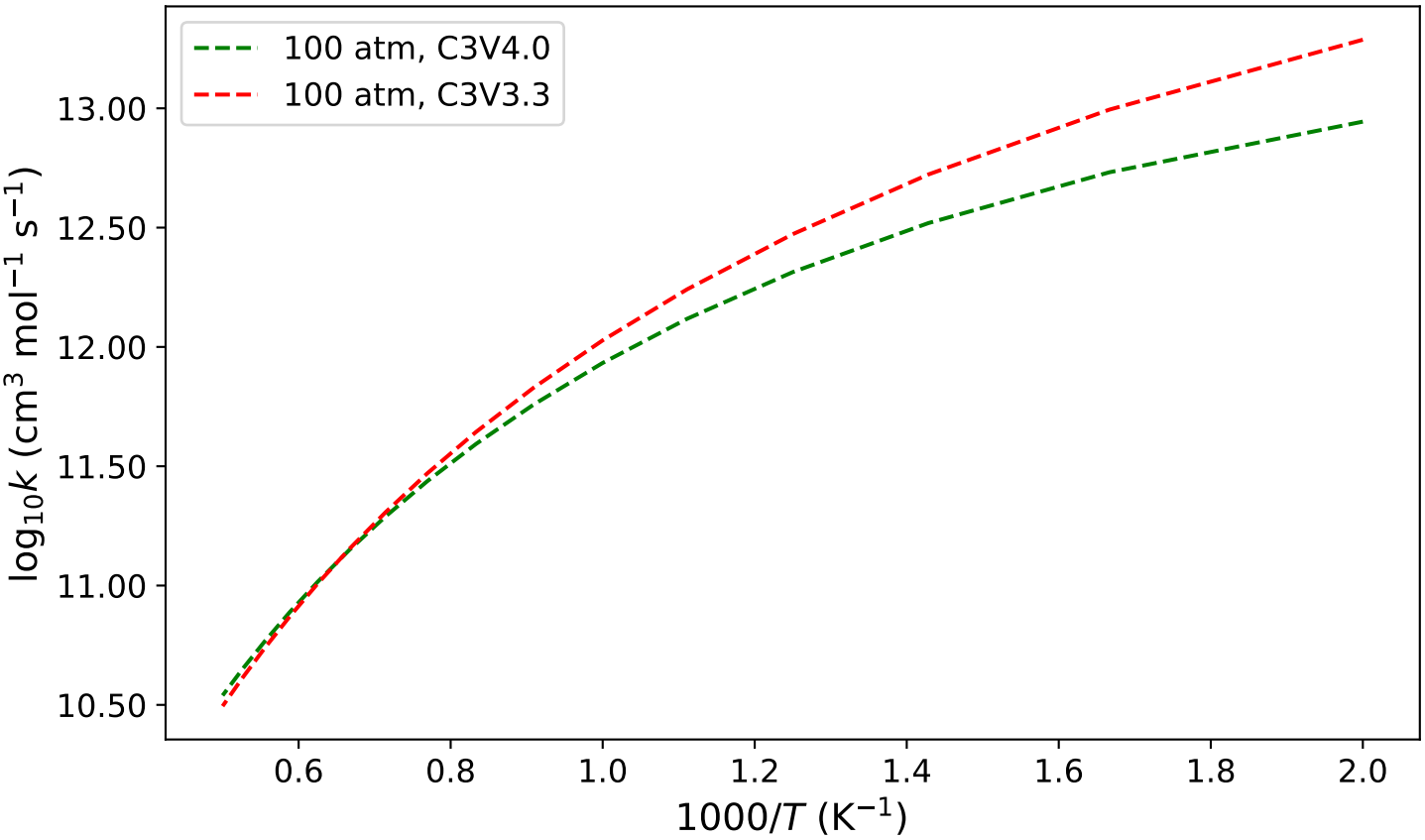


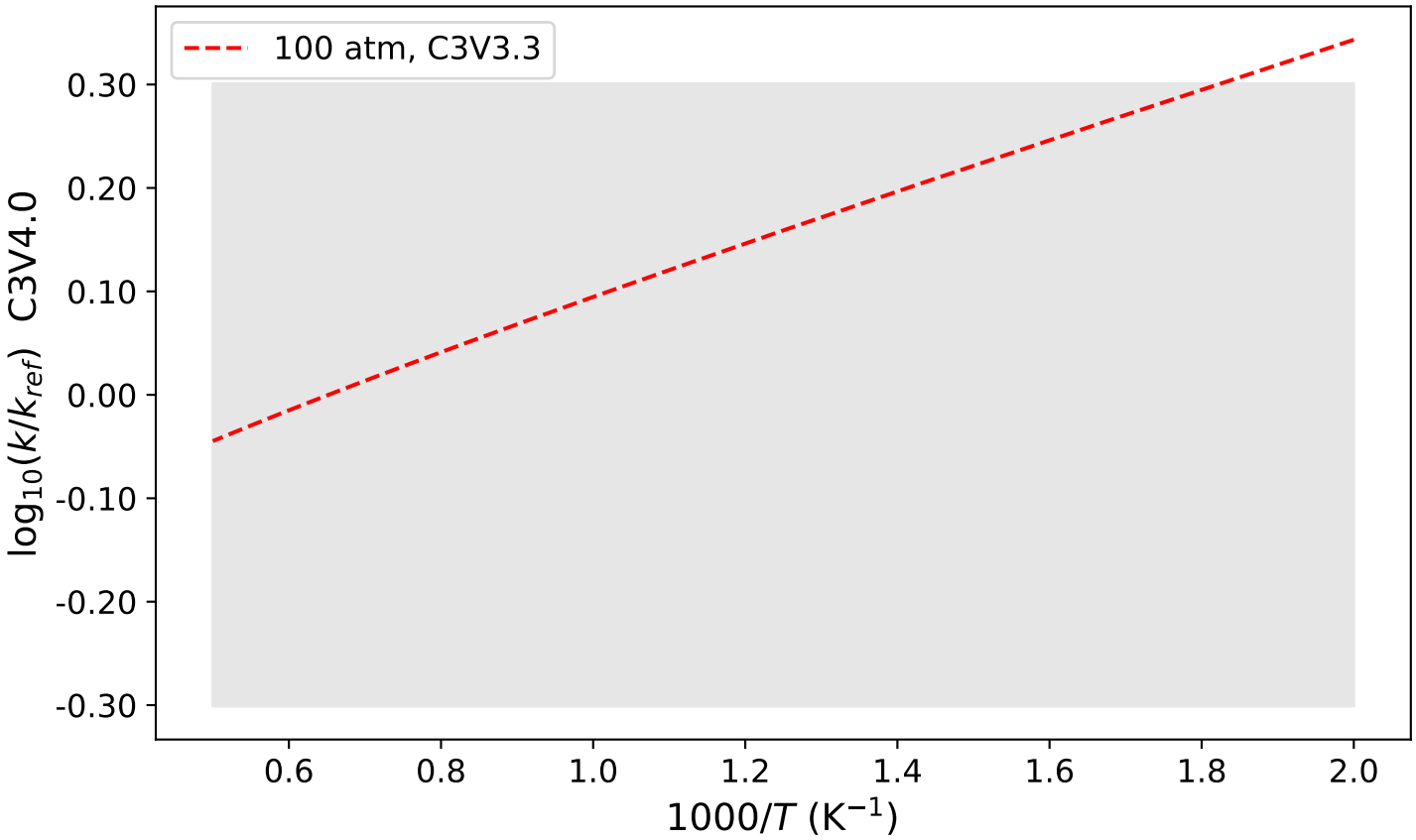
IC4H7 + IC4H7 => H + CH3 + H2 + C6H5CH3



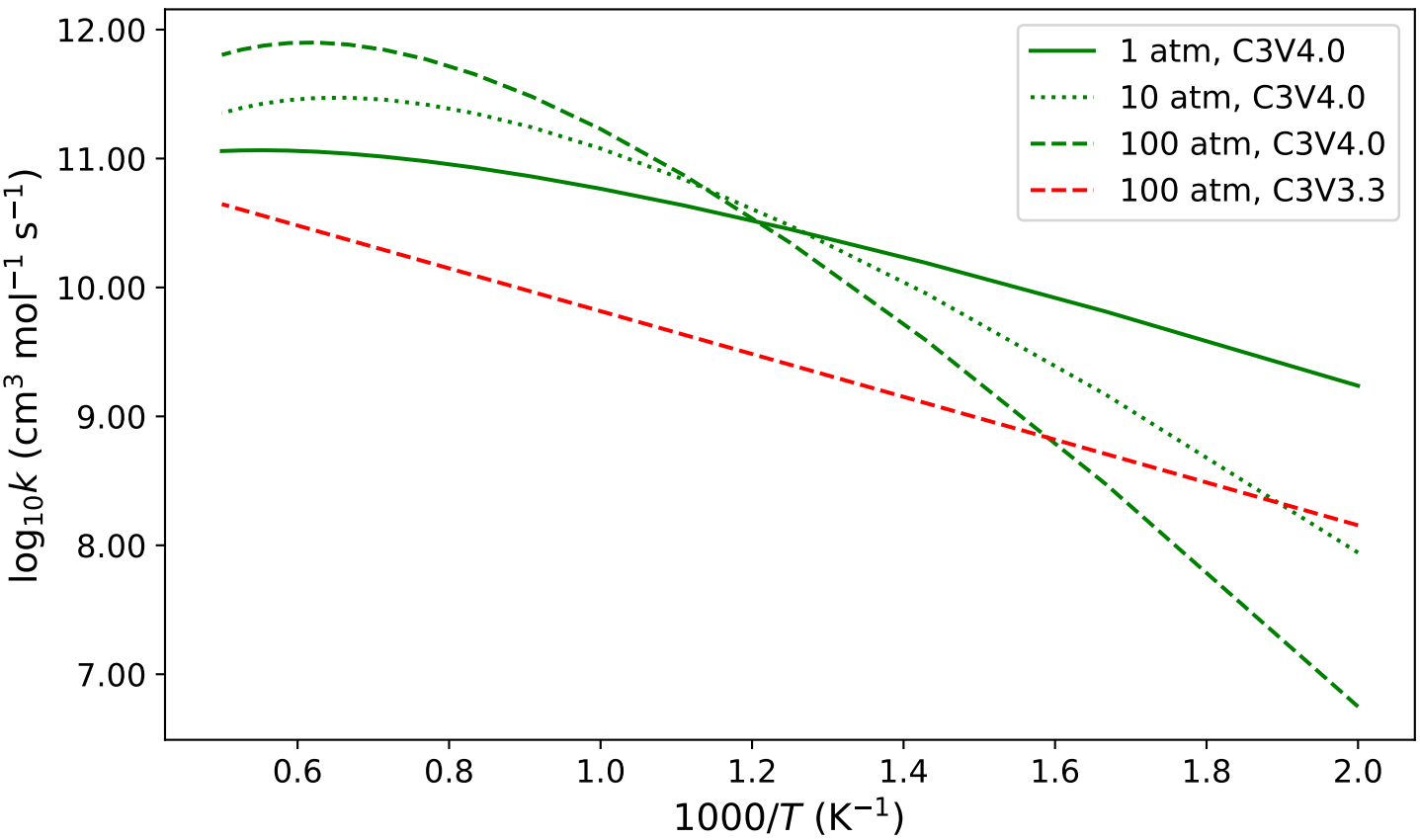


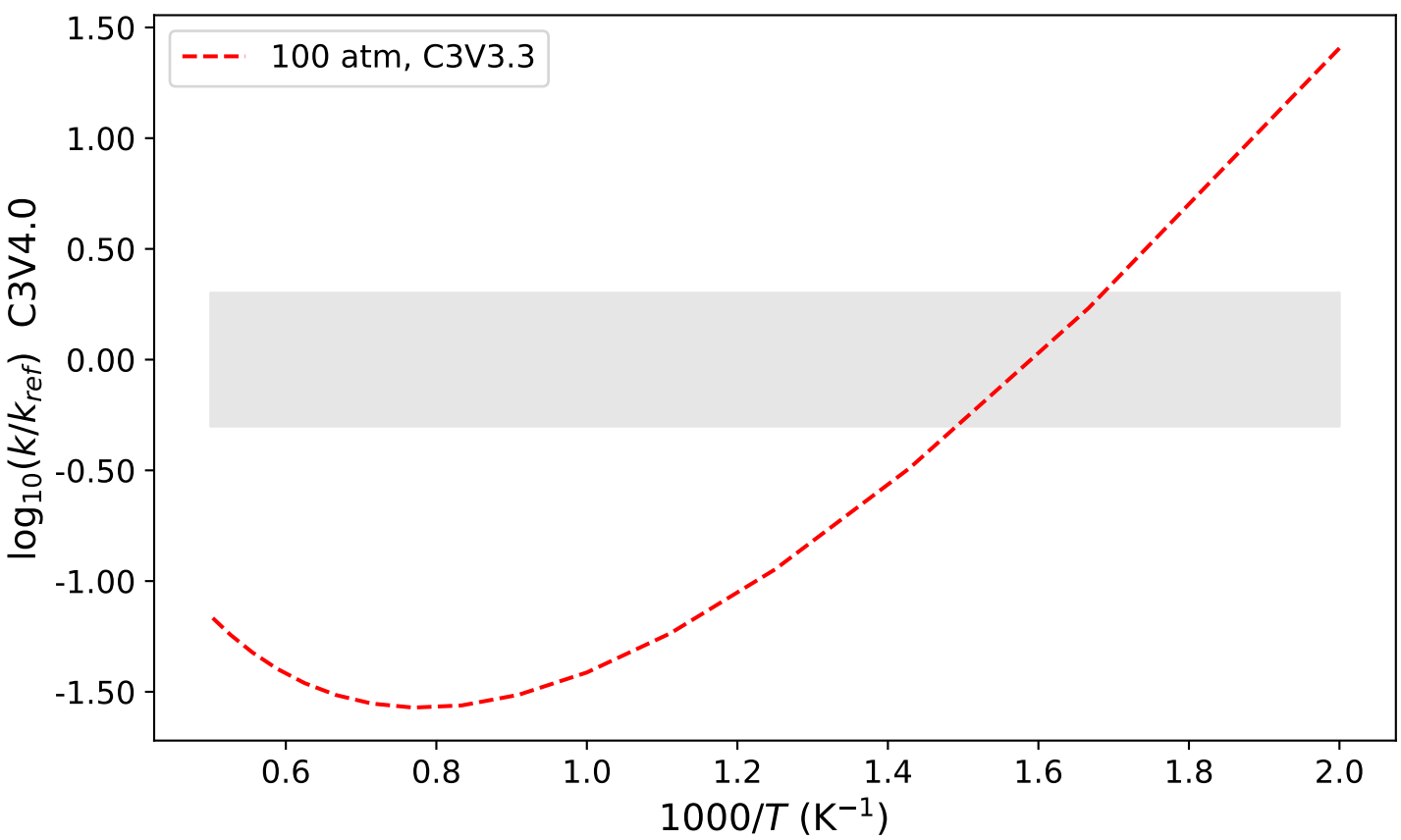
HO2 + C6H5CH2 = BZCOOH



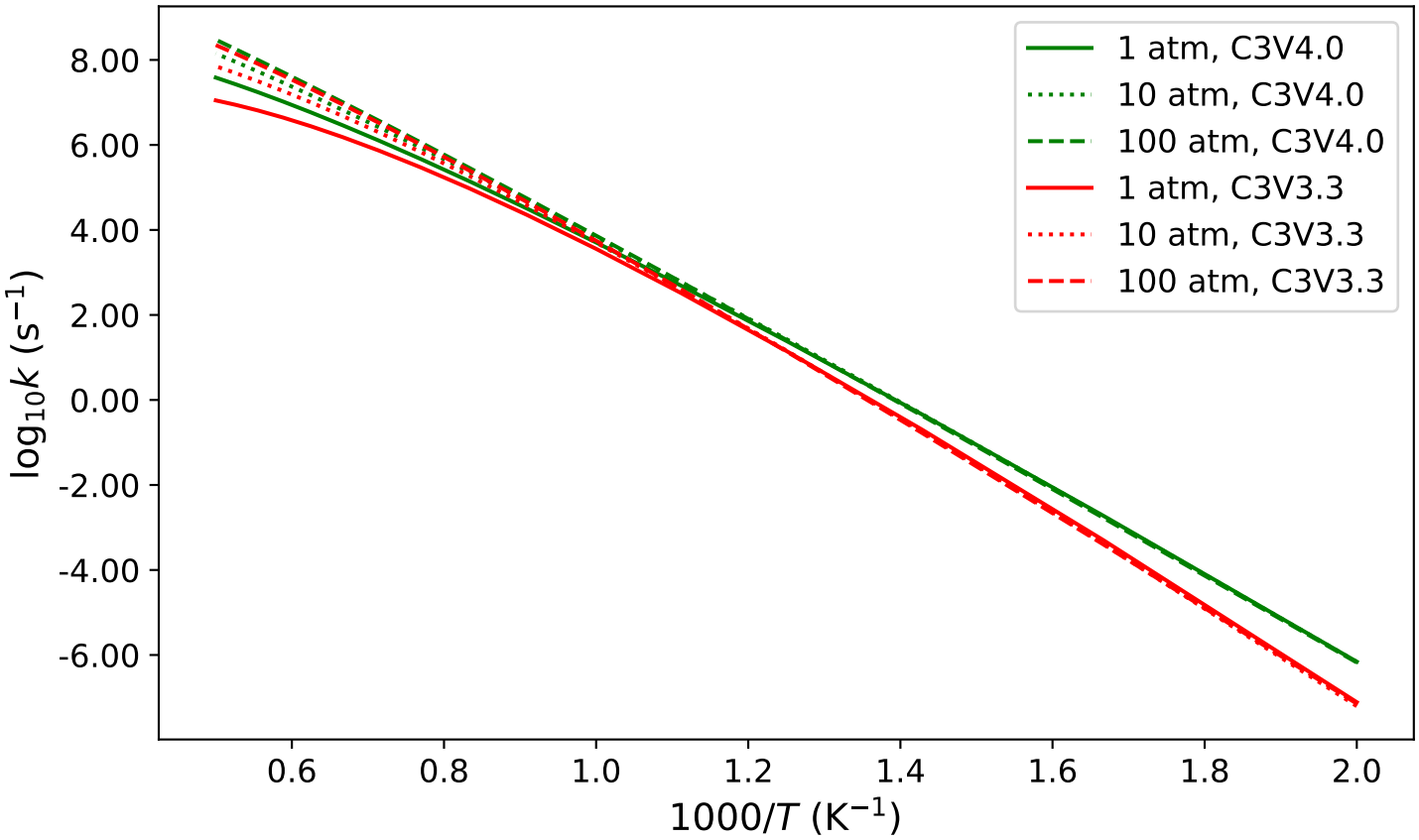


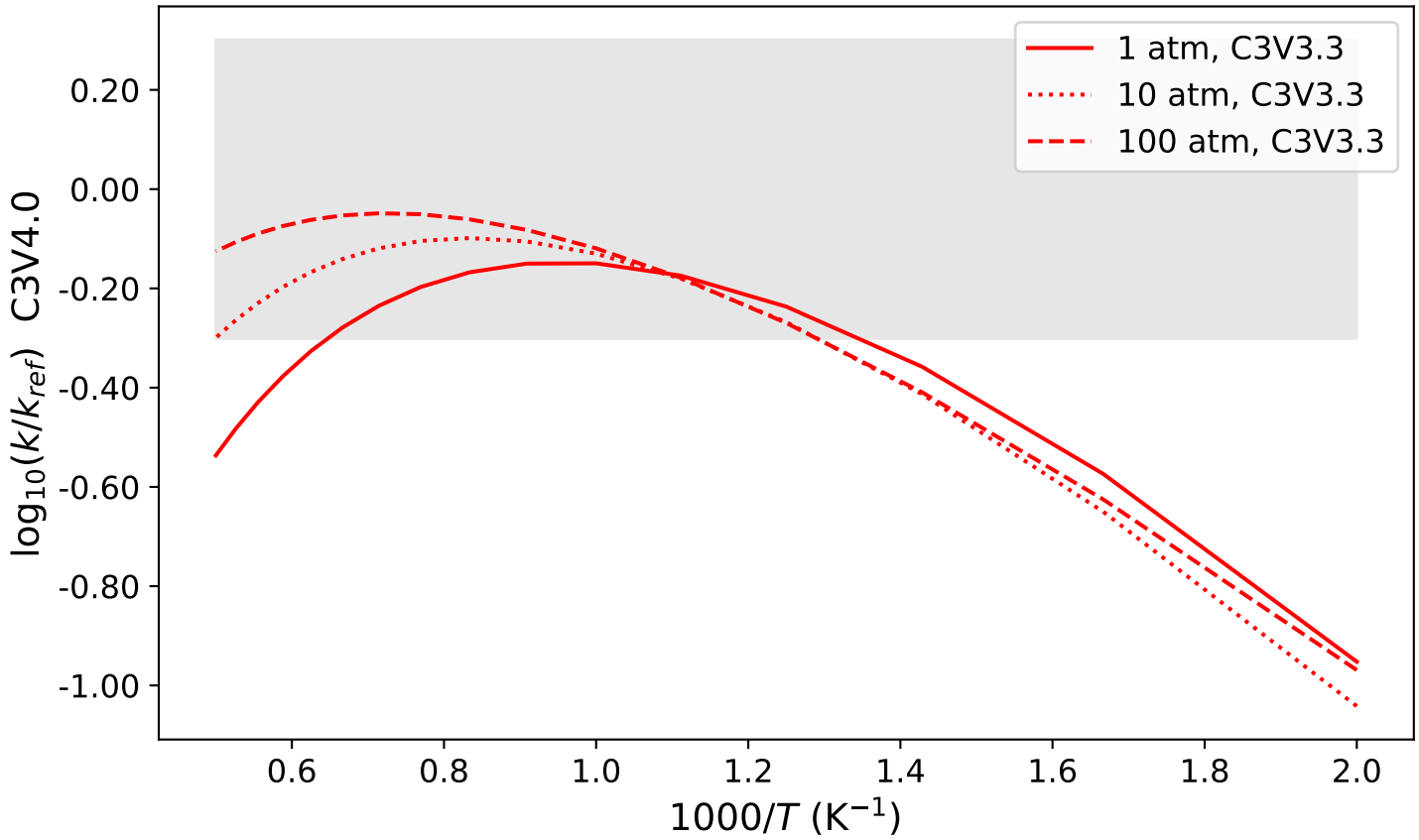
CH3 + C6H5C2H = H + IND



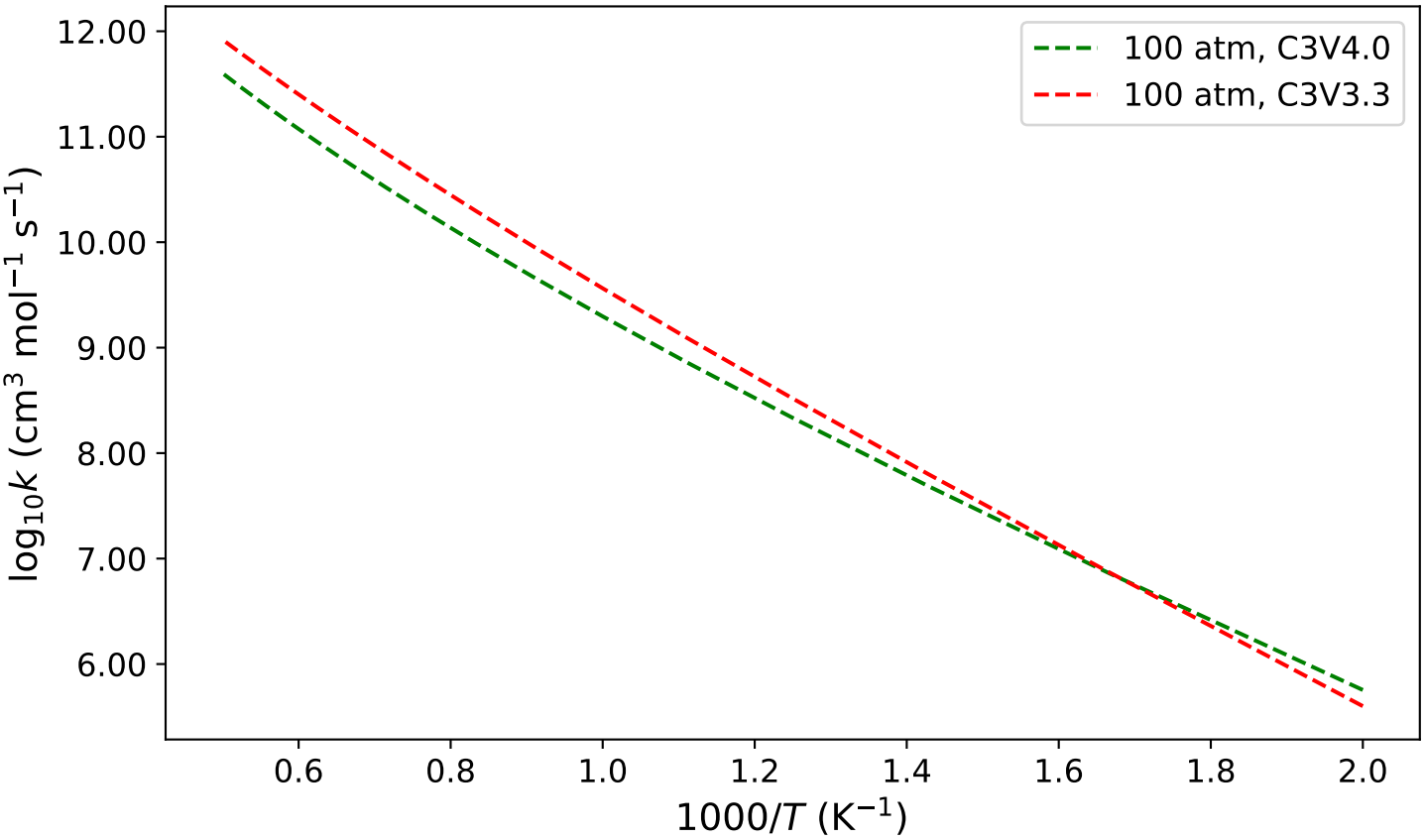


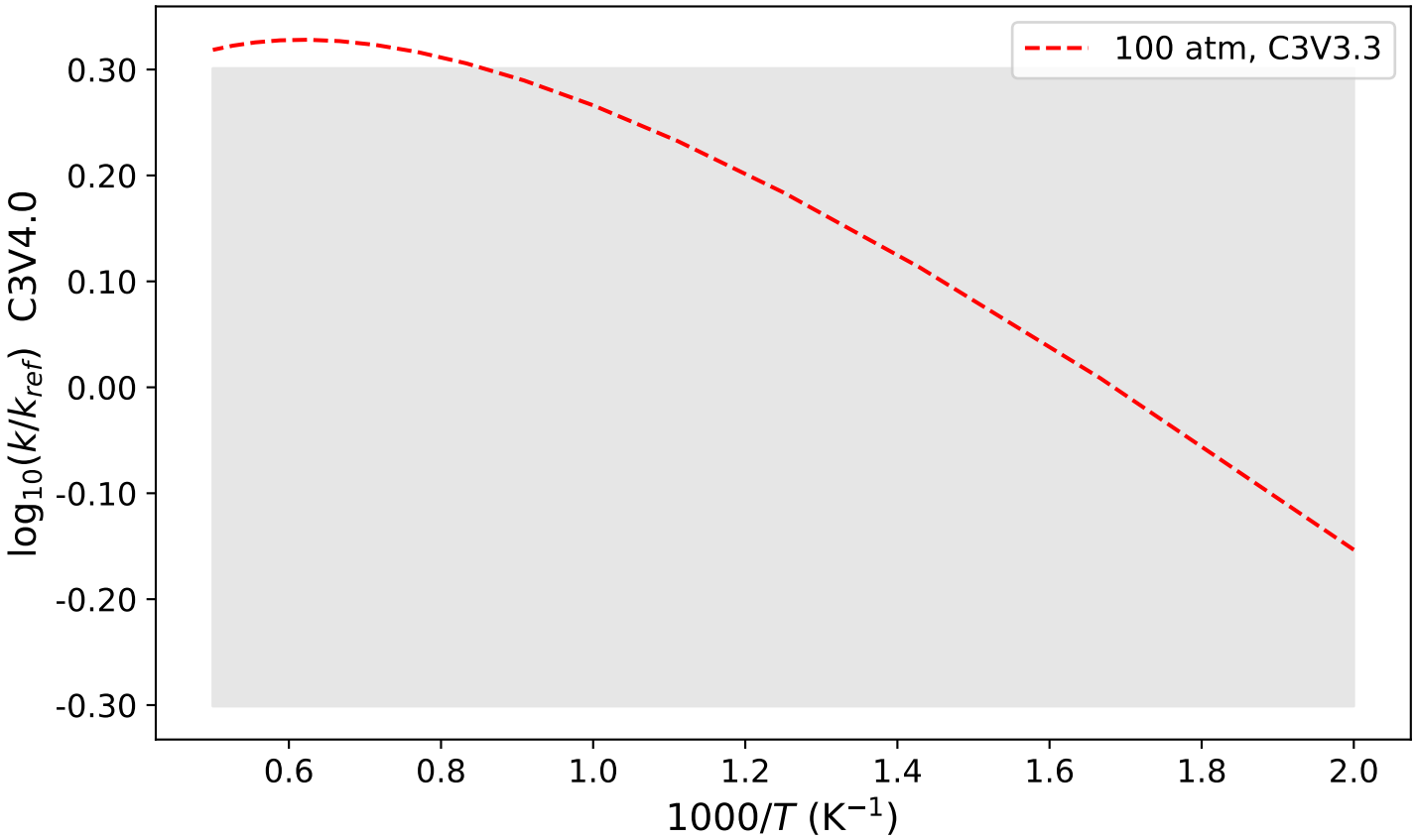
CYC5H71-3 = CVCCJCVC



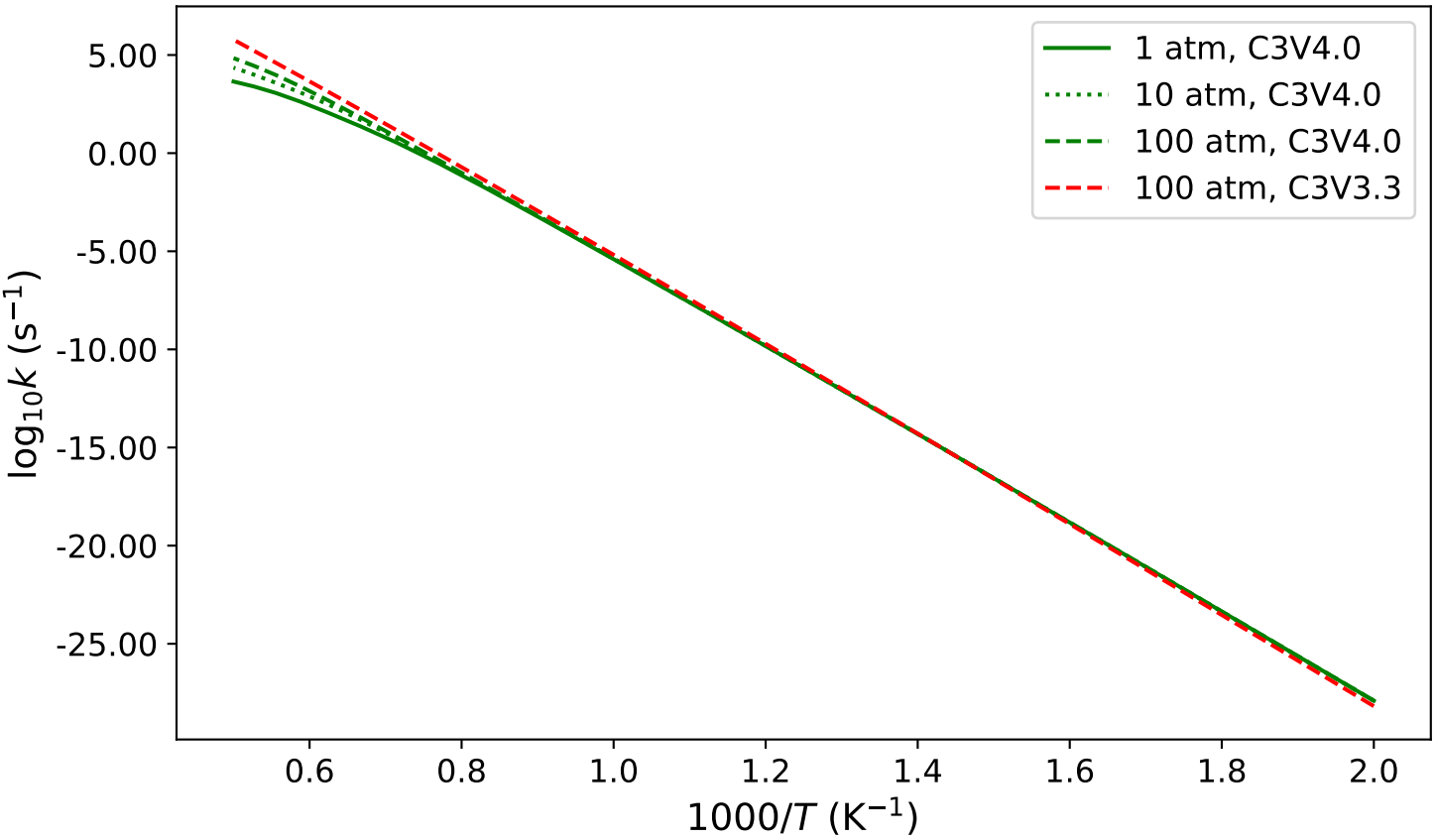


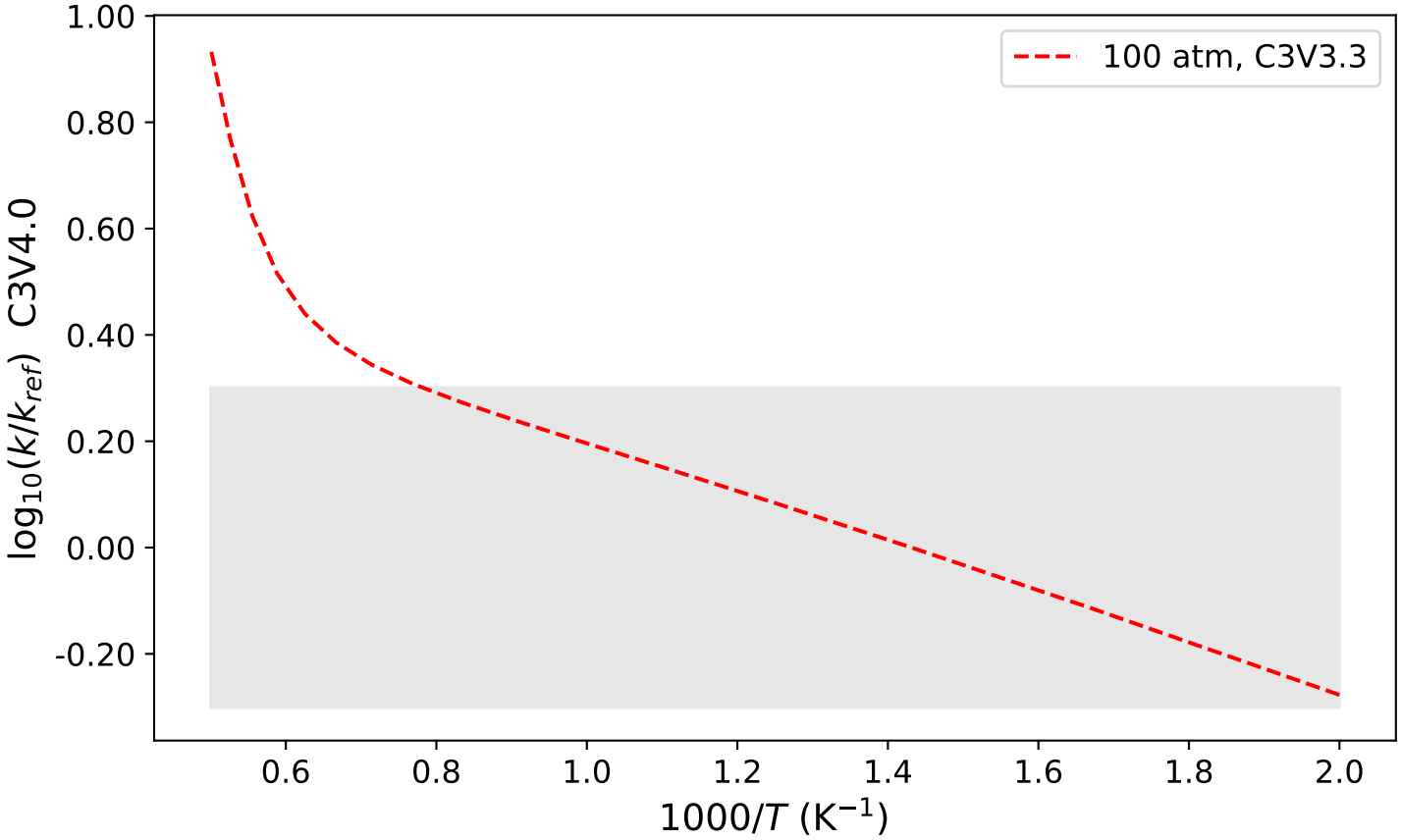
HO2 + XYLENE = H2O2 + RXYLENE



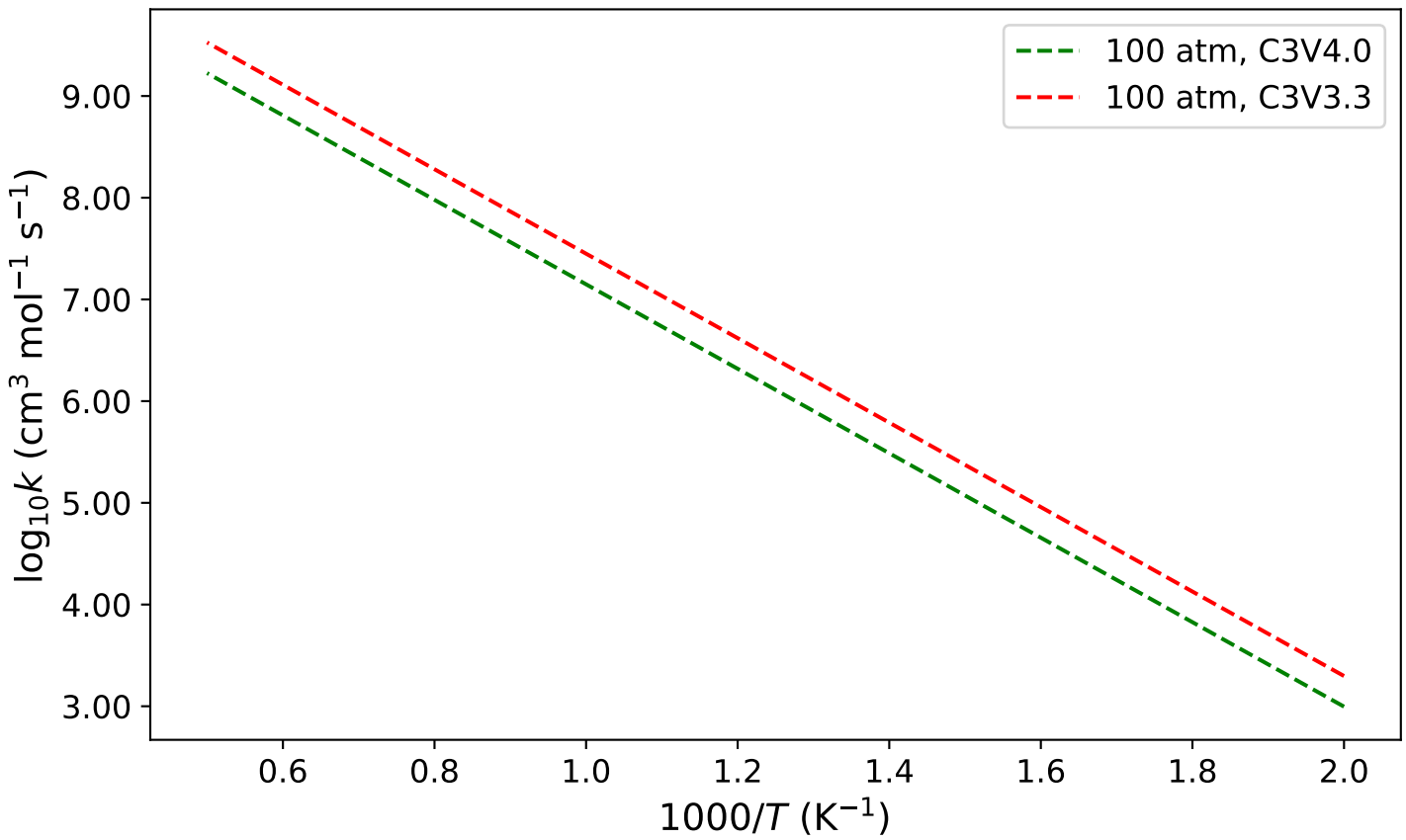


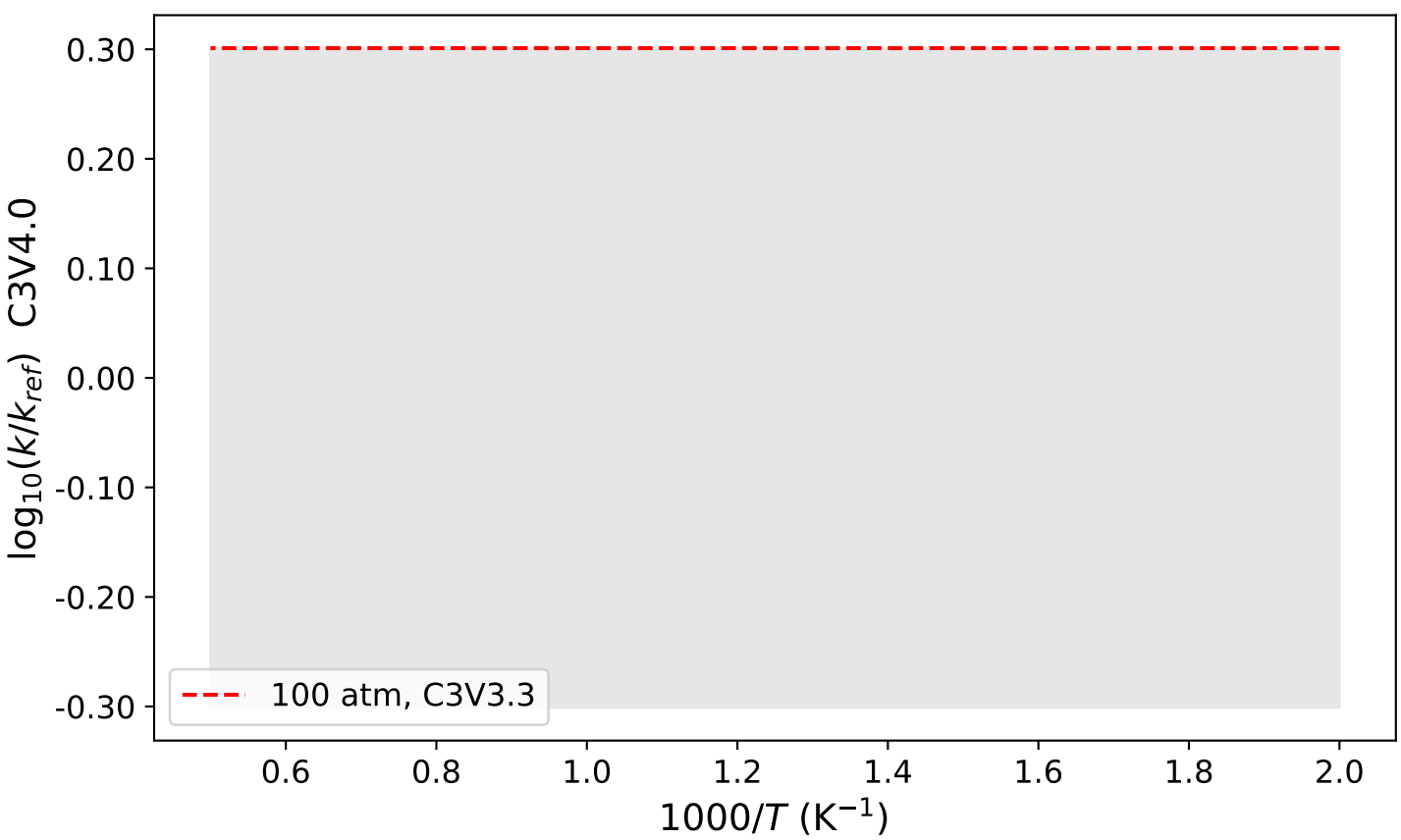
XYLENE (+M) = CH3 + C6H4CH3 (+M)



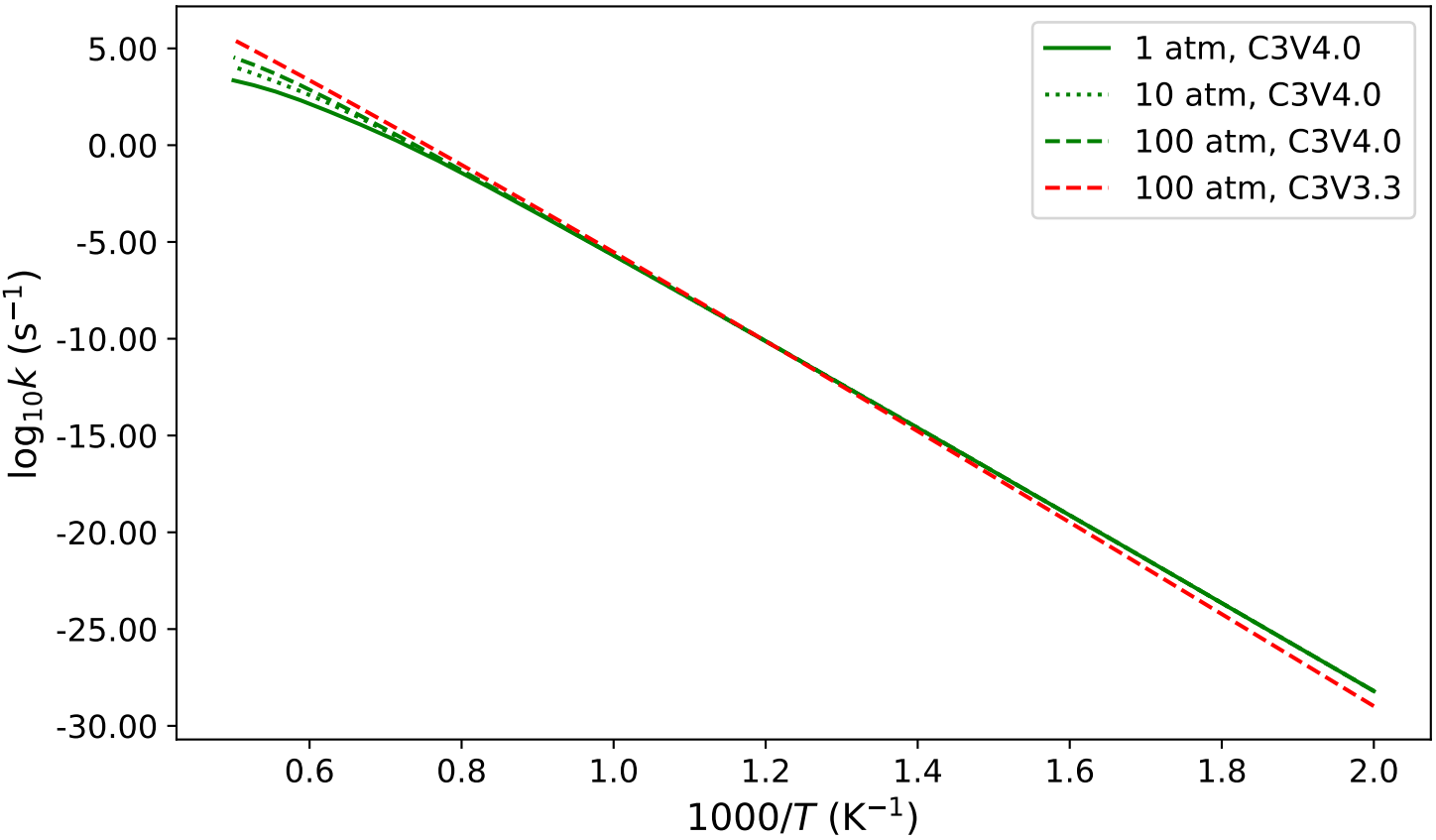


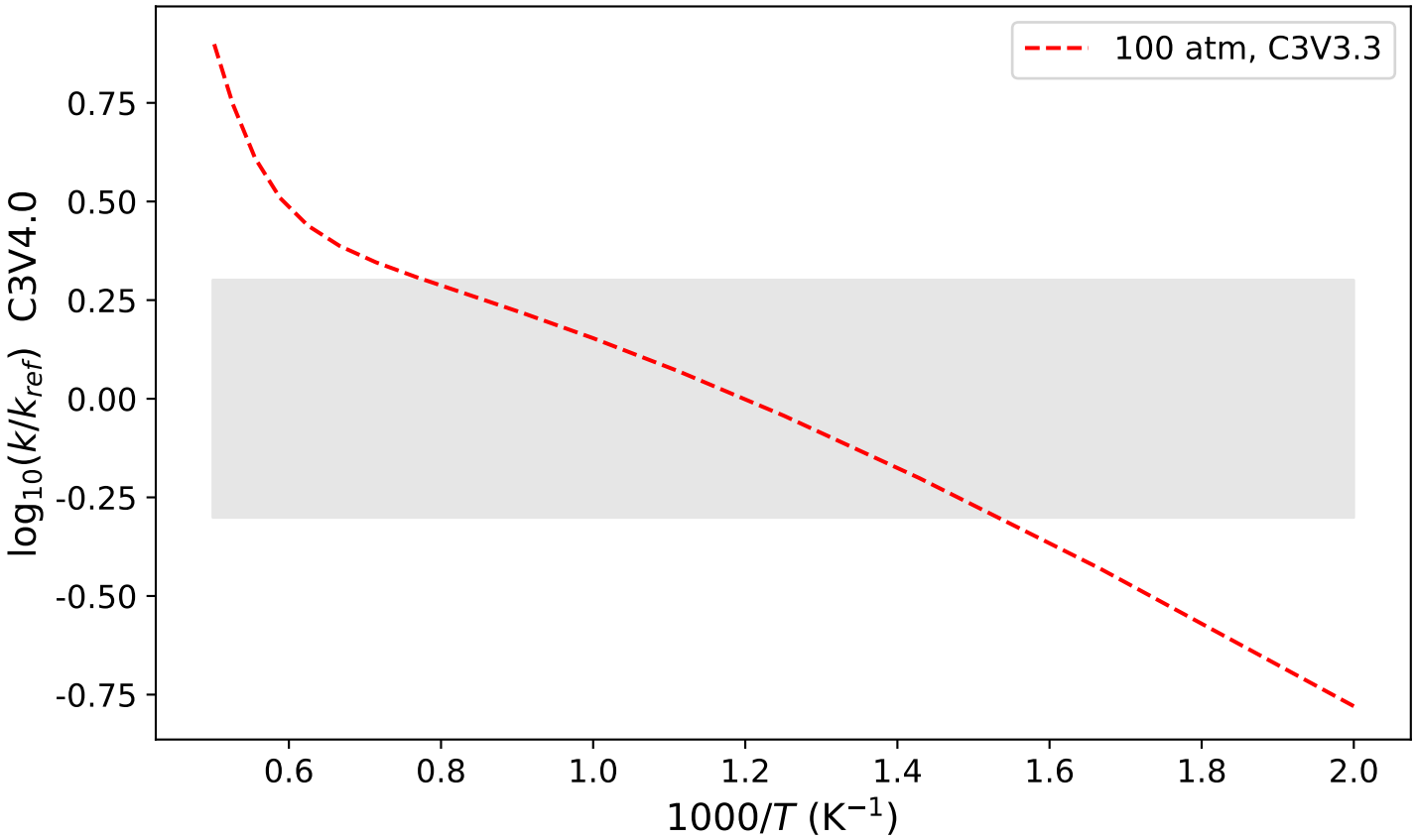
C3H3 + C6H5C2H3 = H + C10H7CH3-A



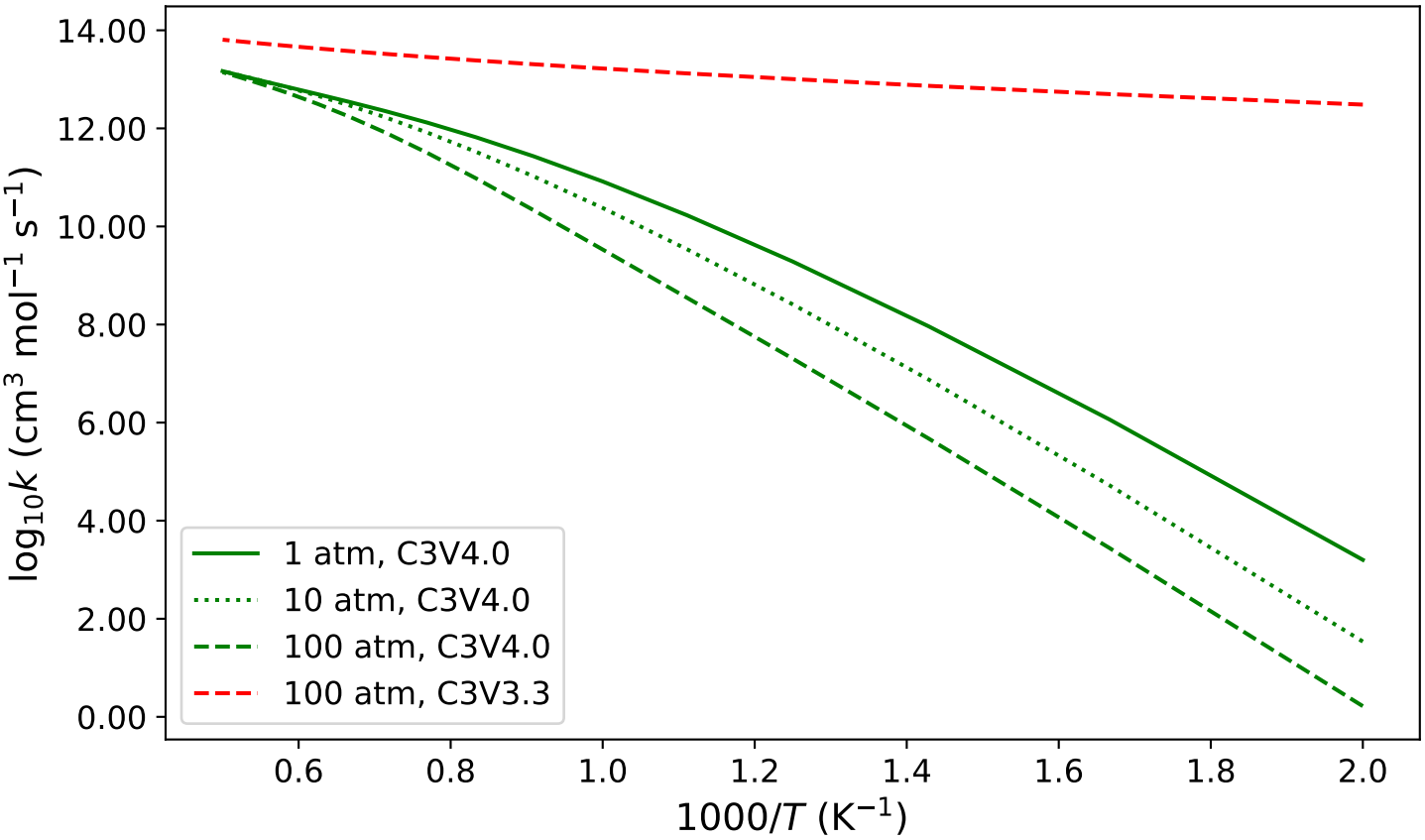


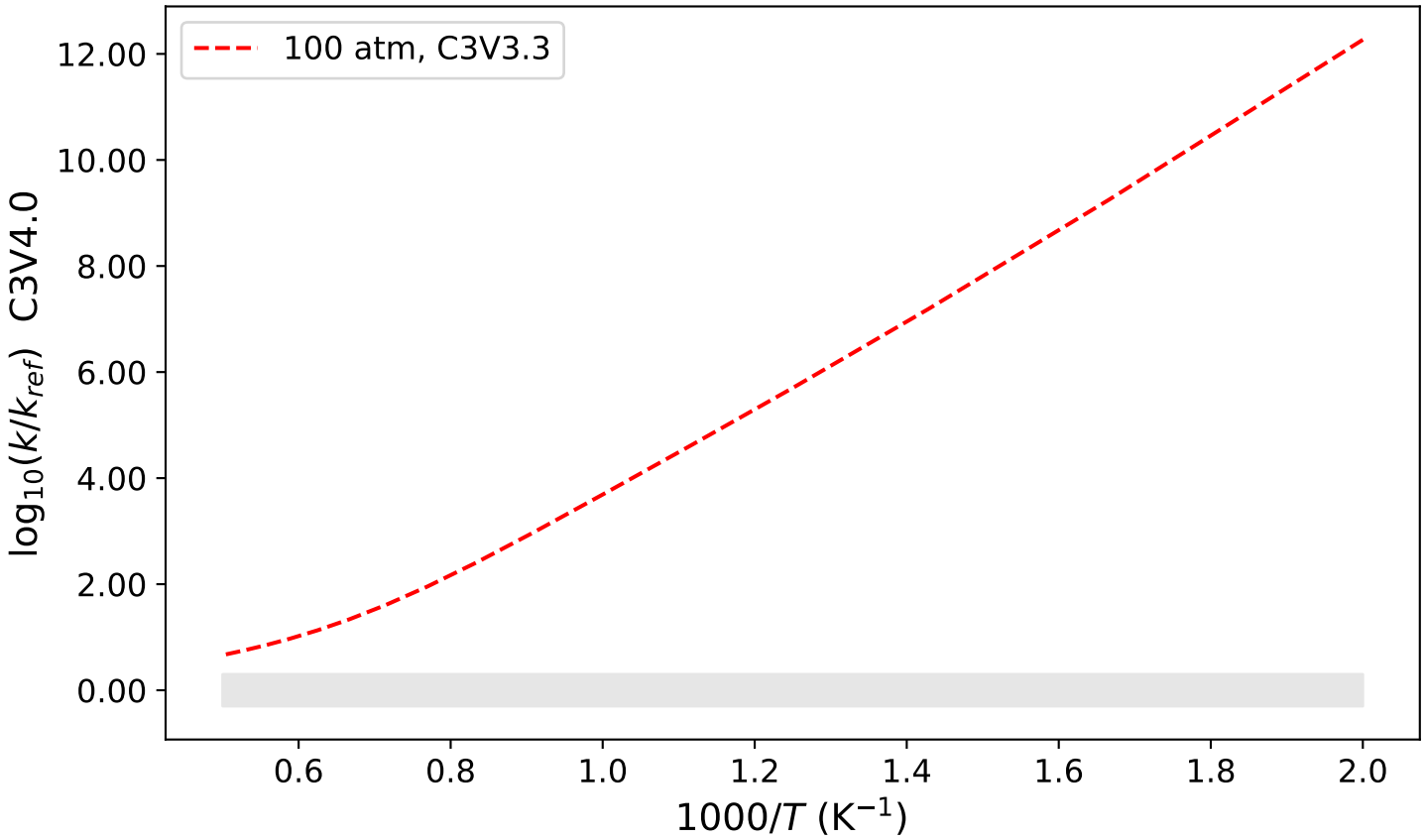
C10H7CH3-A (+M) = CH3 + NAPH-A (+M)



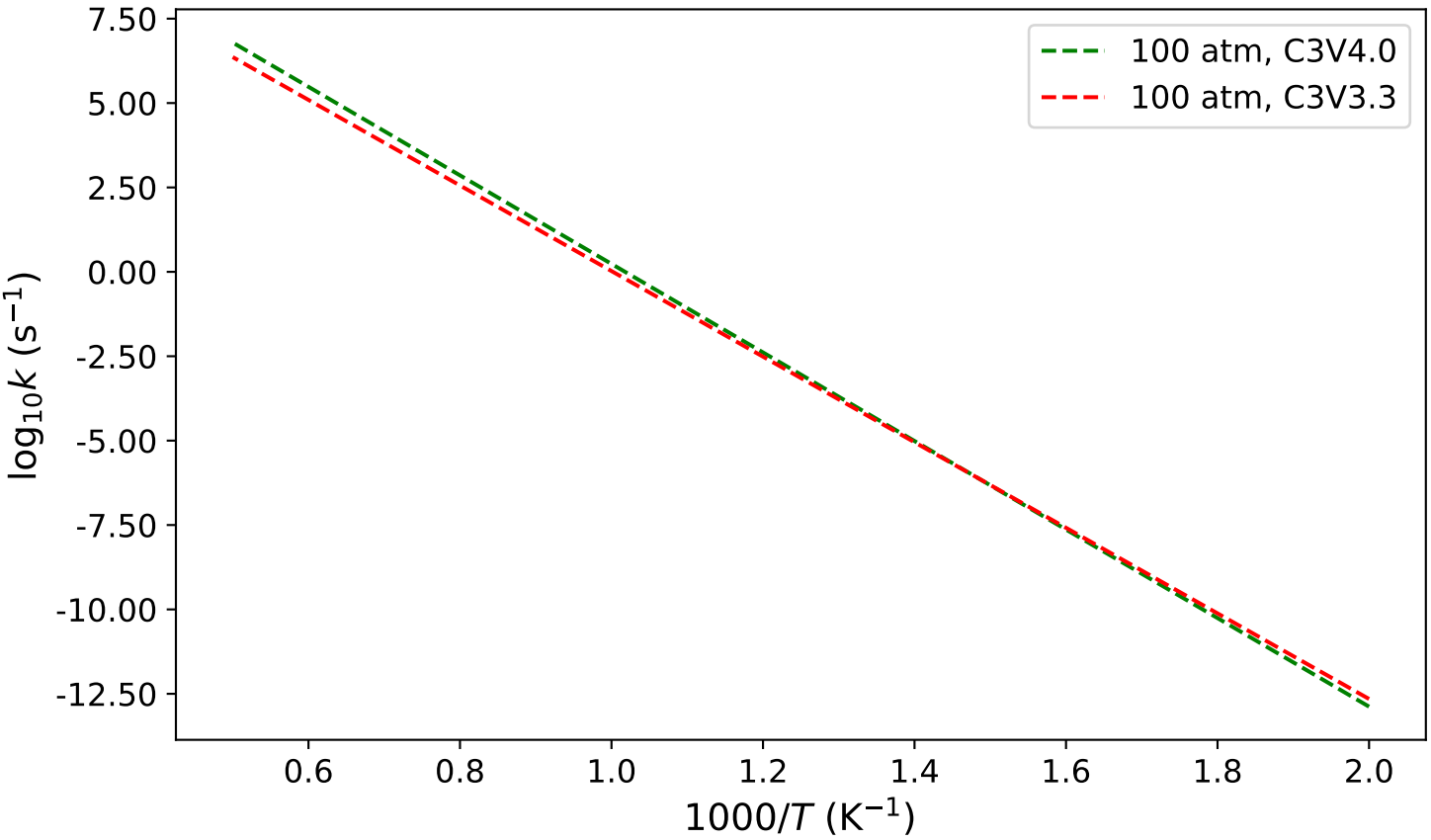


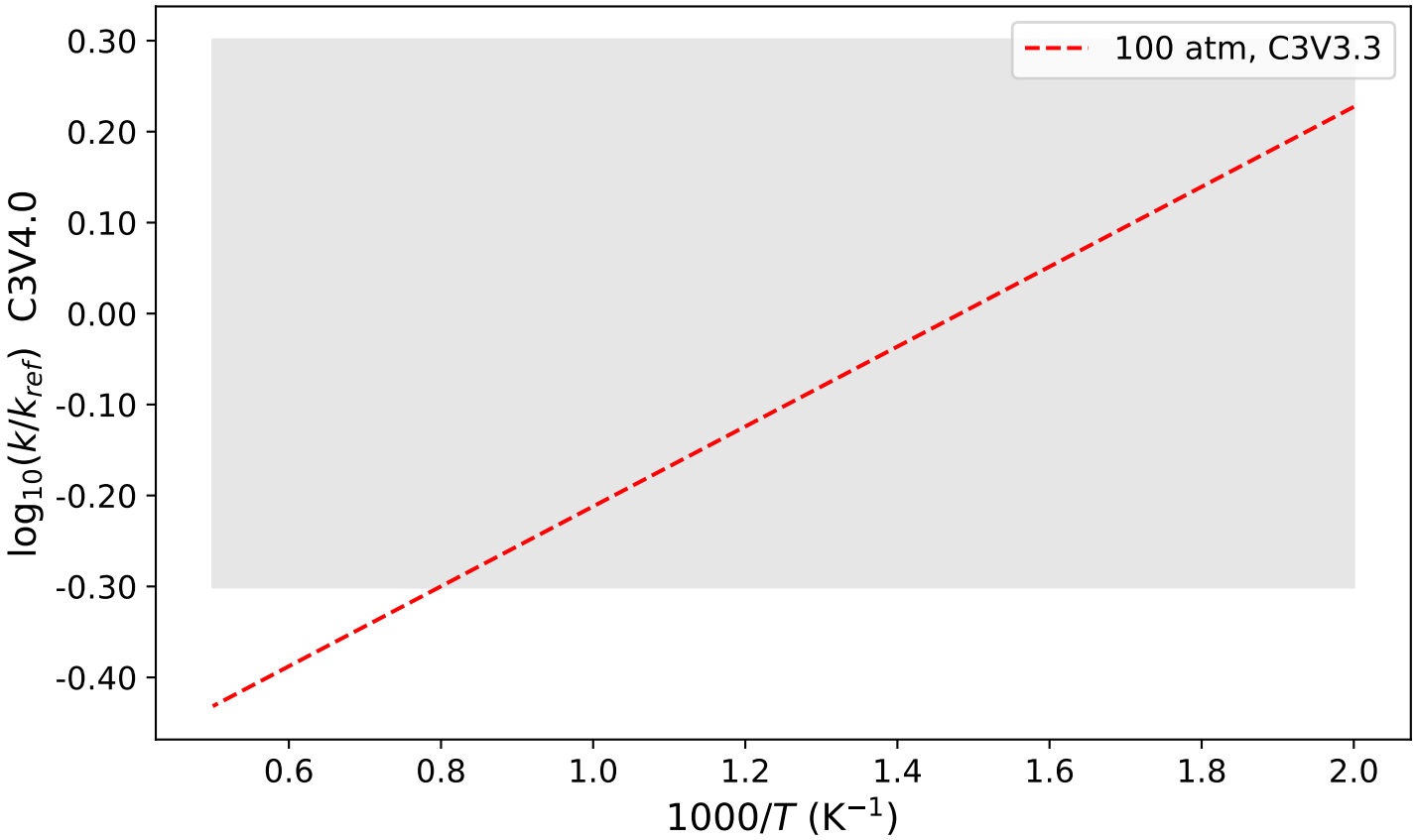
H + C9H6O = CO + C6H4C2H3



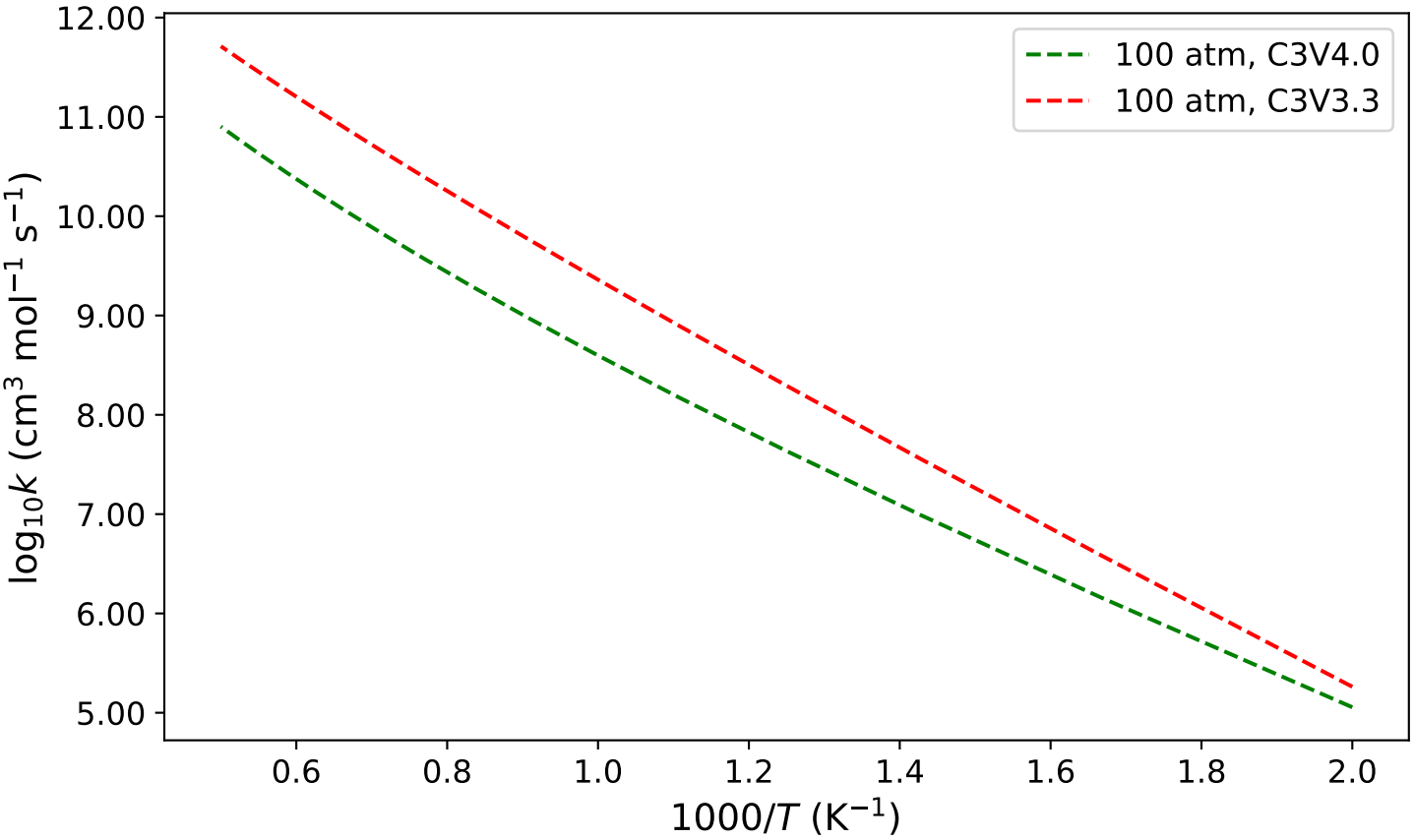


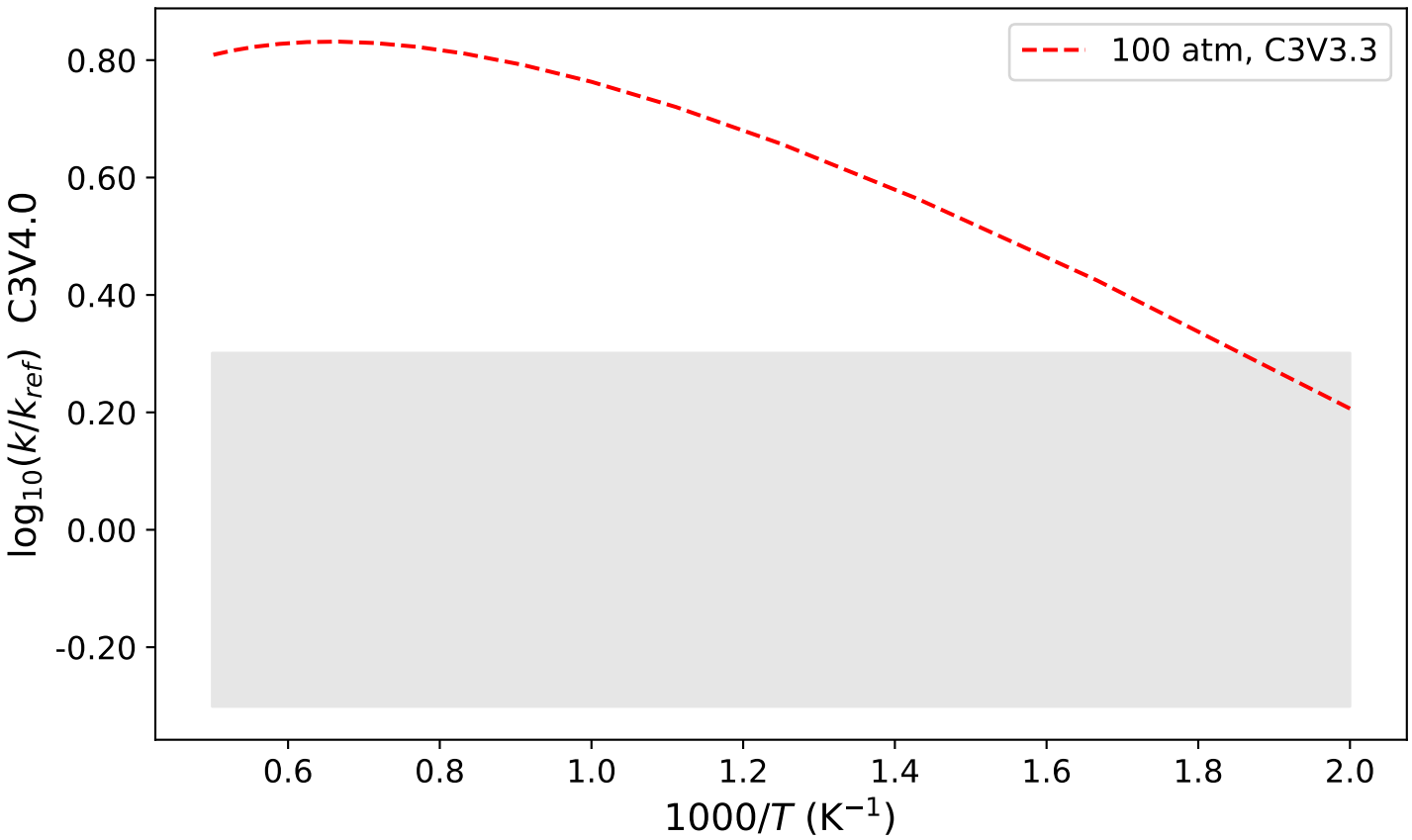
CYC5H8 = H2 + C5H6



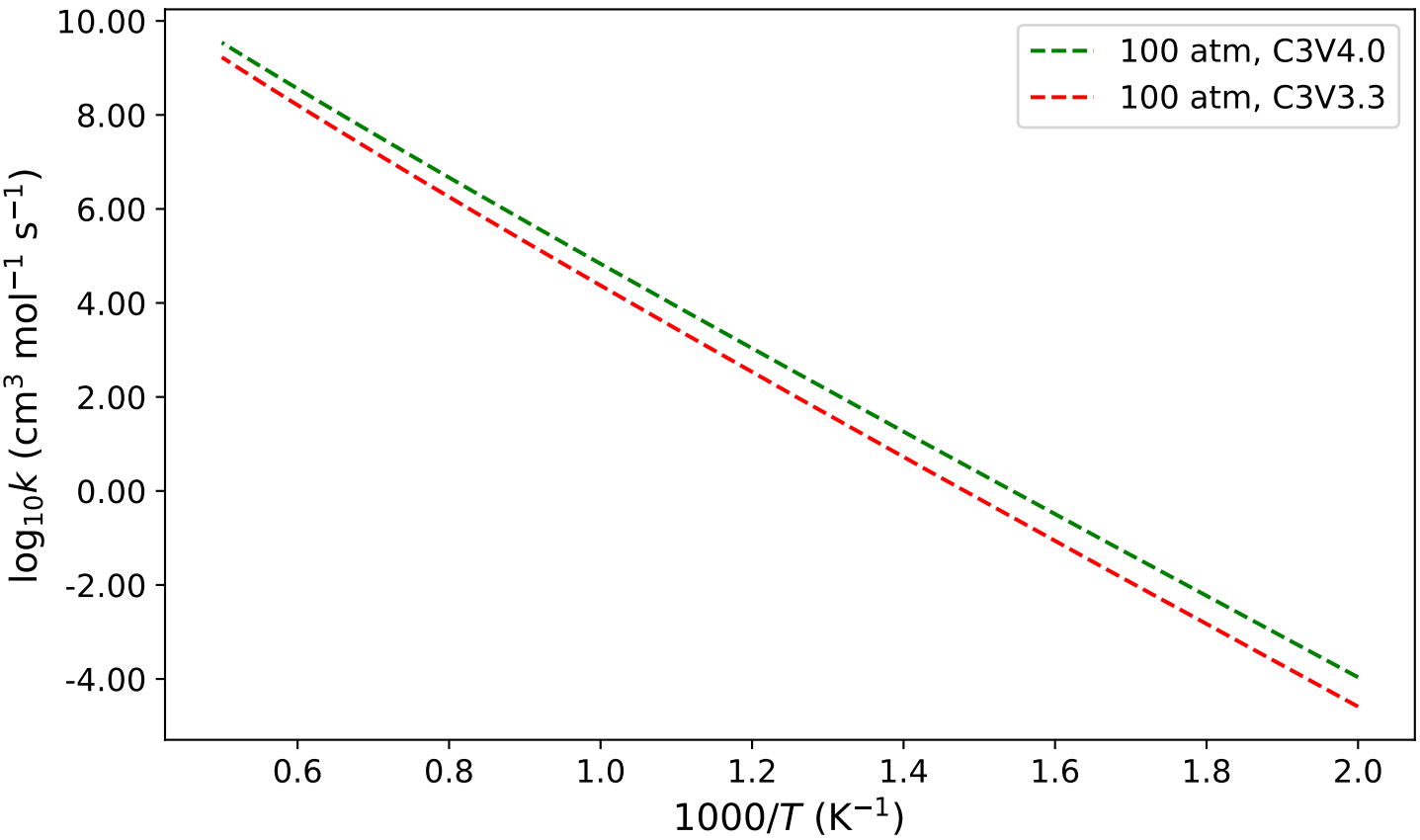


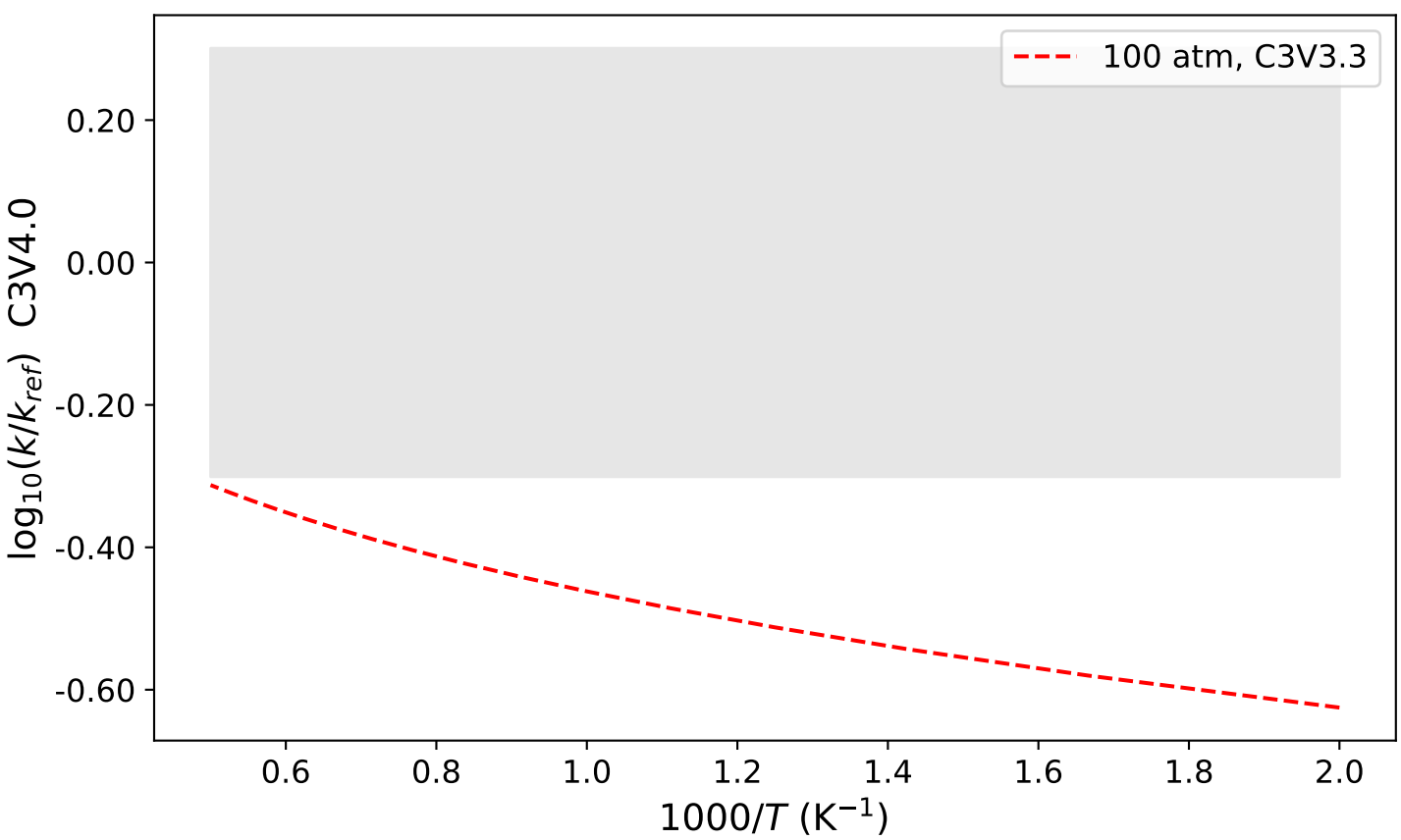
HO2 + C14H14 = H2O2 + C14H13



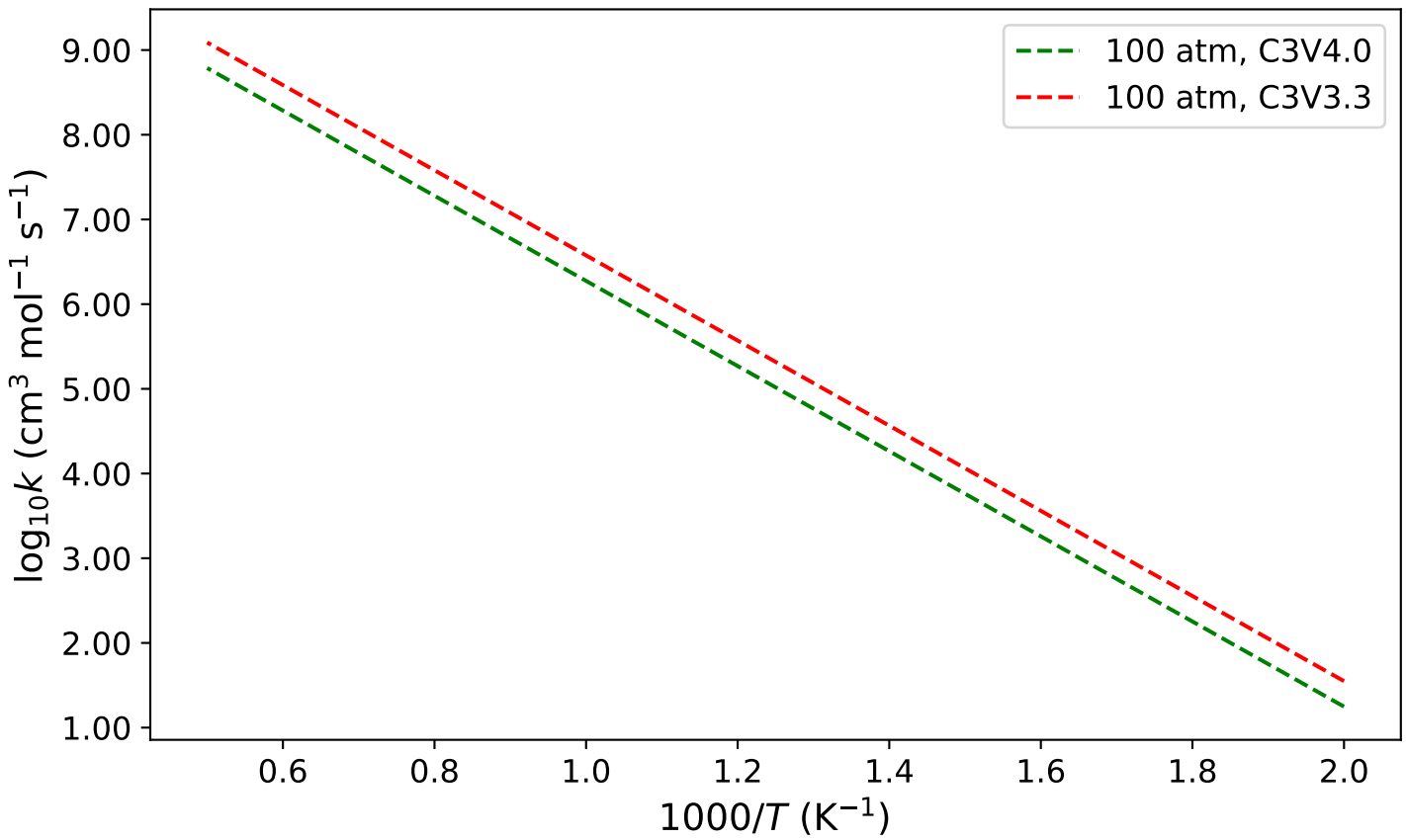


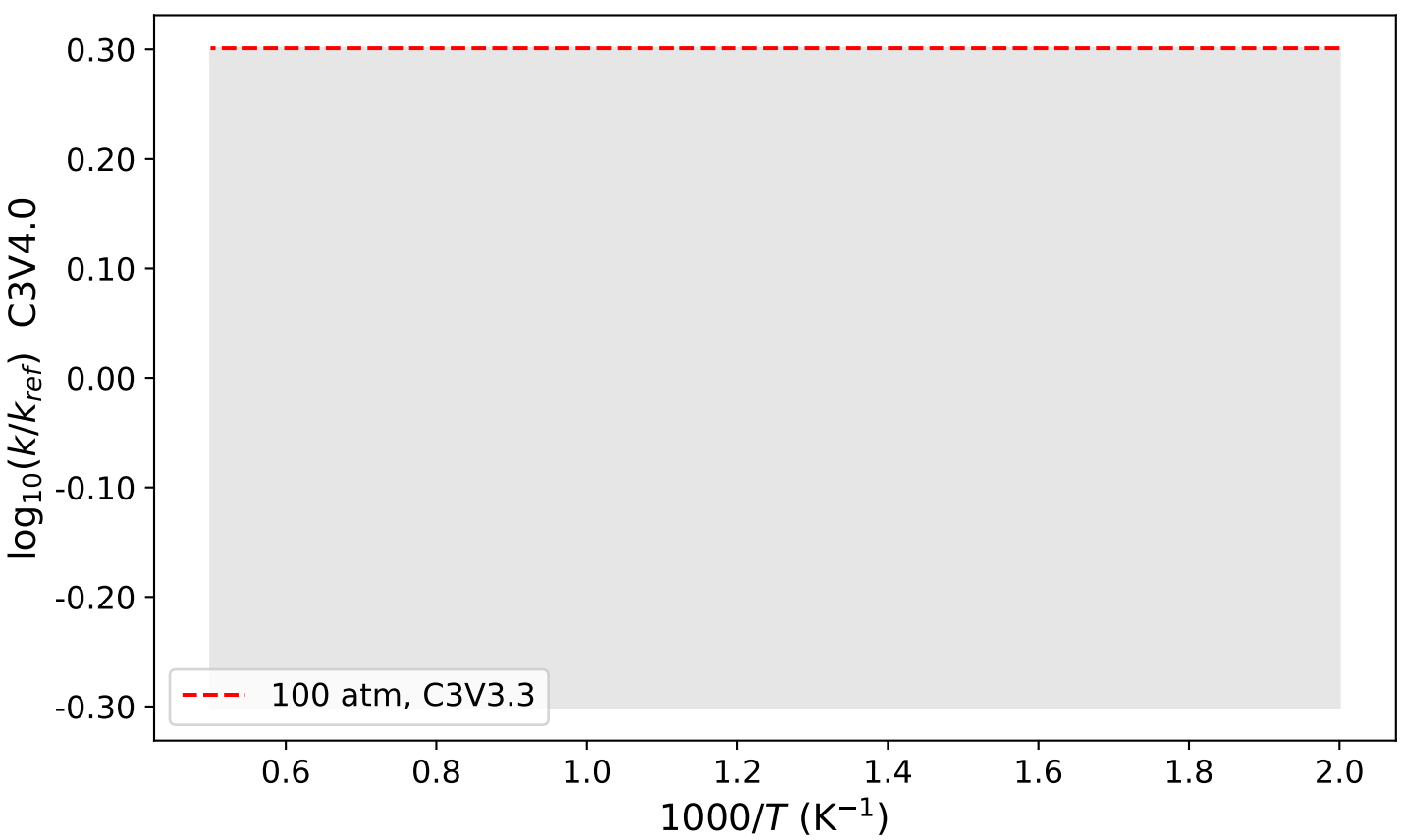
O2 + C6H4CH3CHO => HO2 + CO + C6H4CH3



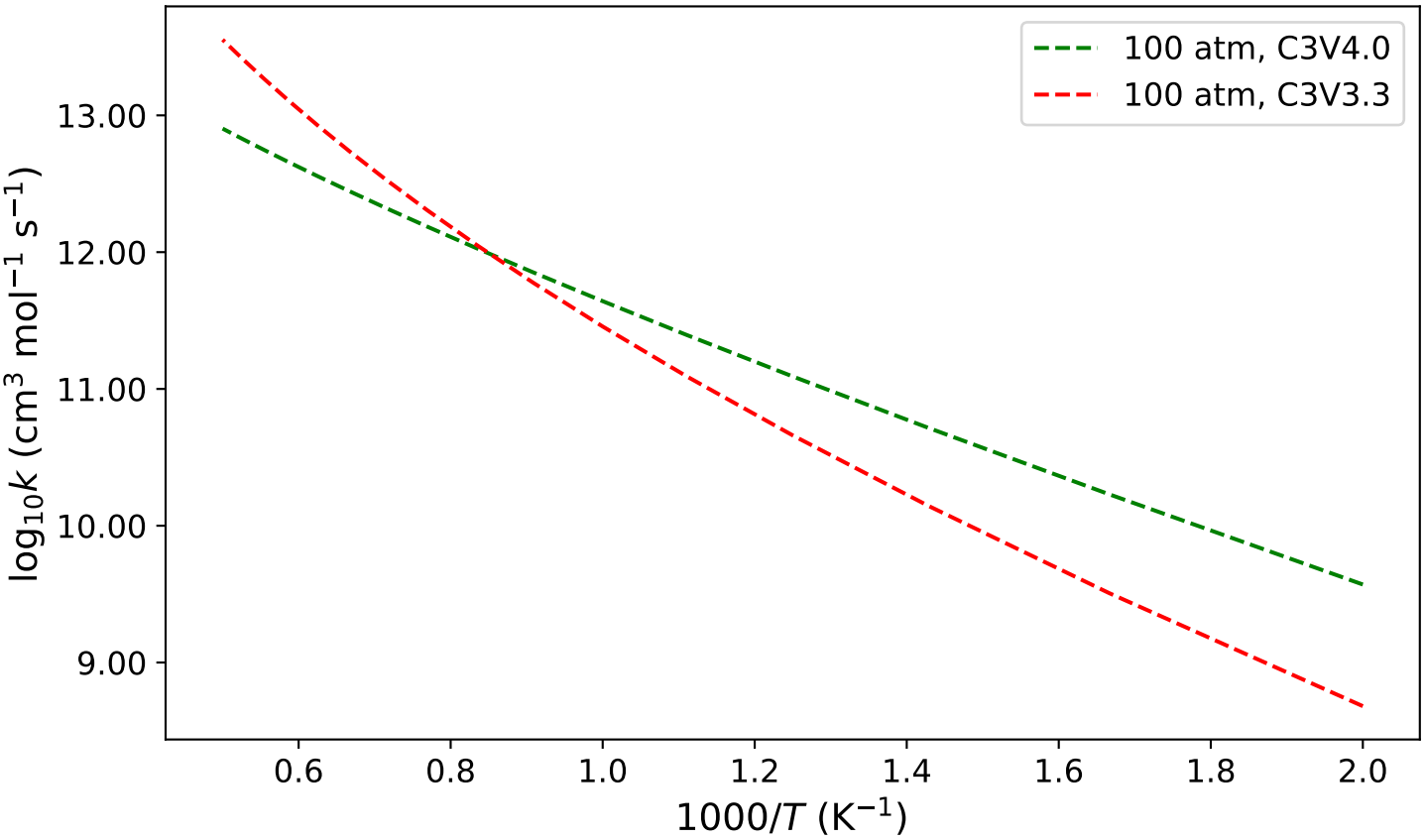


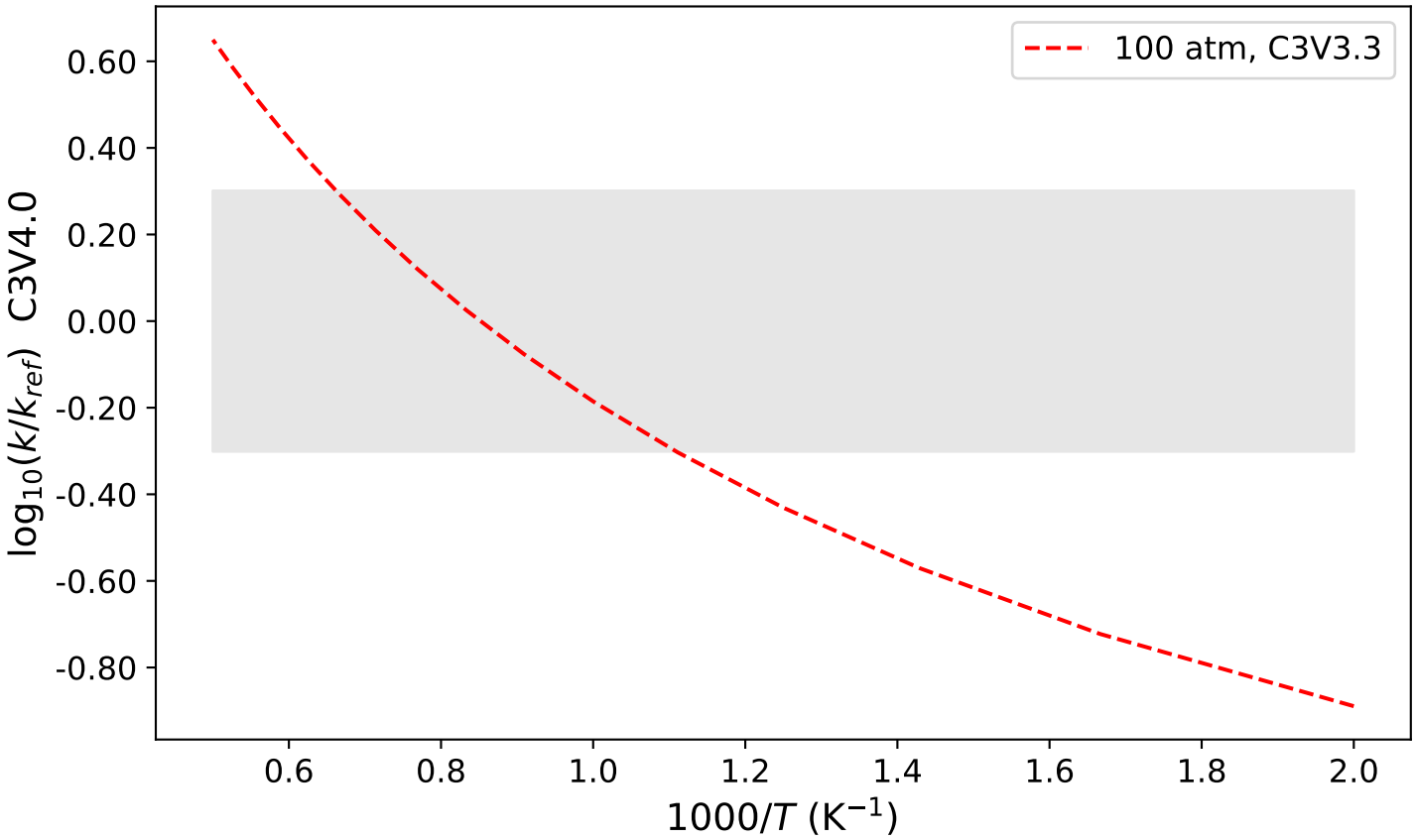
C5H5 + C6H6 = H + C10H7CH3-A



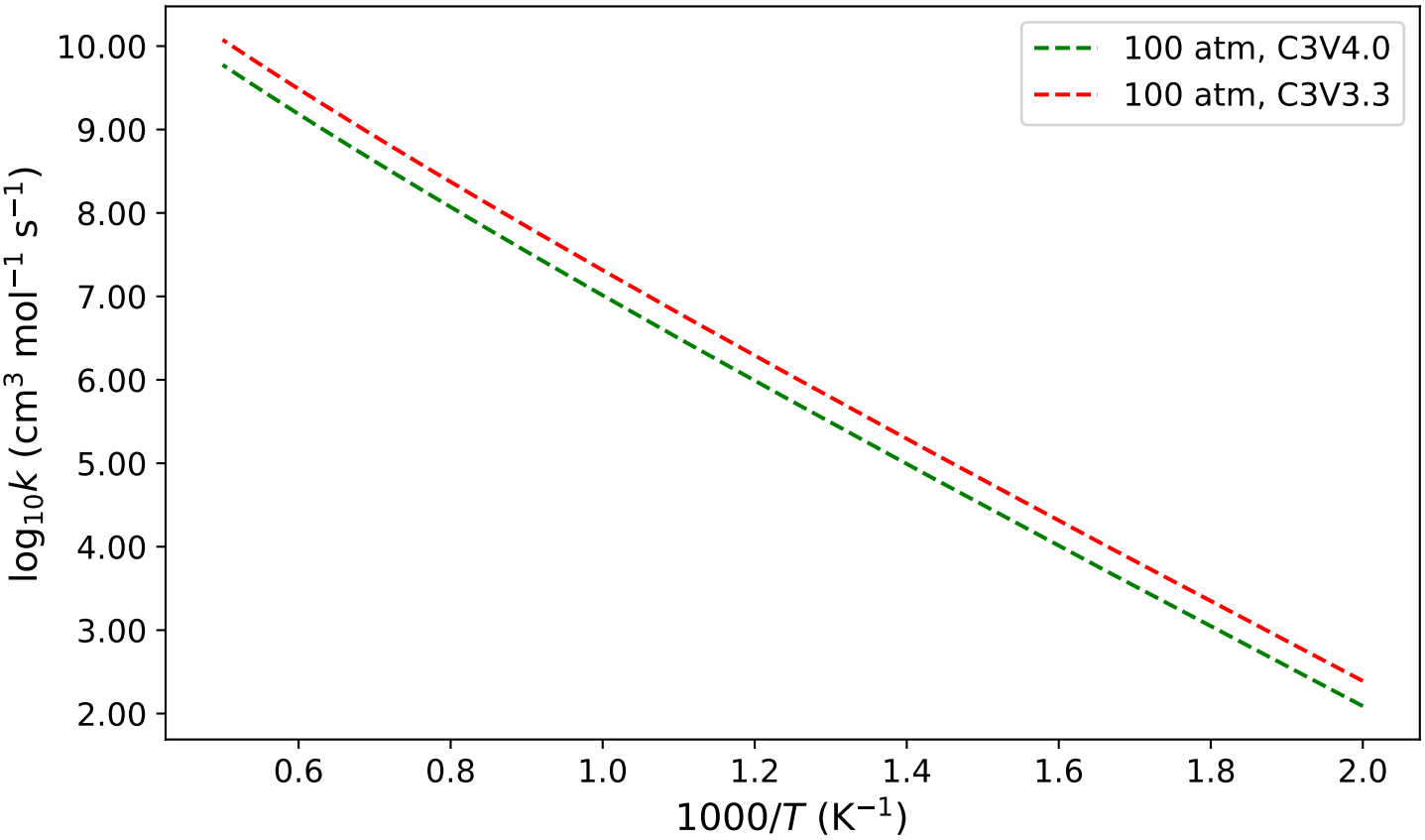


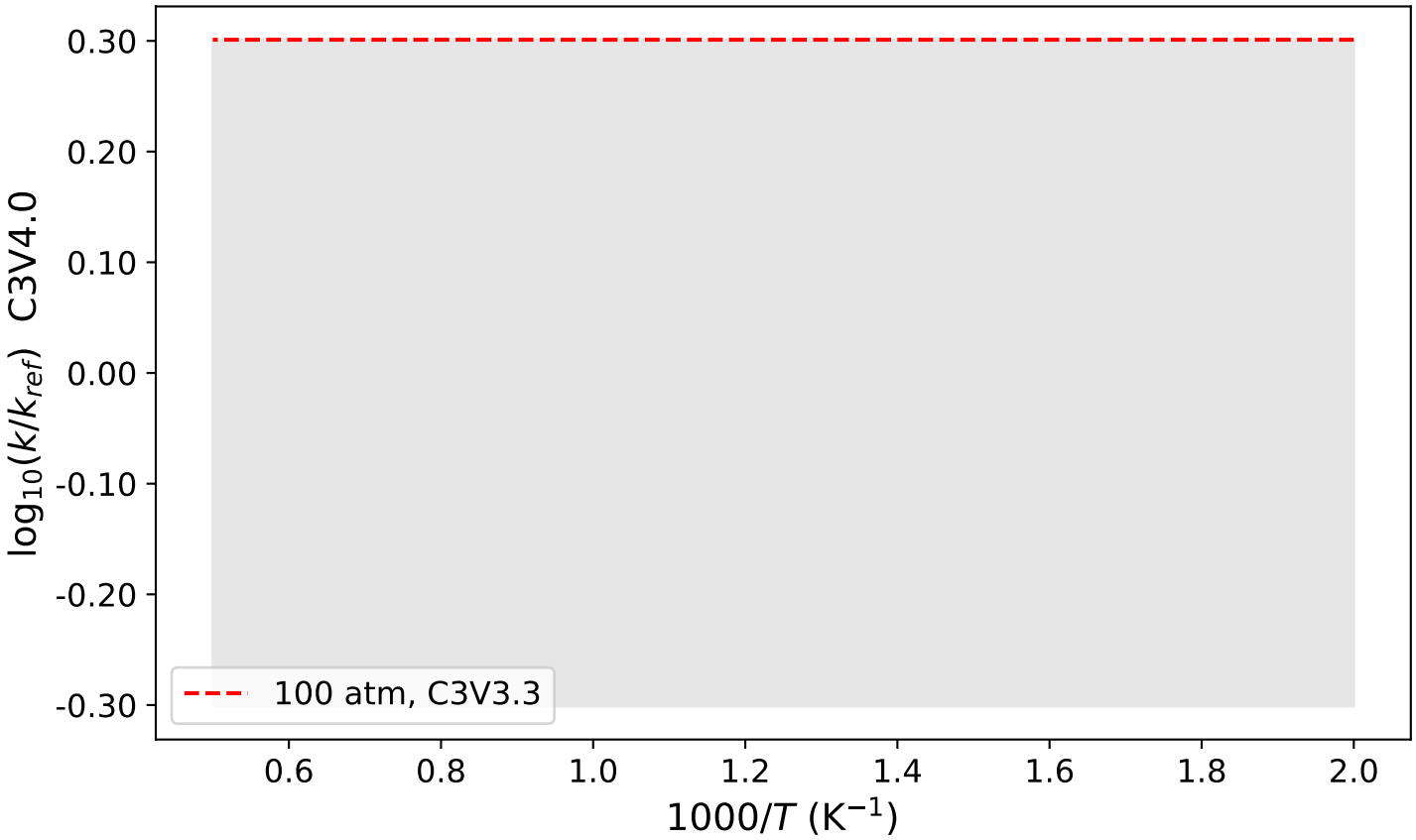
H + C6H5C2H5 = H2 + C6H5CH2CH2



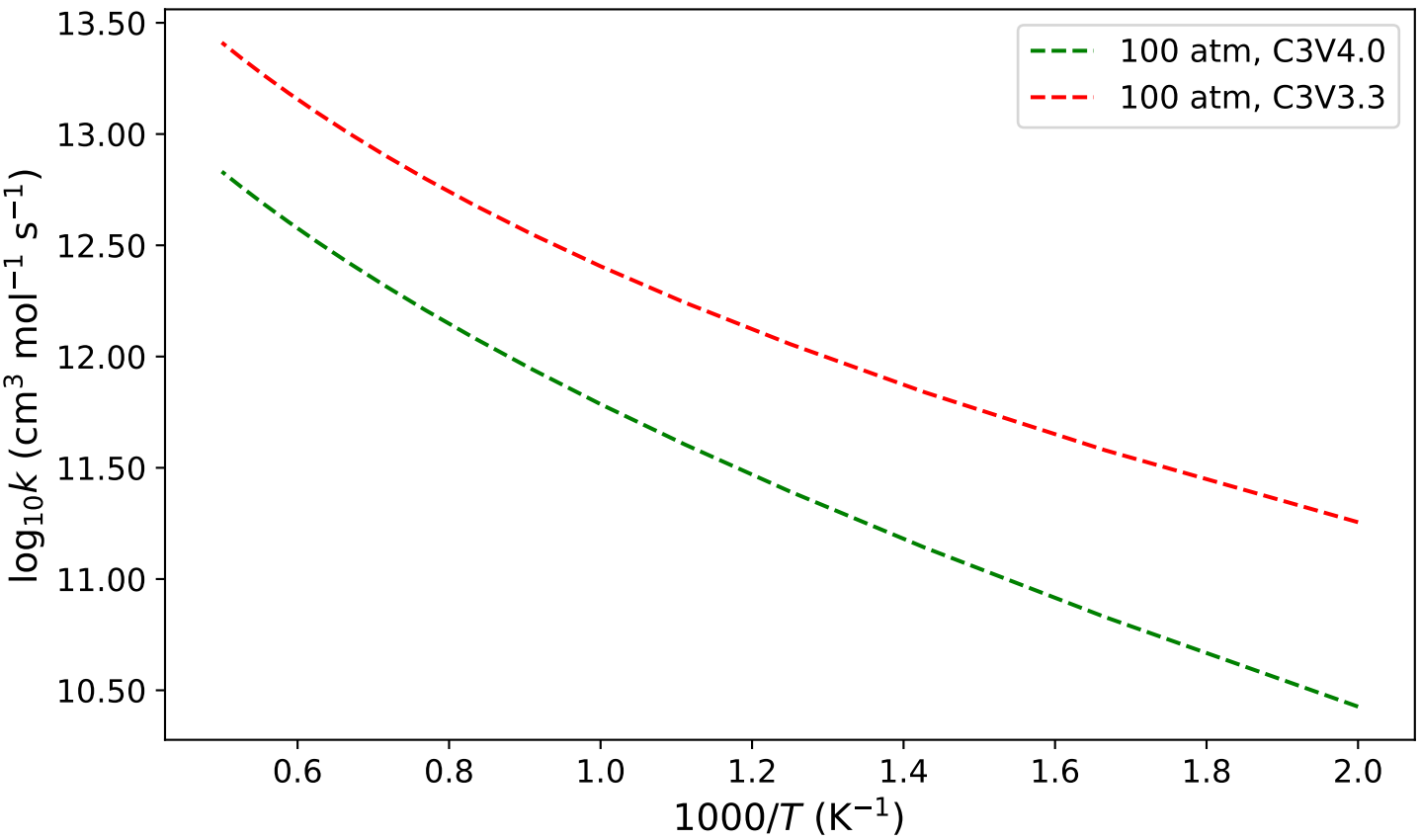


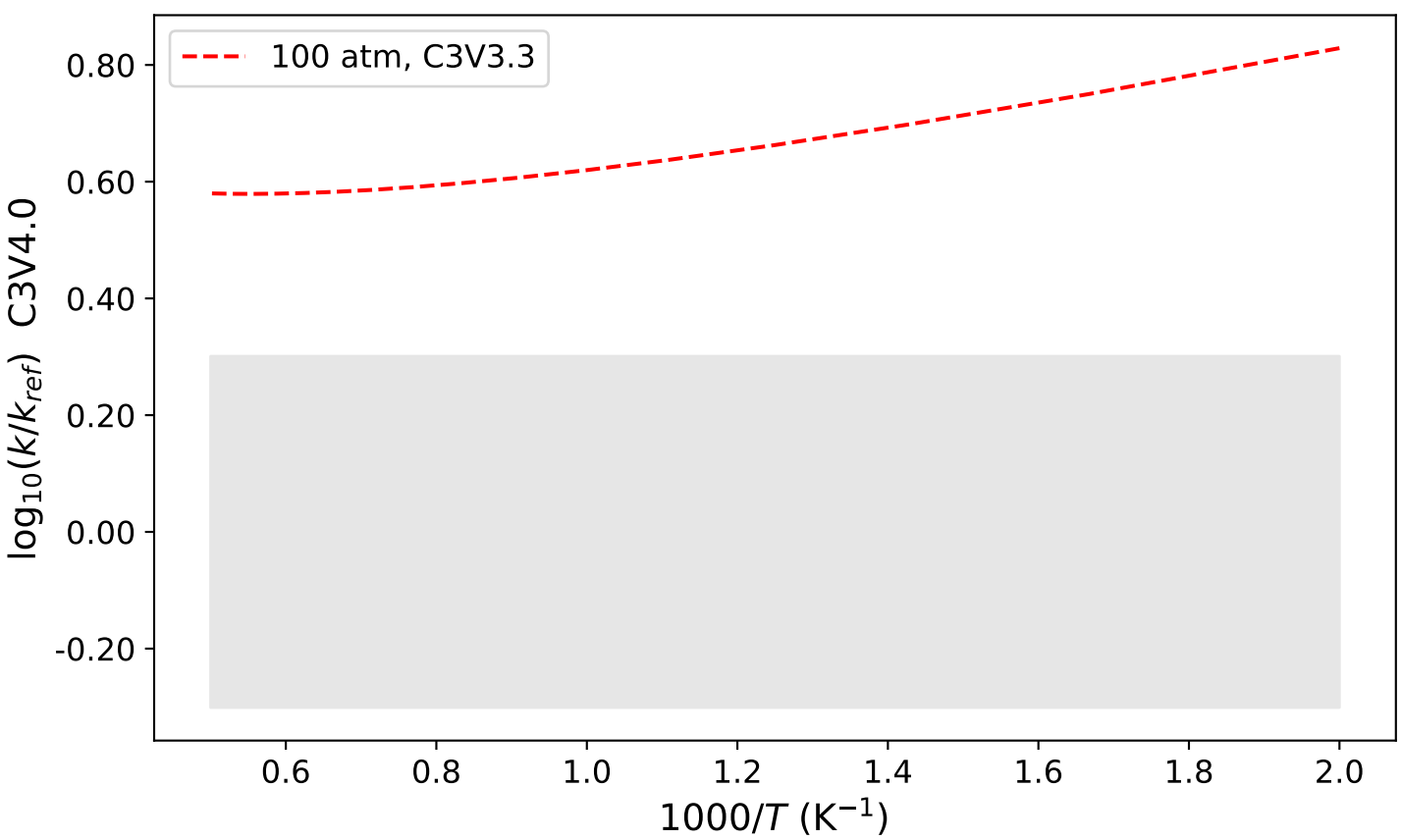
NAPH + OC6H4CH3 = NAPH-A + HOC6H4CH3



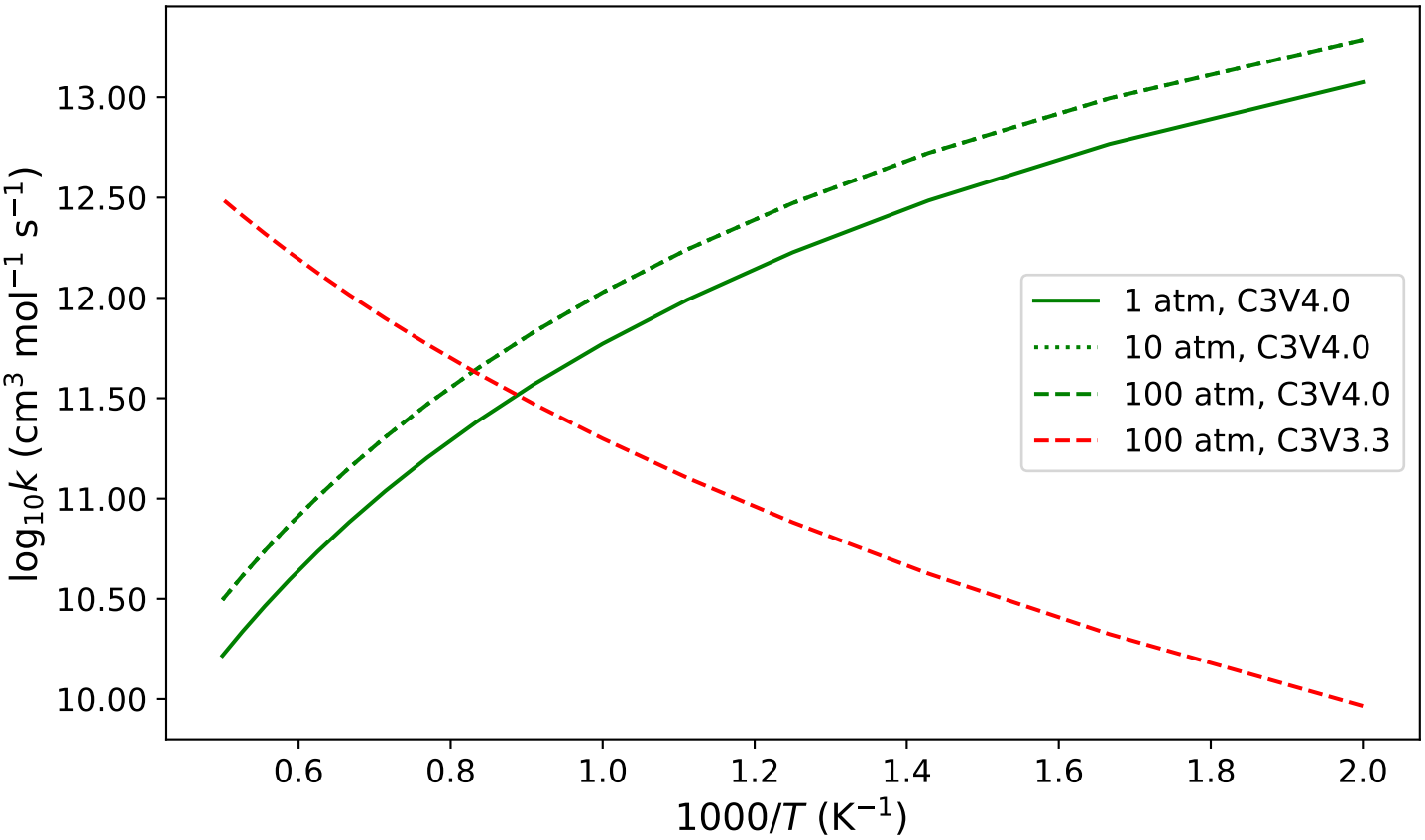


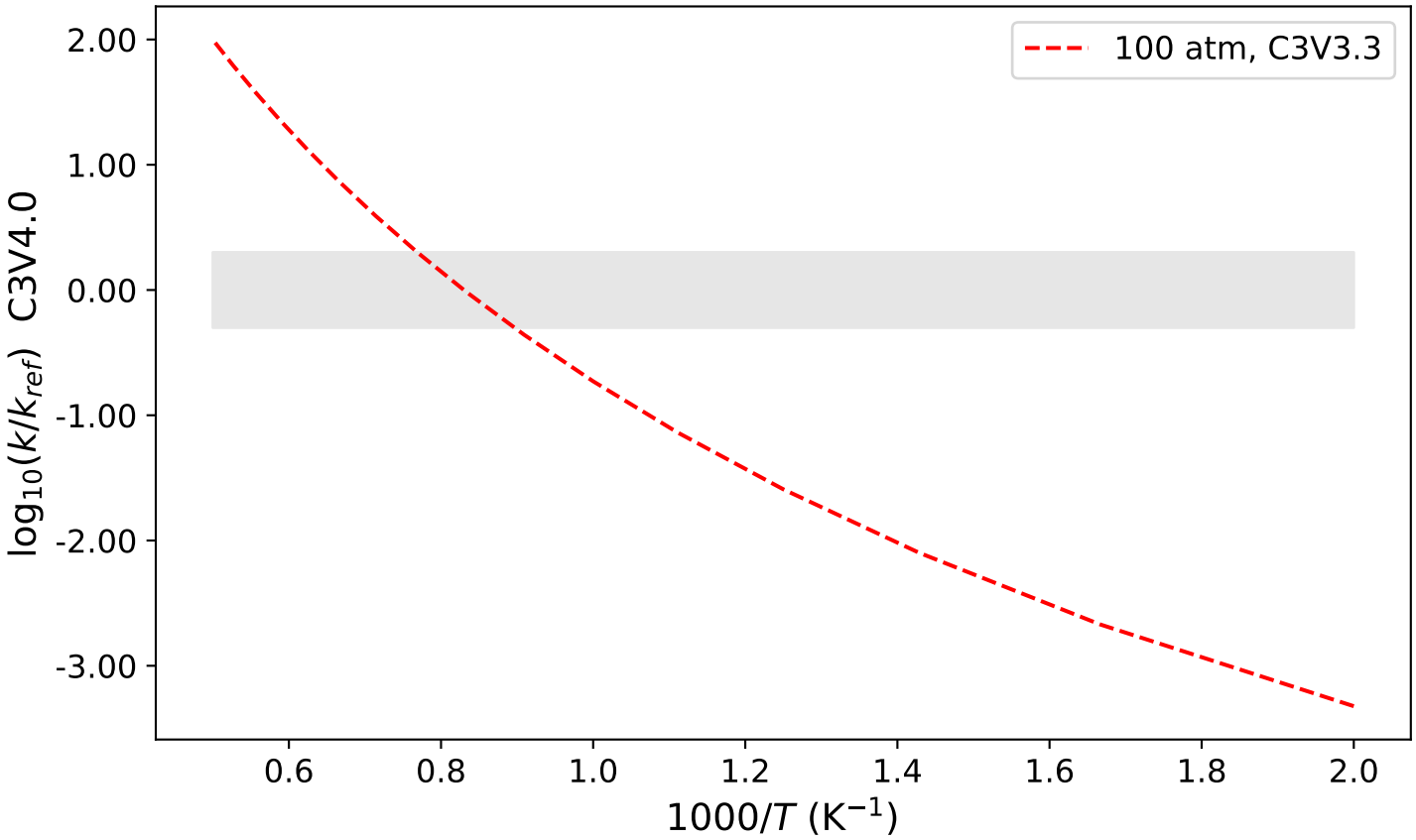
OH + C10H7CH3-A = H2O + C10H6CH3-A



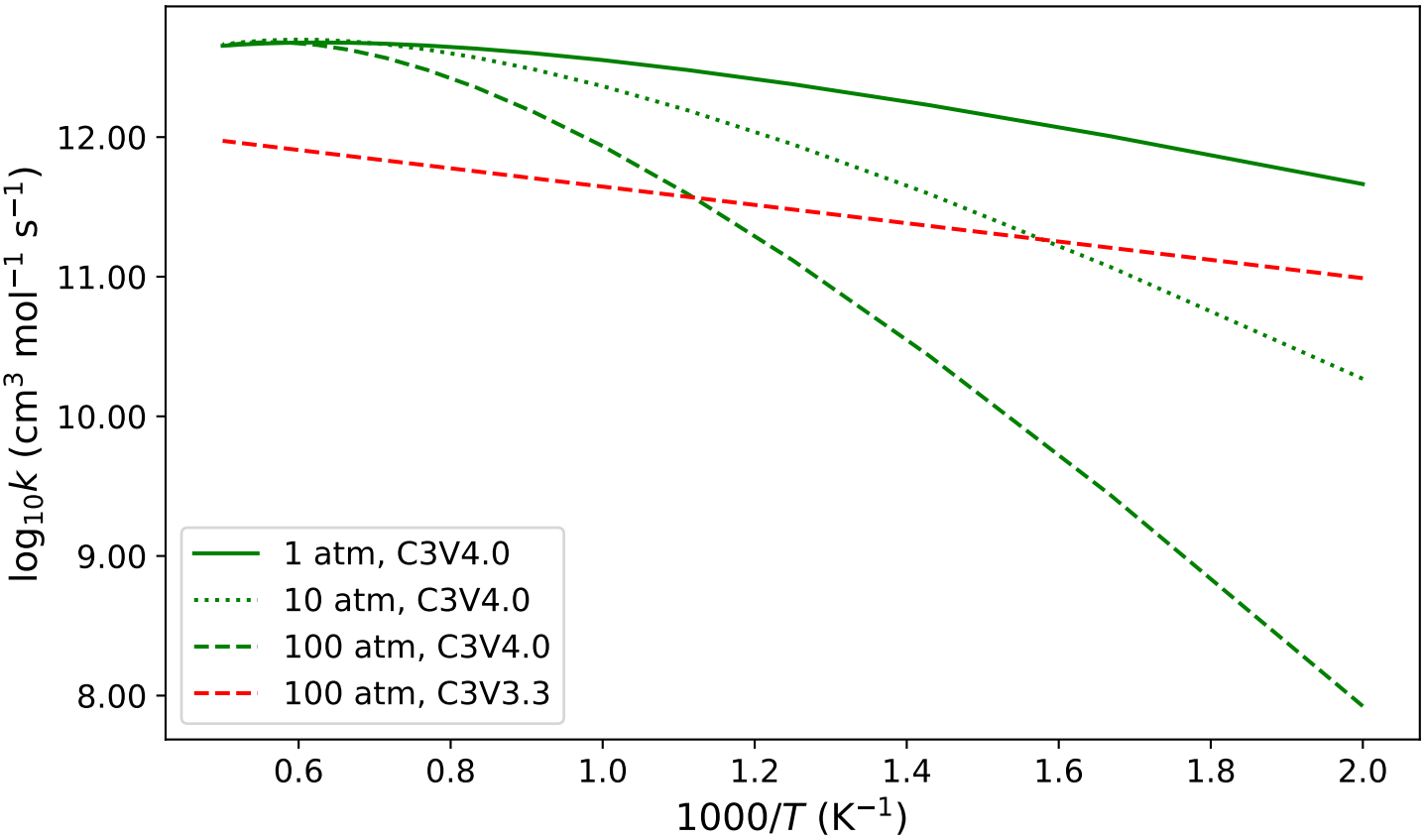


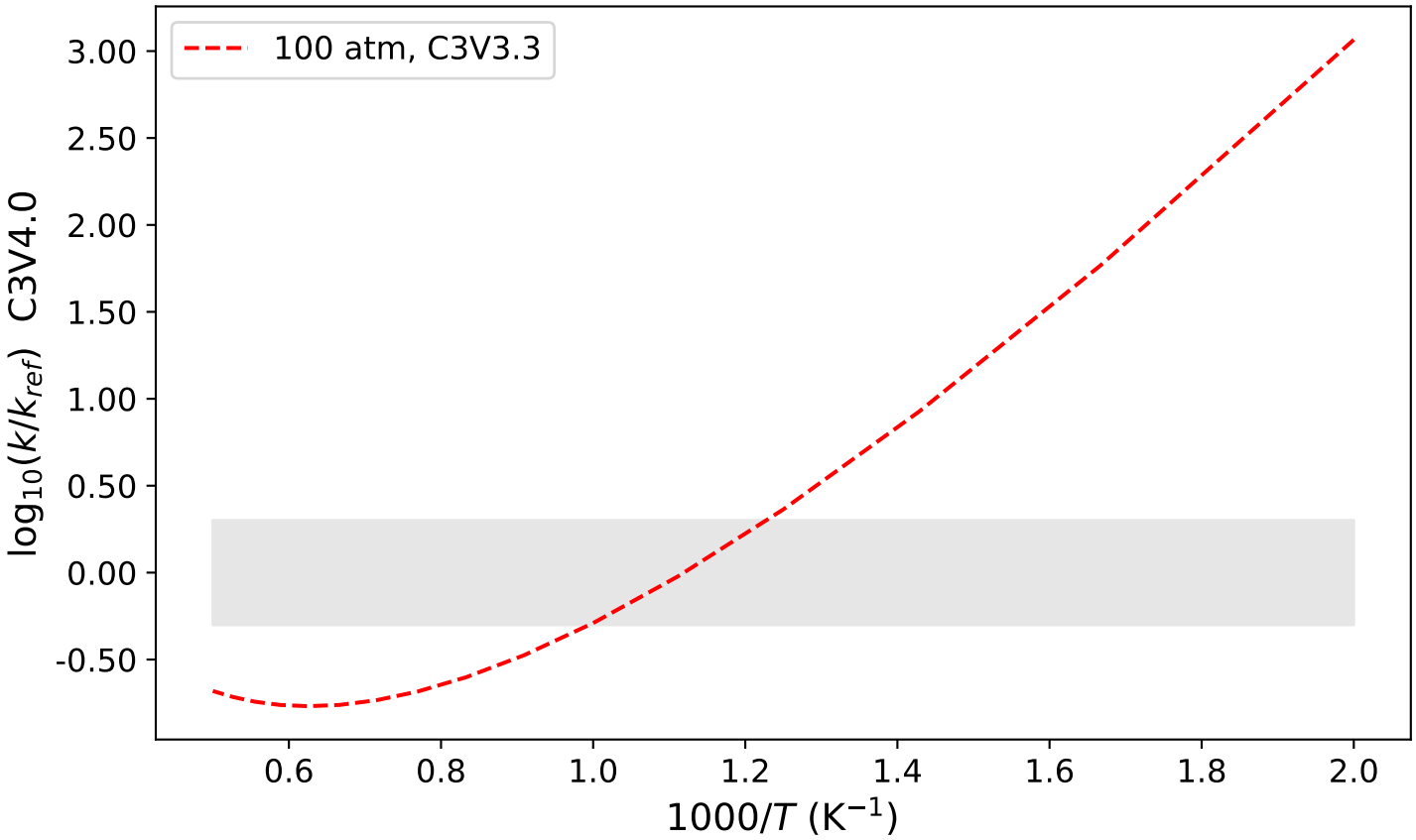
HO2 + C10H7CH2-A = C10H7CH2OOH-A



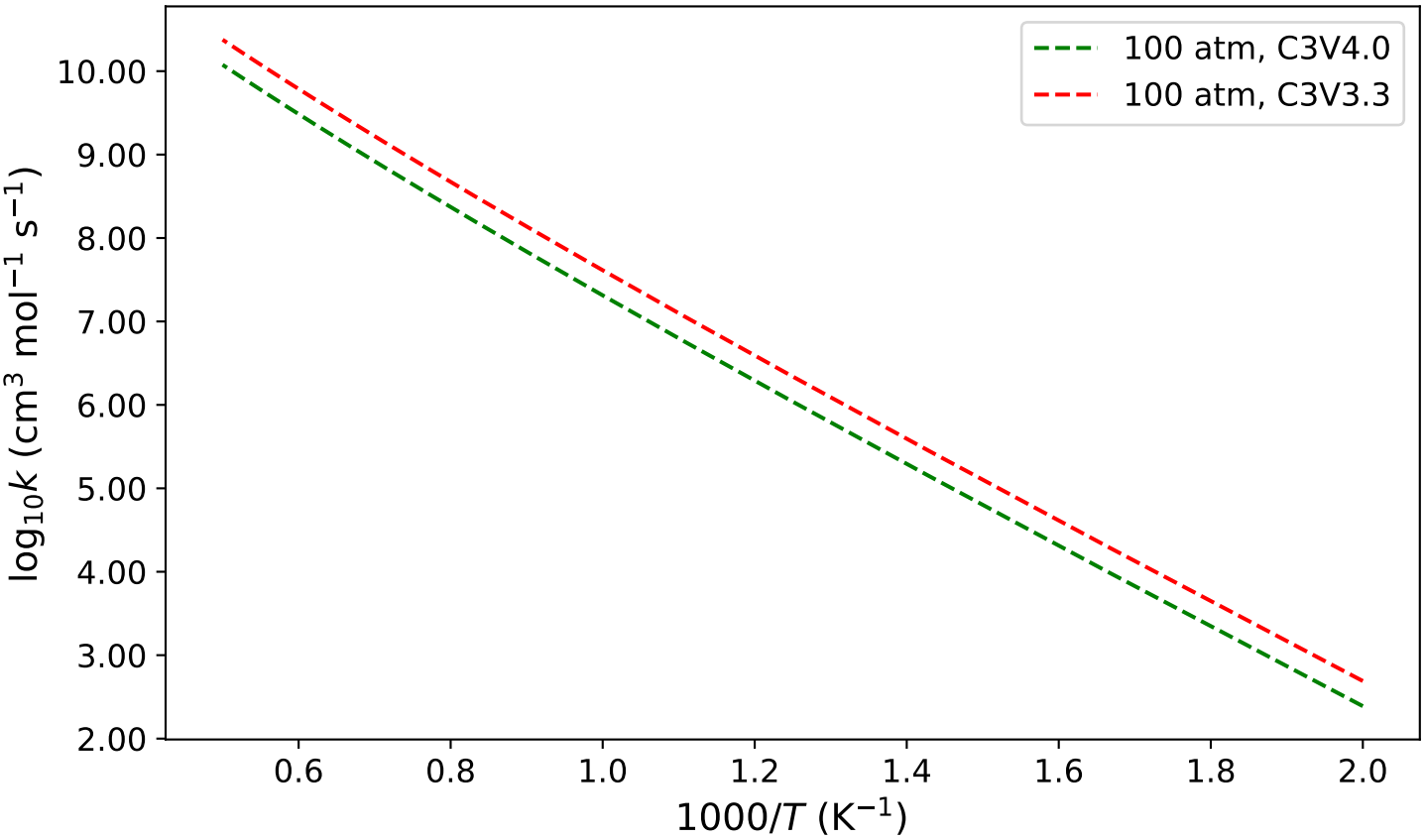


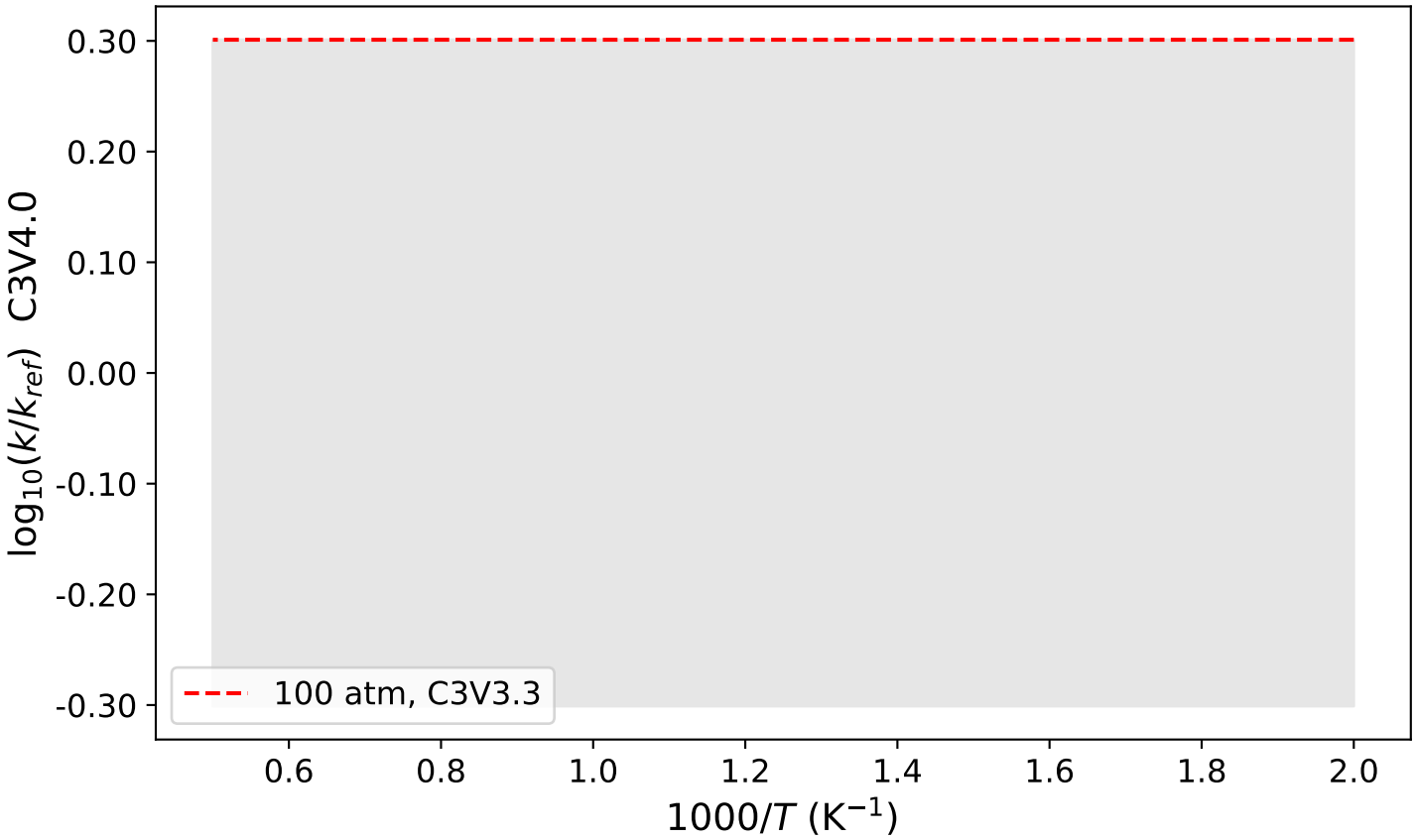
O2 + C12H7 => CO + CO + NAPH-A



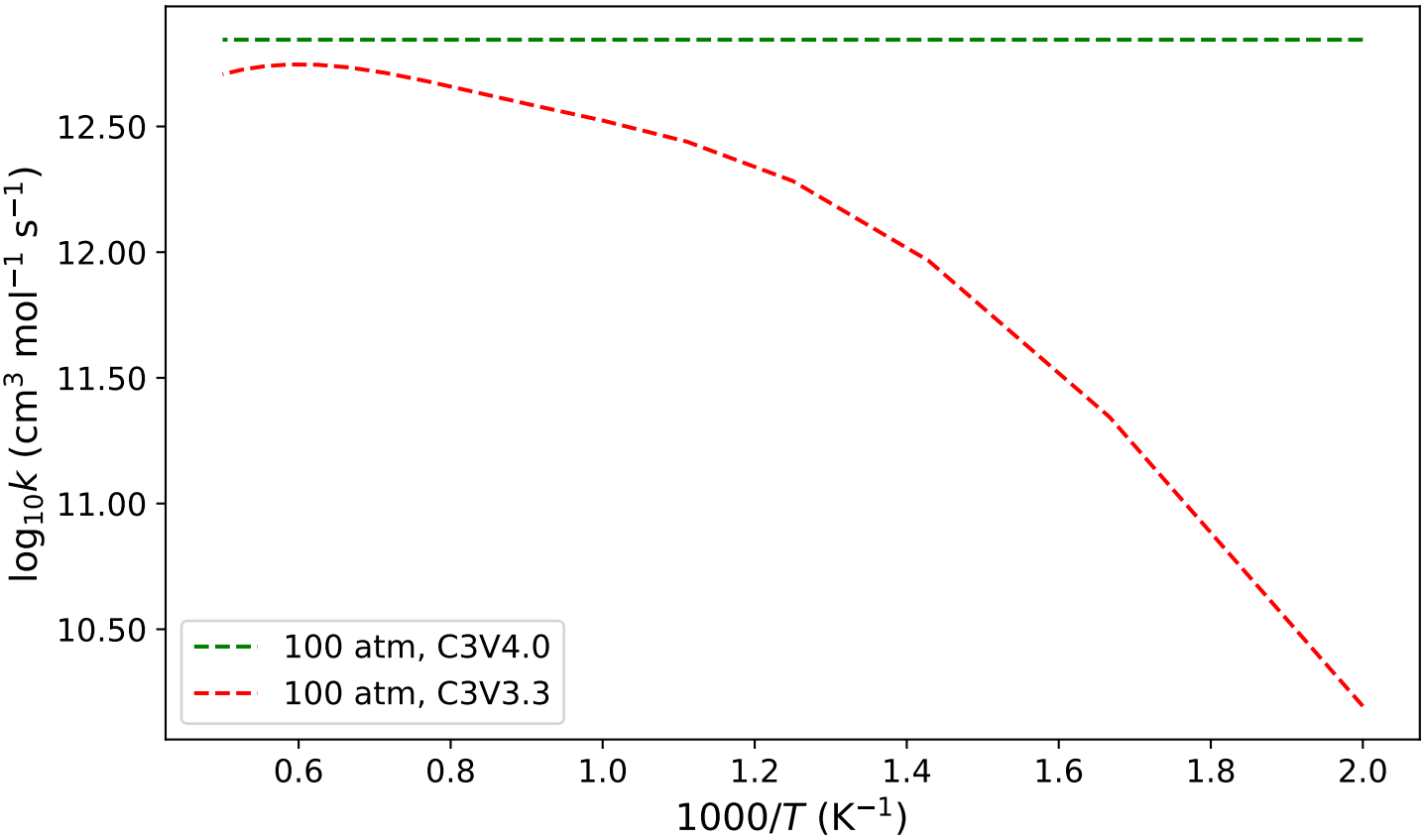


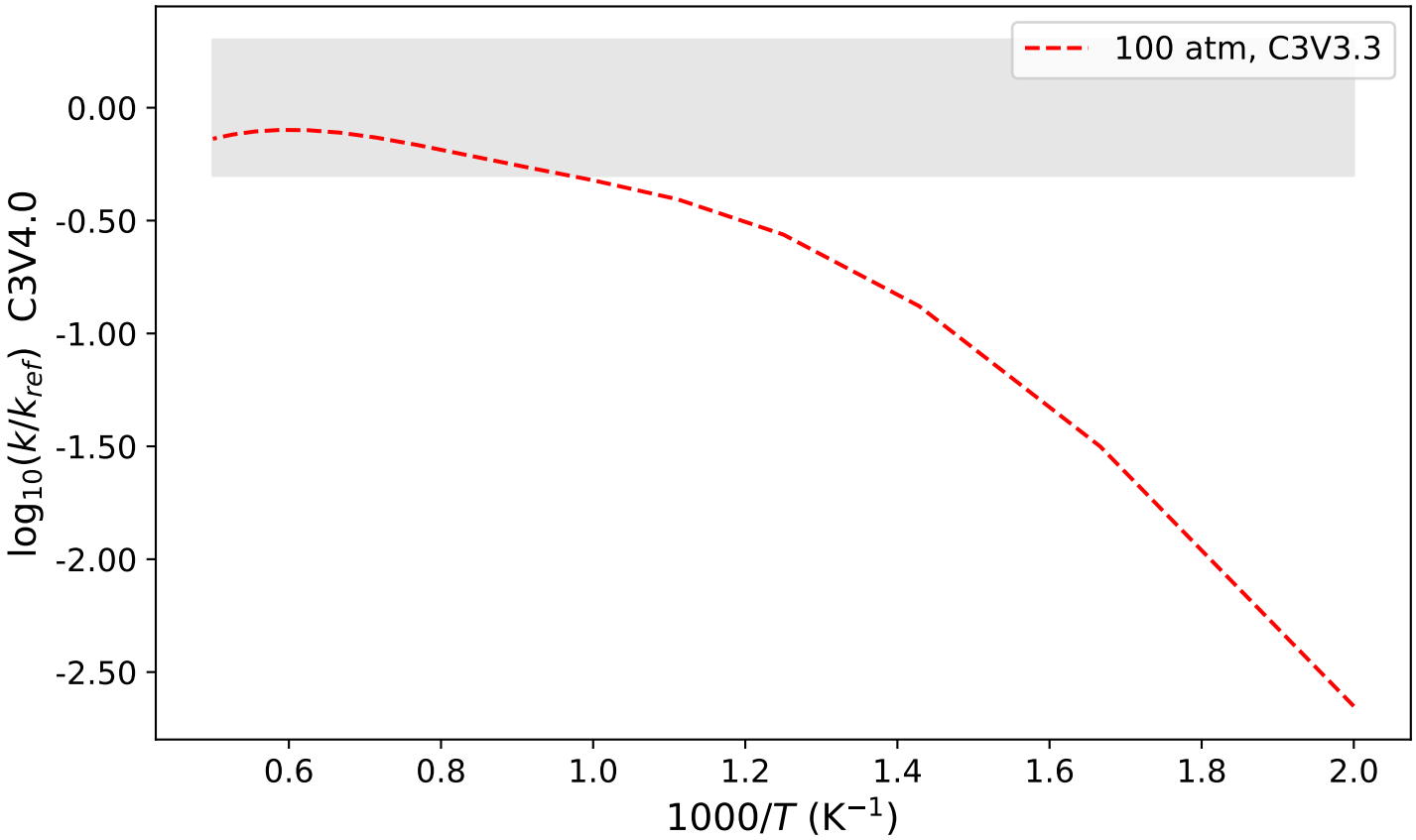
C3H5-A + NAPH = C3H6 + NAPH-A



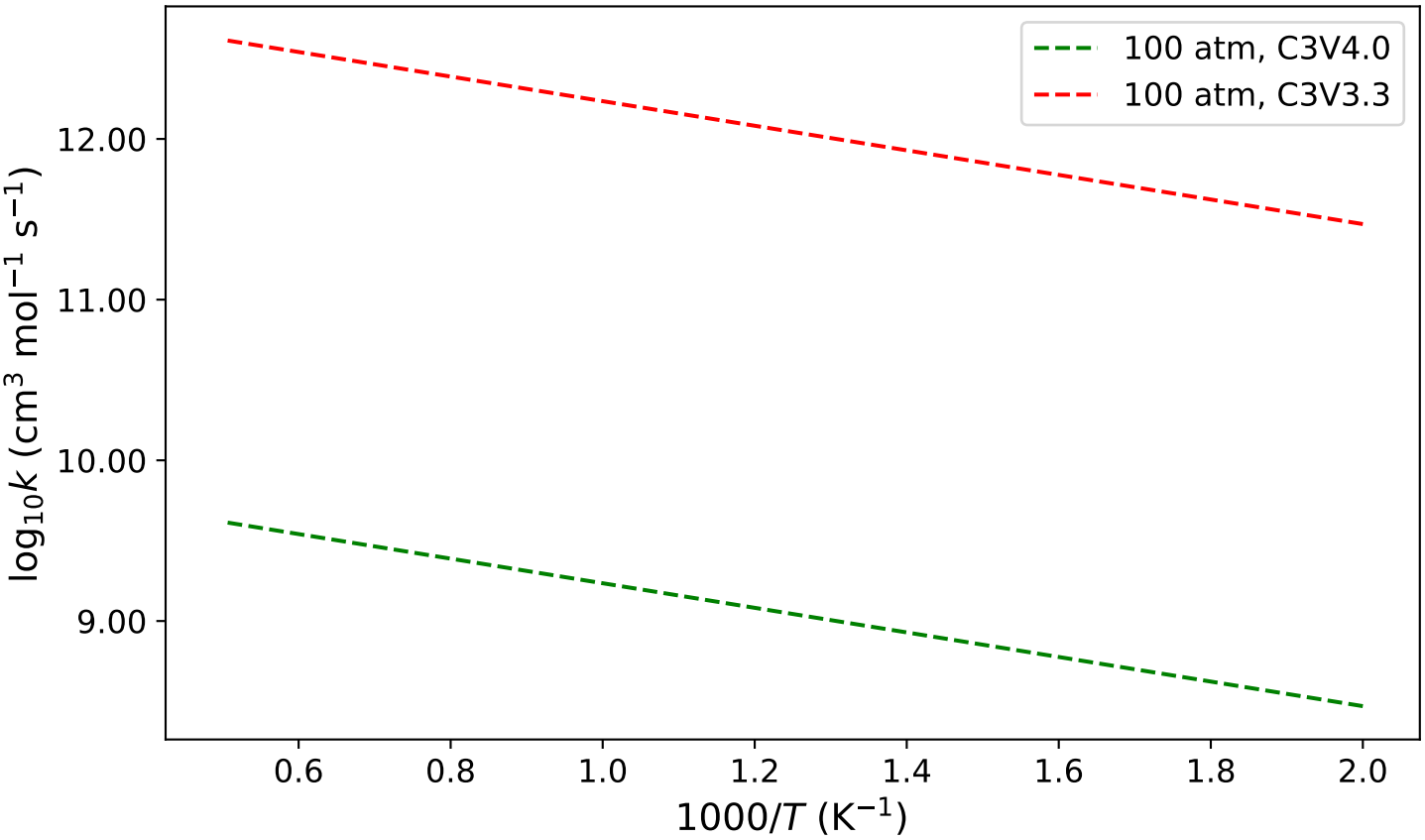


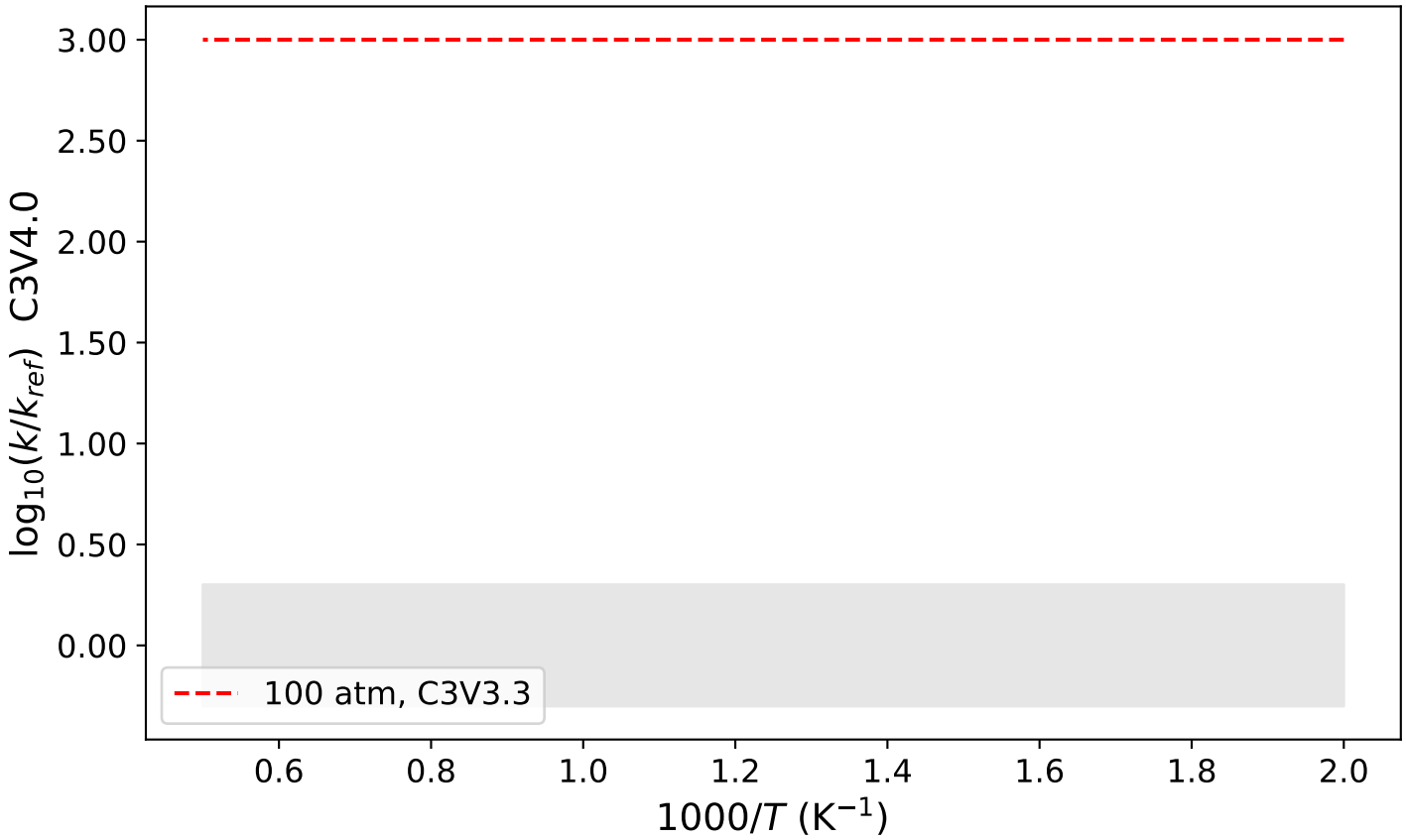
C6H5CH2 + C2H5O2 = C6H5CH2O + C2H5O



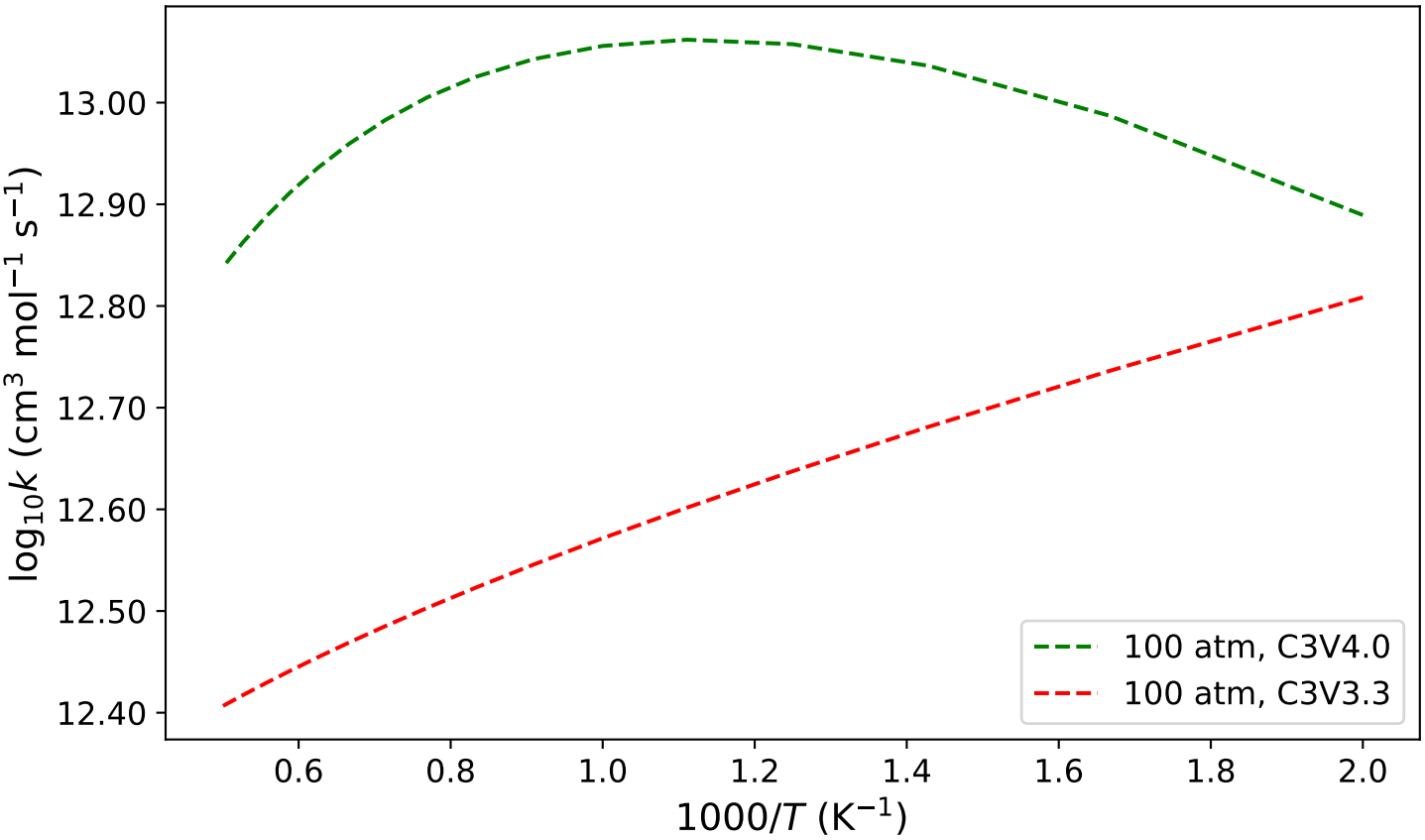


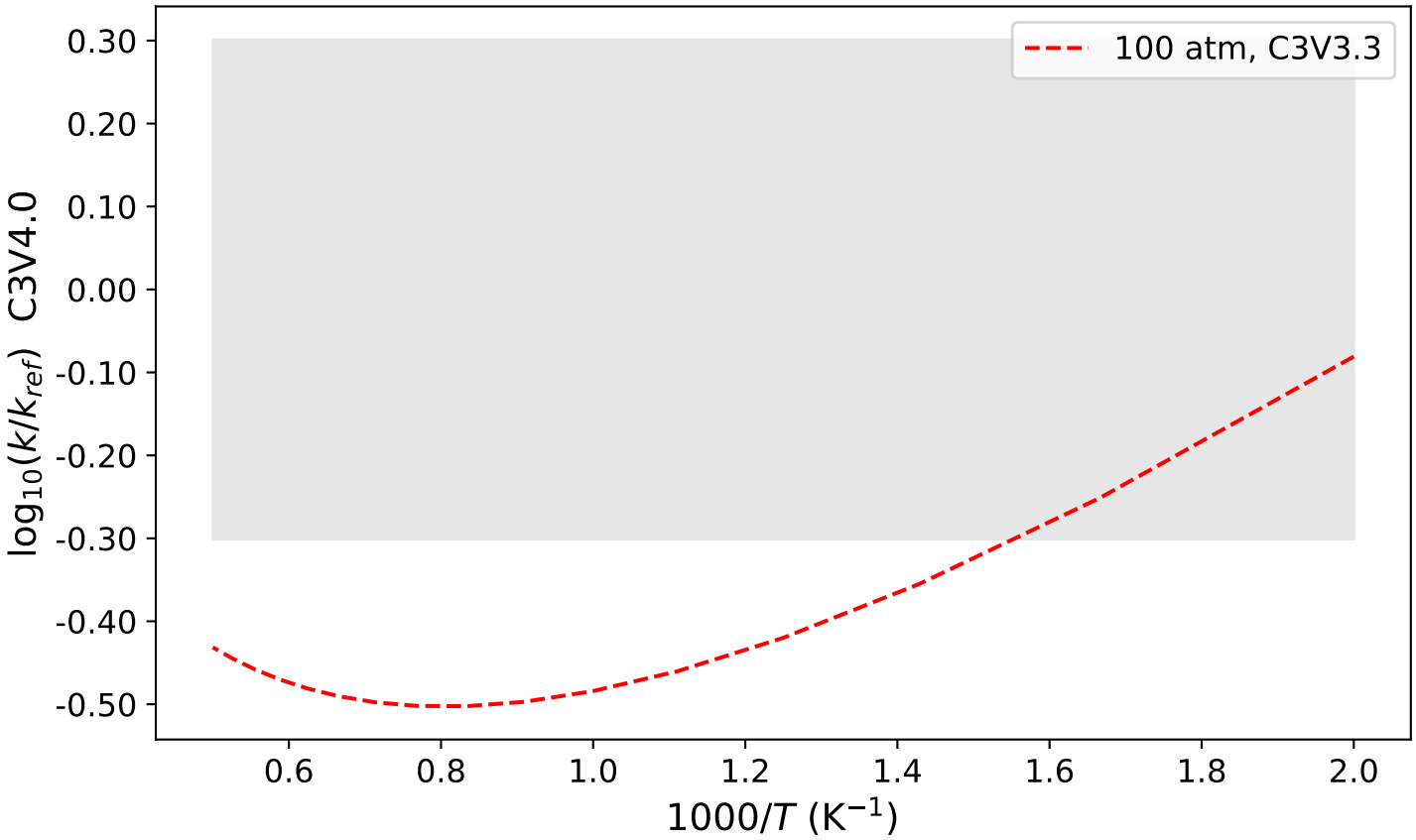
H + PHNTHRN = 0.6875C14H9 + 0.4375RTETRALIN



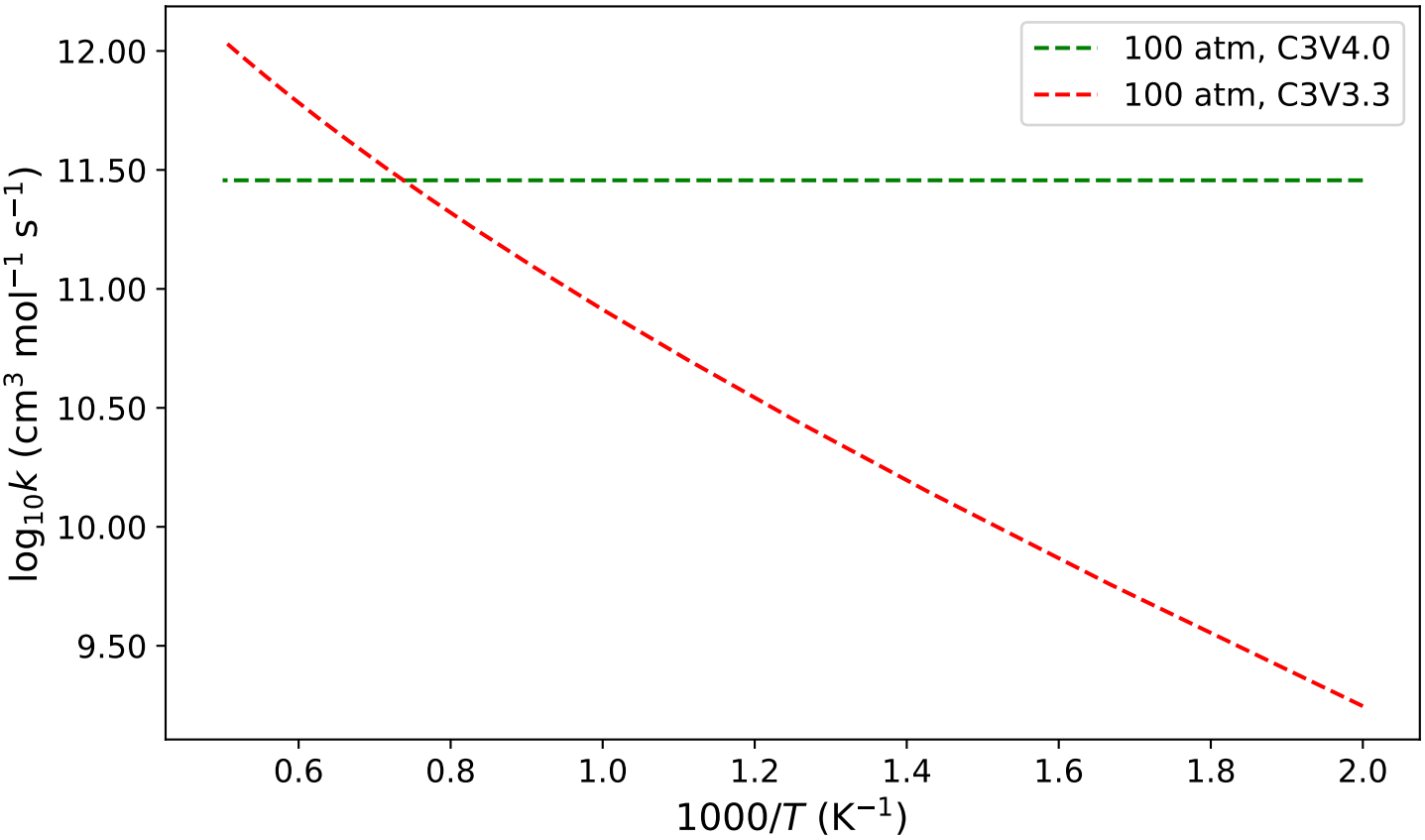


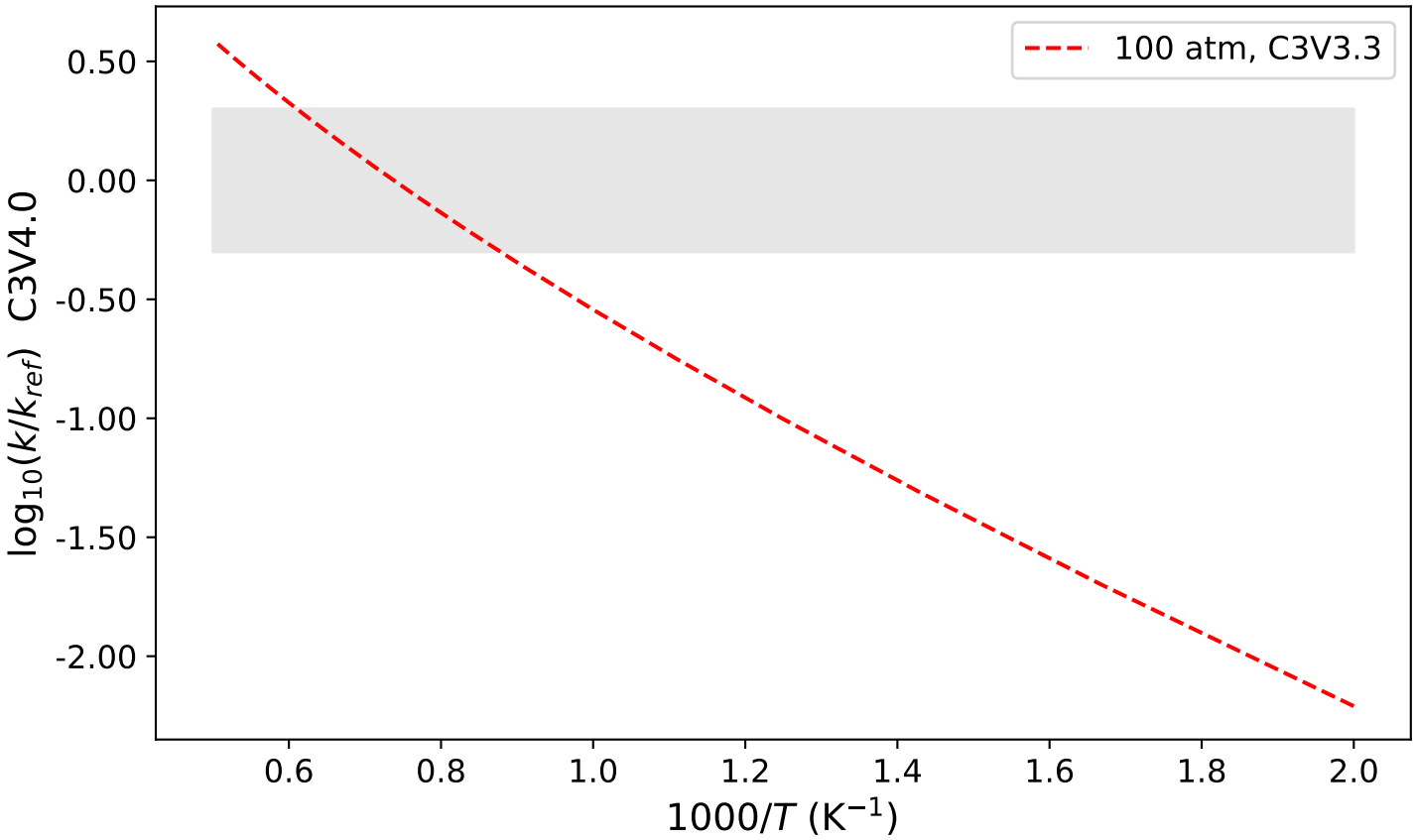
HO2 + RXYLENE => H + OH + C6H4CH3CHO



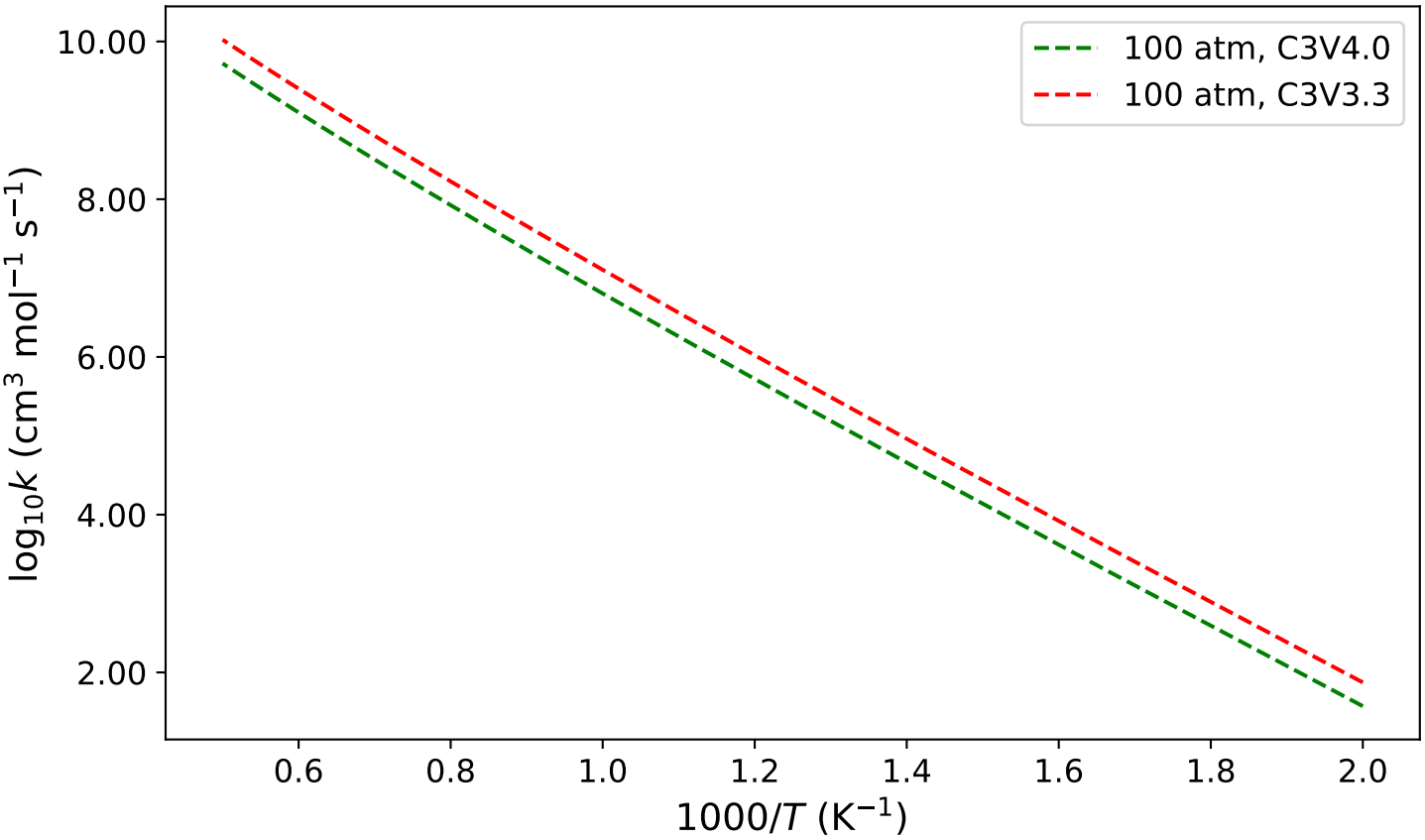


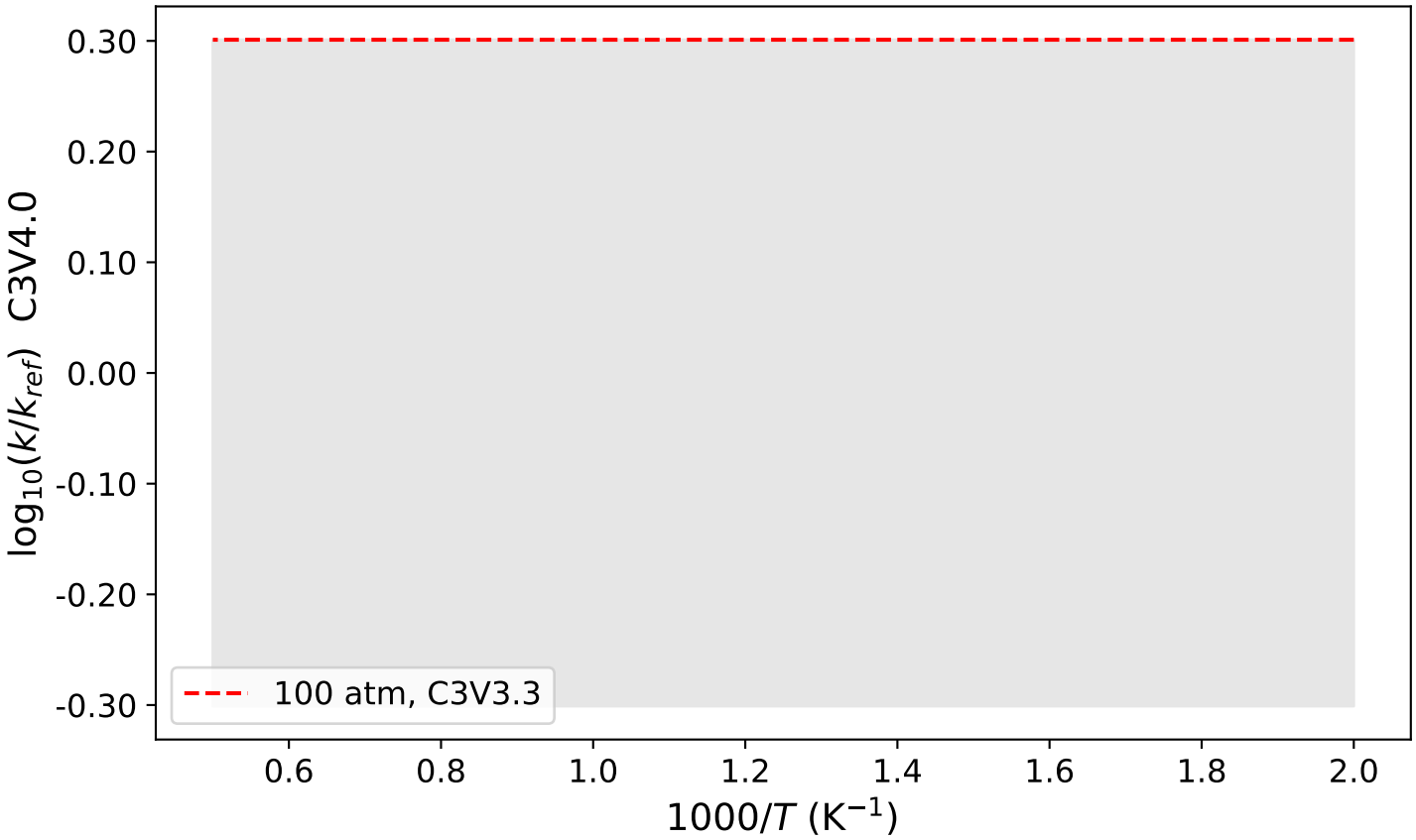
C2H3 + CYC5H8 = C2H4 + CYC5H71-3



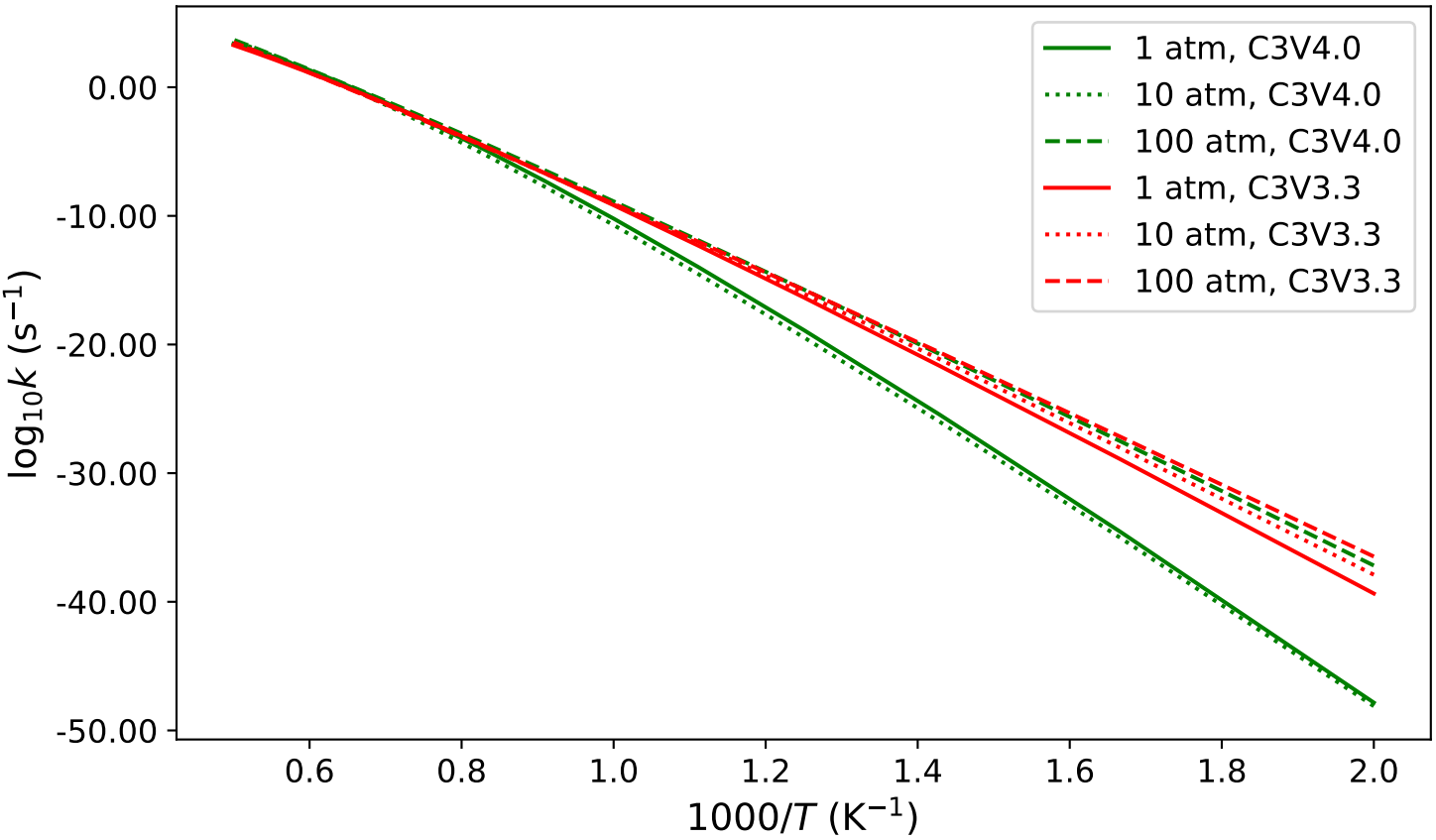


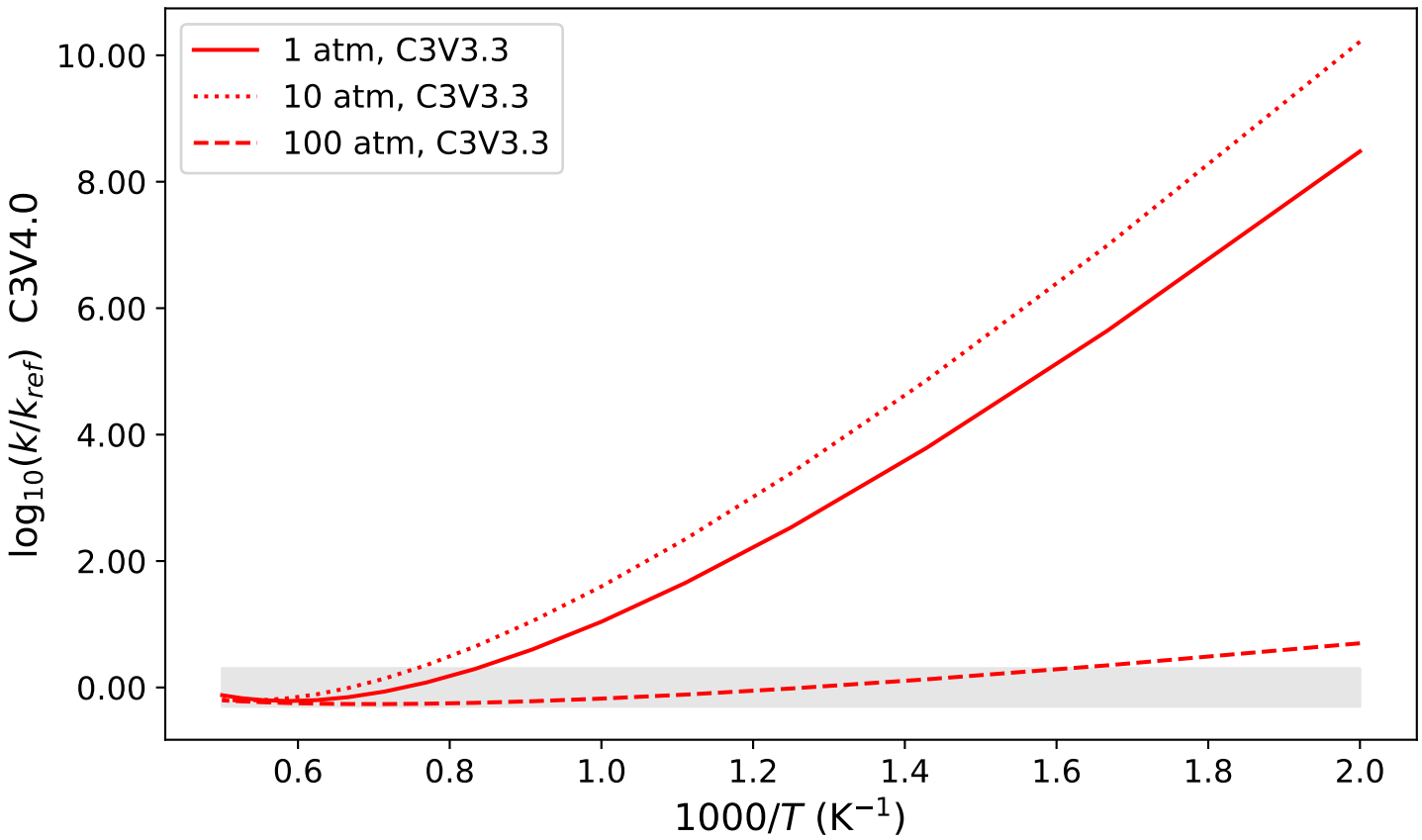
C5H5 + NAPH = C5H6 + NAPH-A



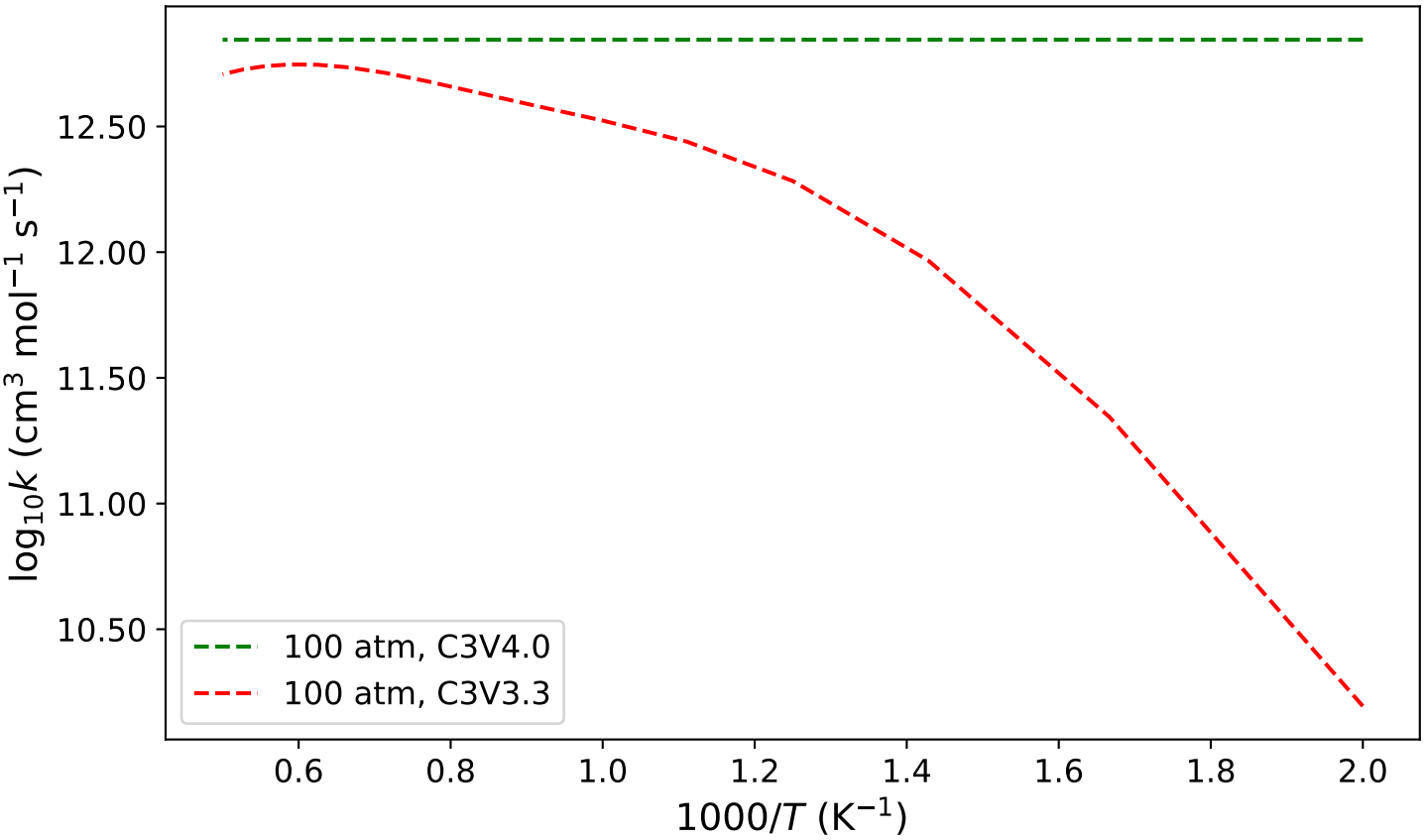


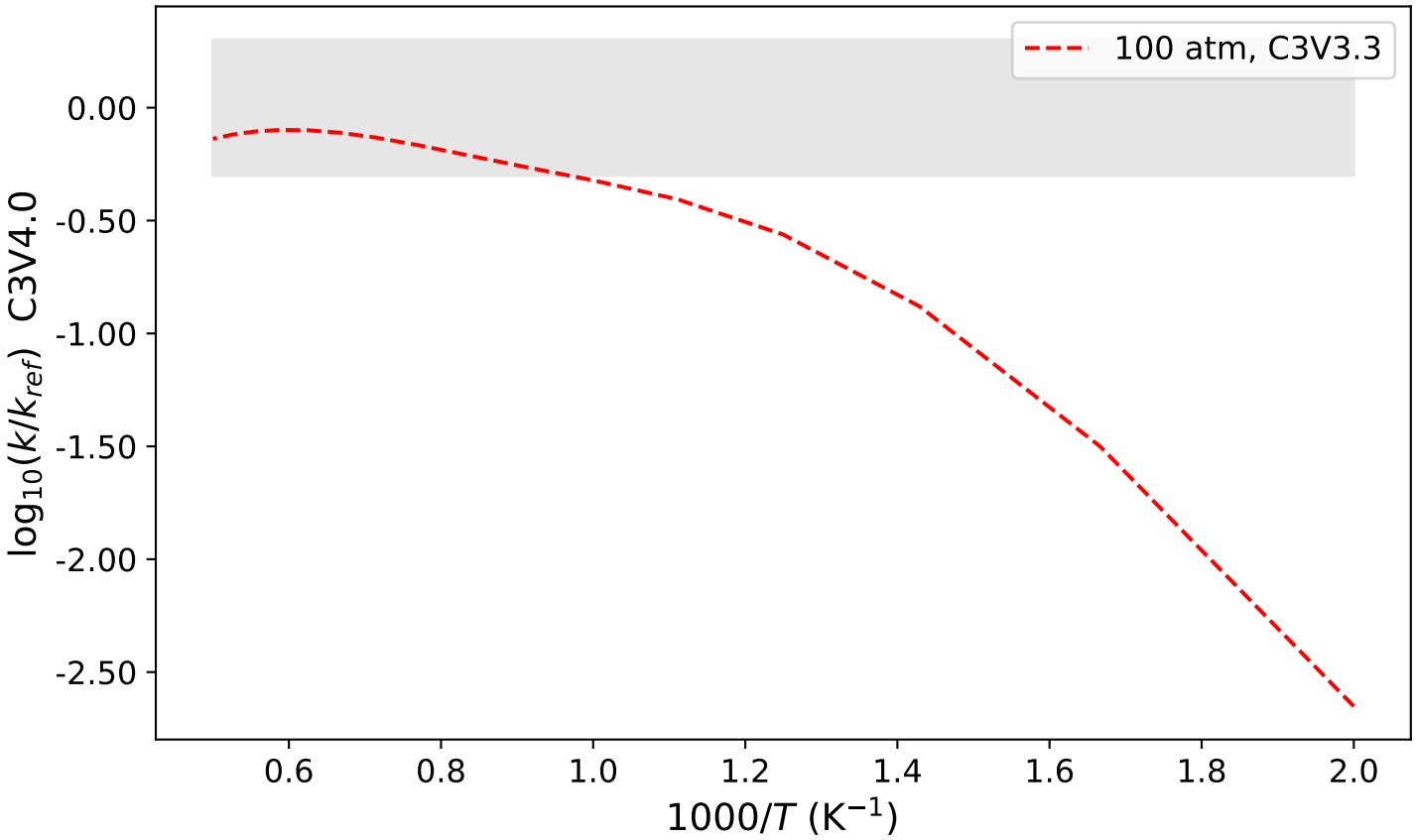
C9H7 = C2H2 + C7H5



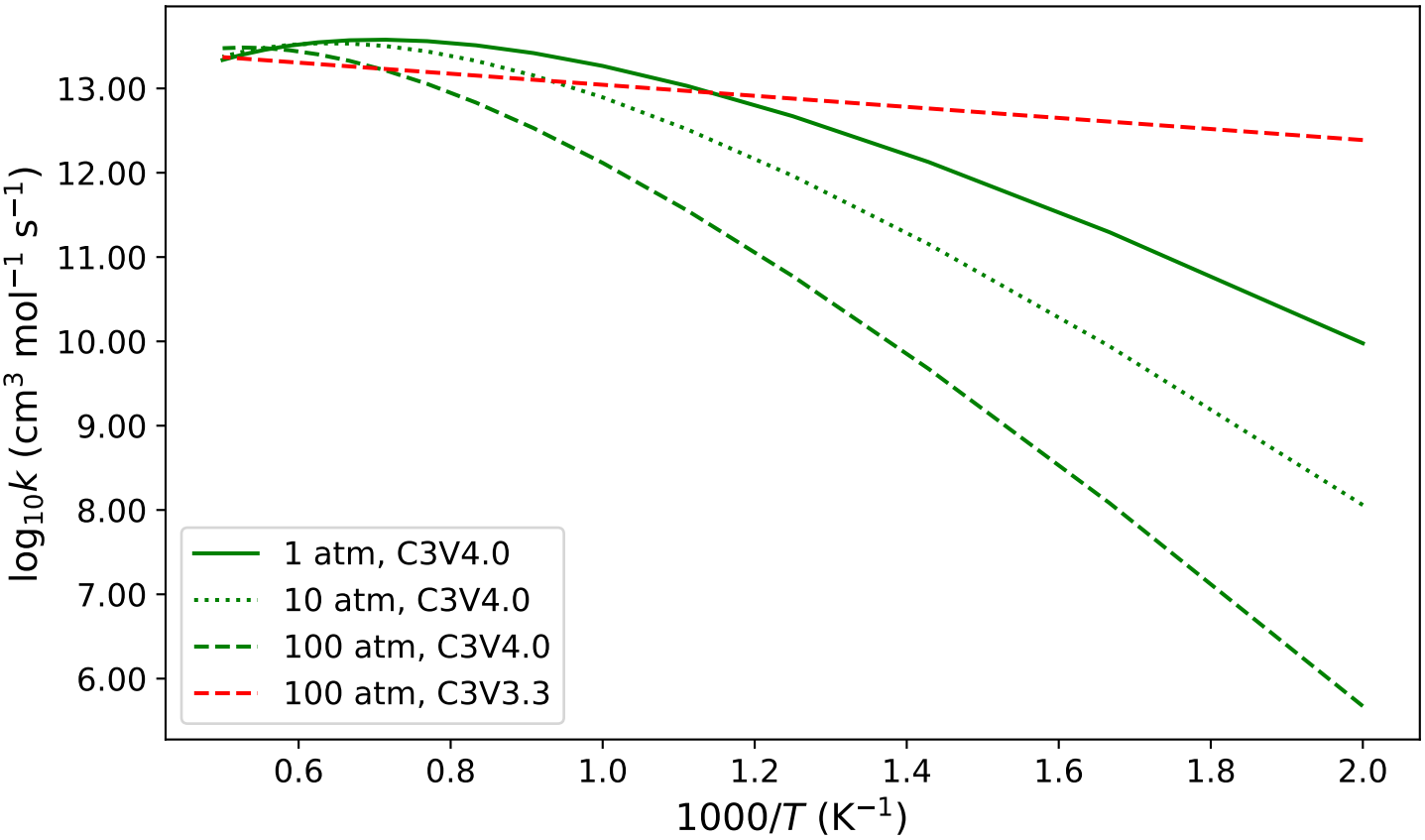


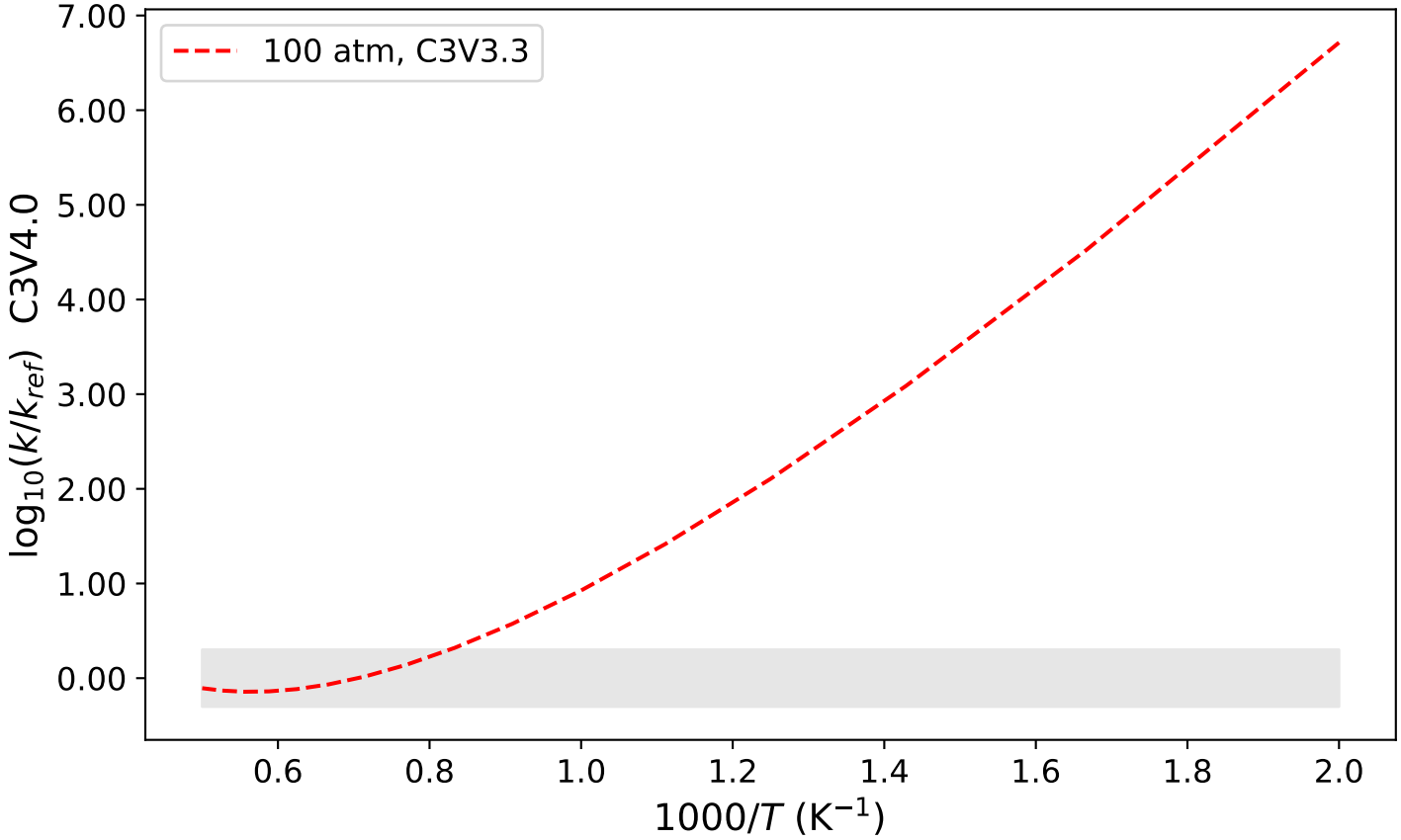
CH3O2 + C6H5CH2 = CH3O + C6H5CH2O



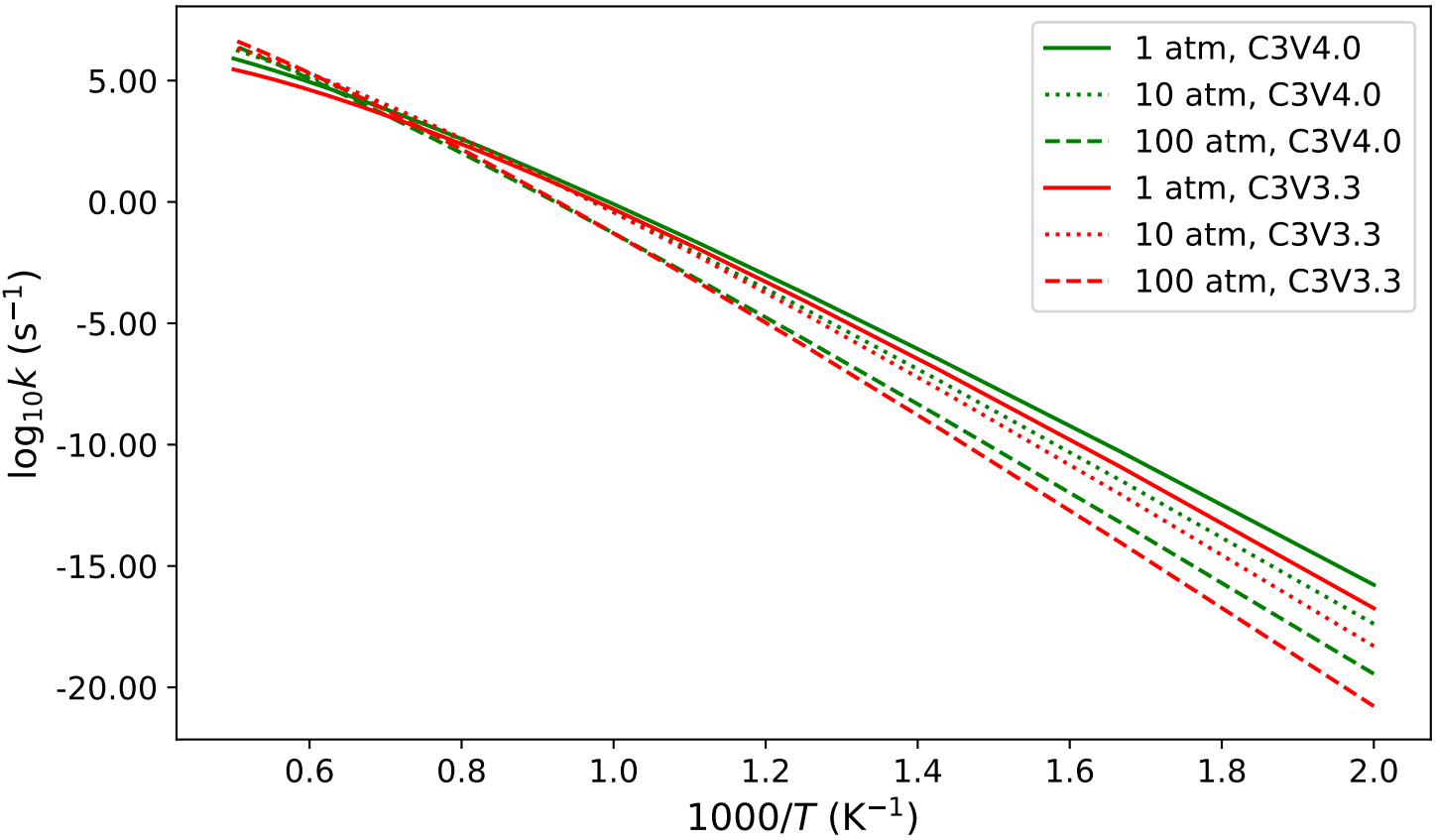


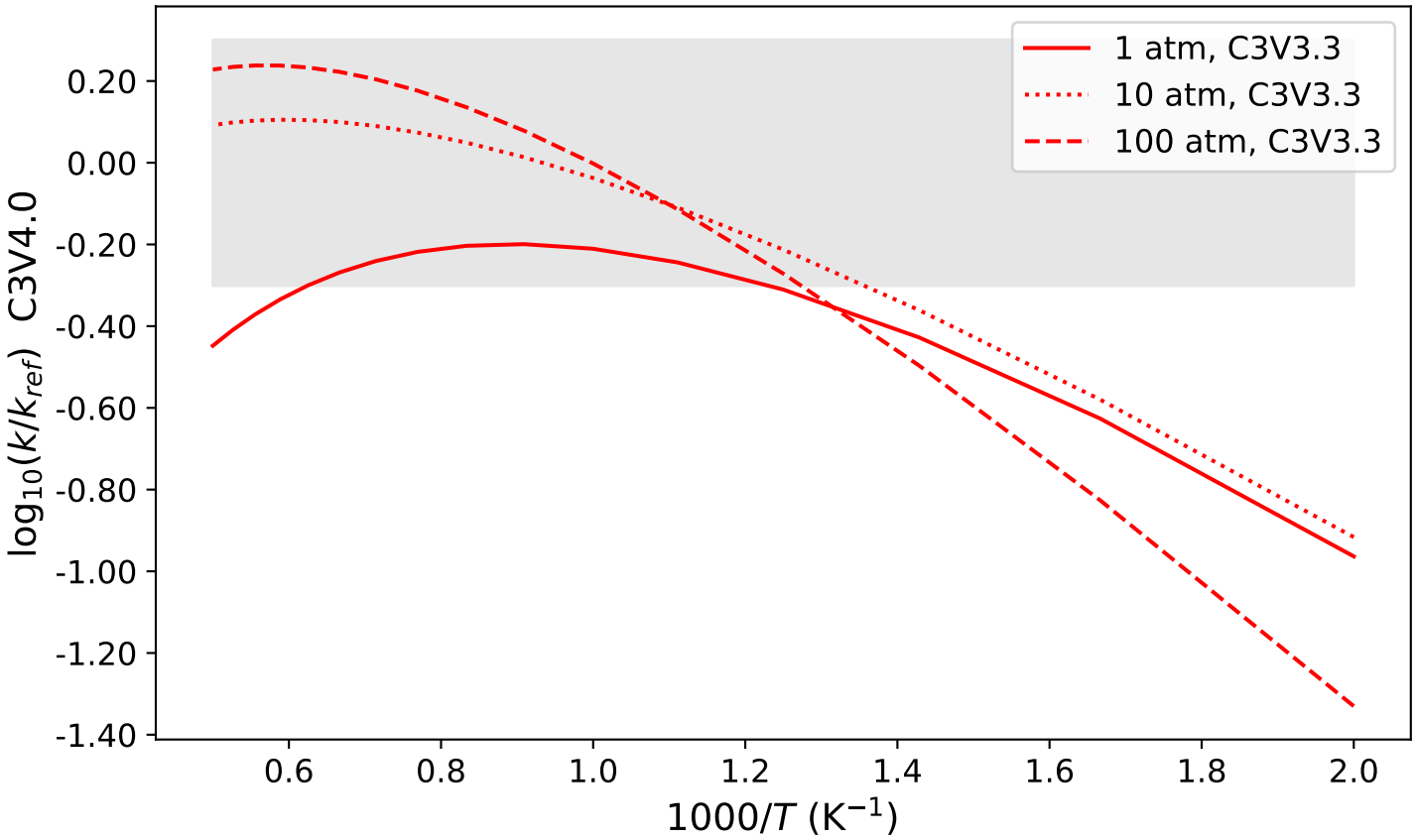
H + CYC5H8 = C3H5-A + C2H4



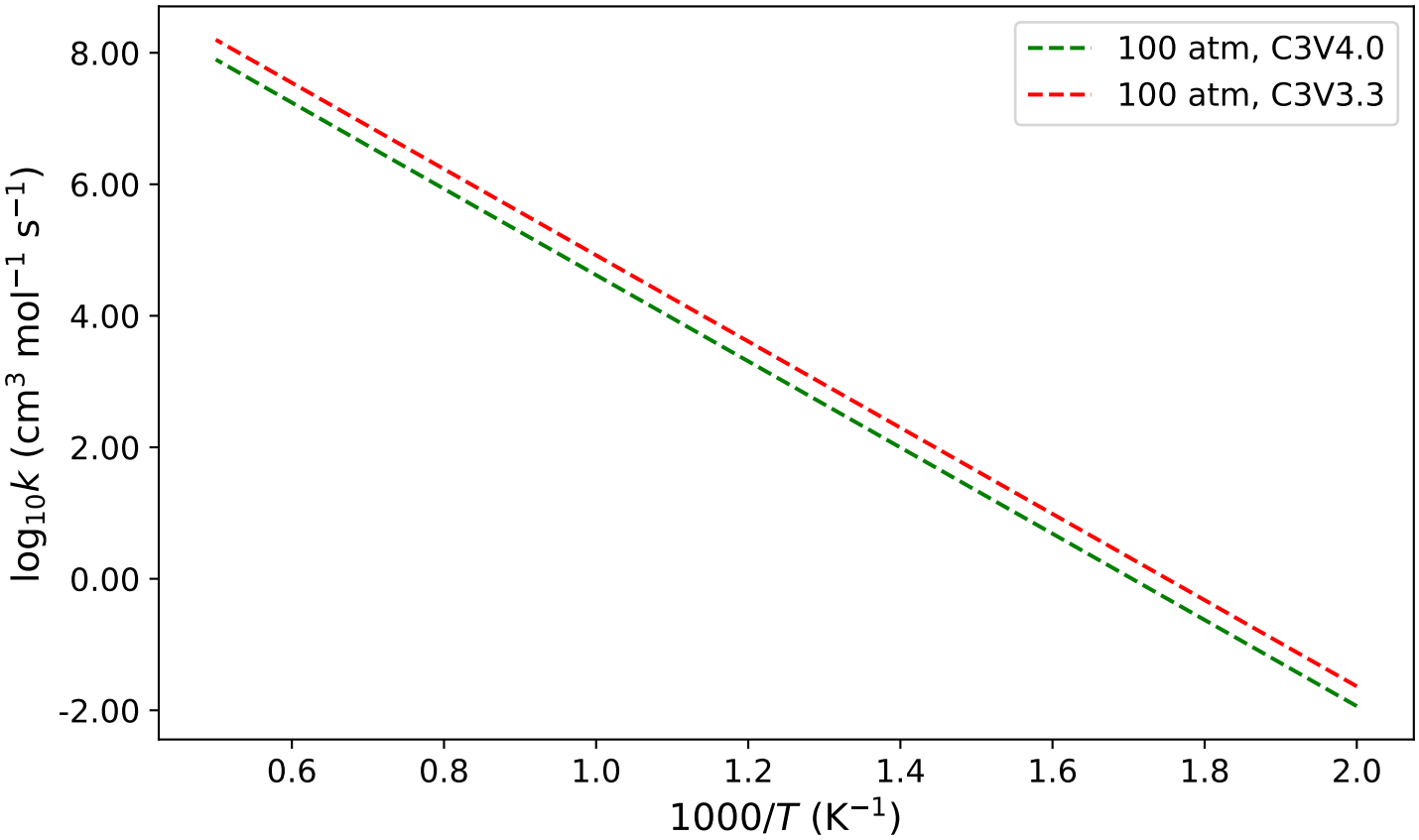


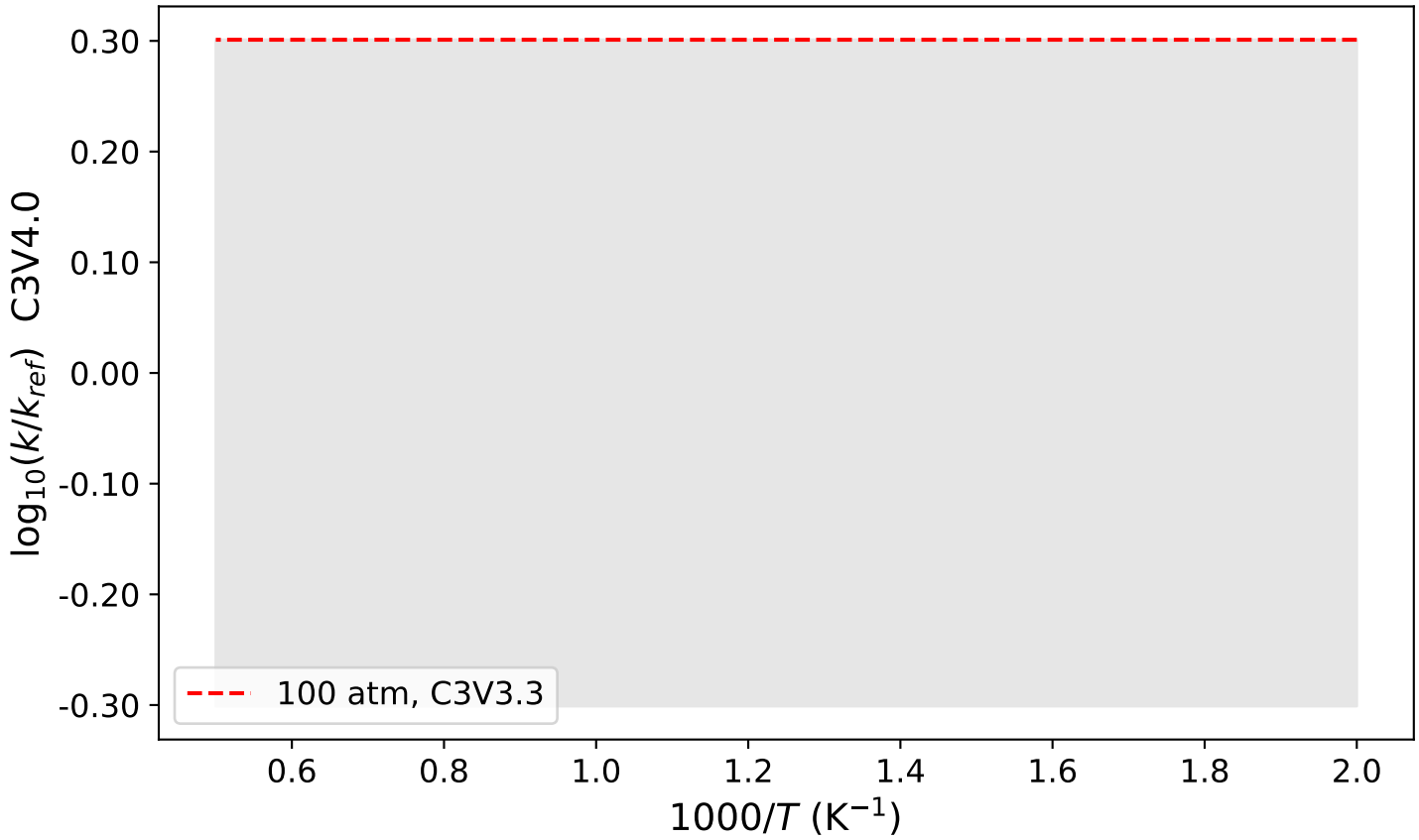
CYC5H71-3 = C2H2 + C3H5-A



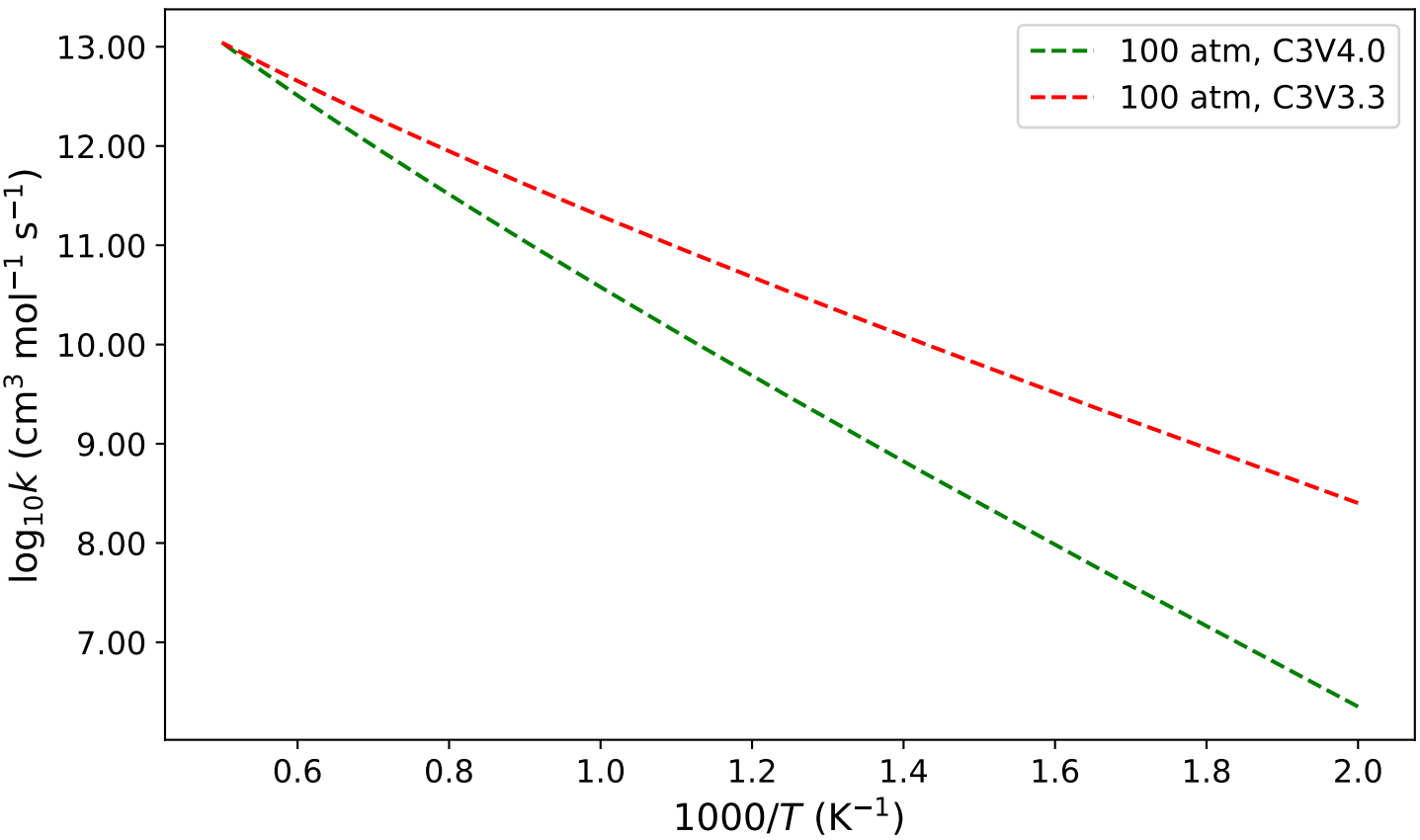


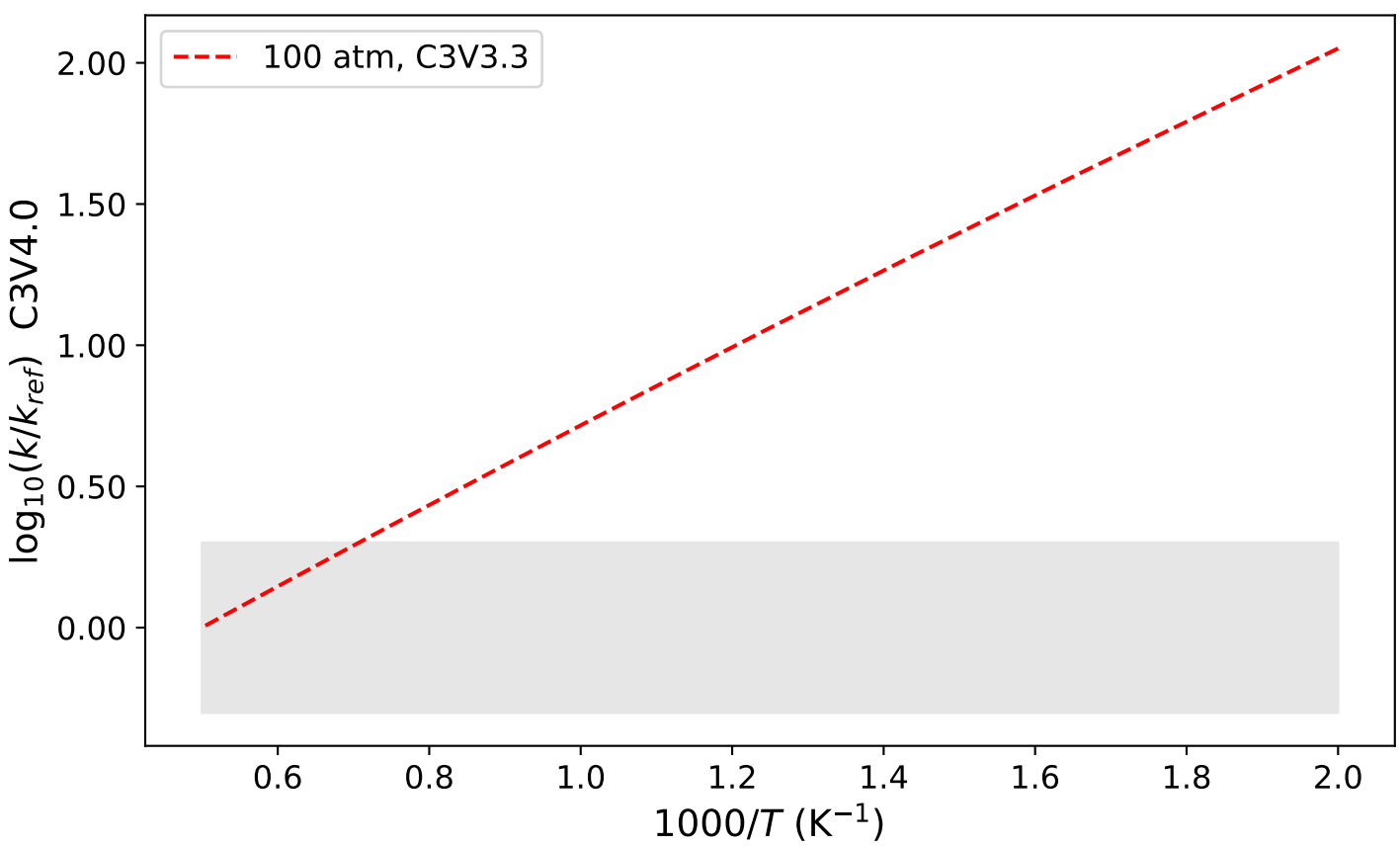
C2H2 + IND = C10H7CH3-A



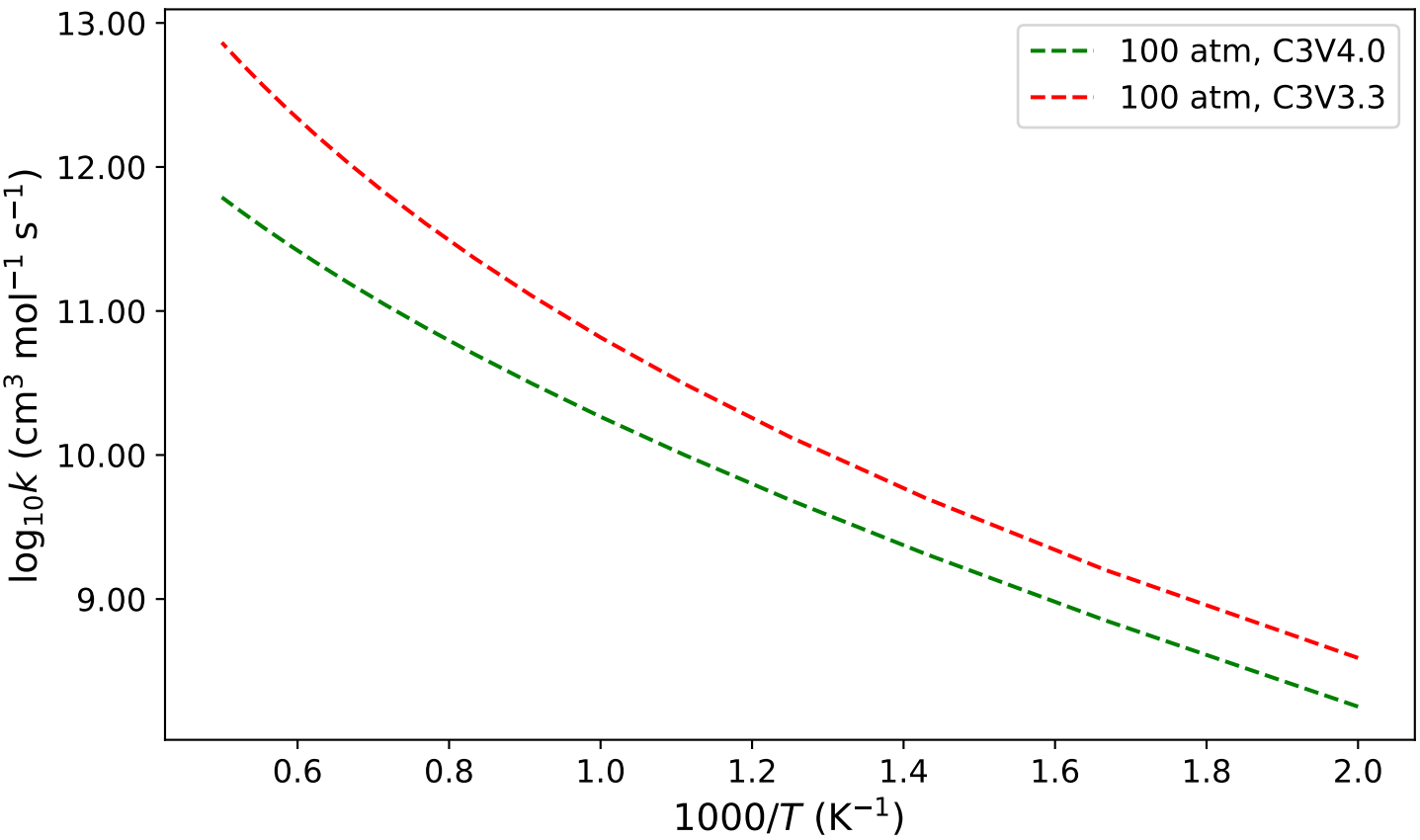


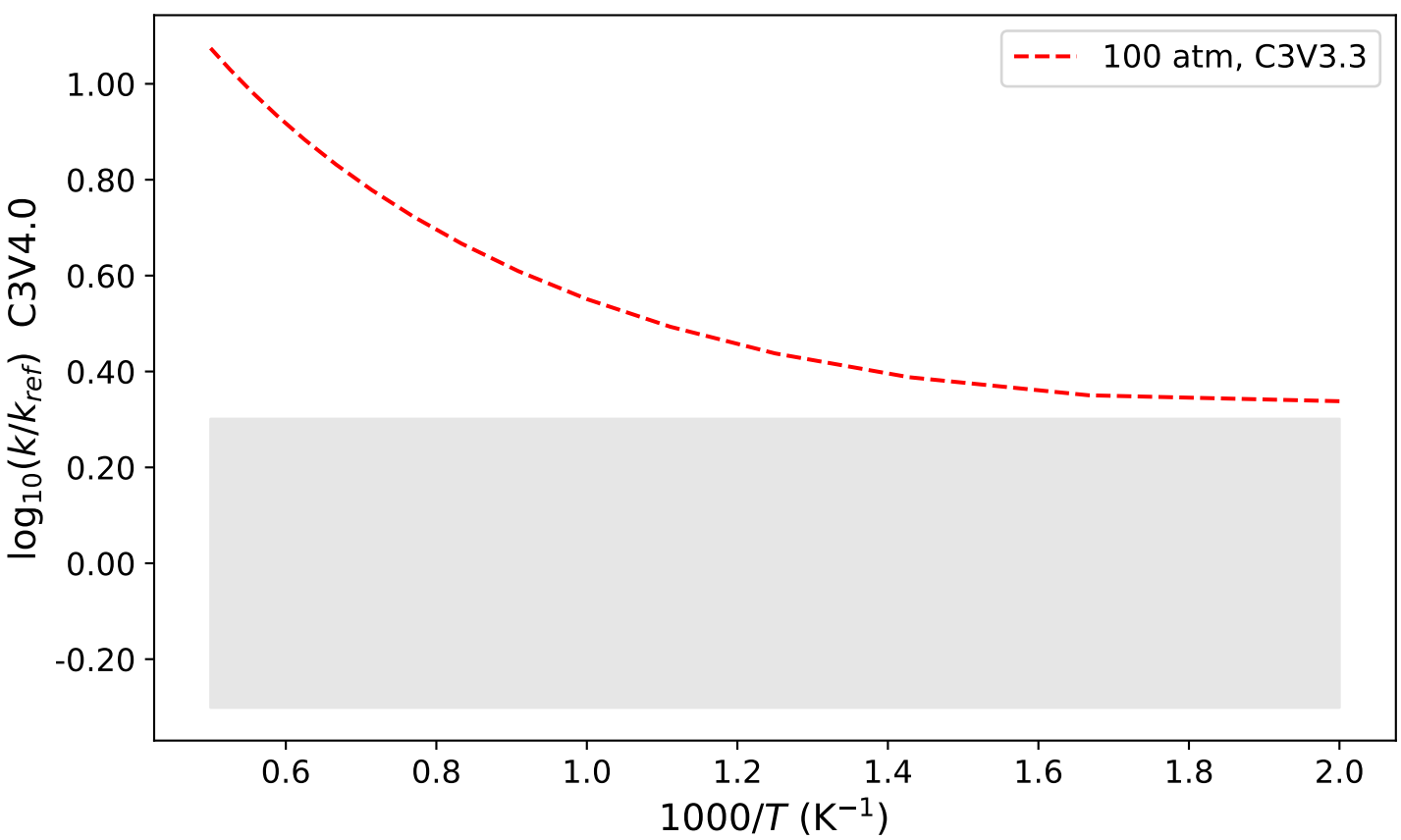
H + C12H8 = H2 + C12H7



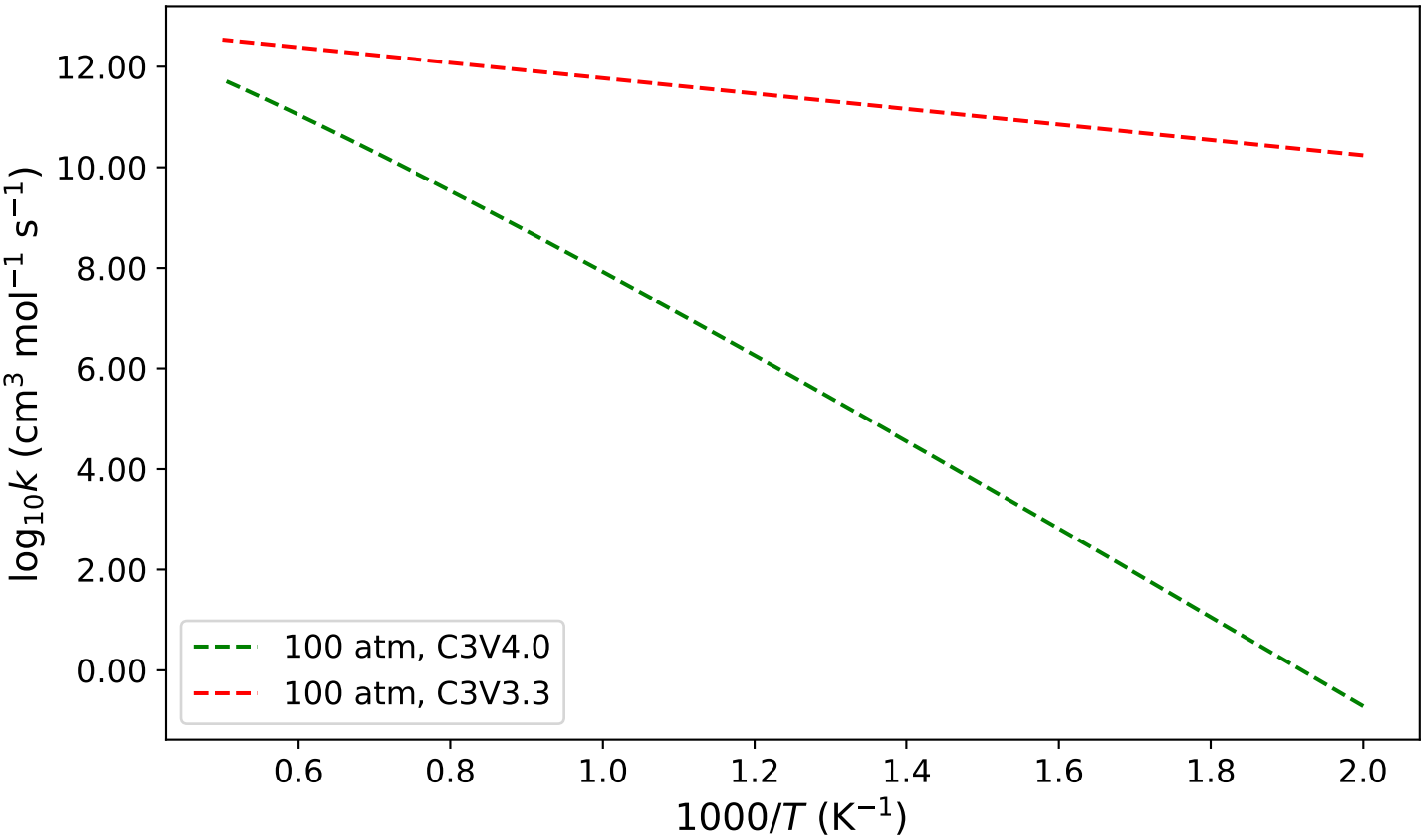


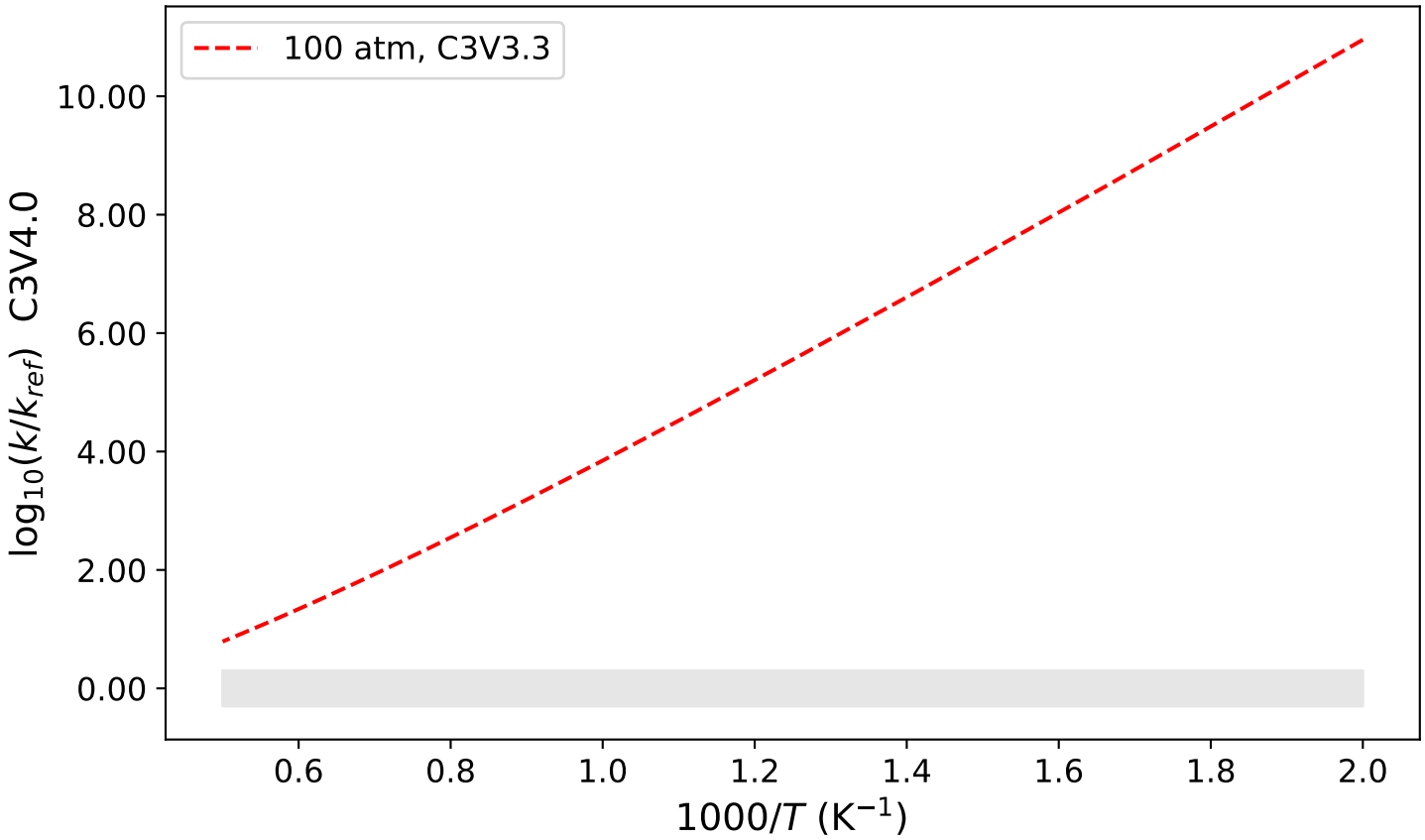
CH3 + C6H4CH3CHO => CH4 + CO + C6H4CH3



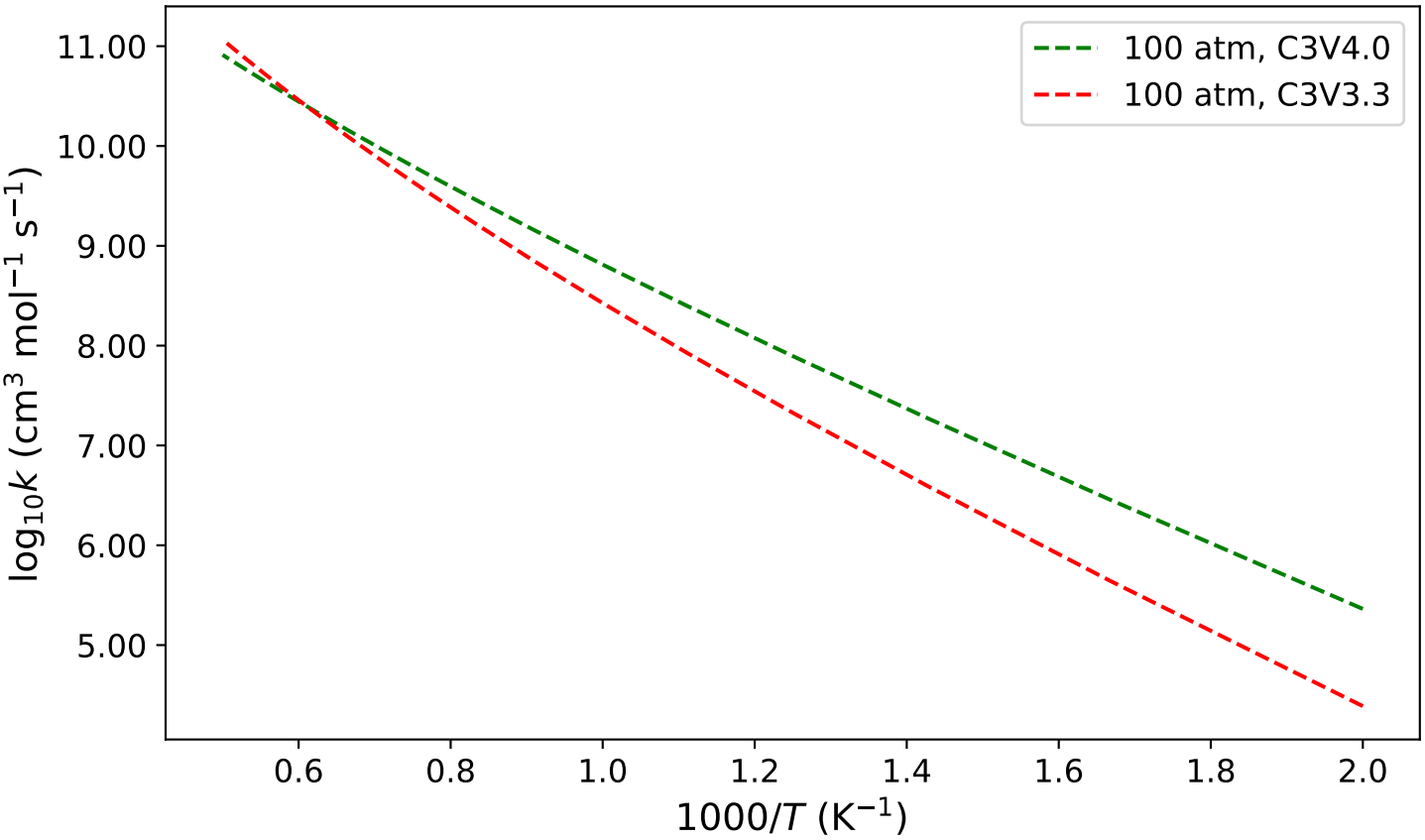


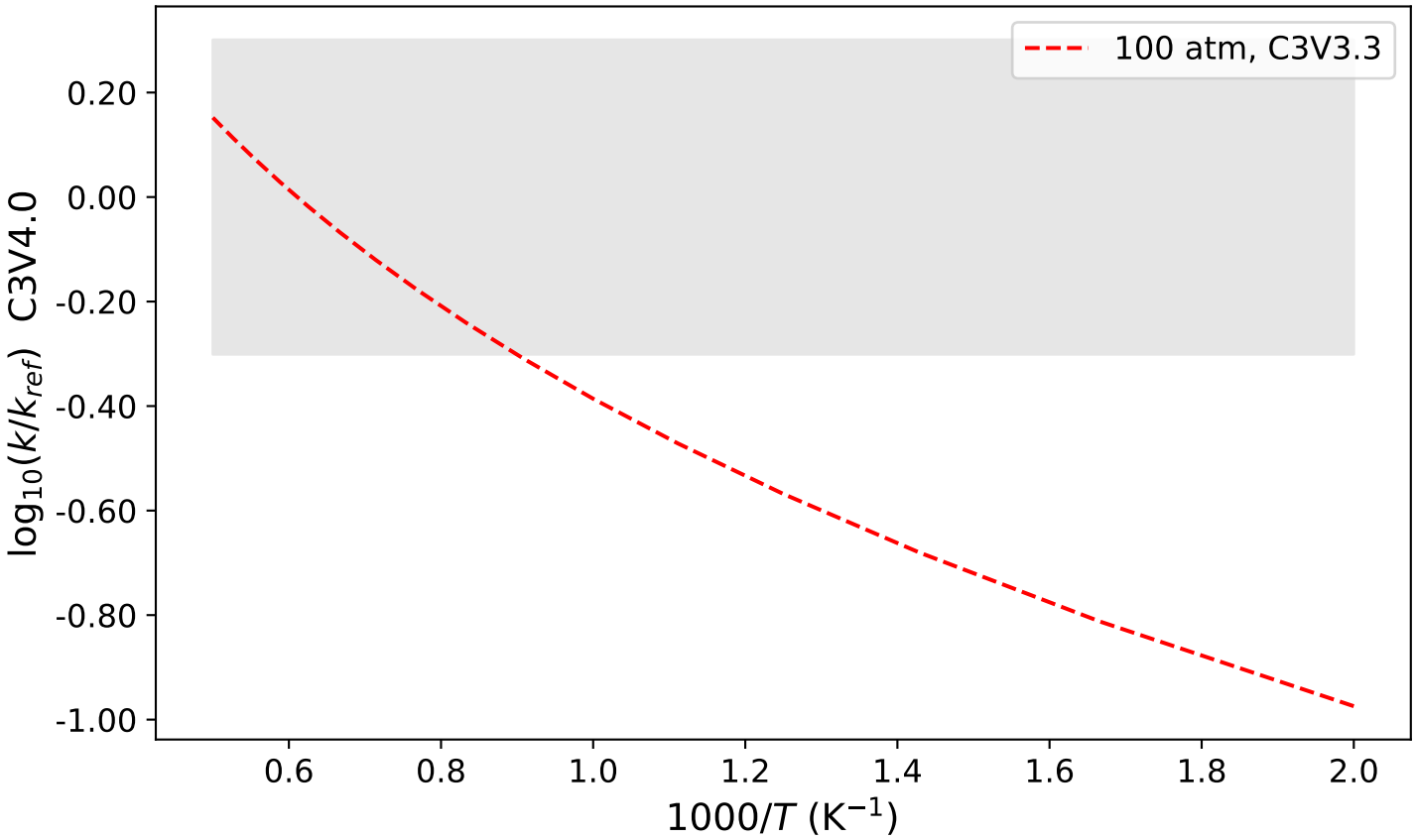
C3H3 + C5H5 = C6H5C2H3



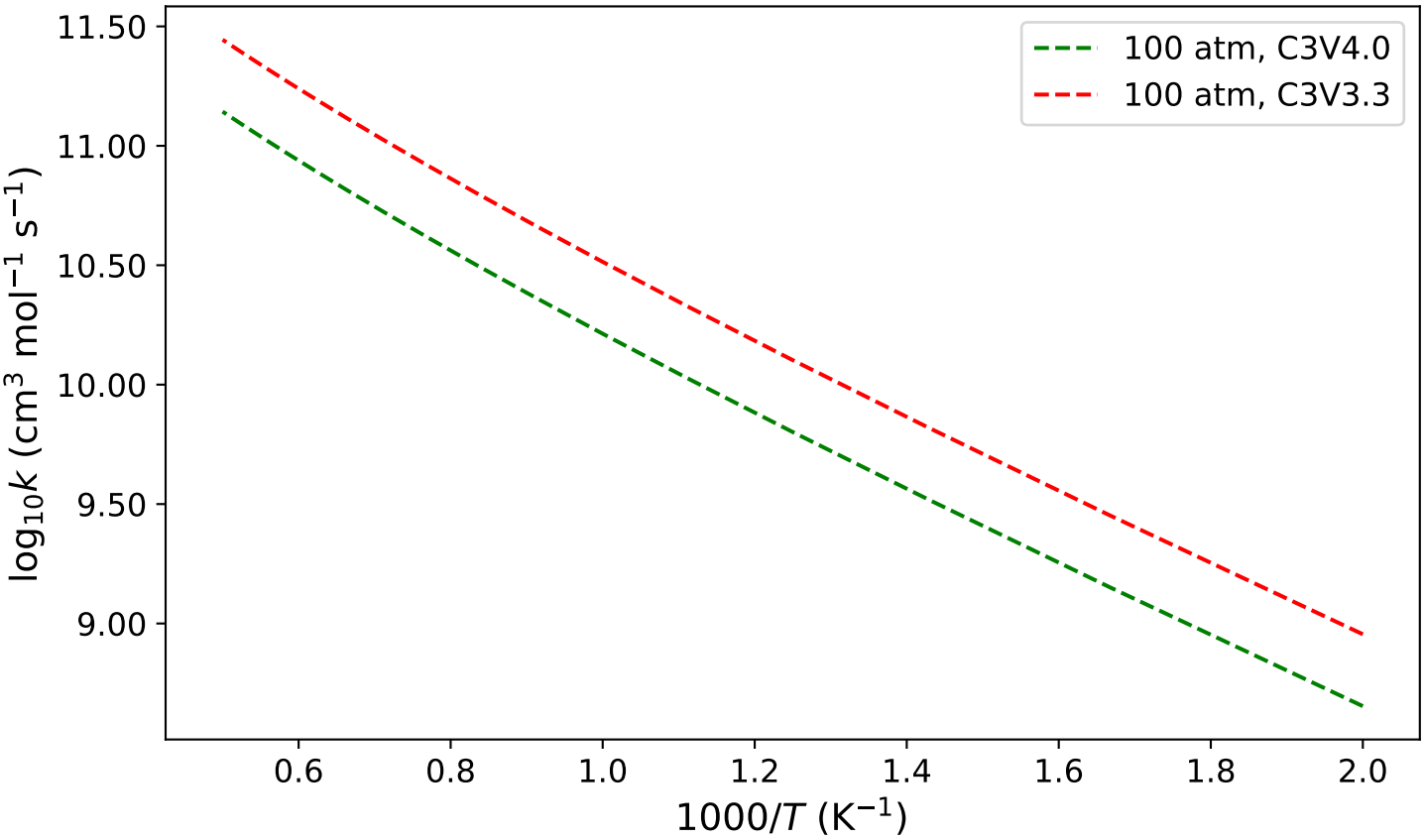


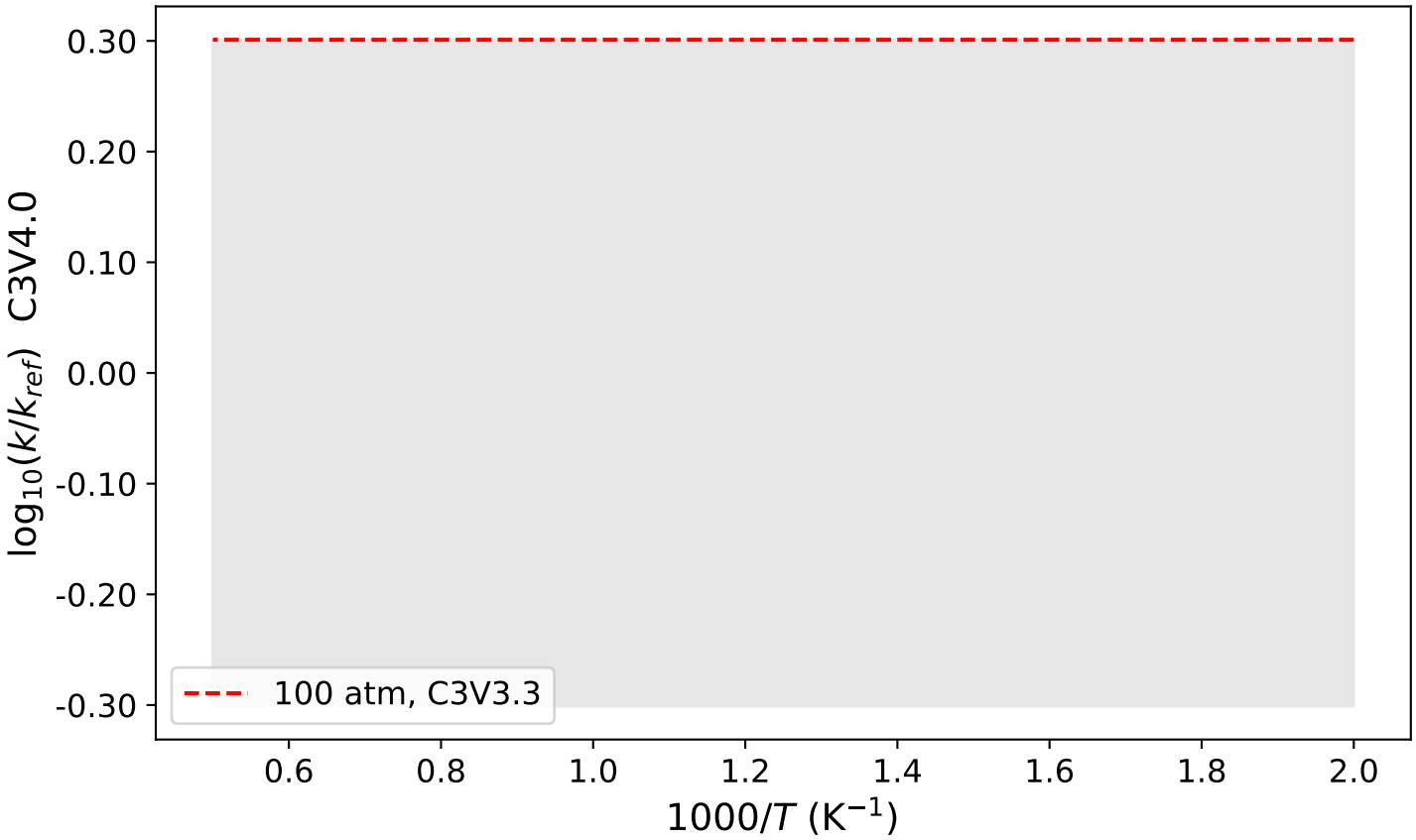
CH3O2 + C6H5CH3 = CH3O2H + C6H5CH2



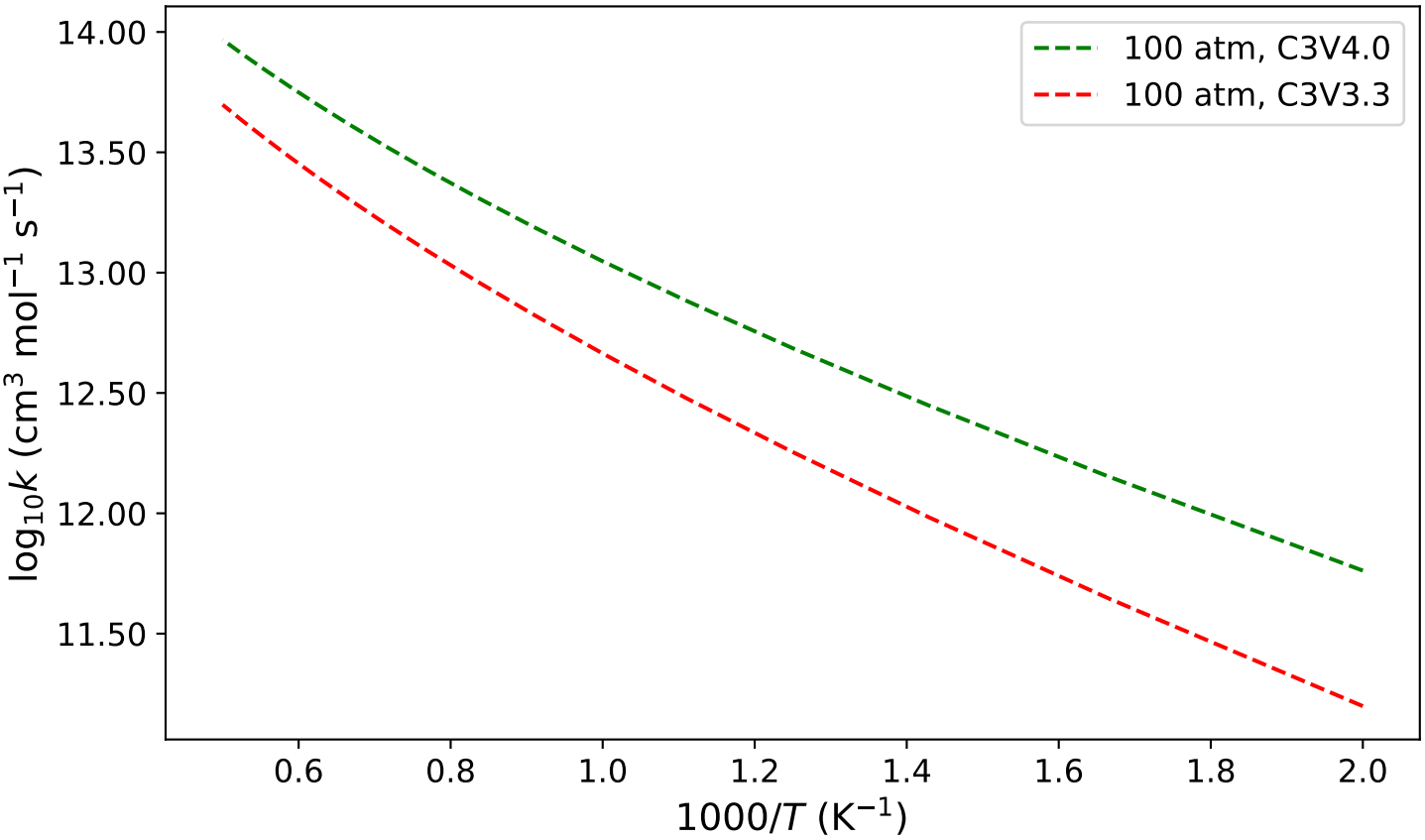


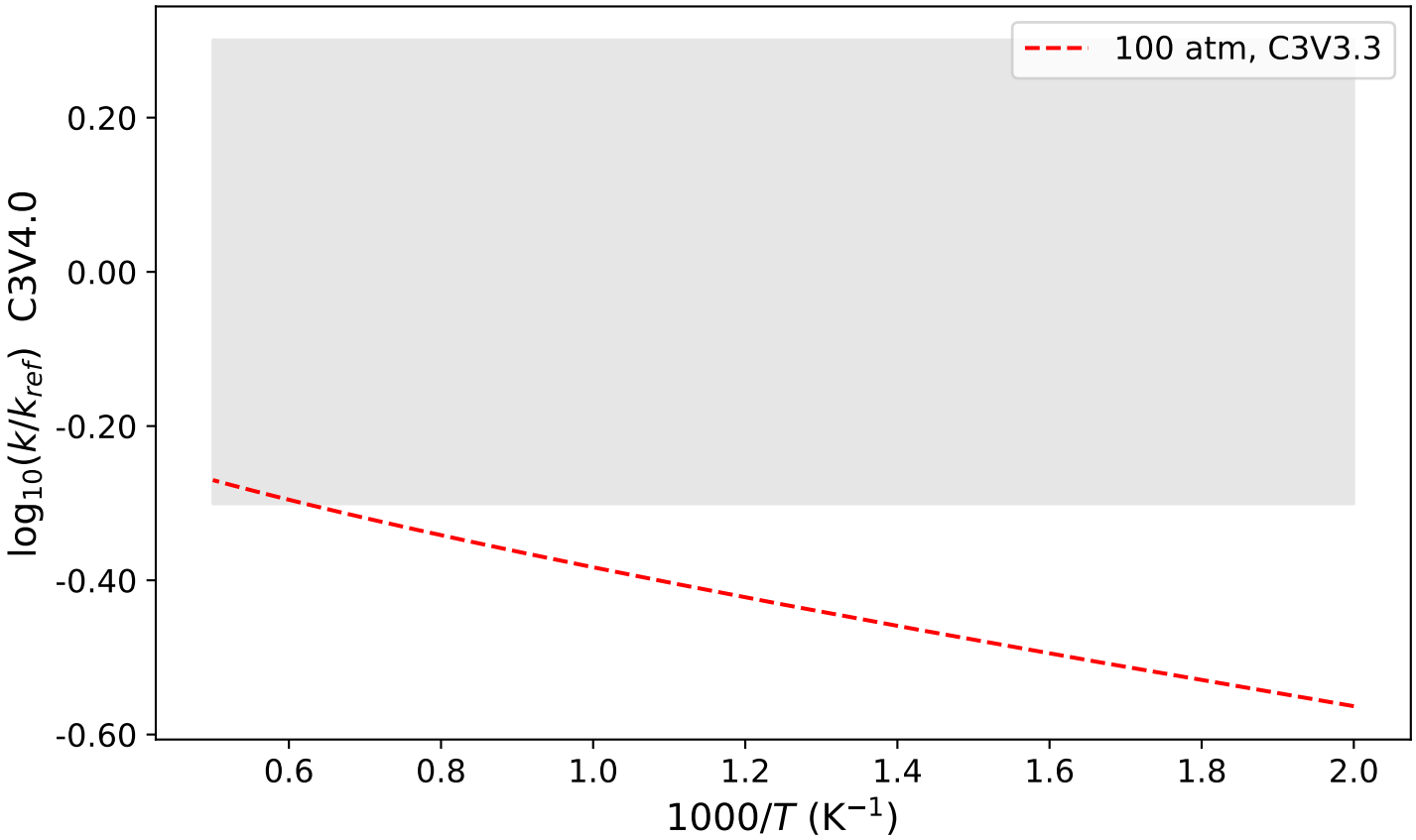
NAPH + C6H4CH3 = NAPH-A + C6H5CH3



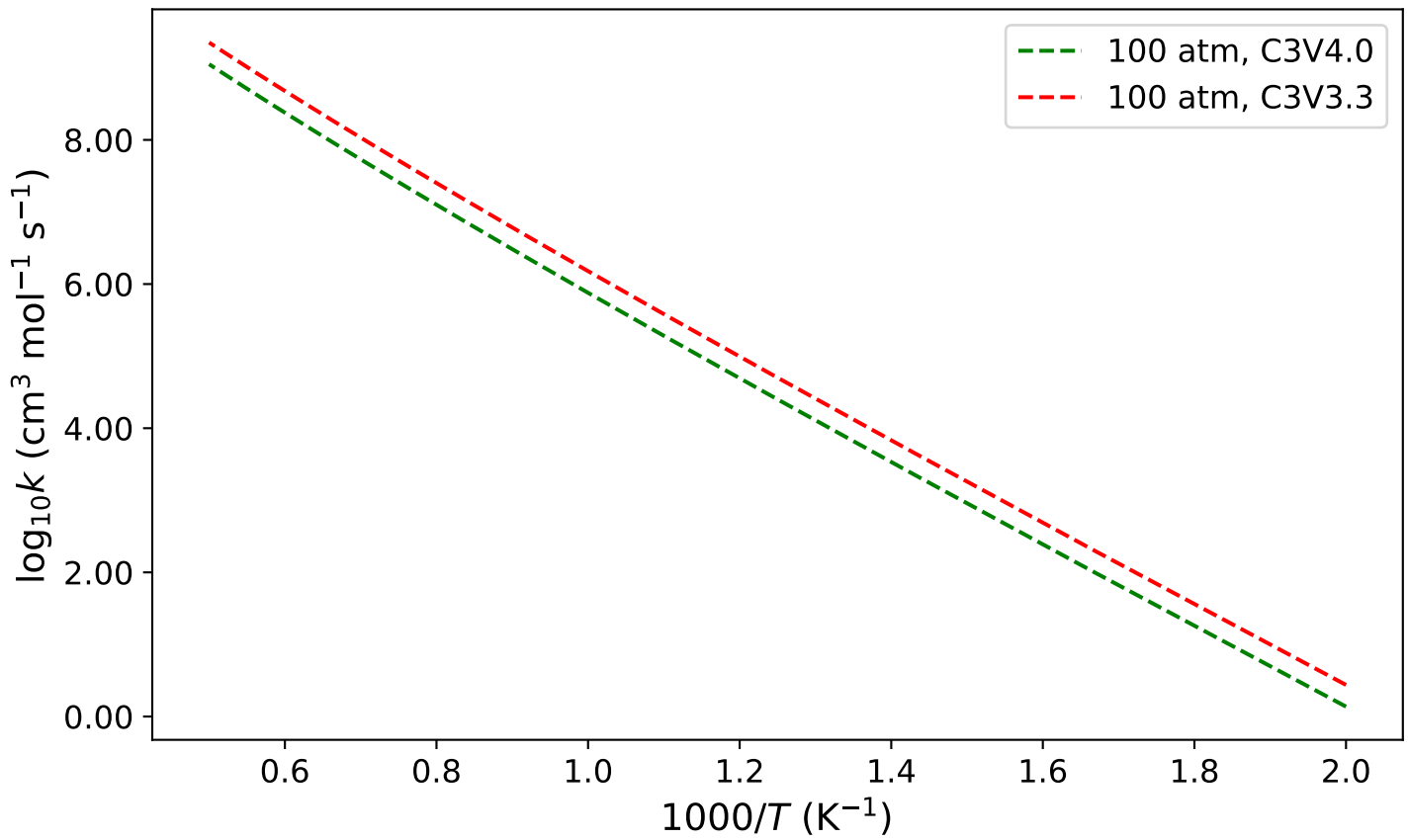


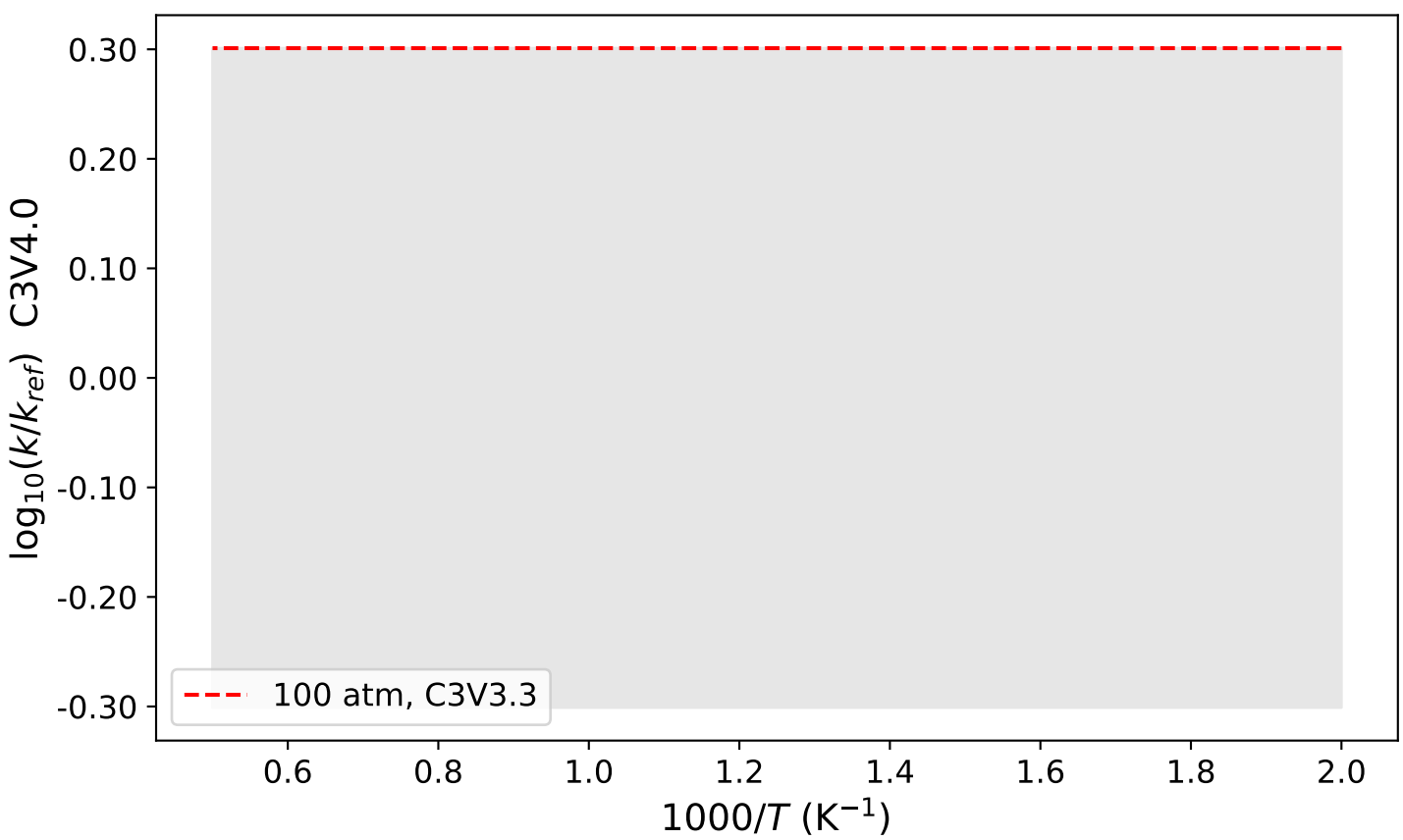
H + CYC5H8 = H2 + CYC5H71-3



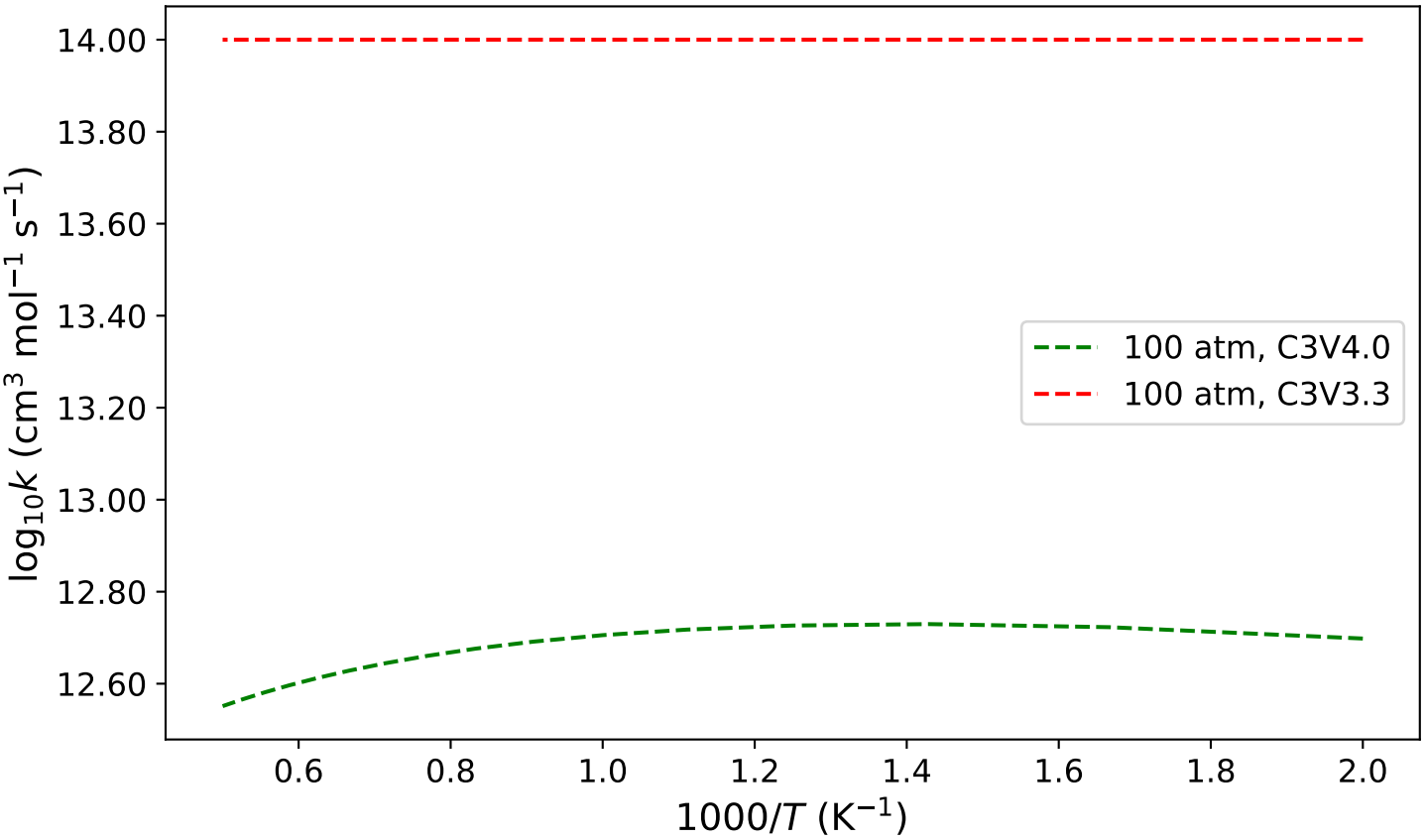


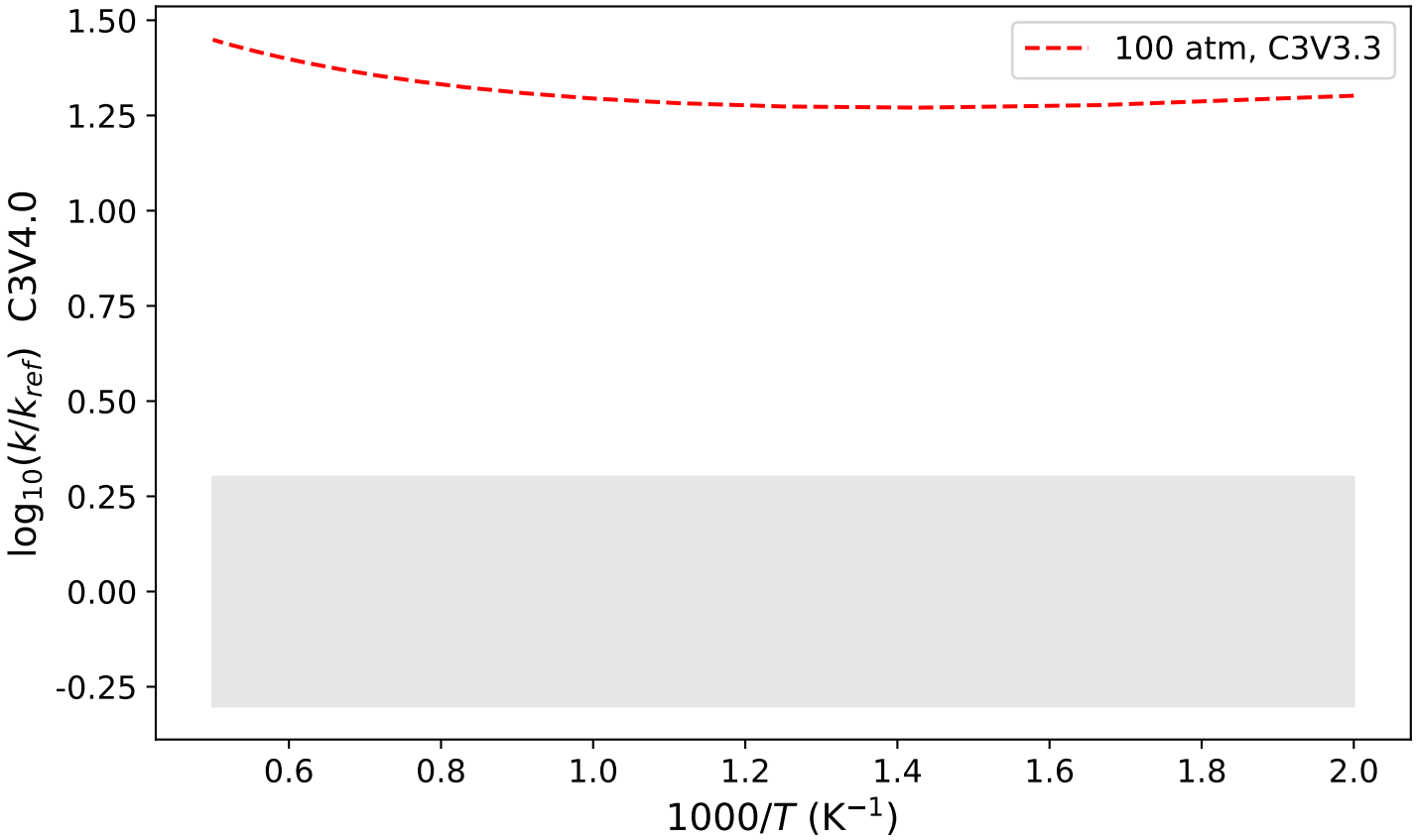
NAPH + C9H7 = NAPH-A + IND



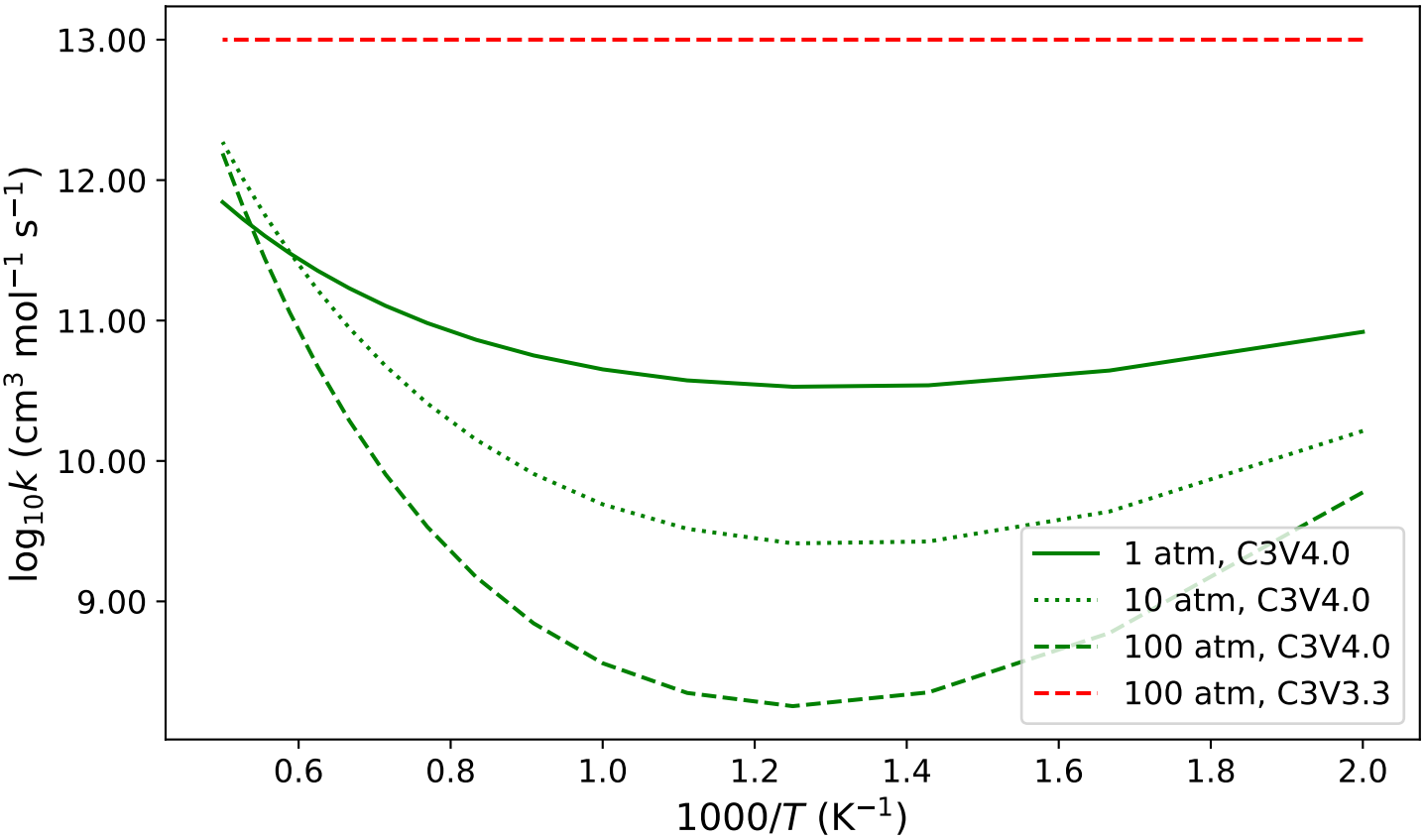


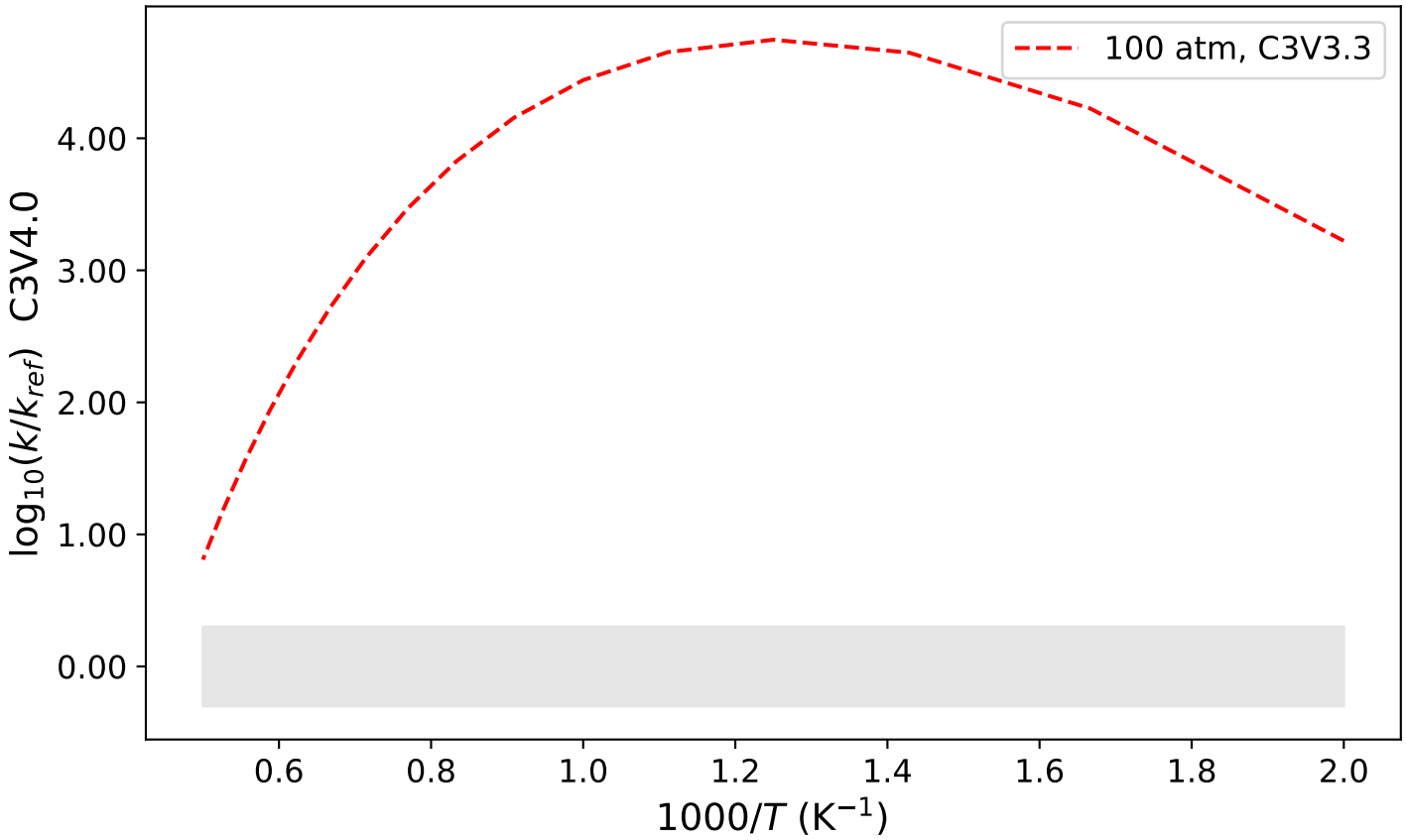
O + C9H7 = CO + C6H5CHCH



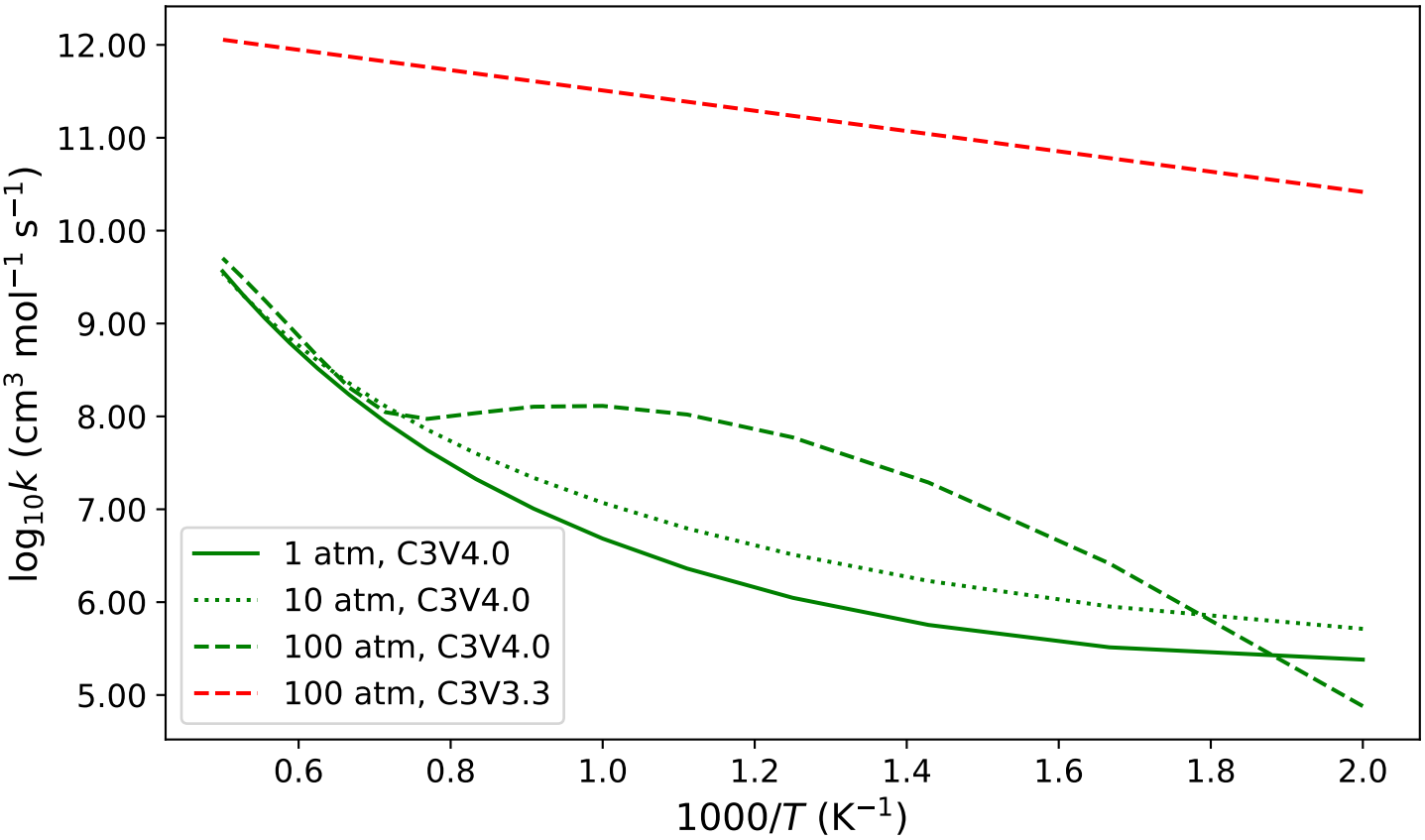


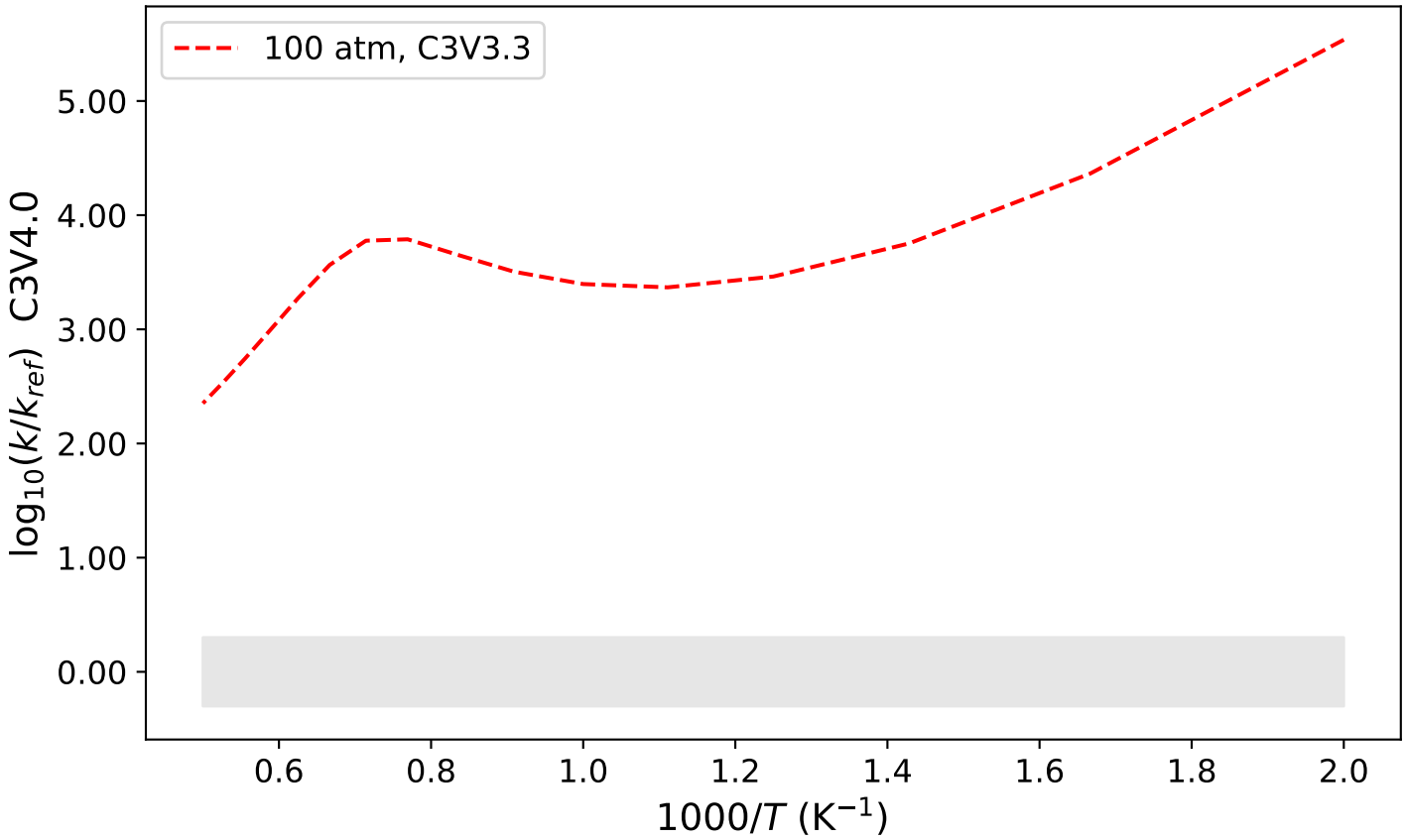
C3H3 + C6H5 = IND



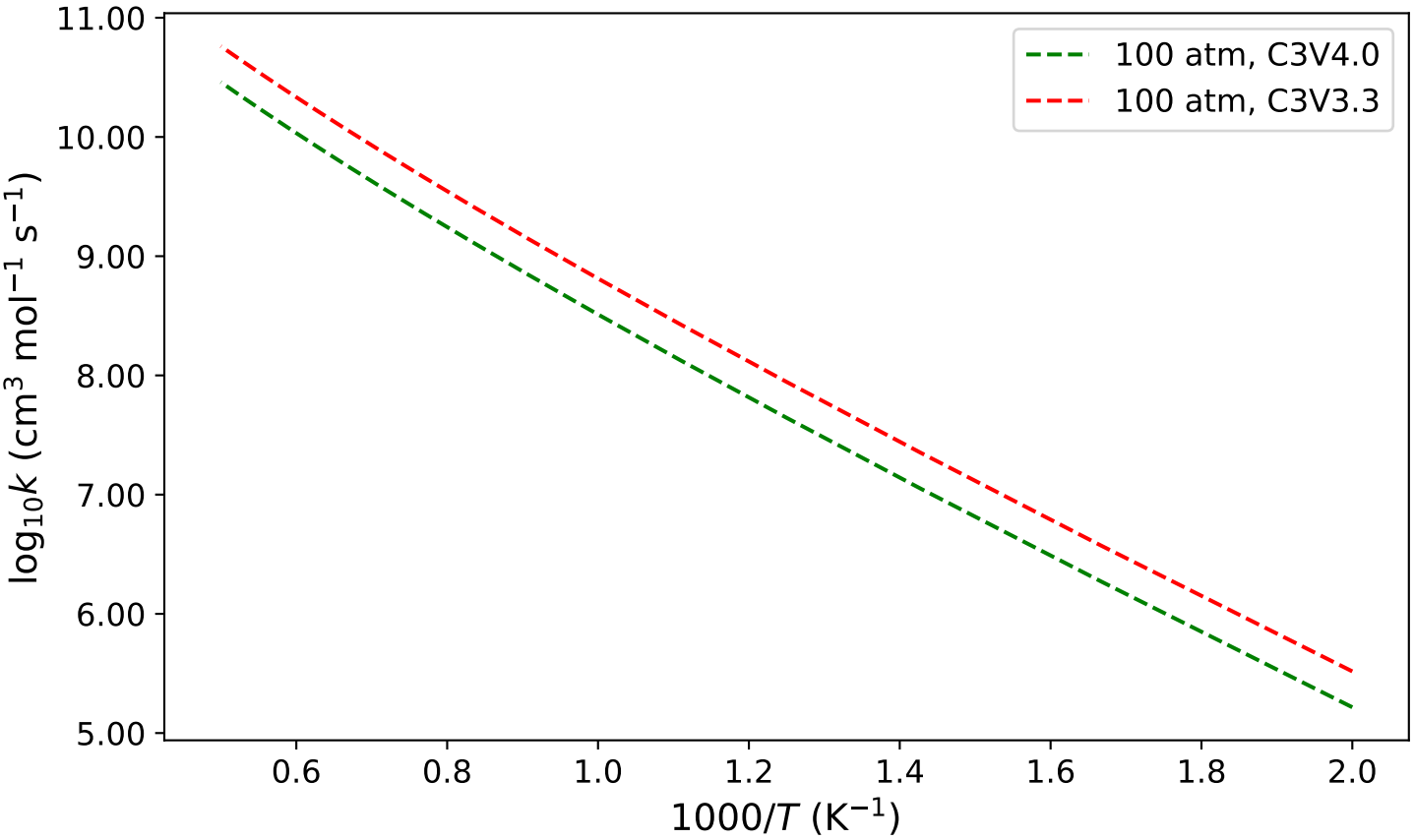


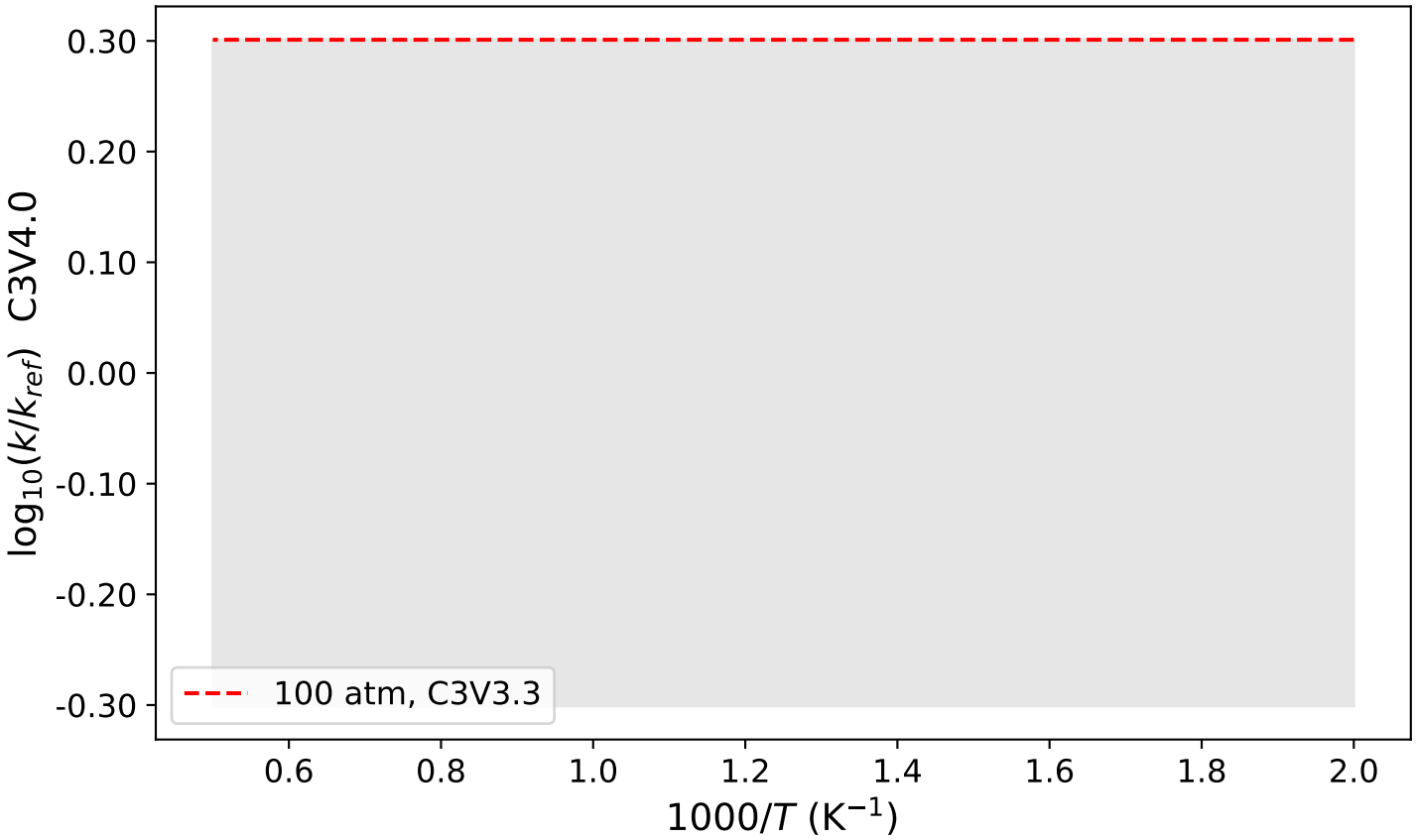
C4H5-I + C6H6 = H + C10H10



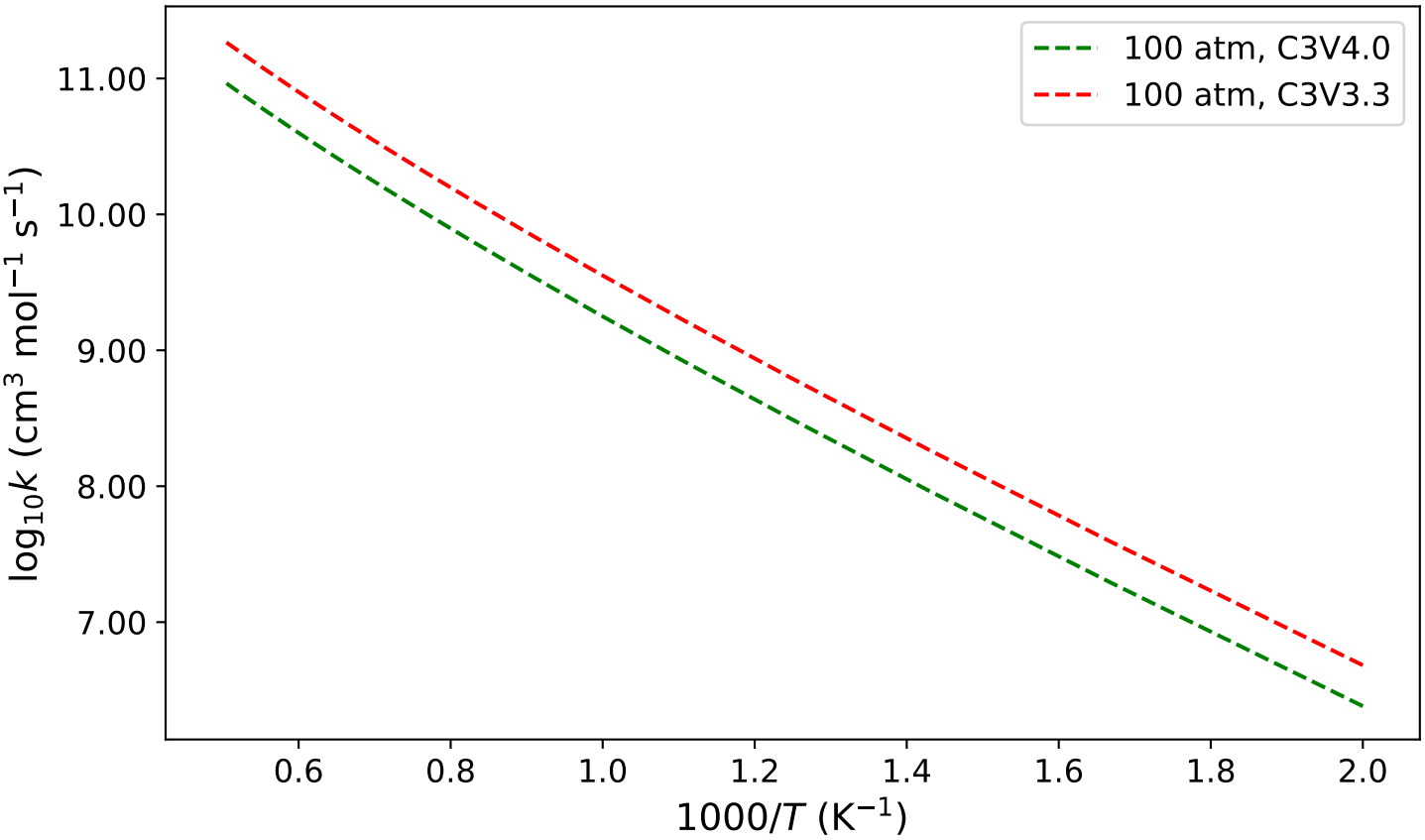


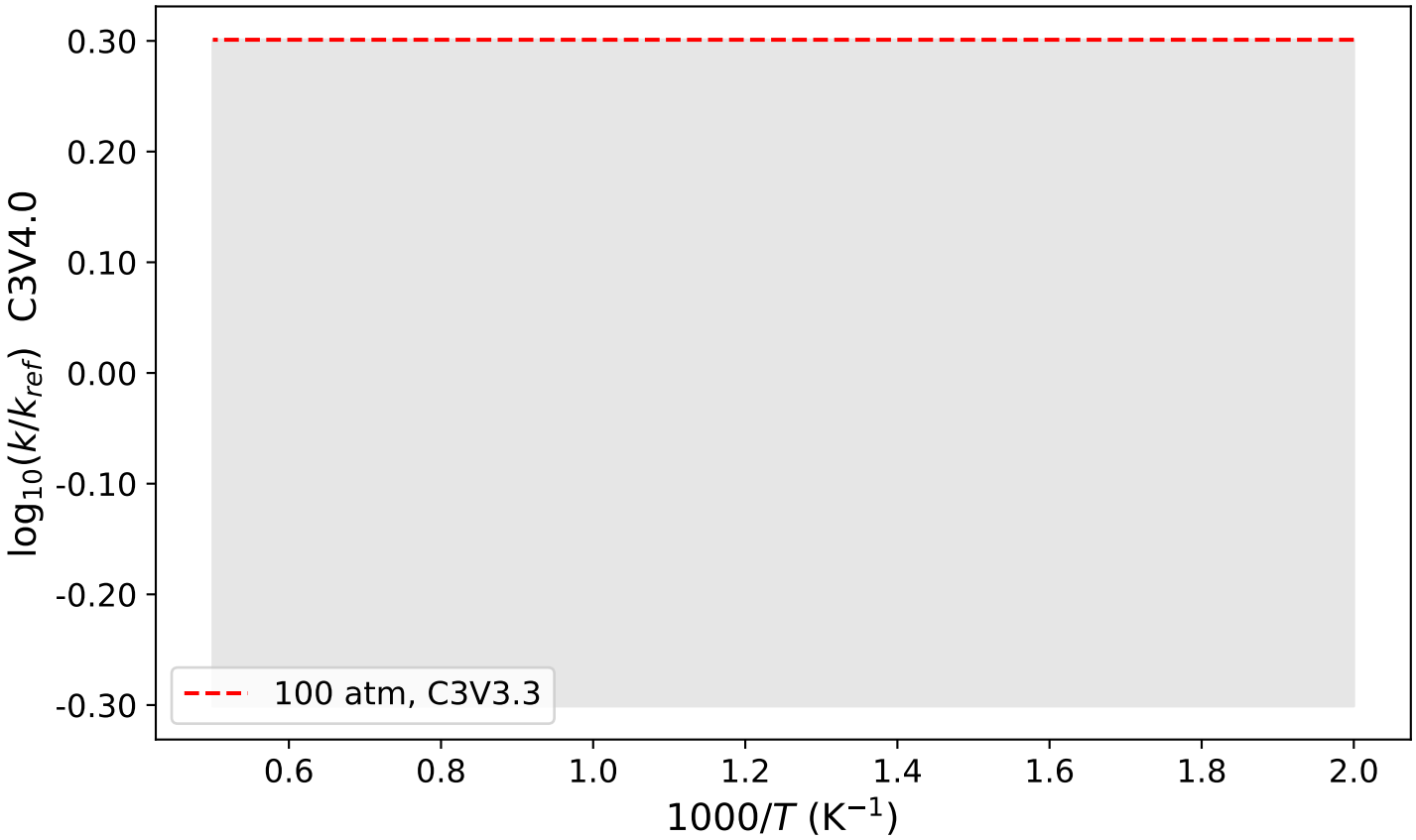
C2H5 + NAPH = C2H6 + NAPH-A



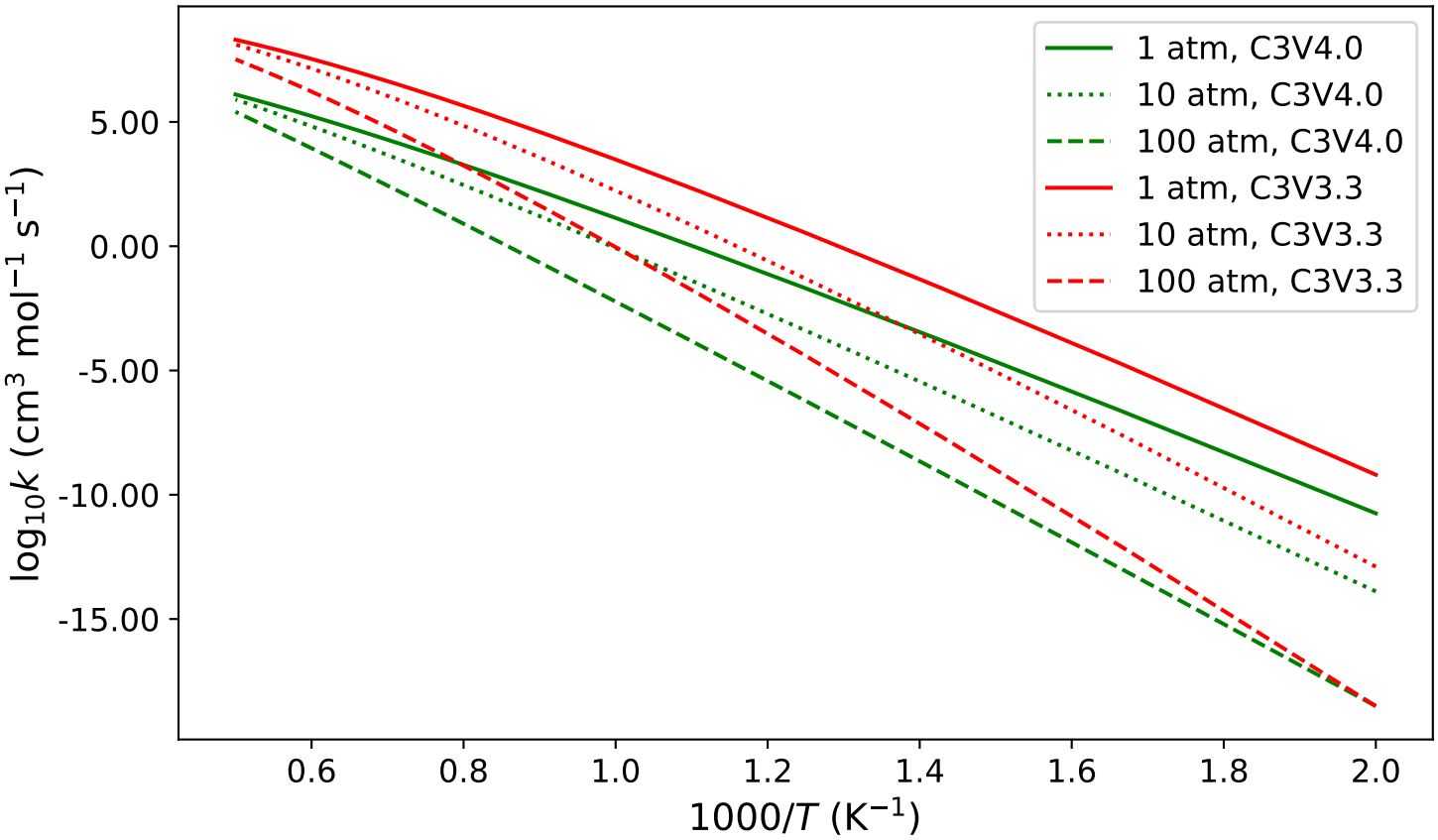


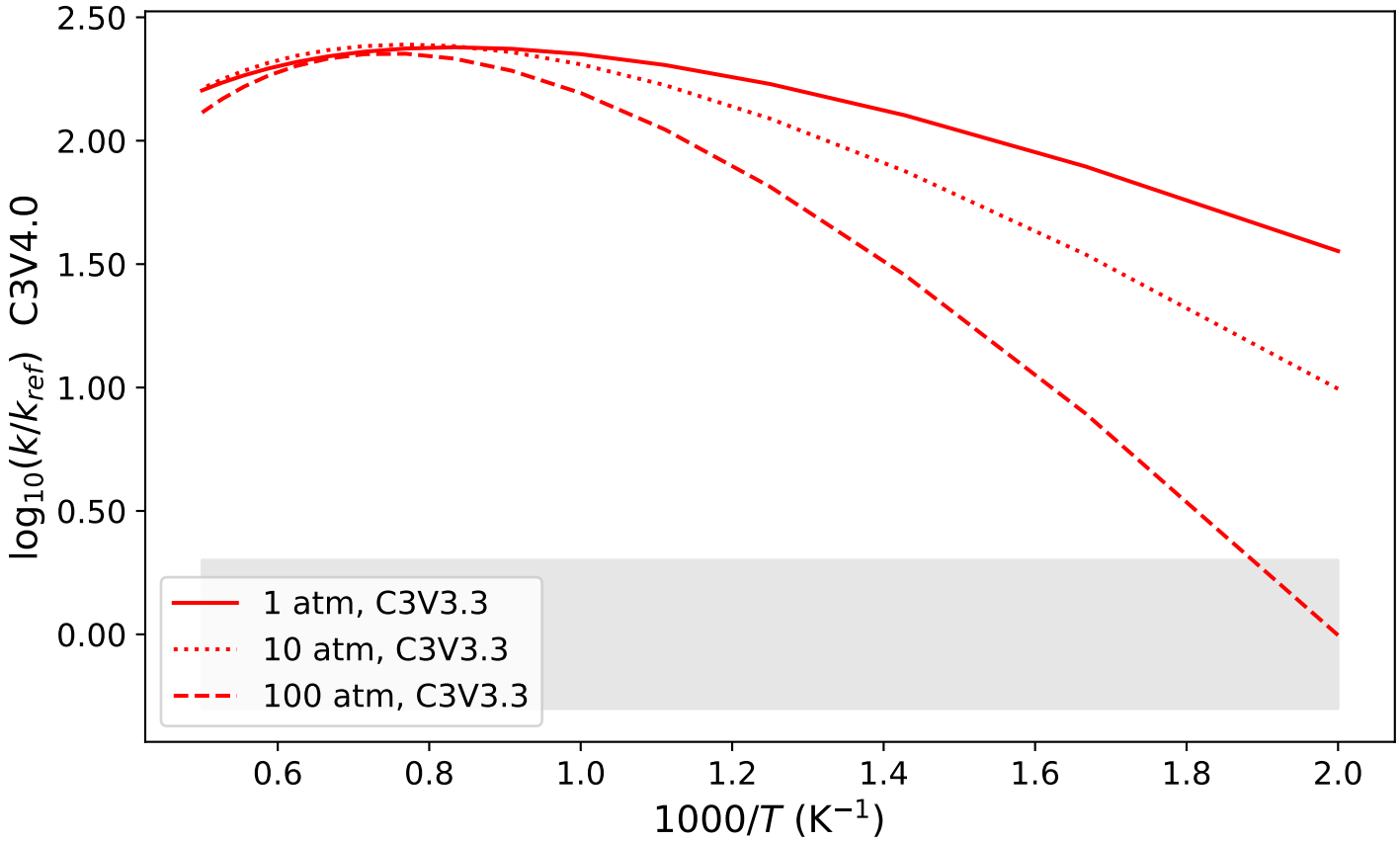
NAPH + C4H3-I = C4H4 + NAPH-A



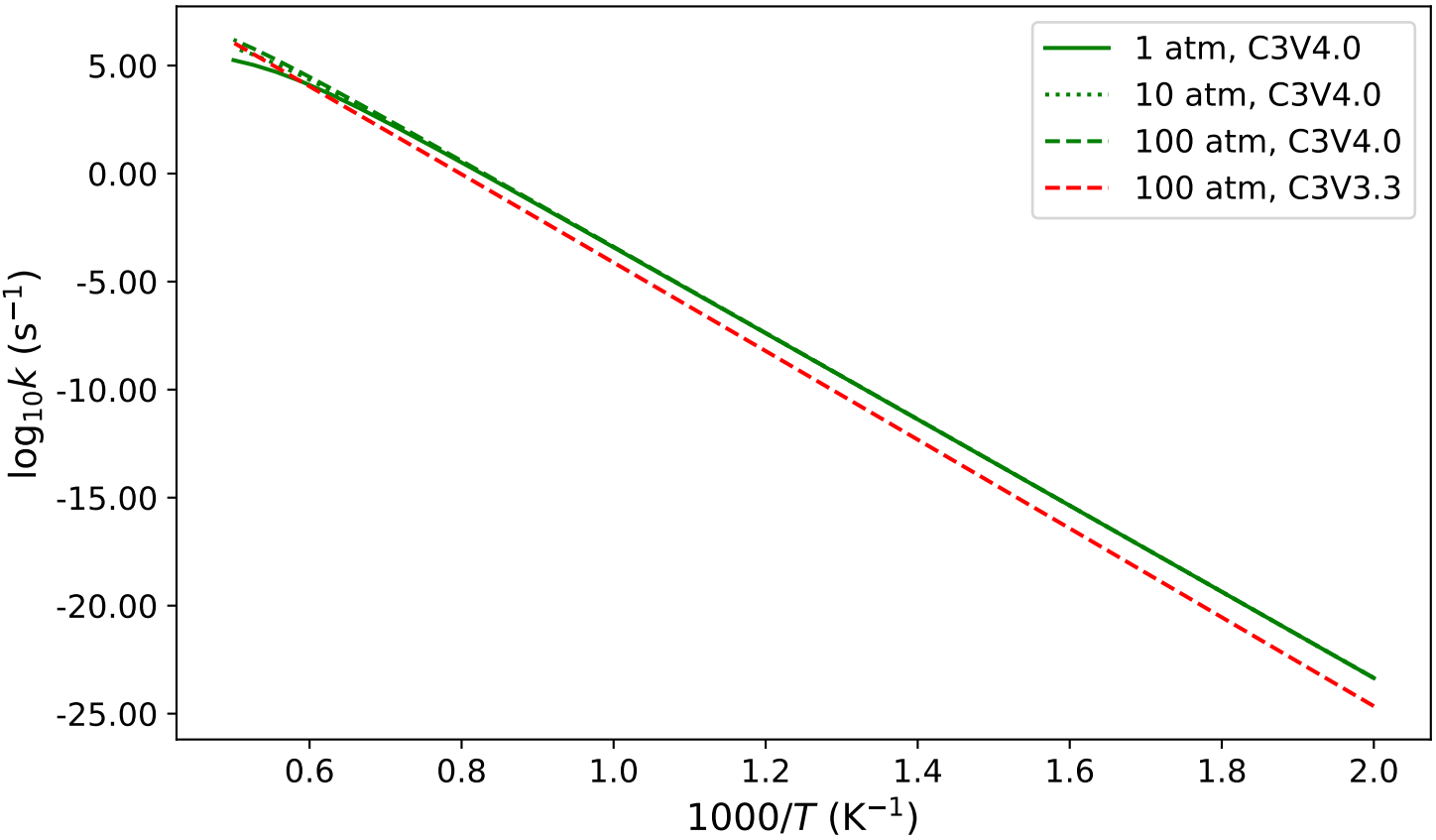


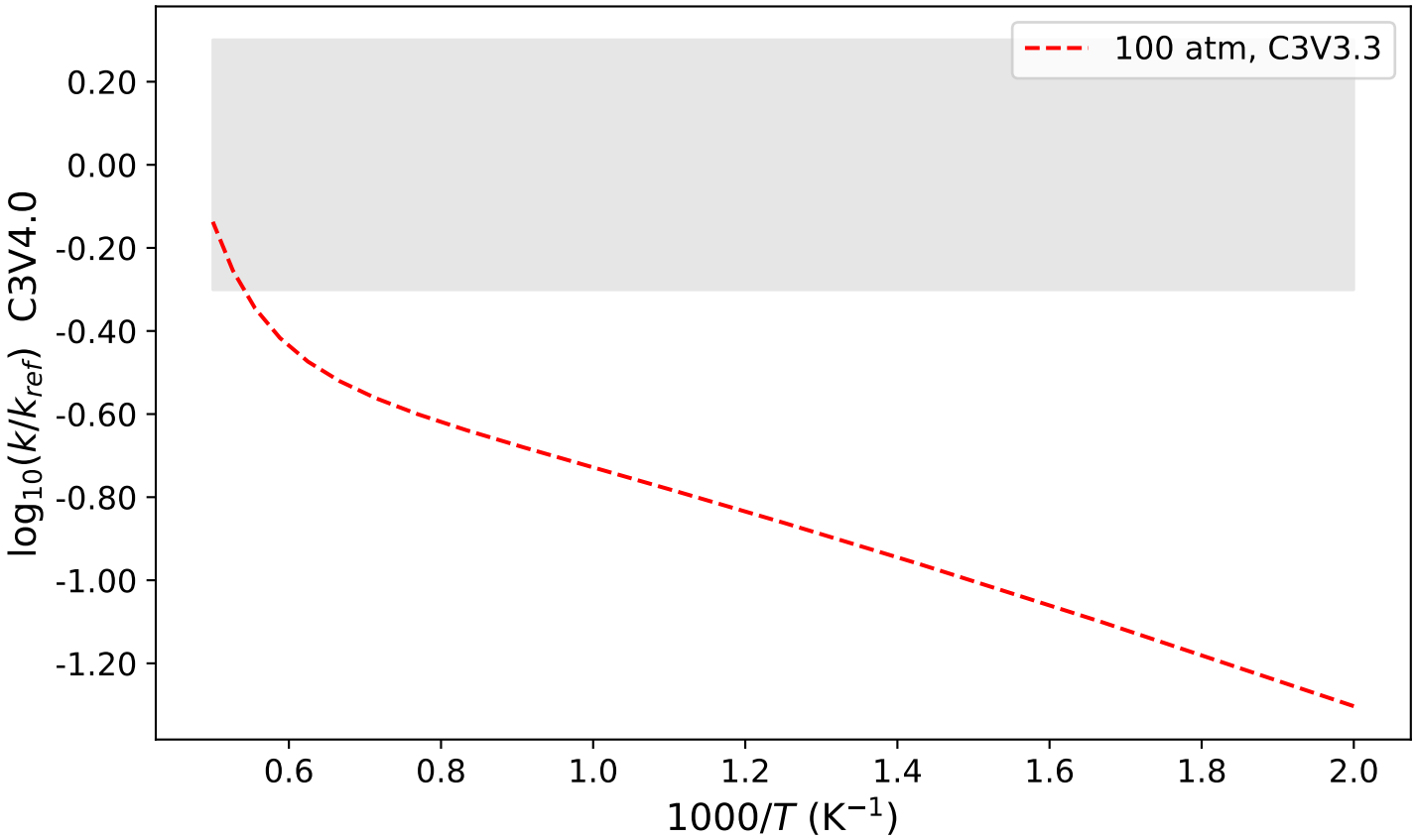
H + C5H6 = CH3 + C4H4



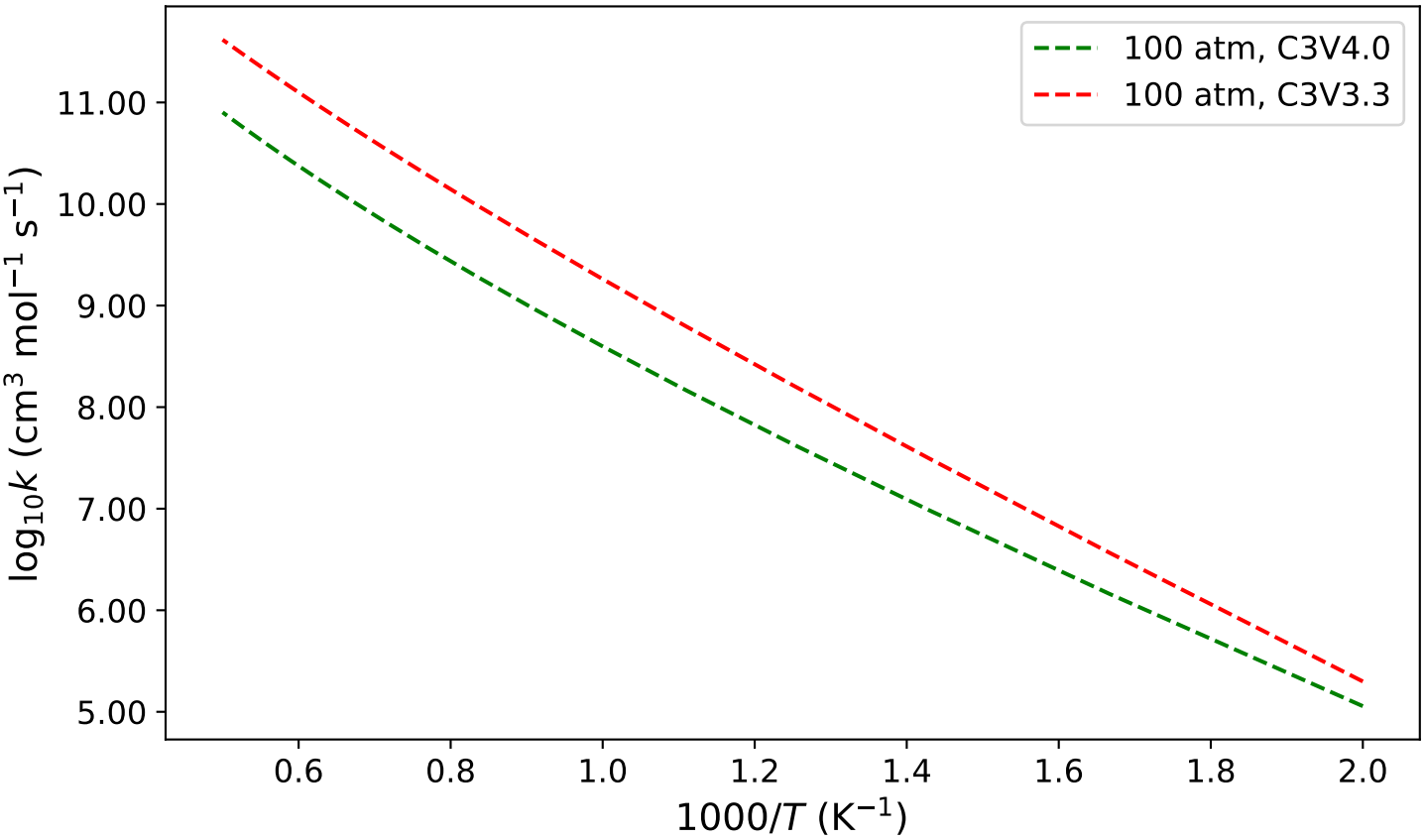


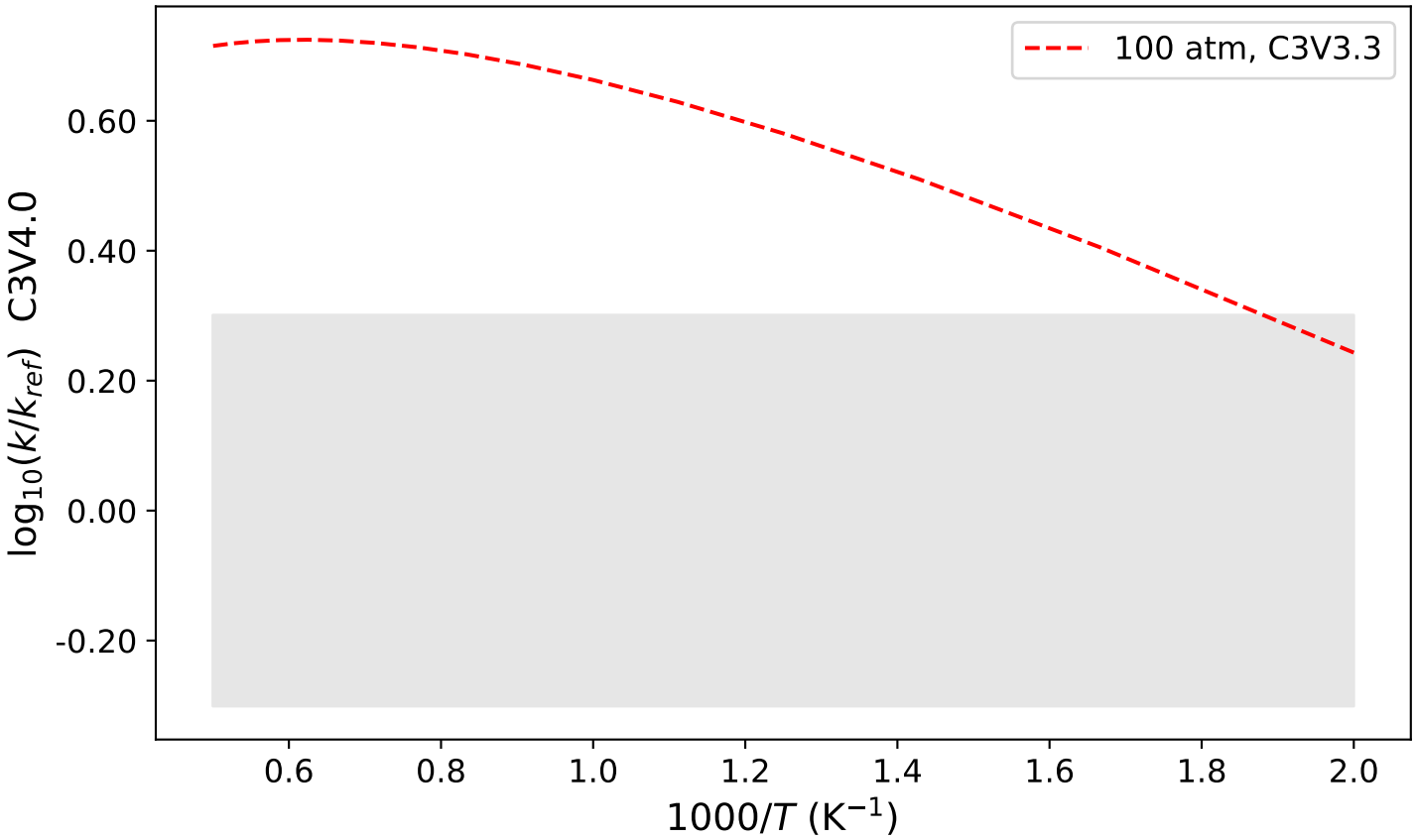
XYLENE (+M) = H + RXYLENE (+M)



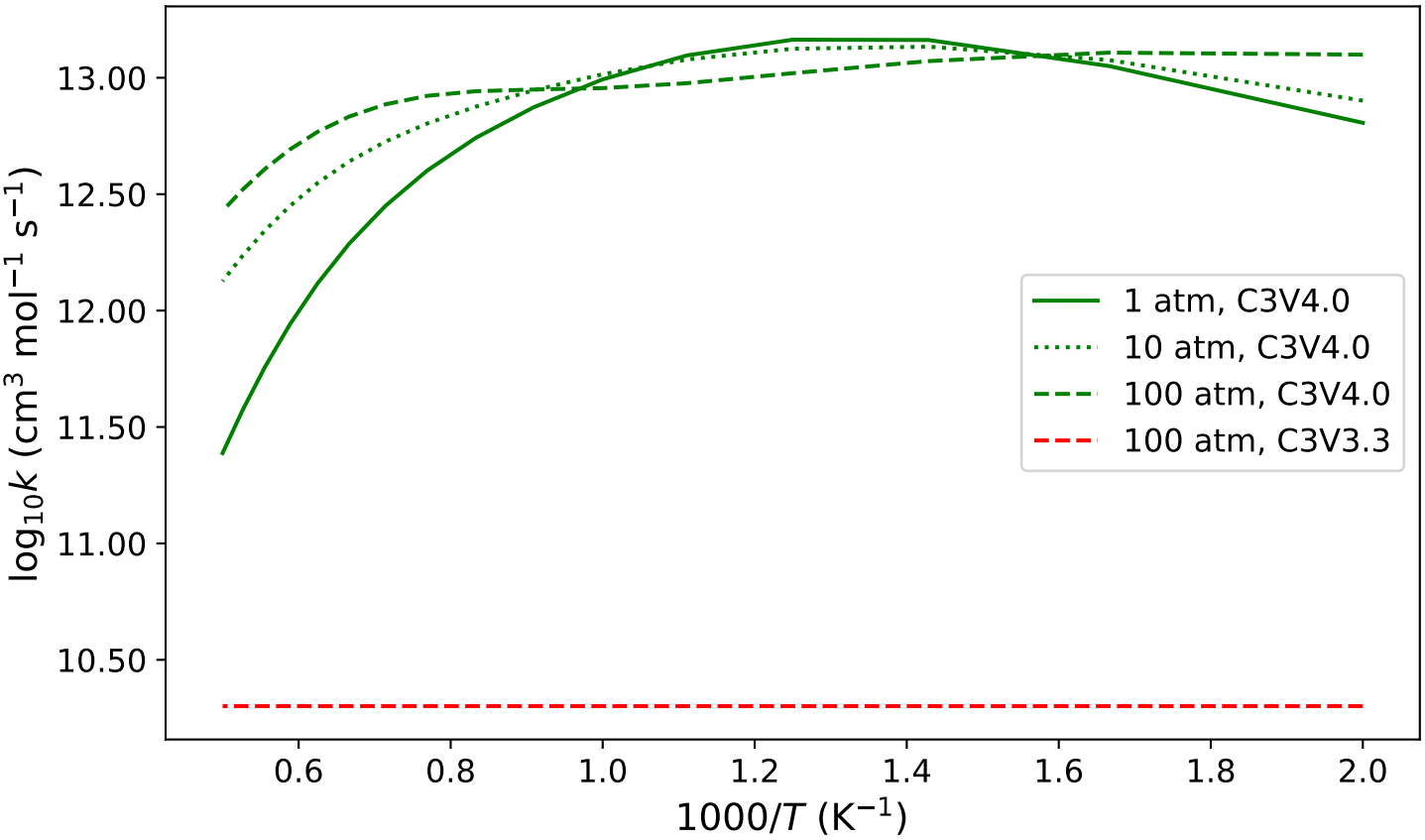


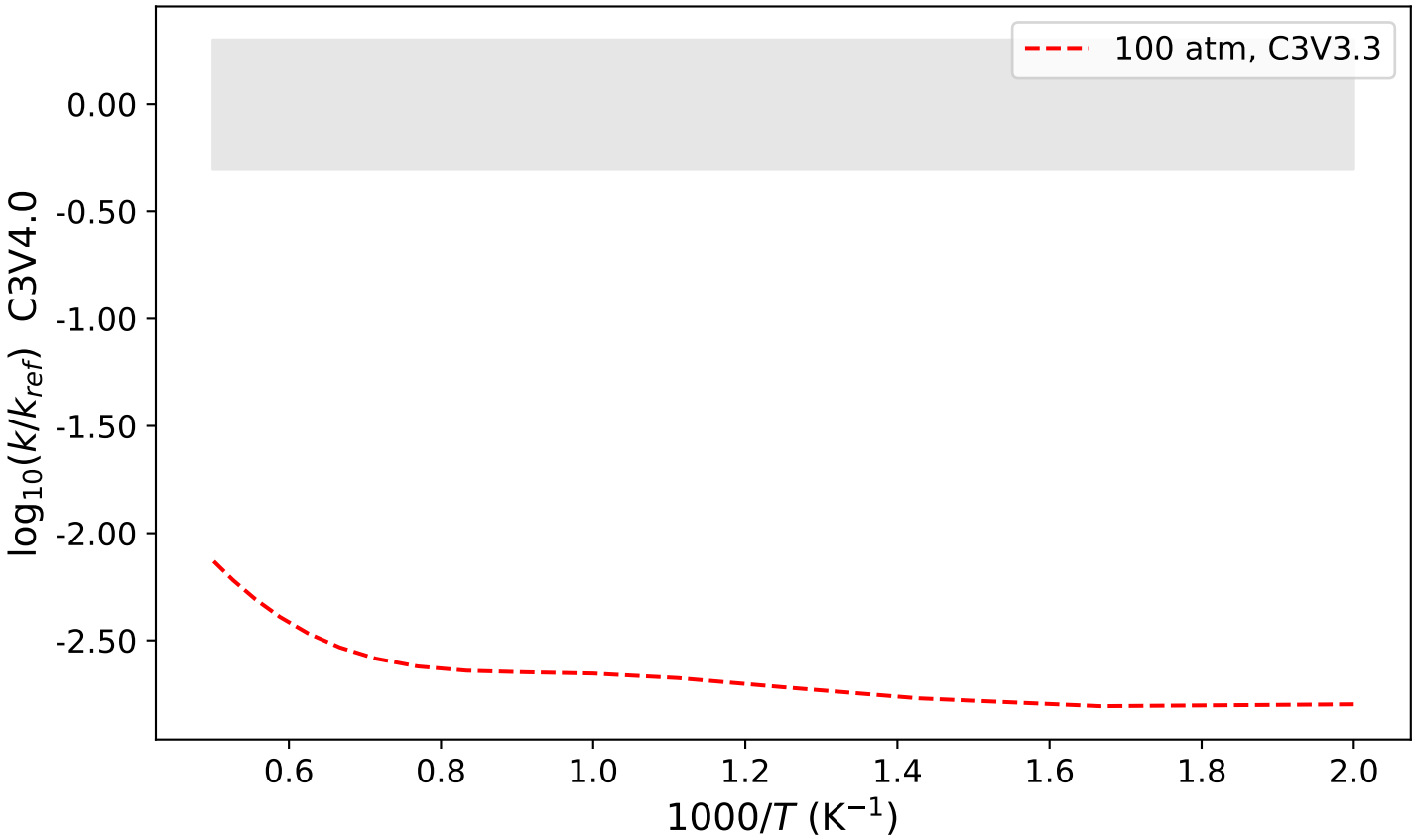
HO2 + C10H7CH3-A = H2O2 + C10H7CH2-A



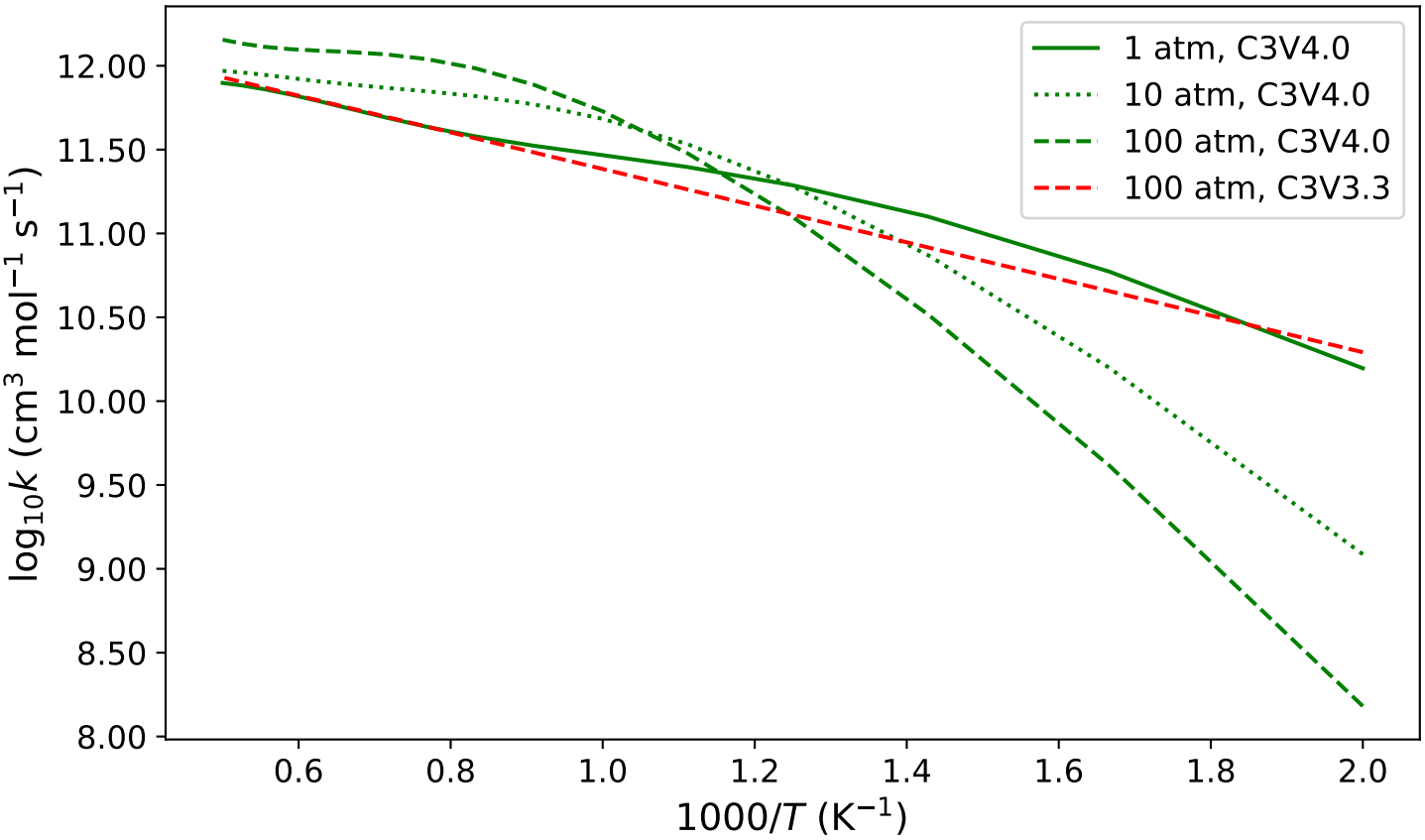


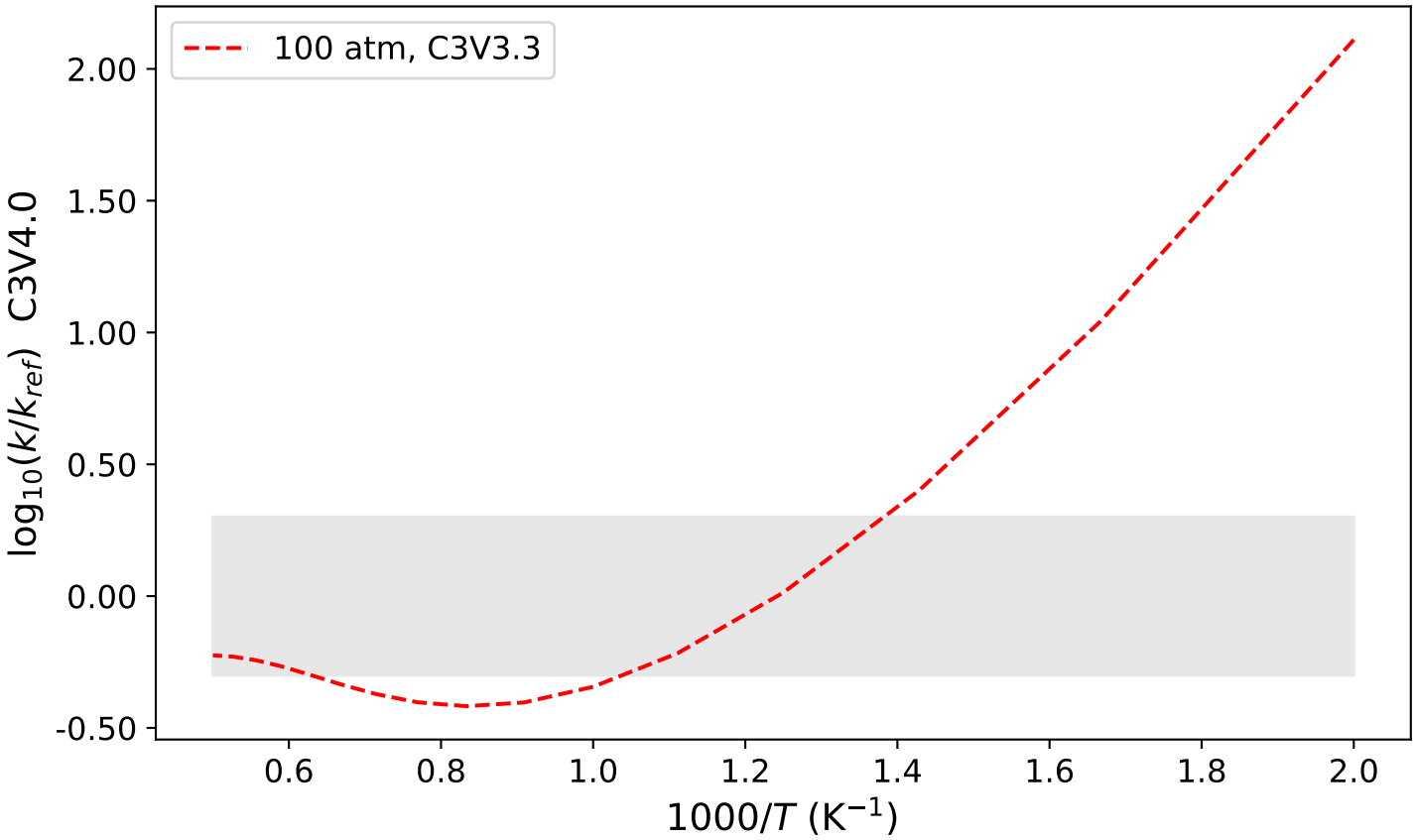
CH3 + C9H7 = MEINDENE



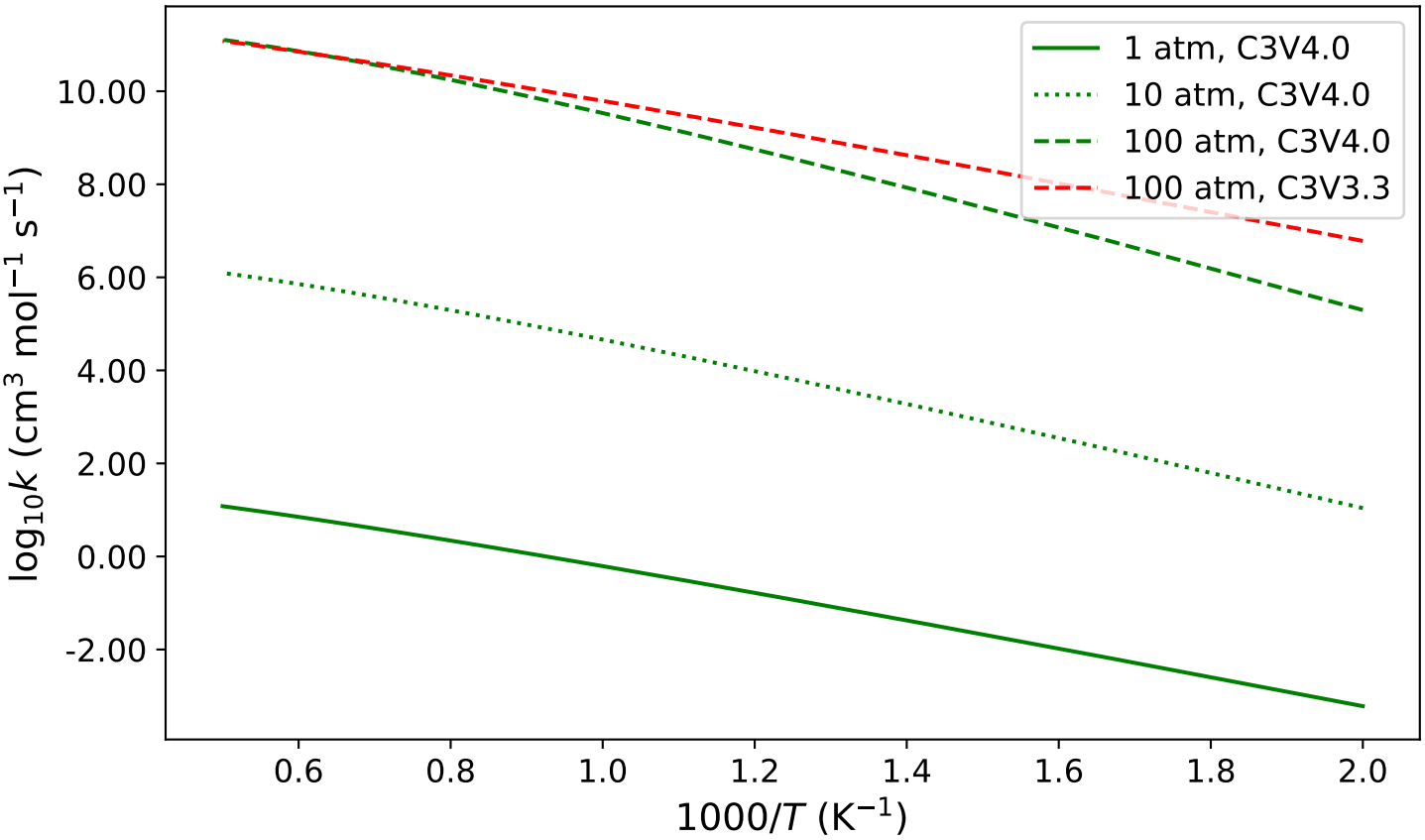


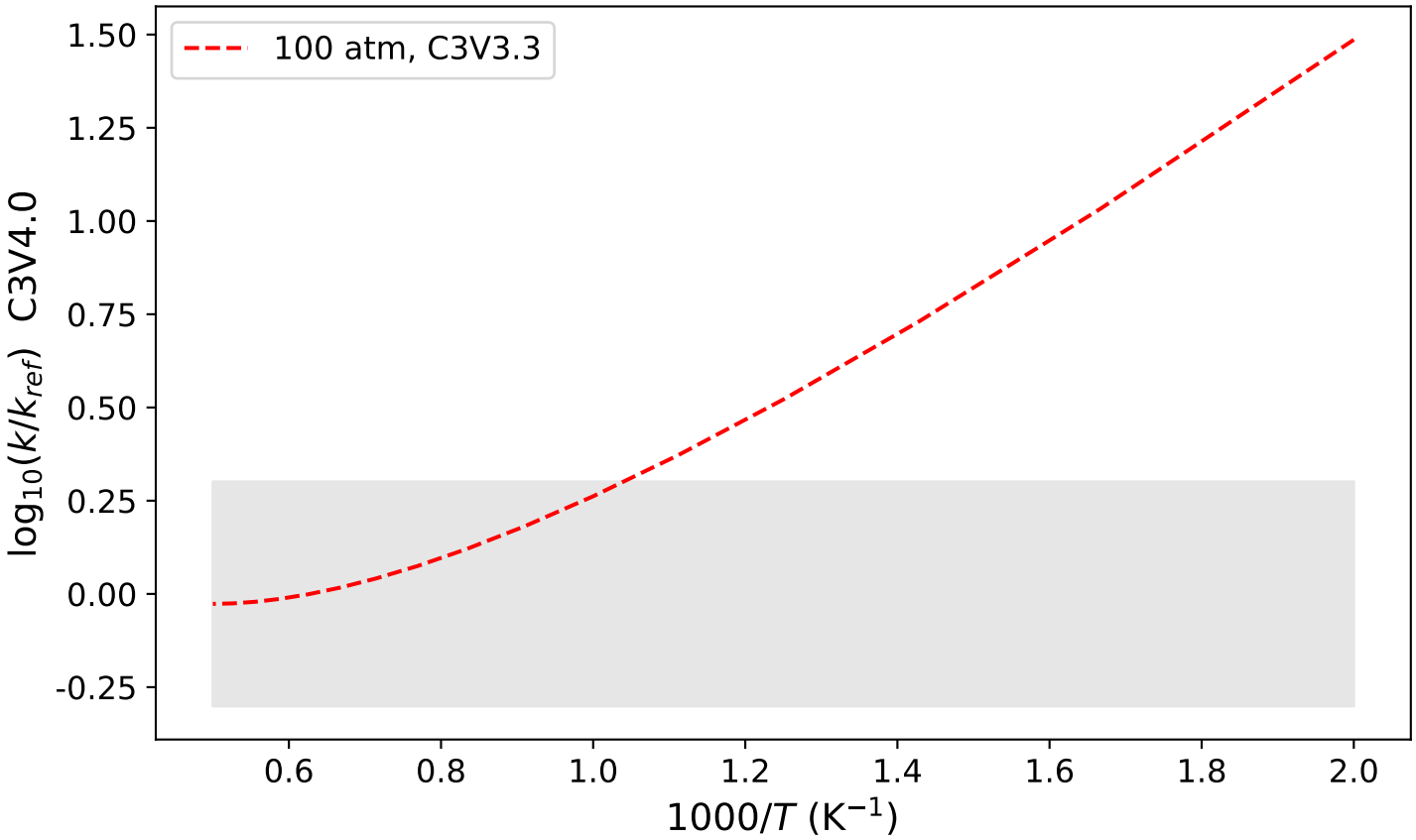
NAPH-A + C3H4-P = H + FLUORENE



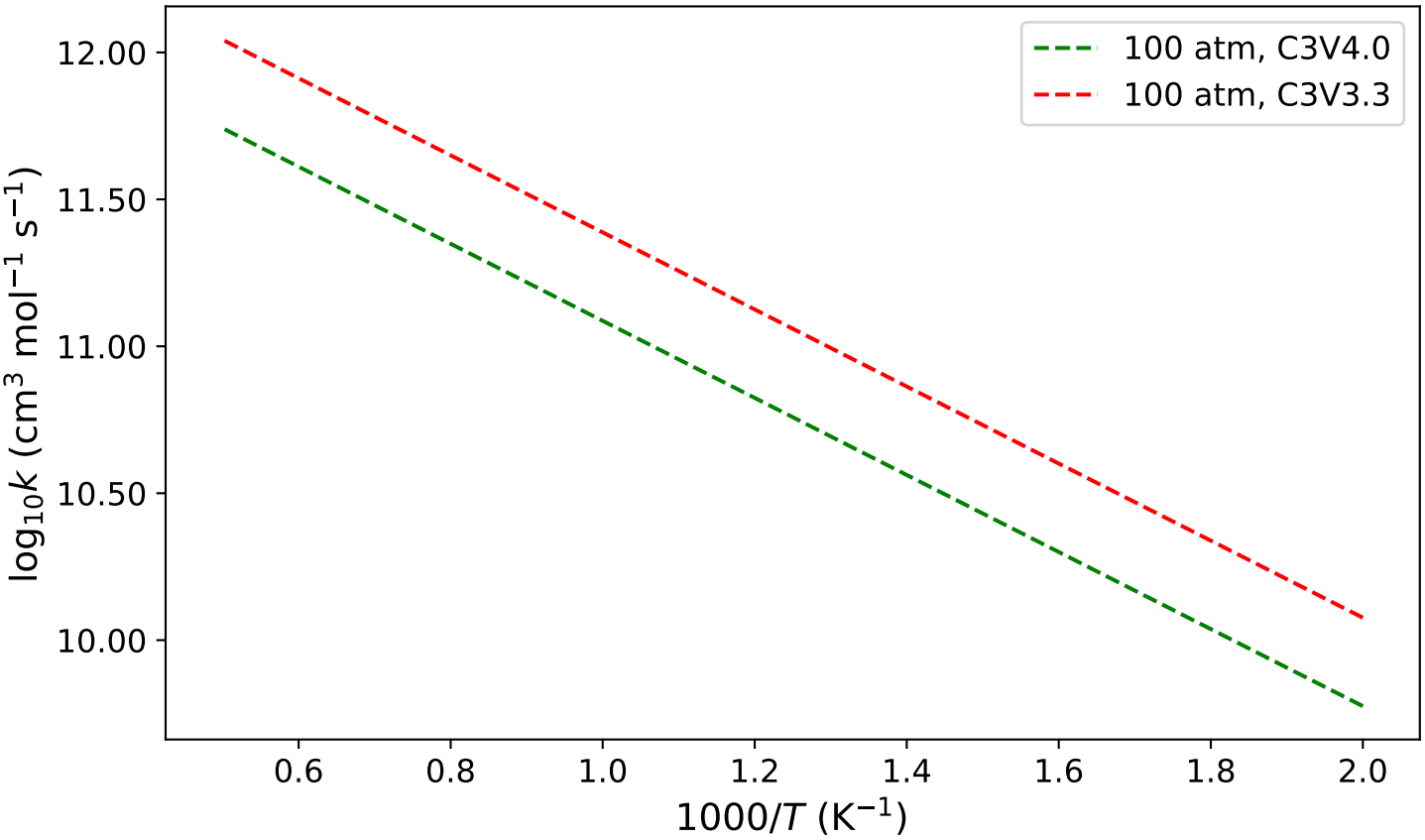


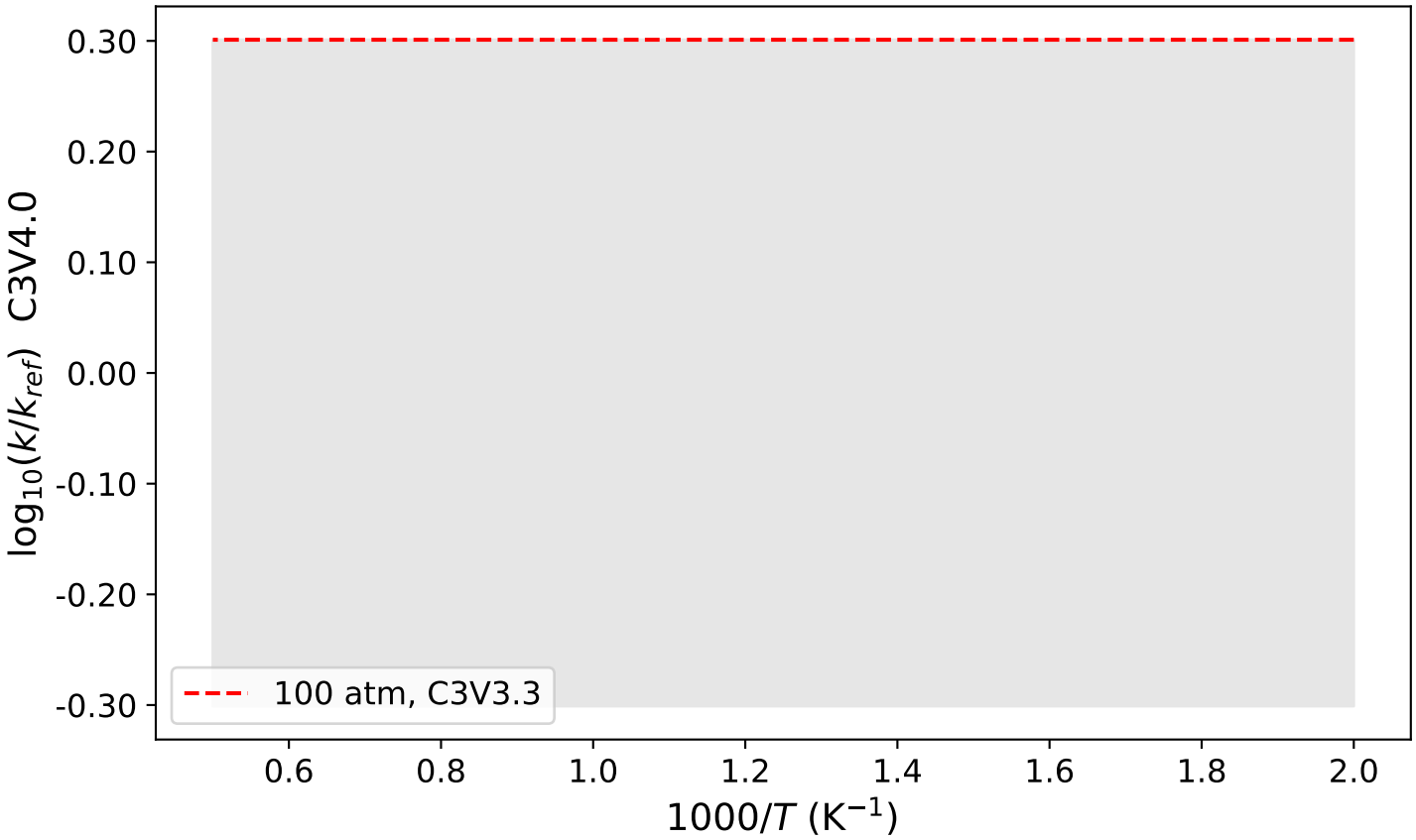
C2H3 + C6H5C2H = H + NAPH



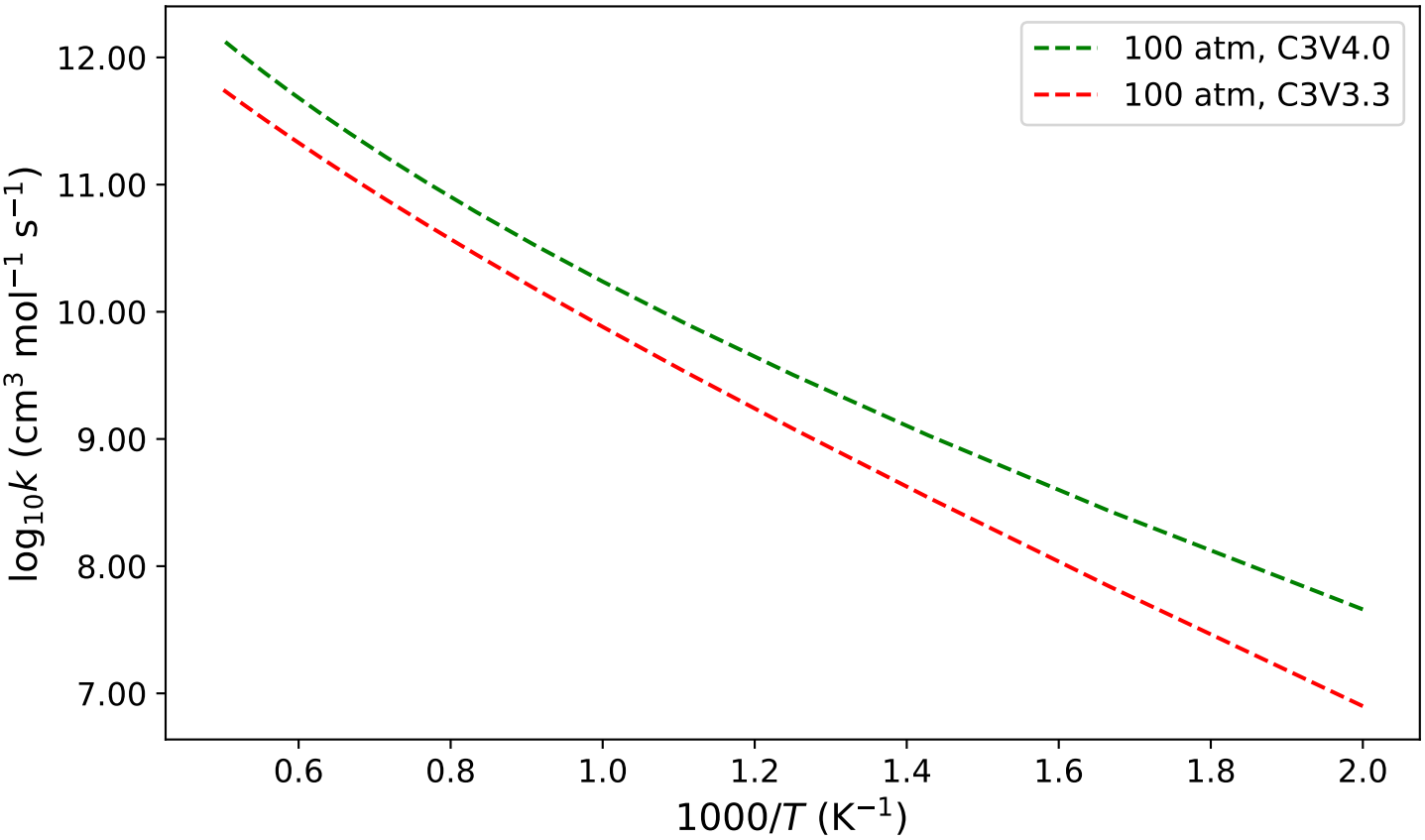


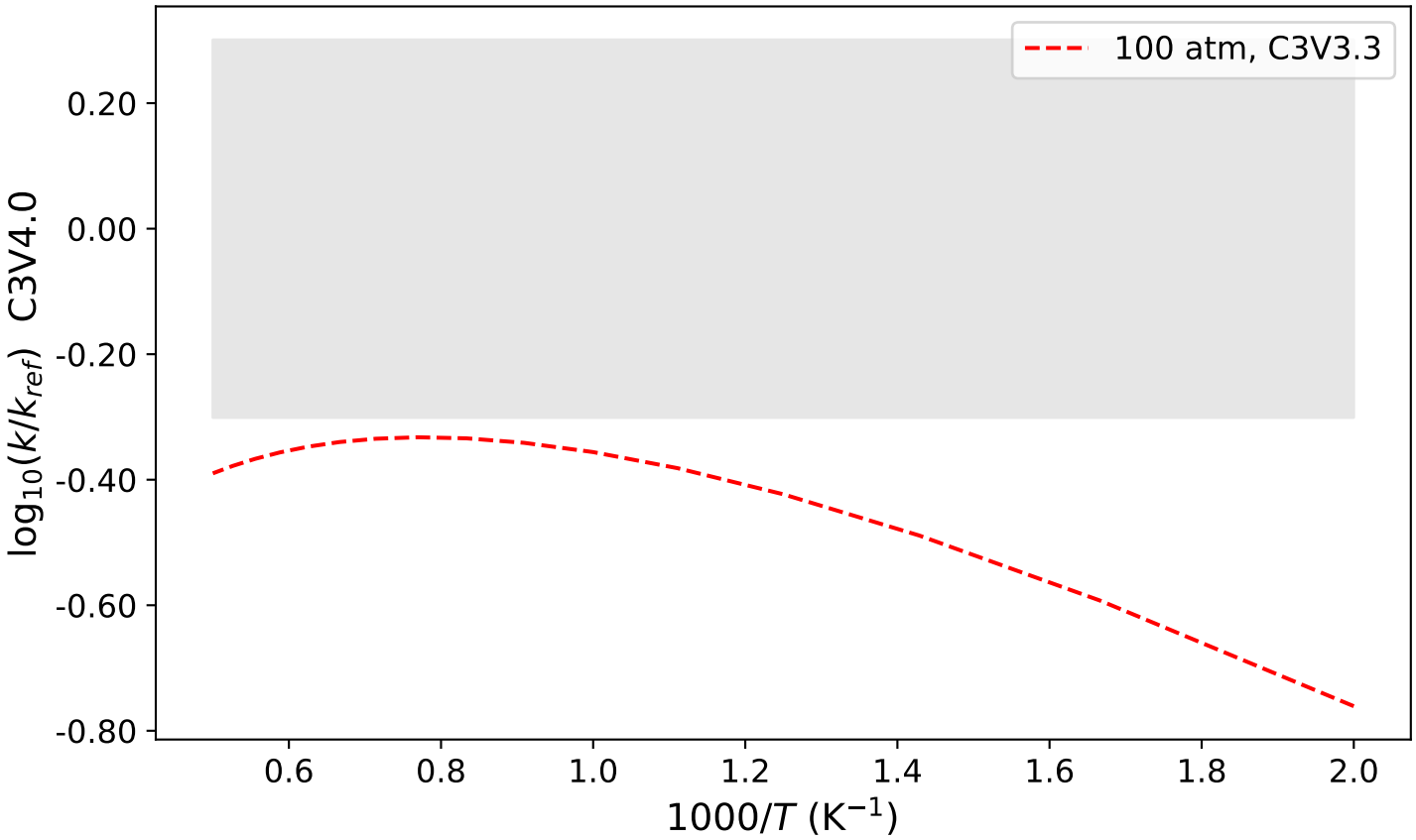
C5H5 + HOC6H4CH2 => H + H + CO + C10H7CH3-A



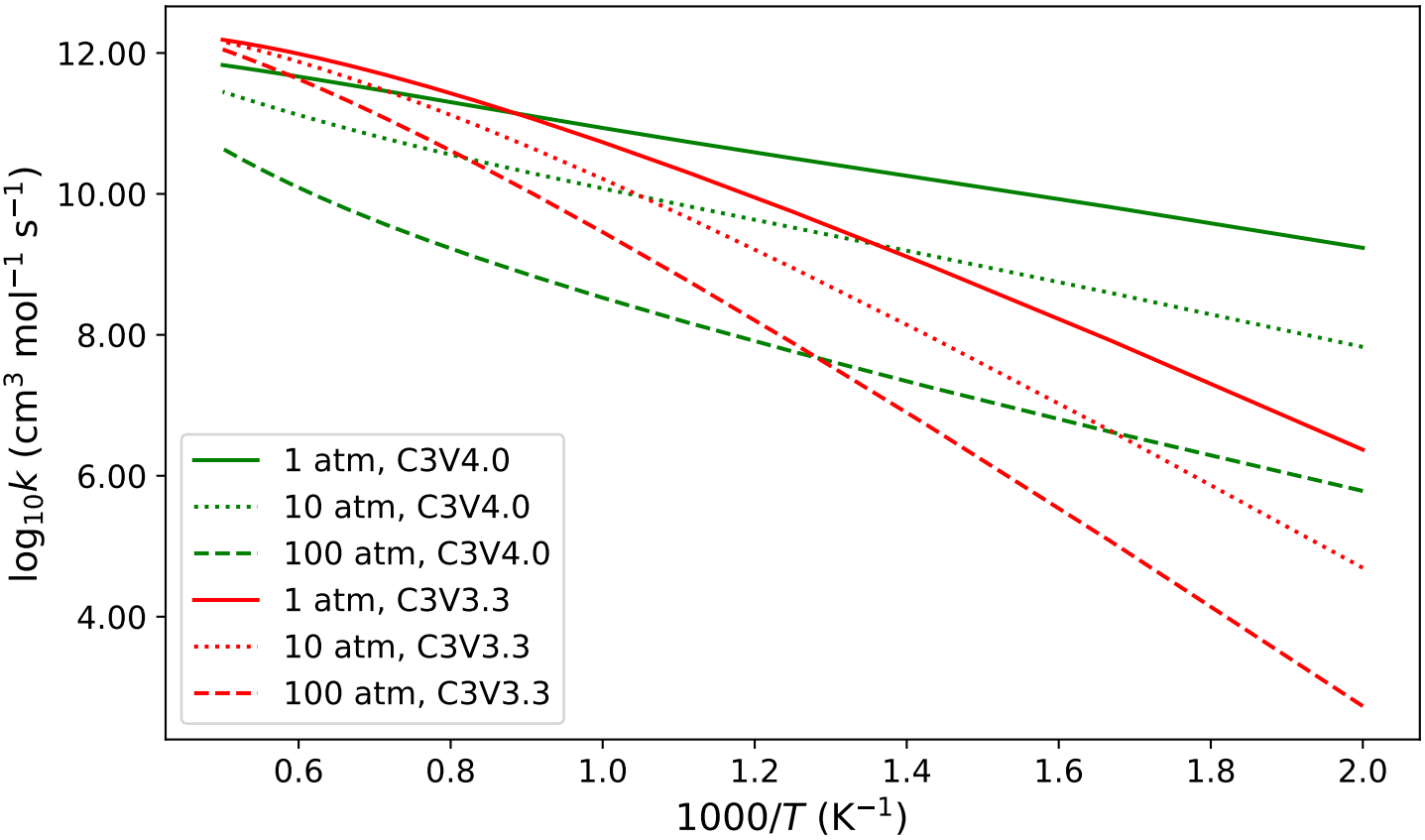


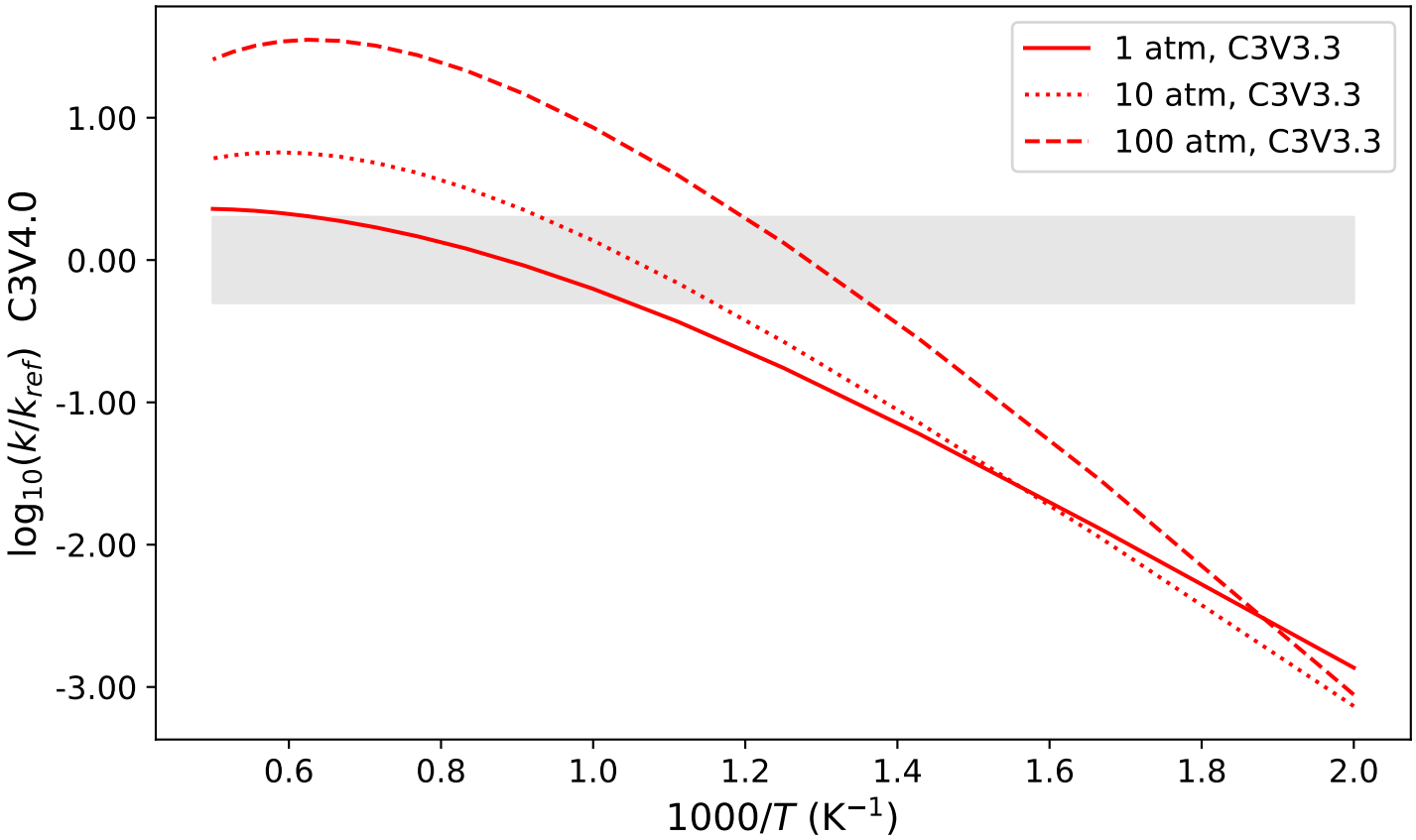
HO2 + C6H4CH3CHO => H2O2 + CO + C6H4CH3



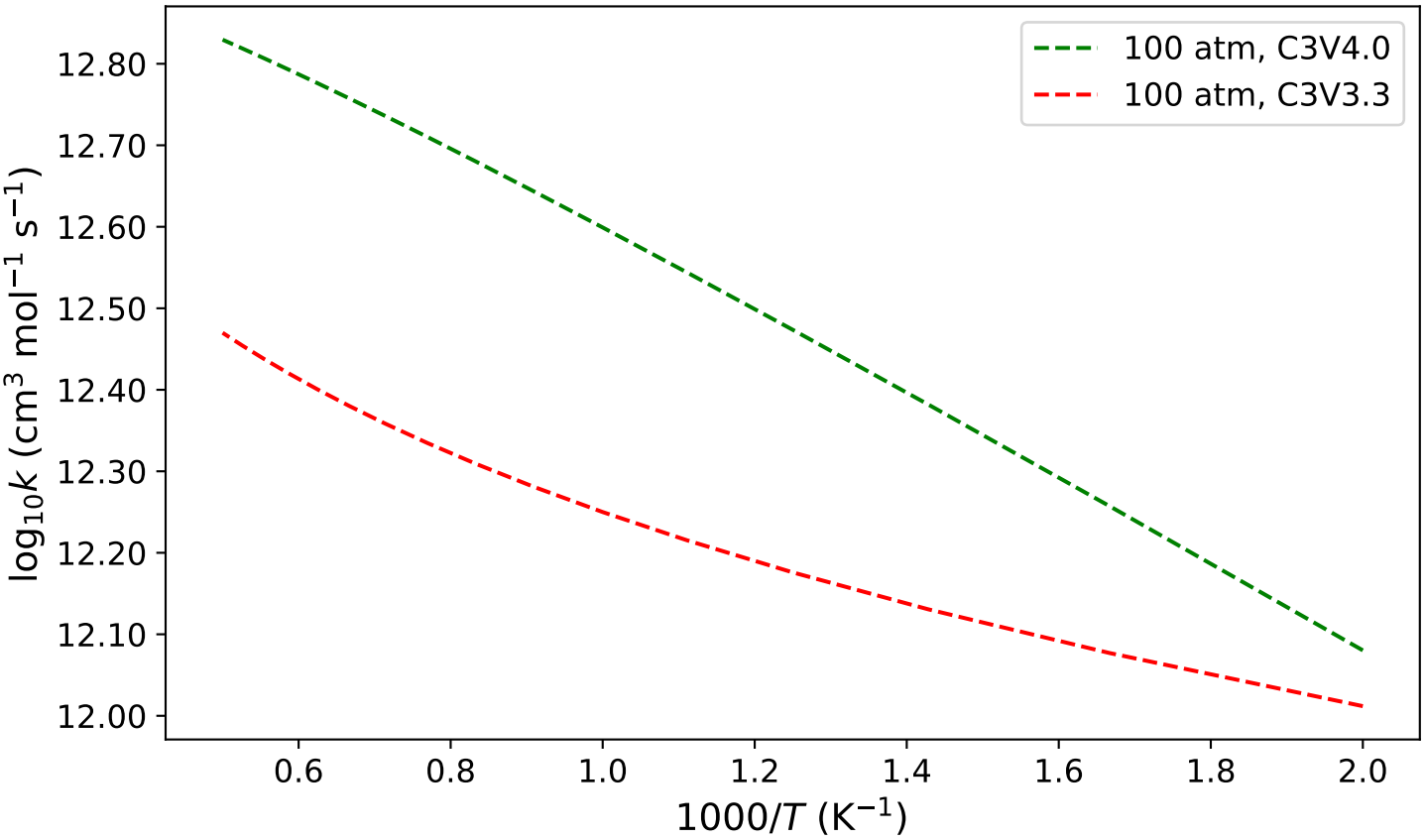


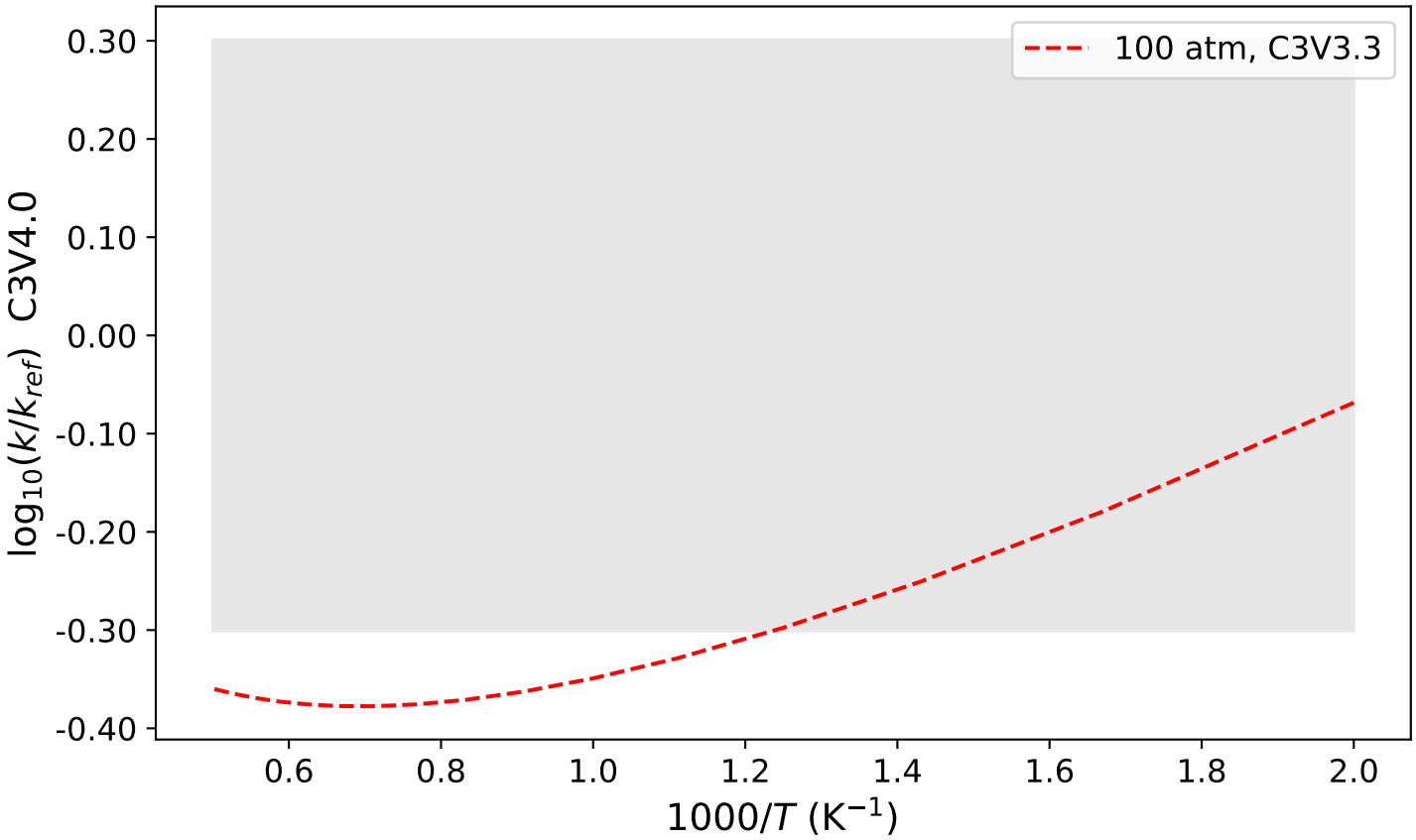
C2H2 + NAPH-A = H + C12H8



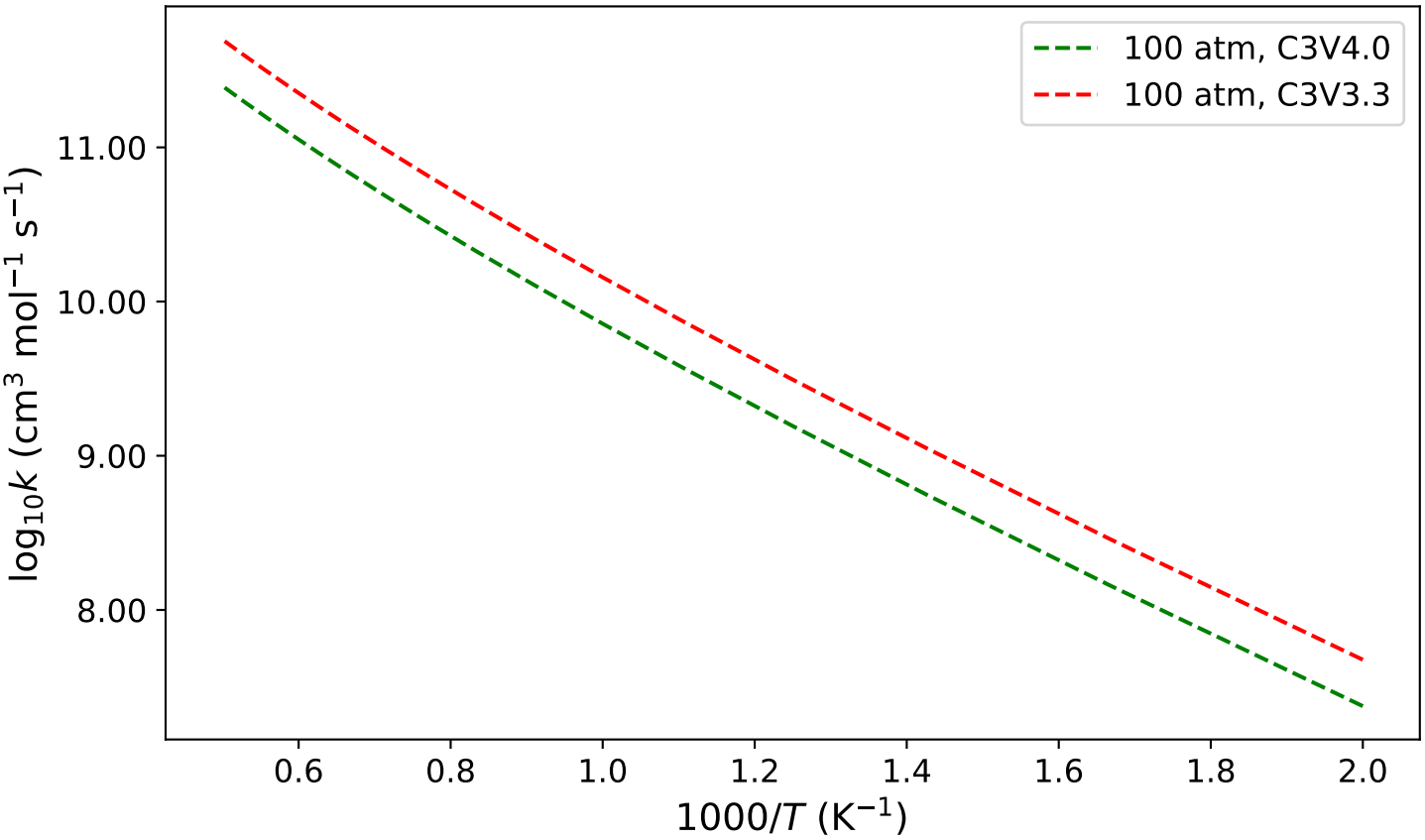


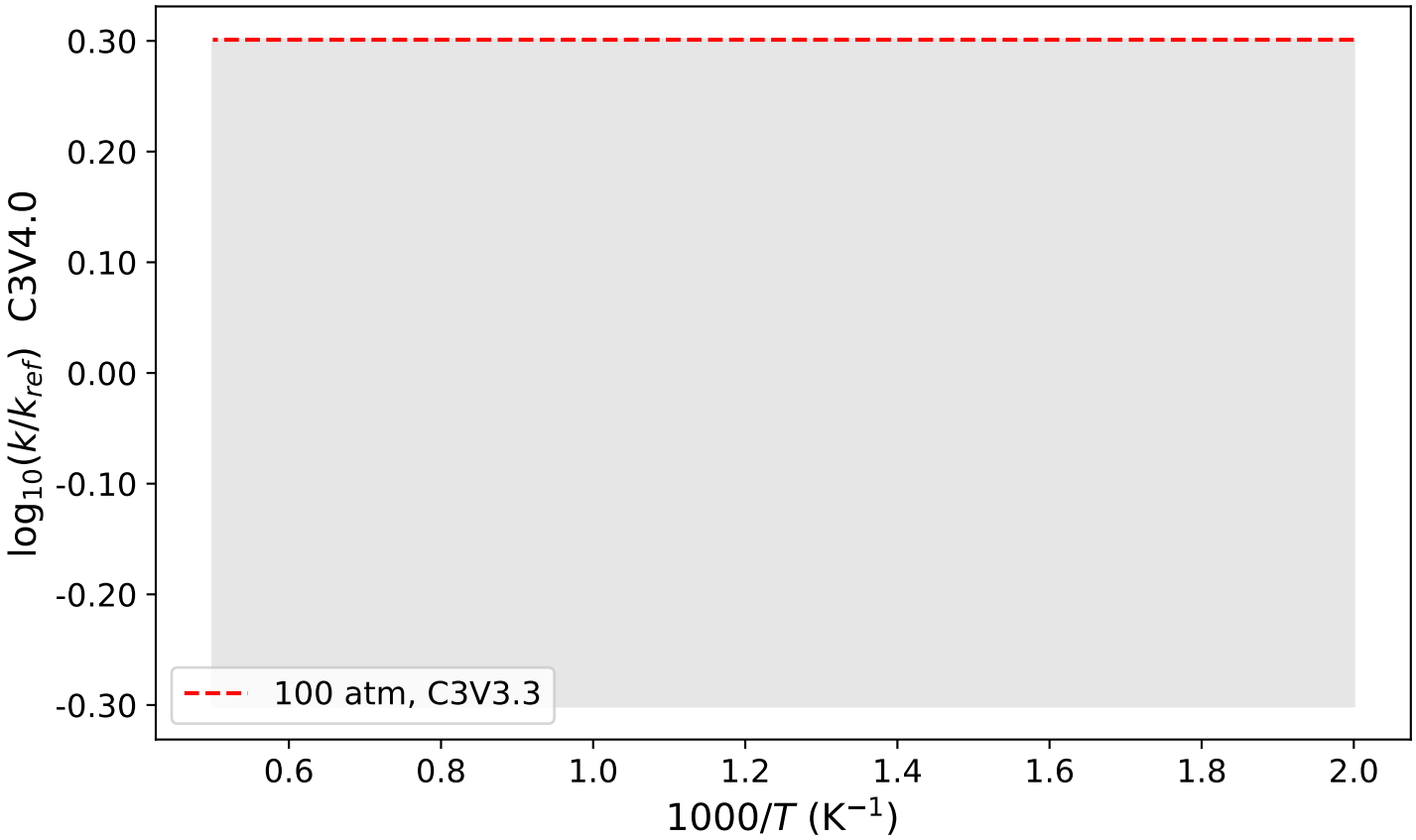
O + C9H7 = H + C9H6O



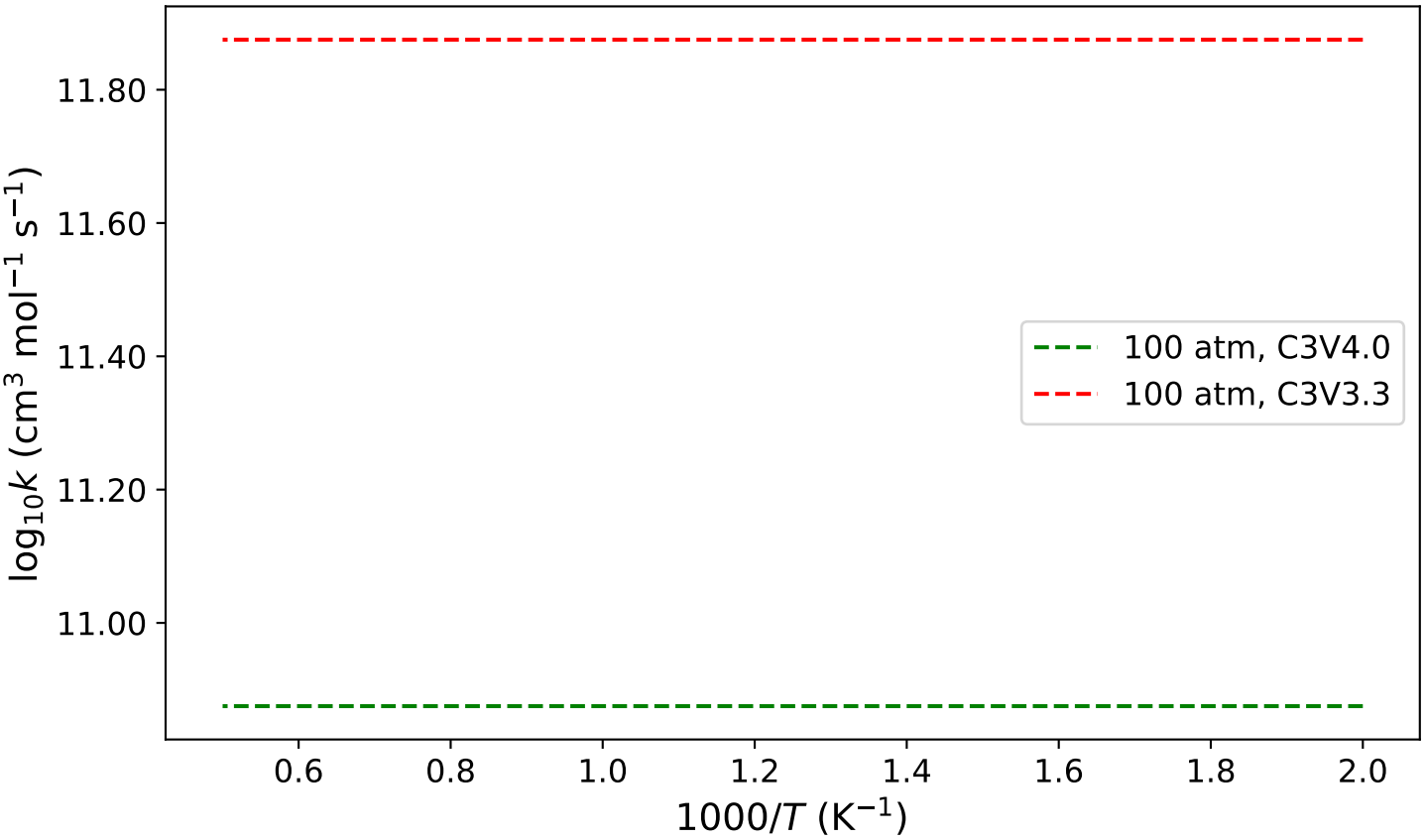


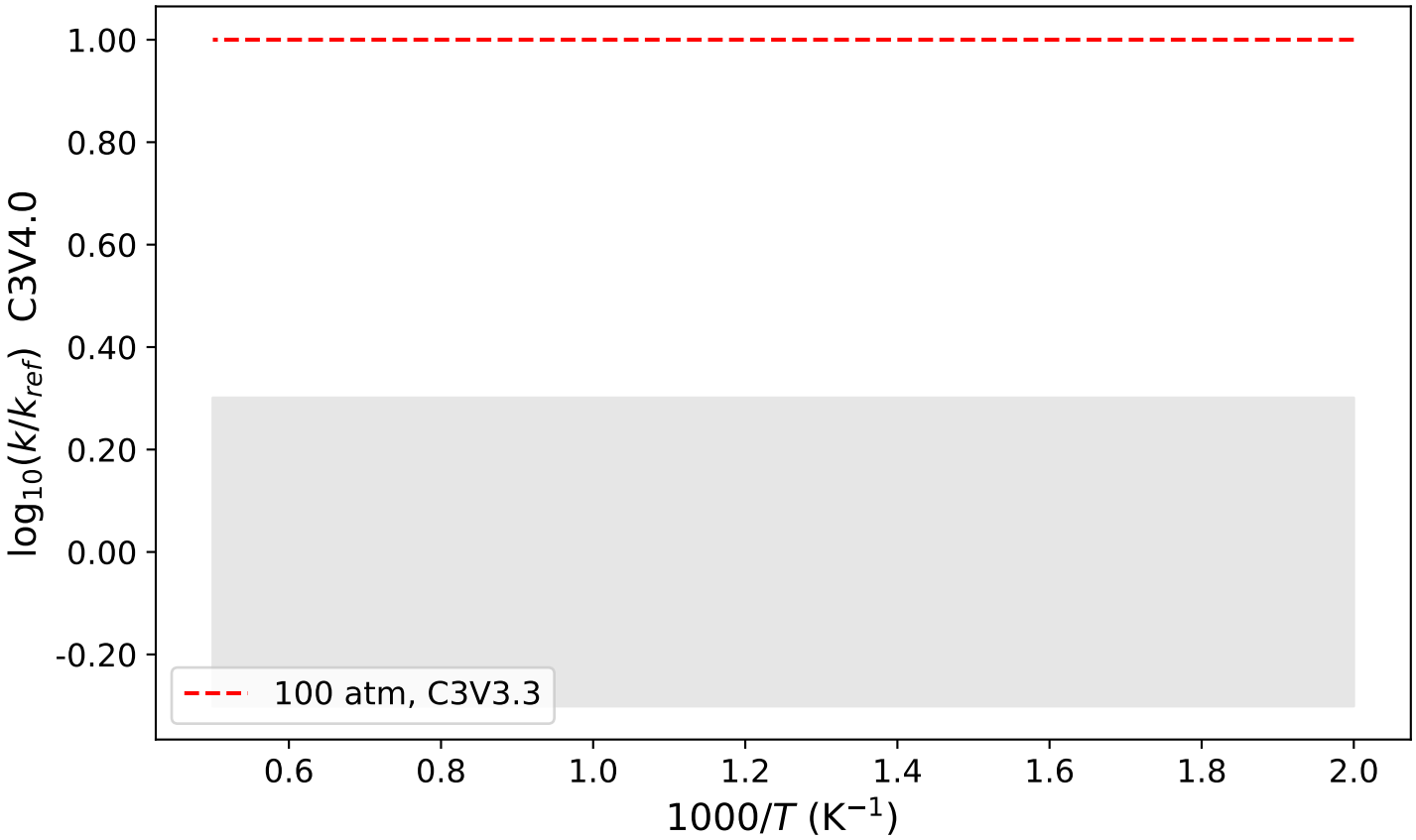
C2H3 + NAPH = C2H4 + NAPH-A



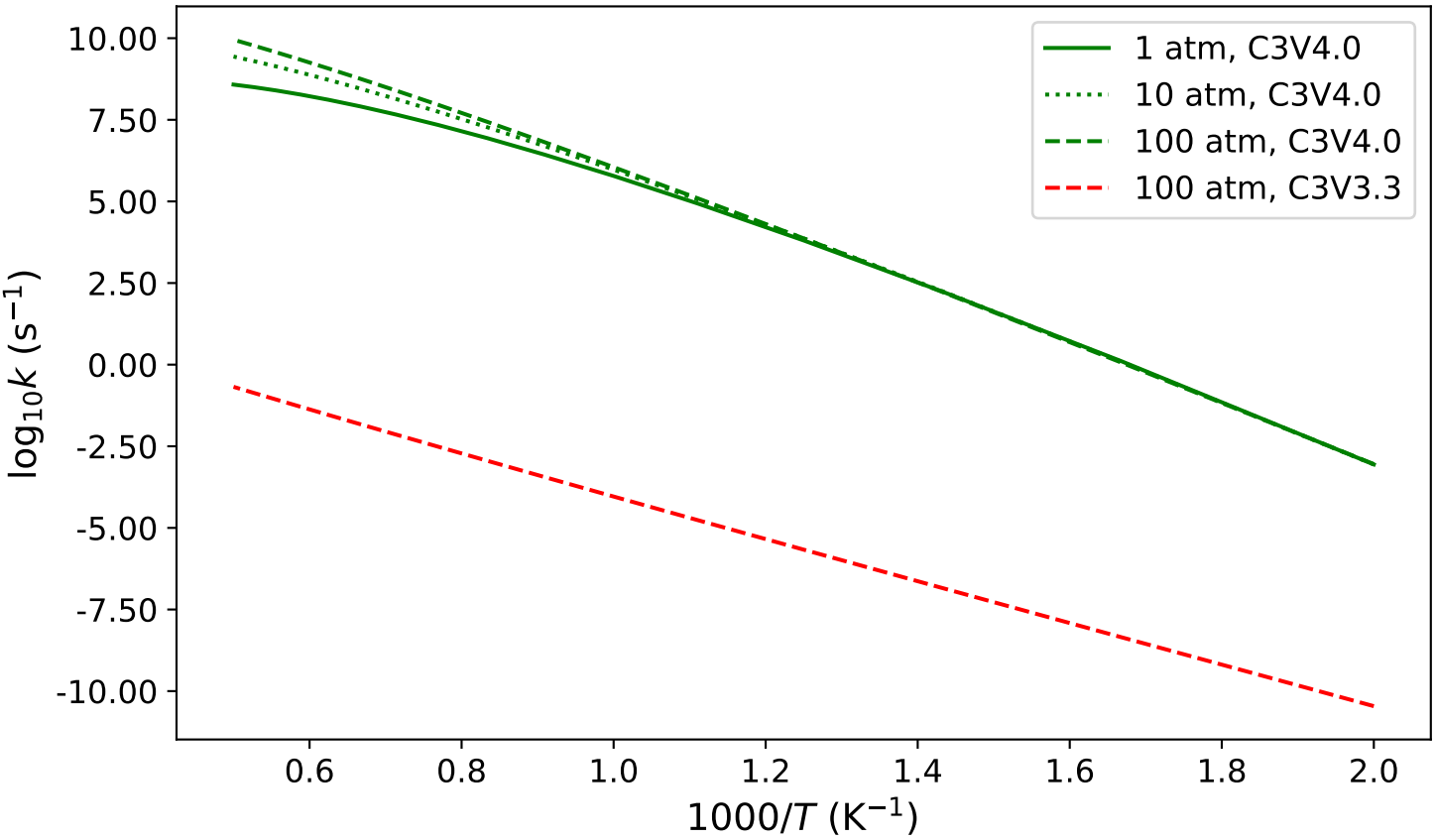


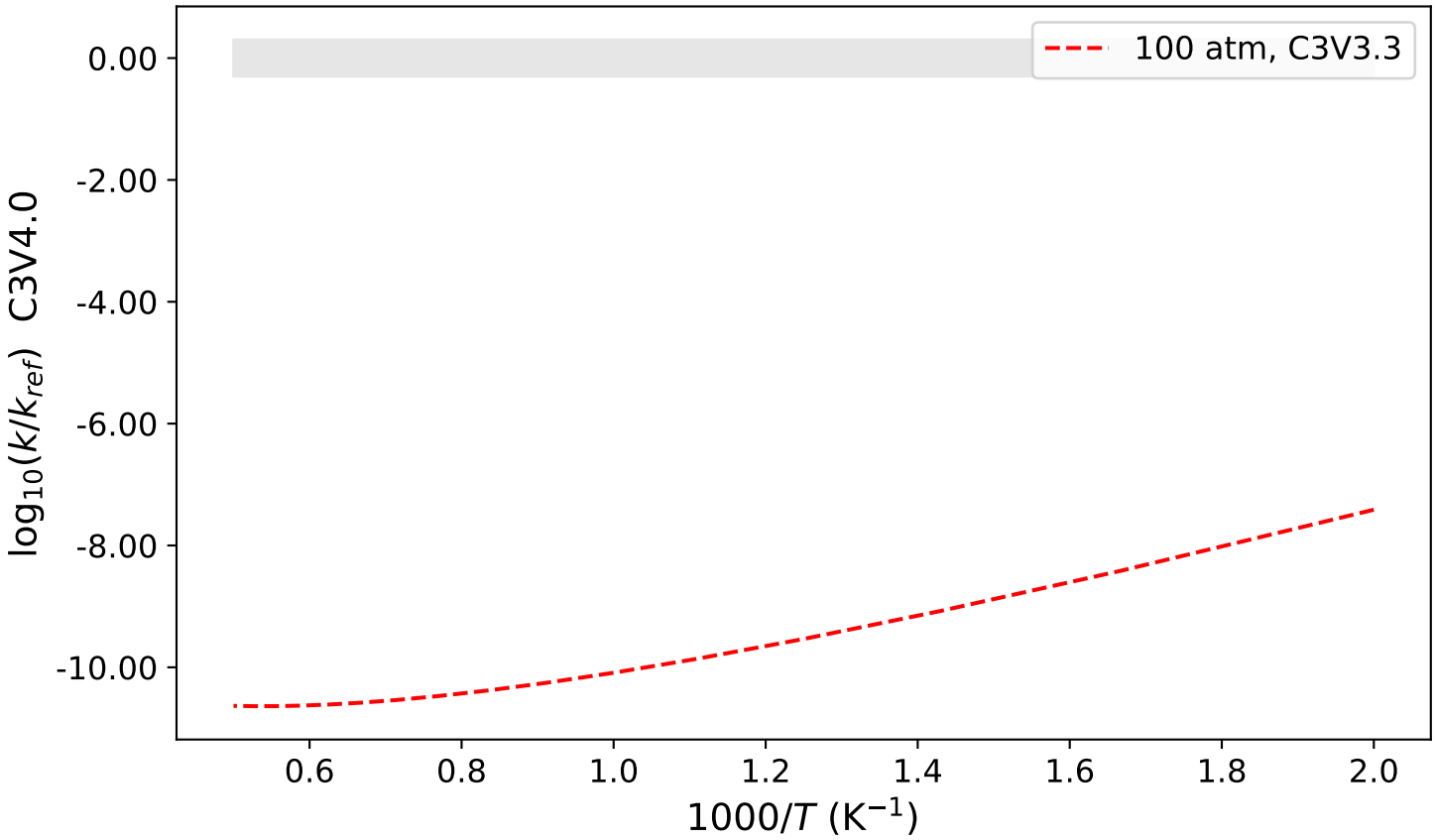
C4H71-3 + C4H71-3 => H + CH3 + H2 + C6H5CH3



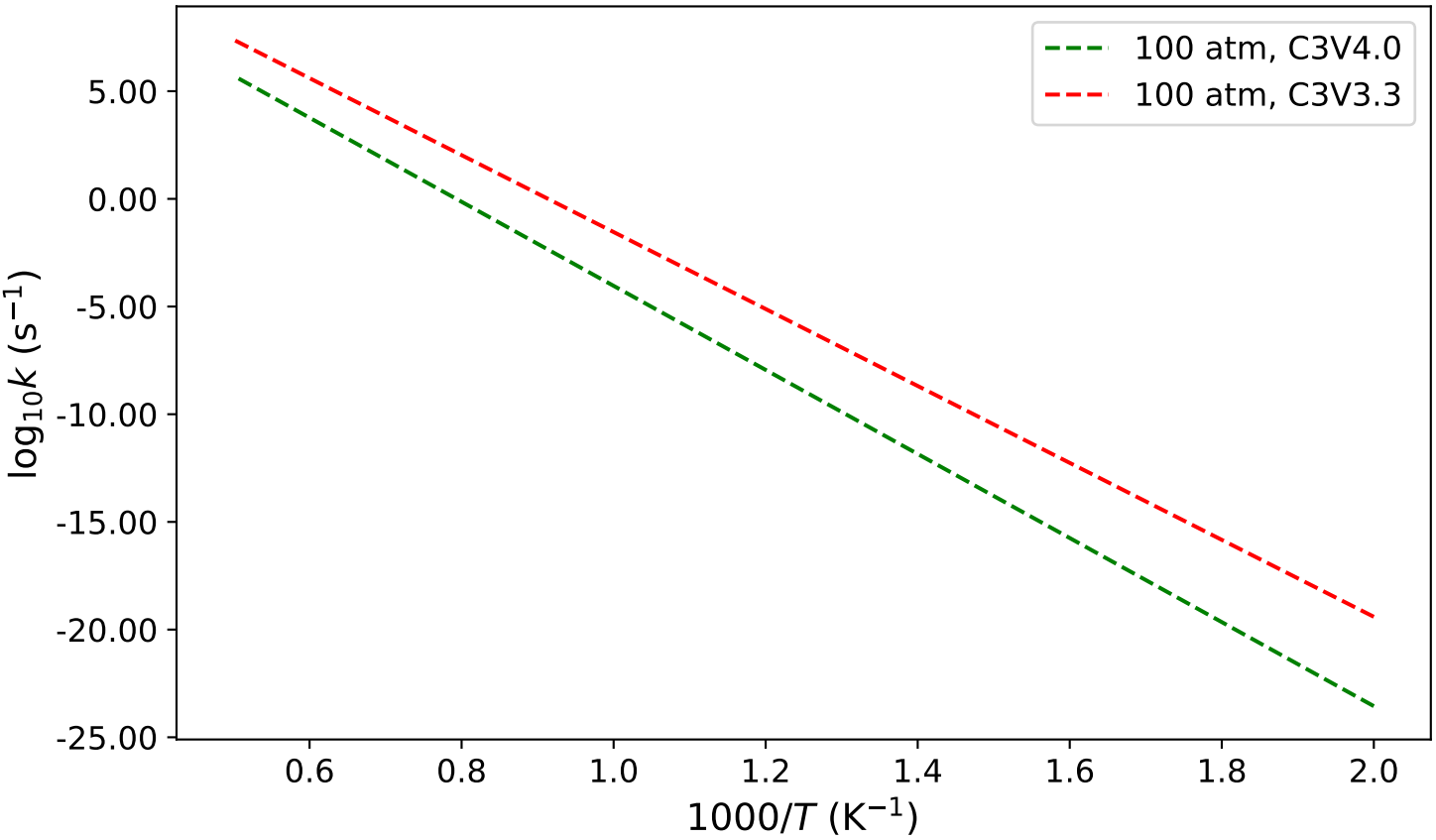


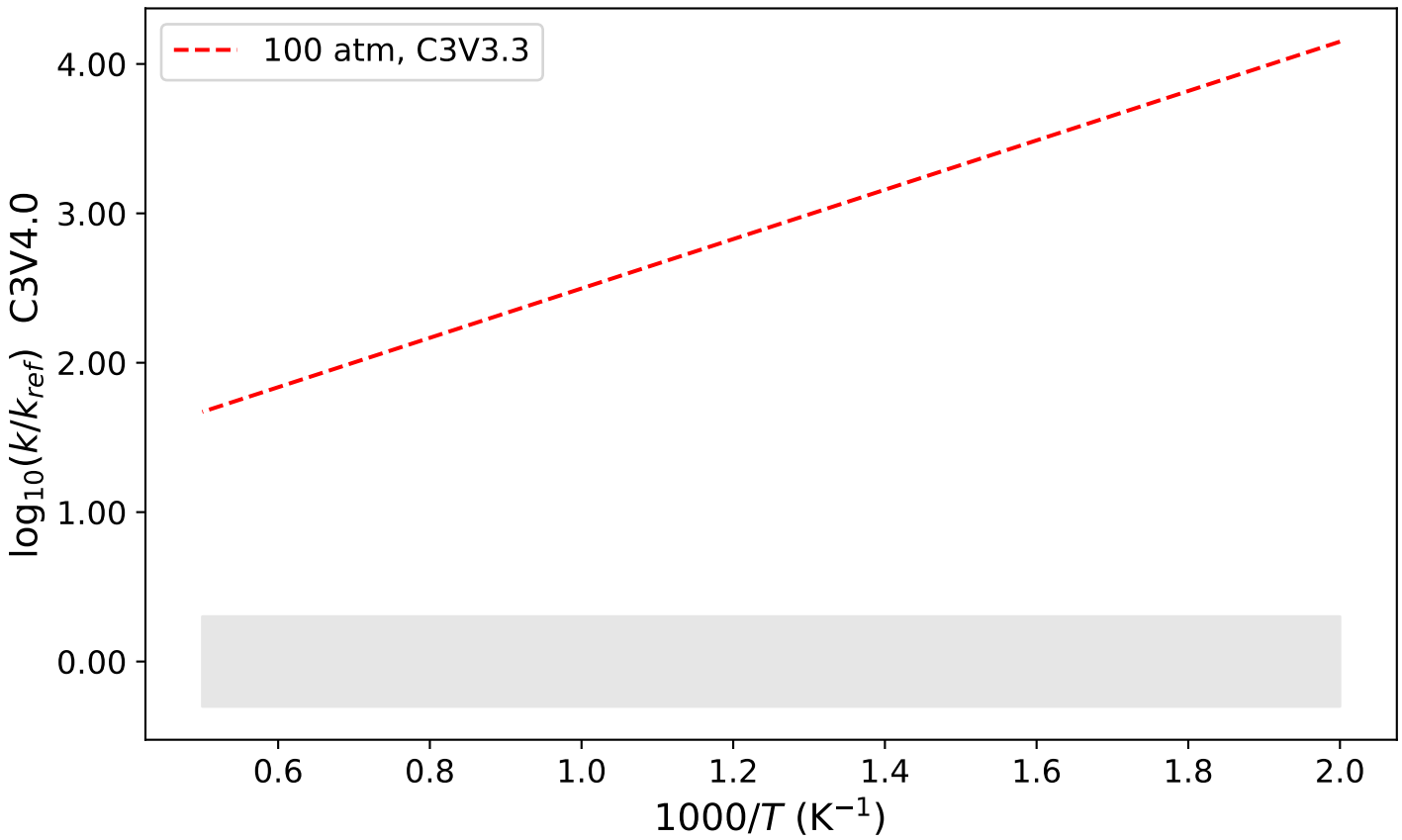
C10H7CH2OOH-A = OH + C10H7CH2O-A



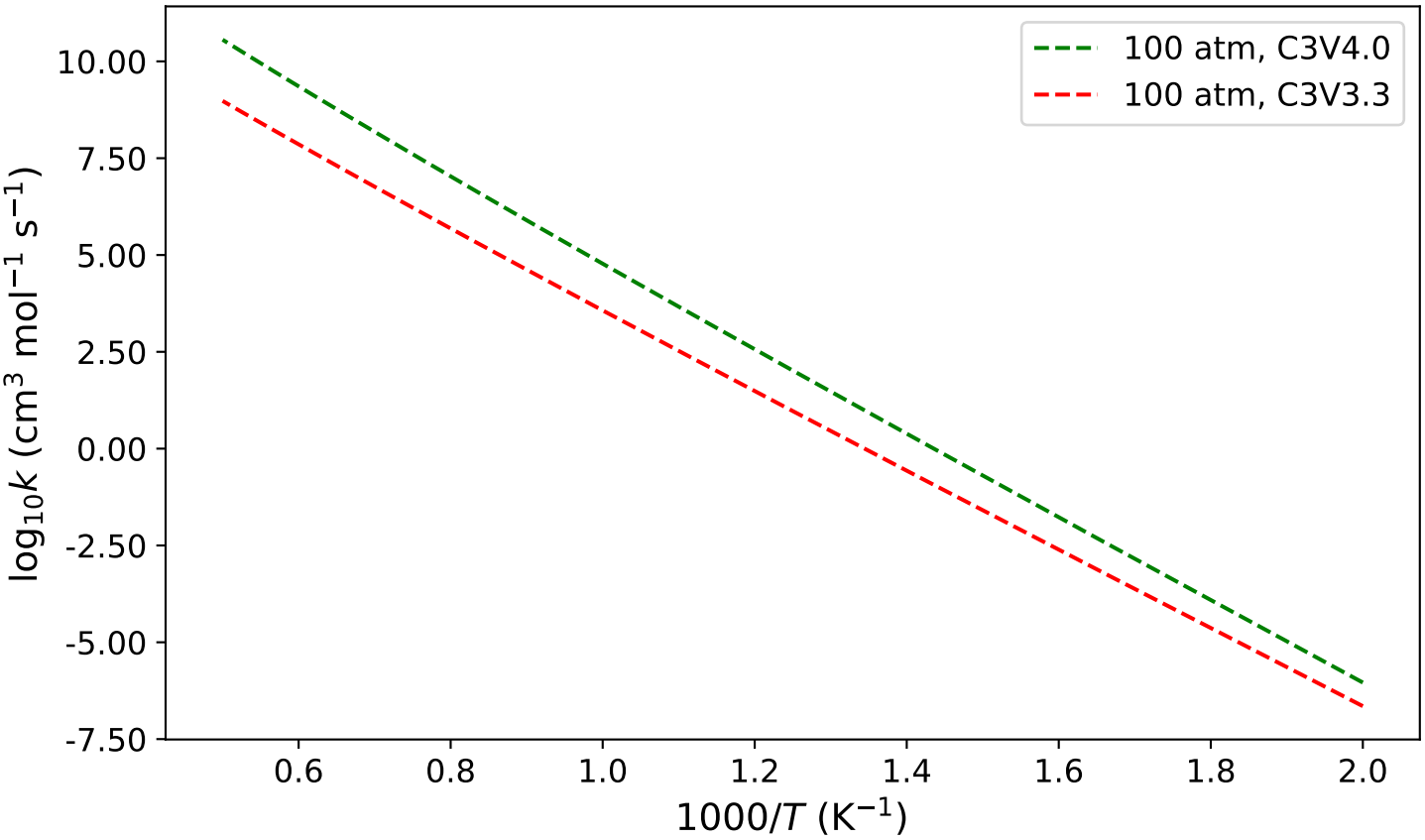


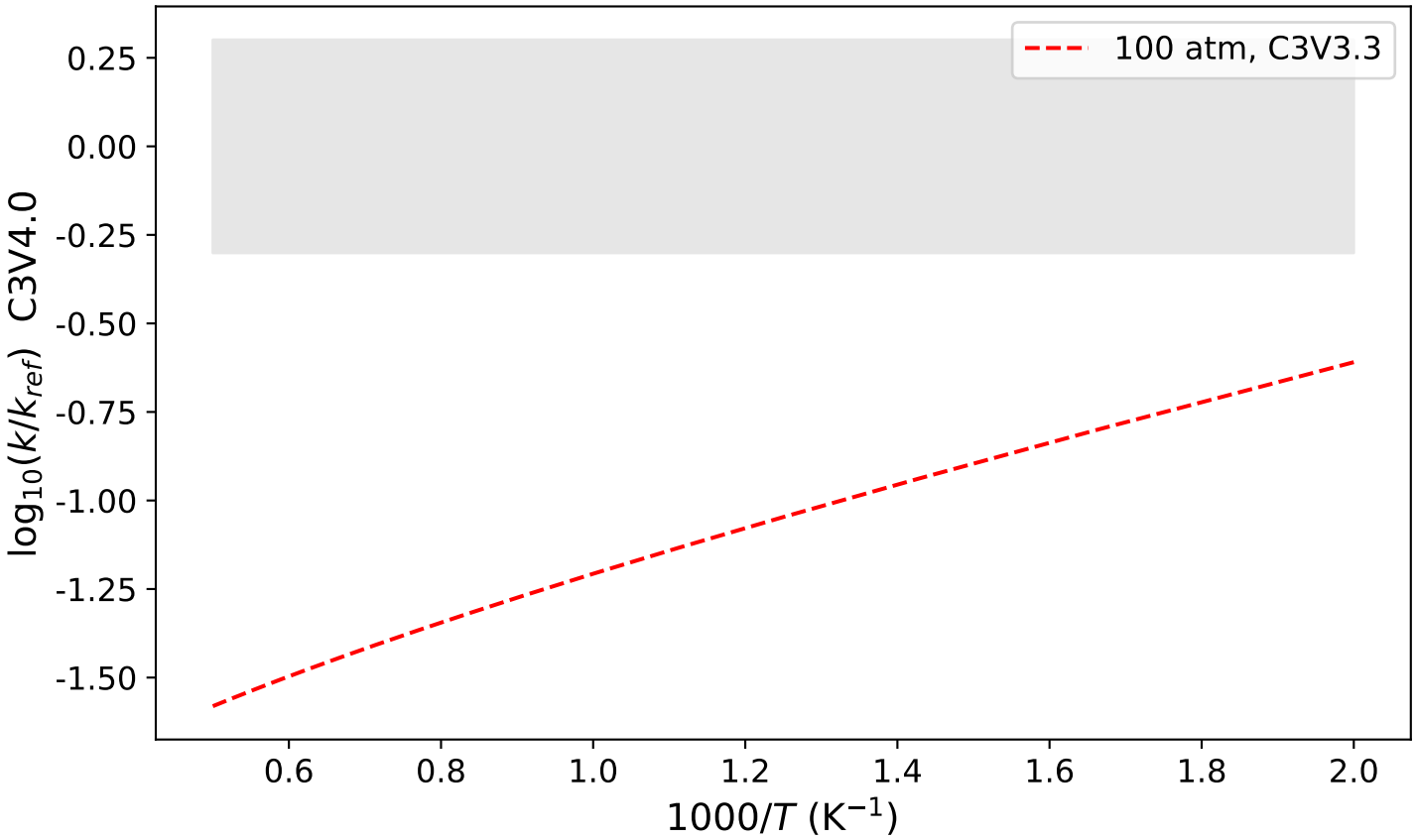
C6H4CH3CHO => H + CO + C6H4CH3



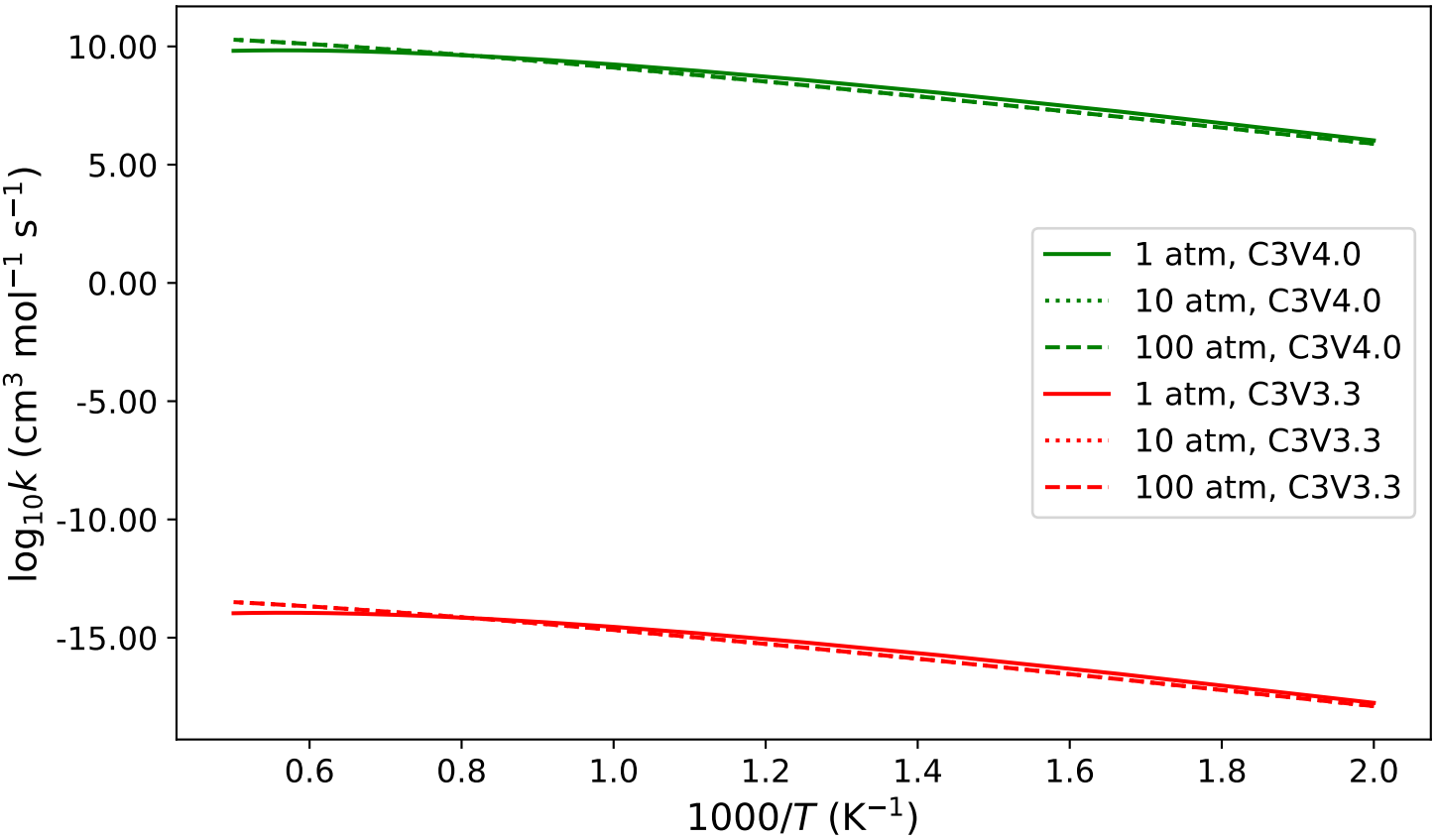


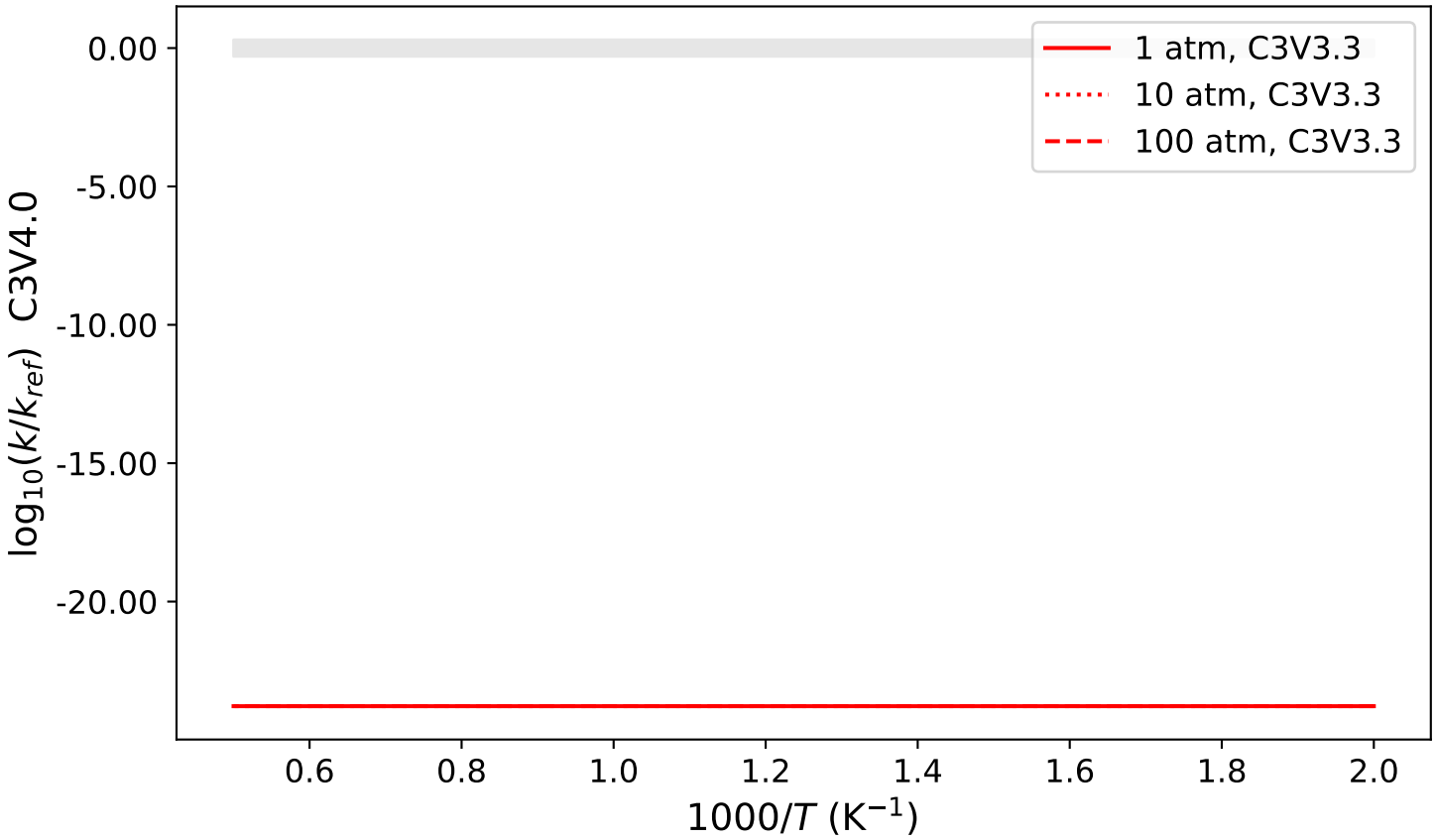
O2 + C14H14 = HO2 + C14H13



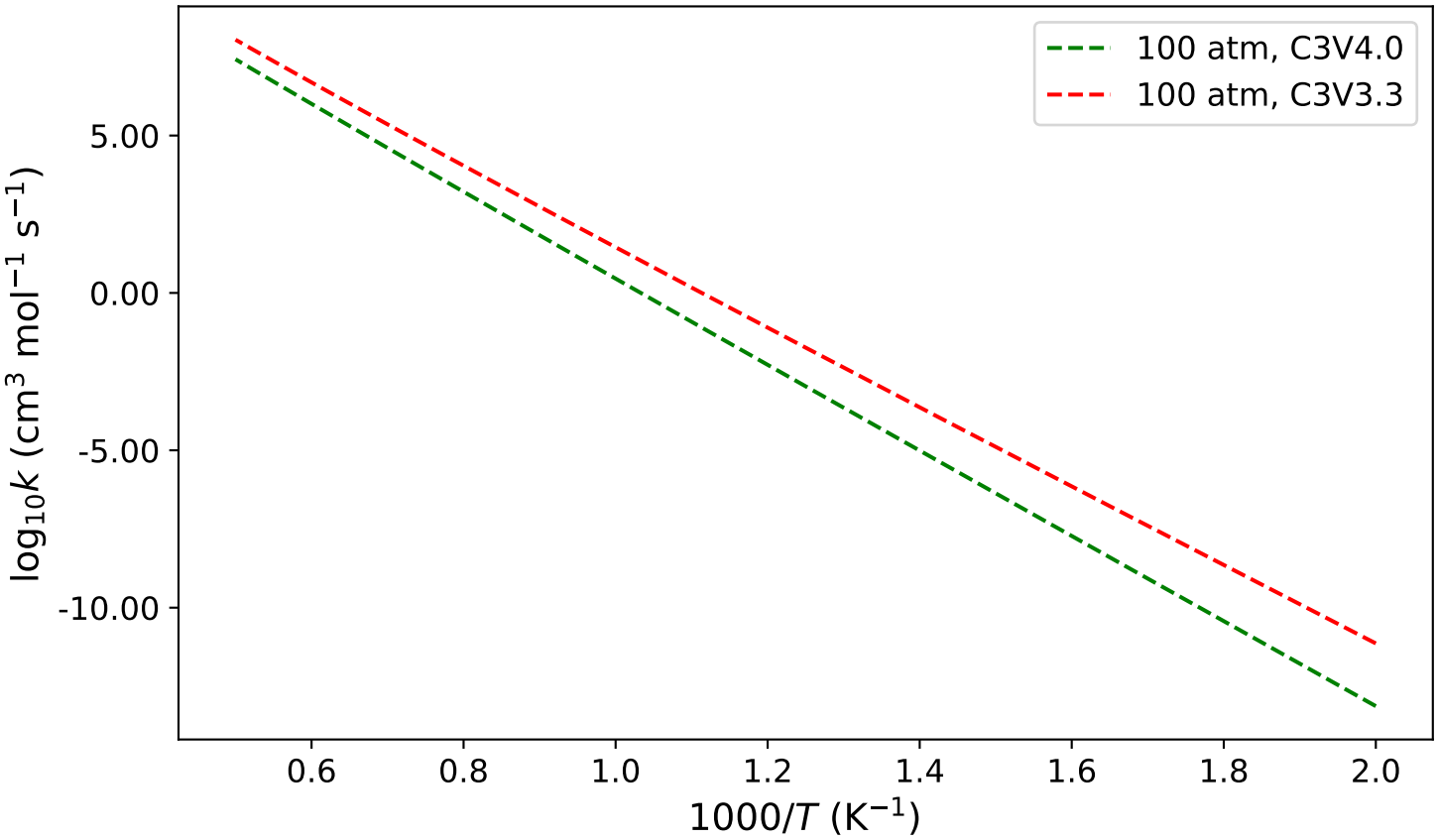


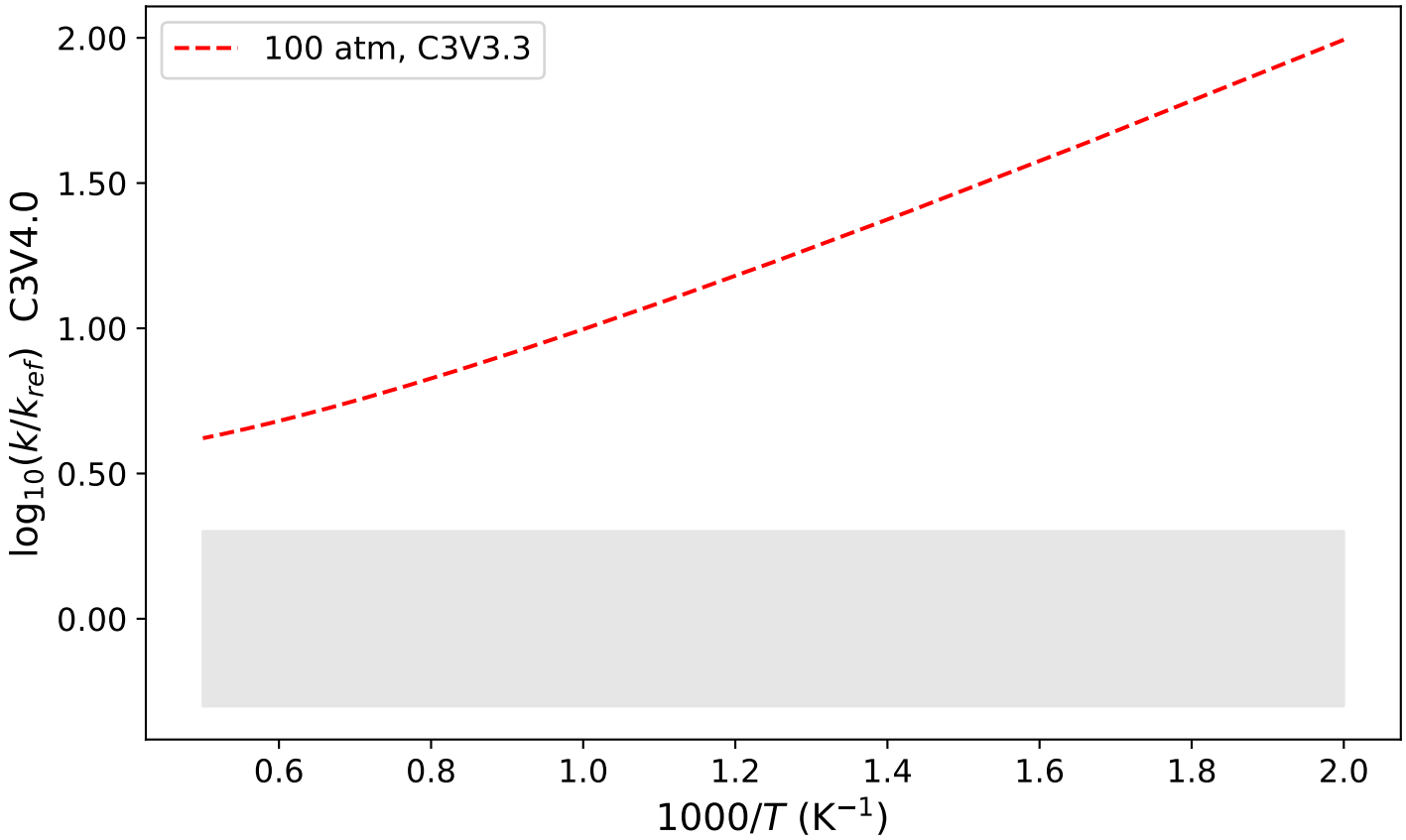
C2H2 + C5H5 = C6H5CH2



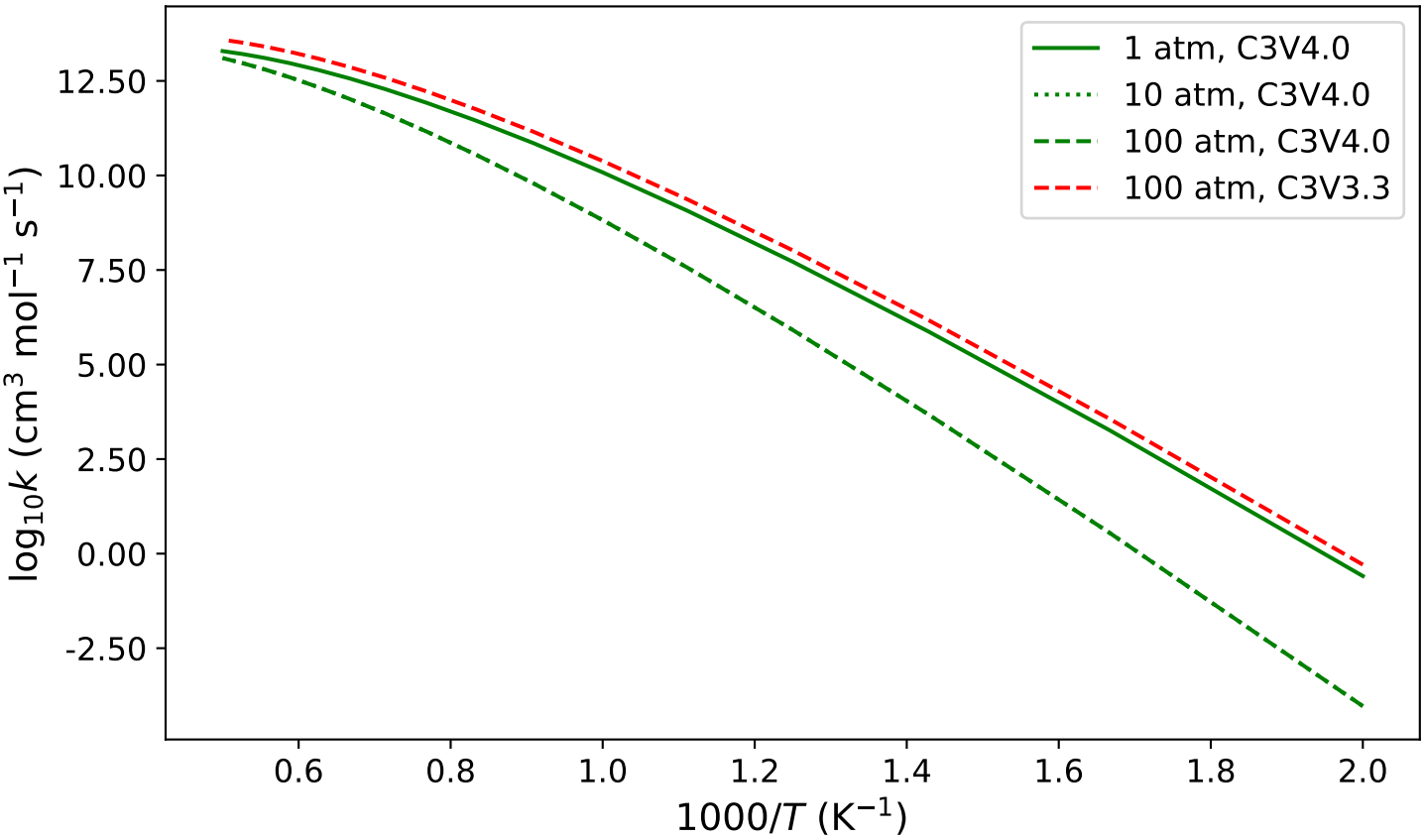


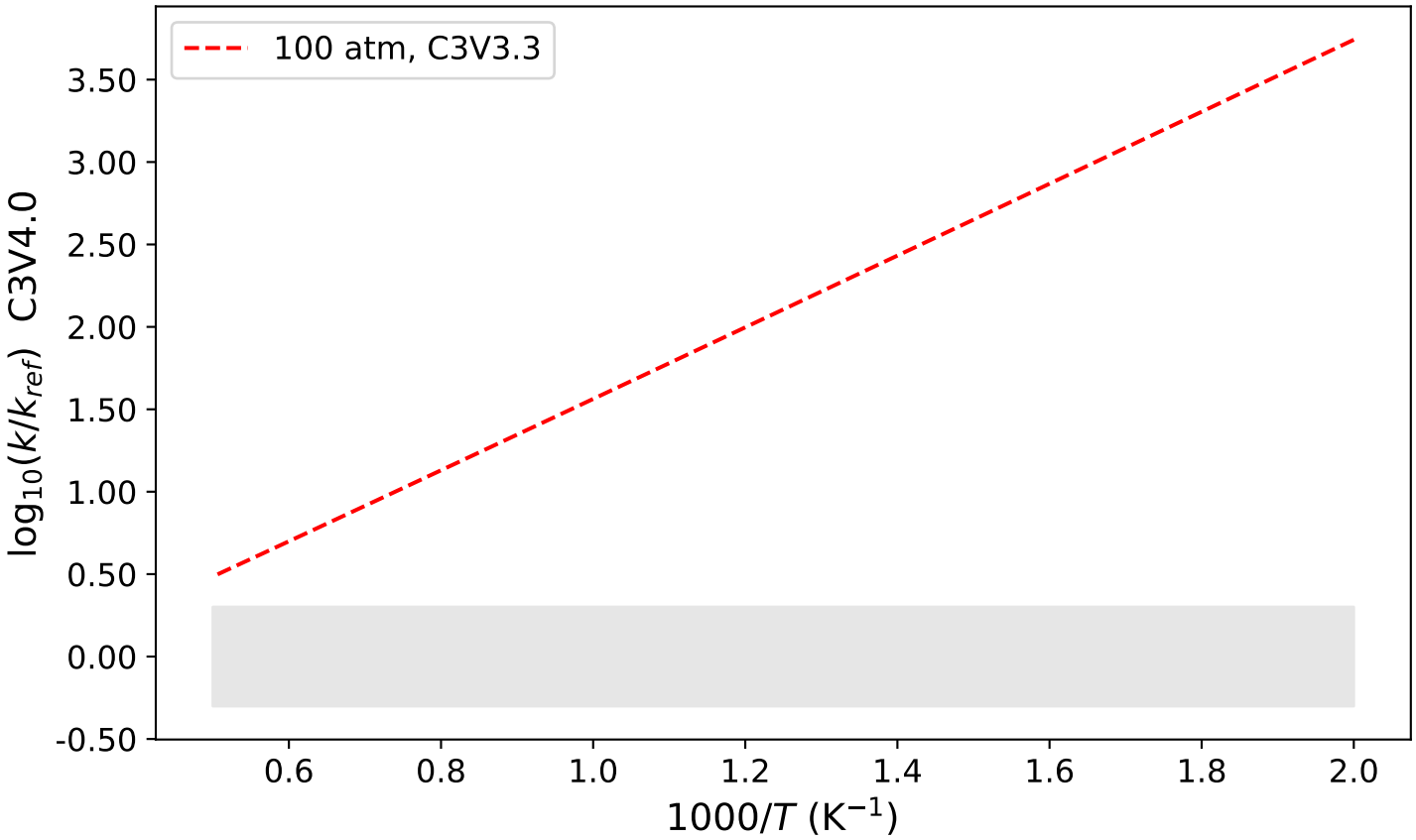
O2 + NAPH = HO2 + NAPH-A



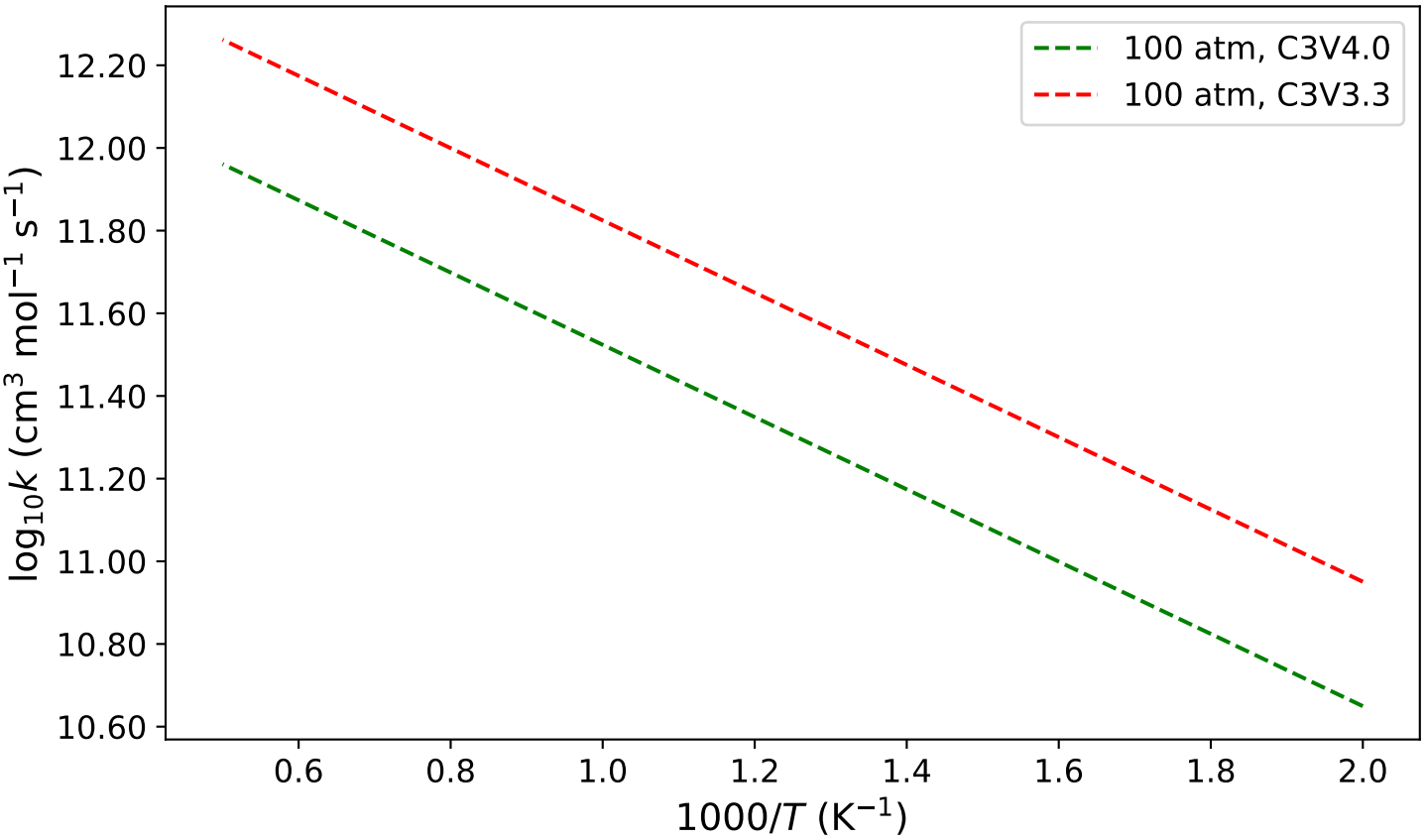


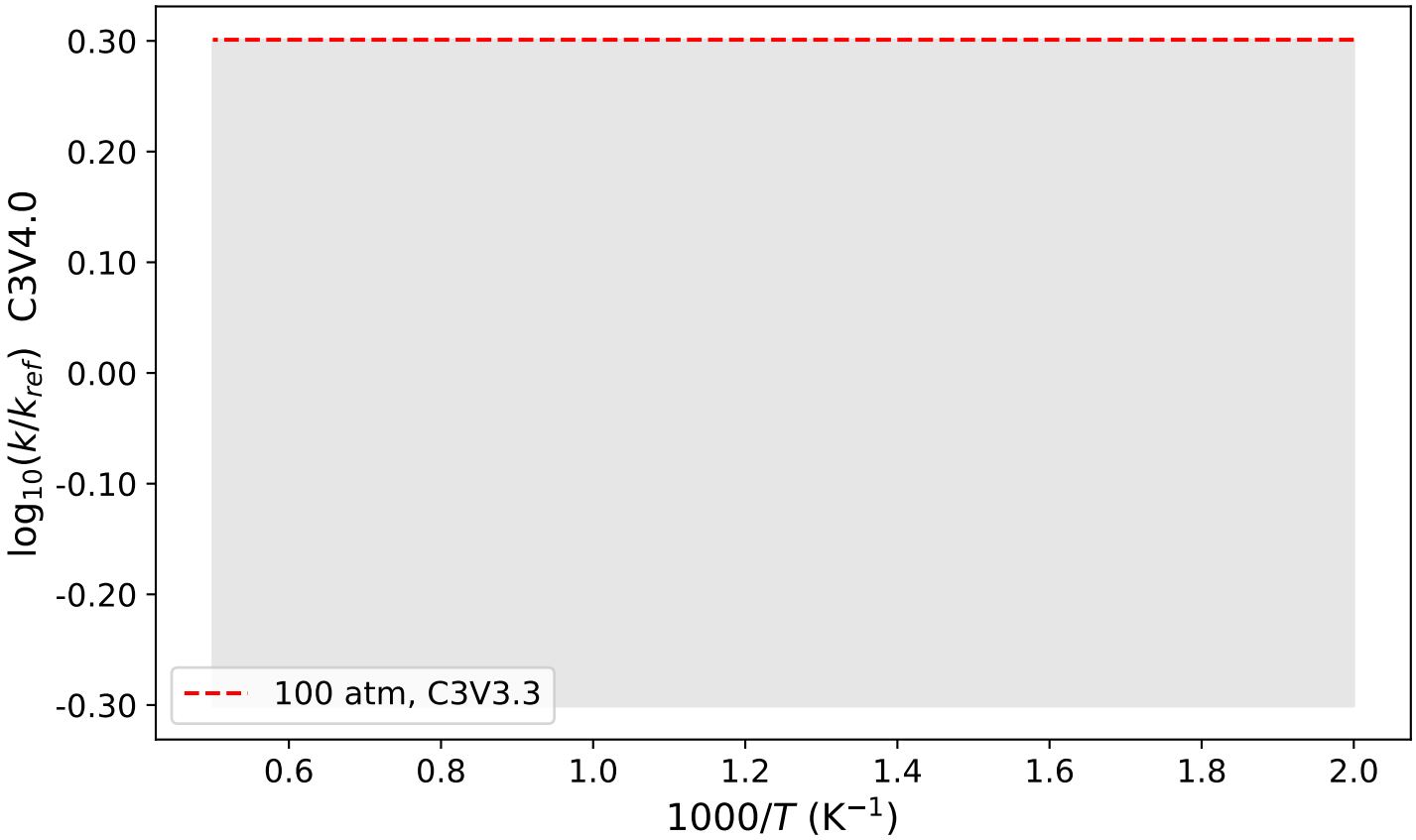
H + RXYLENE = CH3 + C6H4CH3



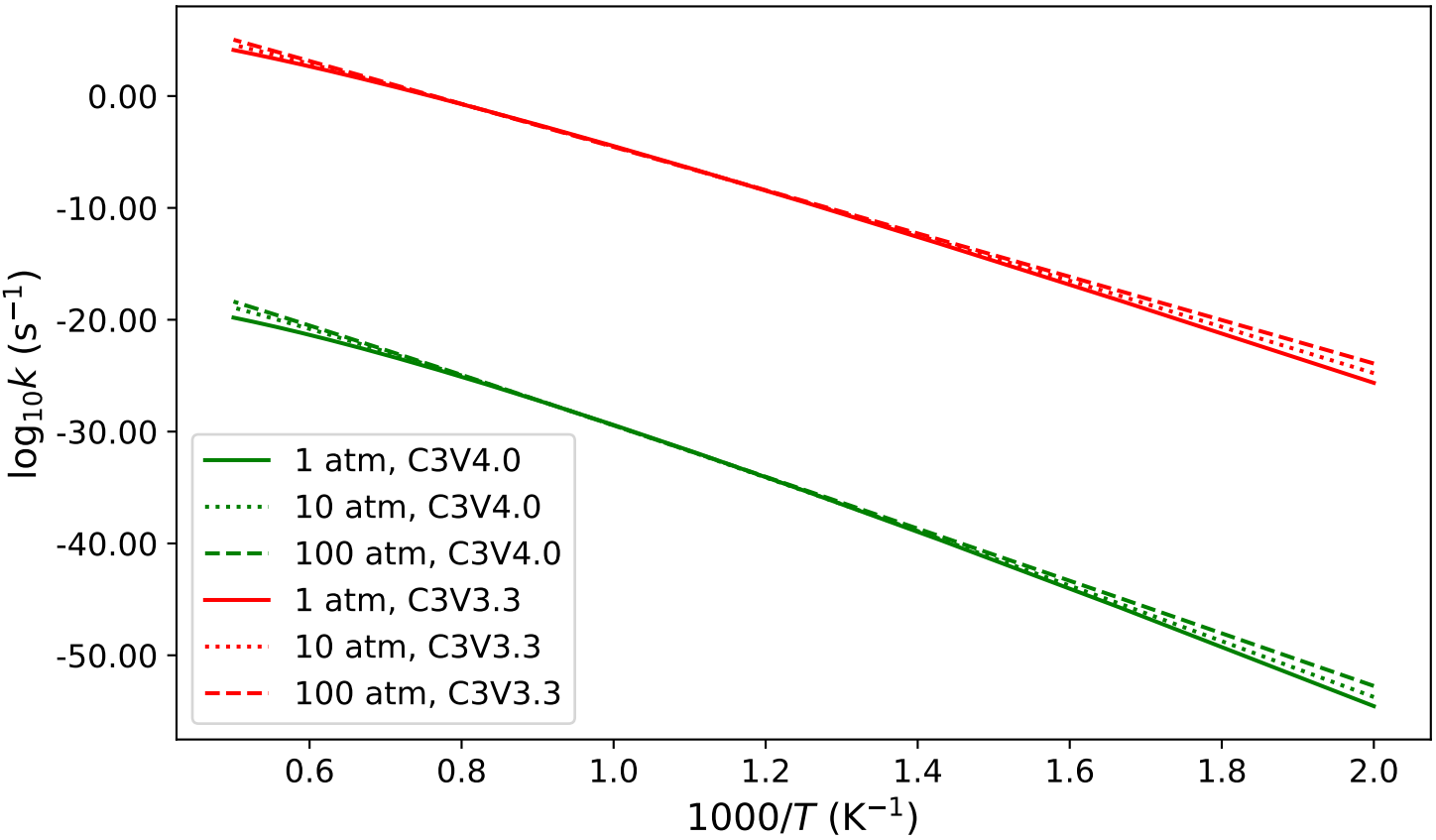


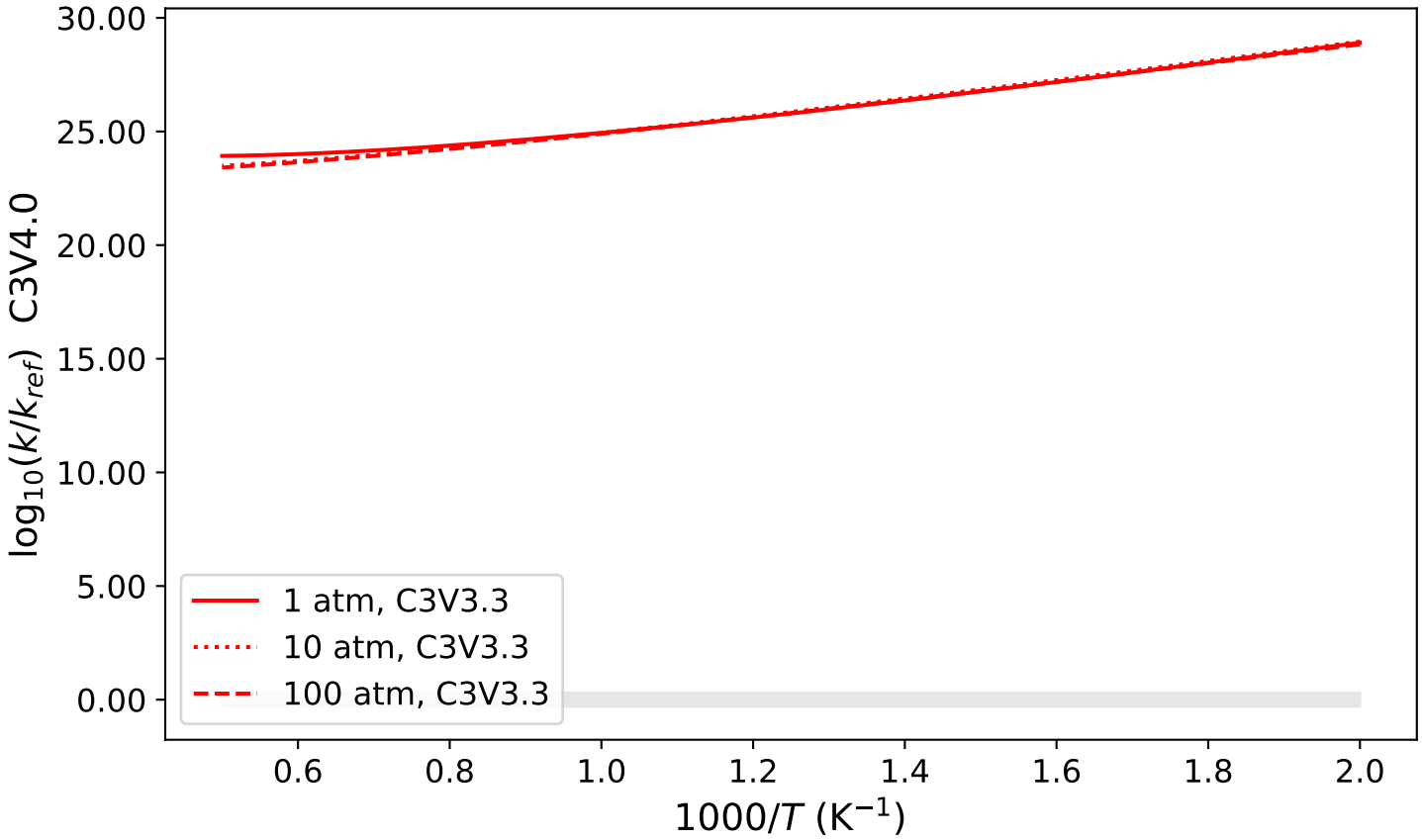
C4H5-I + C6H5CHCH3 => H + CH3 + C10H7CH3-A



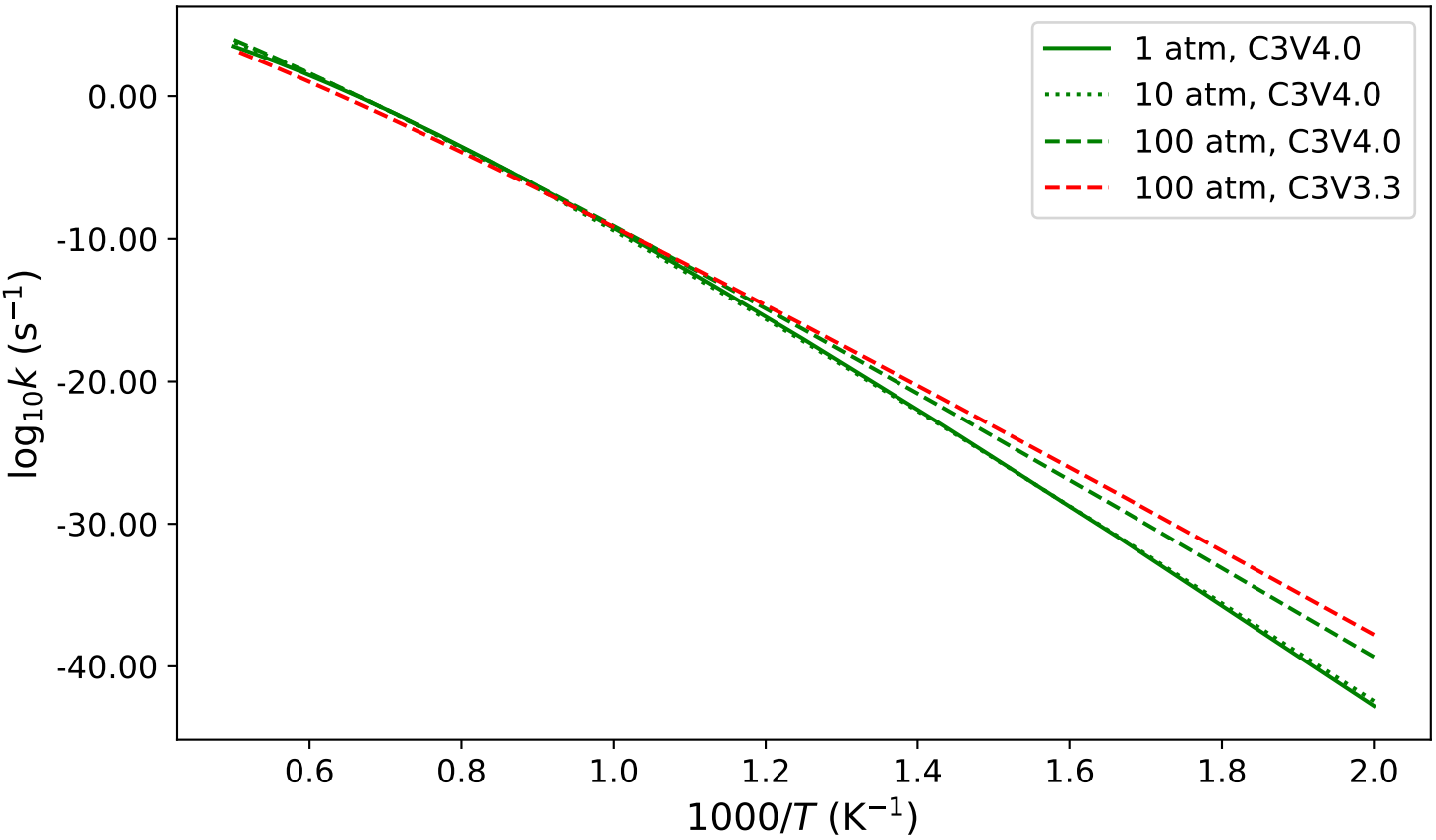


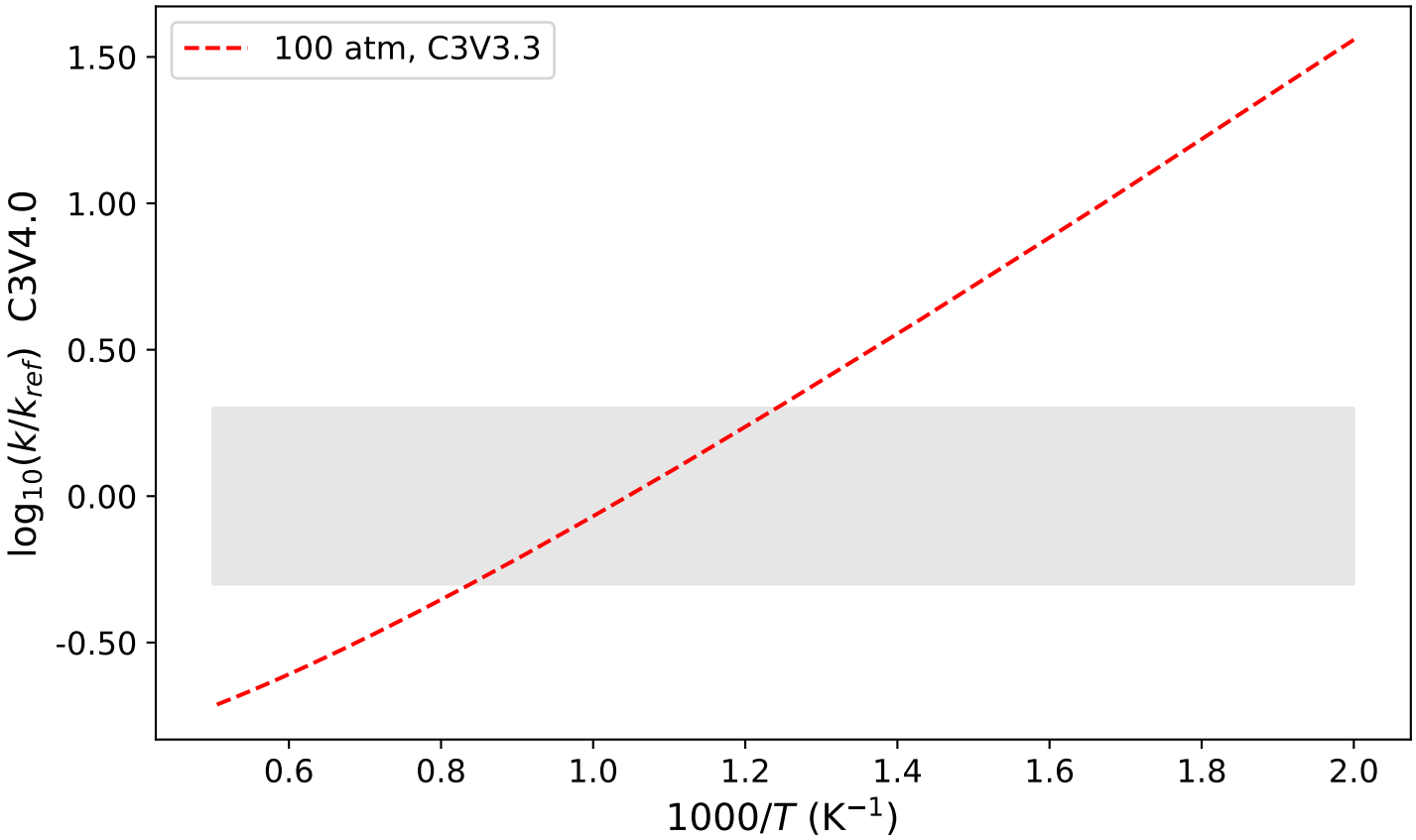
C6H5C2H5 = H + C6H5CH2CH2



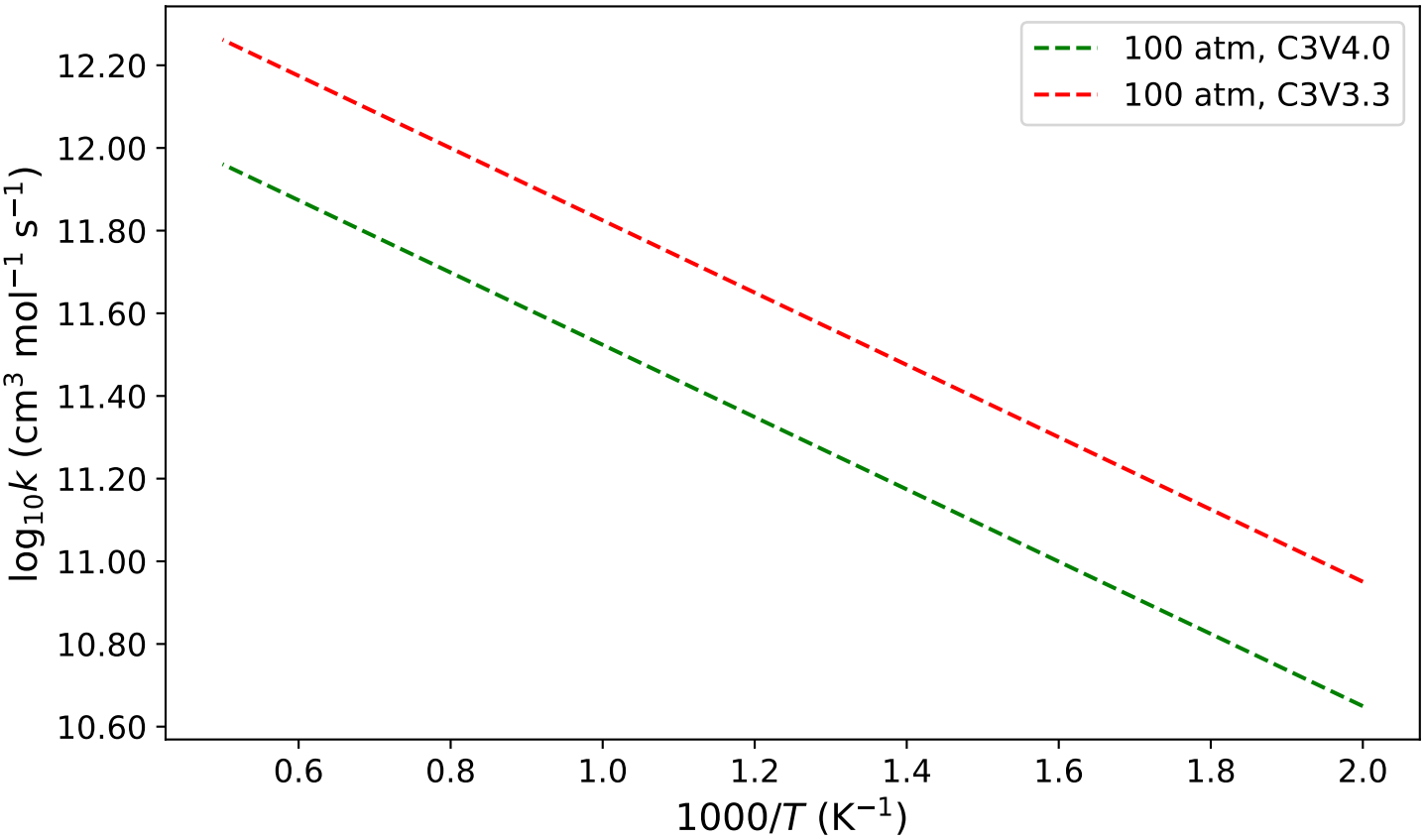


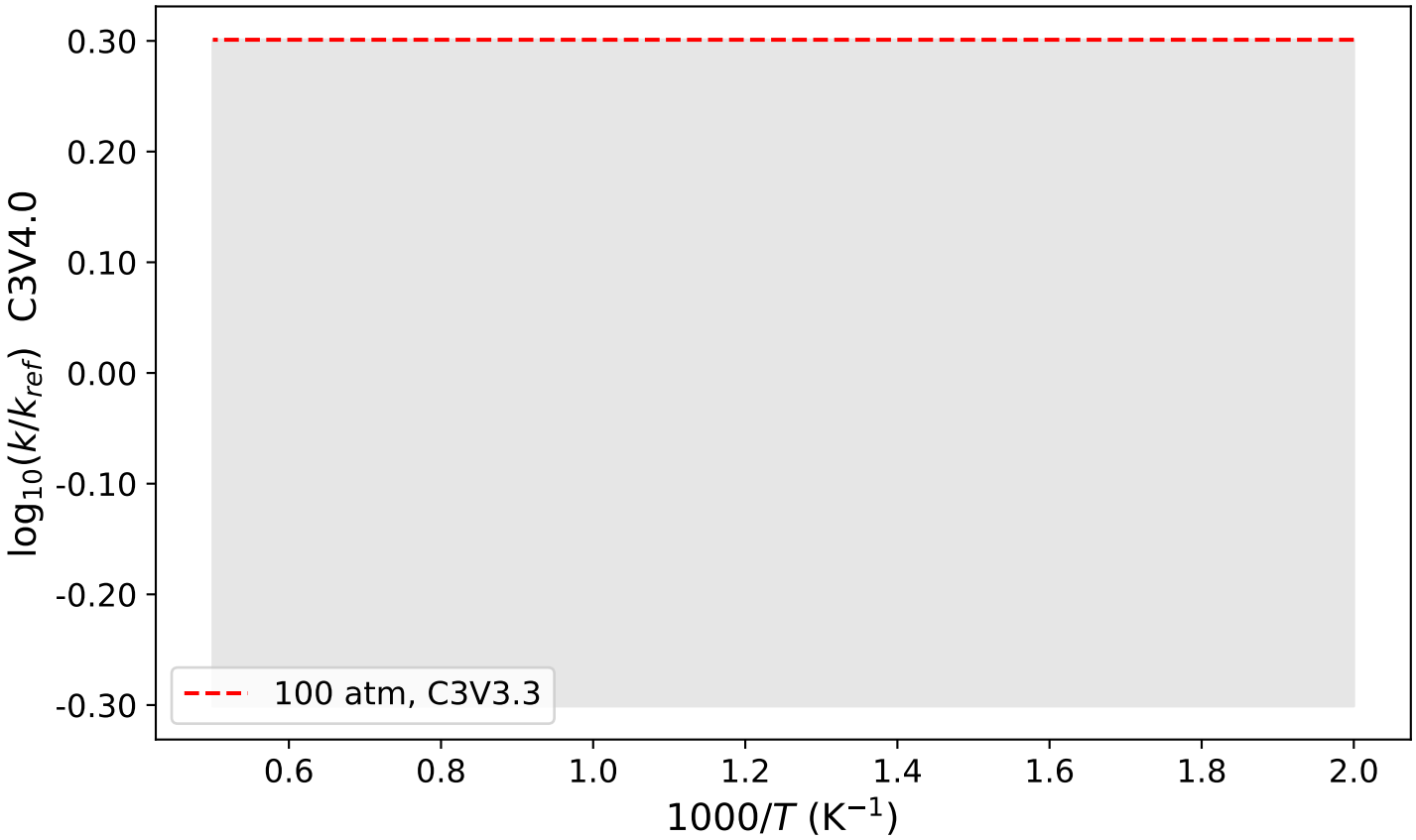
C6H5C2H3 = H + C6H4C2H3



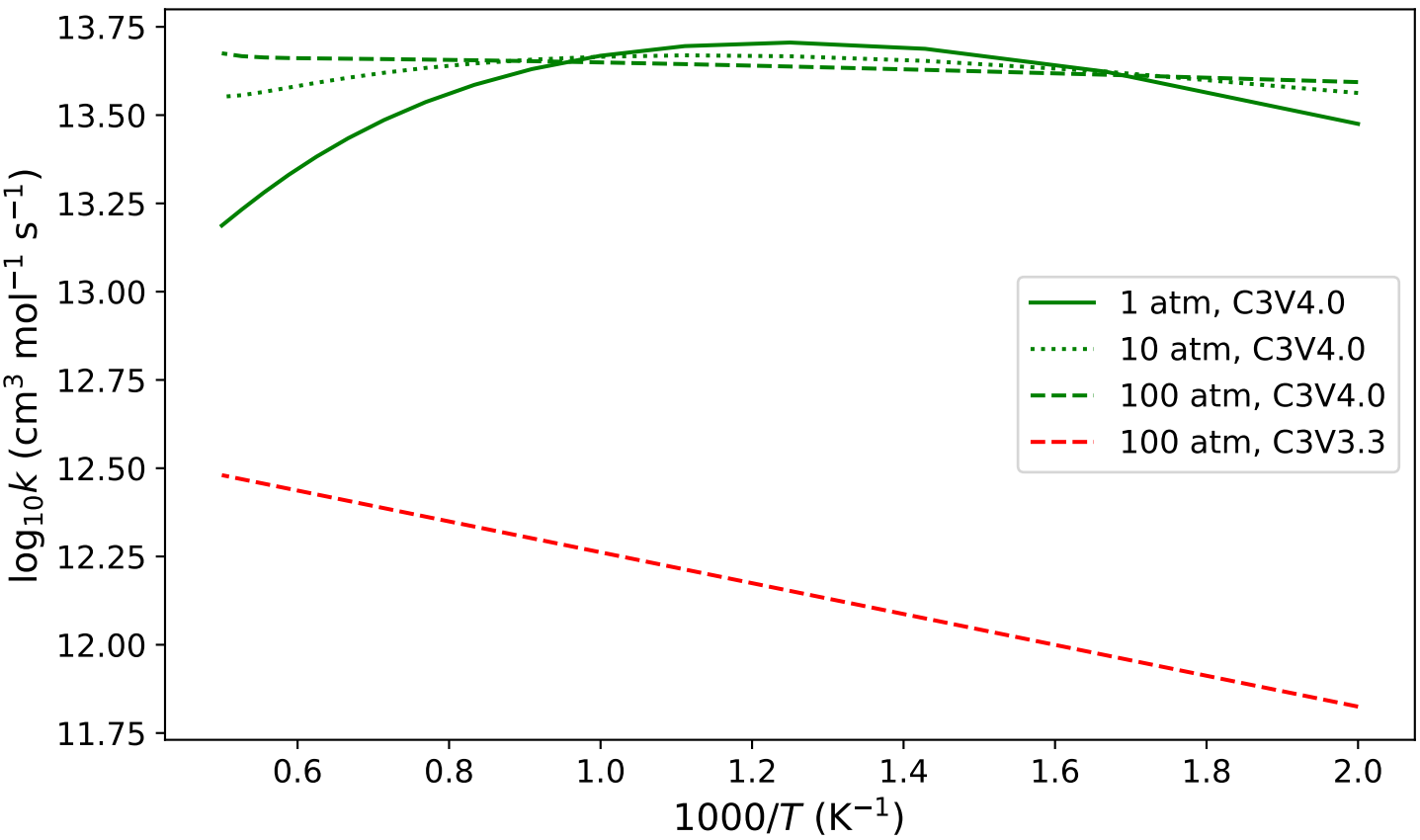


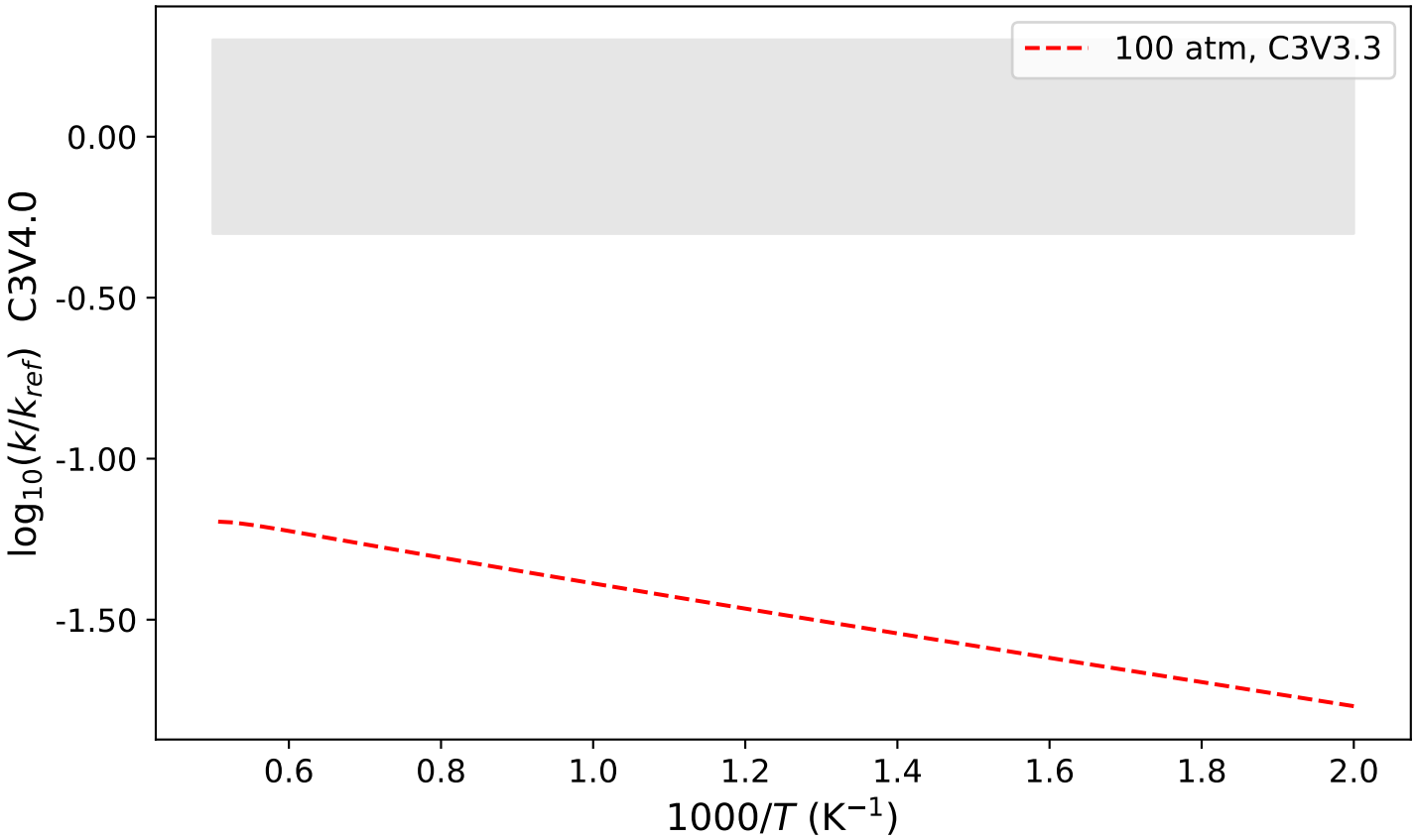
C4H5-I + C6H5CH2 => H + H + C10H7CH3-A



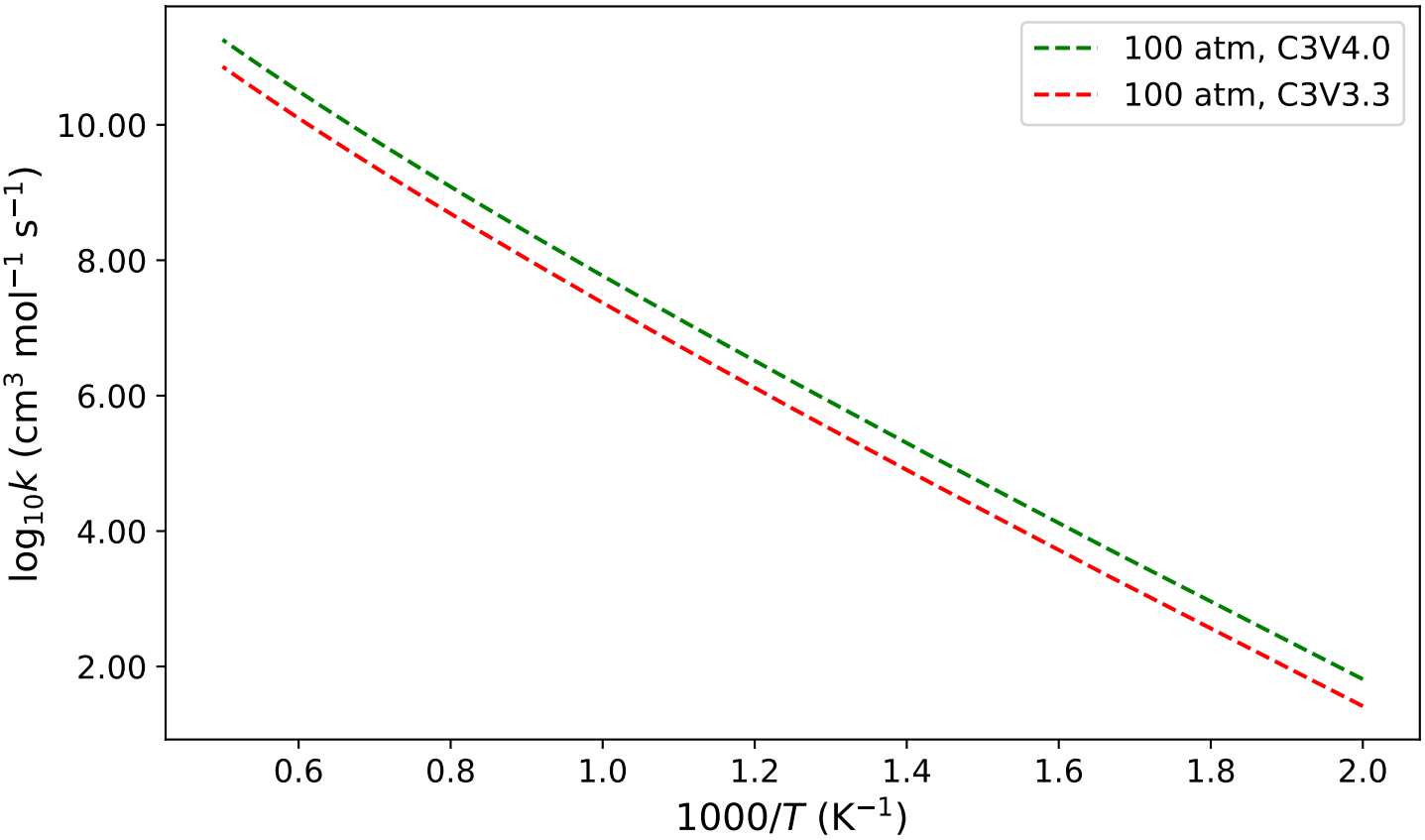


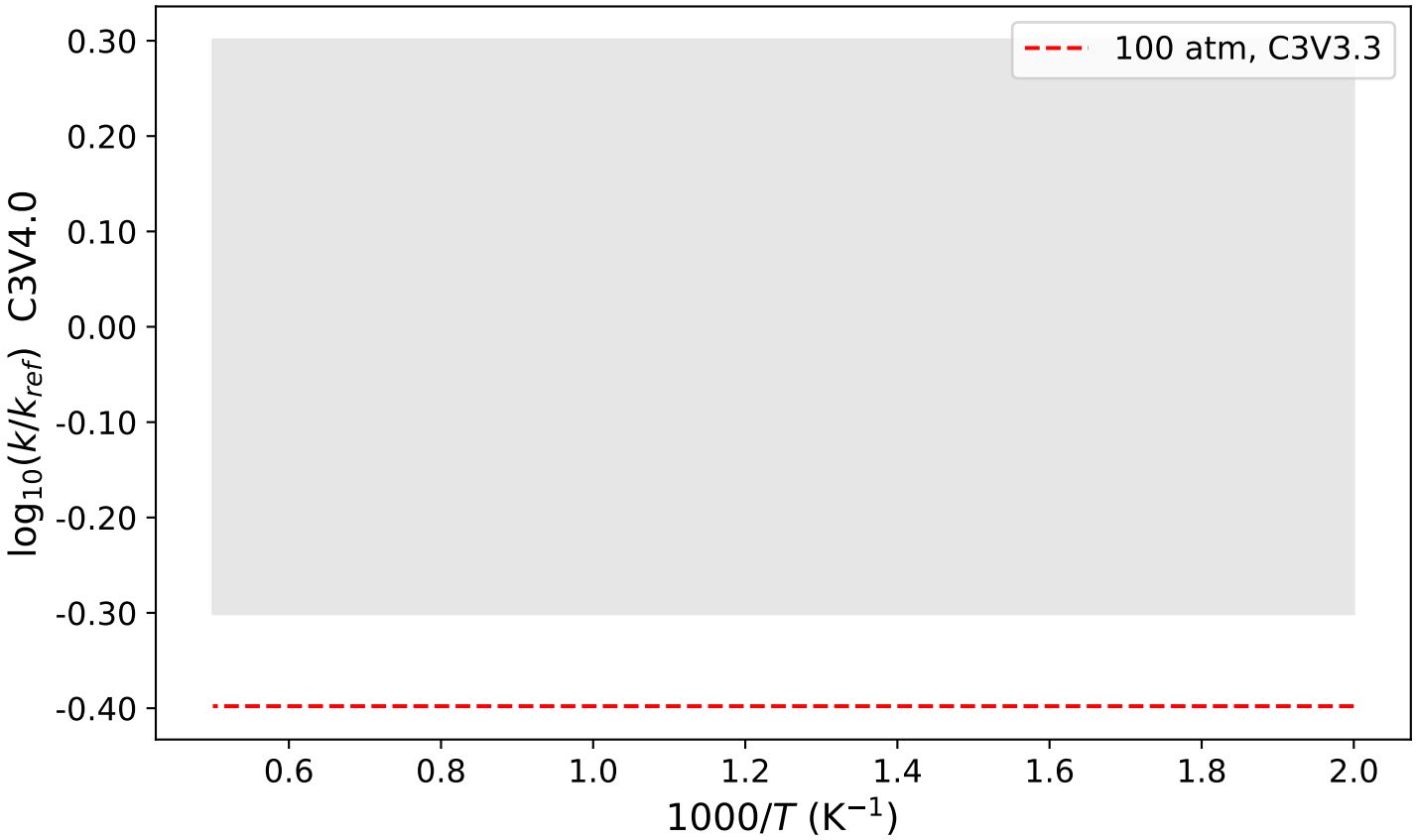
C3H3 + NAPH-A = FLUORENE



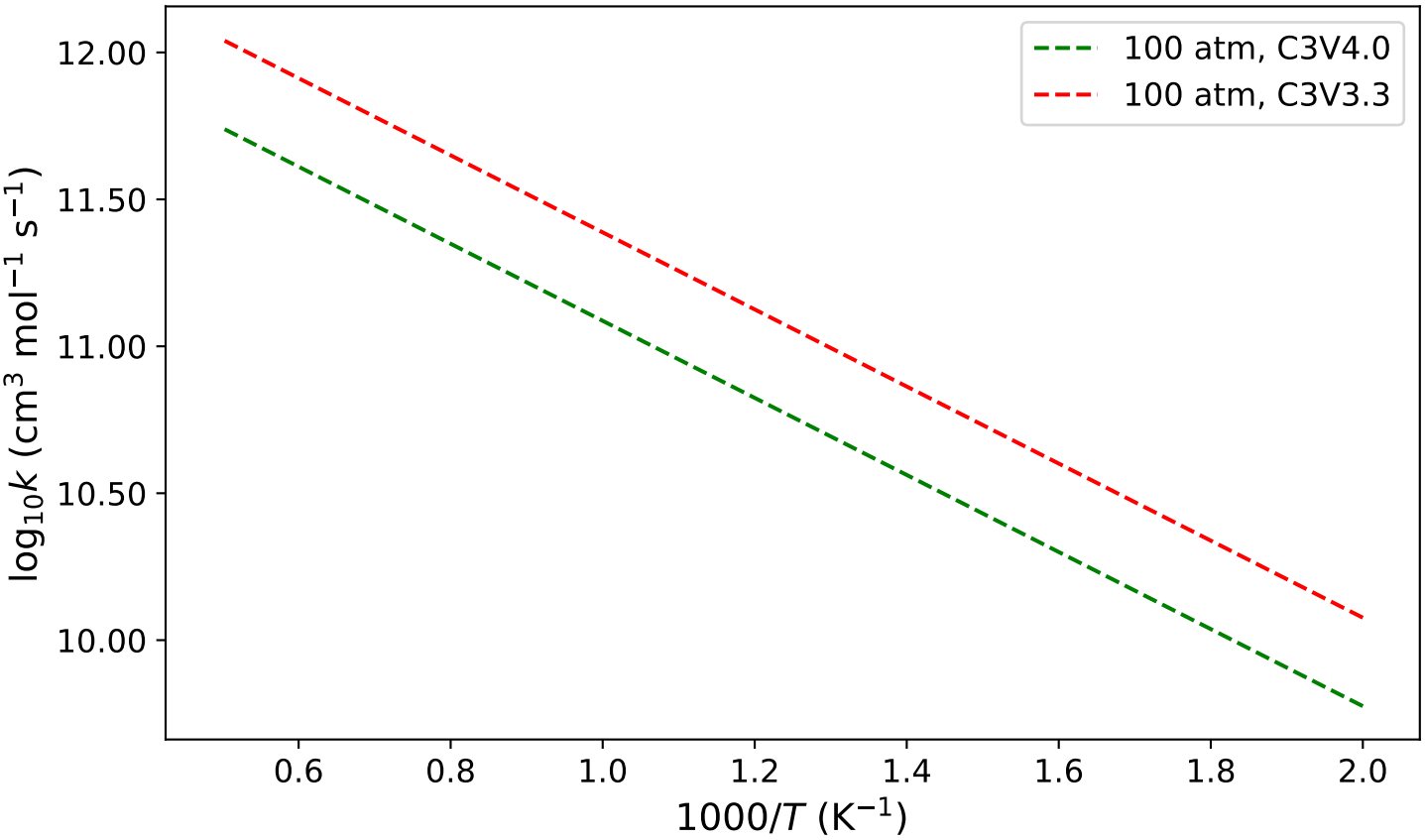


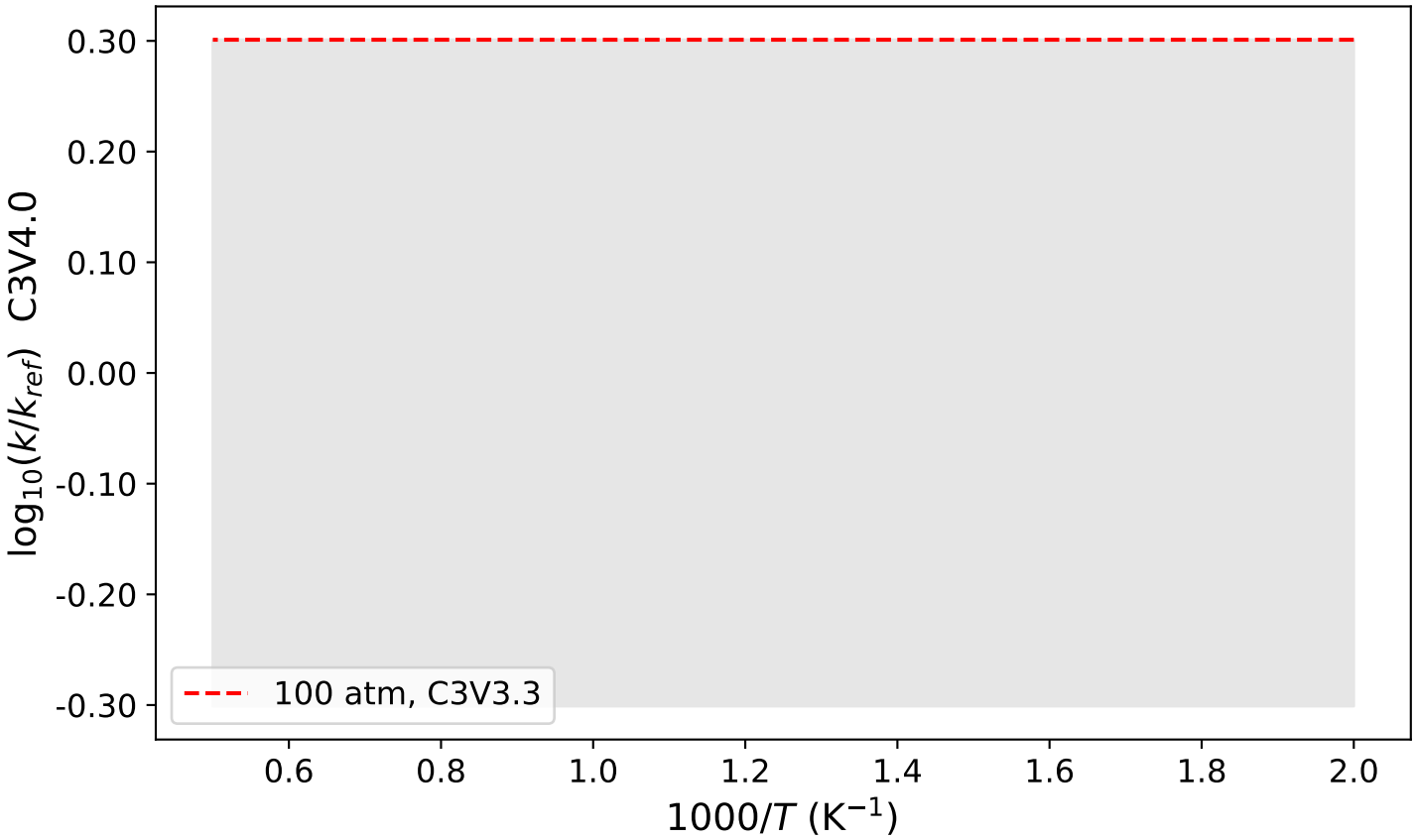
HO2 + C6H5CH3 = H2O2 + C6H4CH3



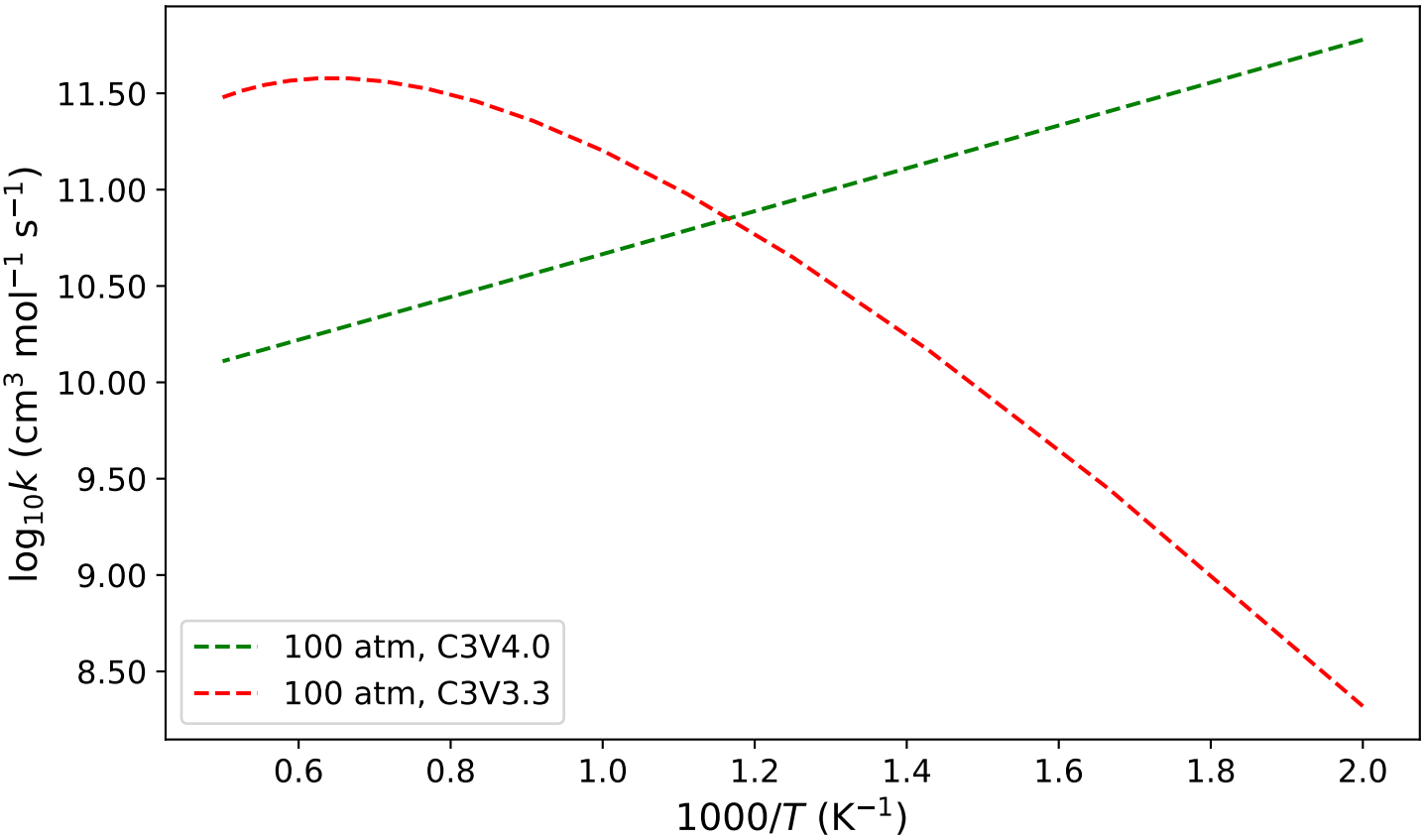


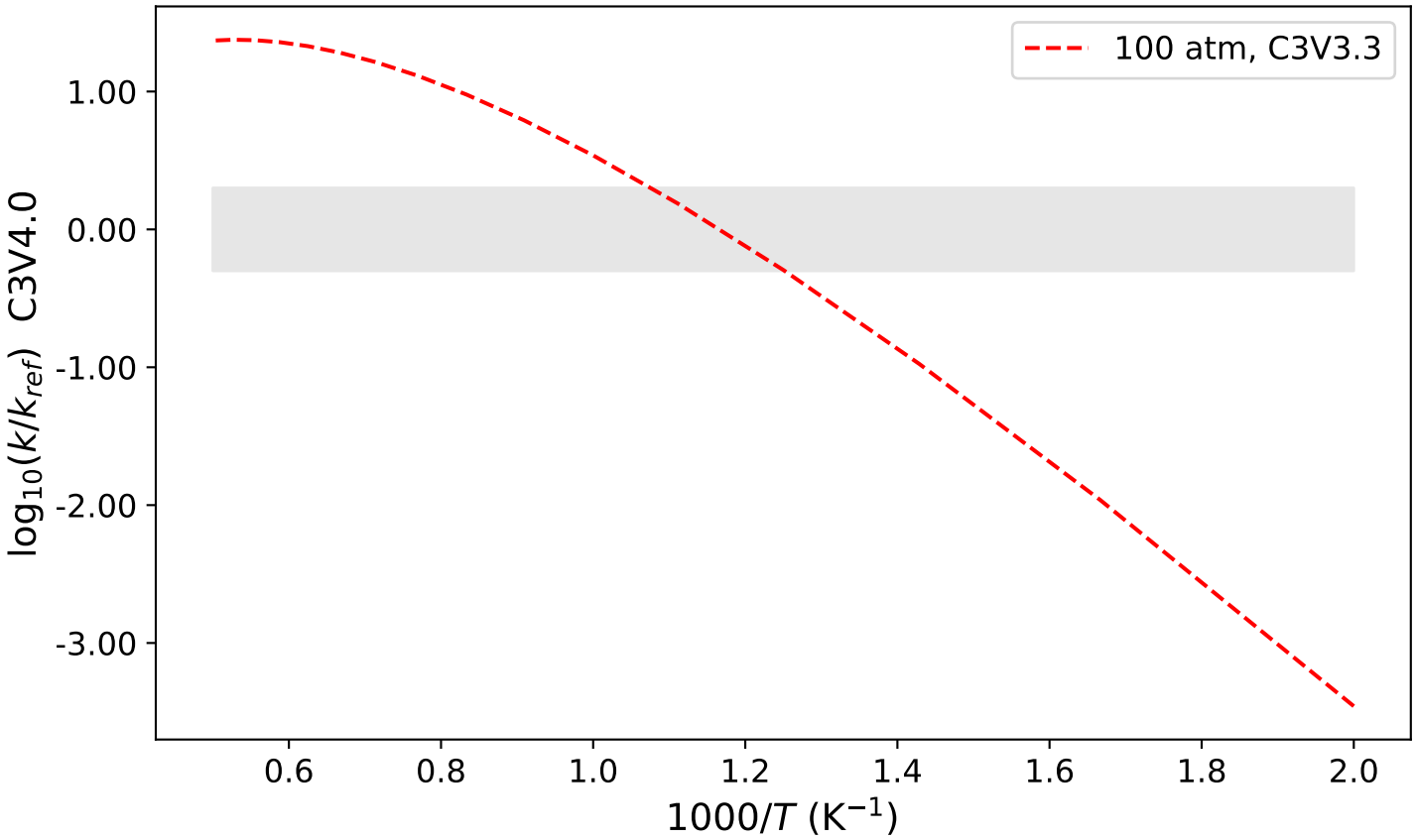
C3H5-A + HOC6H4CH2 => H + H + H2 + NAPHOH-A



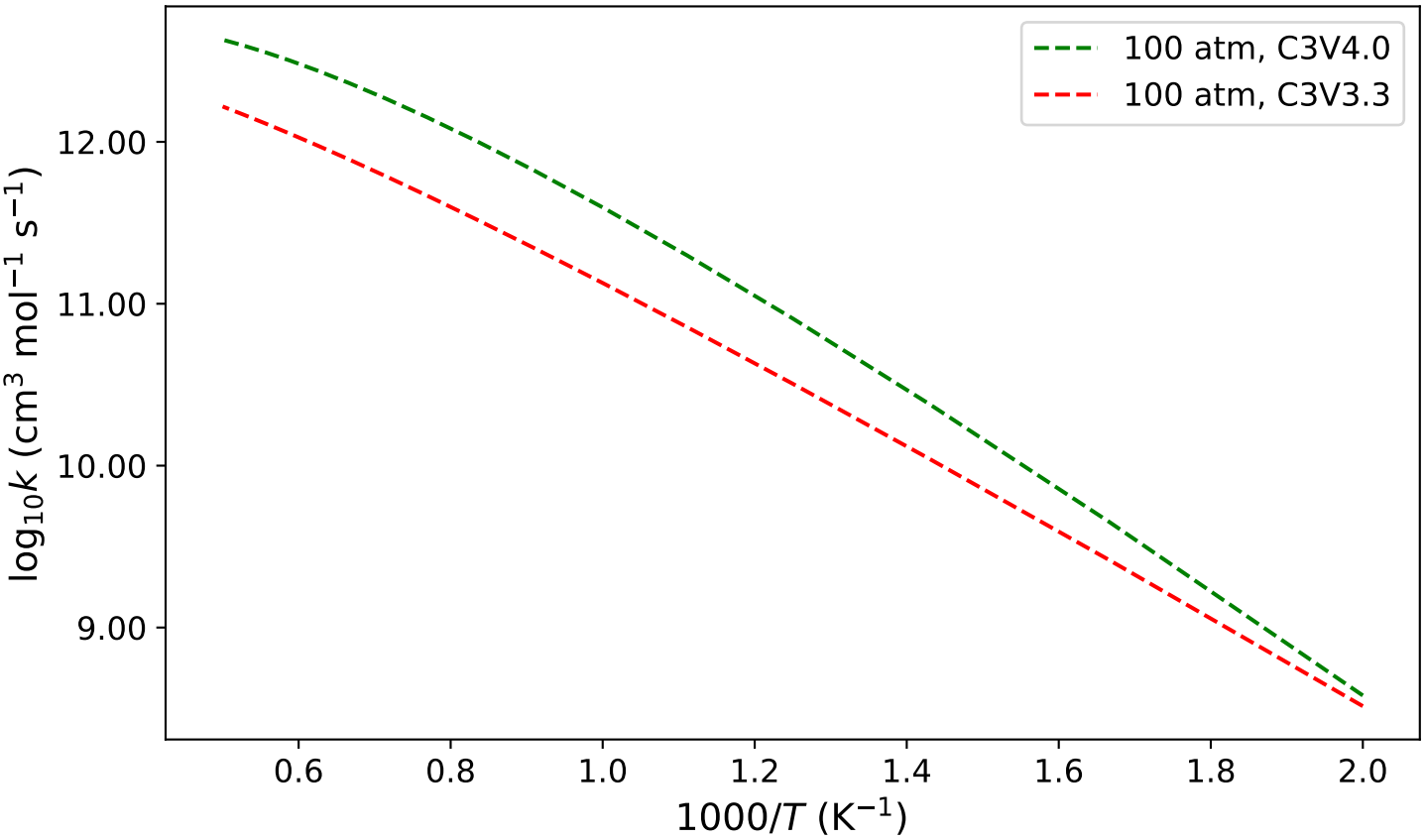


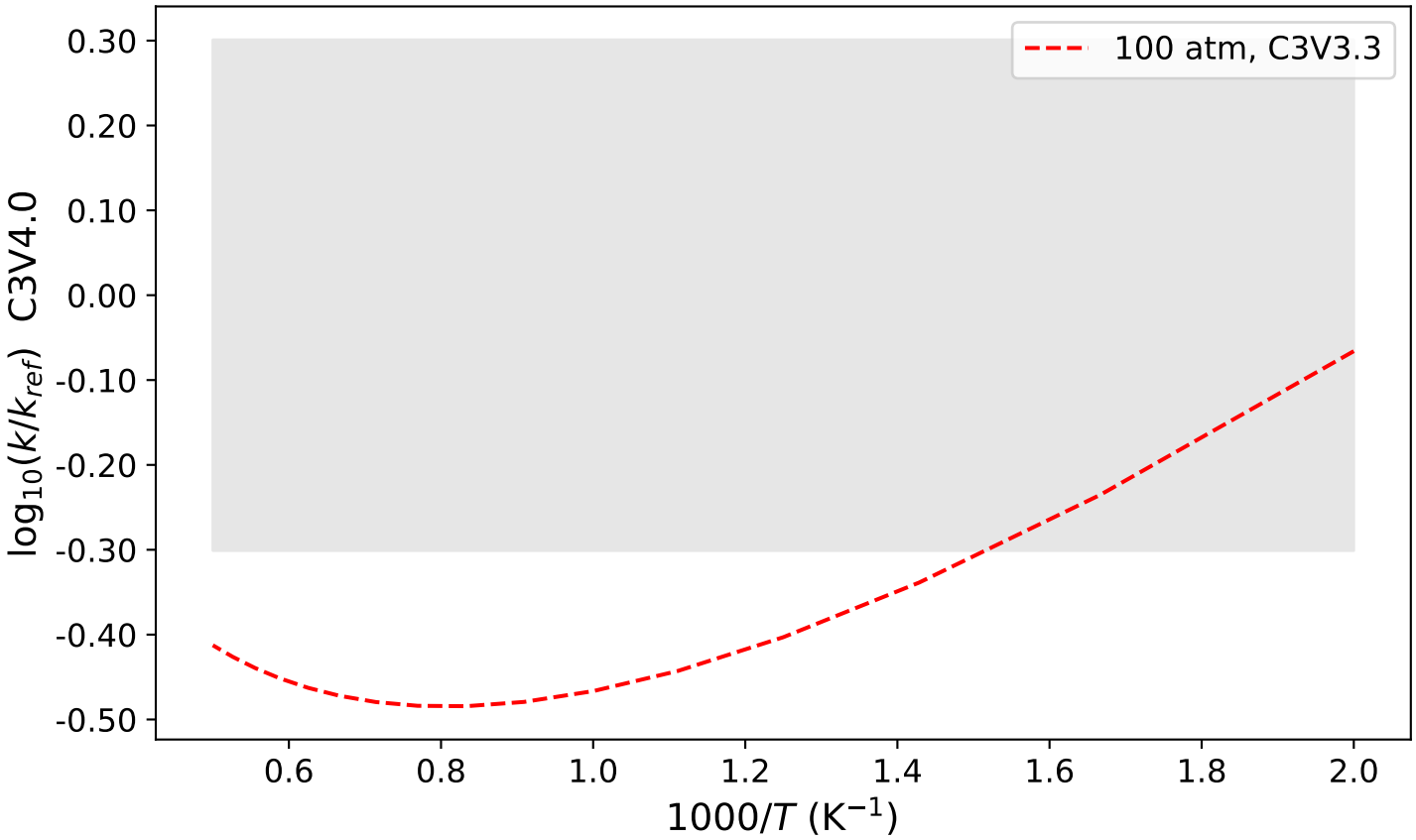
C3H3 + RXYLENE => H + H + C10H7CH3-A



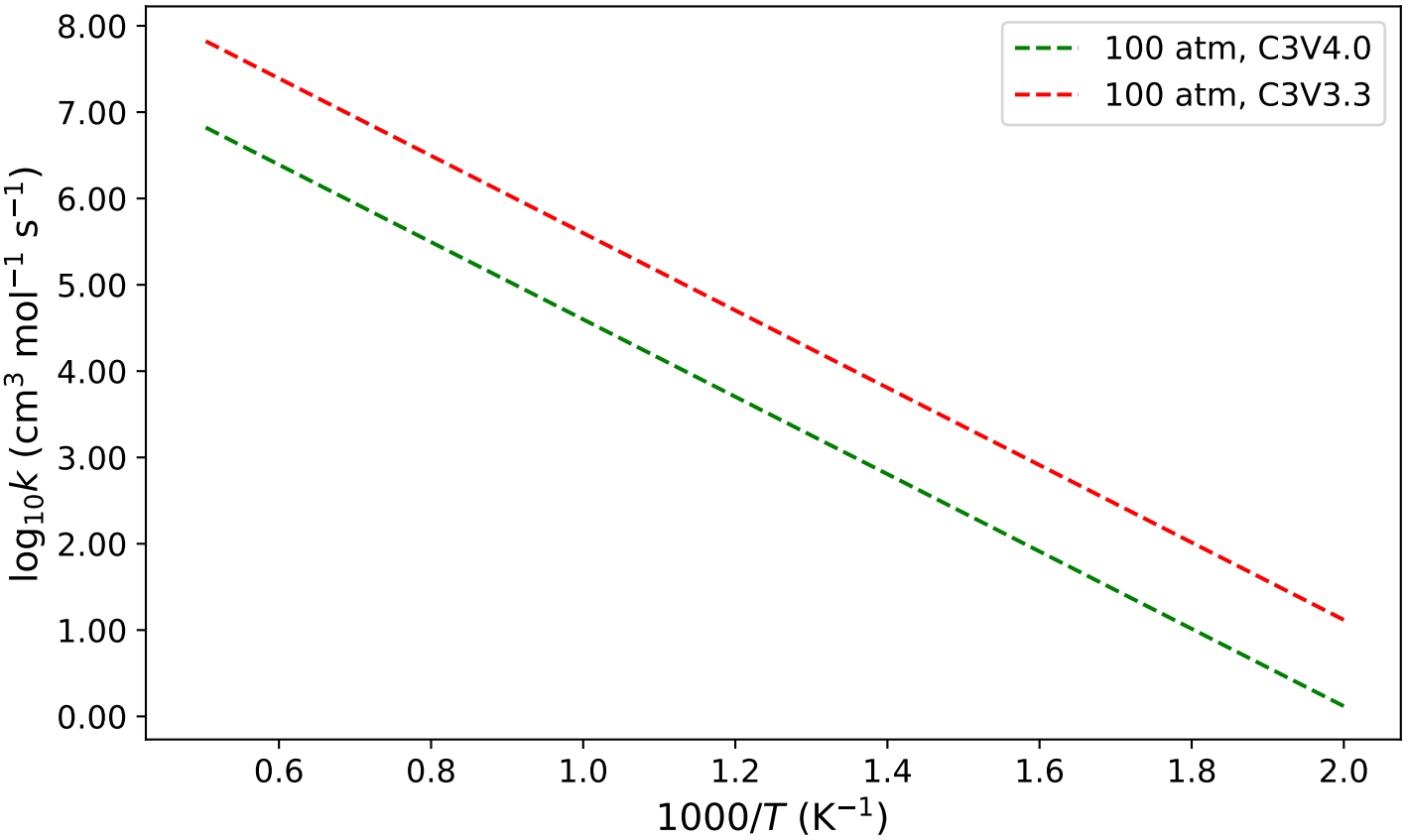


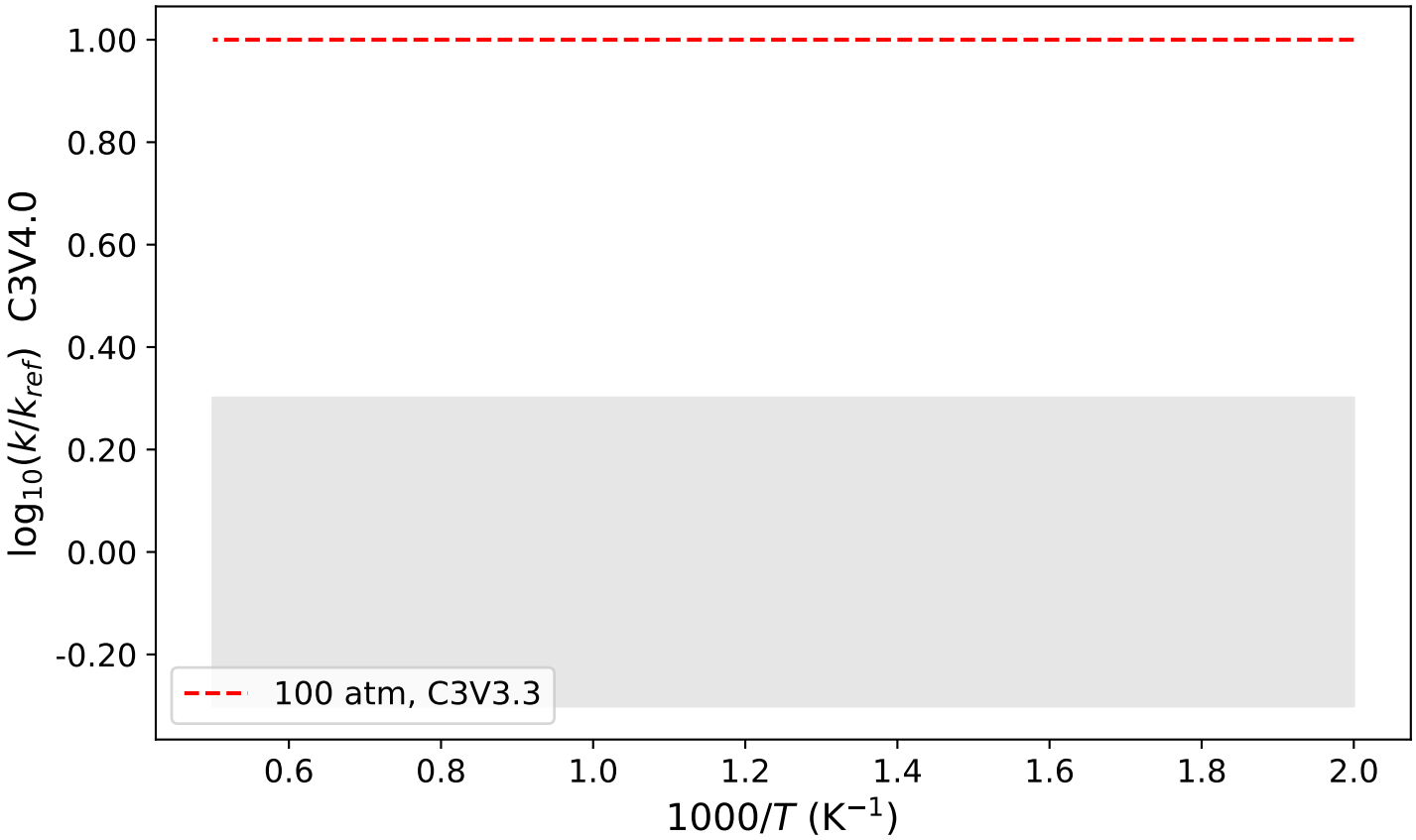
HO2 + RXYLENE => OH + CH2O + C6H4CH3



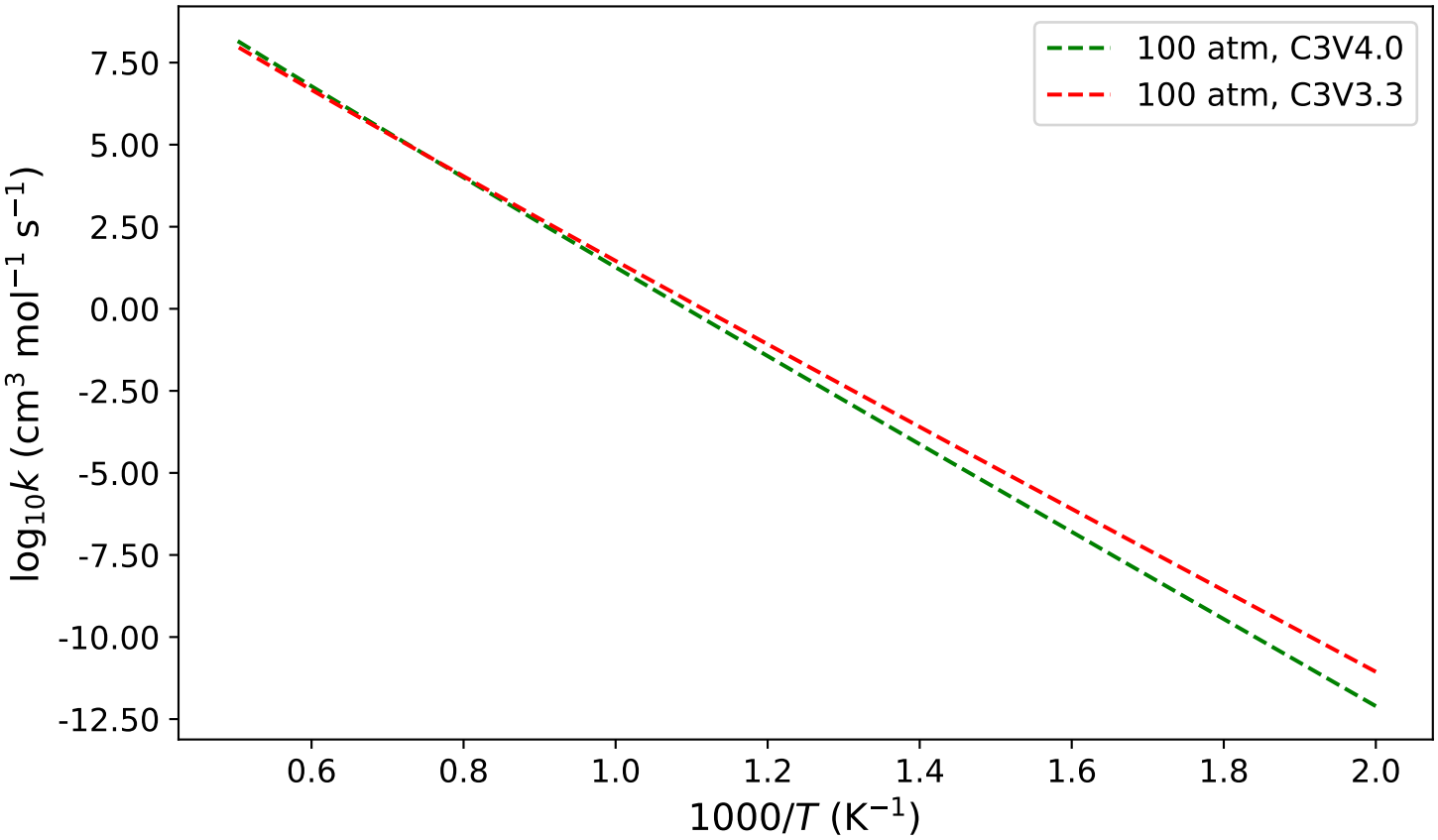


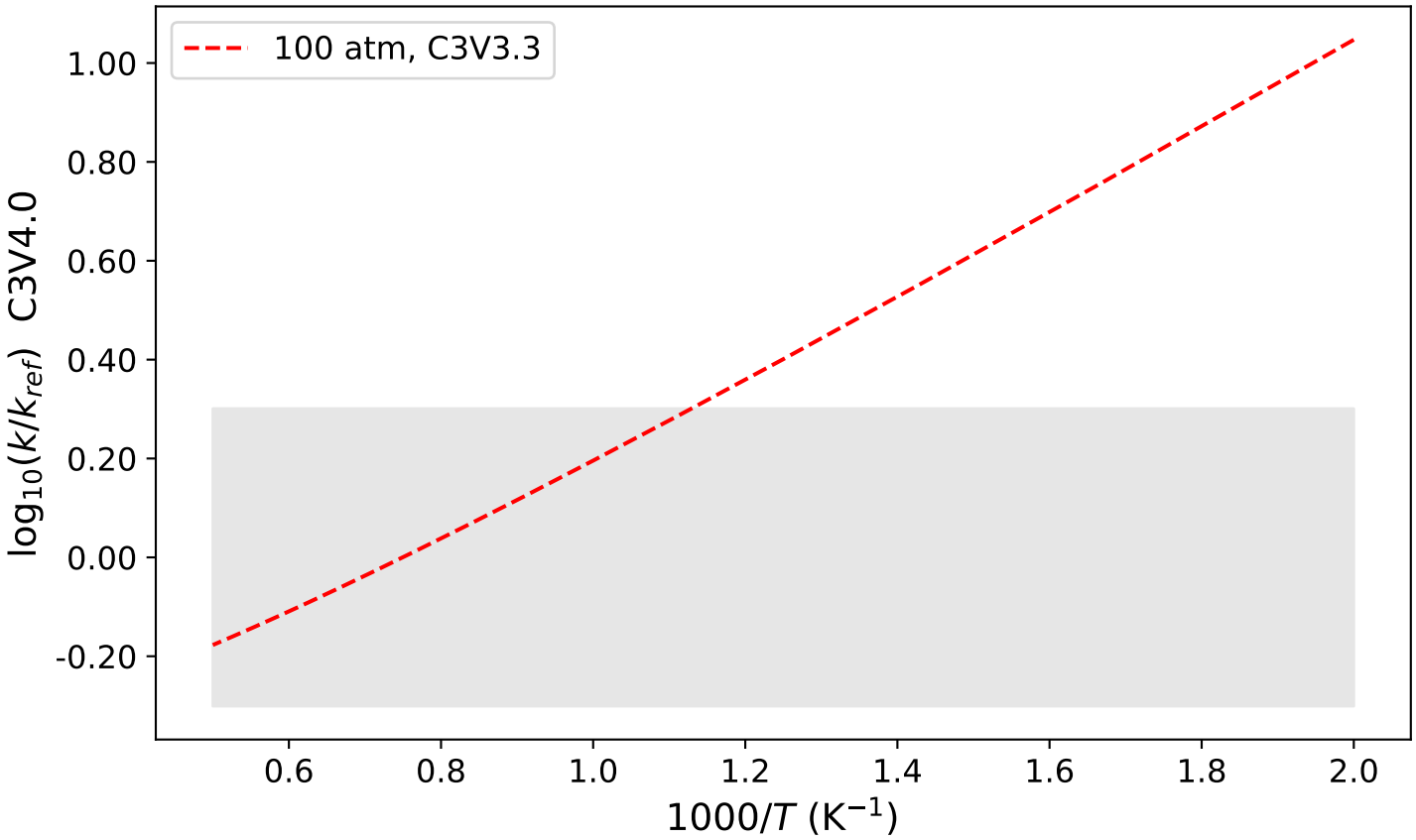
IC4H7 + IC4H8 => CH3 + H2 + H2 + C6H5CH3



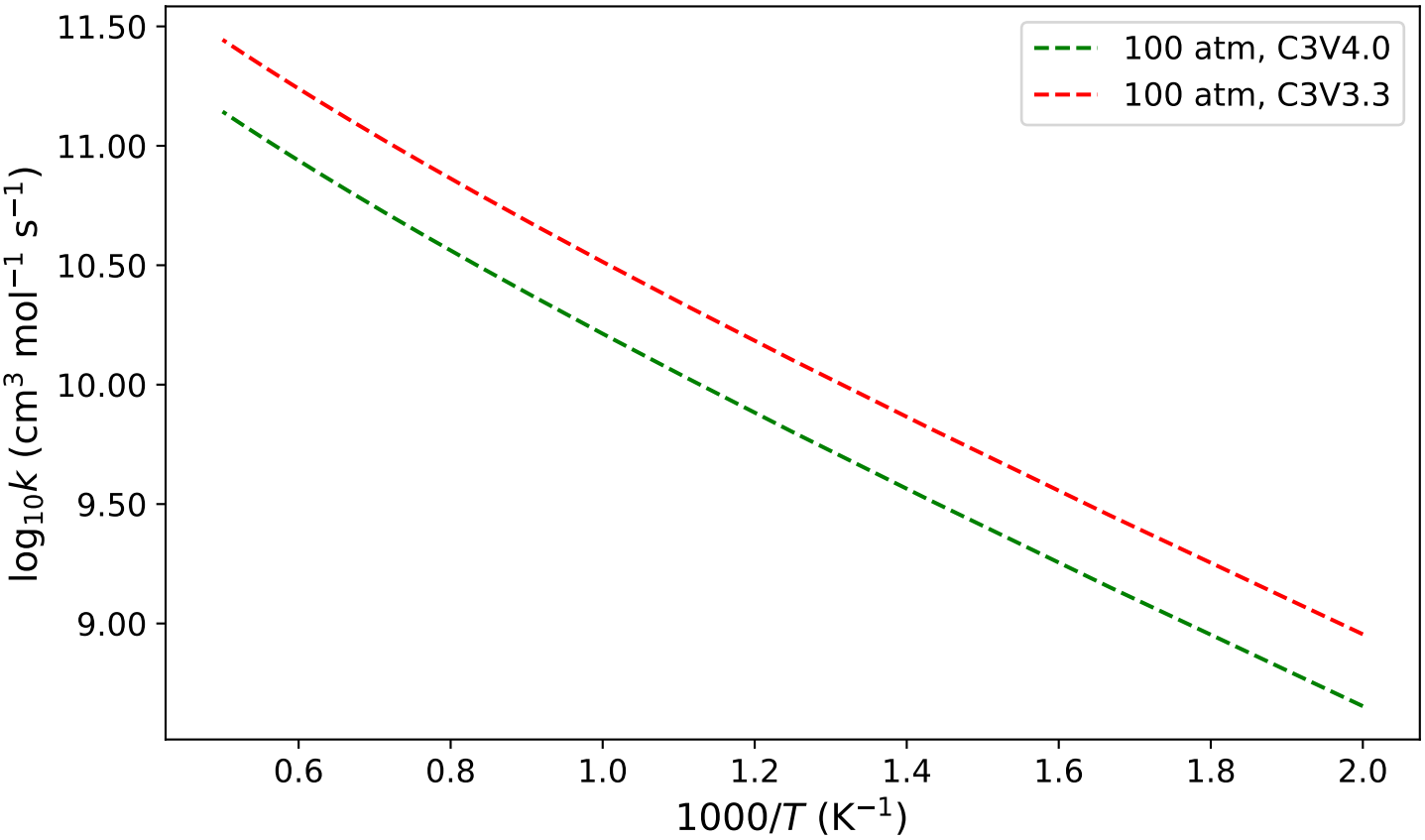


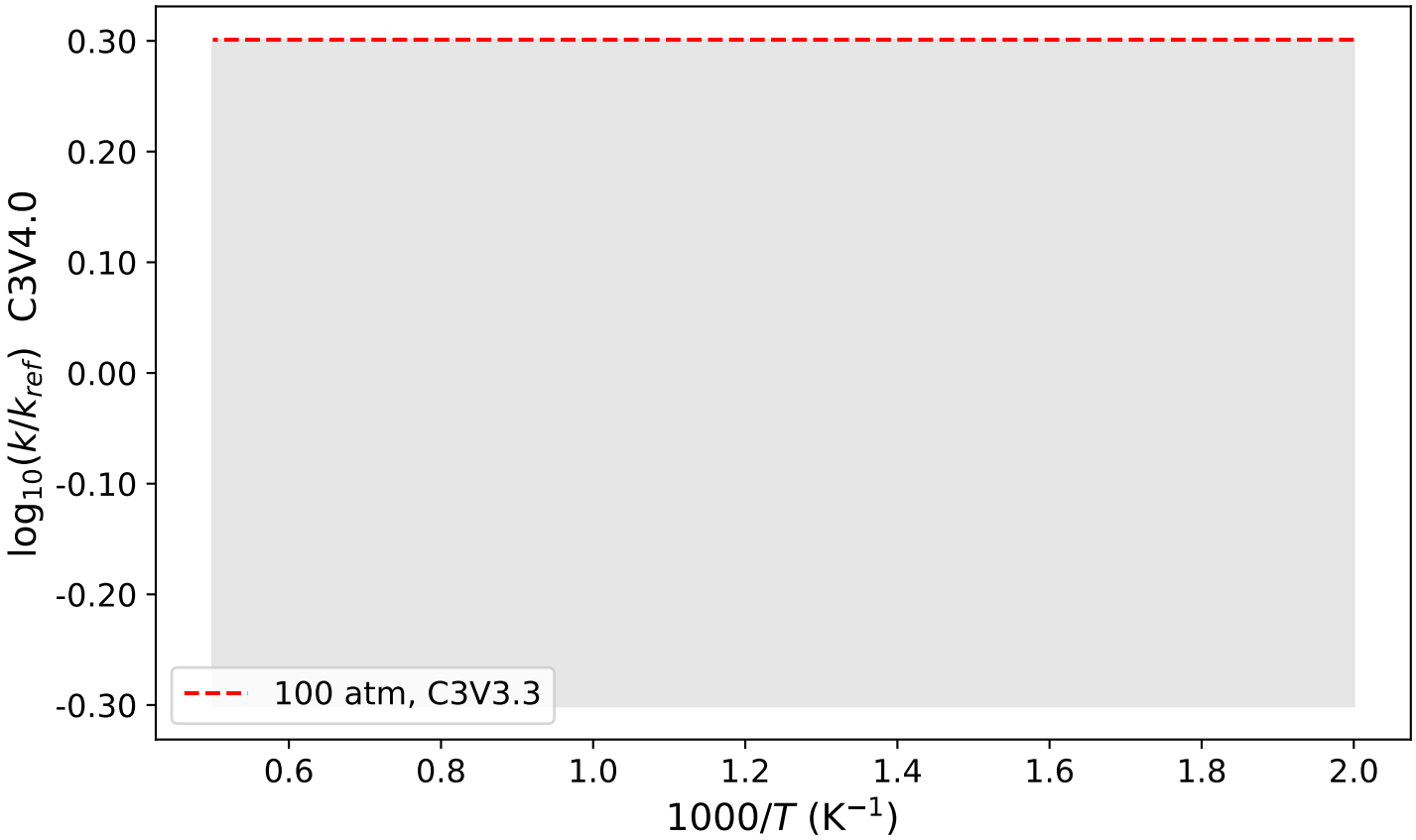
O2 + C10H7CH3-A = HO2 + C10H6CH3-A



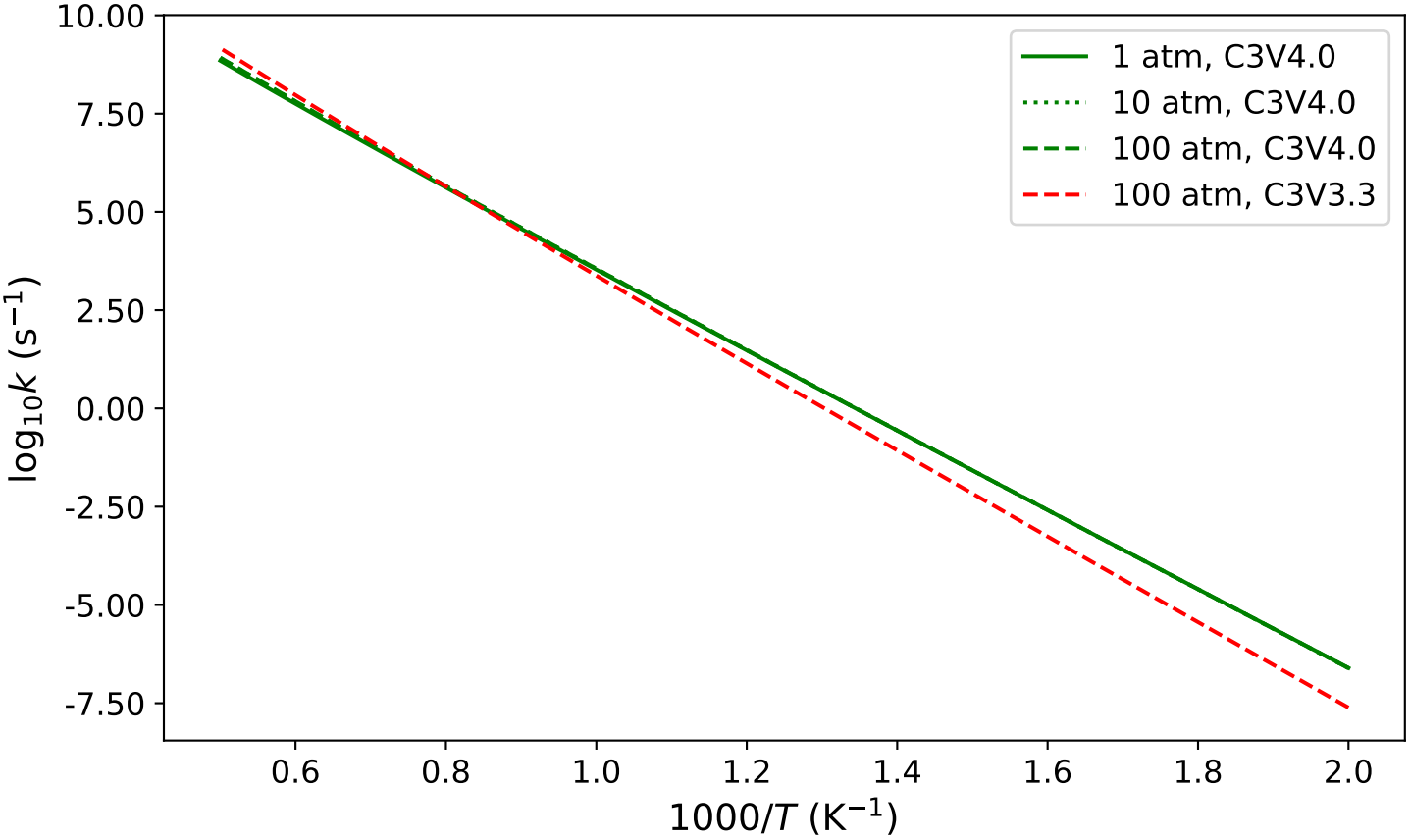


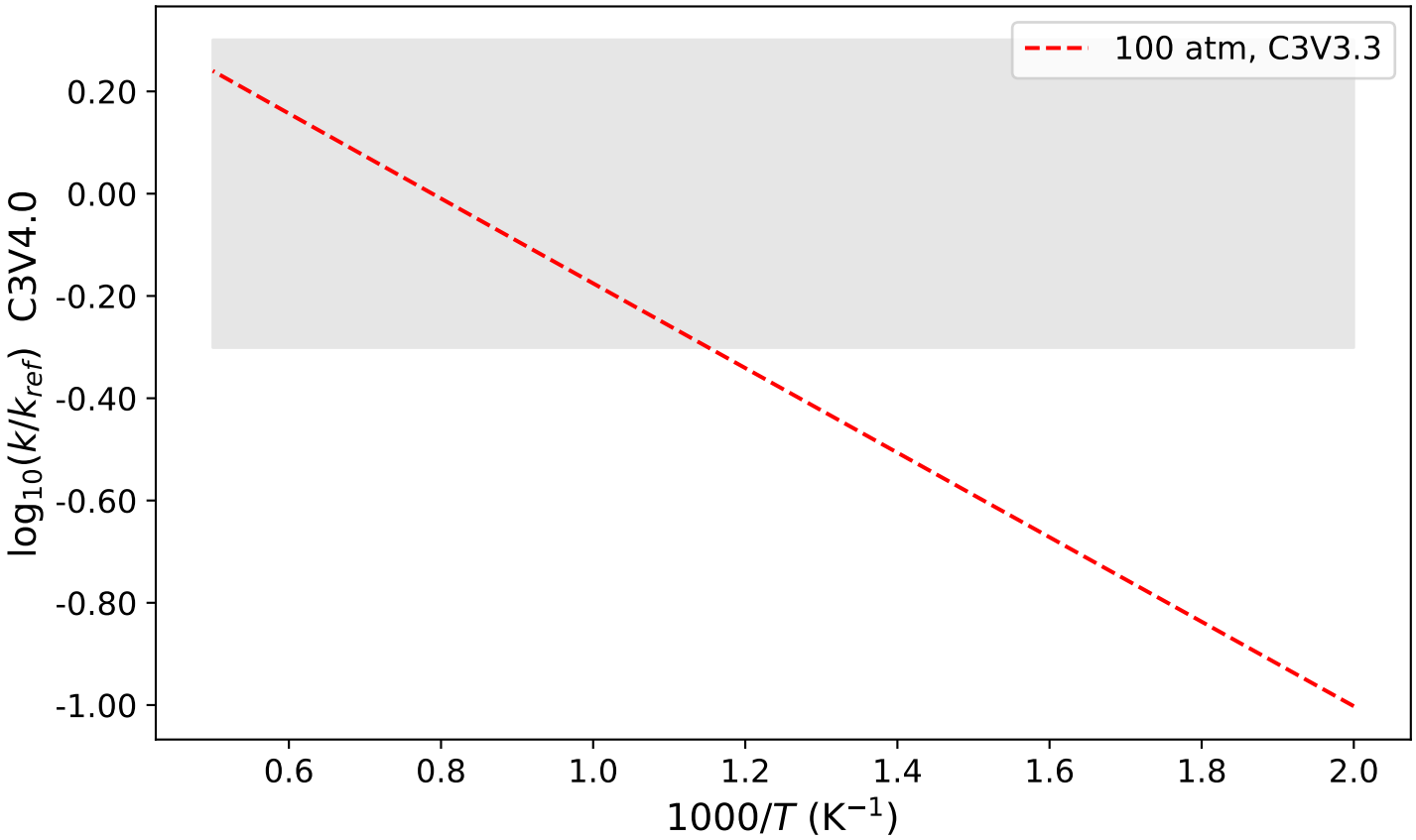
C6H5 + NAPH = C6H6 + NAPH-A



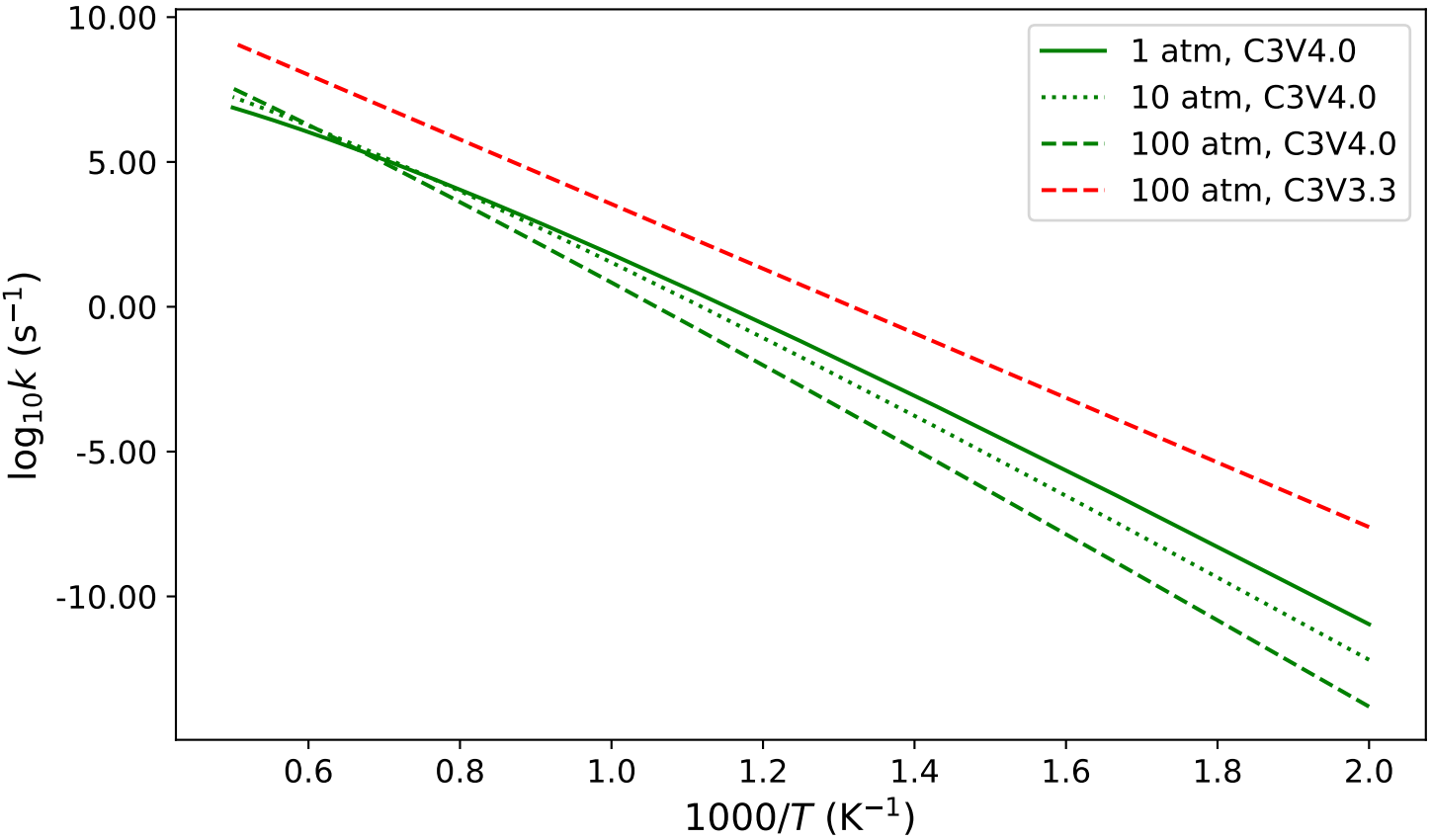


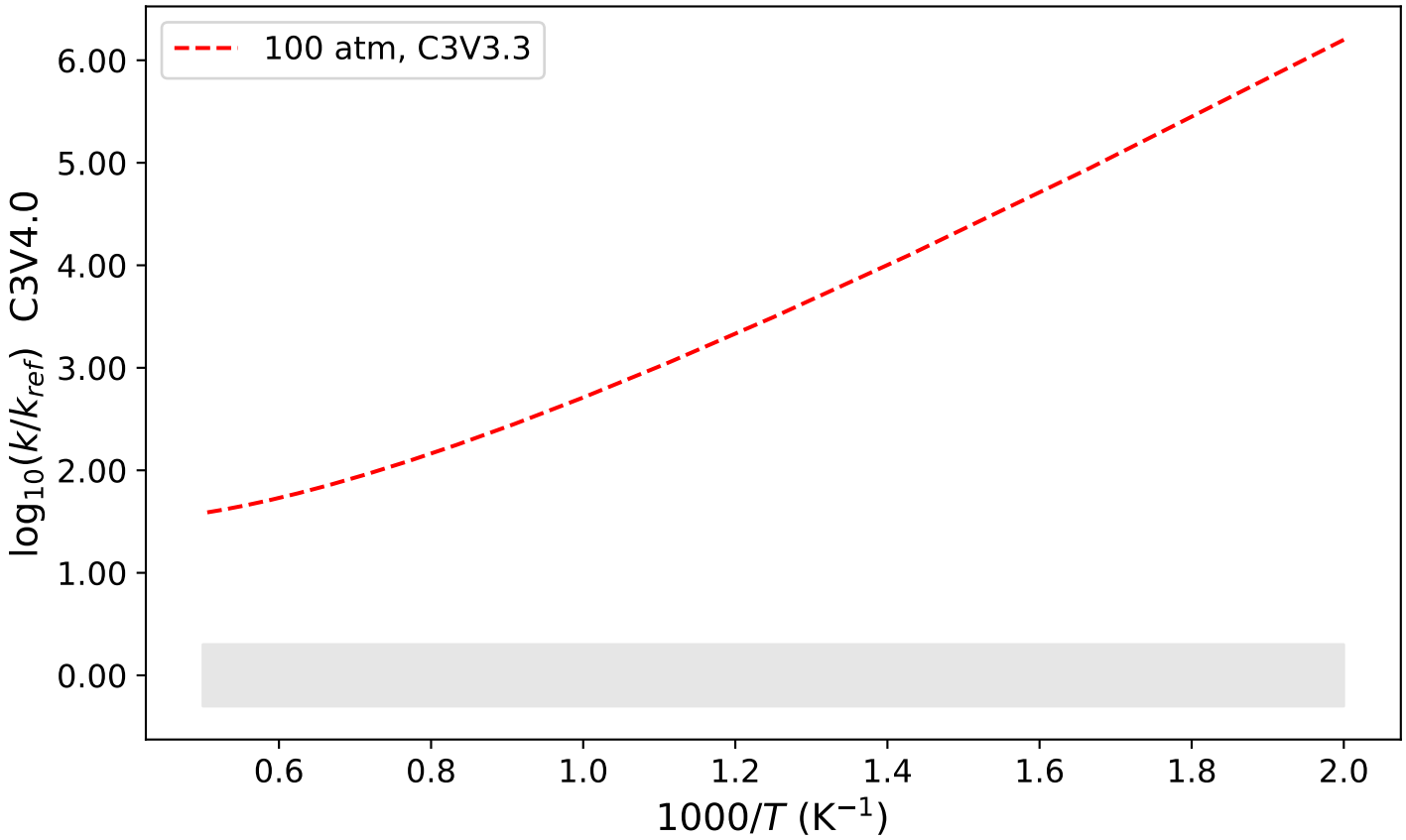
C6H5CHCH3 = H + C6H5C2H3



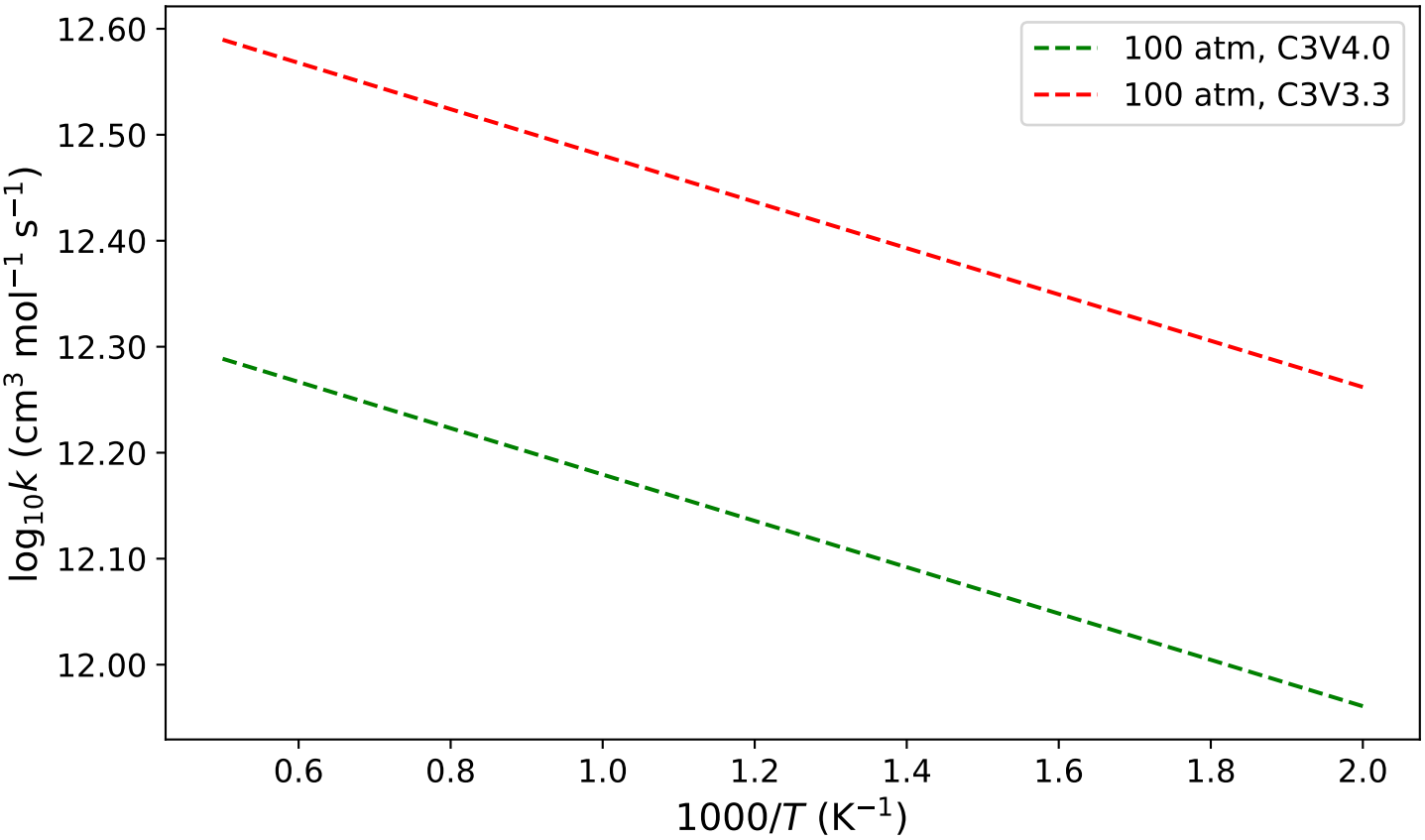


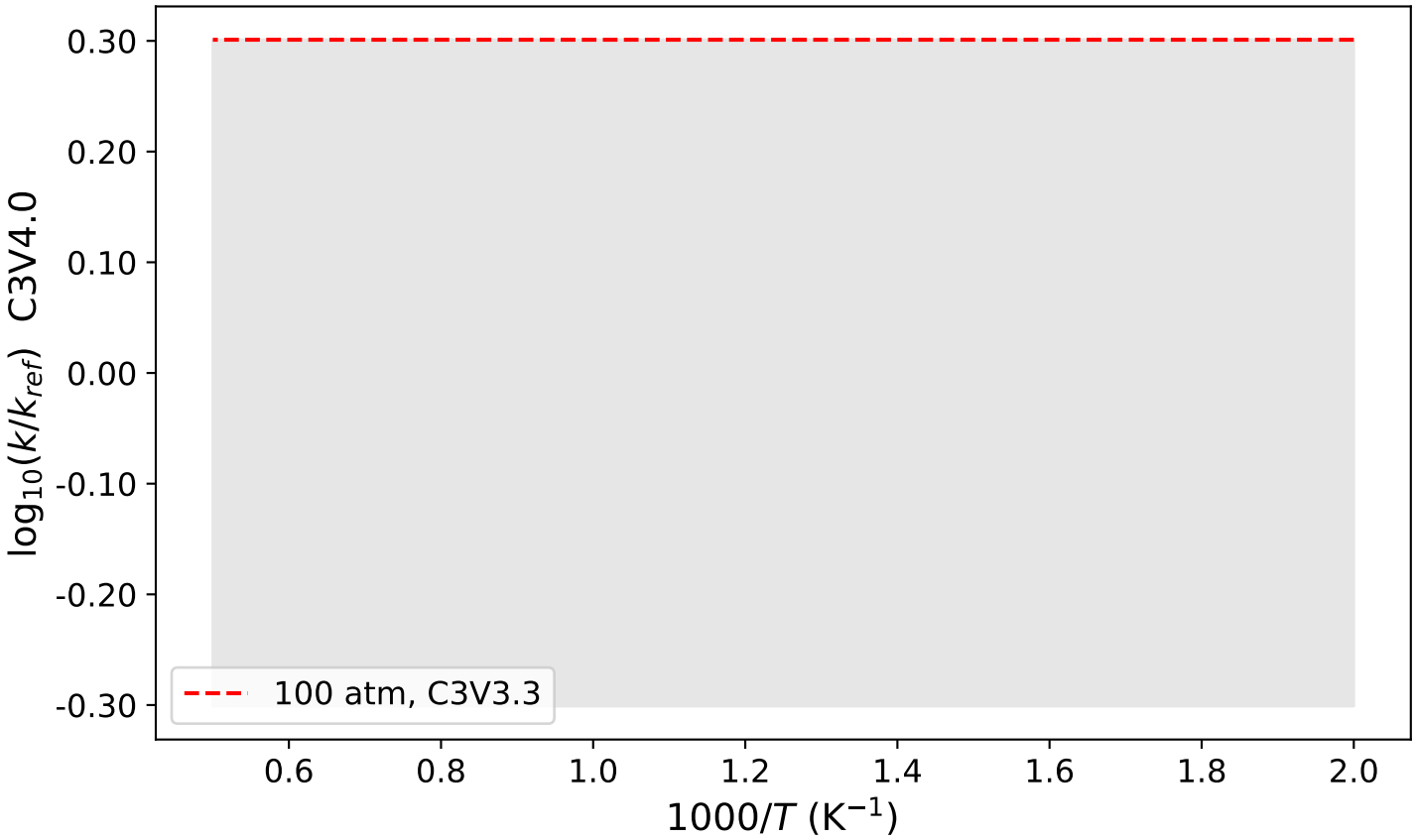
MEINDENYL = H + NAPH



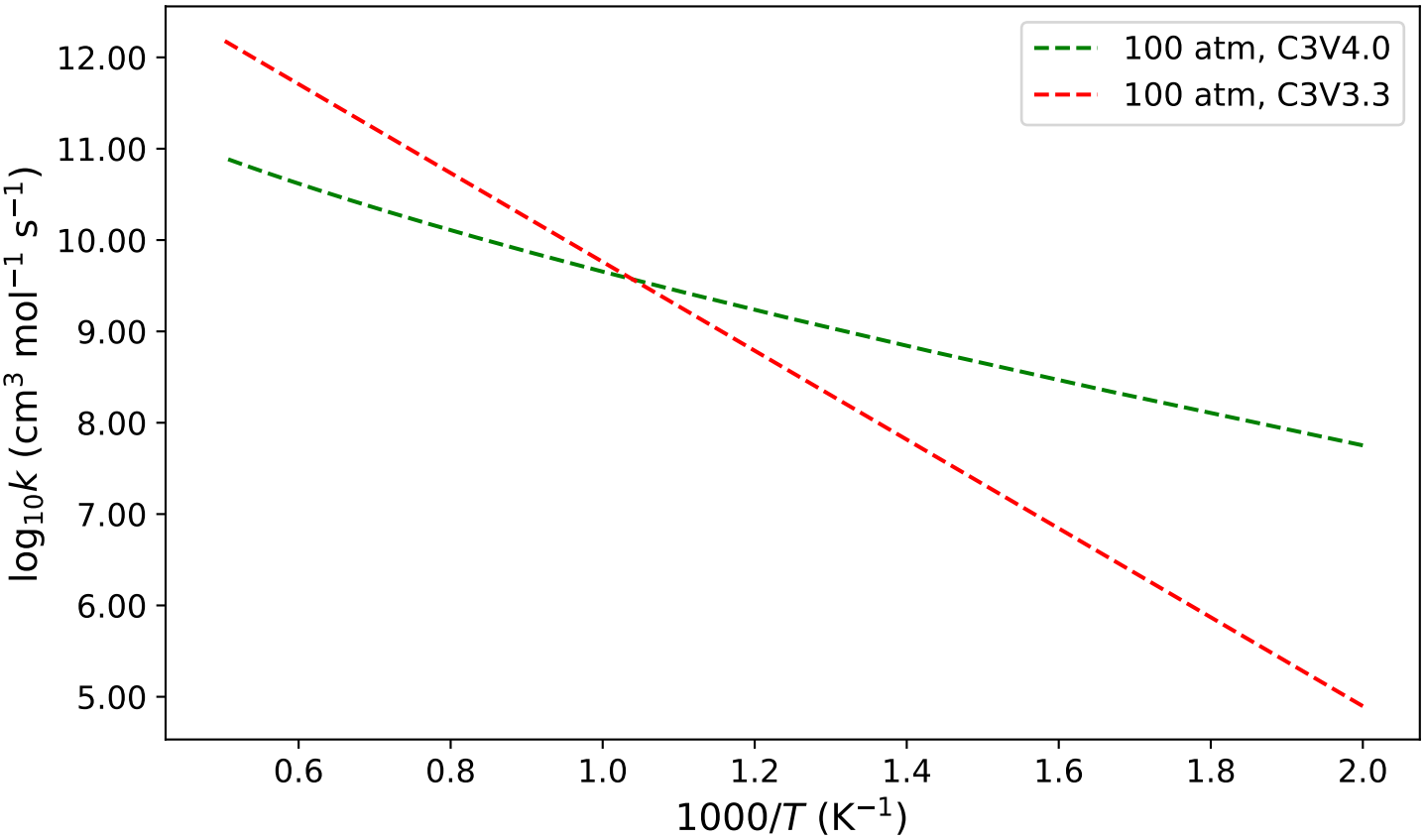


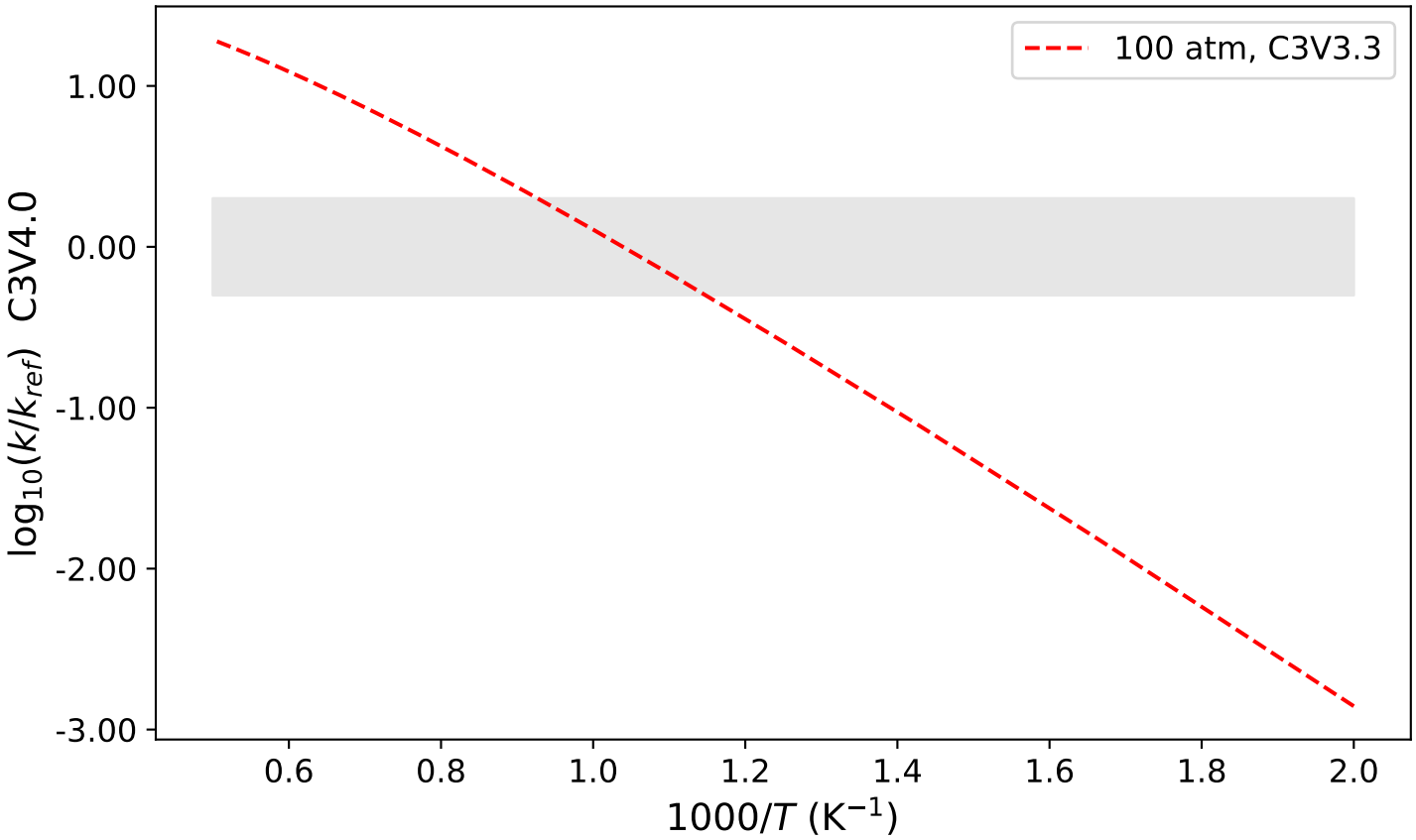
C4H5-I + C6H4CH3 => H + H + C10H7CH3-A



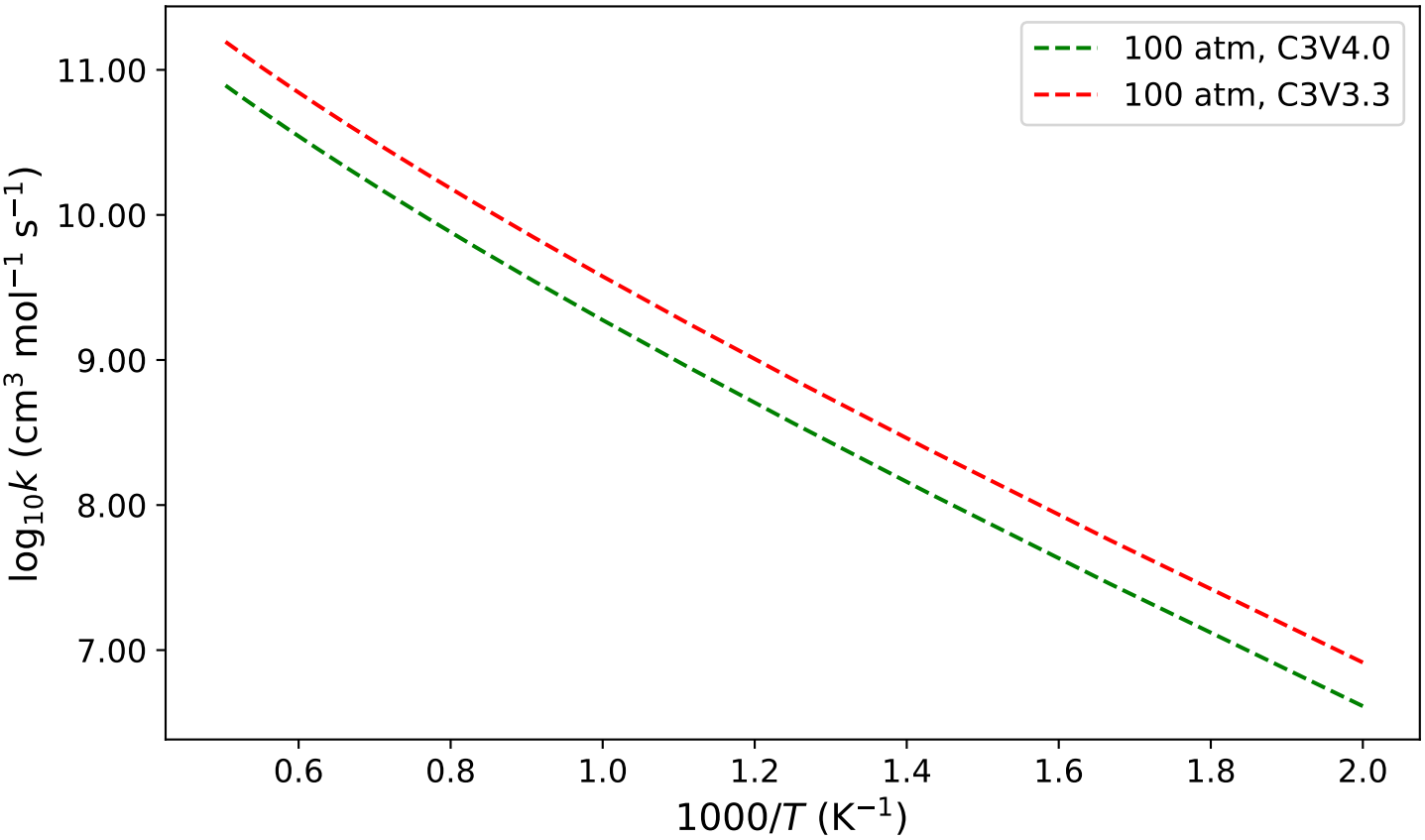


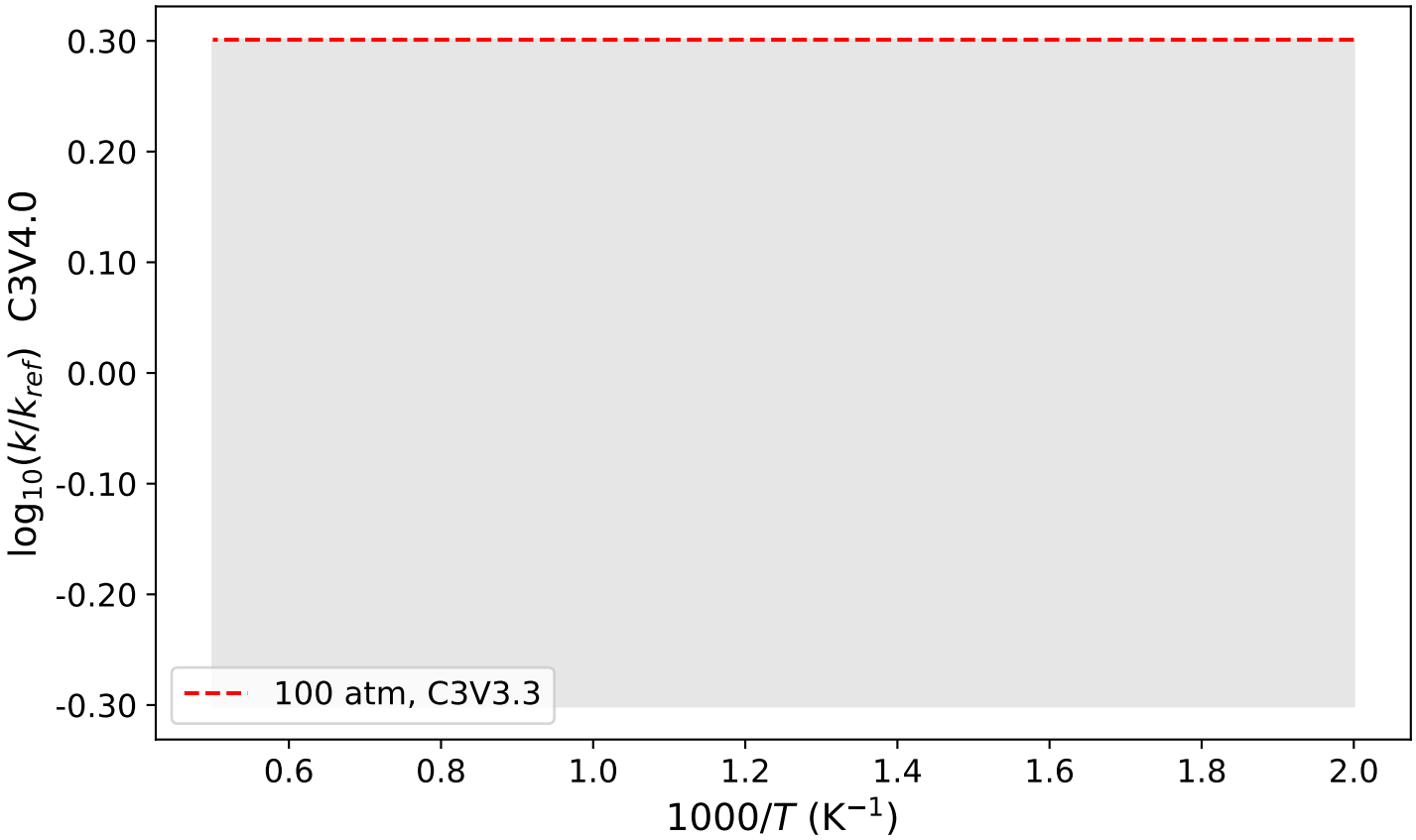
CH3 + C10H7CH3-A = CH4 + C10H7CH2-A



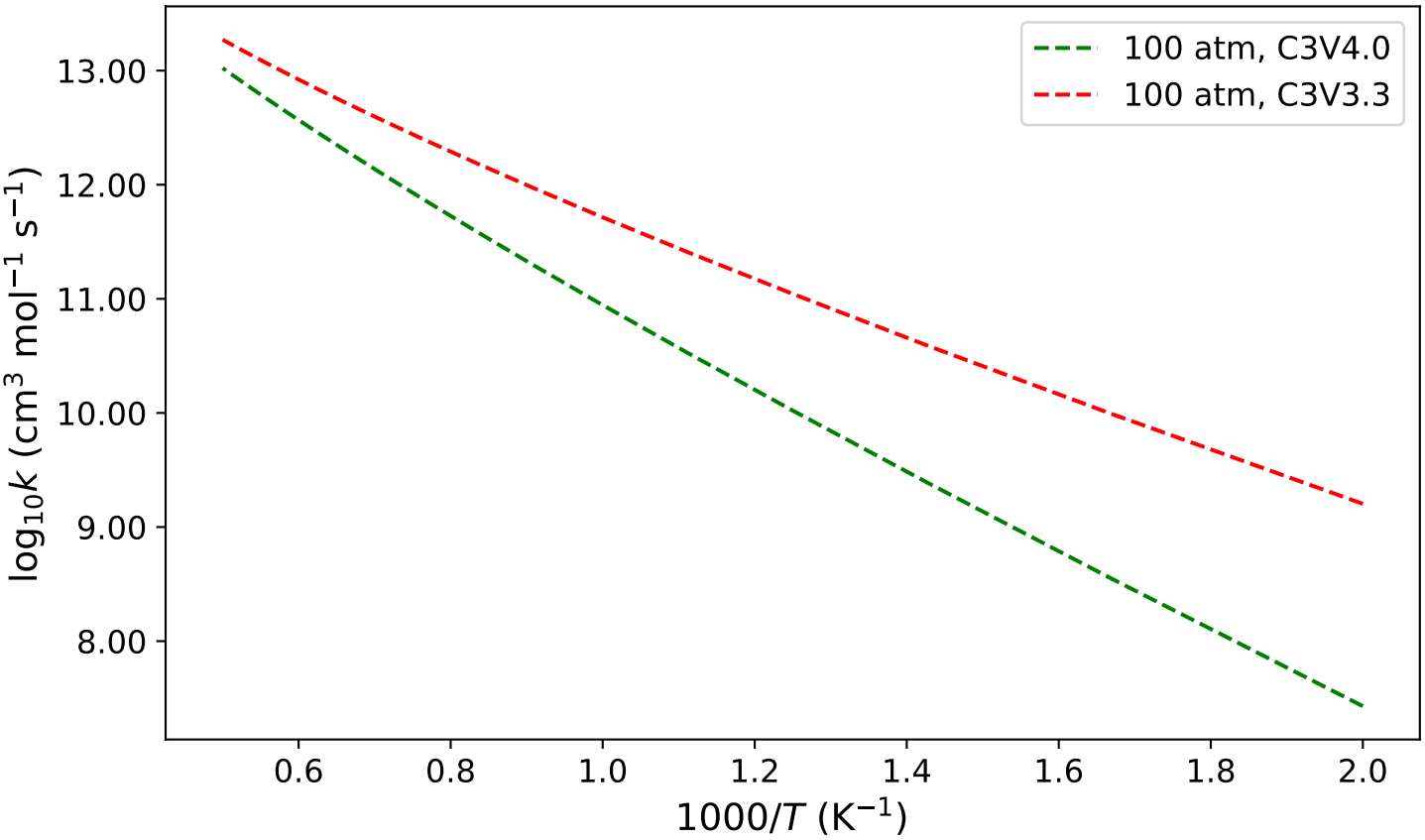


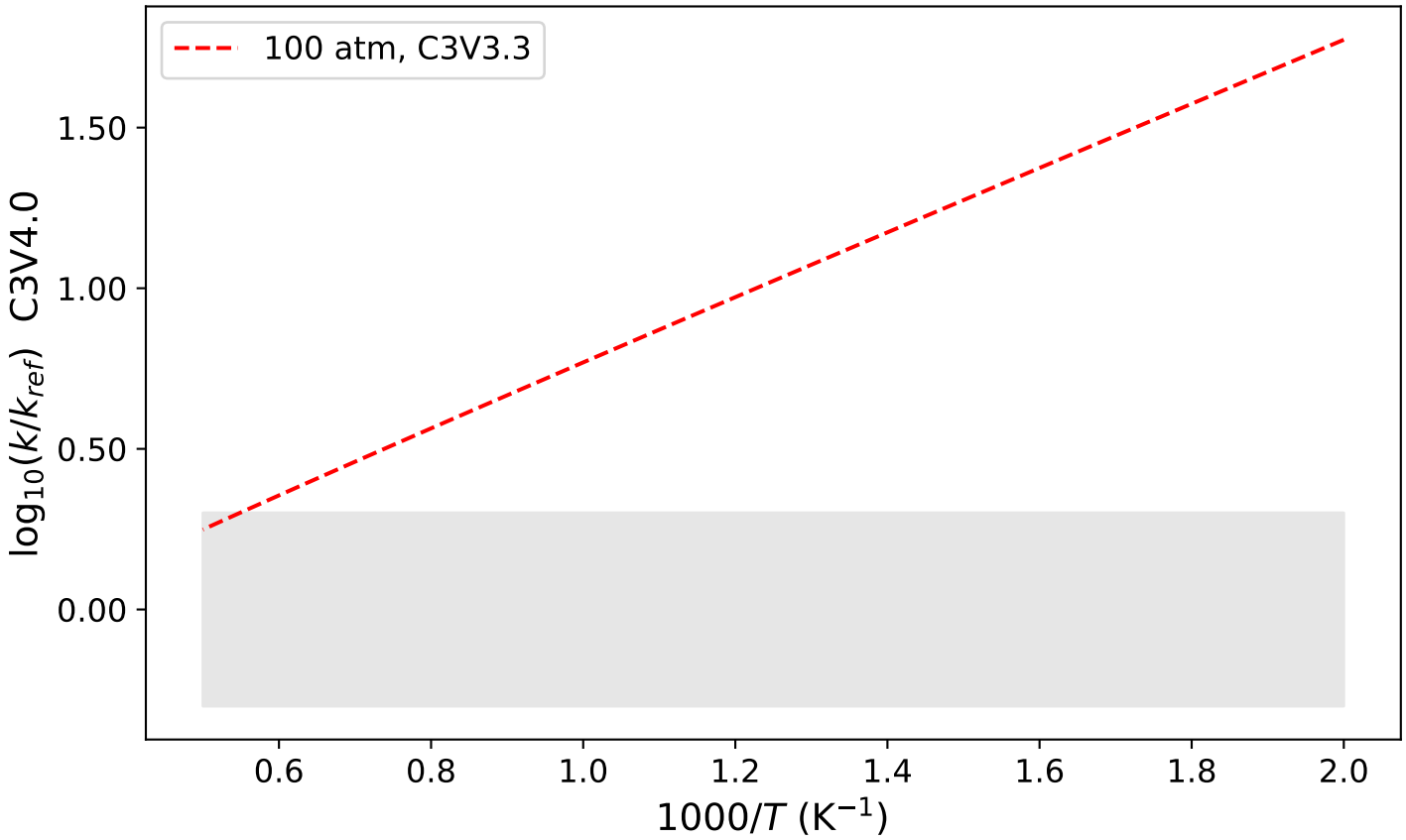
NAPH + C3H5-T = C3H6 + NAPH-A



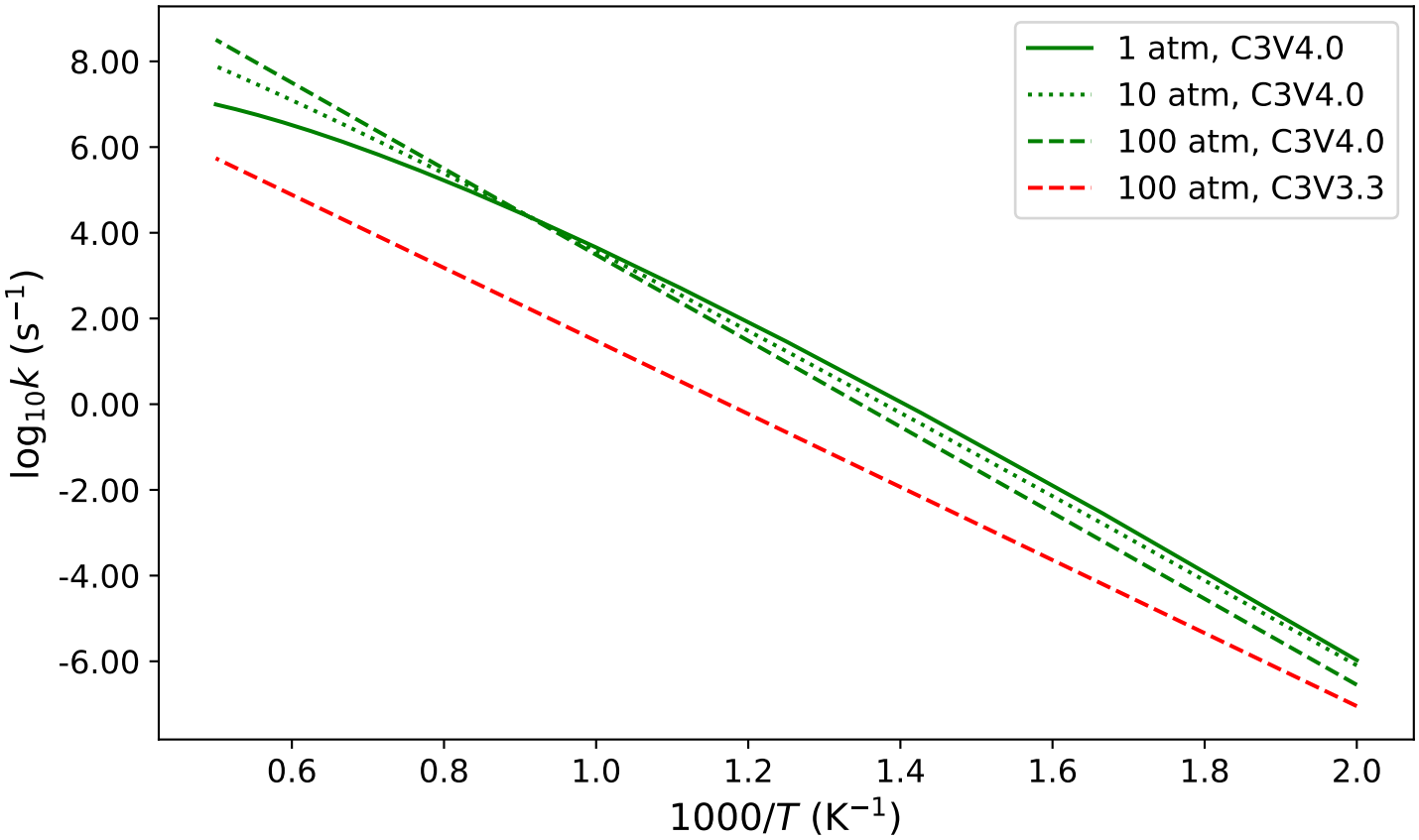


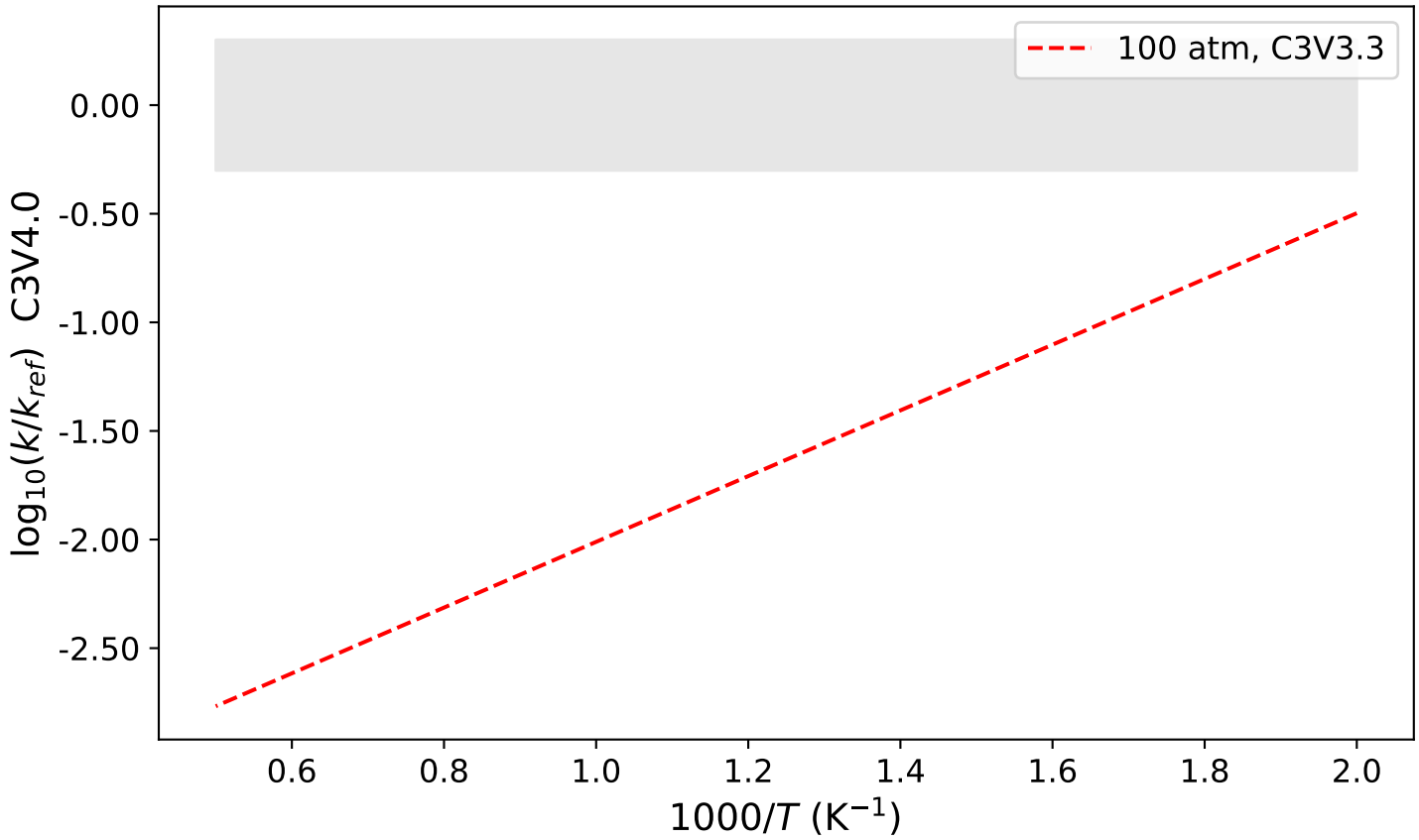
H + C10H7CH3-A = H2 + C10H6CH3-A



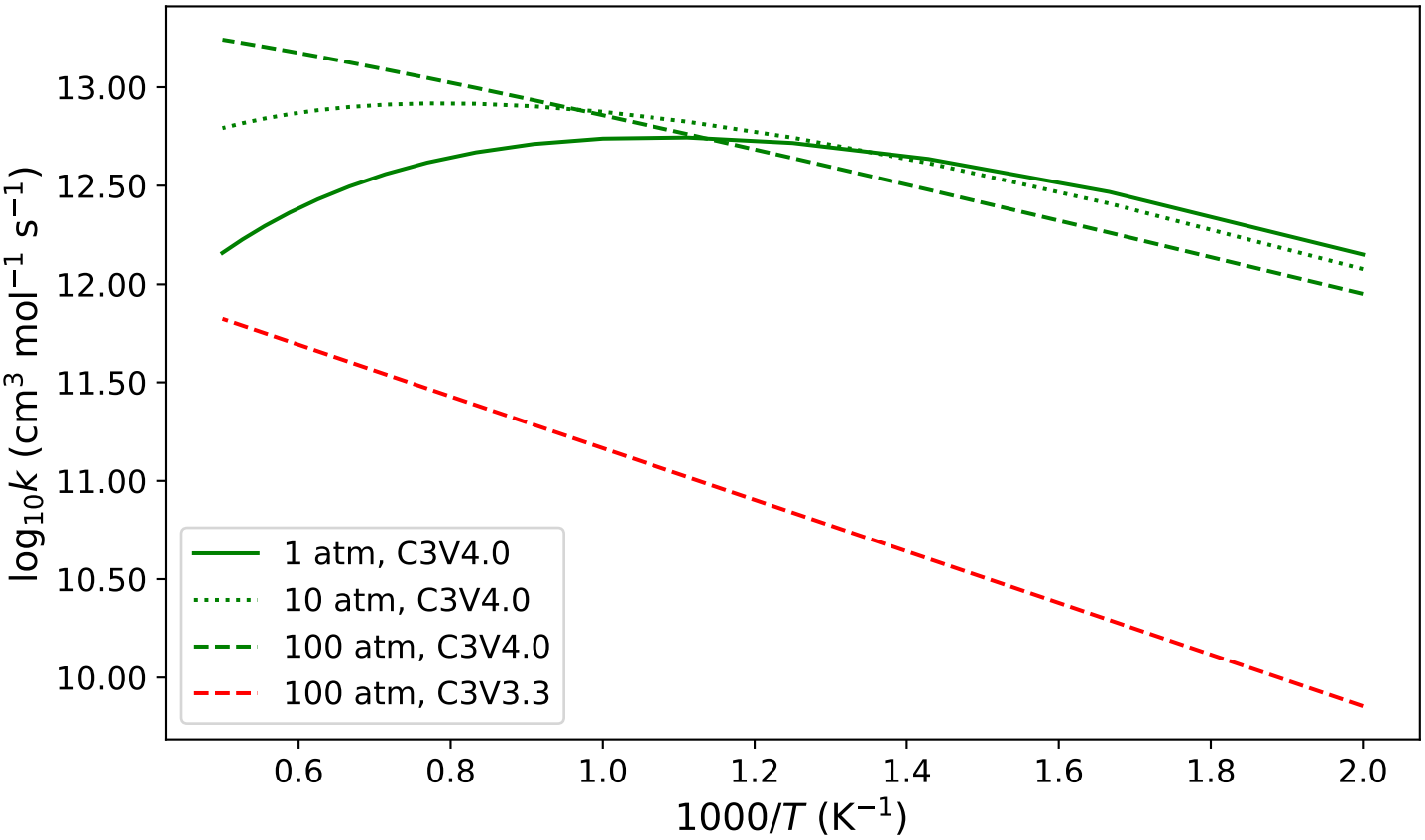


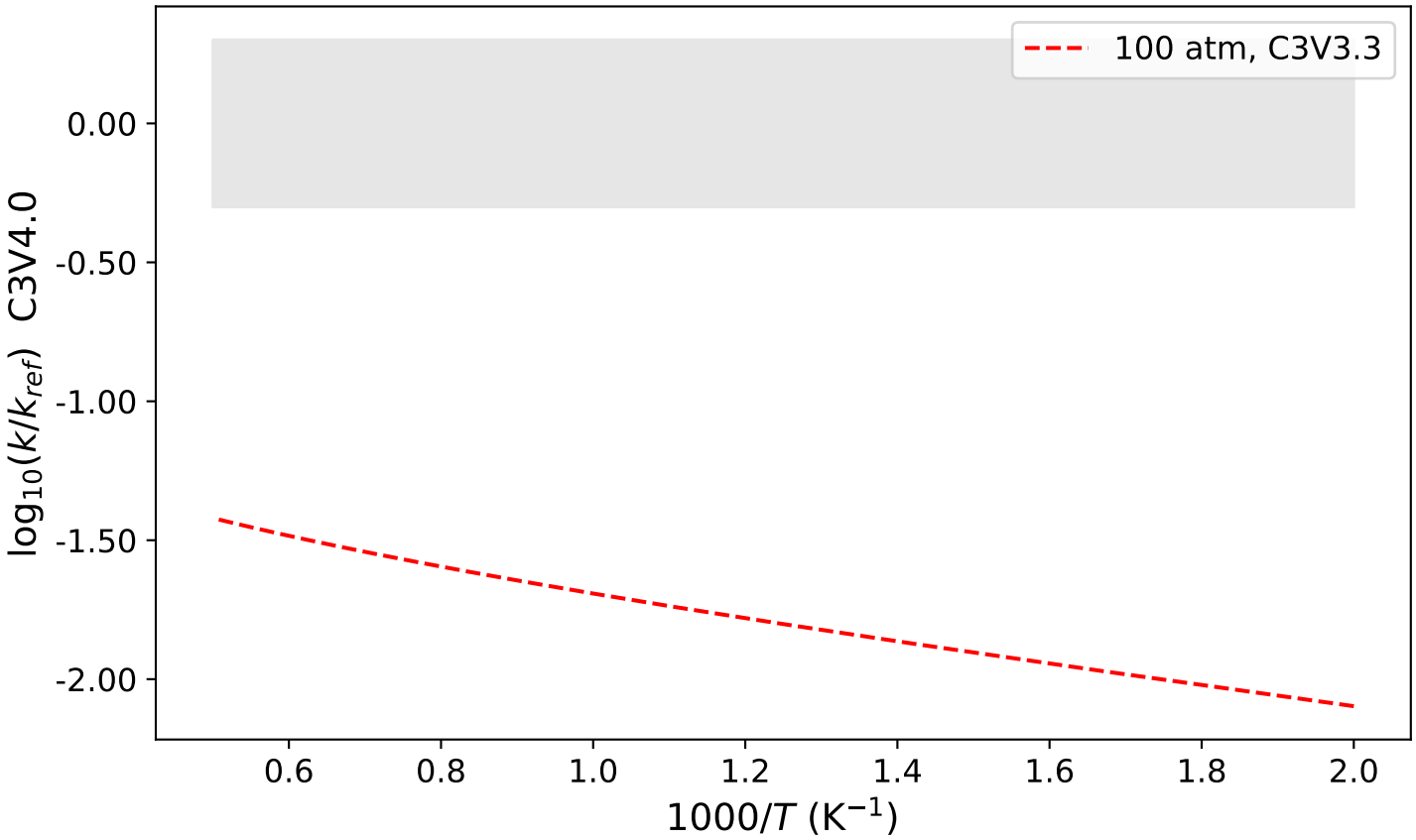
C10H6CH3-A = C10H7CH2-A



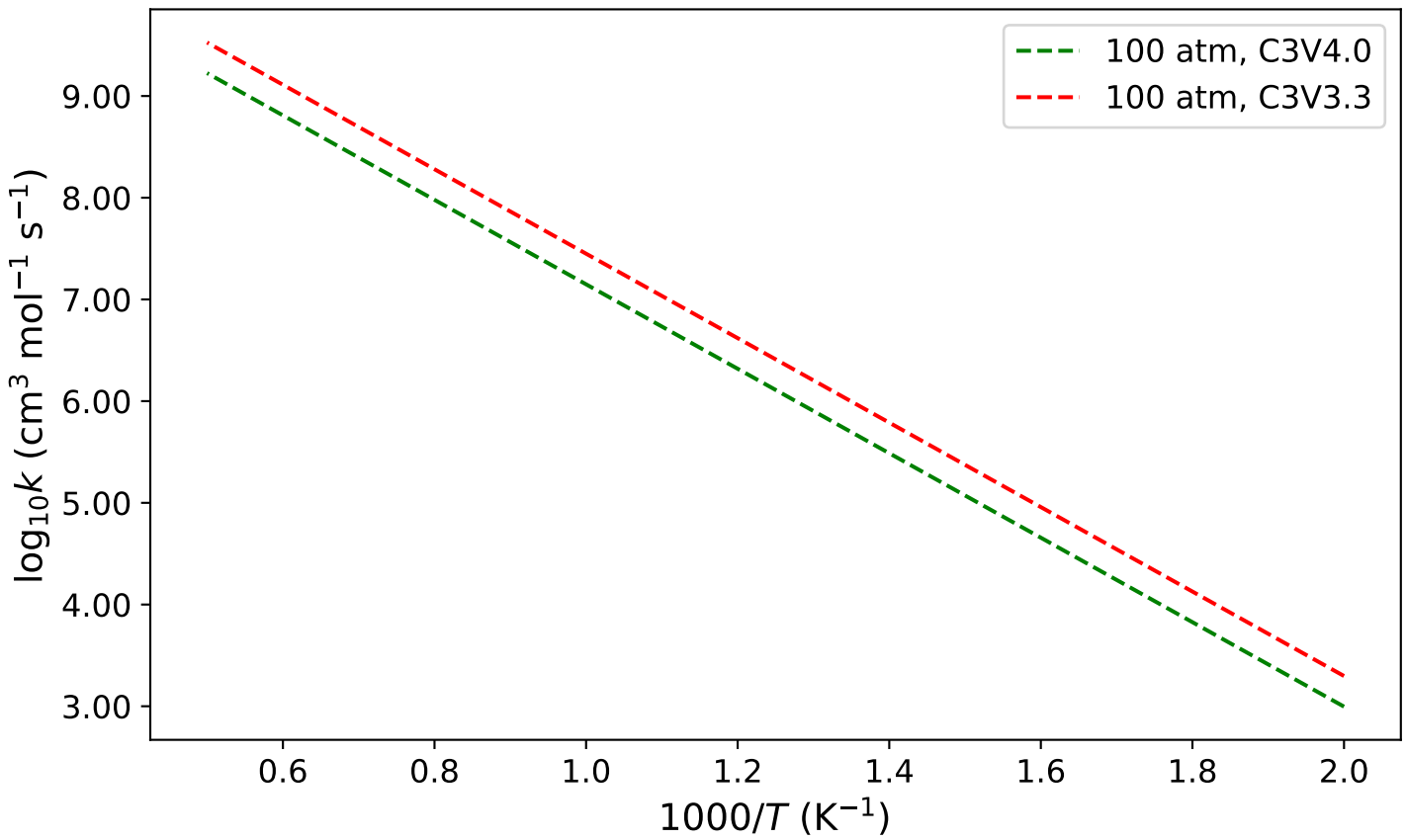


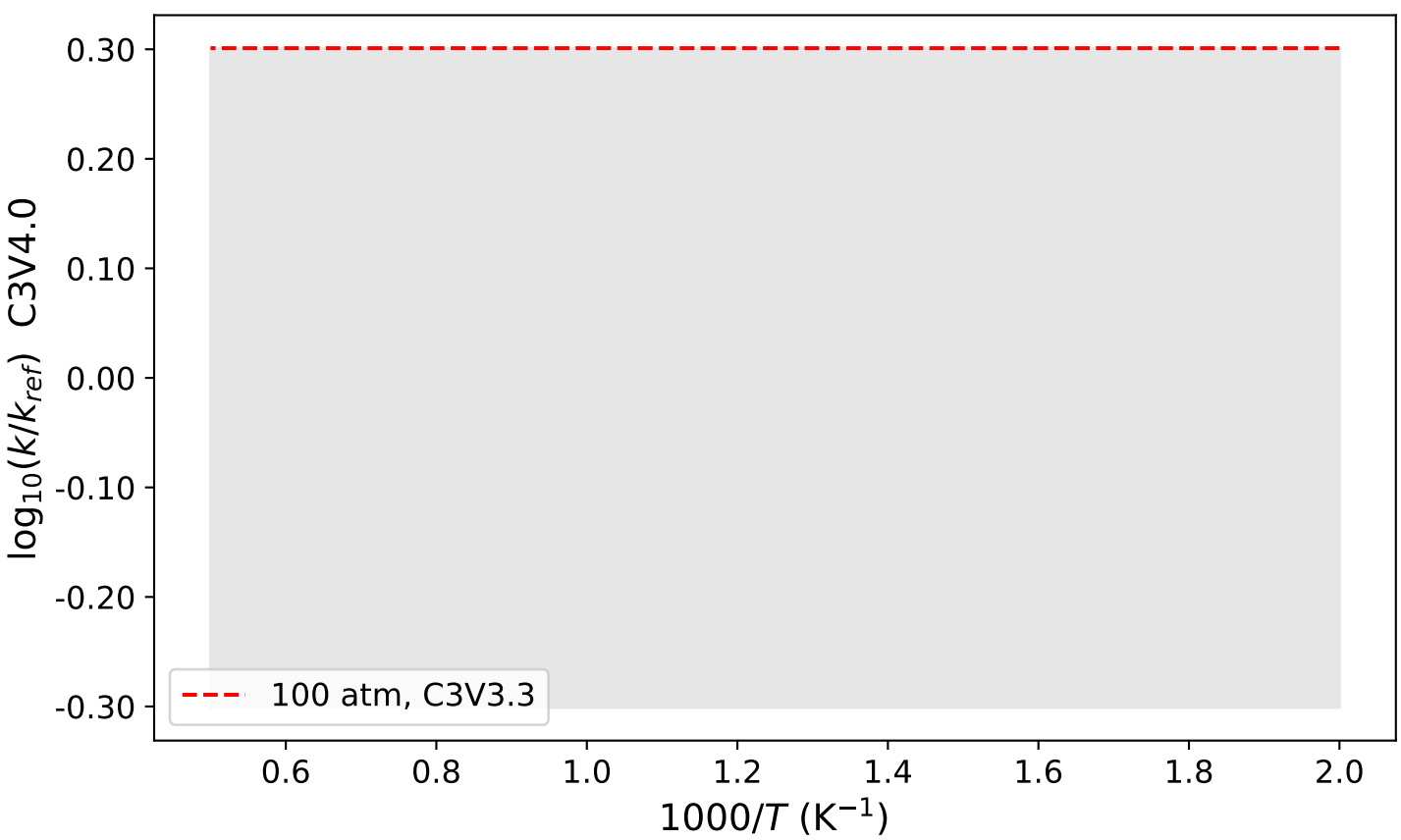
C2H2 + C12H7 = C14H9



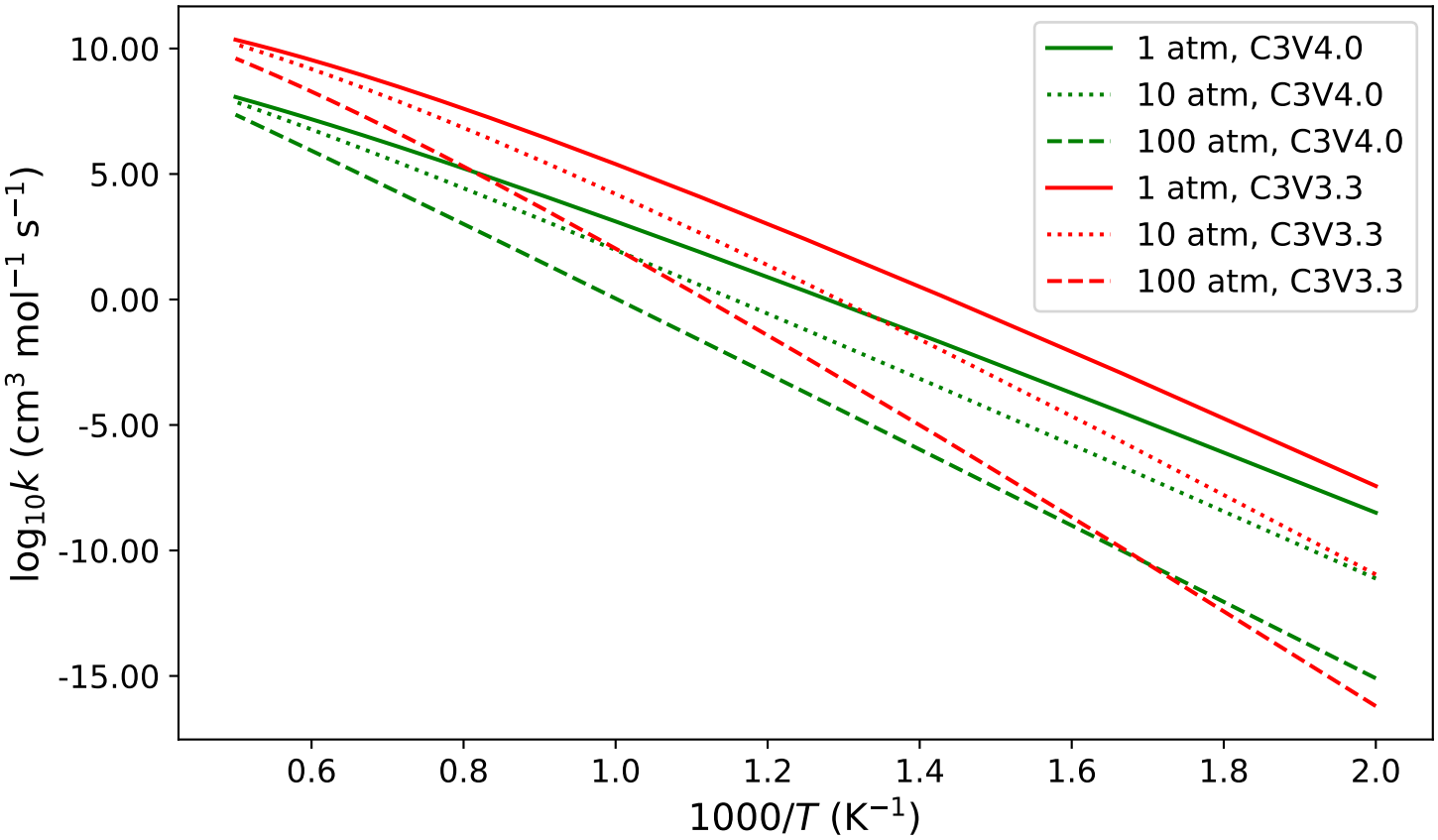


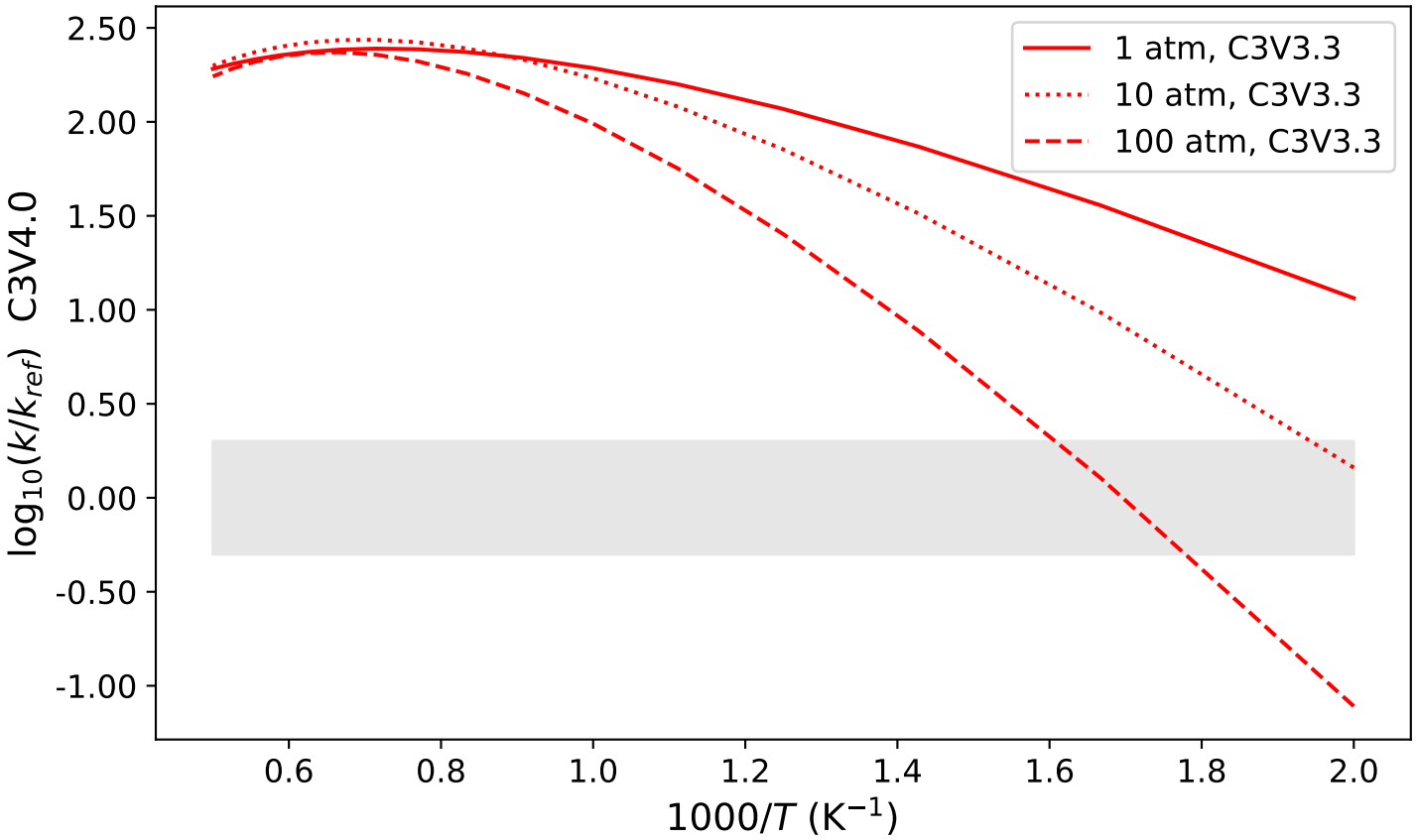
C3H3 + XYLENE => H + H2 + C10H7CH3-A



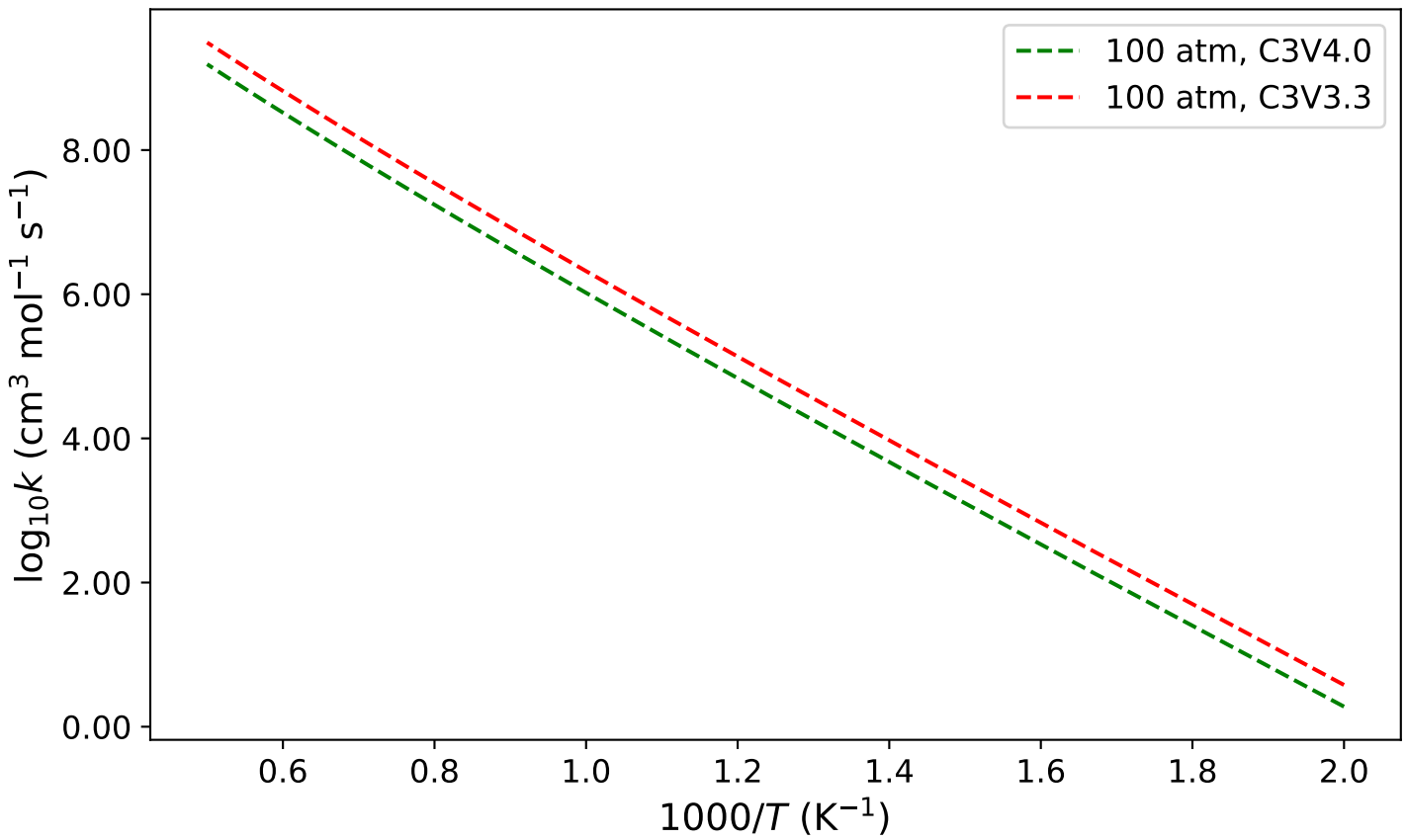


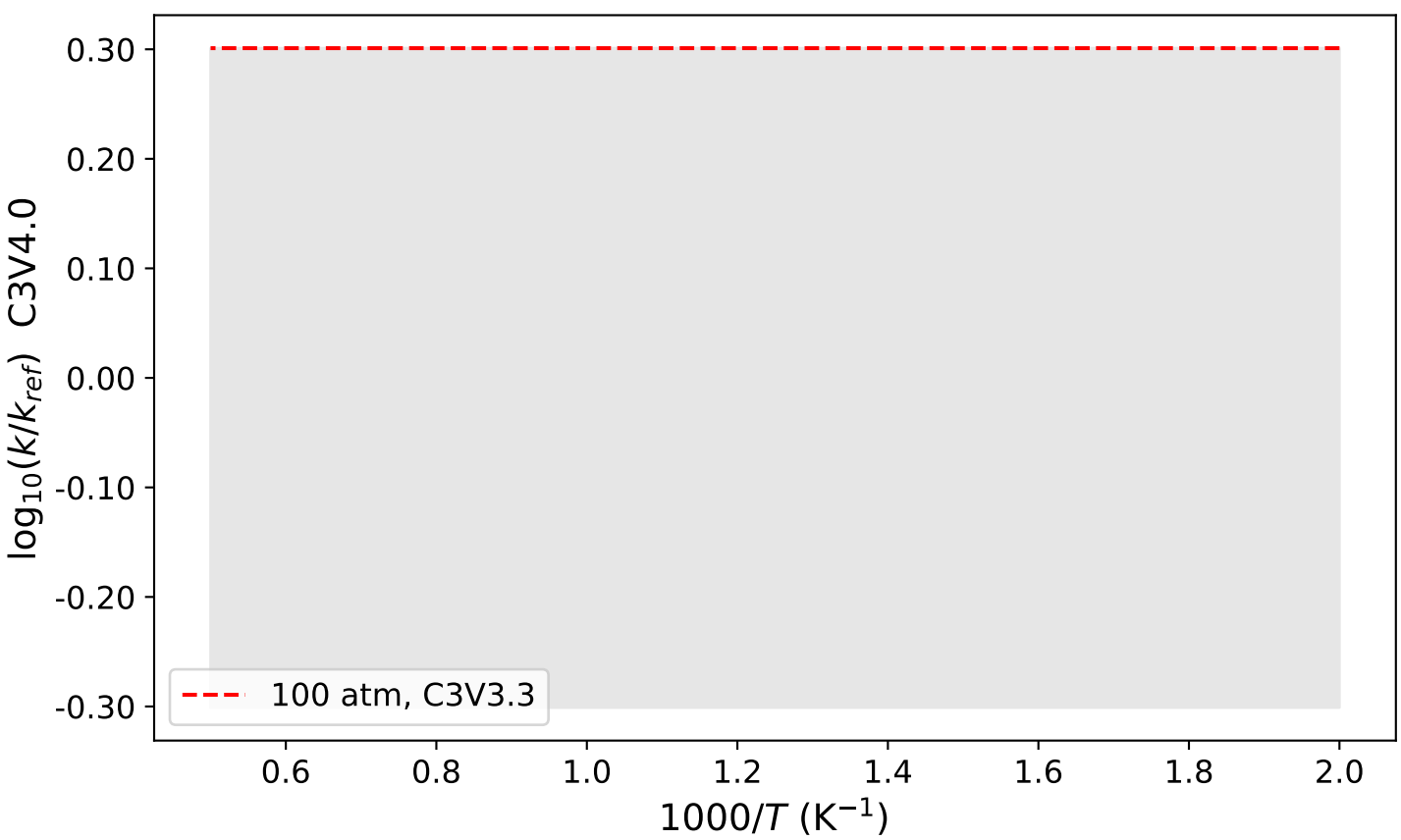
H + C5H6 = C2H3 + C3H4-A



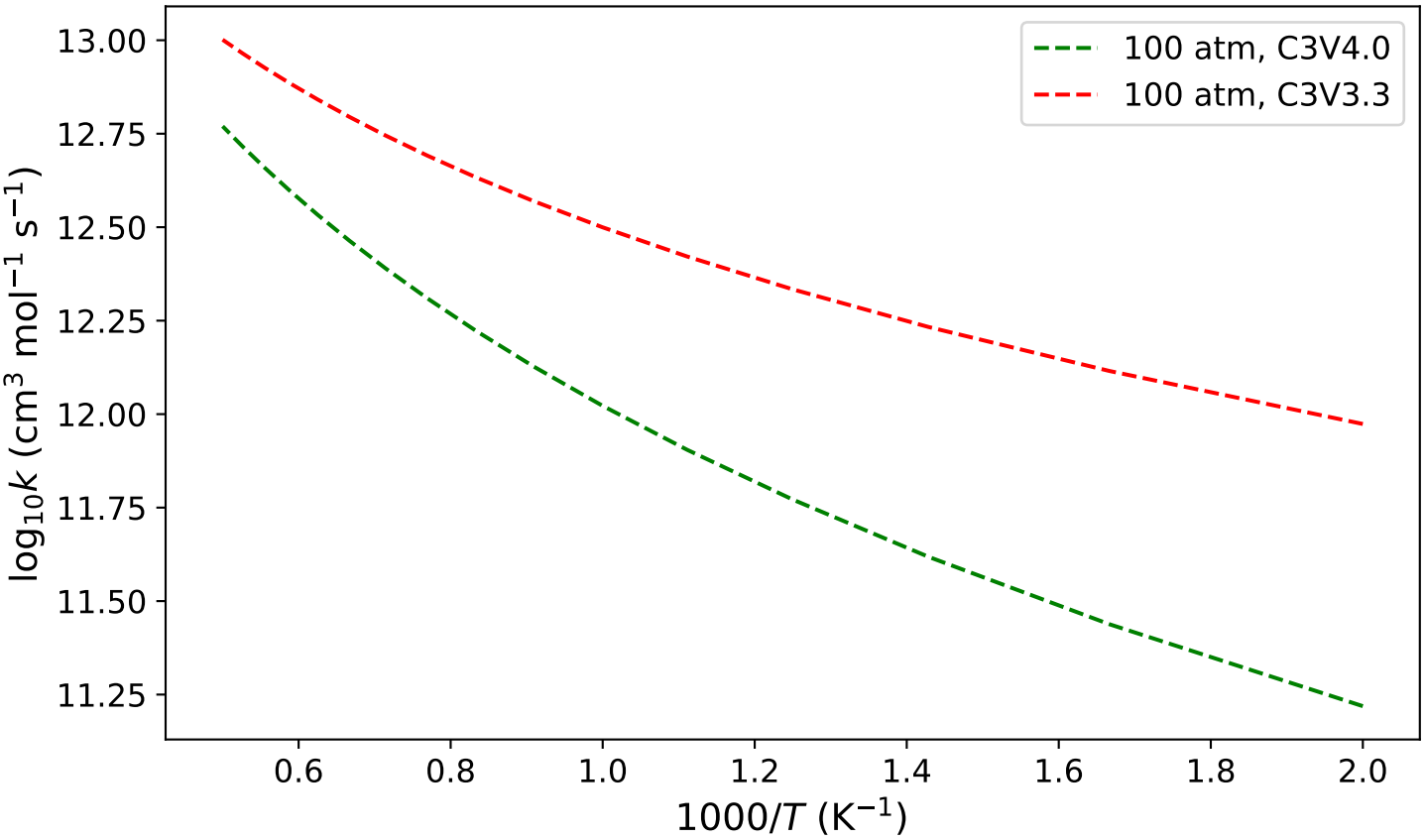


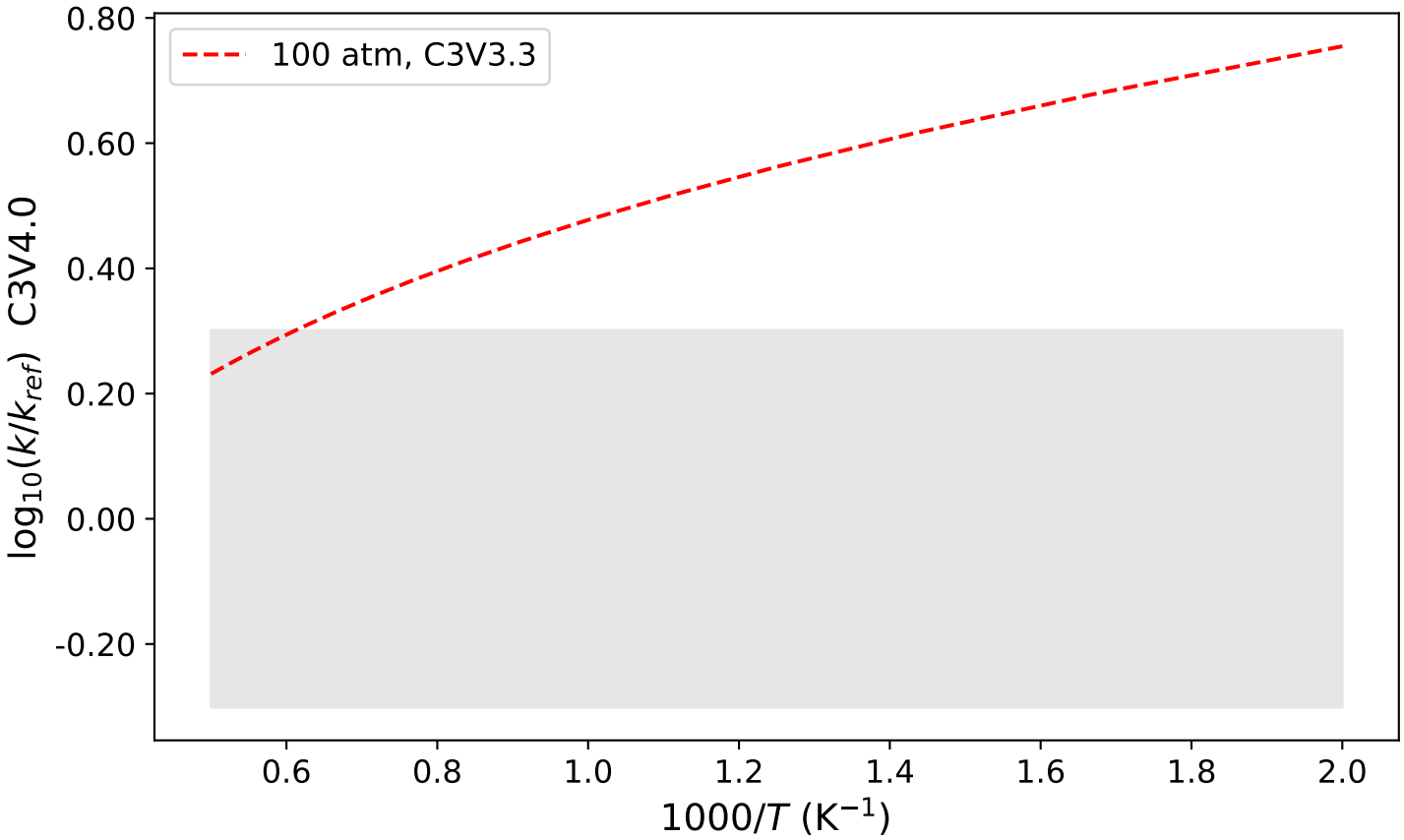
NAPH + C6H5CHCH3 = NAPH-A + C6H5C2H5



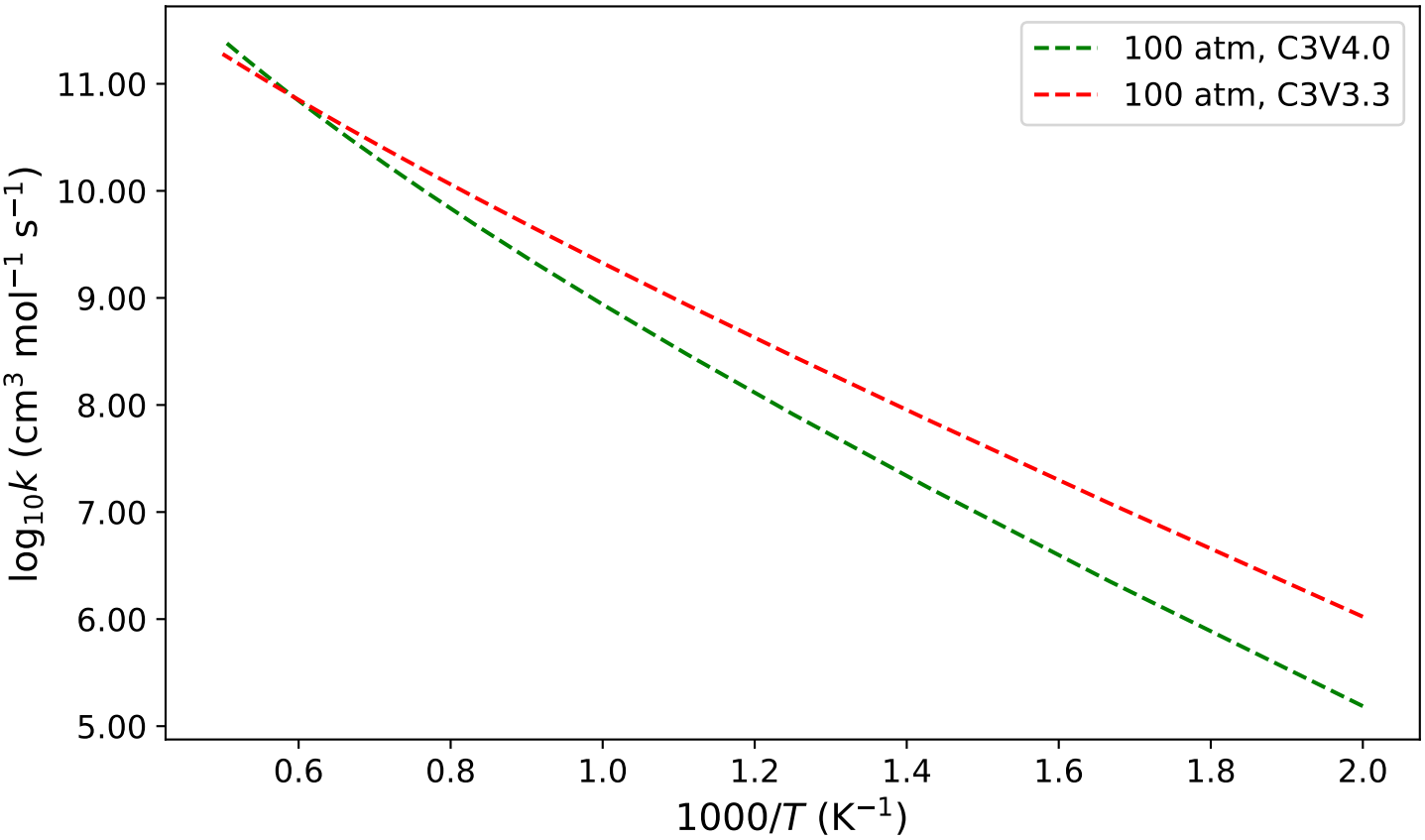


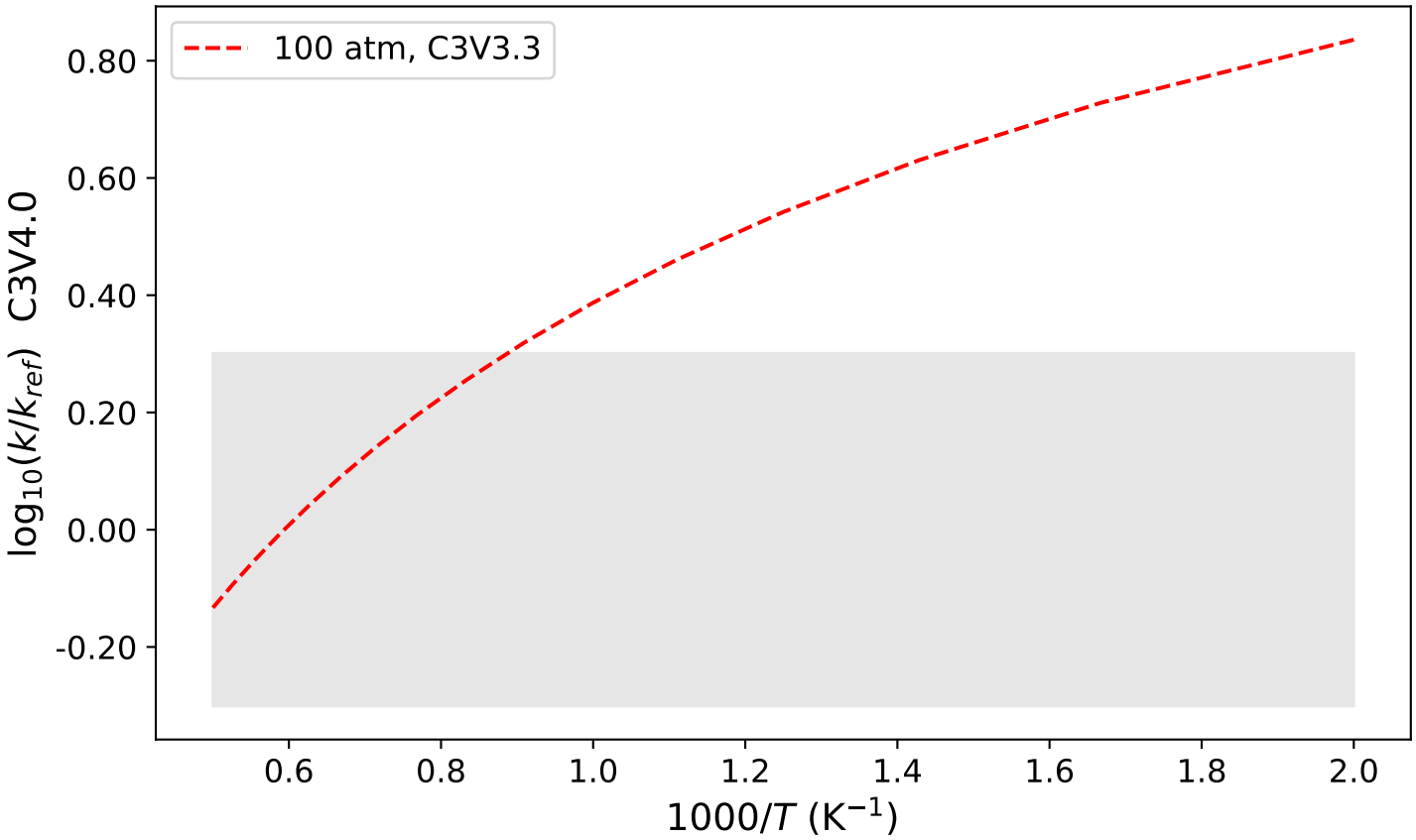
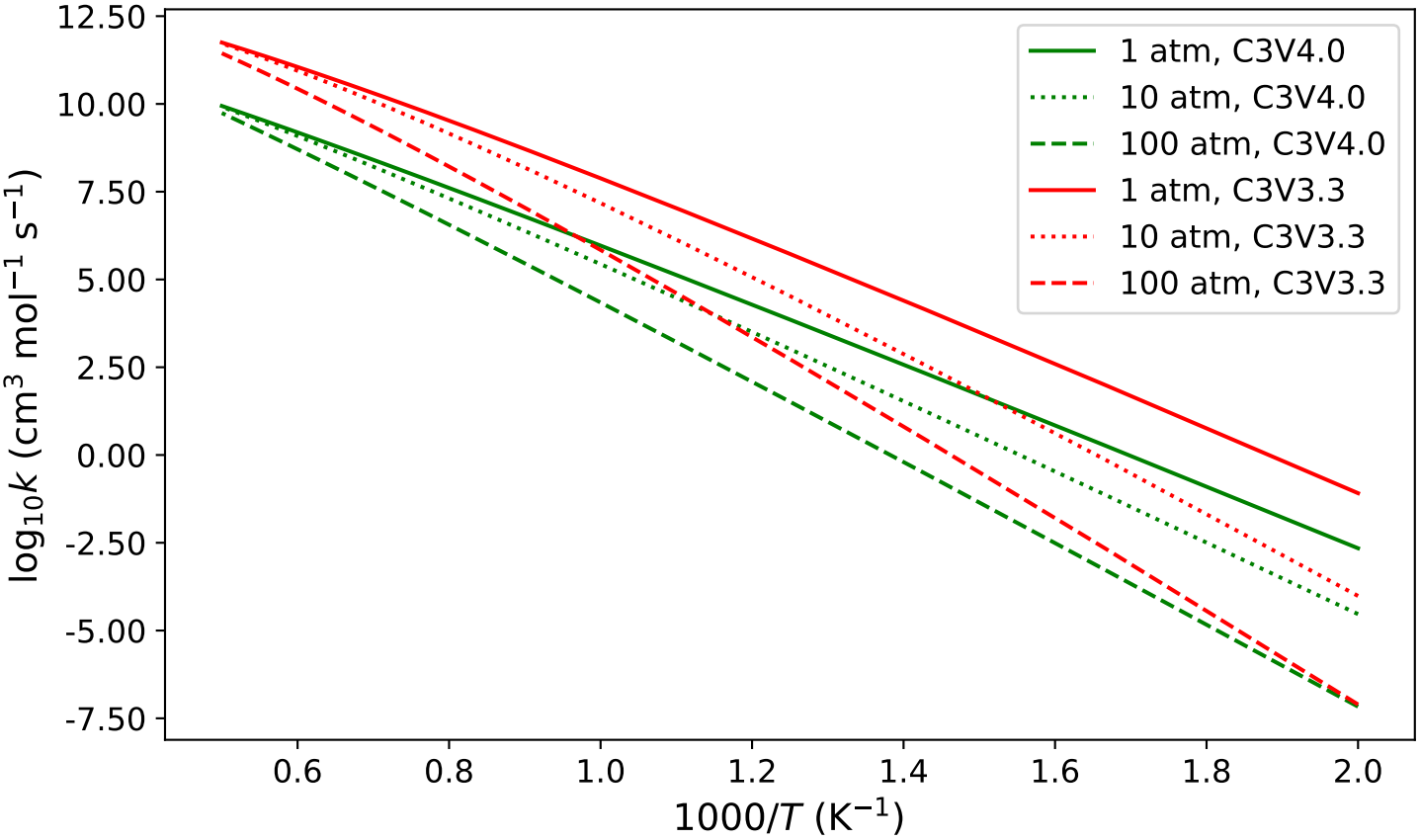
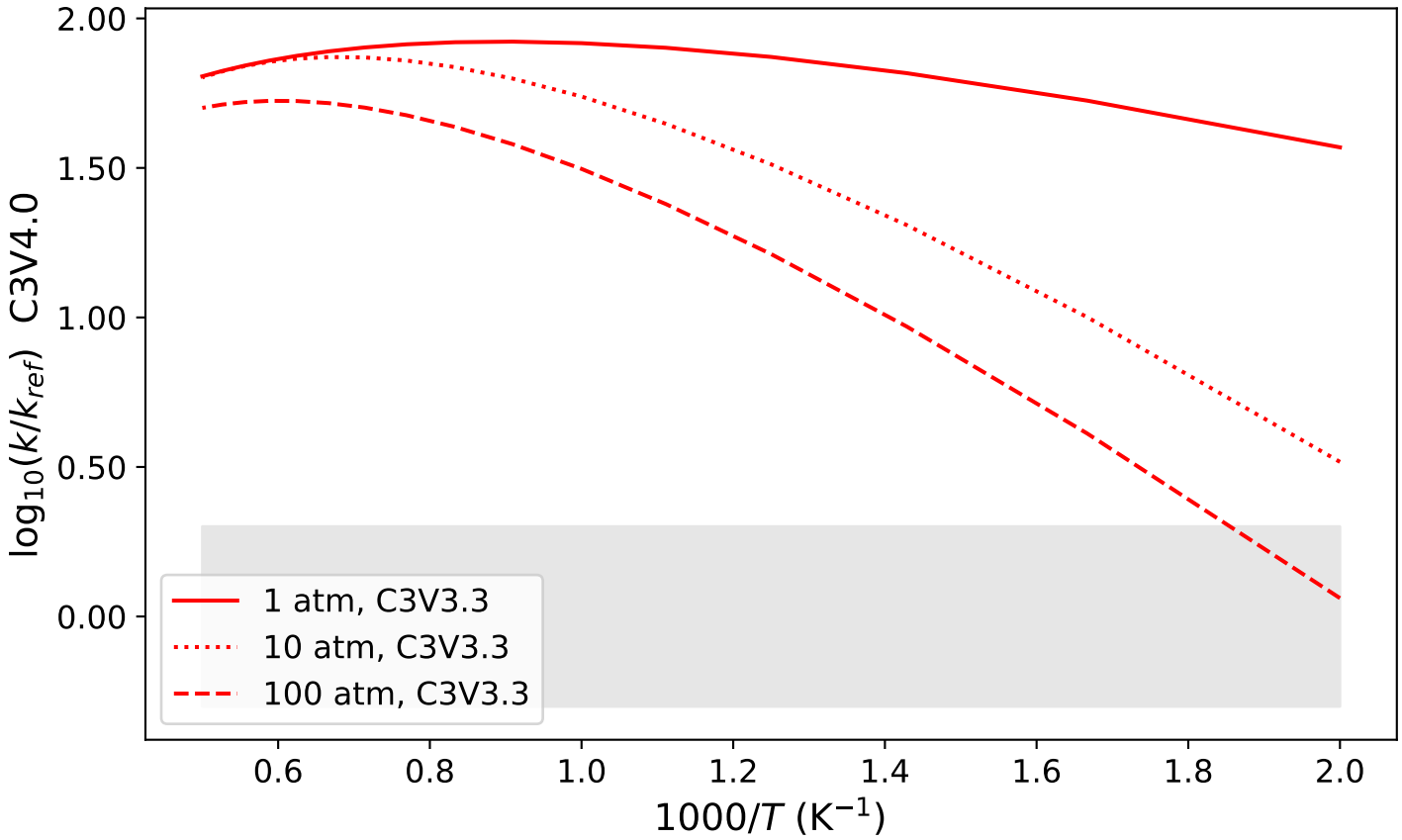
OH + C10H7CH3-A = H2O + C10H7CH2-A





CH3O2 + CYC5H8 = CYC5H71-3 + CH3O2H





H + C5H6 = C3H3 + C2H4